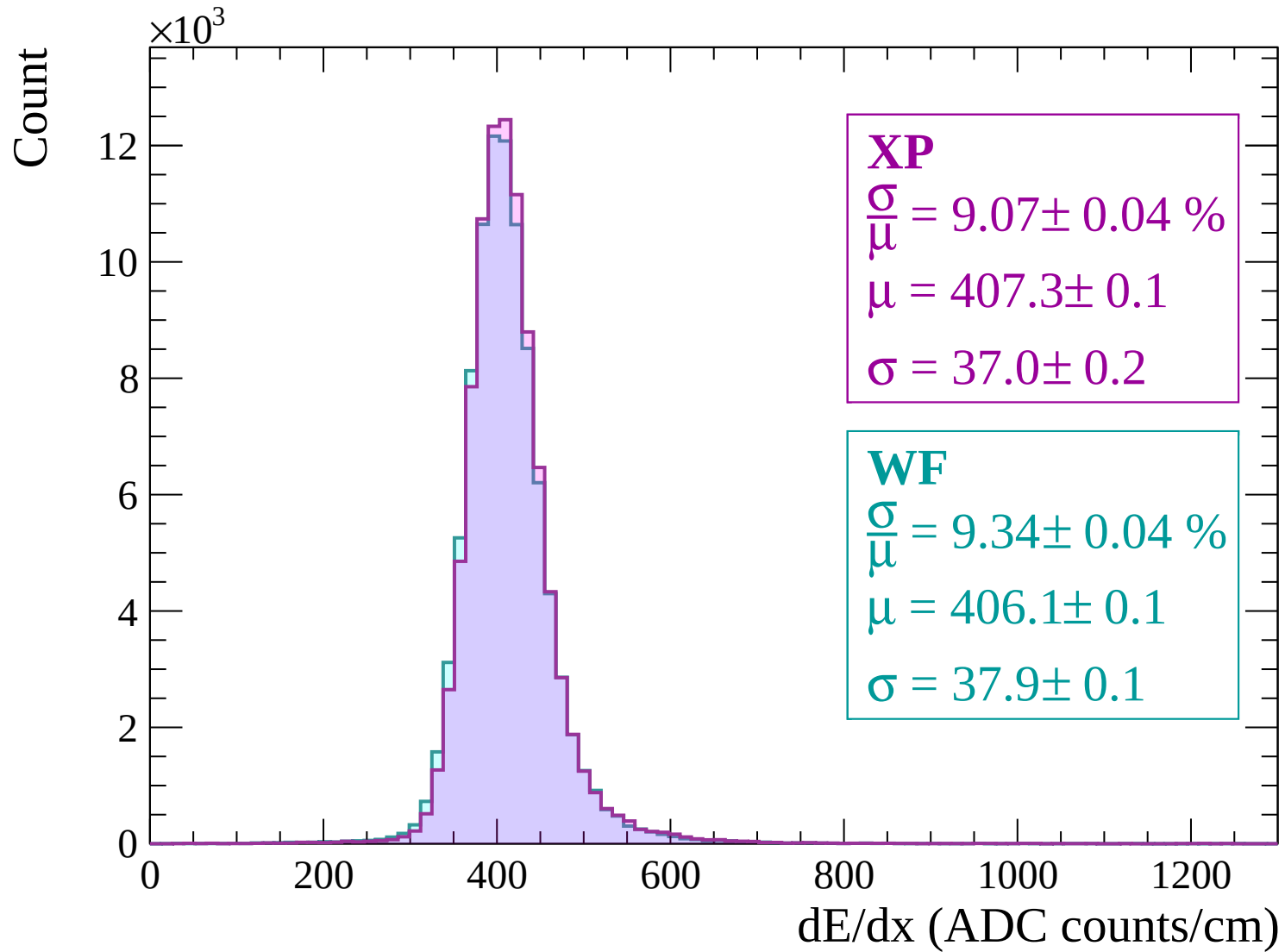
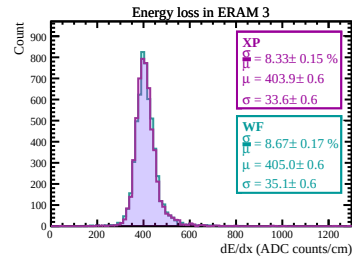
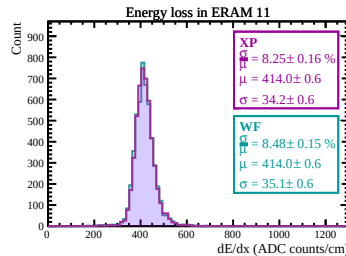
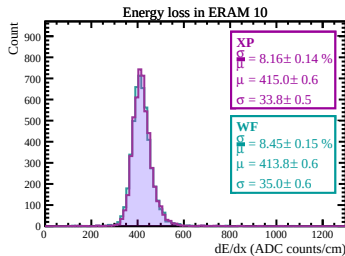
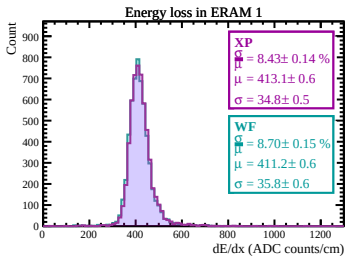
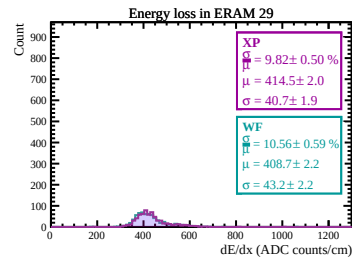
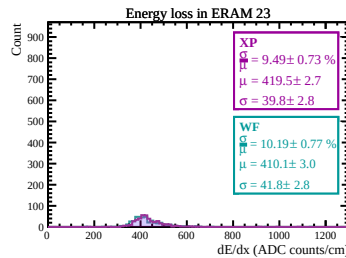
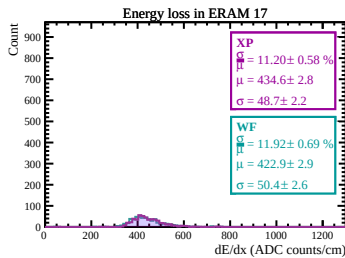
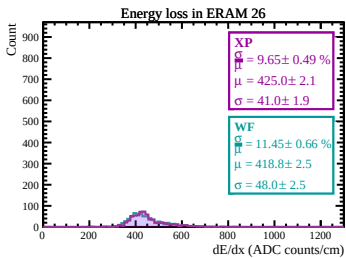
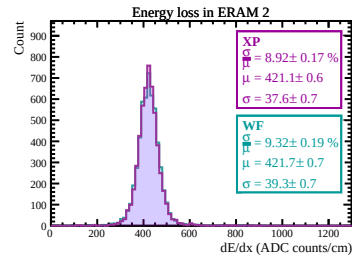
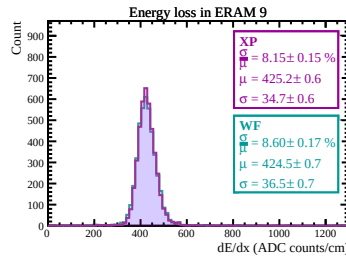
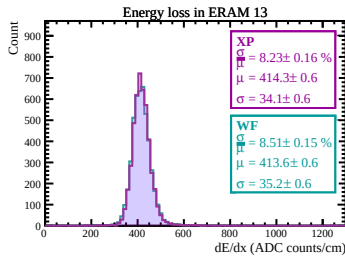
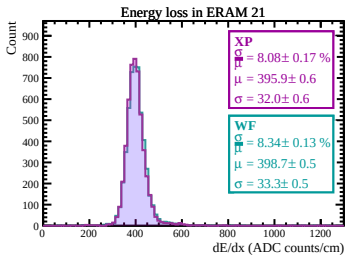
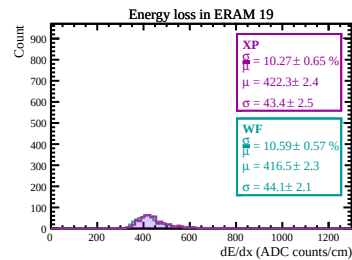
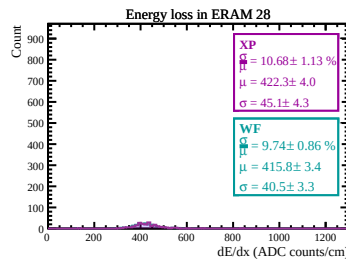
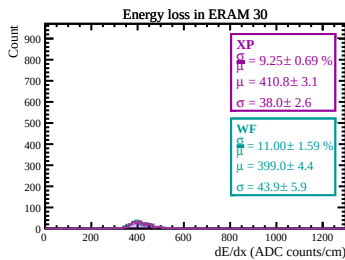
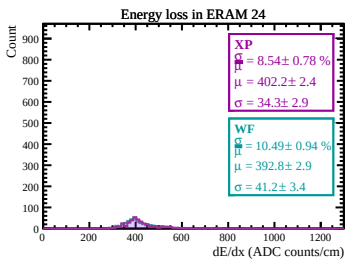
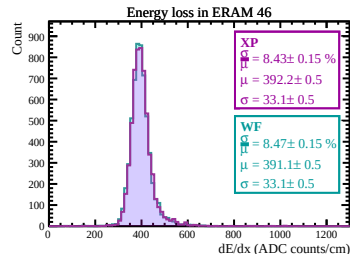
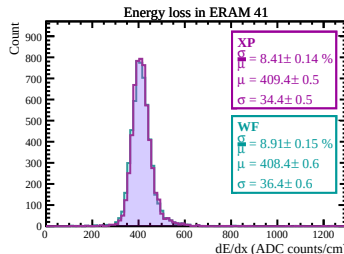
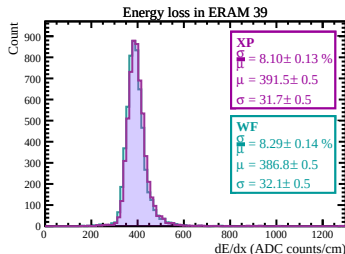
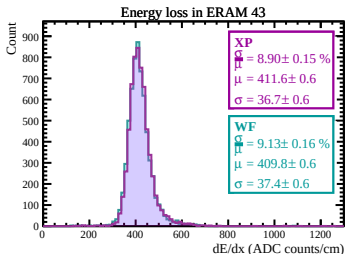
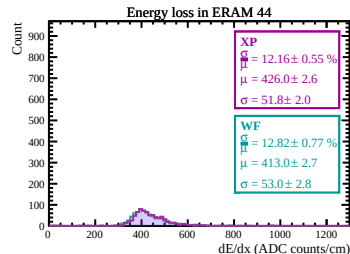
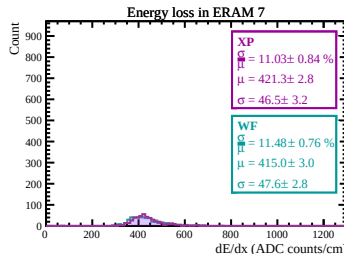
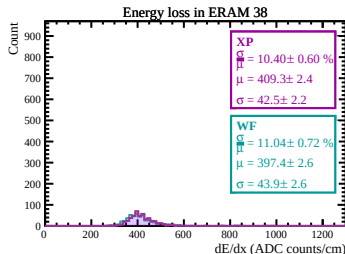
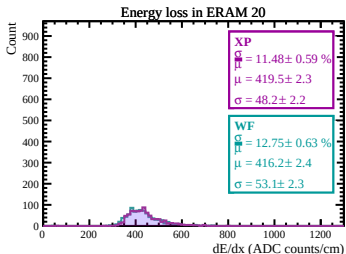
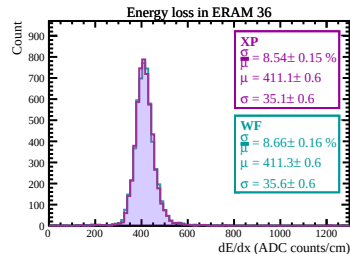
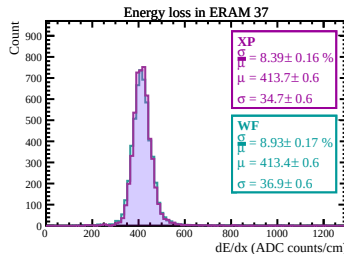
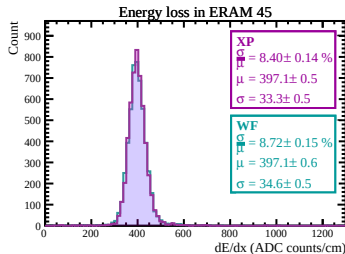
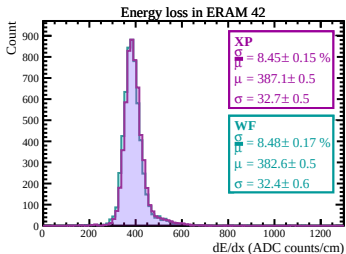
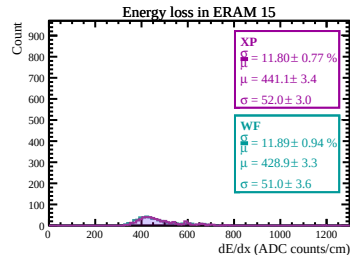
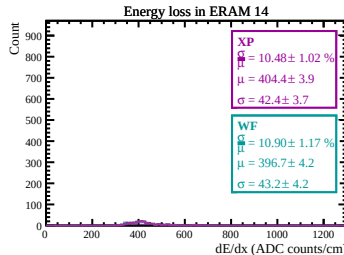
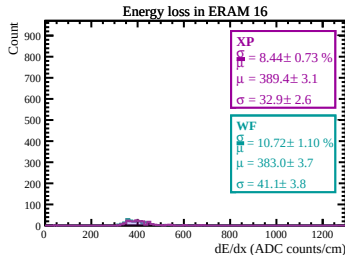
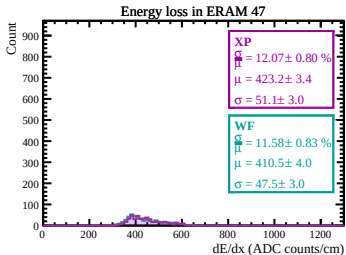
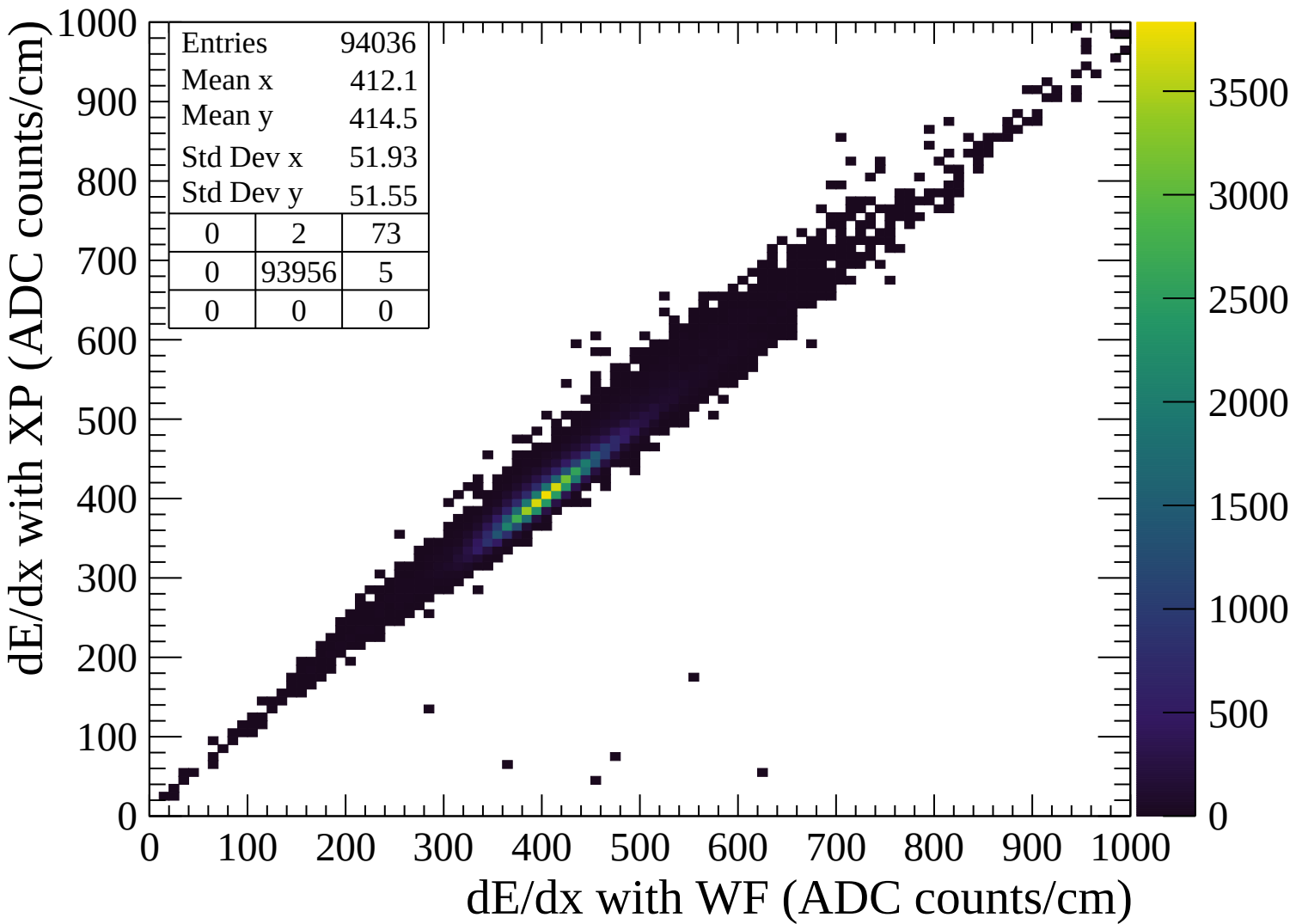
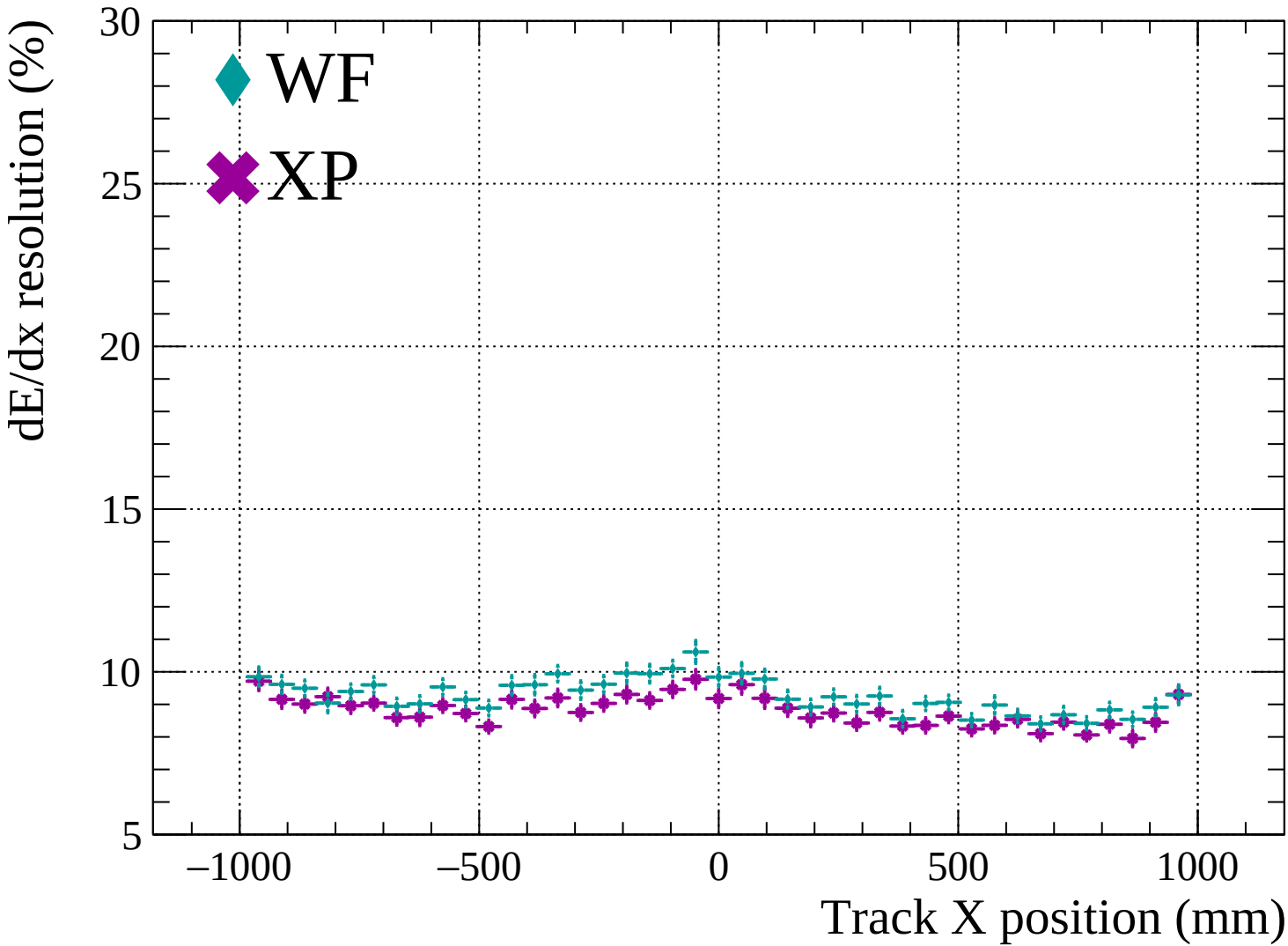


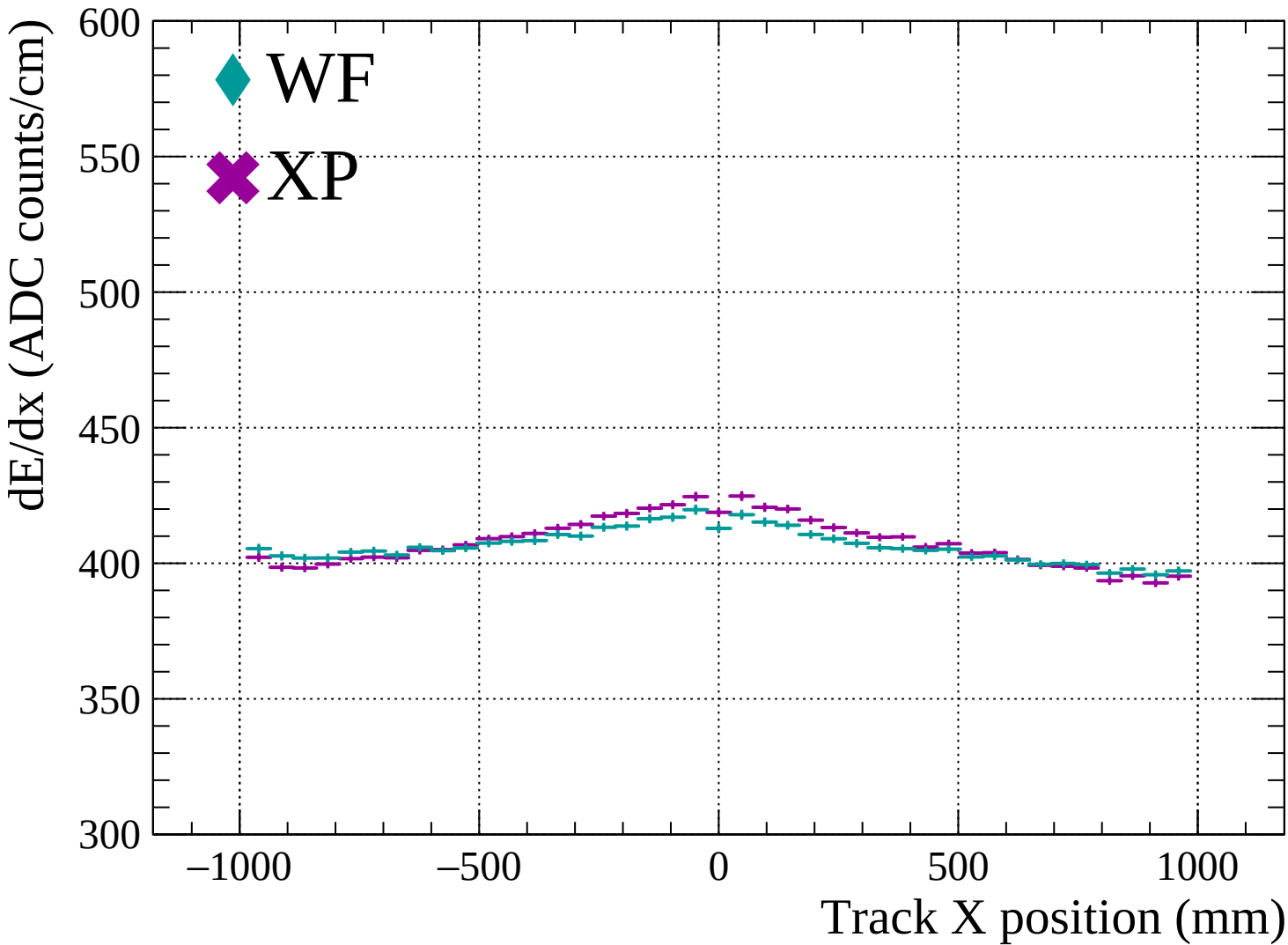


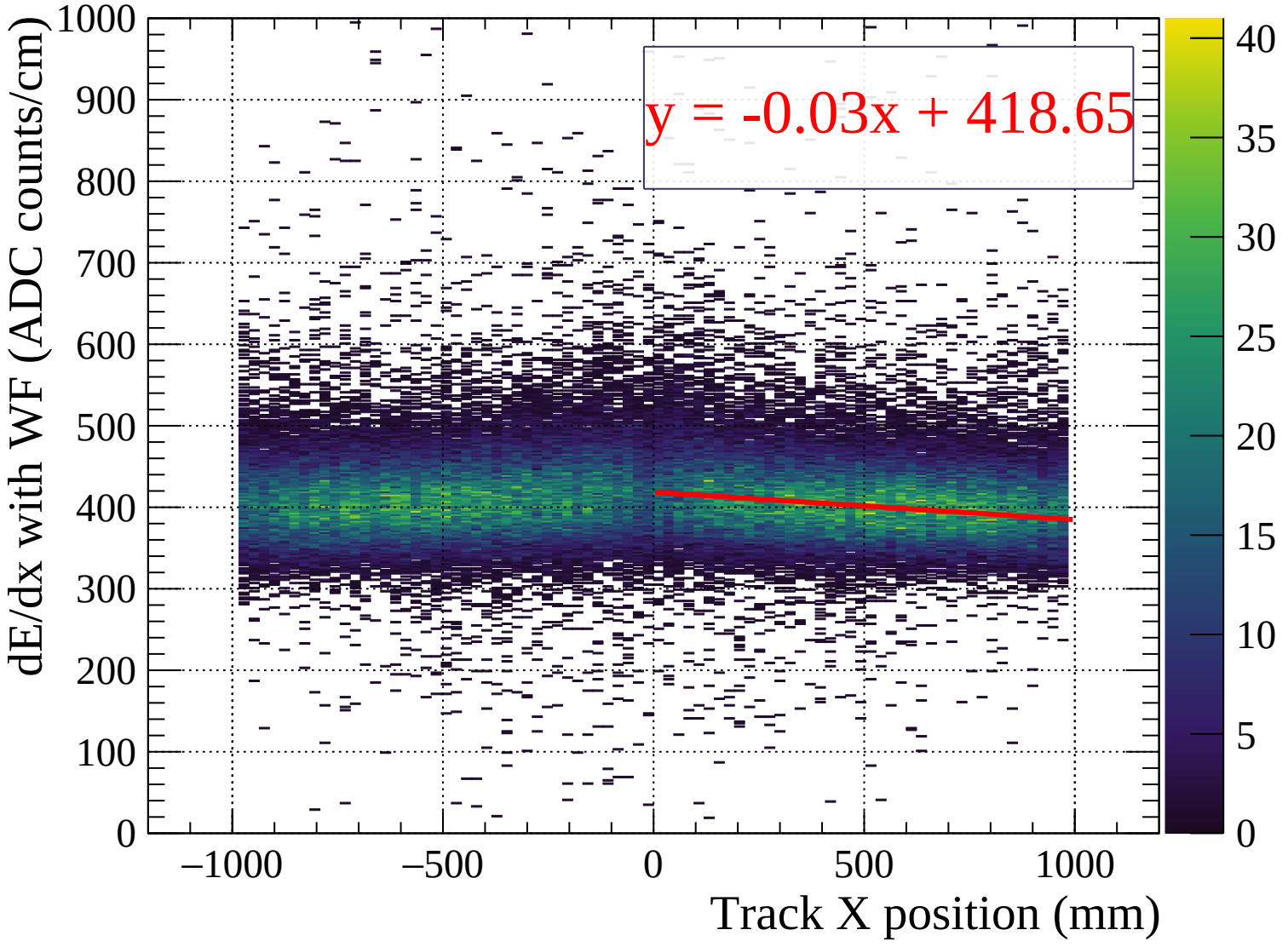


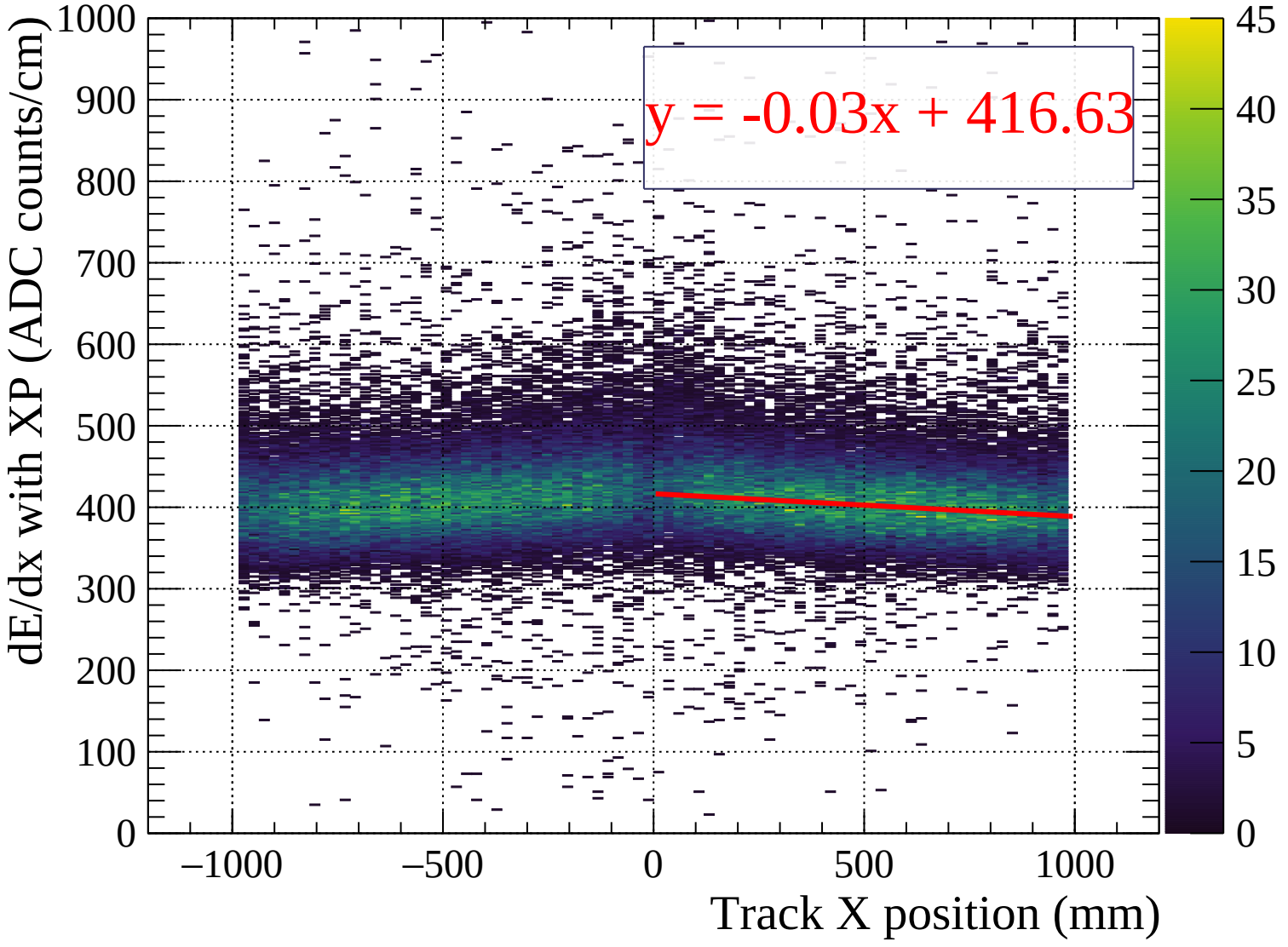


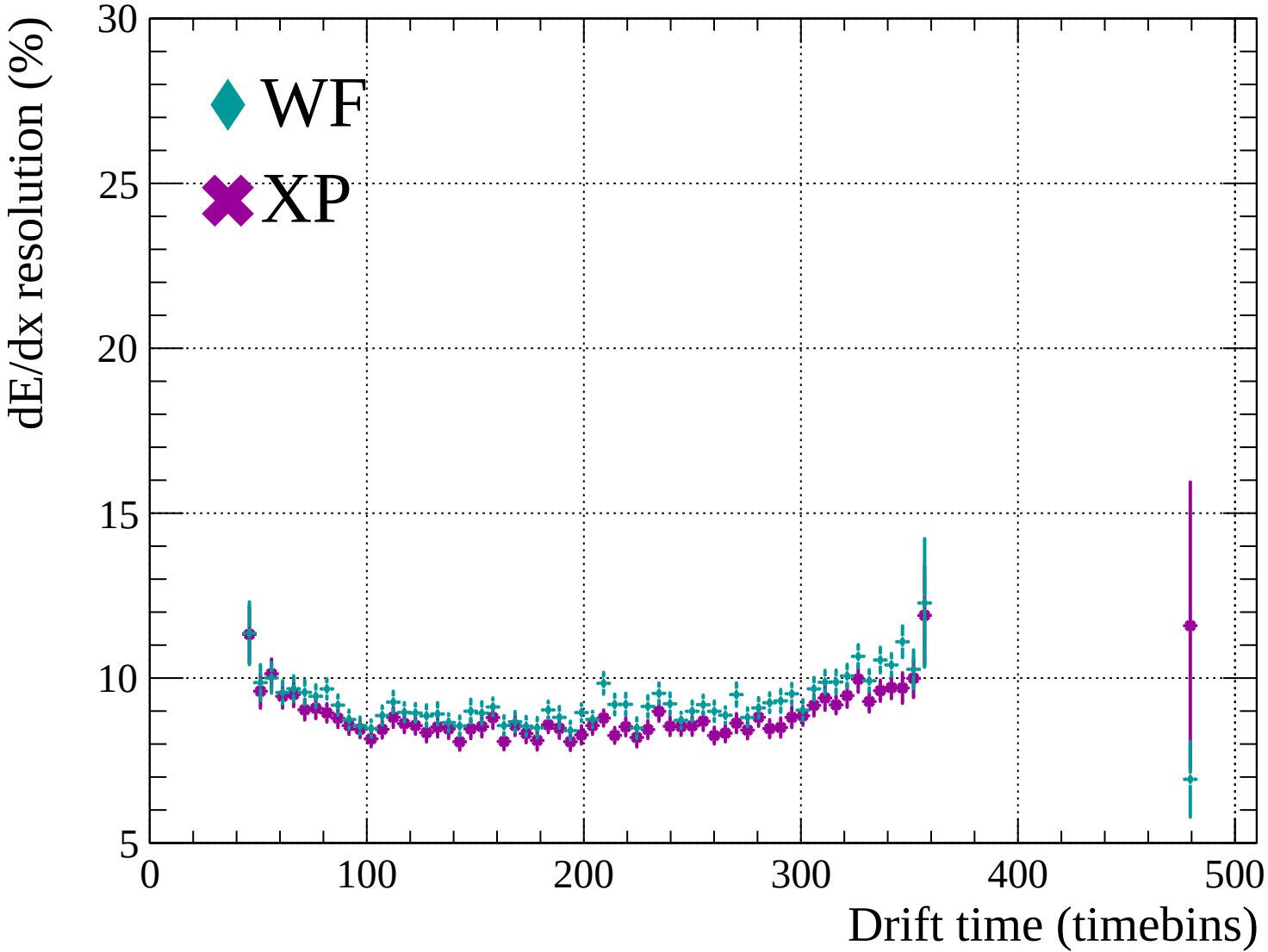


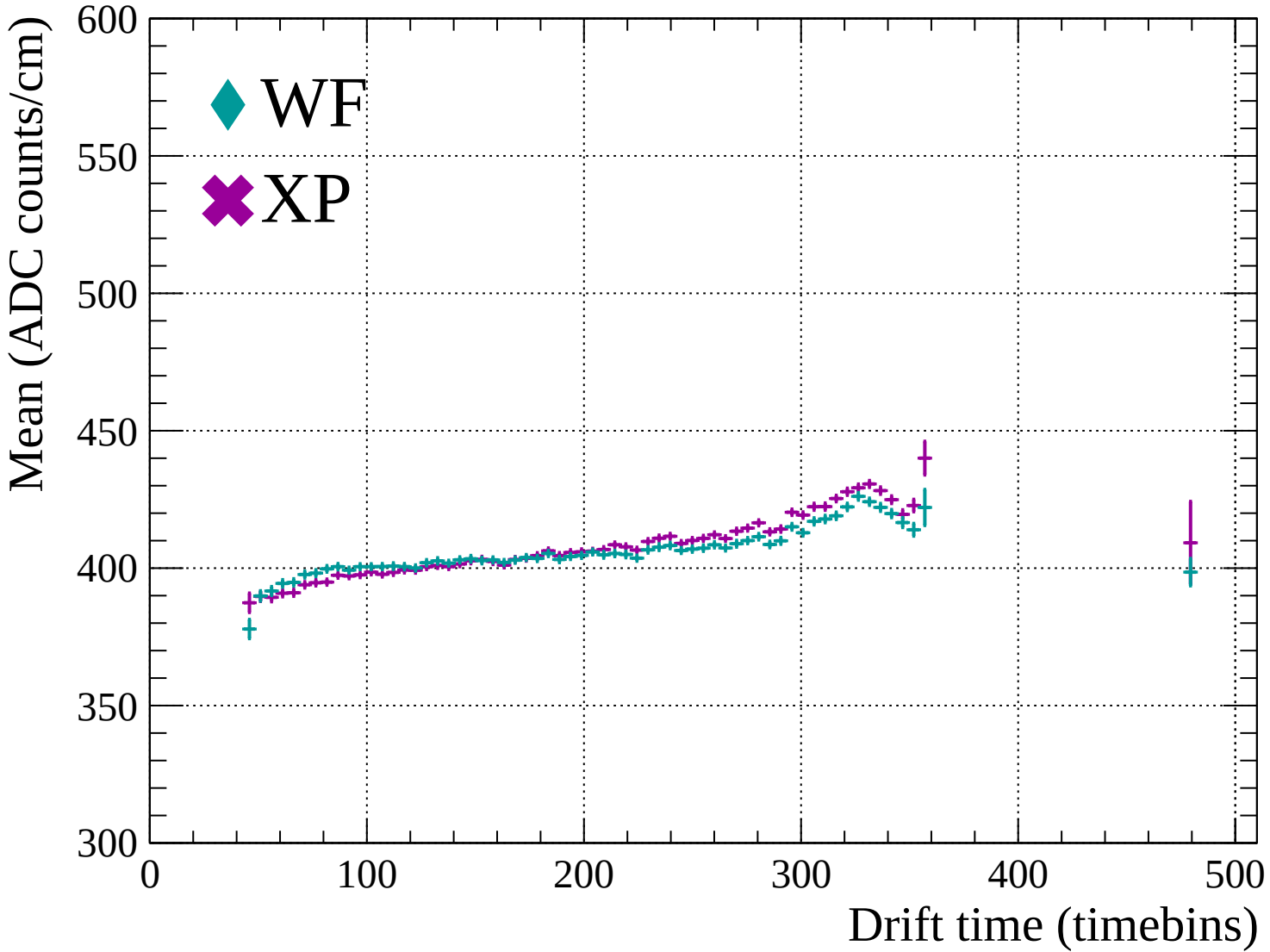


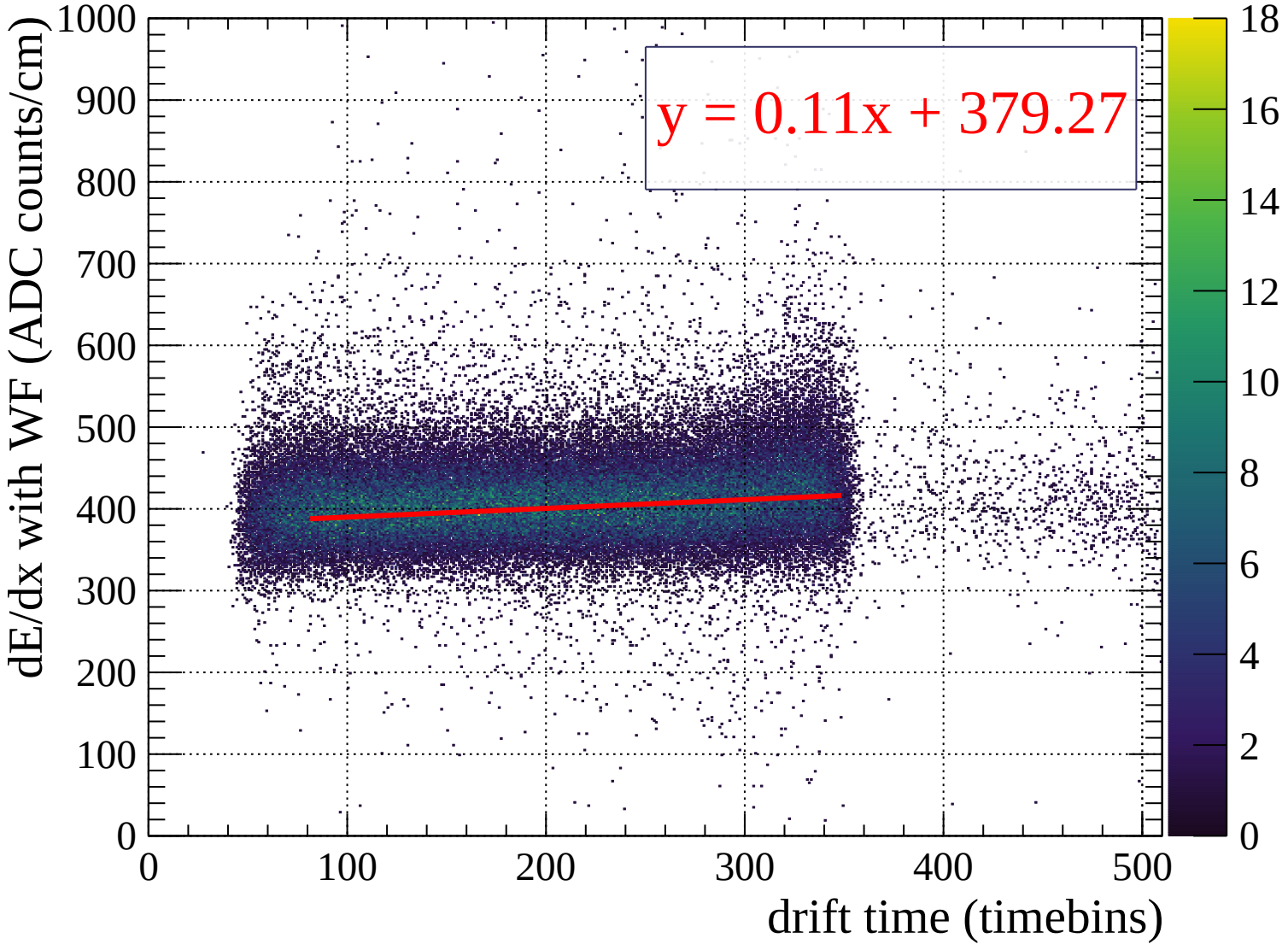


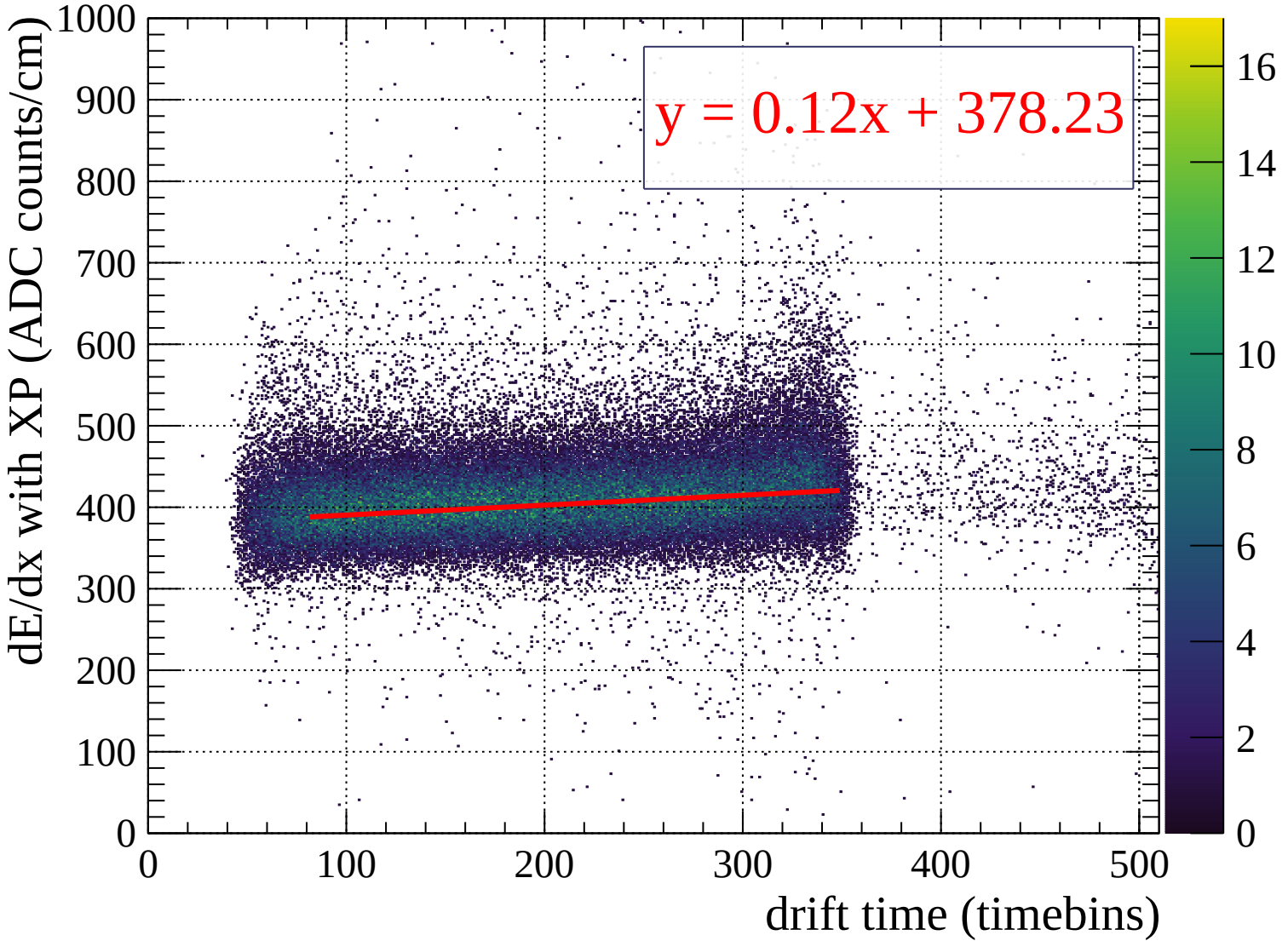


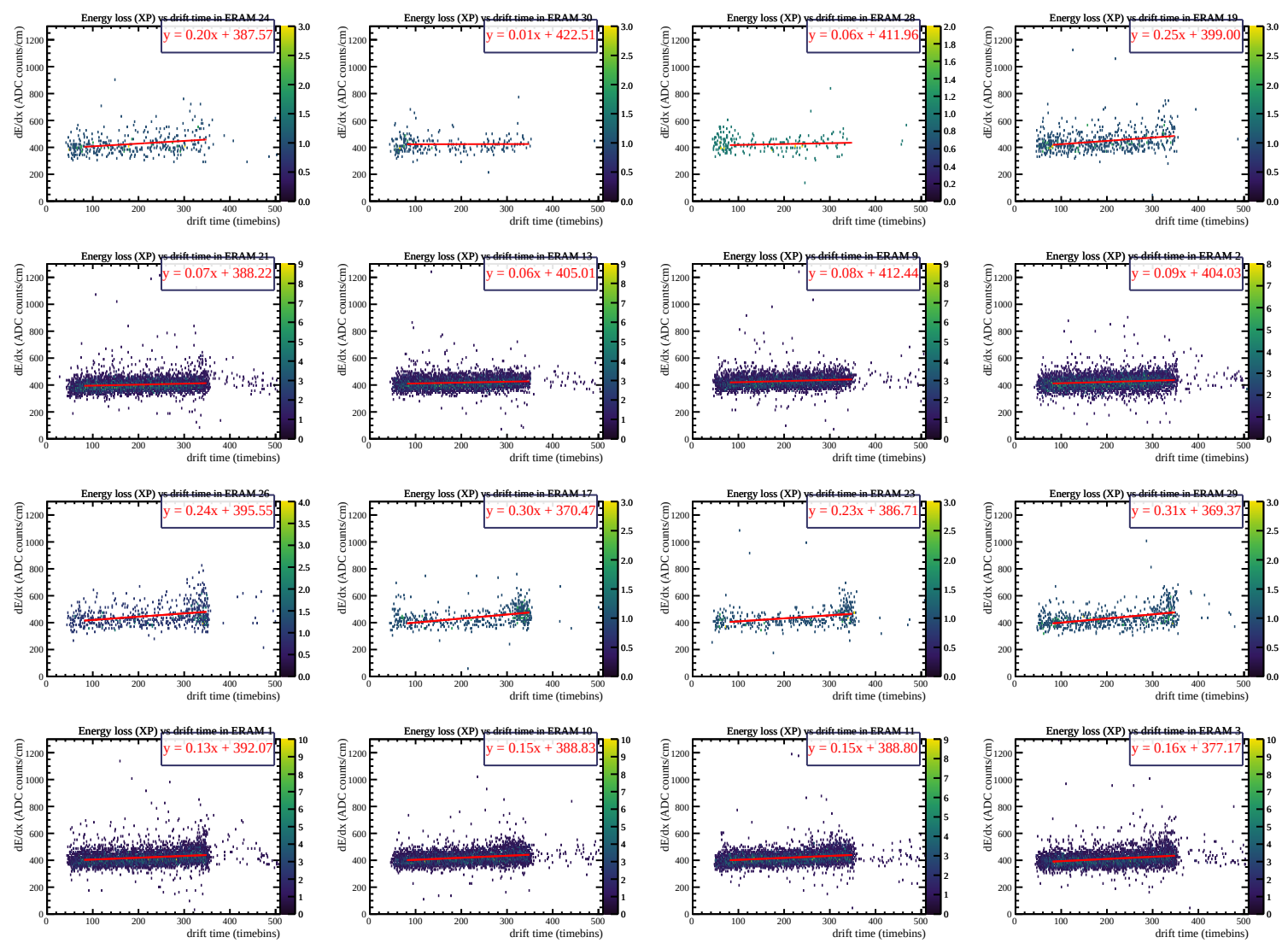


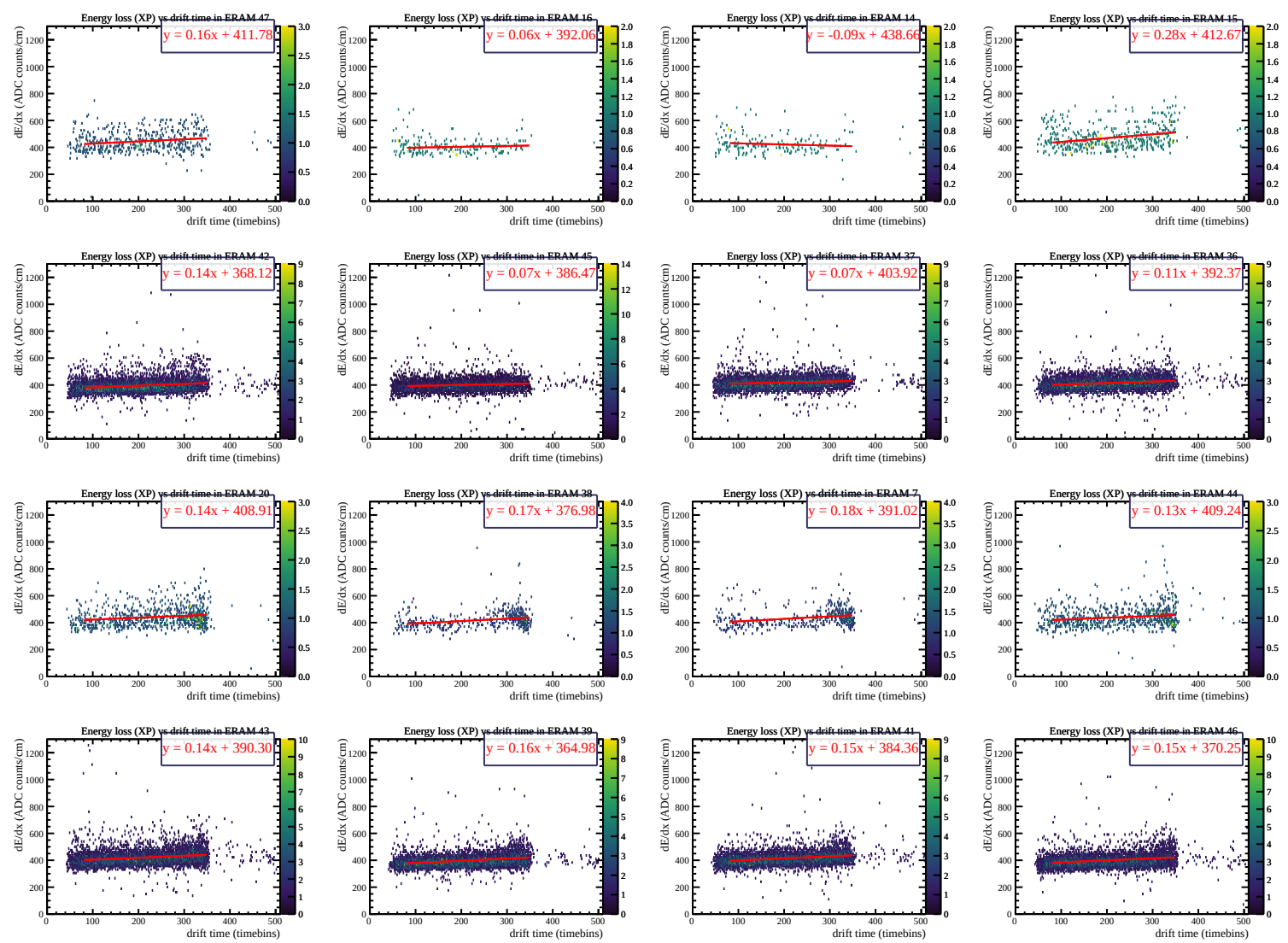


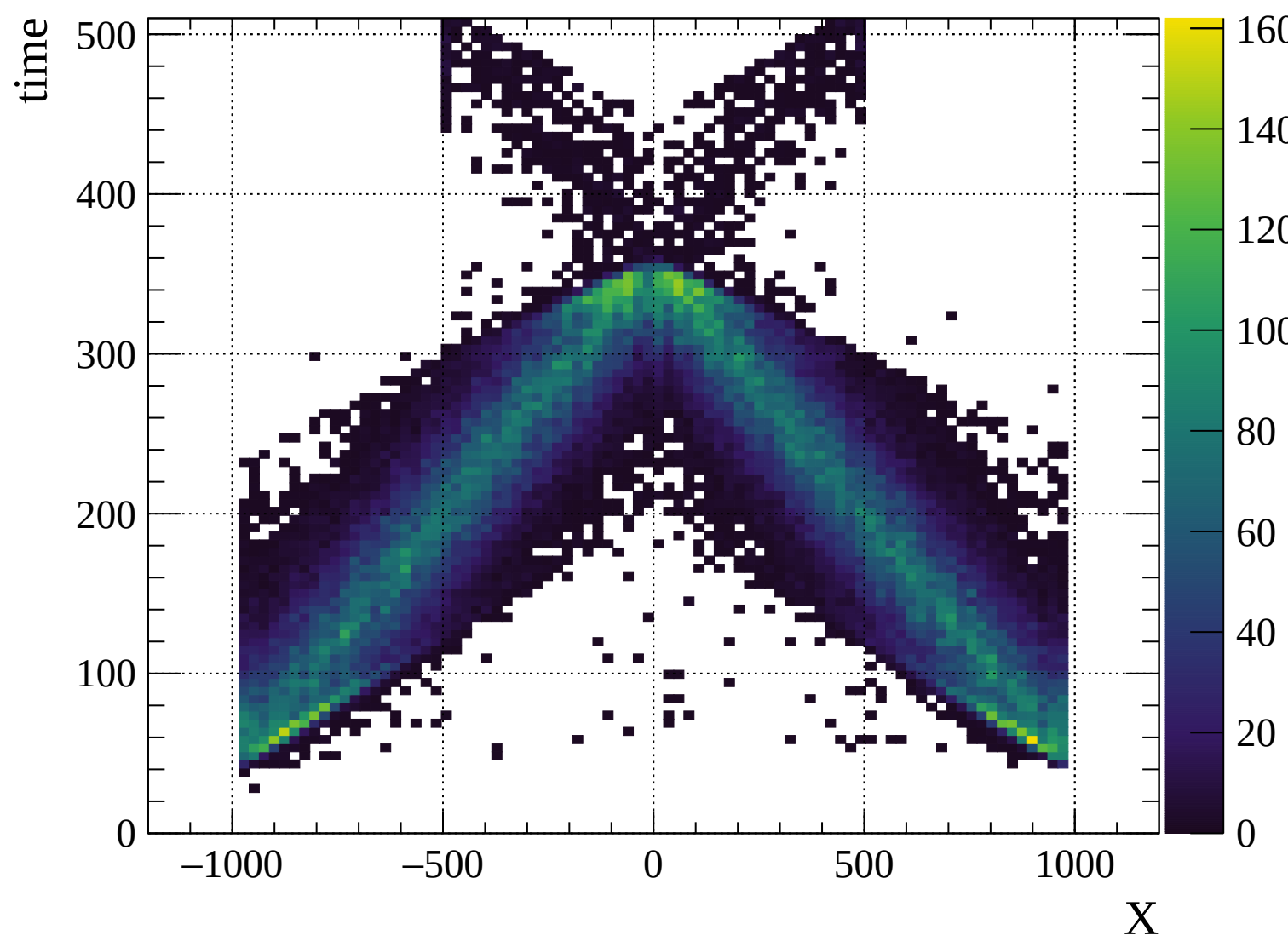


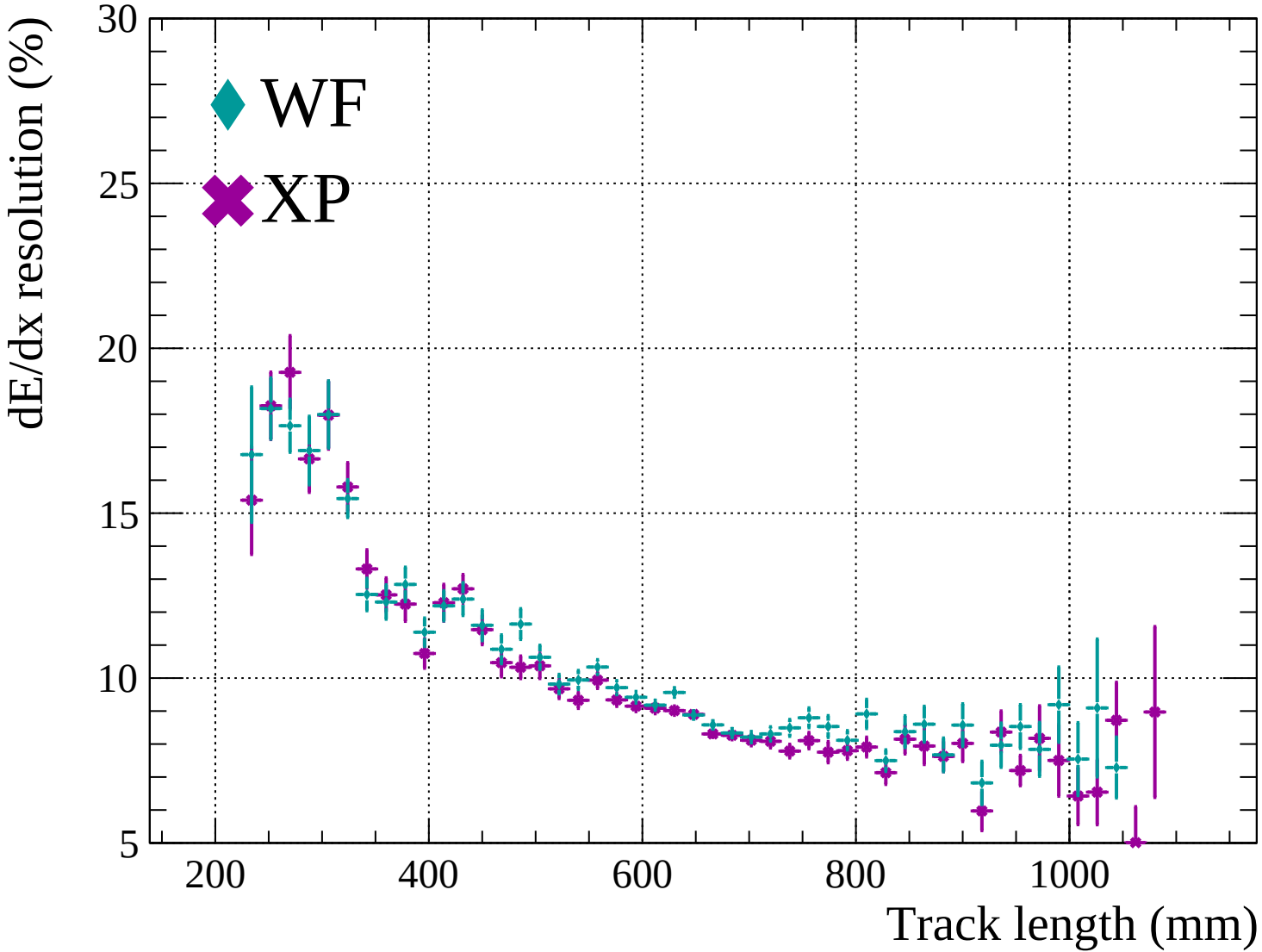


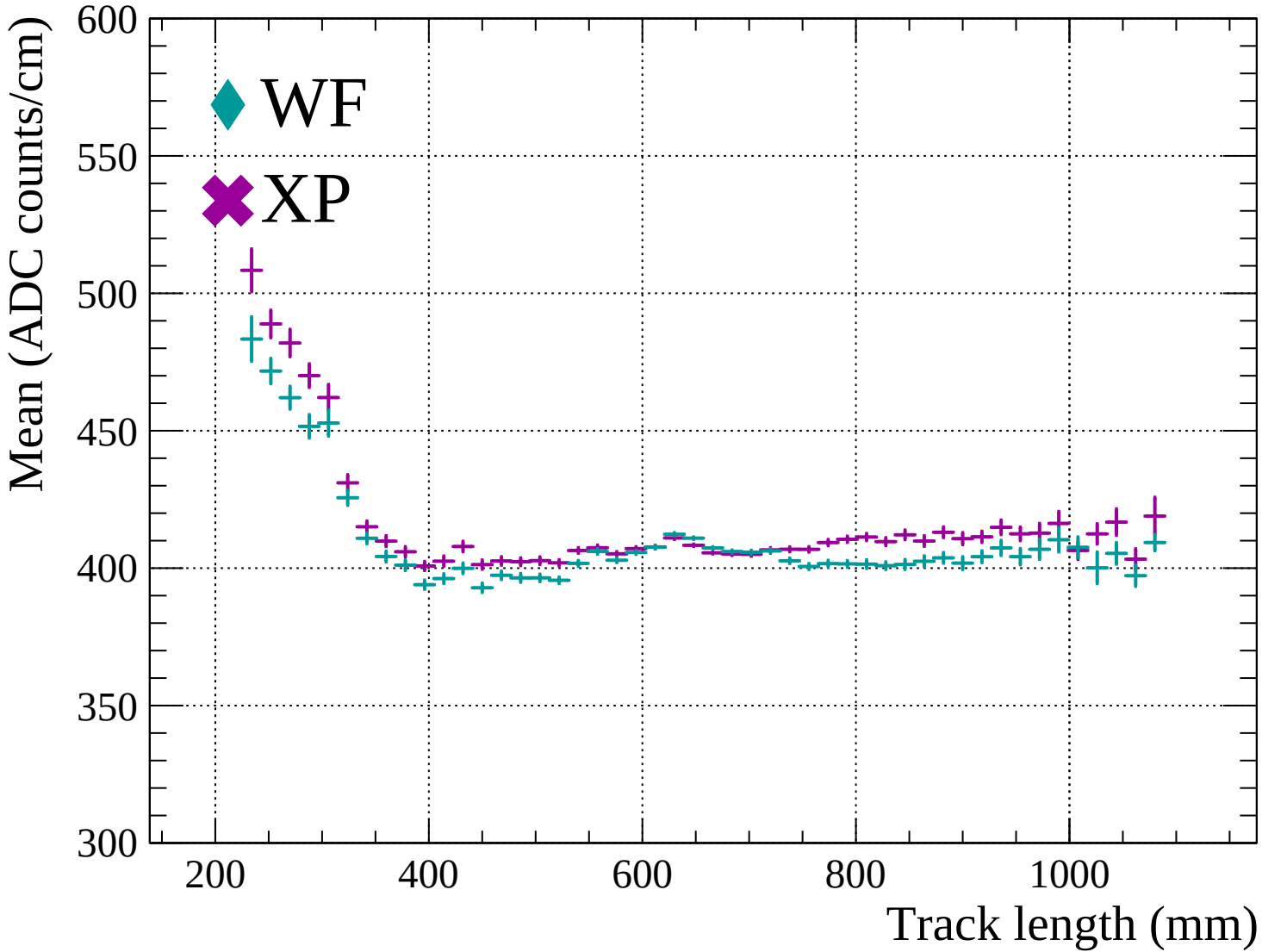


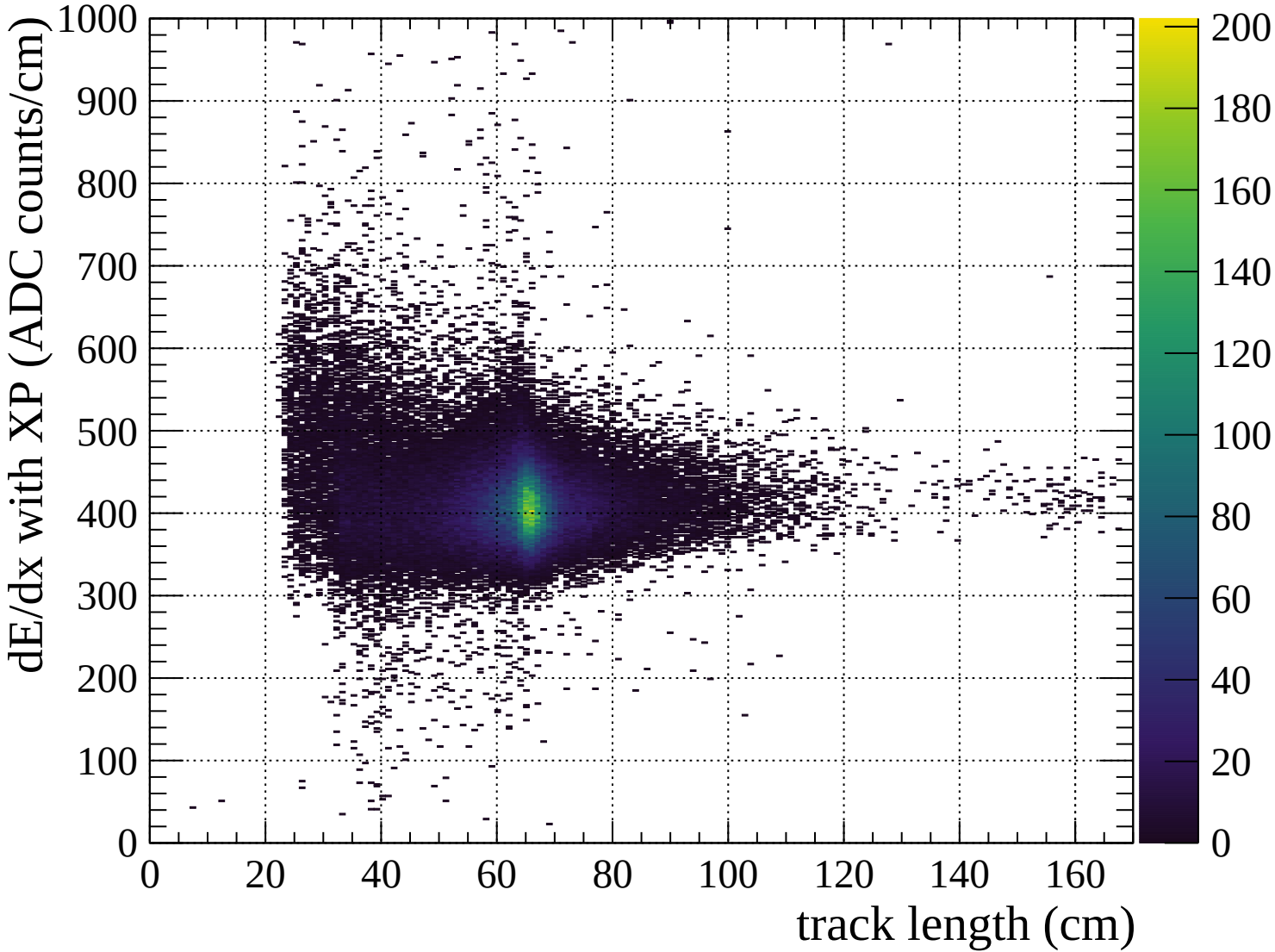


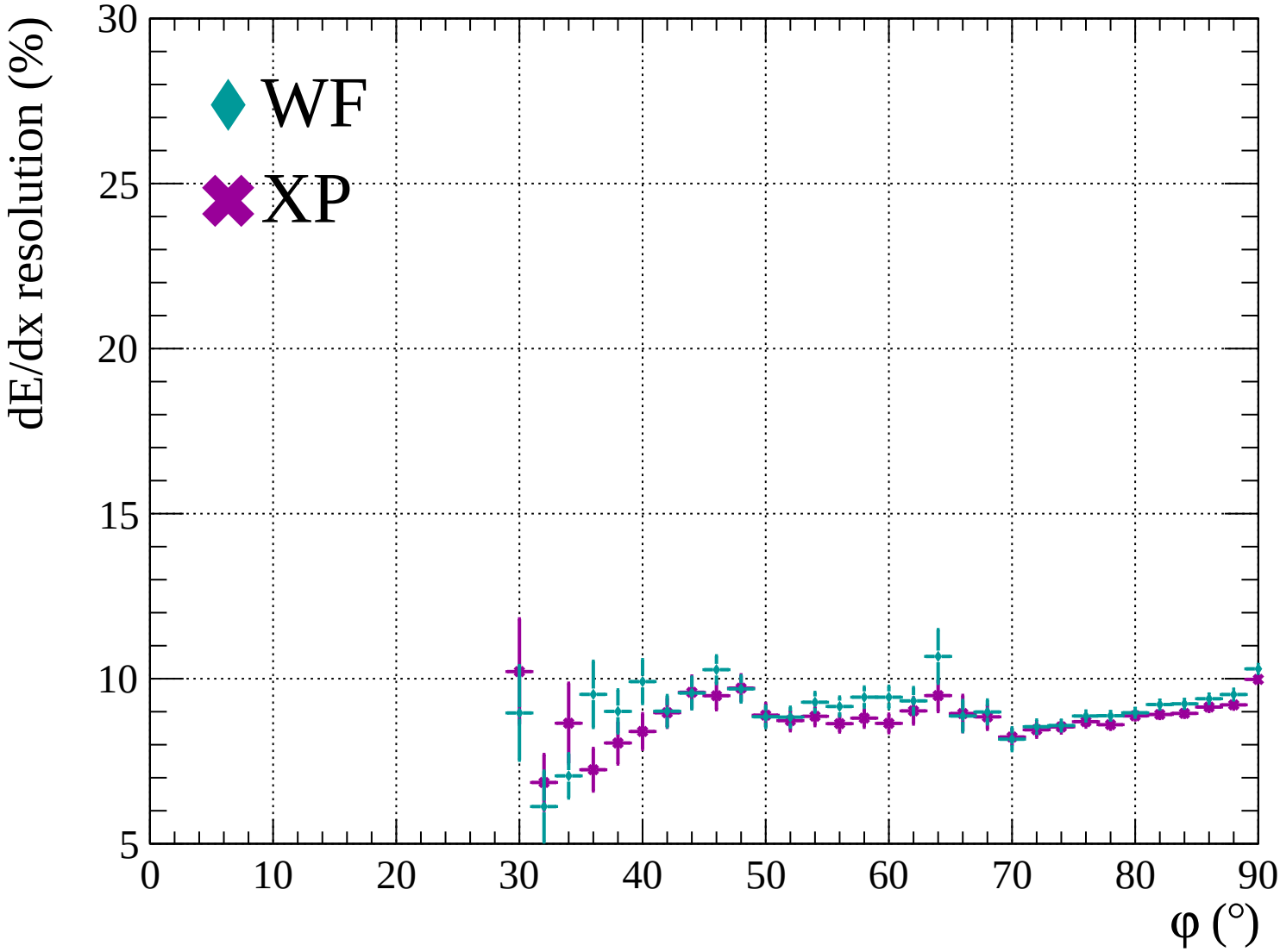


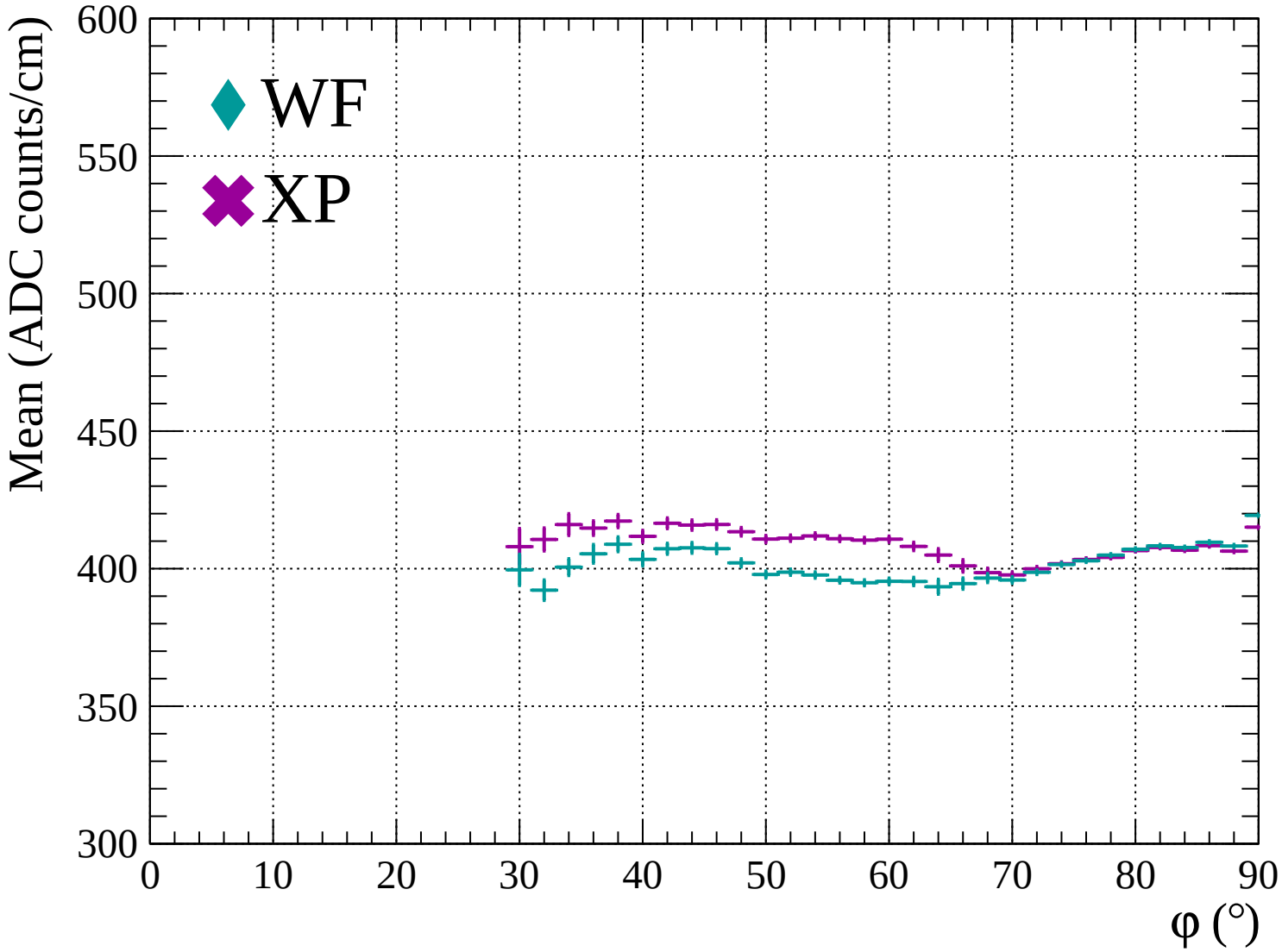


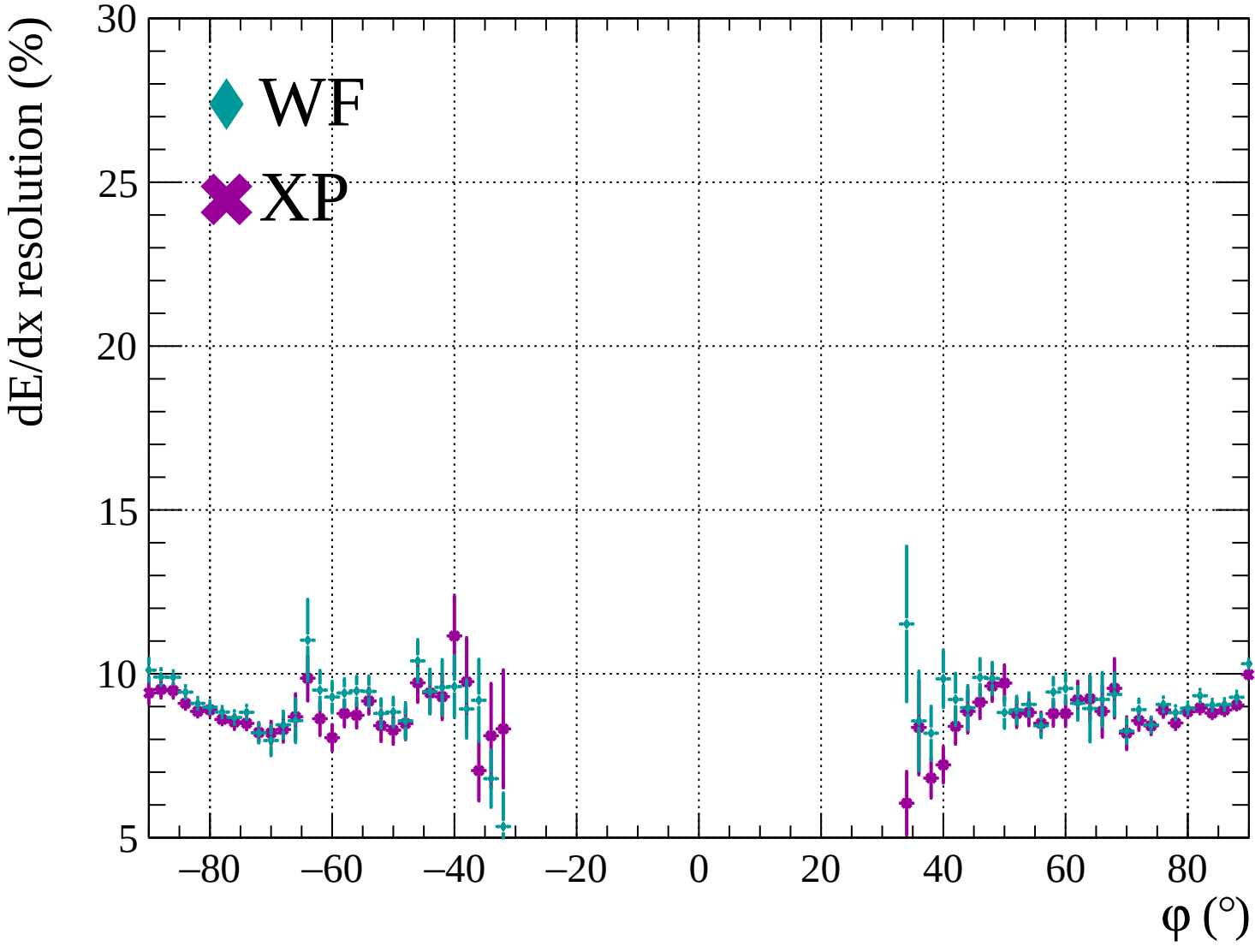


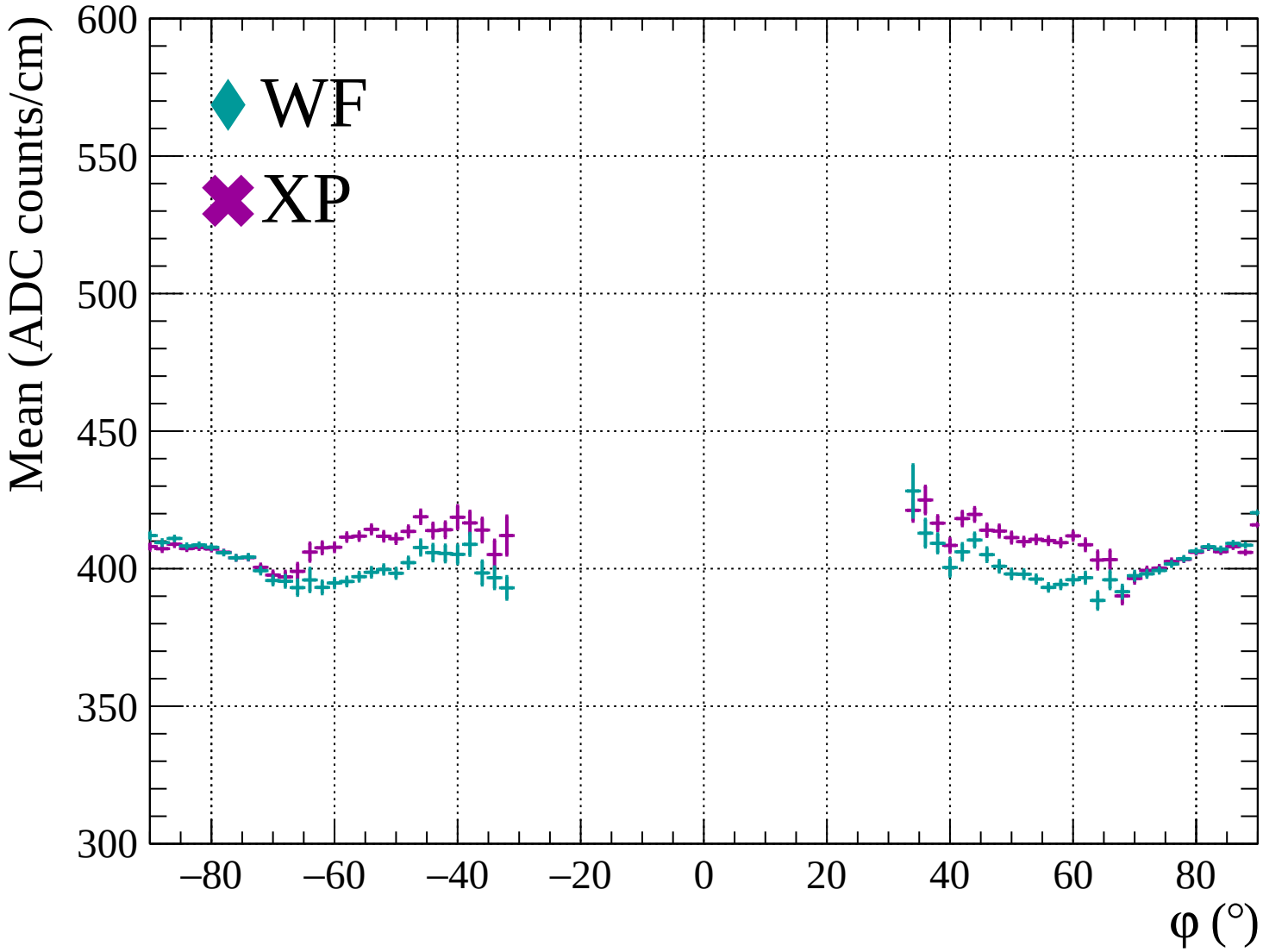


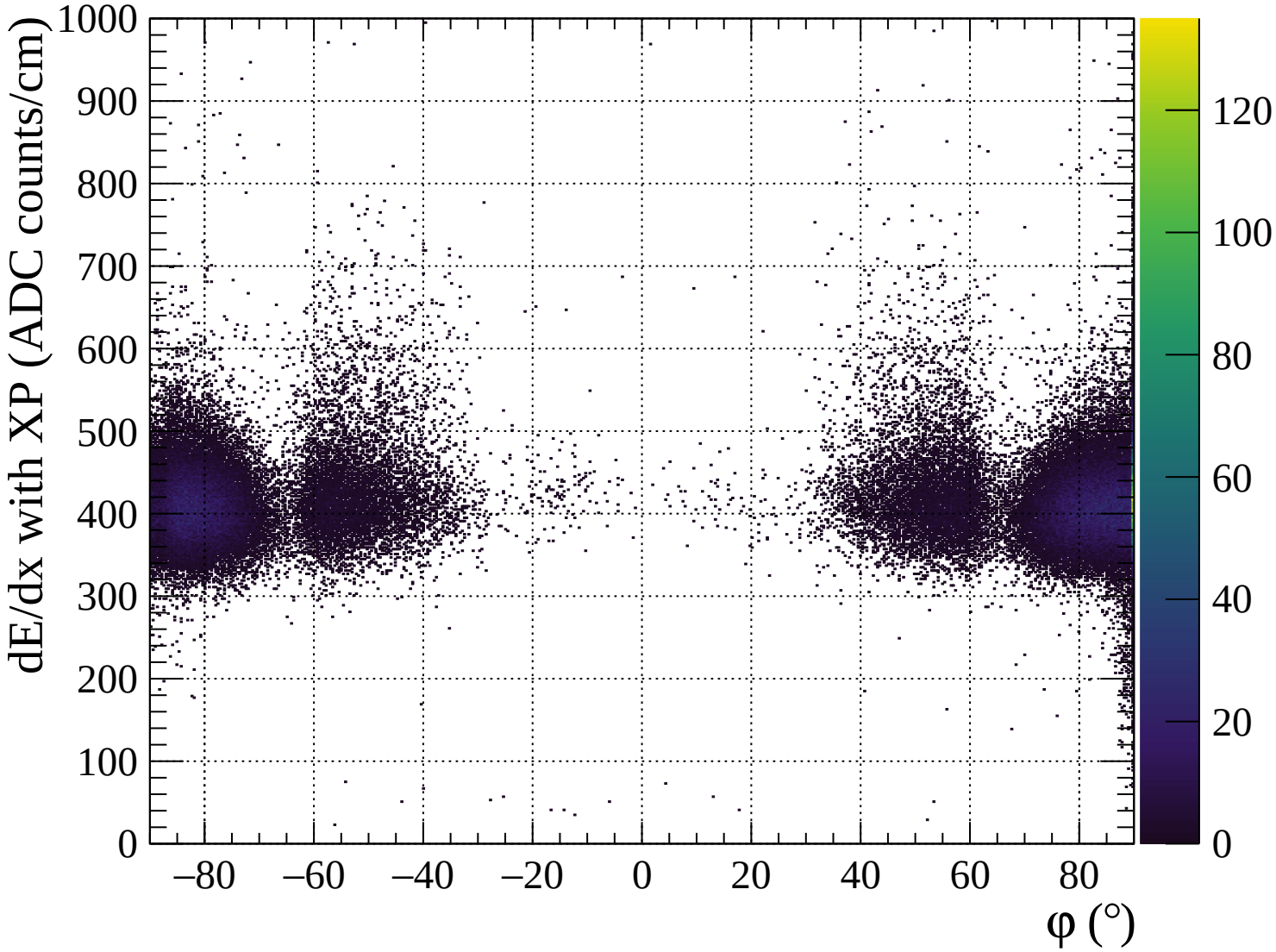


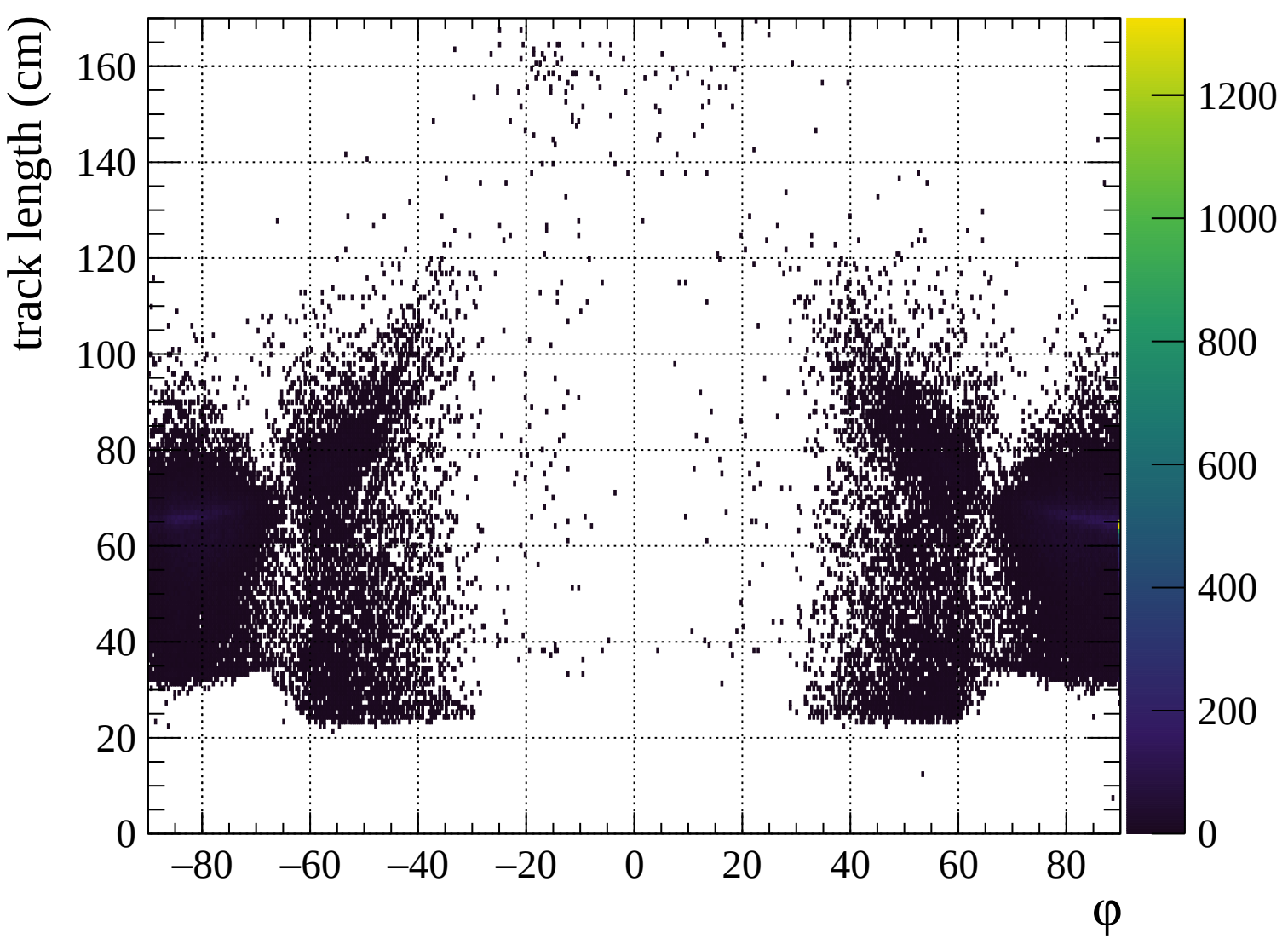


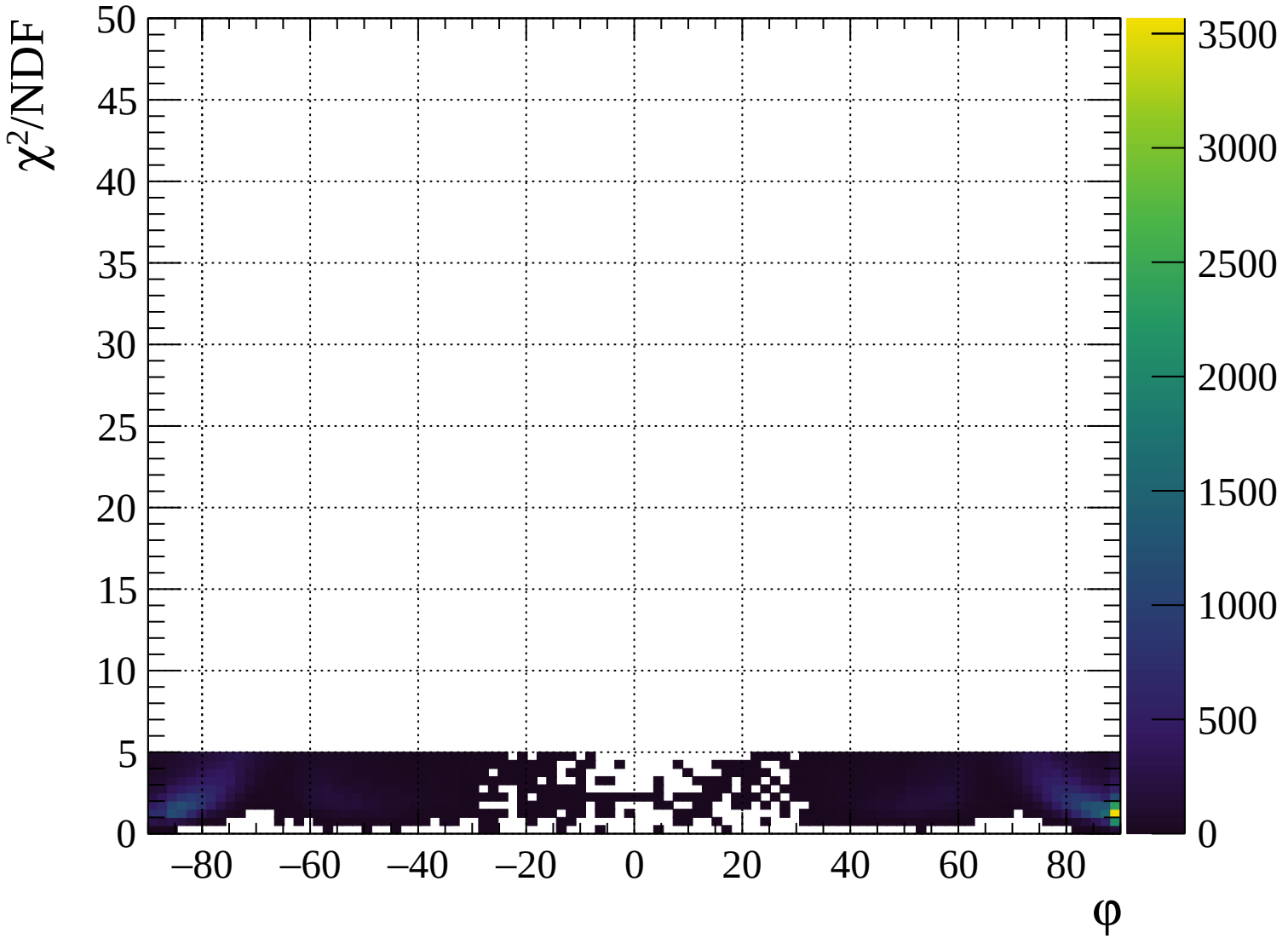


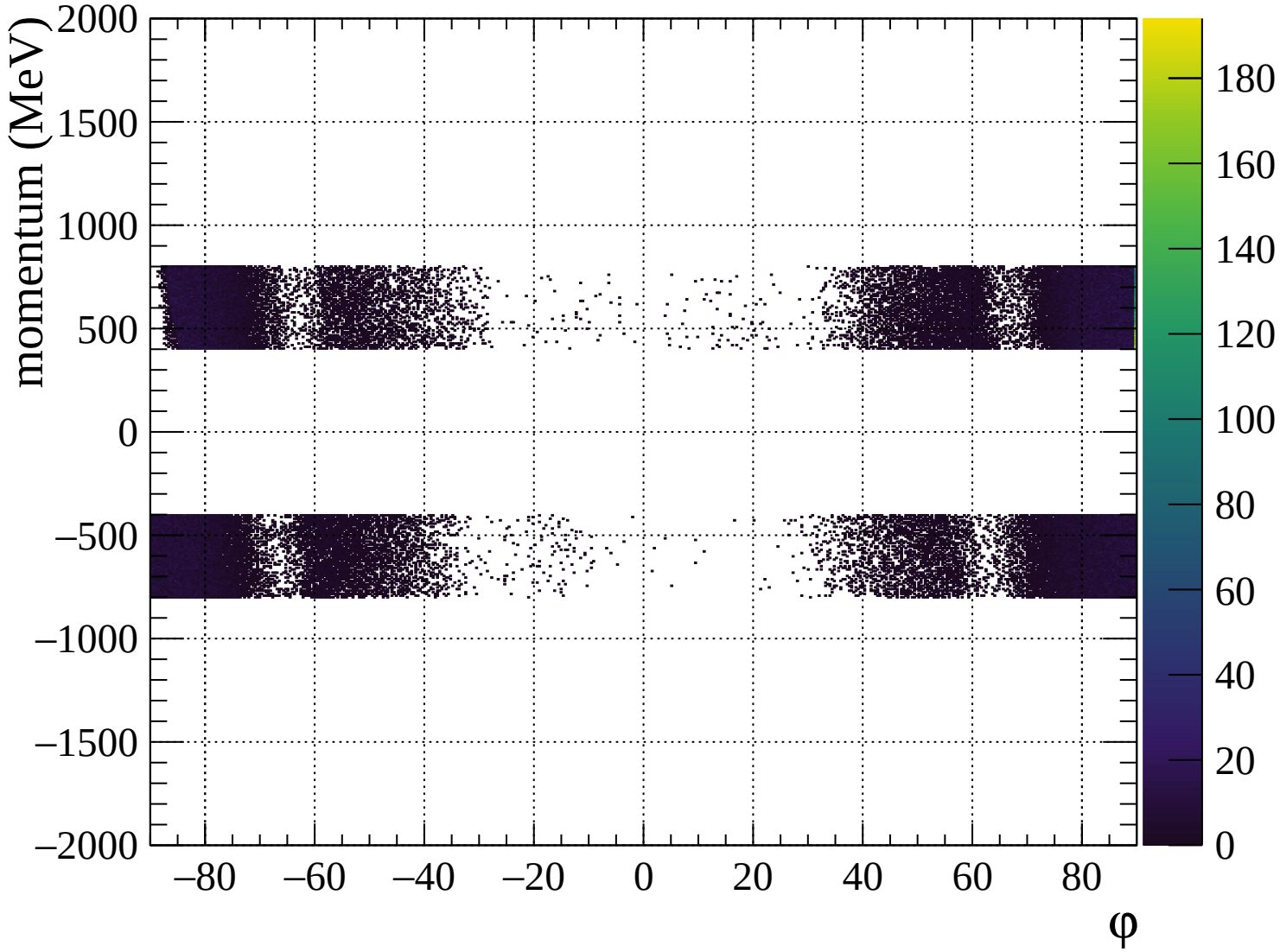


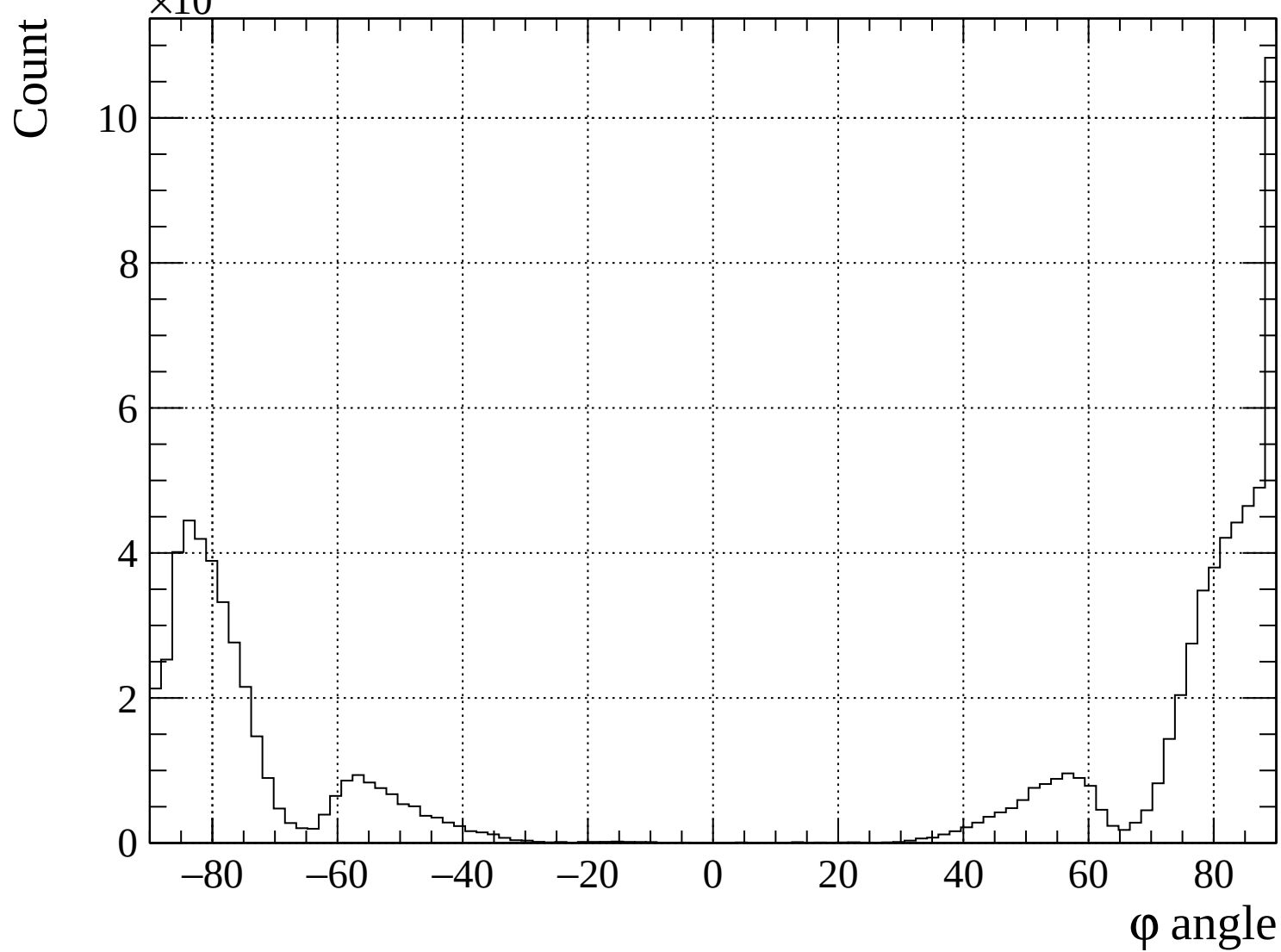


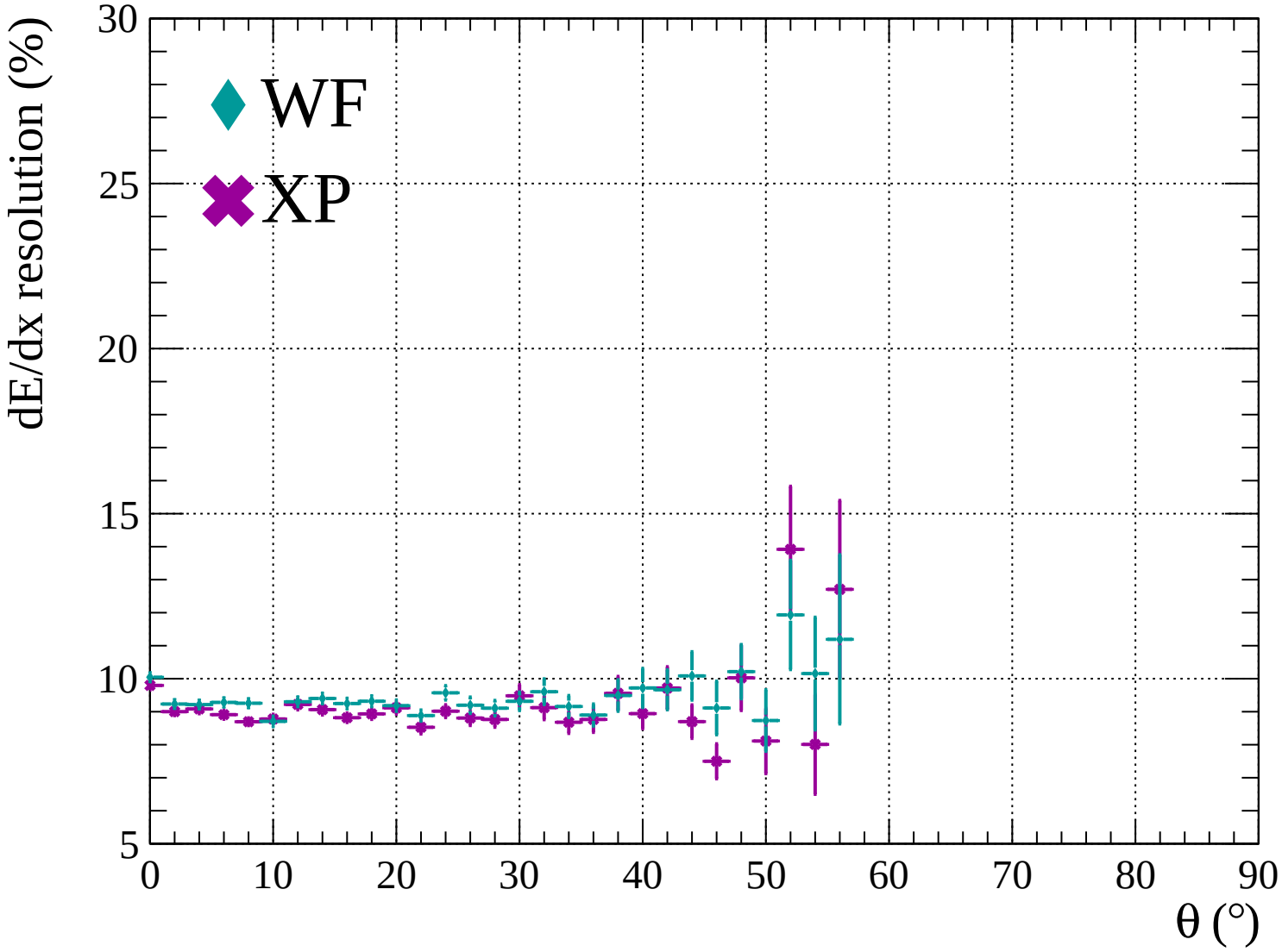


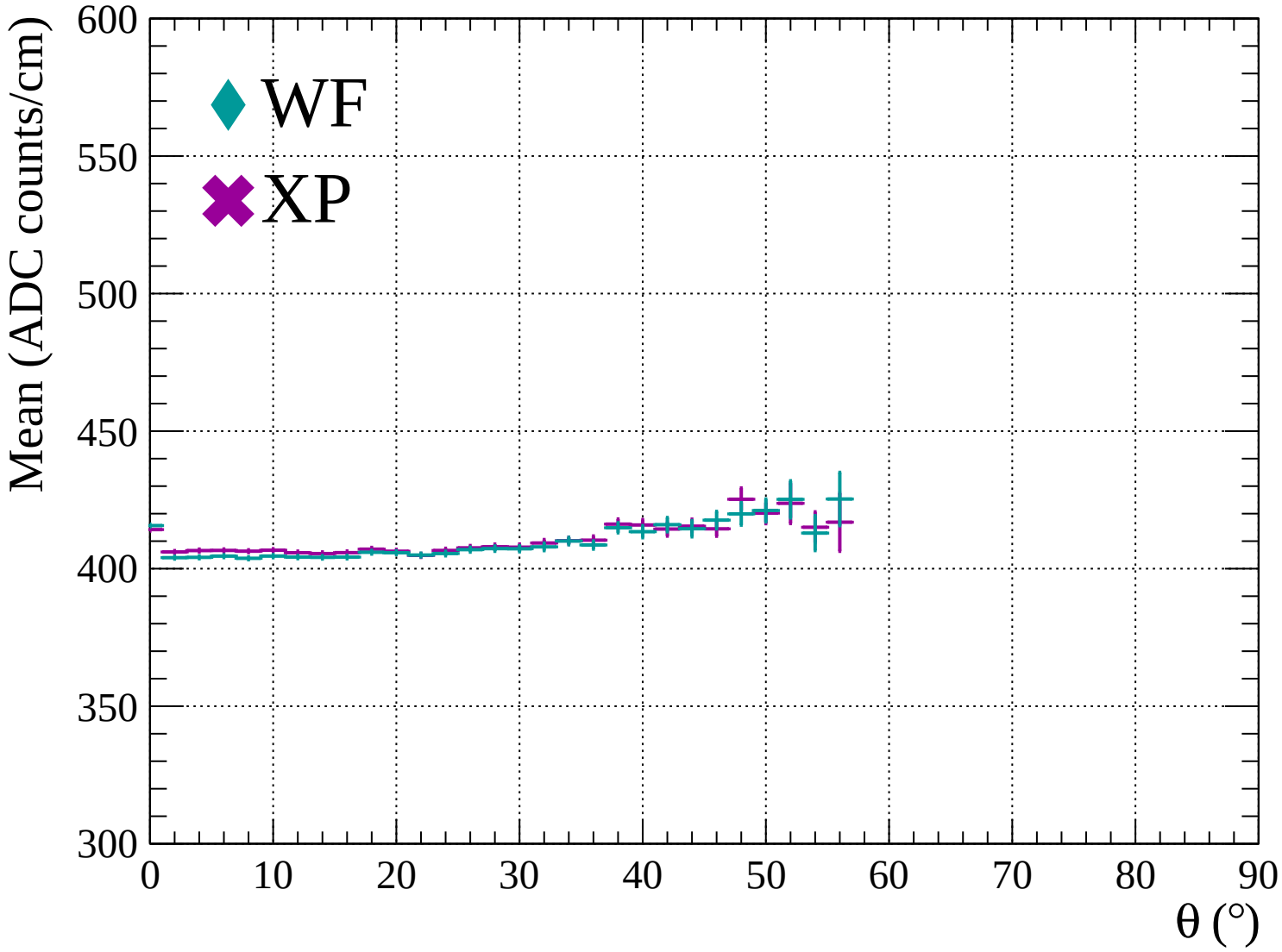


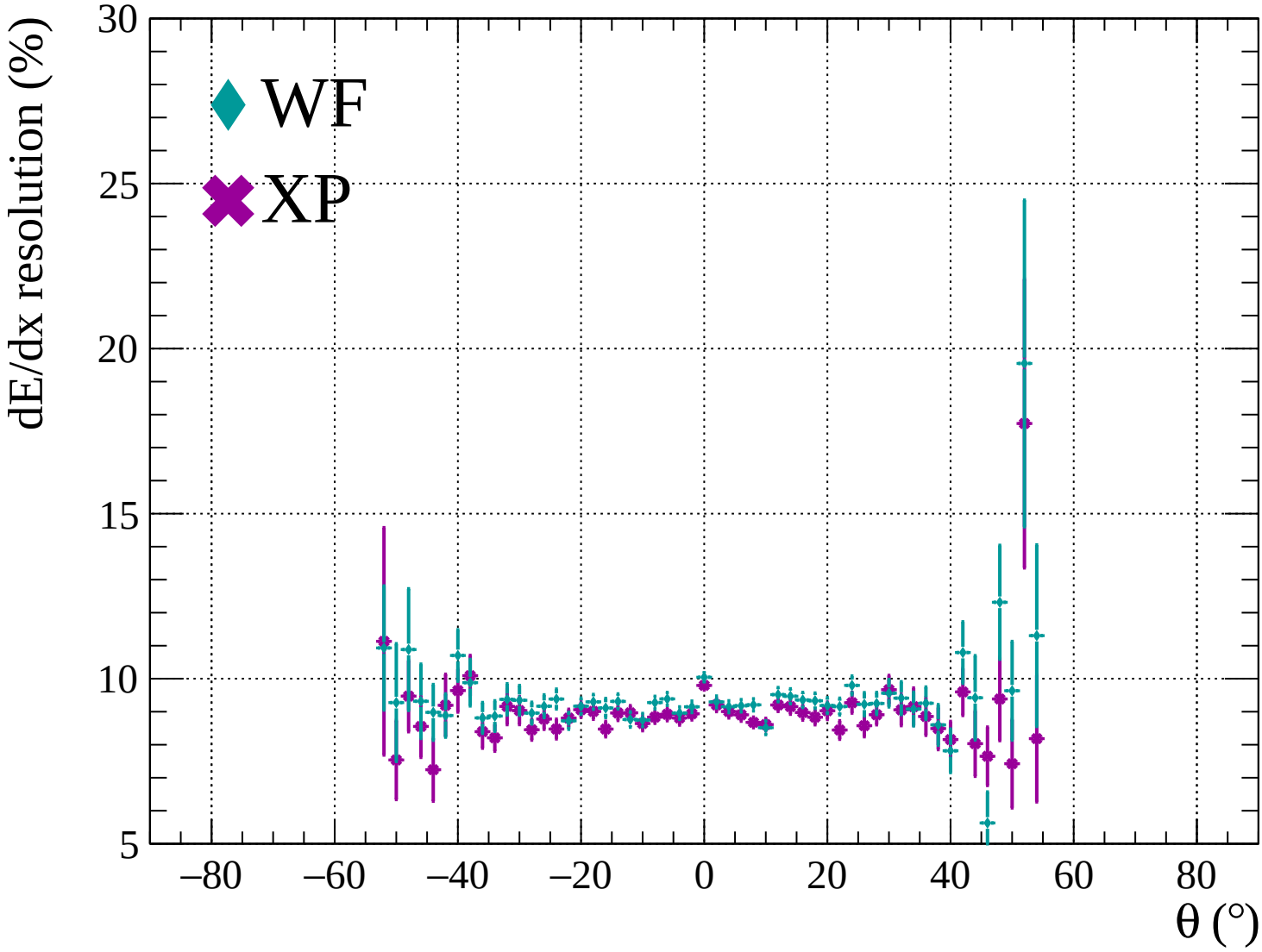


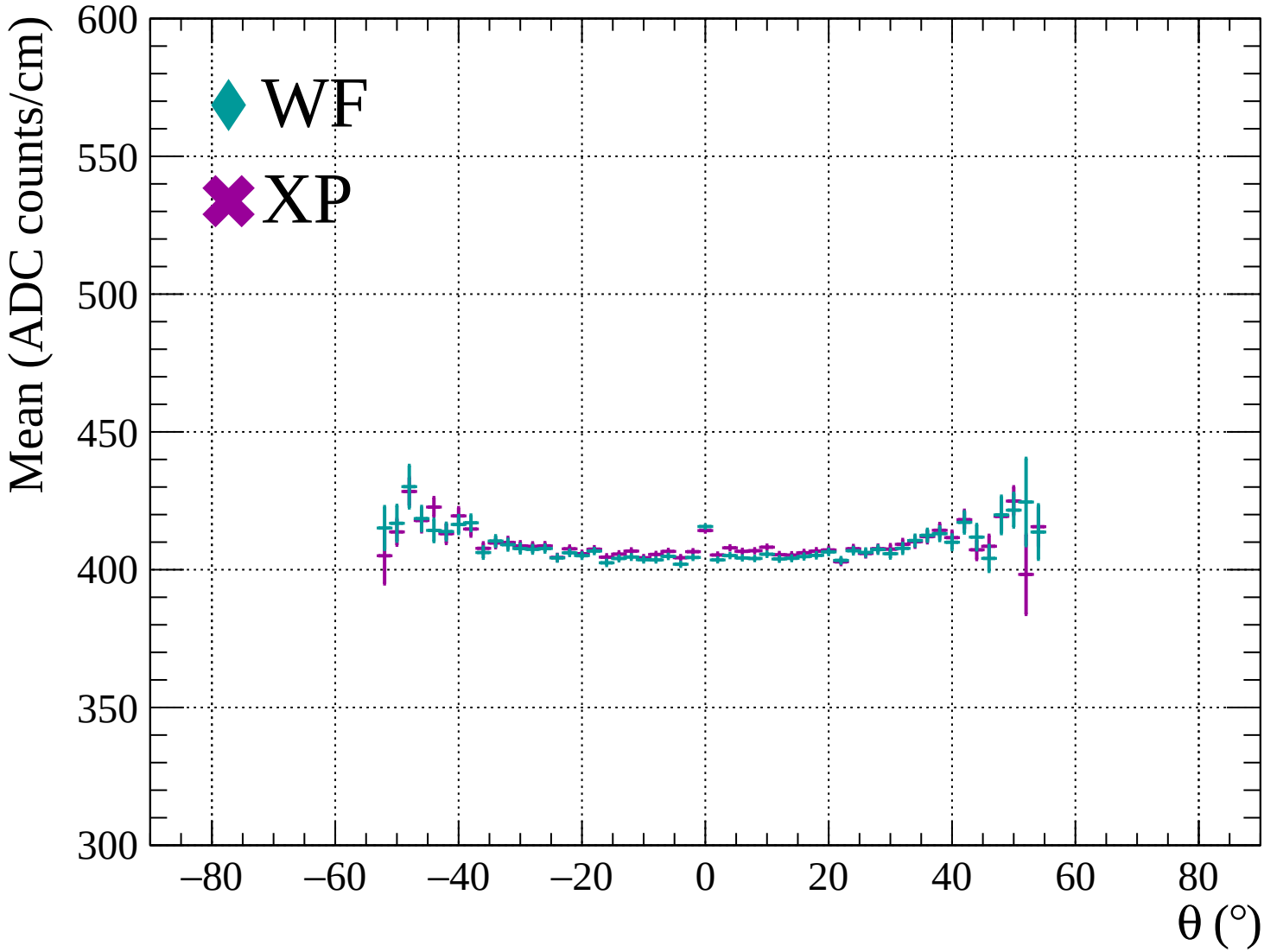


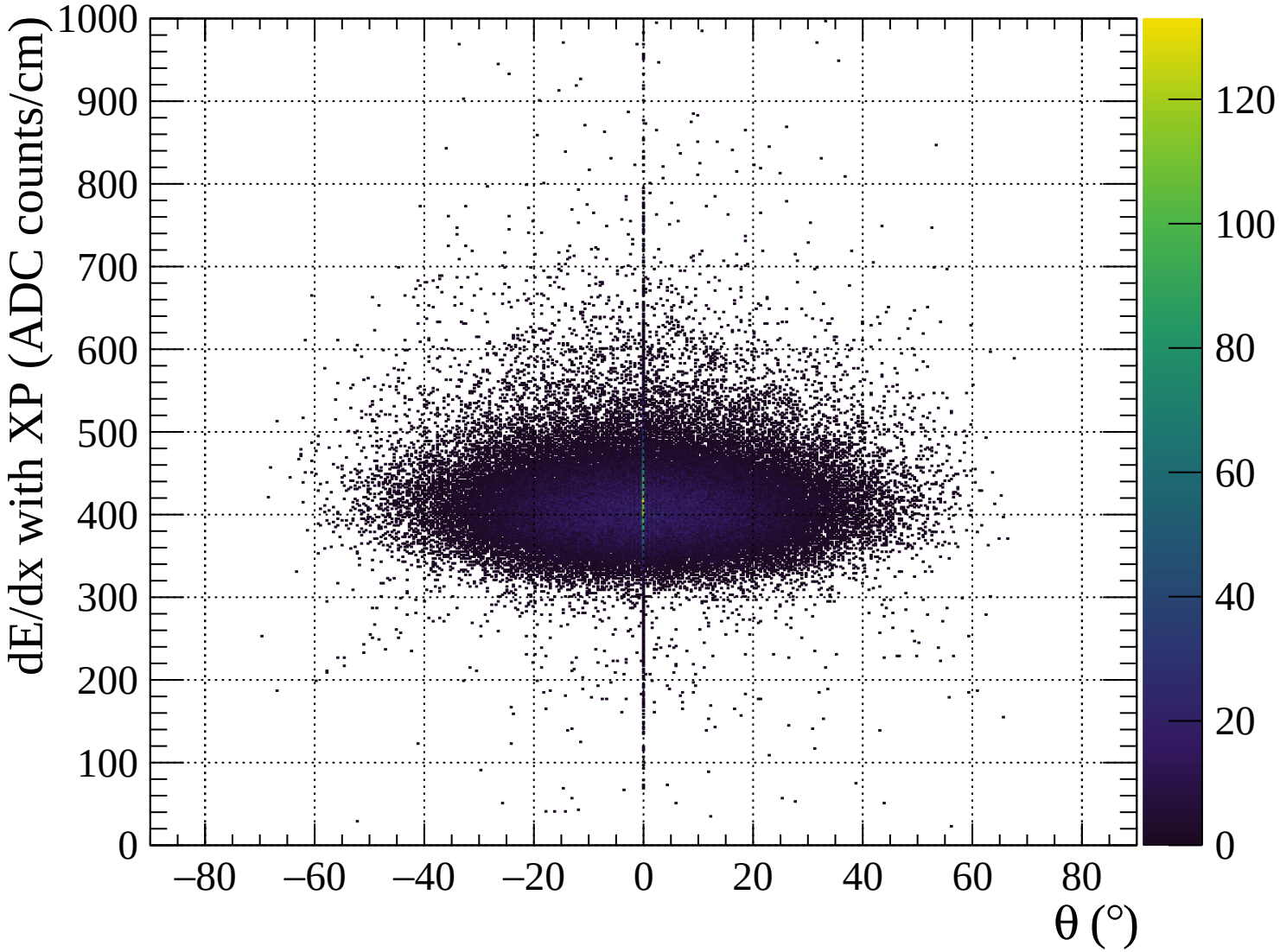


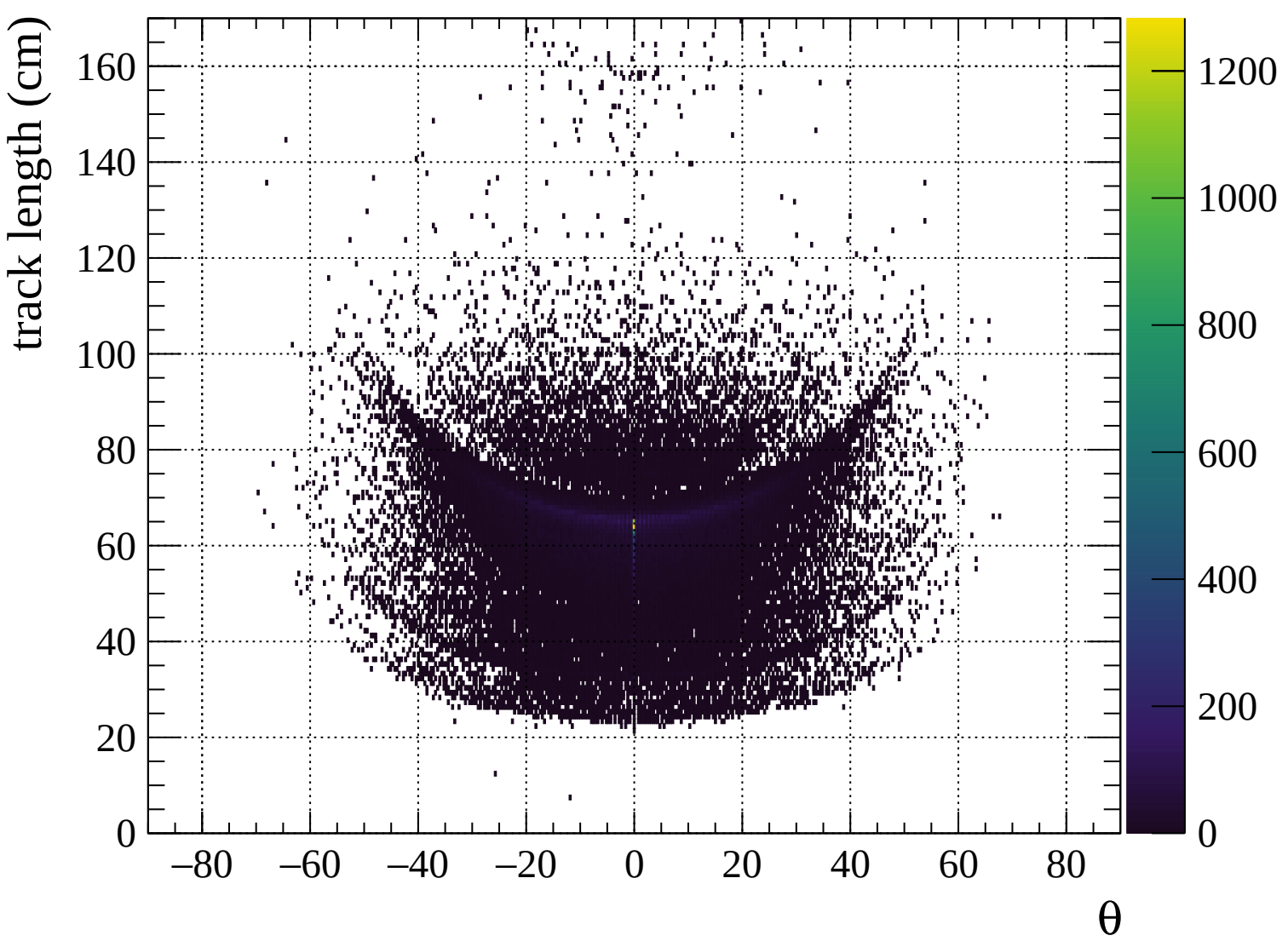


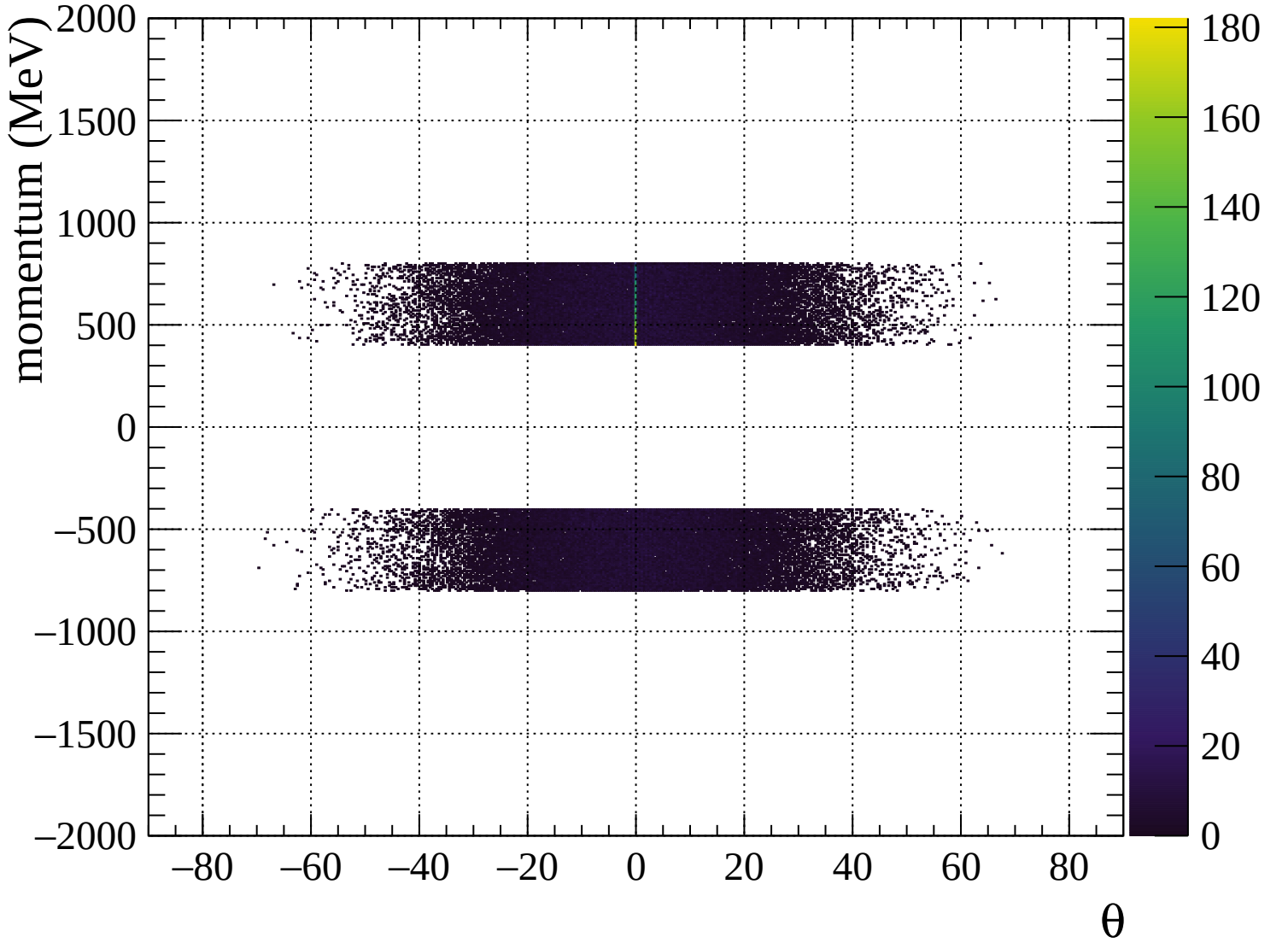


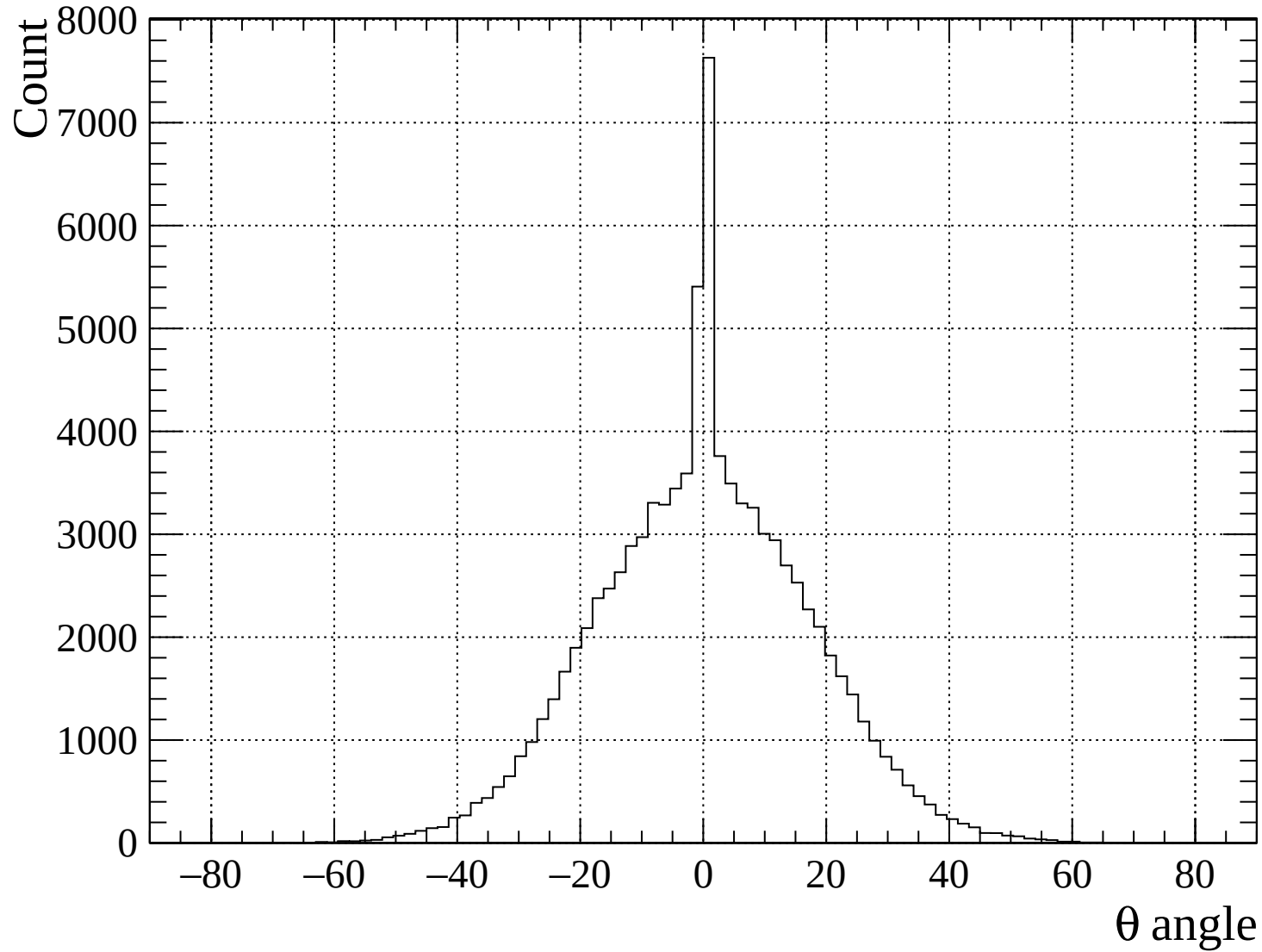


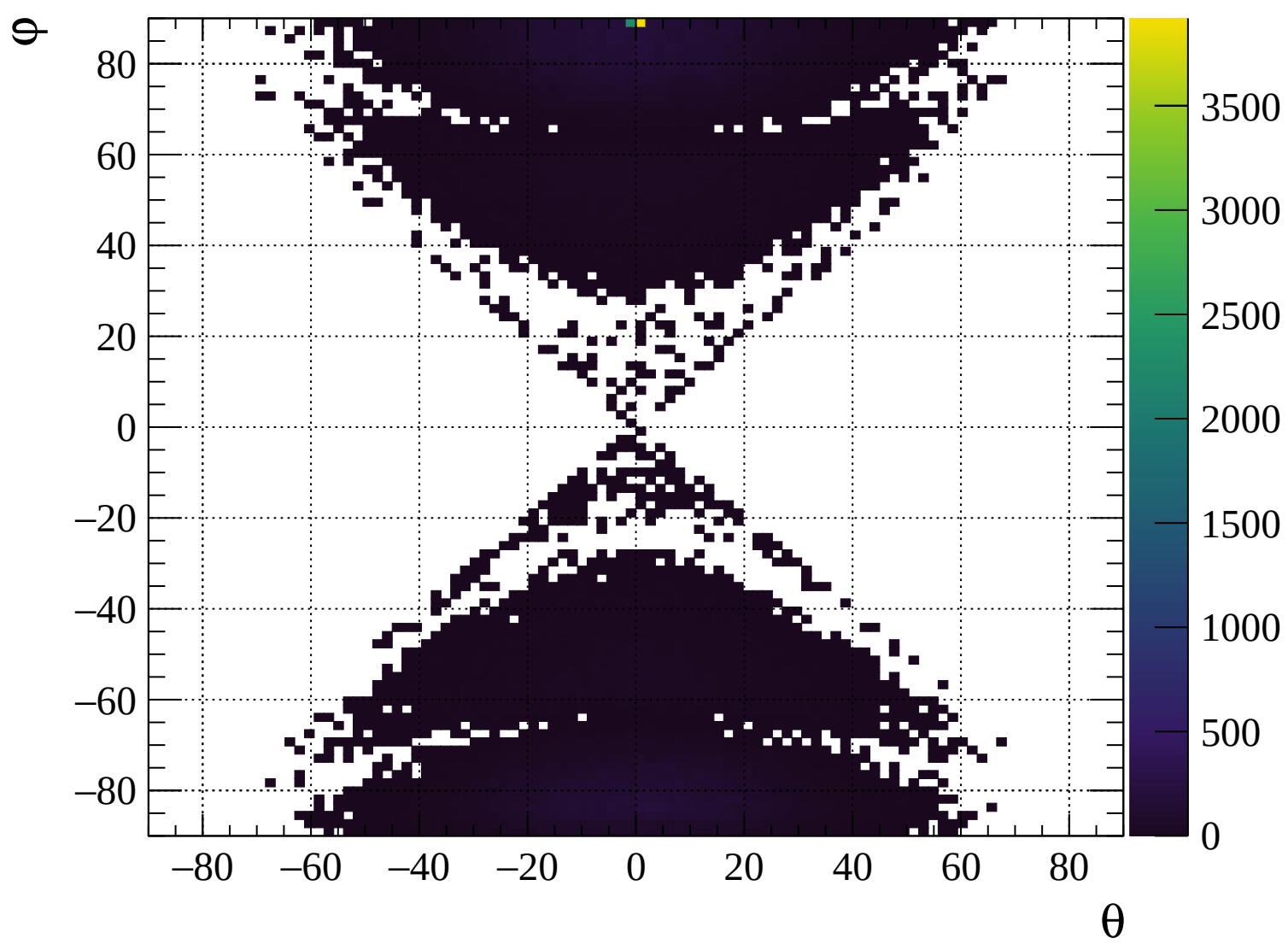


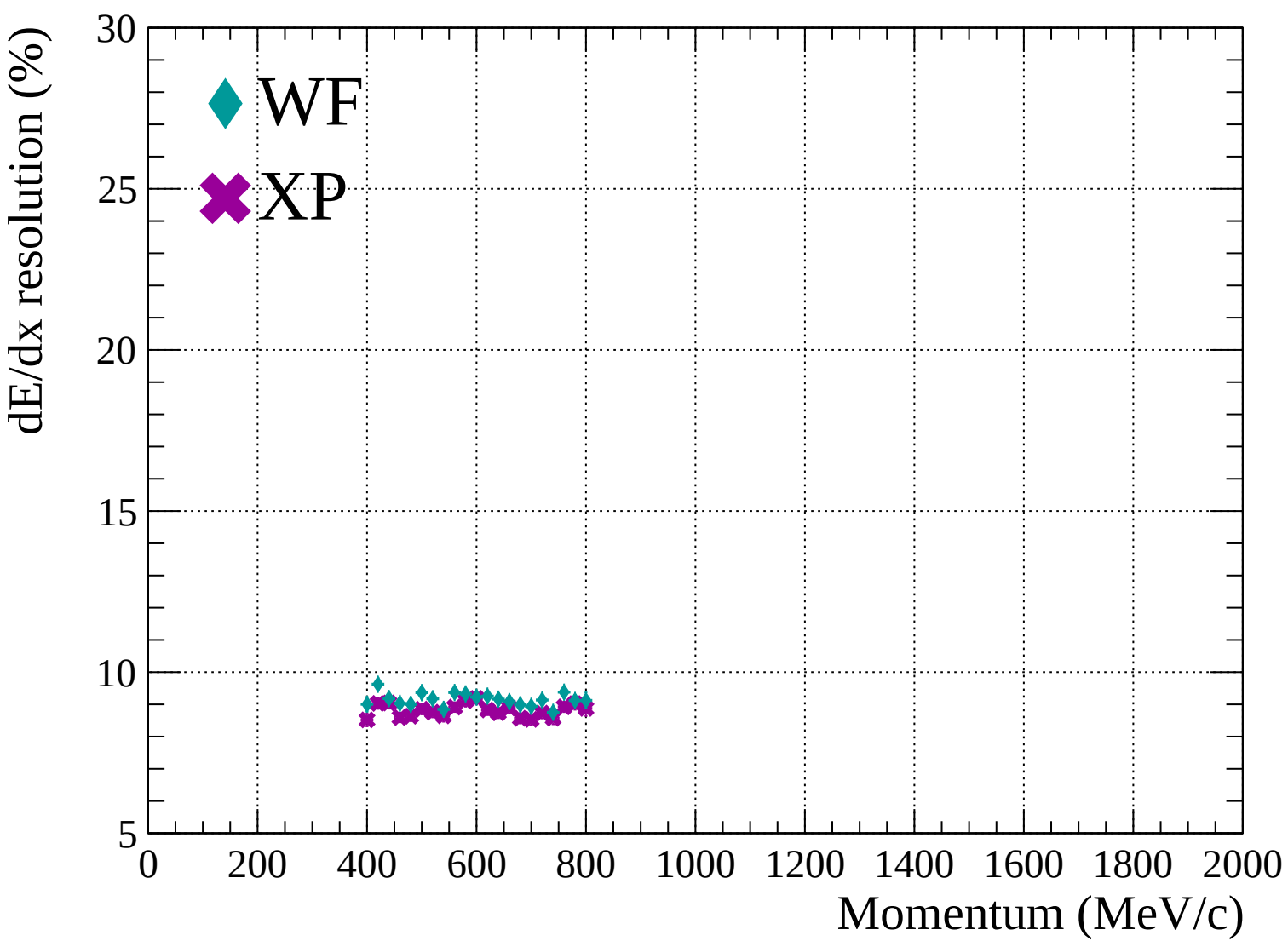


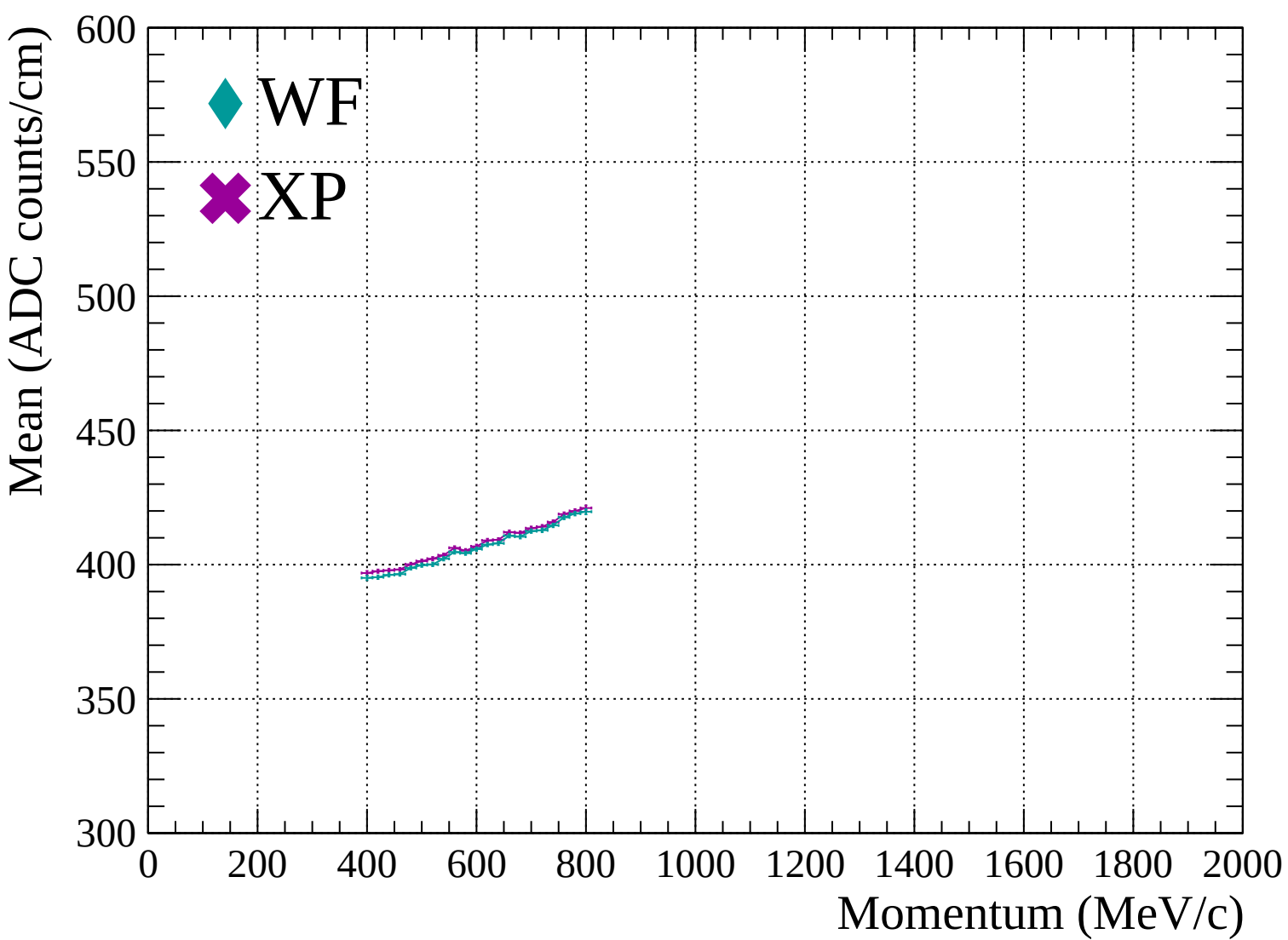


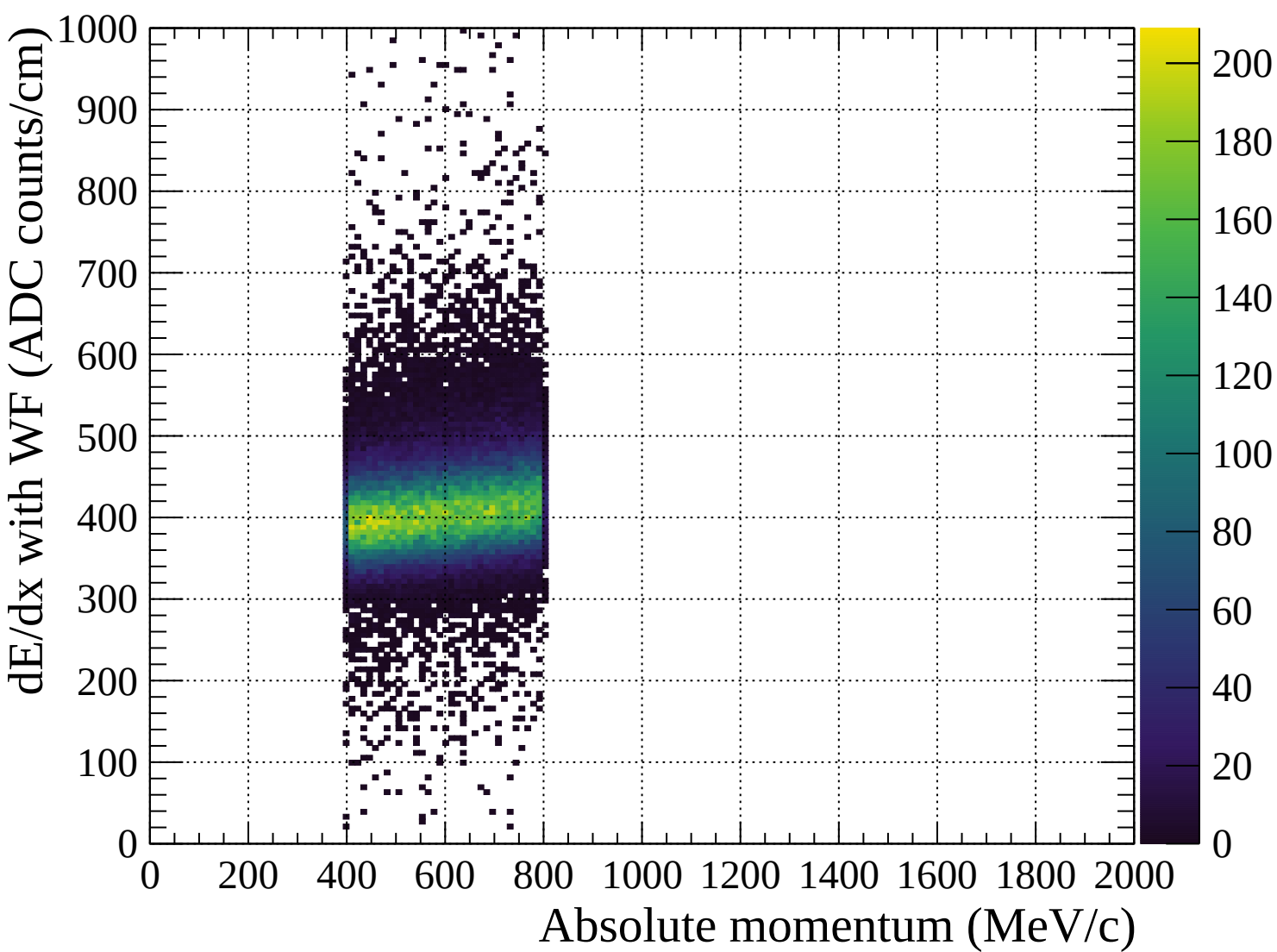


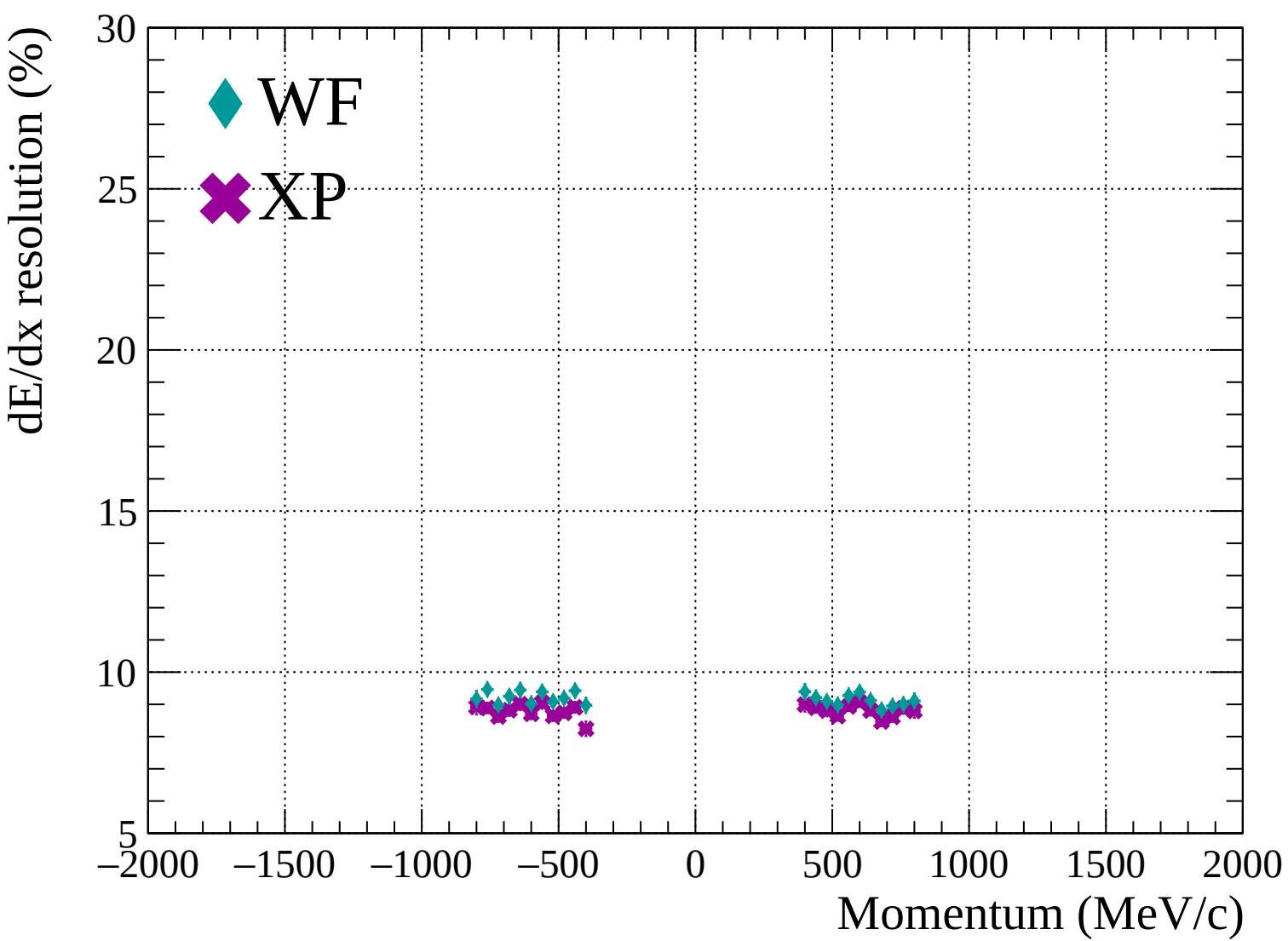


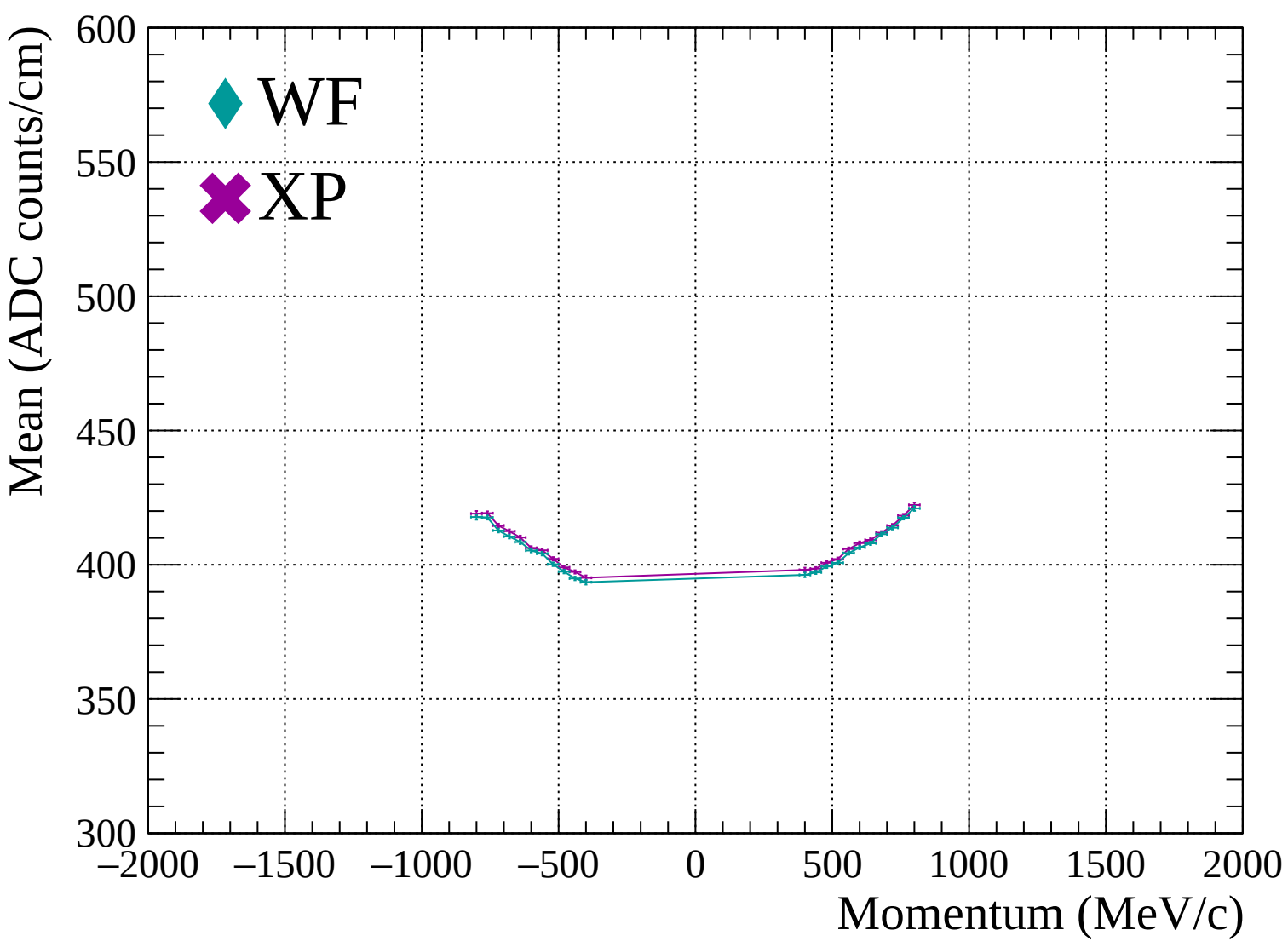


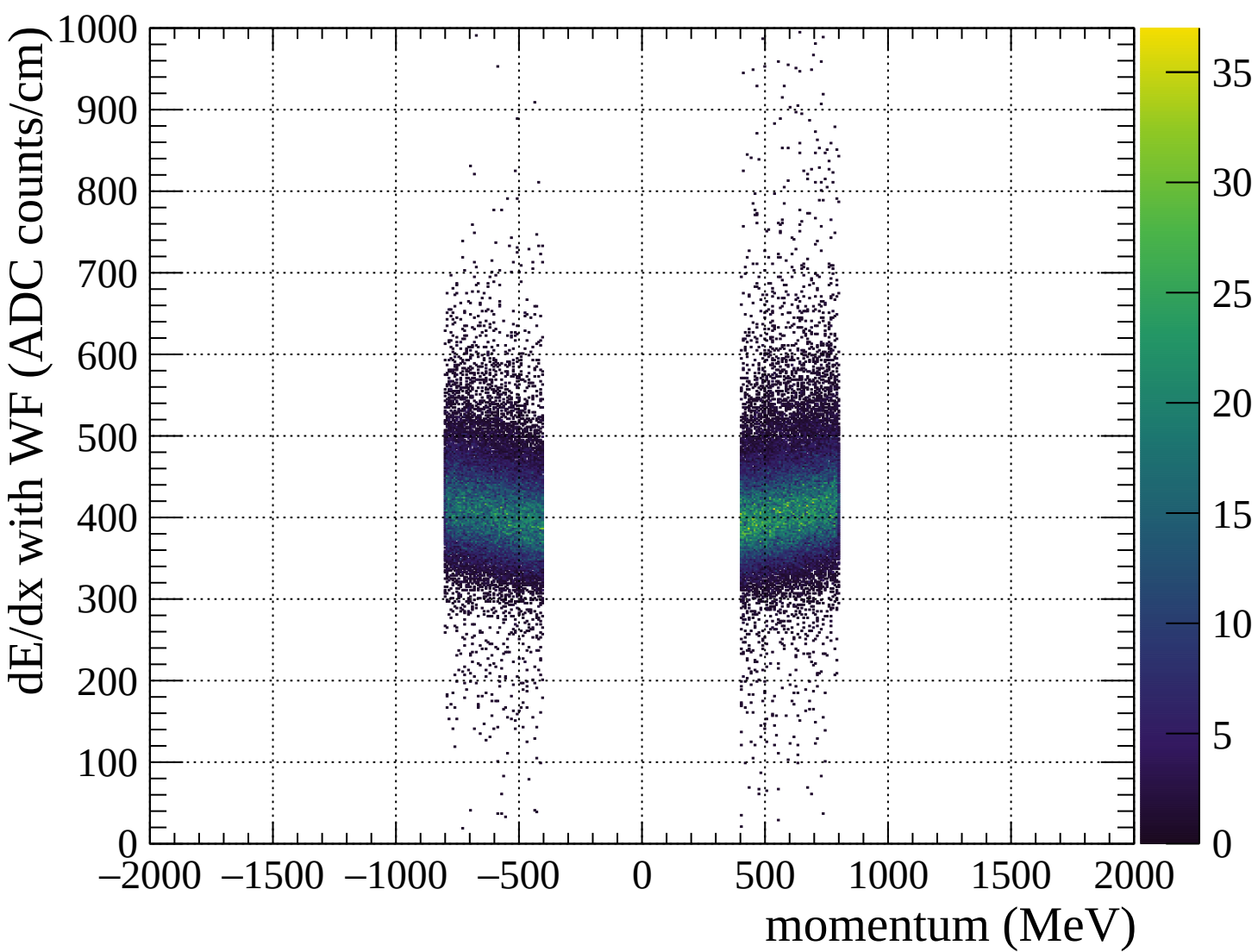


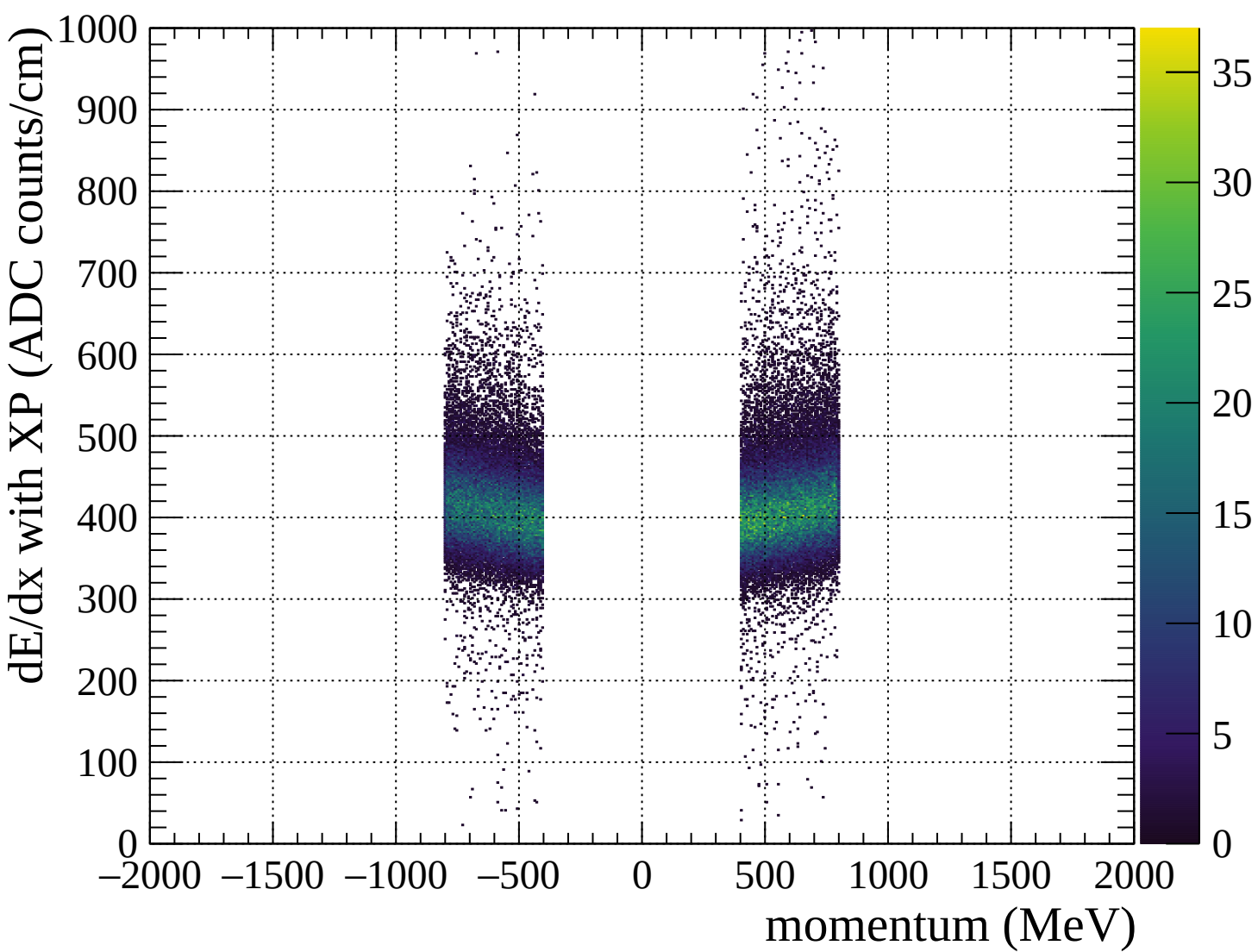


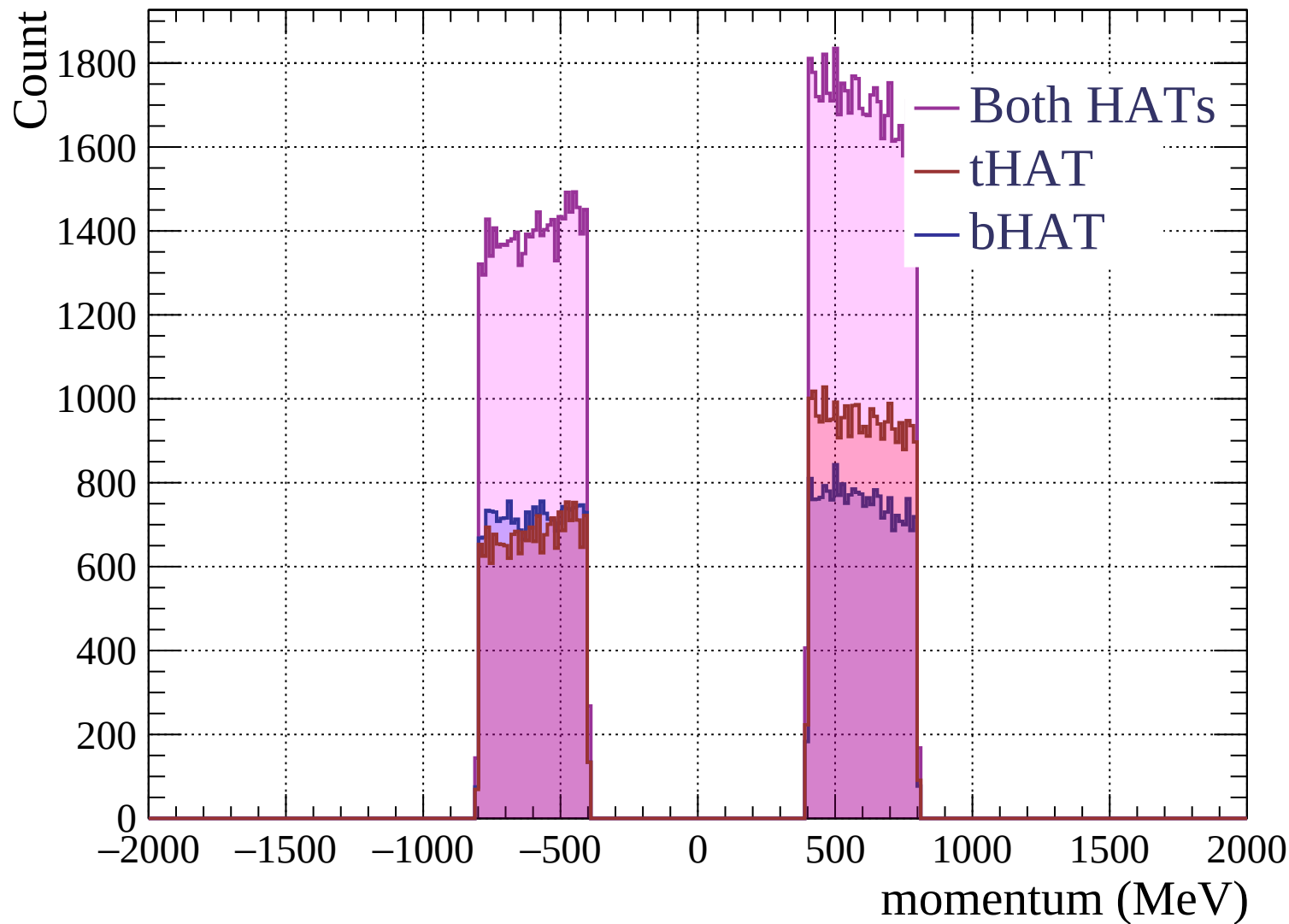


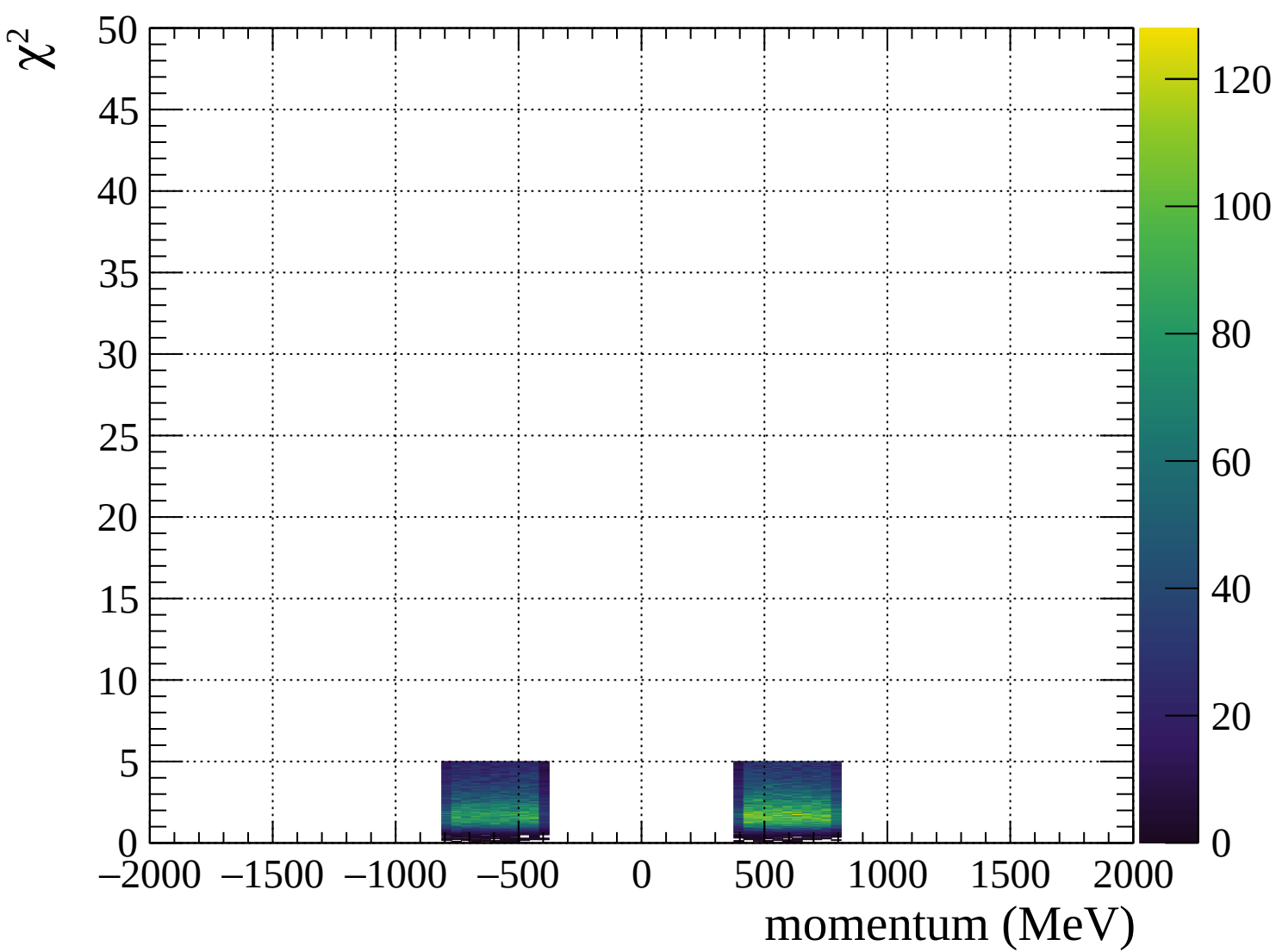




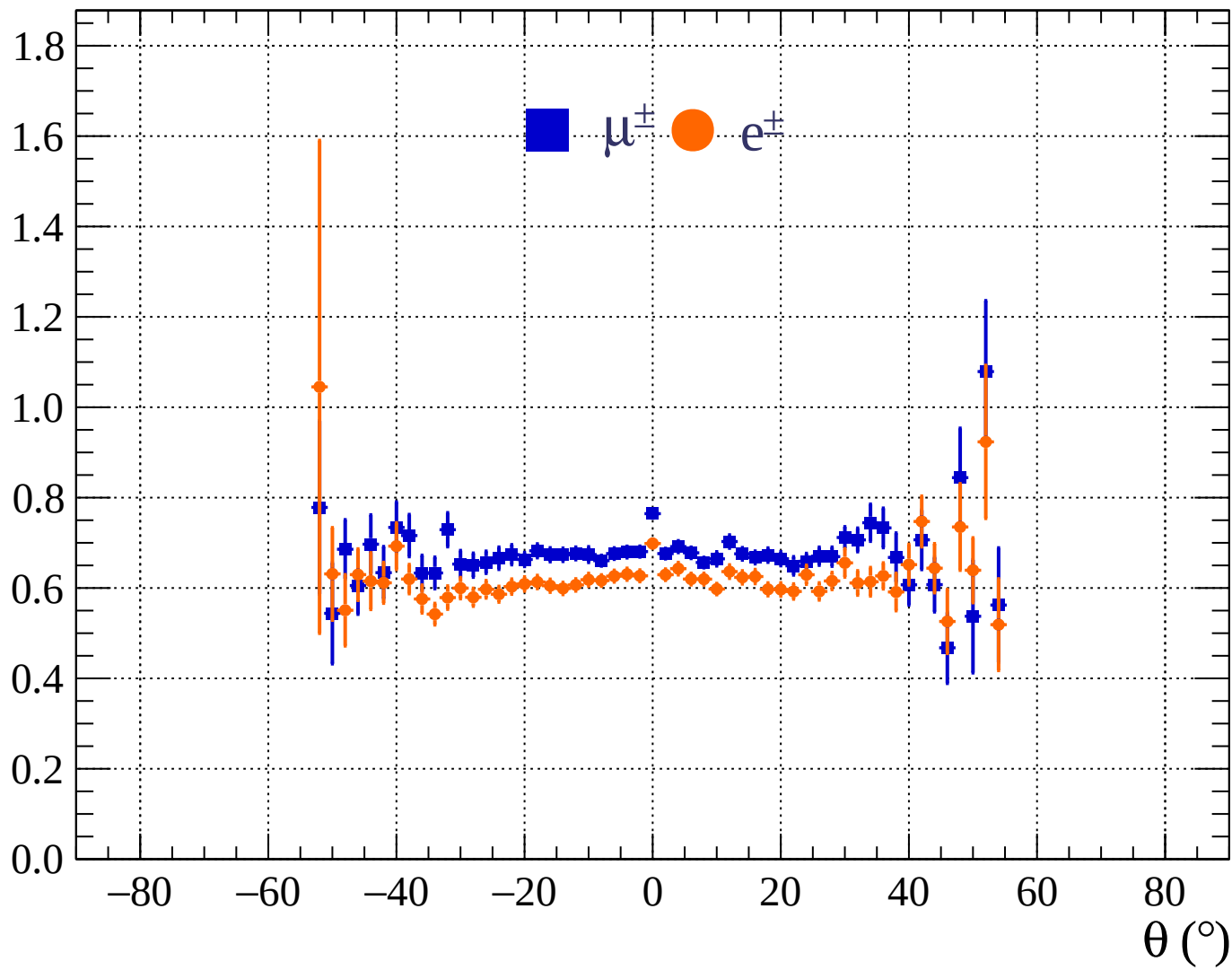


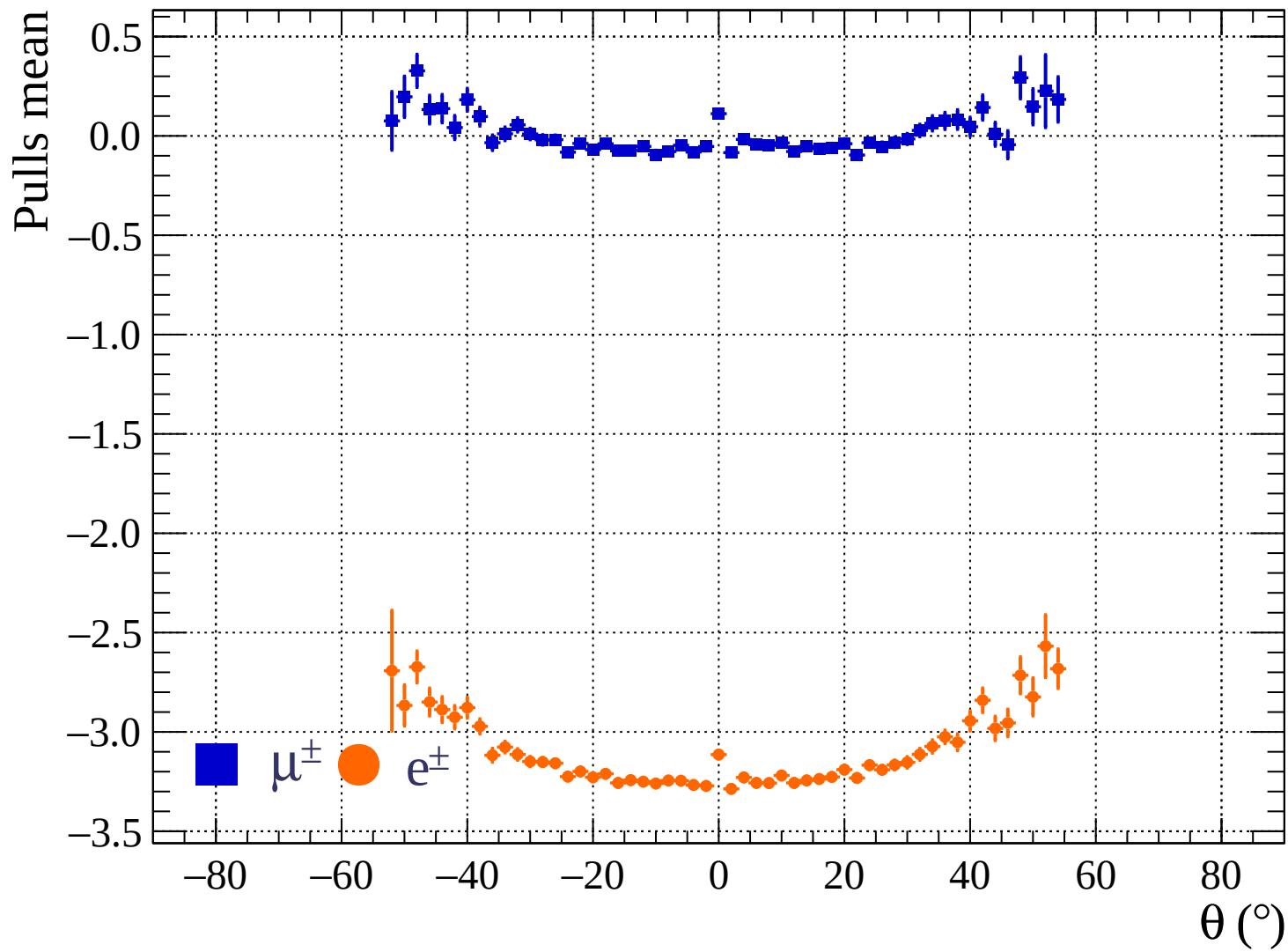




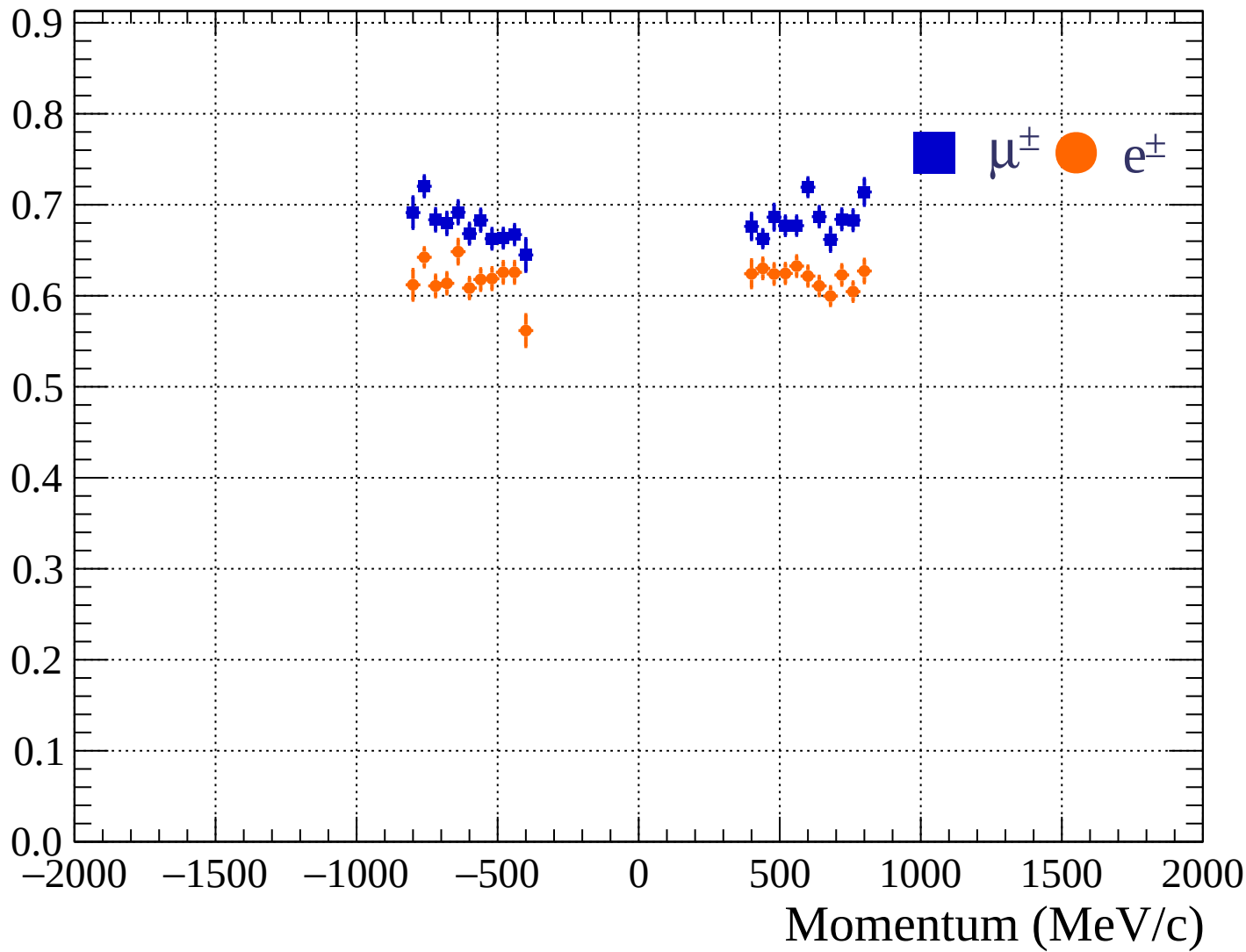


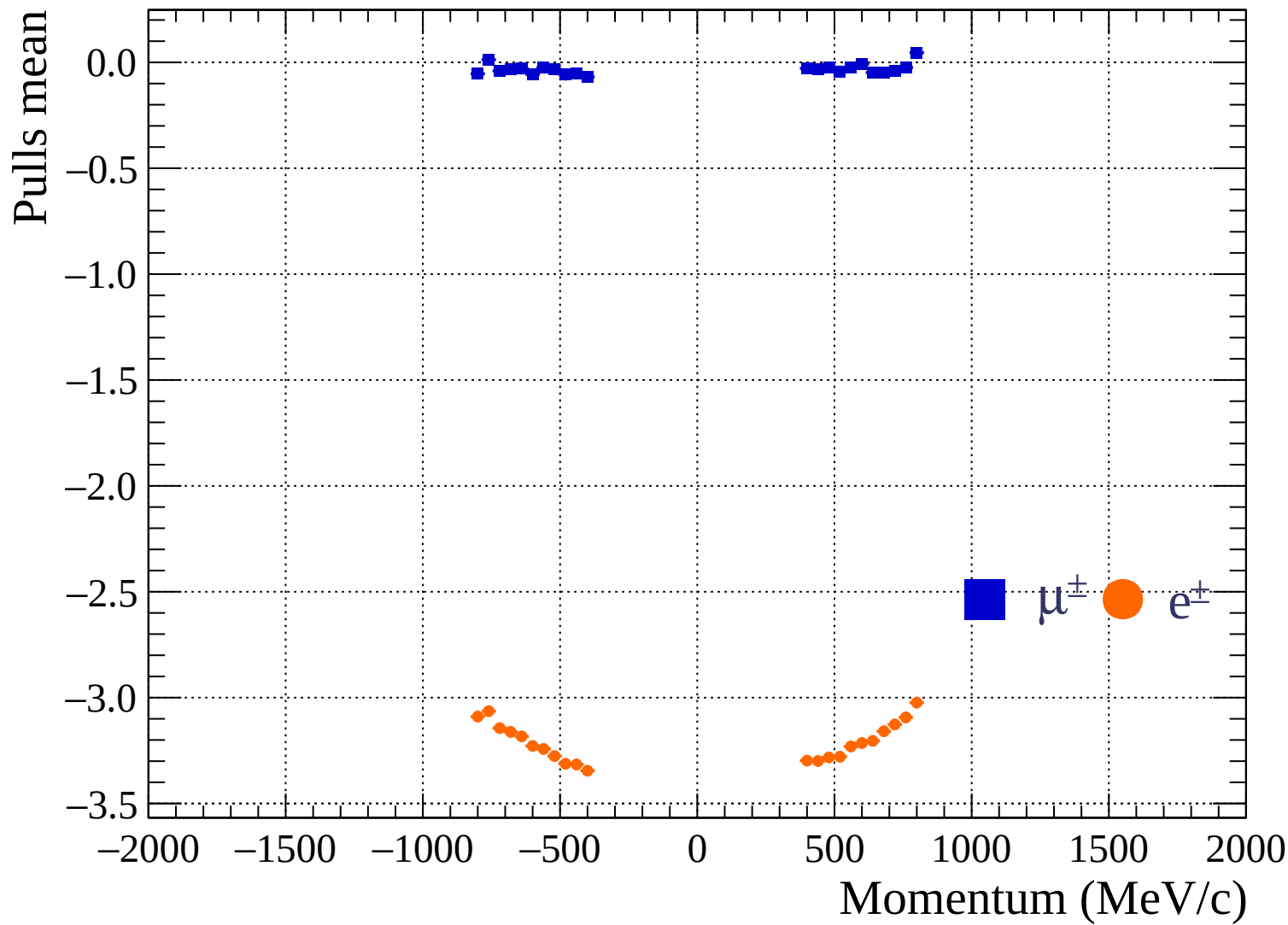
Pulls standard deviation

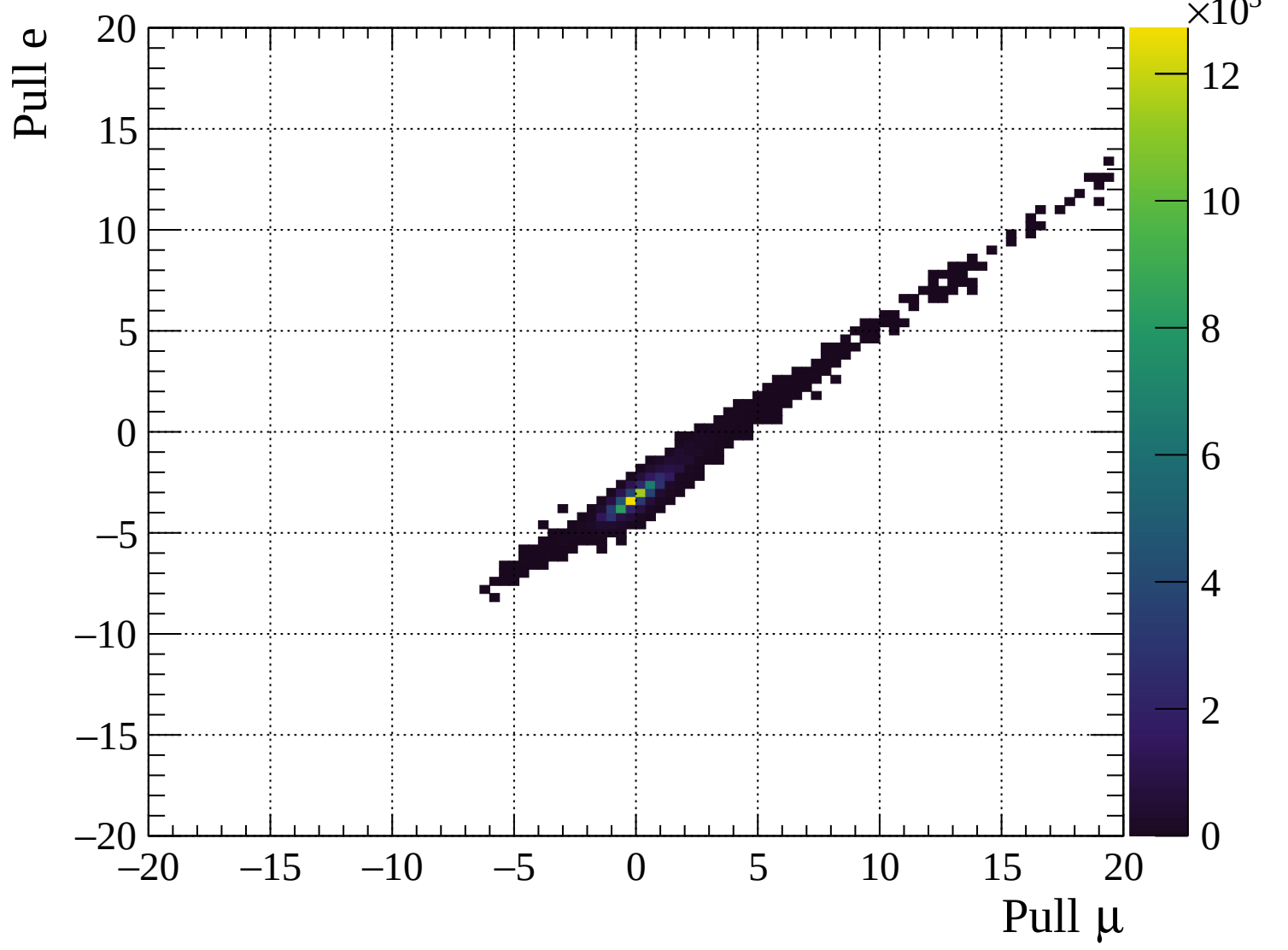


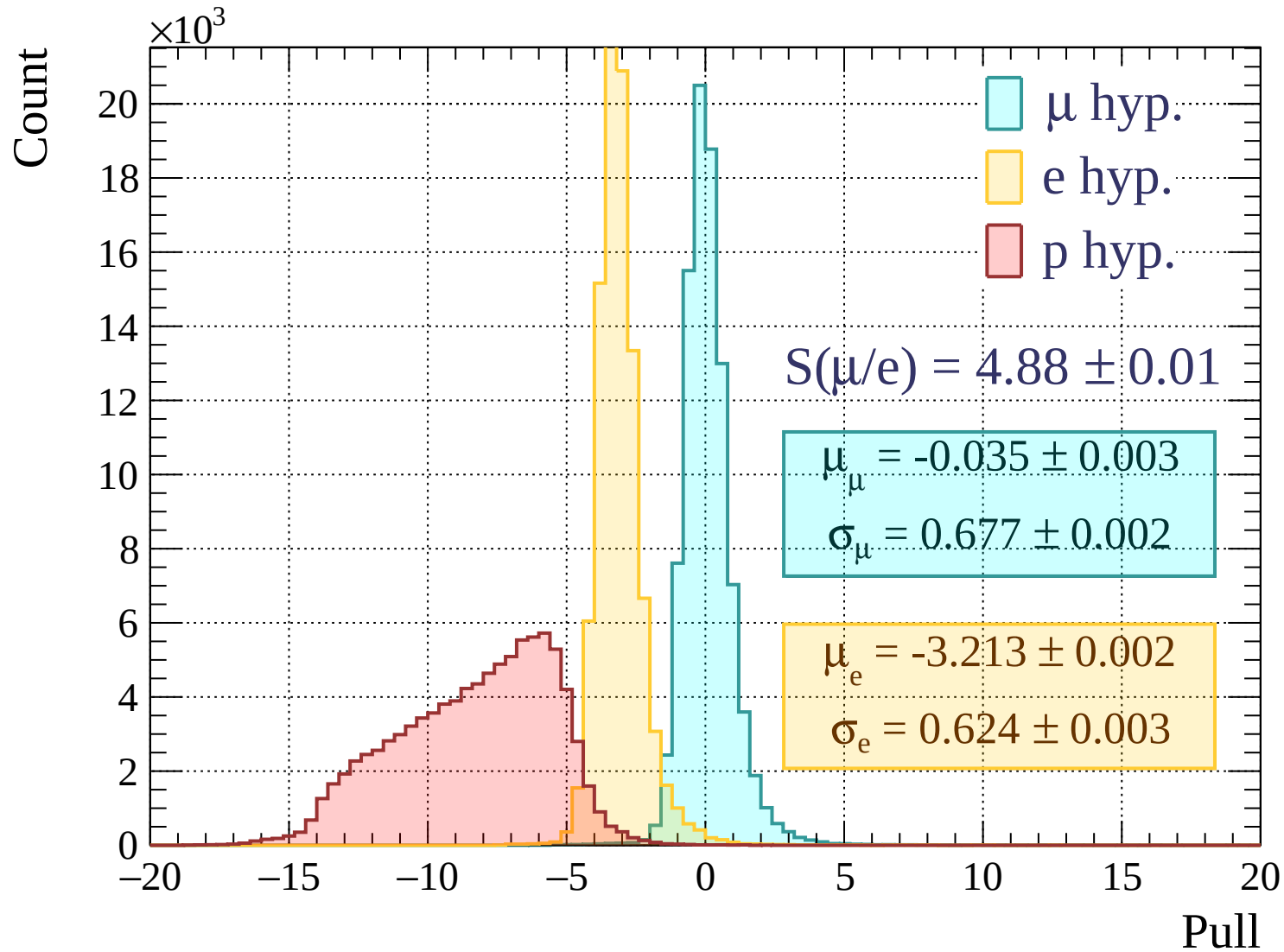


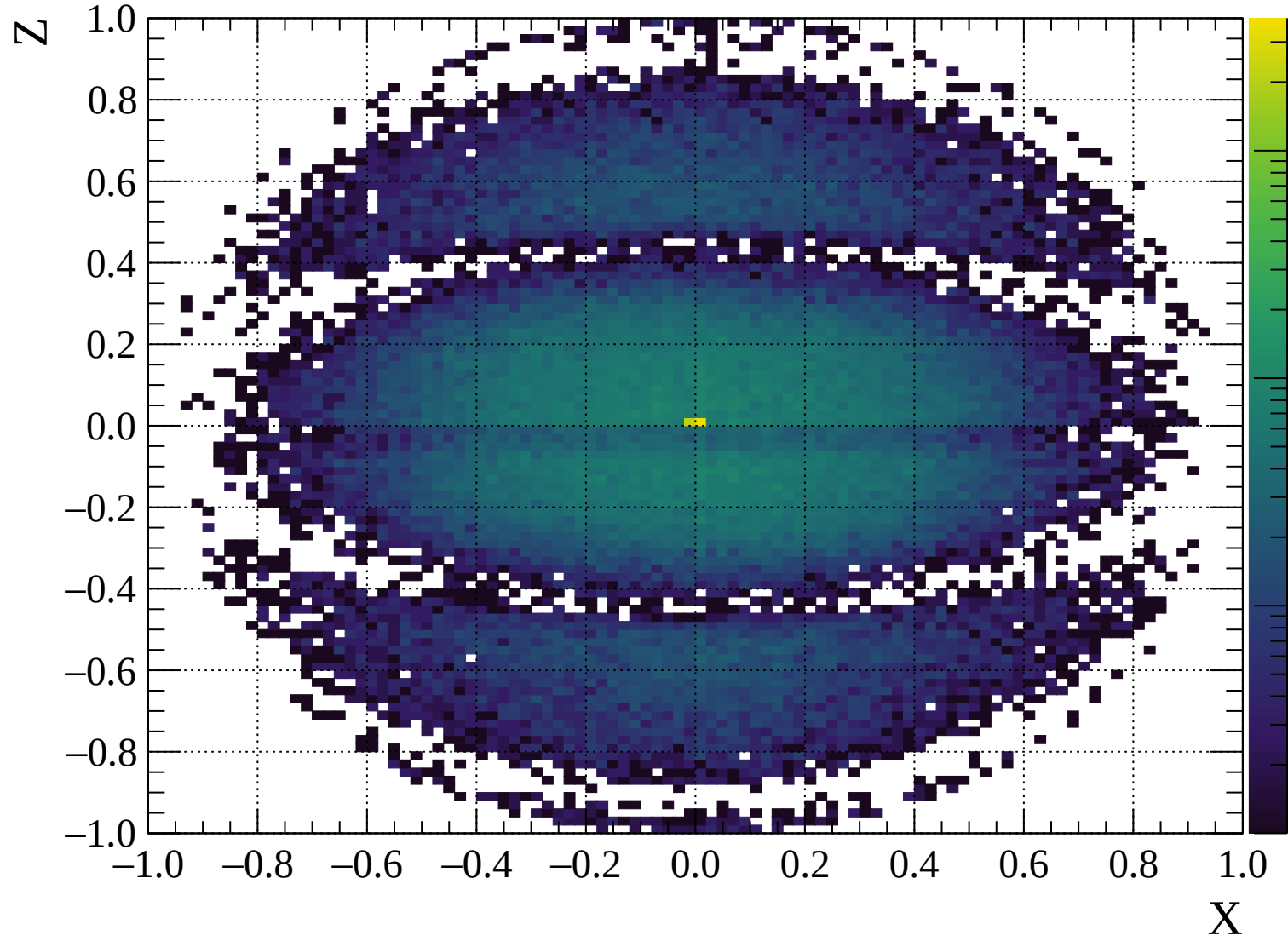
Pulls standard deviation

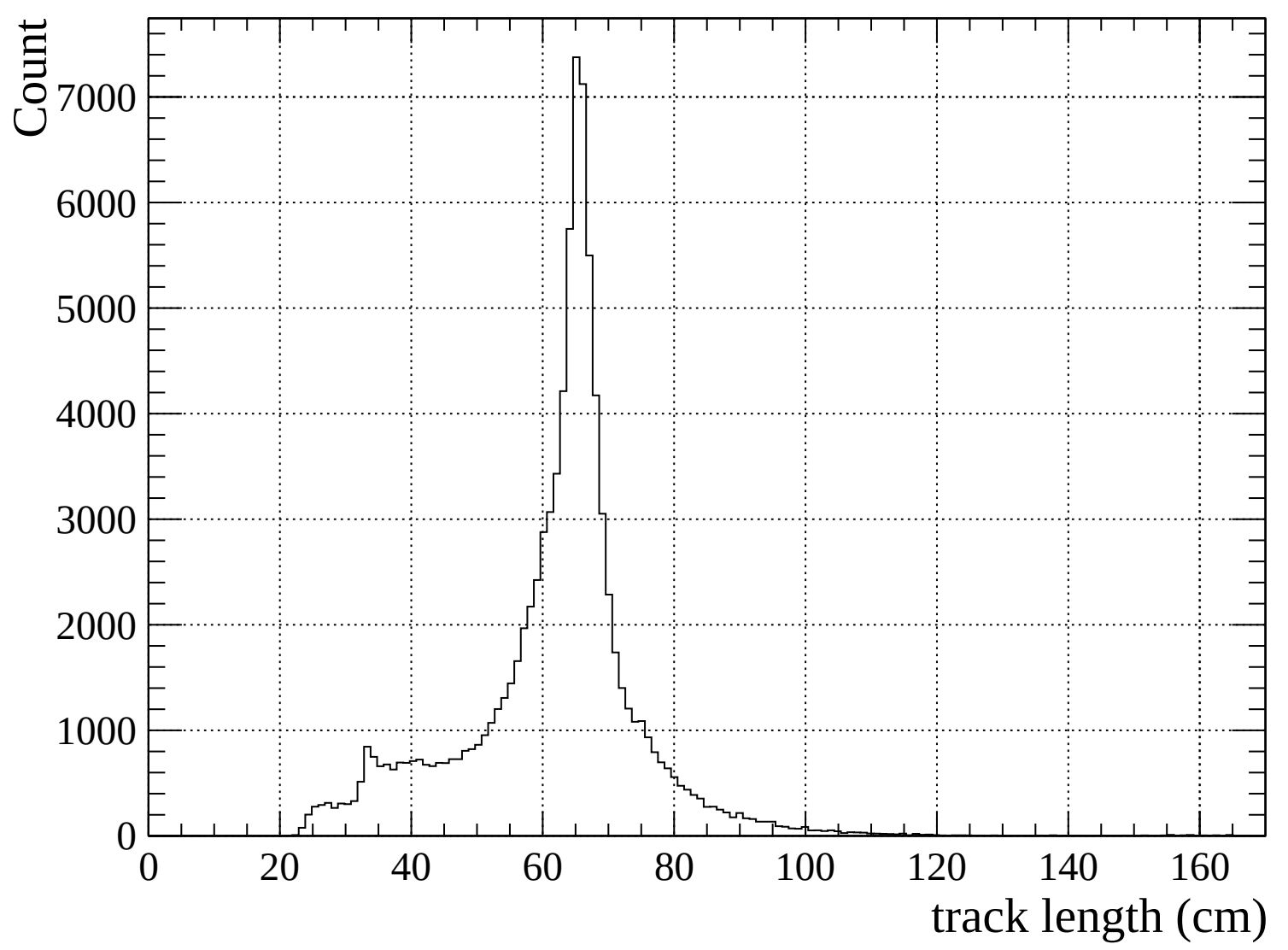


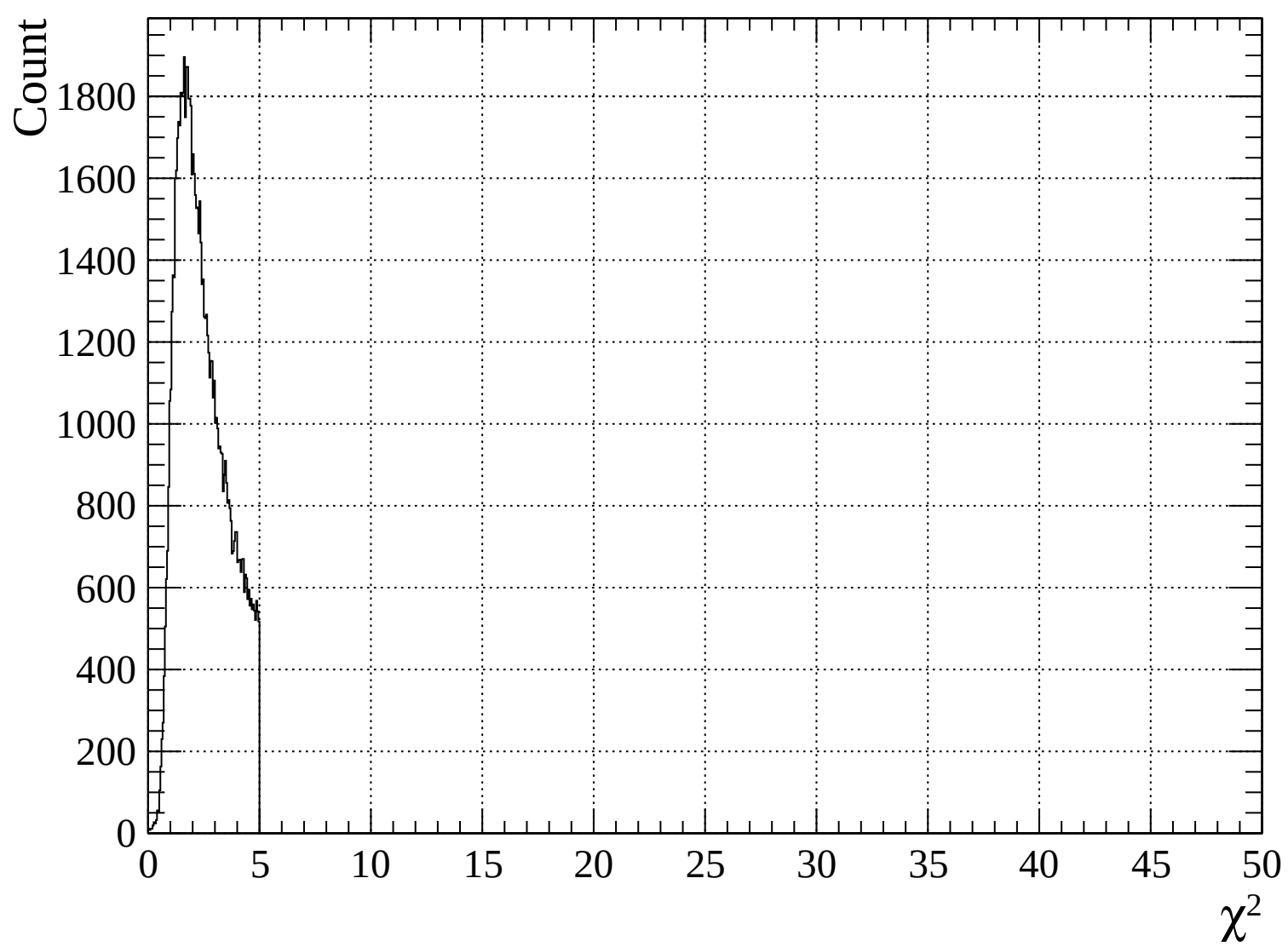


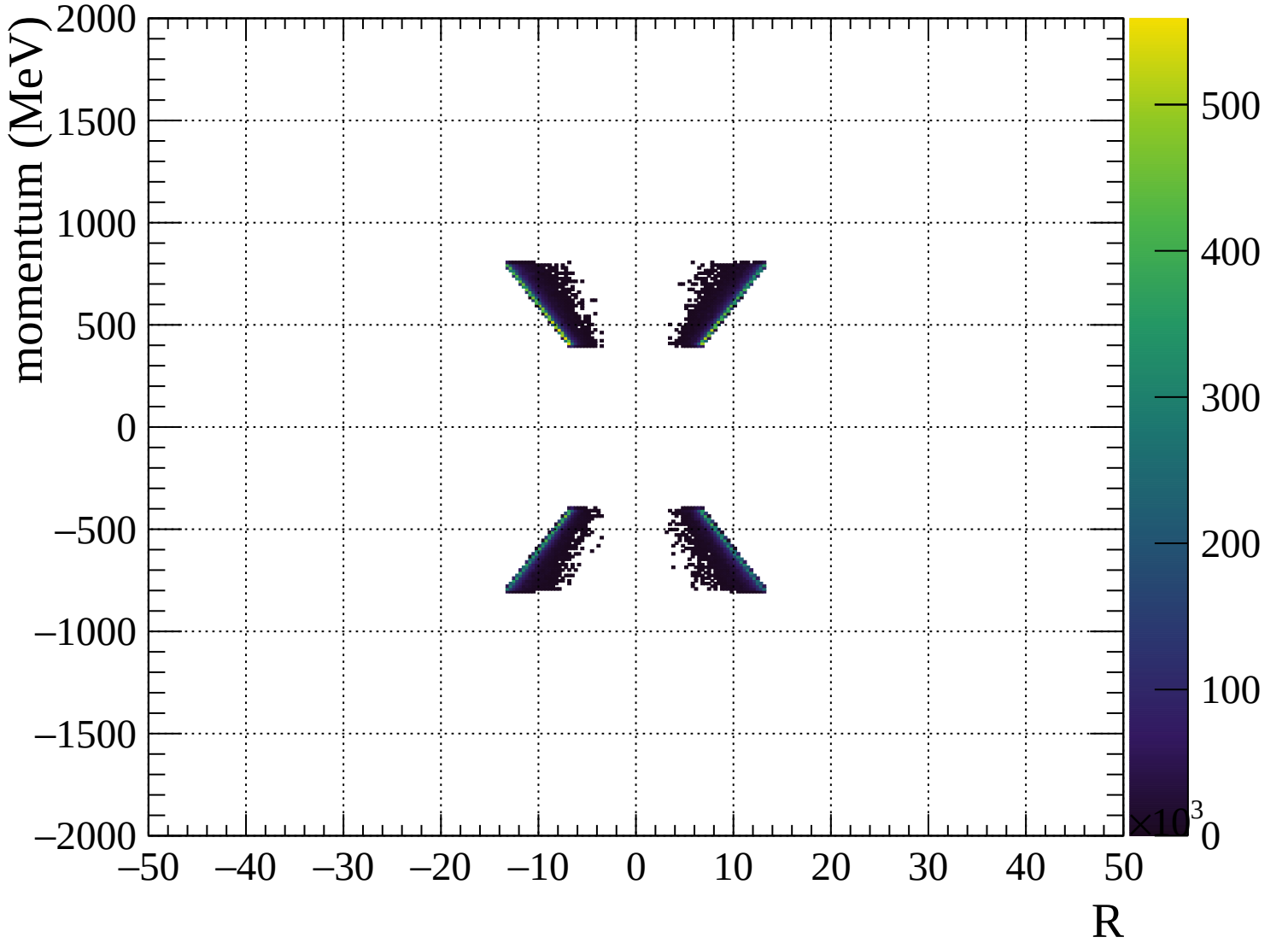


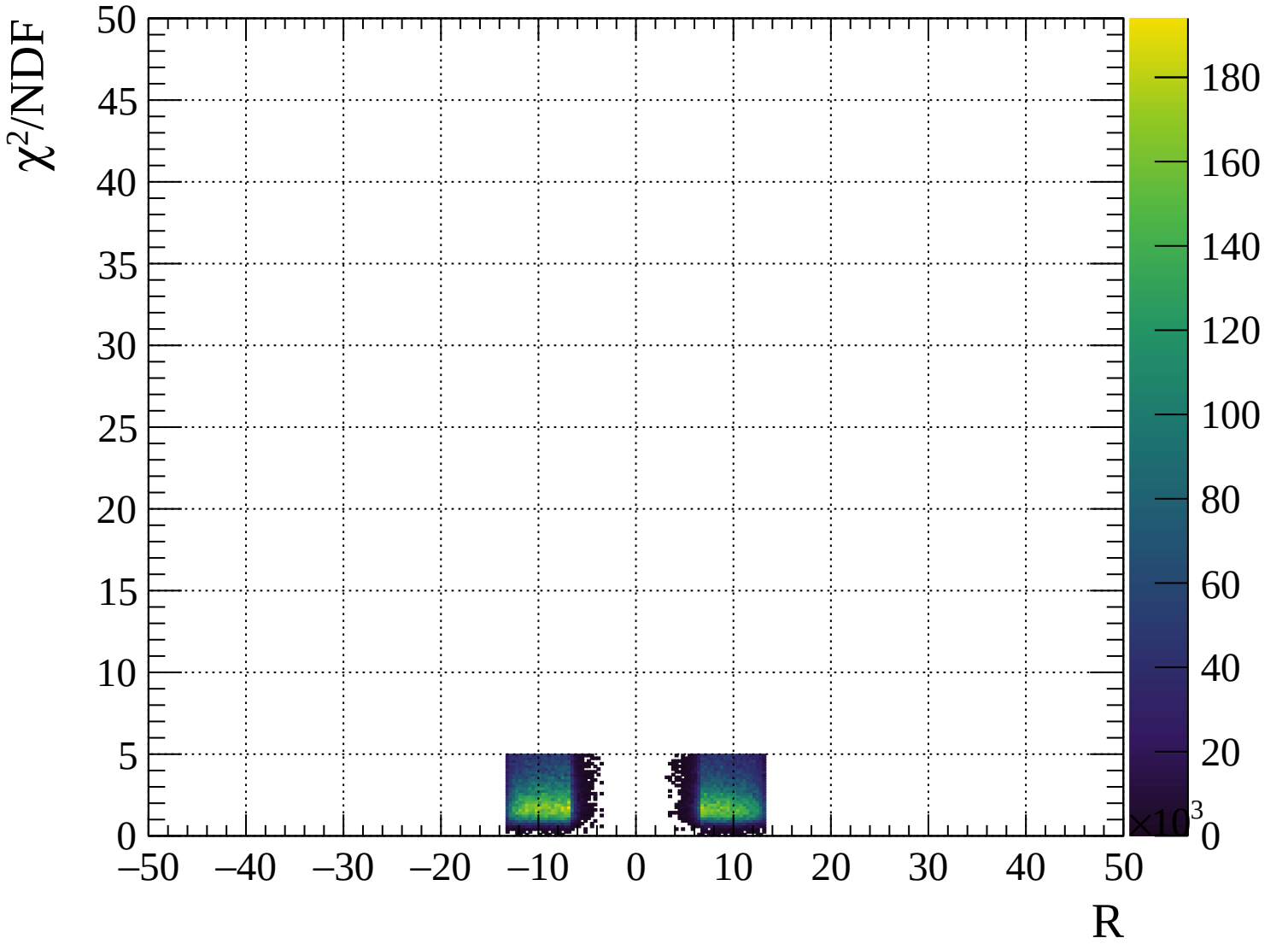






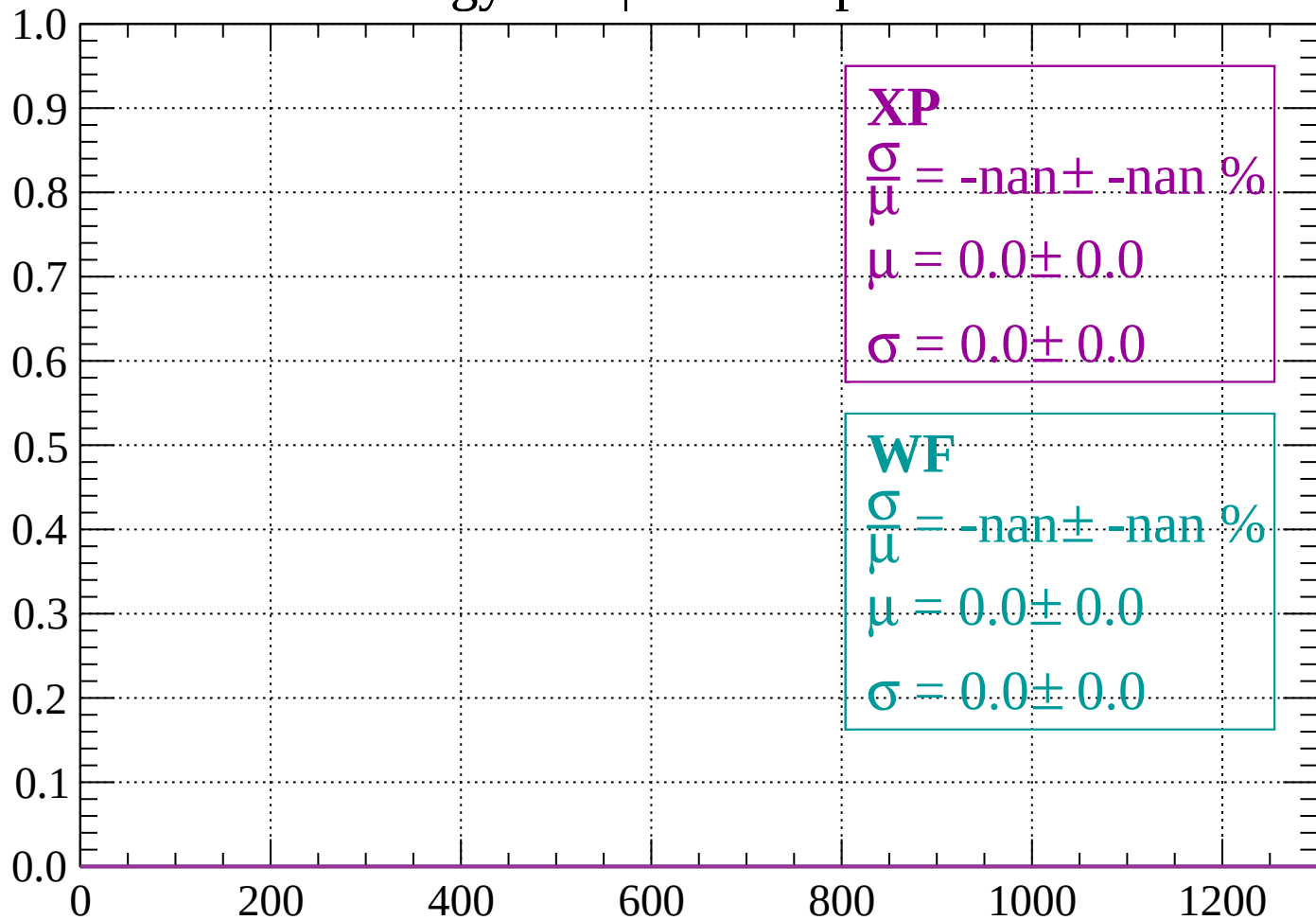






Energy loss | -2000 < p < -1960

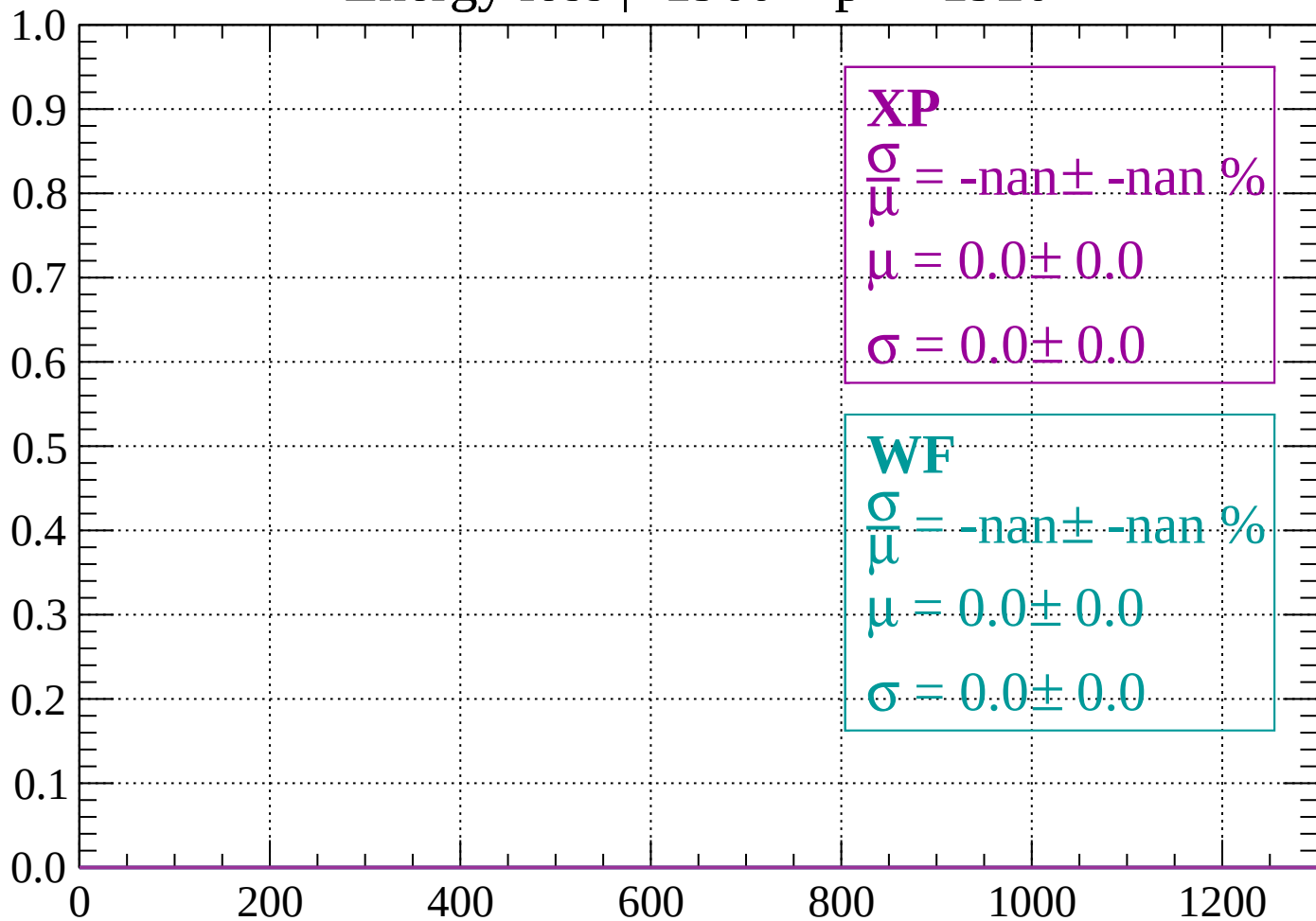
Count



dE/dx (ADC counts/cm)

Energy loss | -1960 < p < -1920

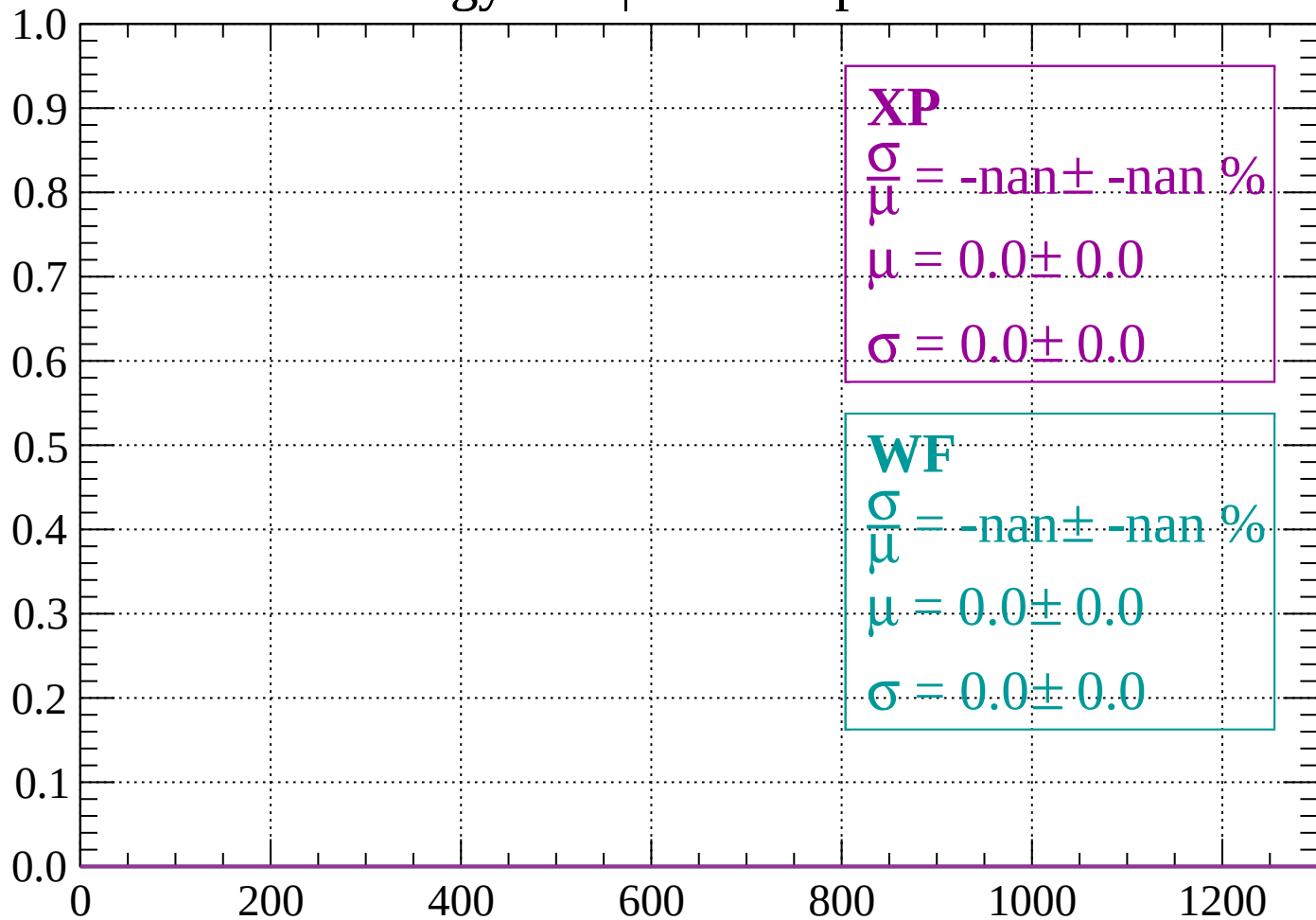
Count



dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1880

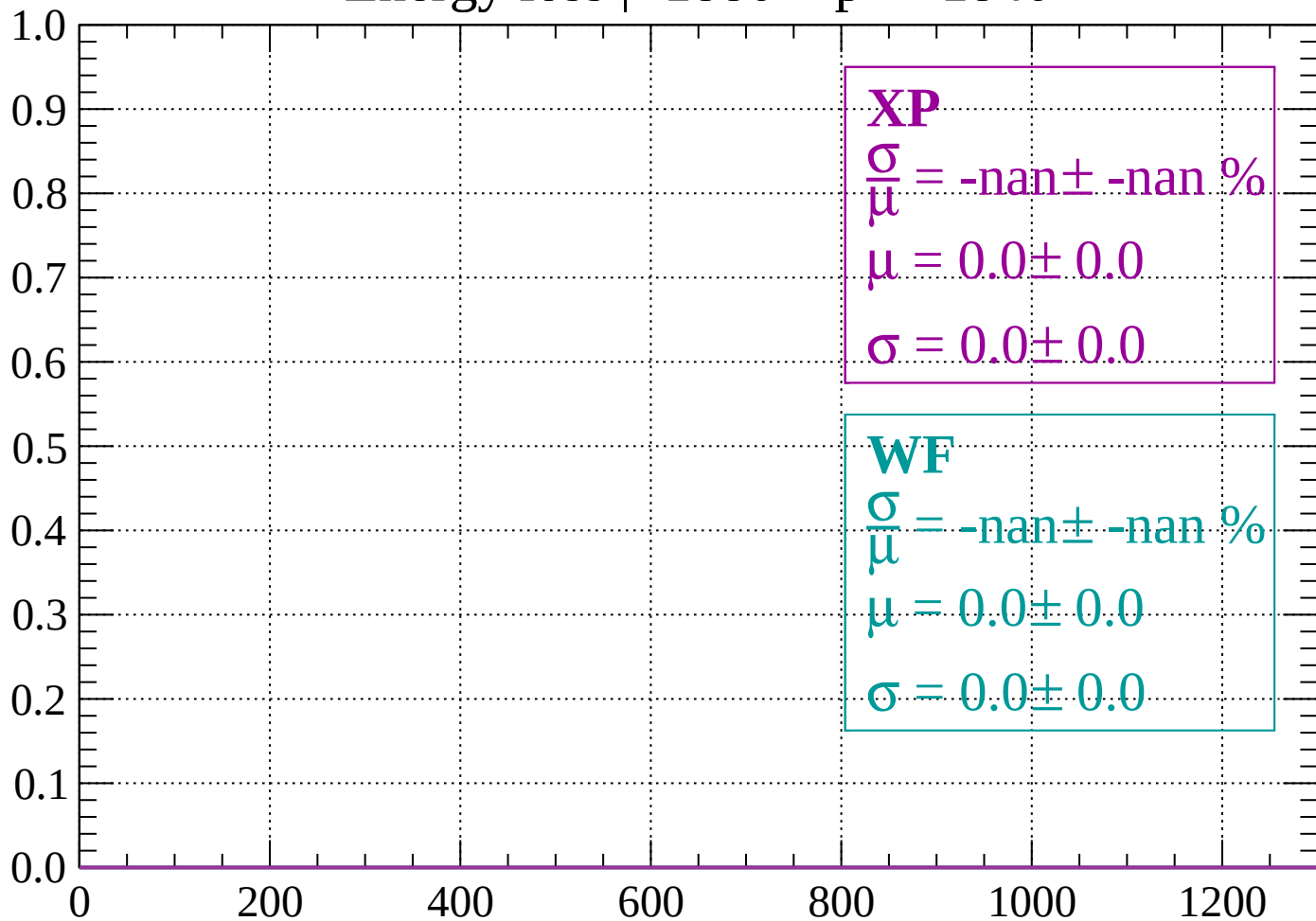
Count



dE/dx (ADC counts/cm)

Energy loss | $-1880 < p < -1840$

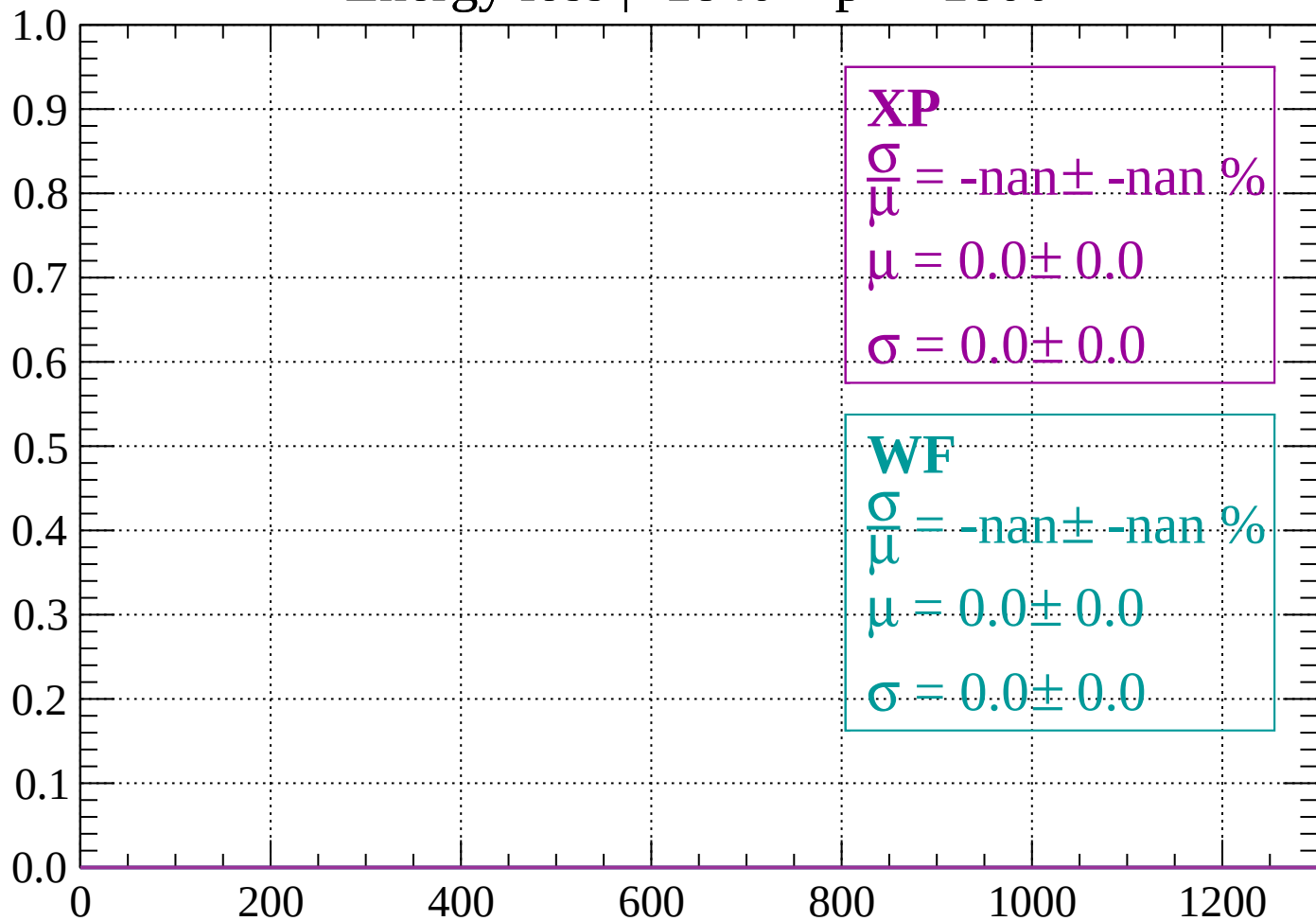
Count



dE/dx (ADC counts/cm)

Energy loss | -1840 < p < -1800

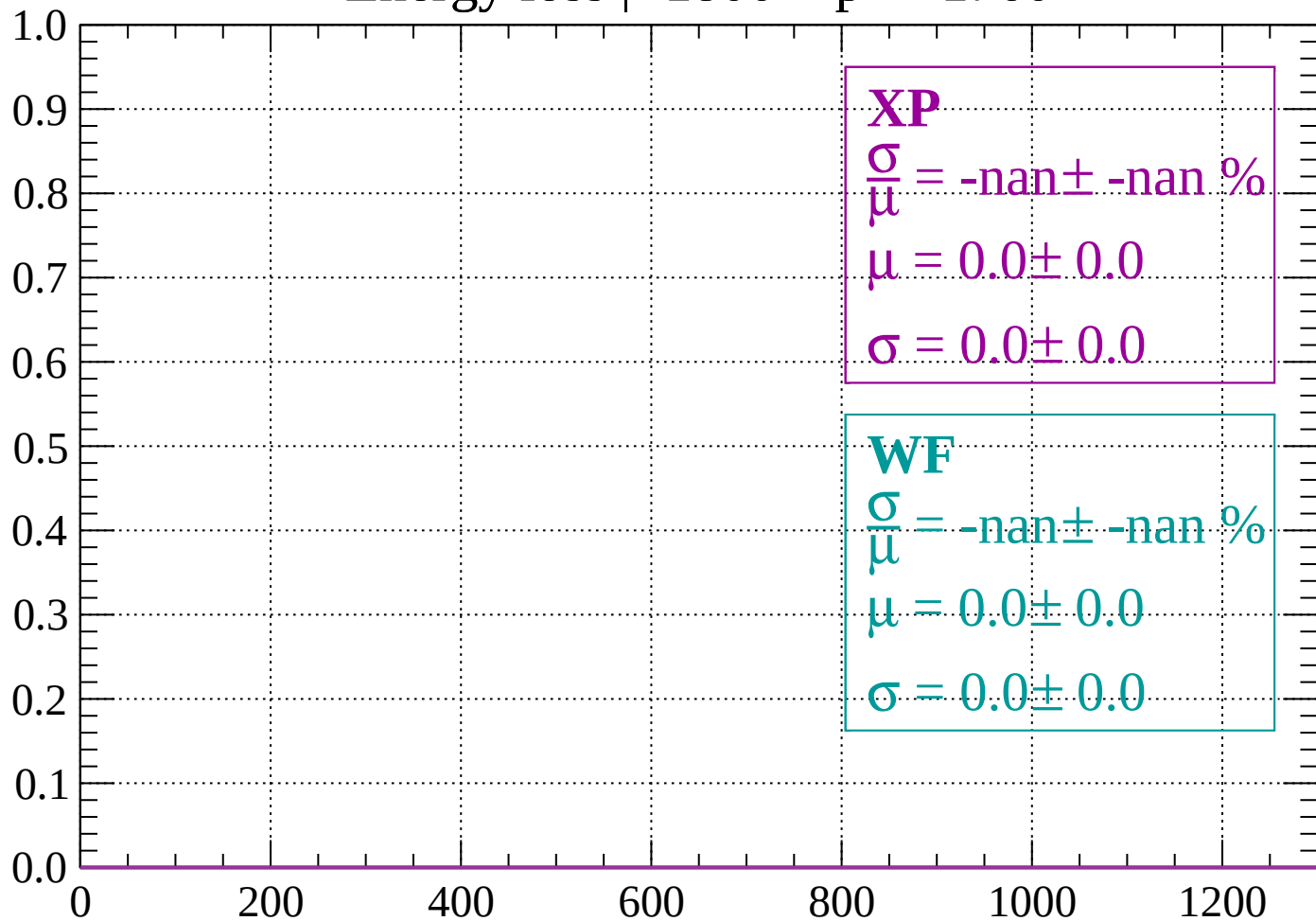
Count



dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1760$

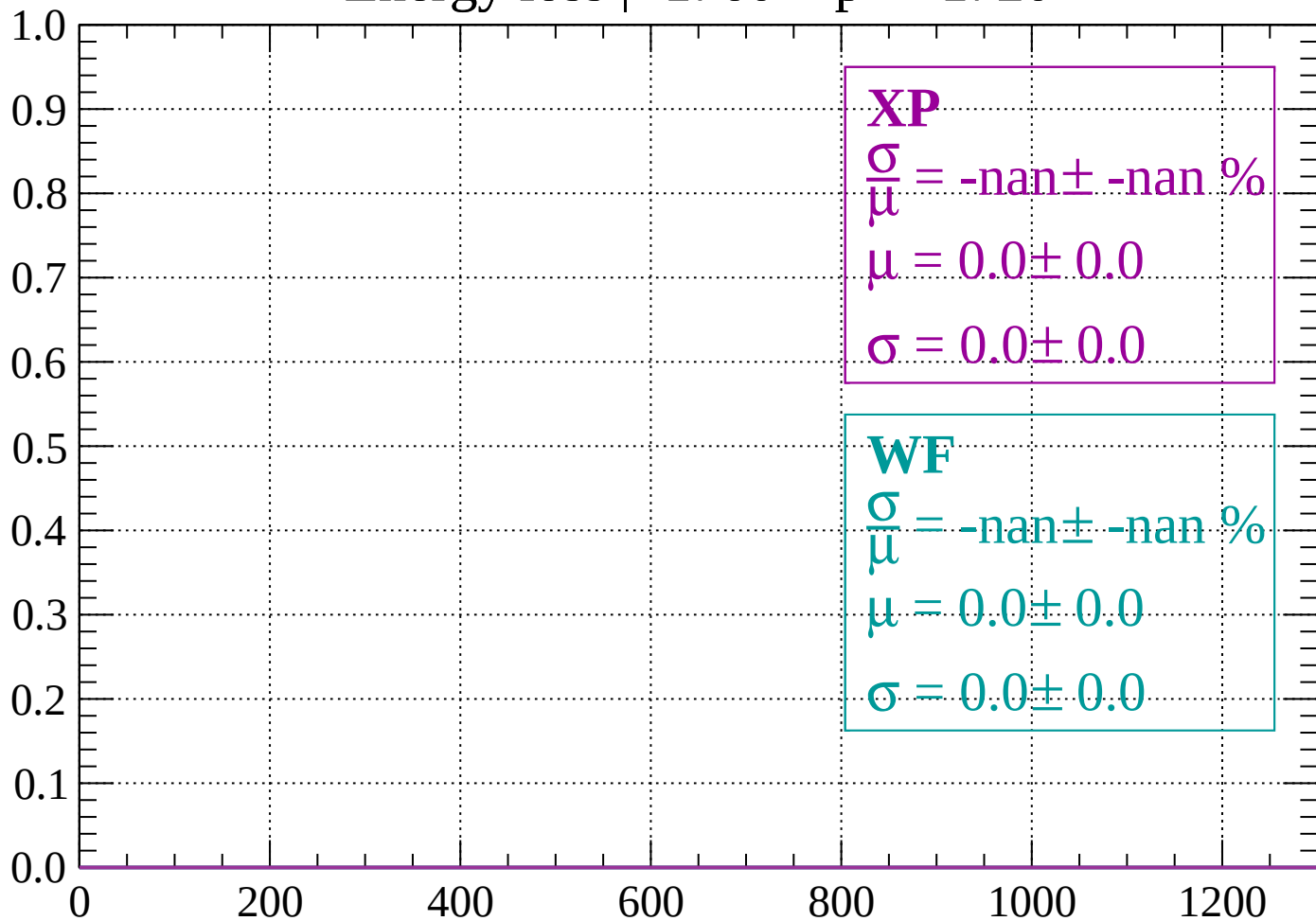
Count



dE/dx (ADC counts/cm)

Energy loss | $-1760 < p < -1720$

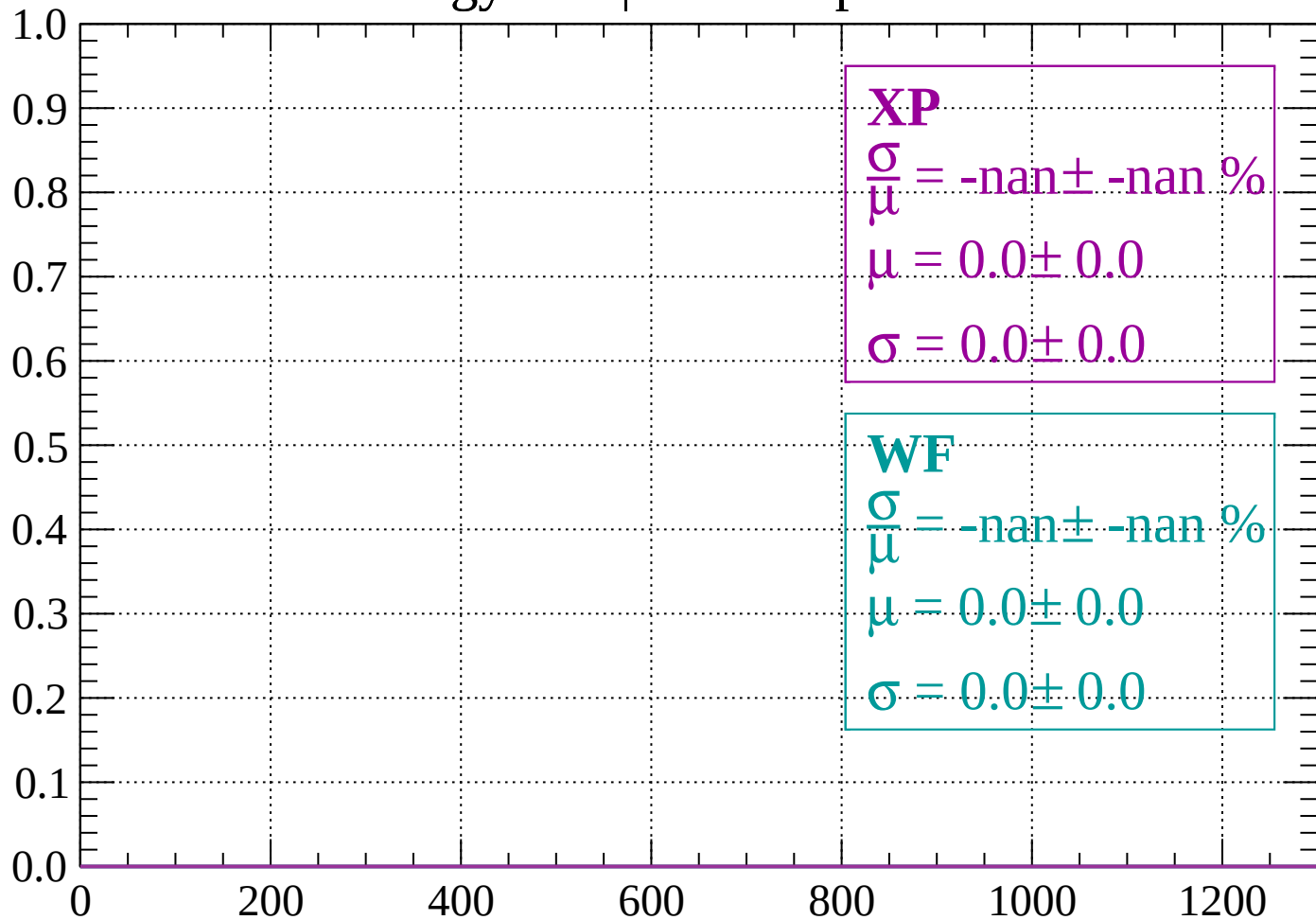
Count



dE/dx (ADC counts/cm)

Energy loss | -1720 < p < -1680

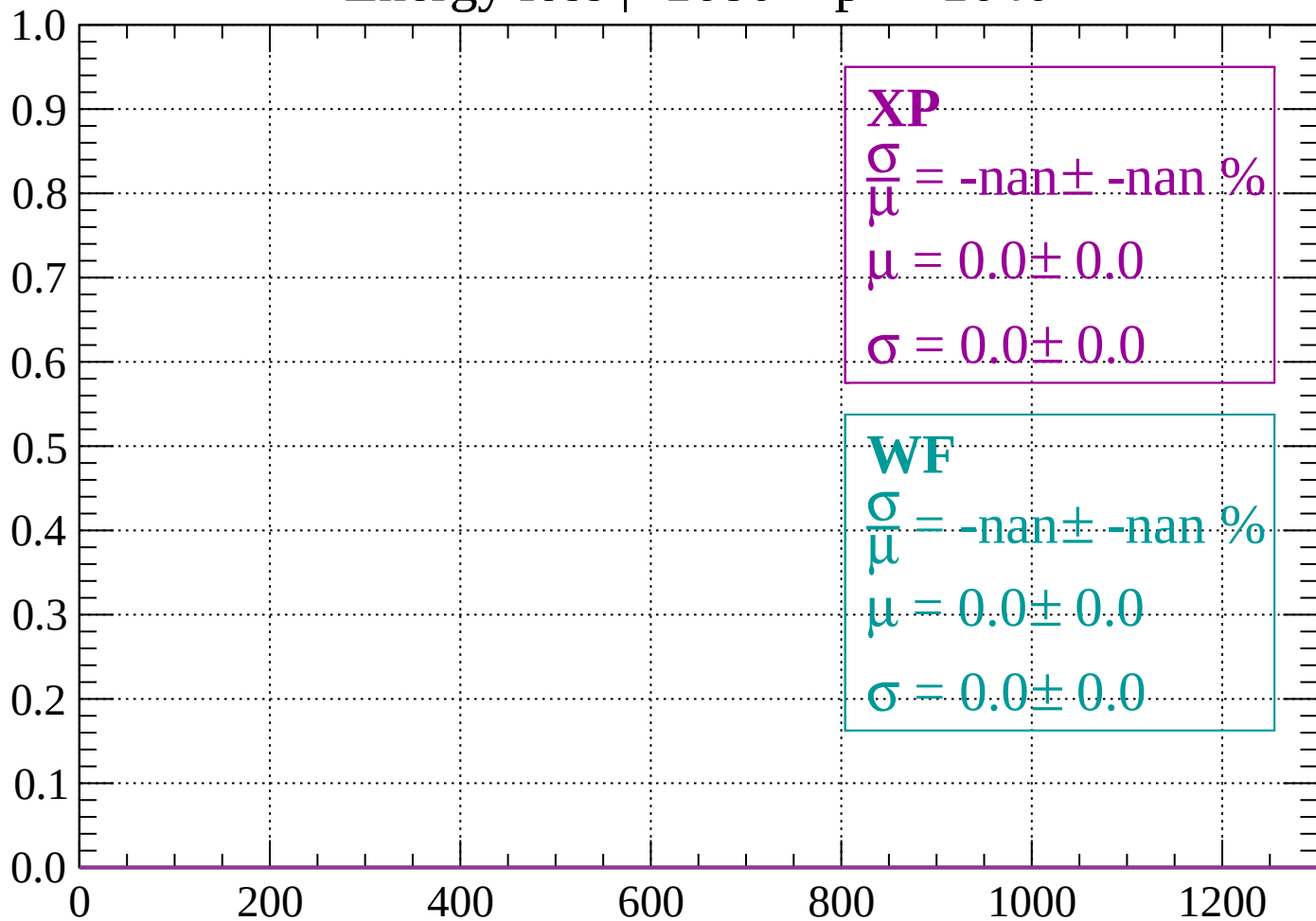
Count



dE/dx (ADC counts/cm)

Energy loss | -1680 < p < -1640

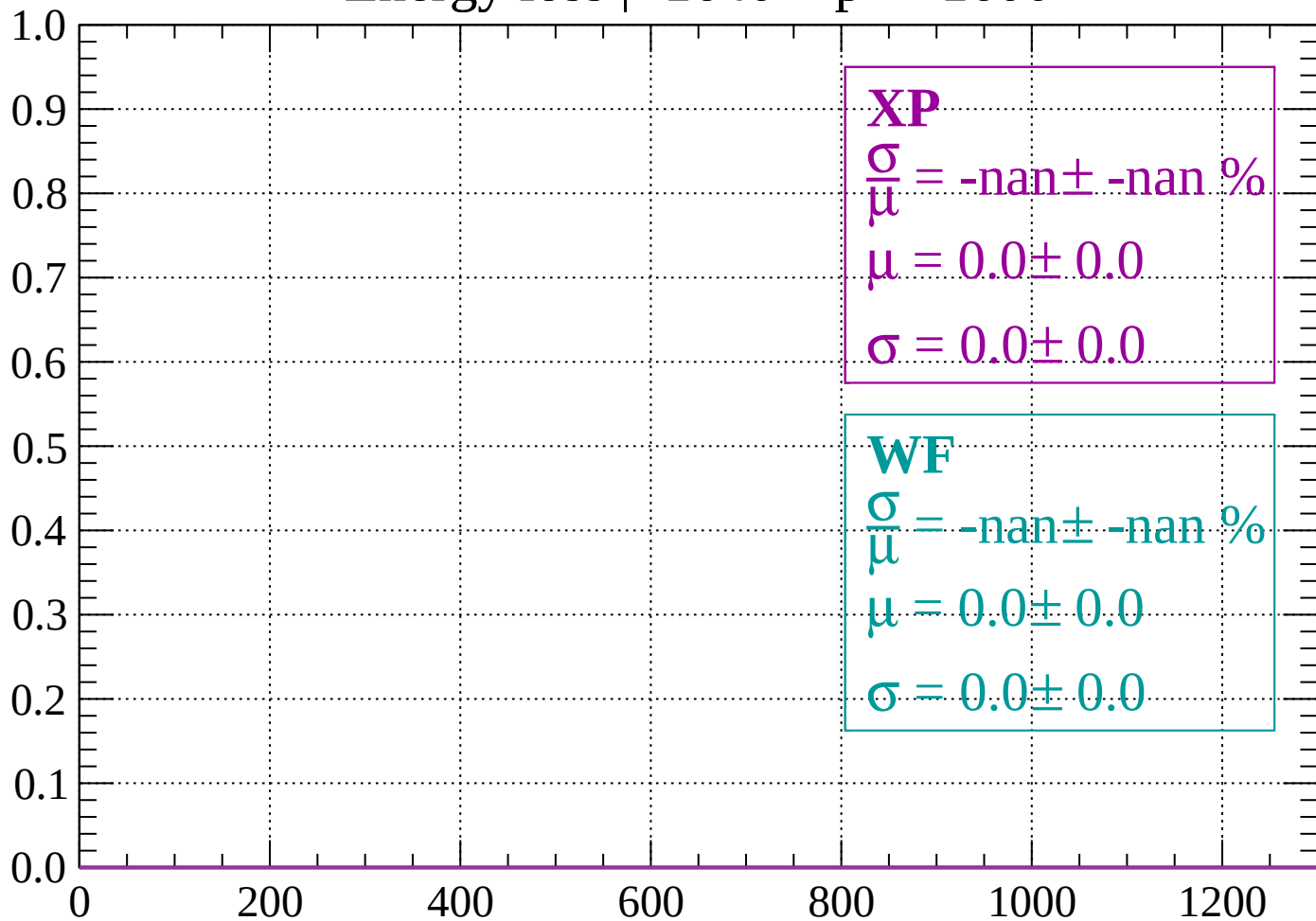
Count



dE/dx (ADC counts/cm)

Energy loss | -1640 < p < -1600

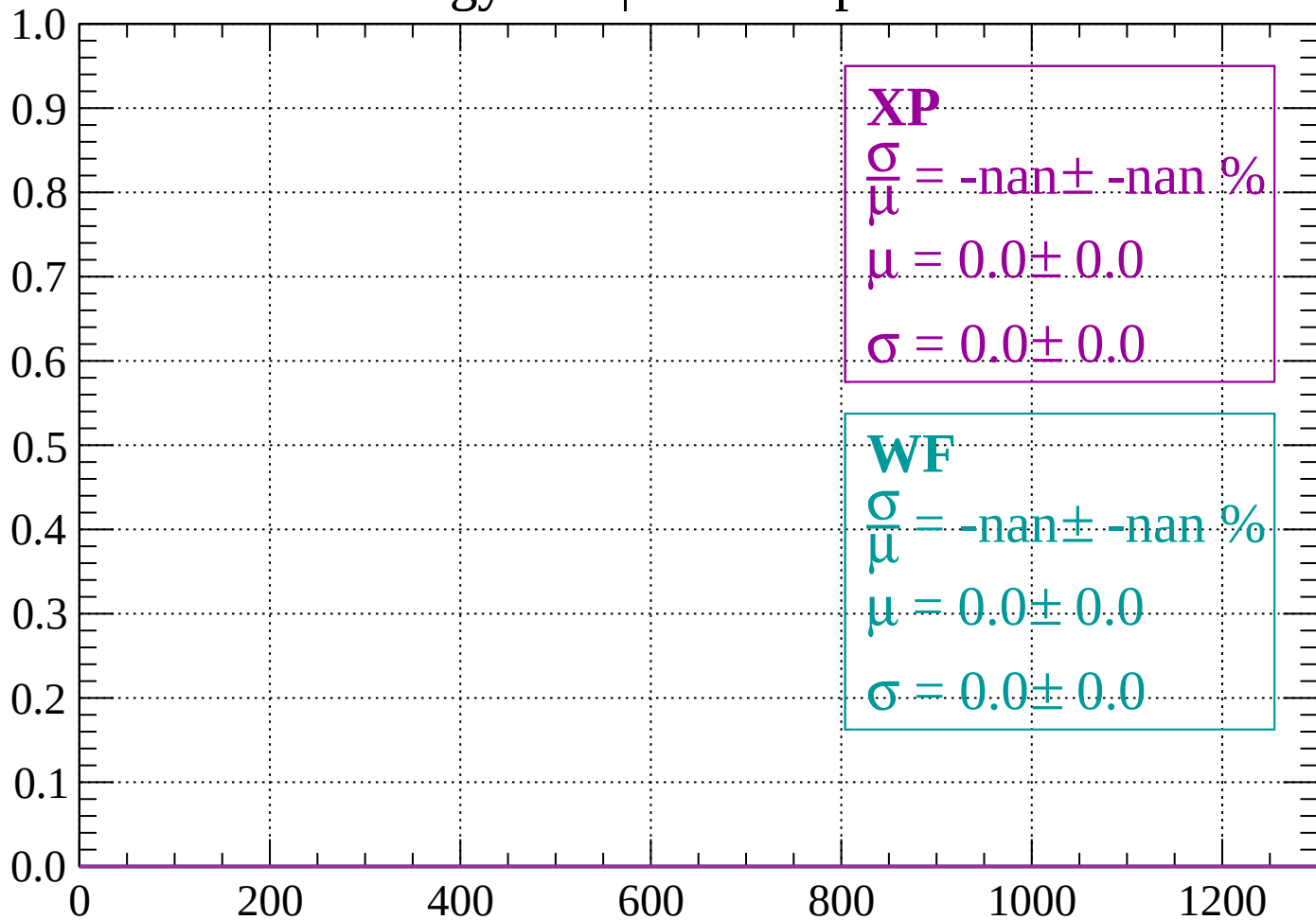
Count



dE/dx (ADC counts/cm)

Energy loss | -1600 < p < -1560

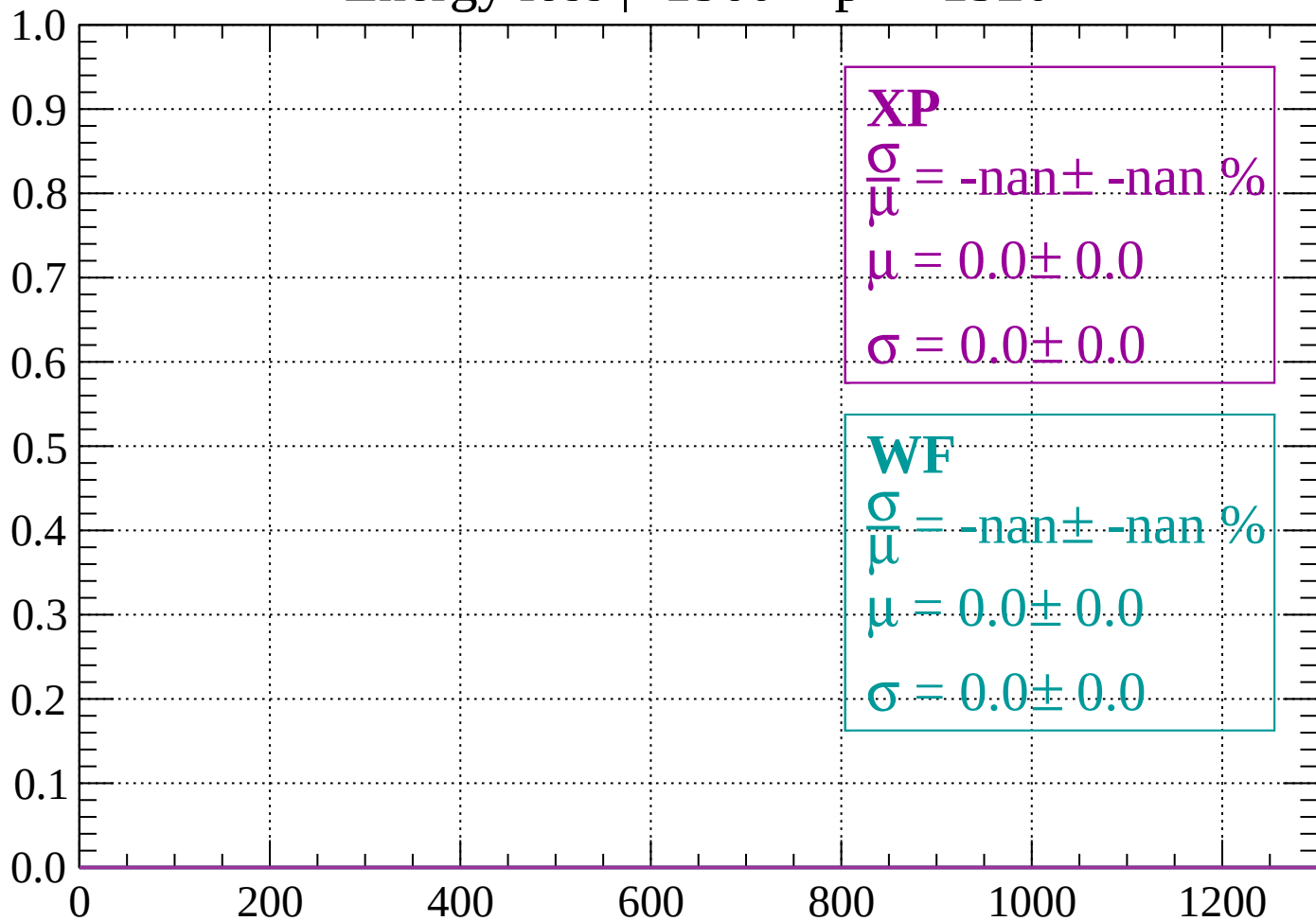
Count



dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1520

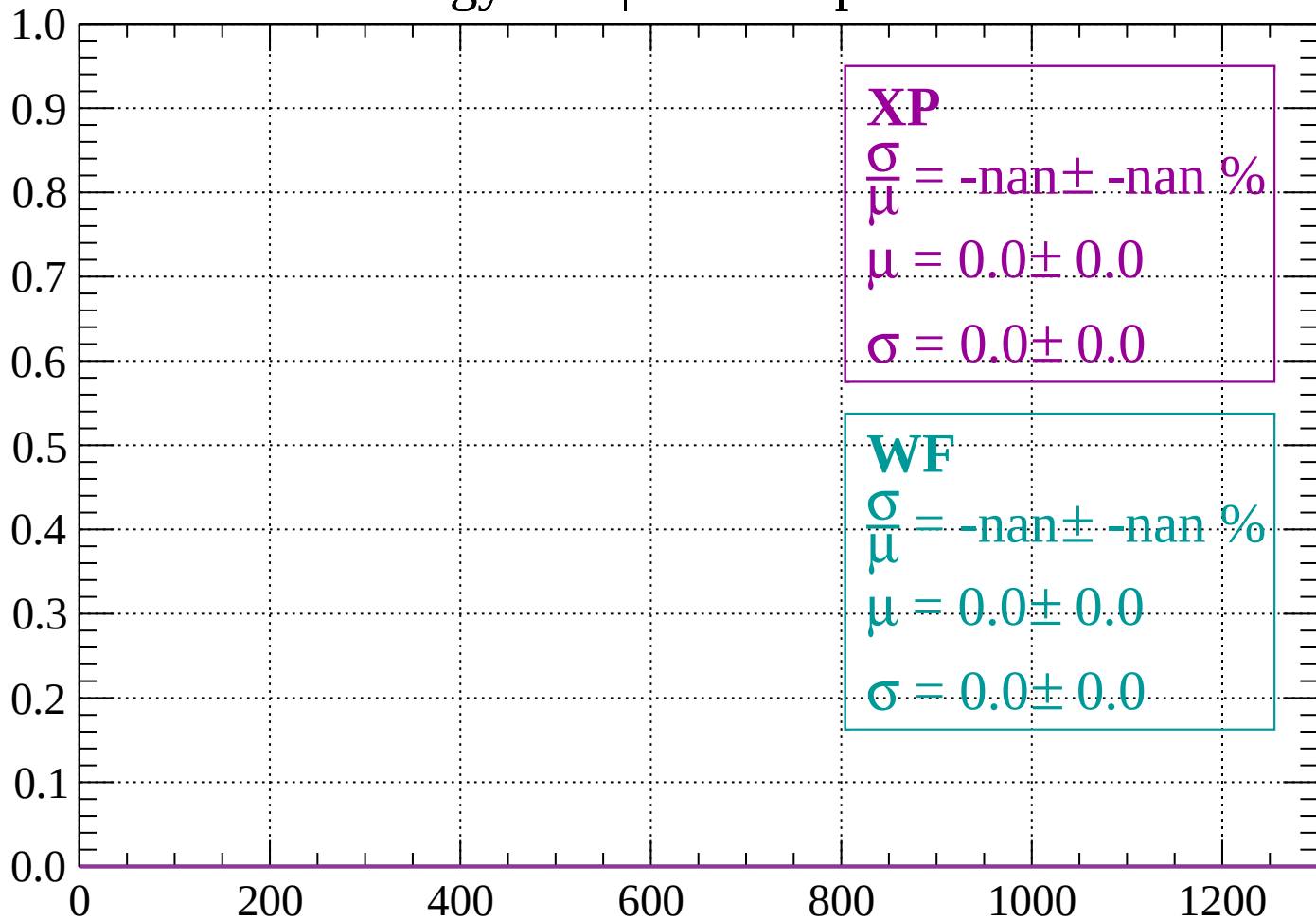
Count



dE/dx (ADC counts/cm)

Energy loss | -1520 < p < -1480

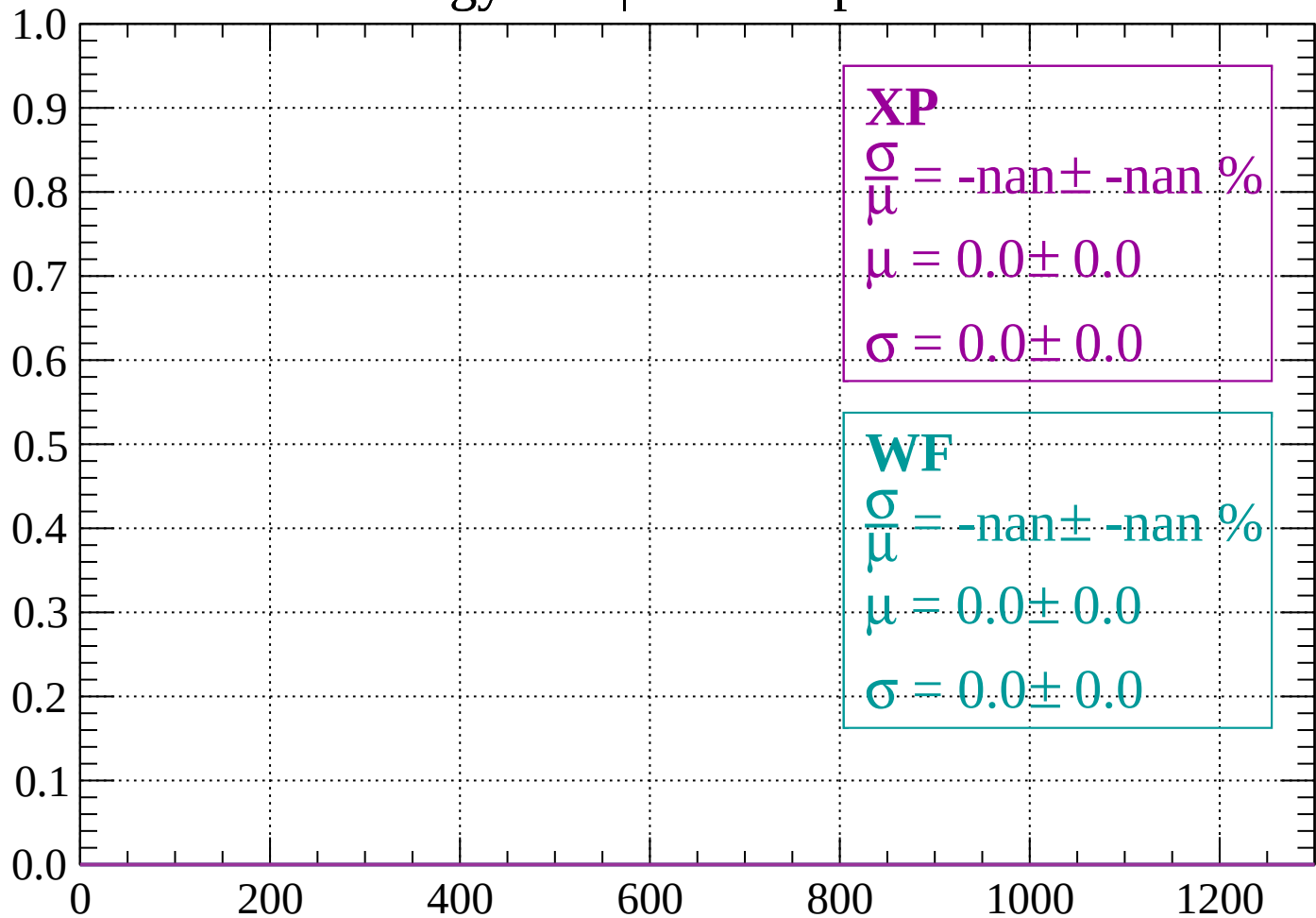
Count



dE/dx (ADC counts/cm)

Energy loss | $-1480 < p < -1440$

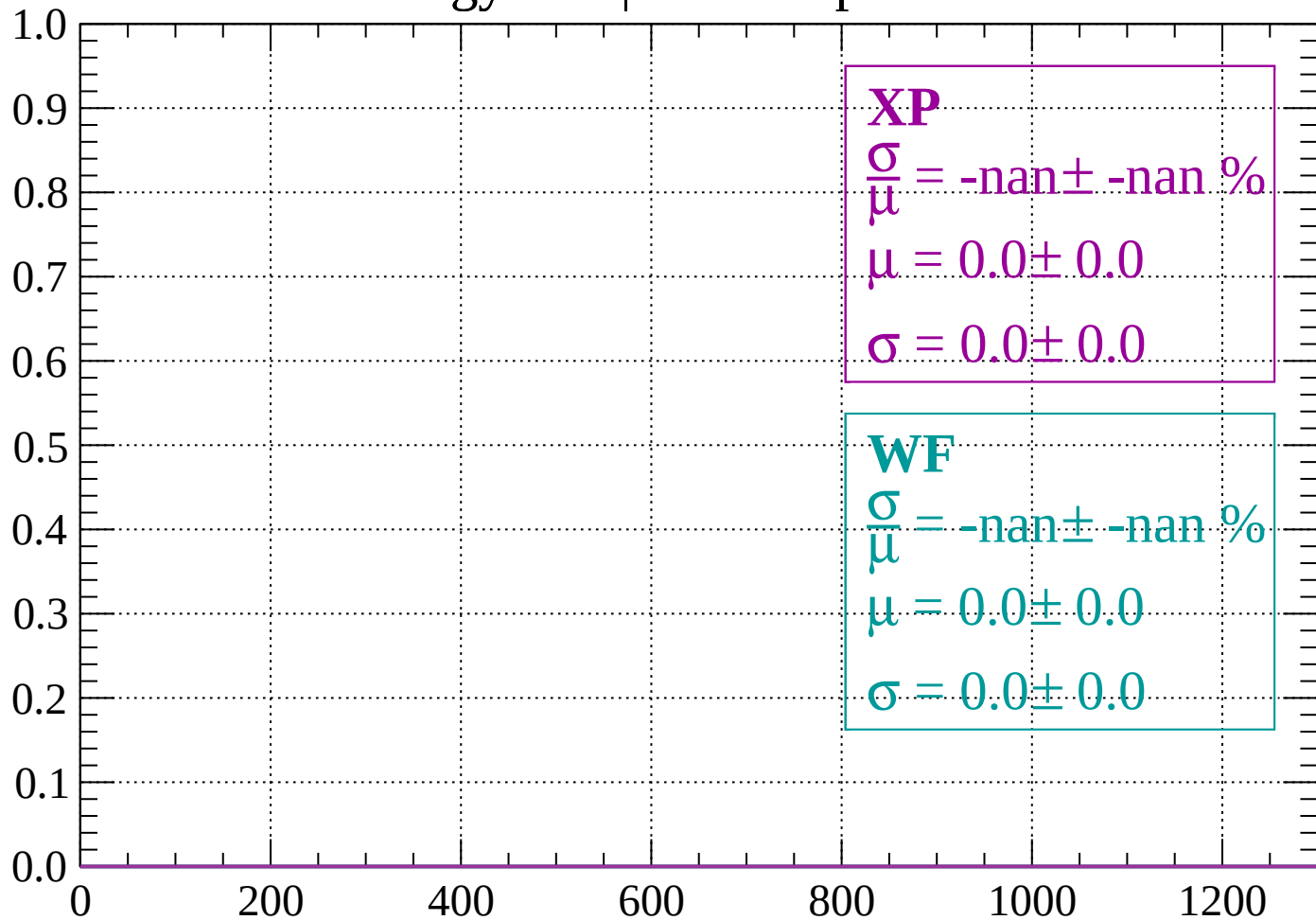
Count



dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1400$

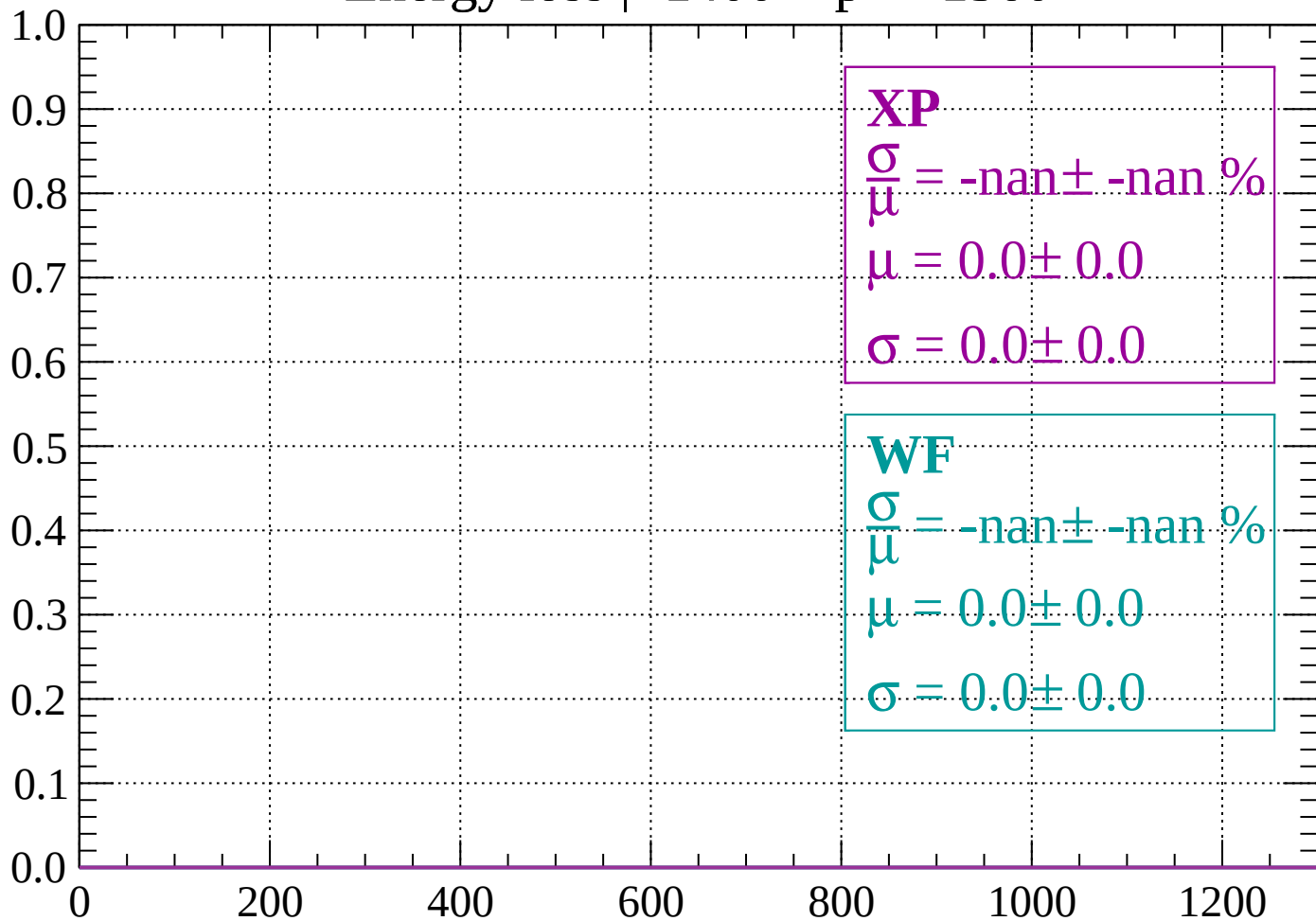
Count



dE/dx (ADC counts/cm)

Energy loss | $-1400 < p < -1360$

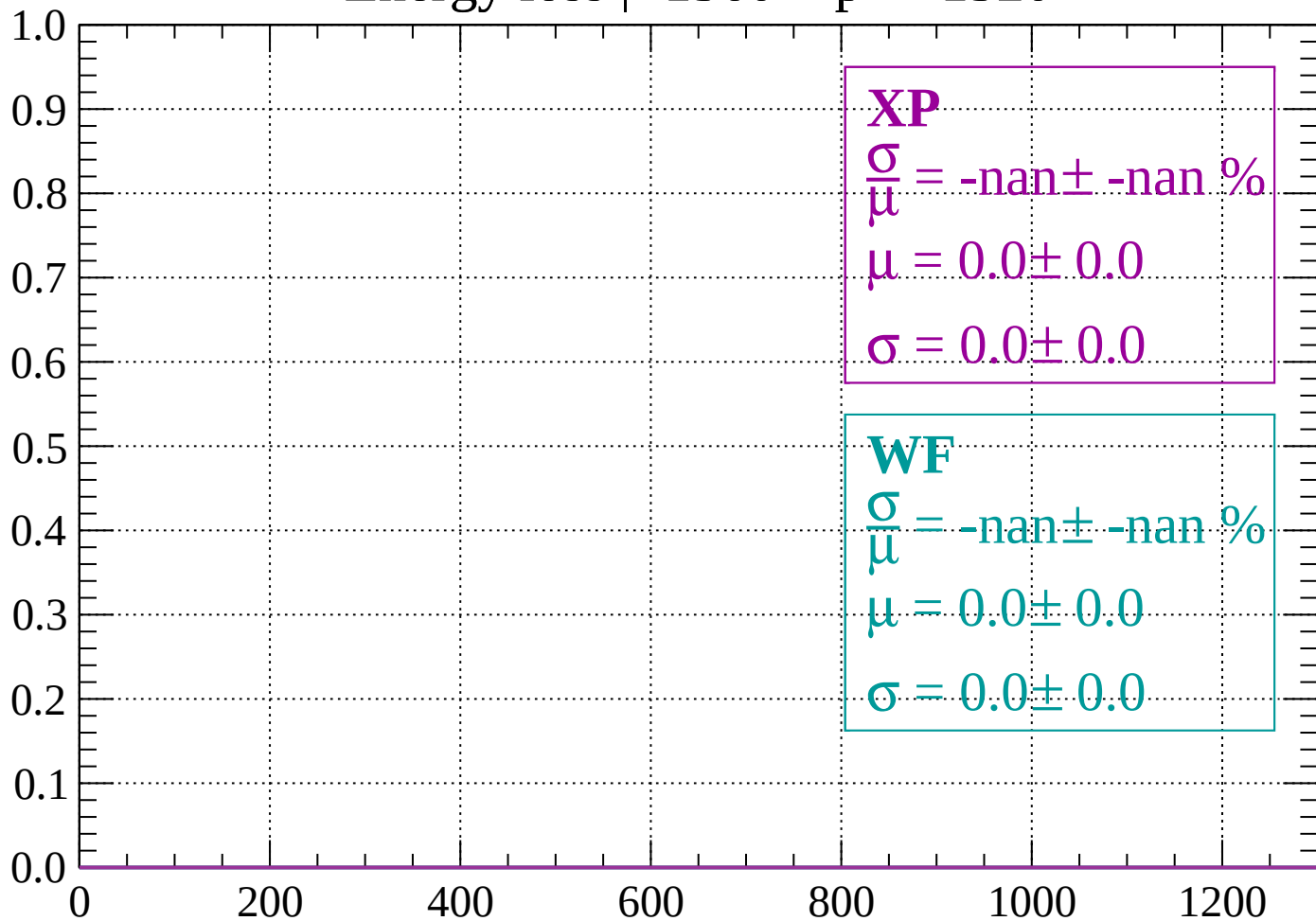
Count



dE/dx (ADC counts/cm)

Energy loss | -1360 < p < -1320

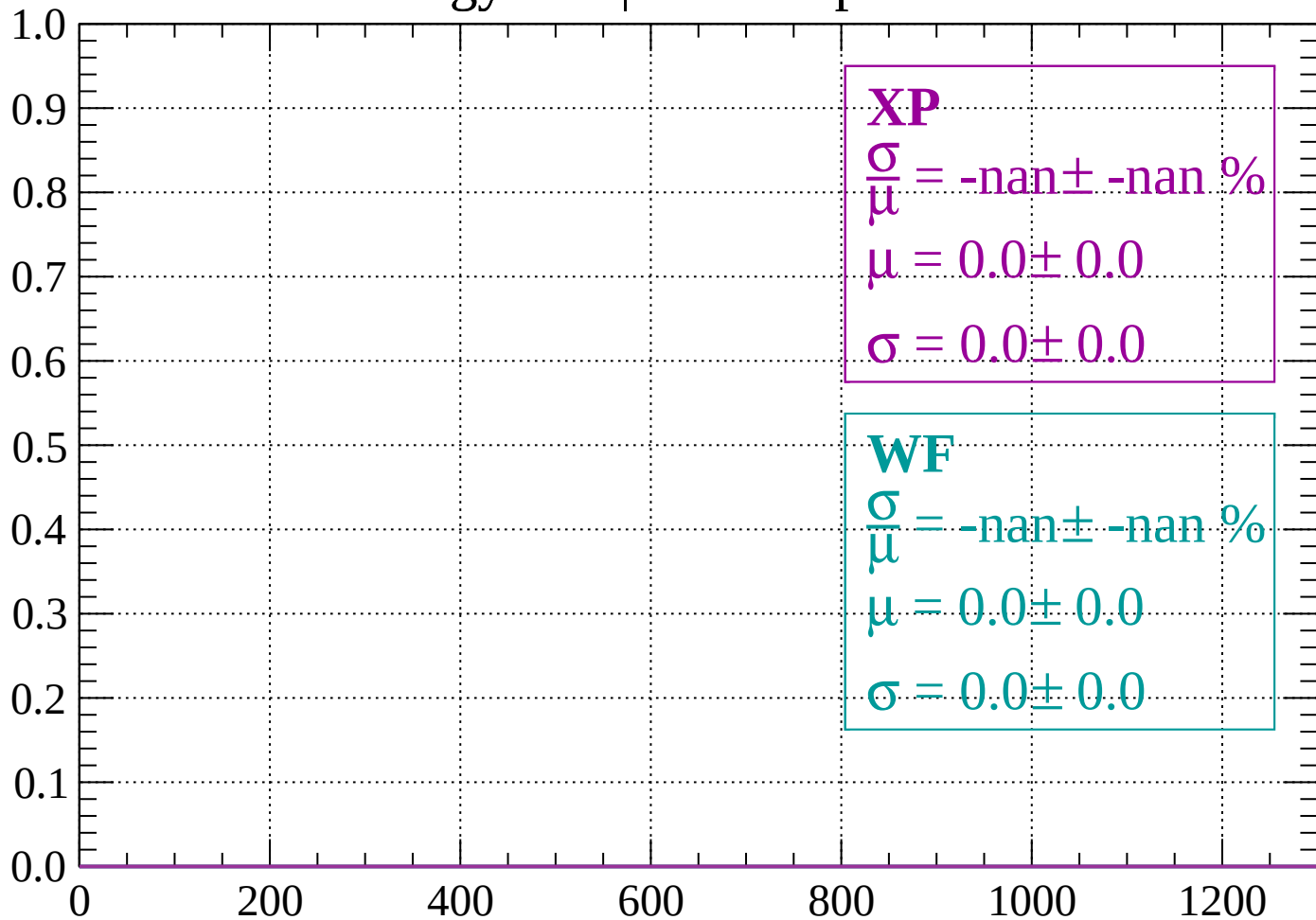
Count



dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1280

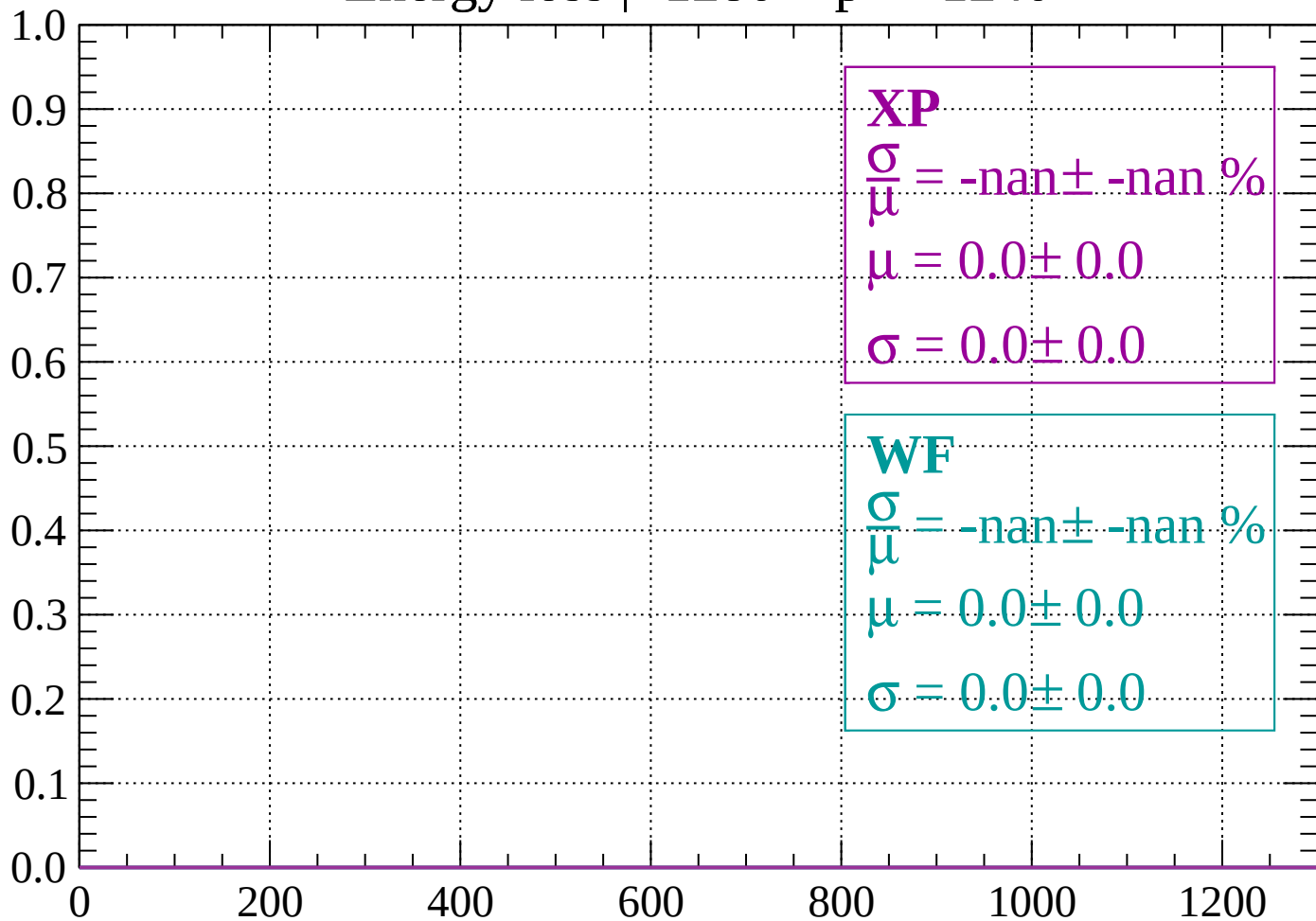
Count



dE/dx (ADC counts/cm)

Energy loss | $-1280 < p < -1240$

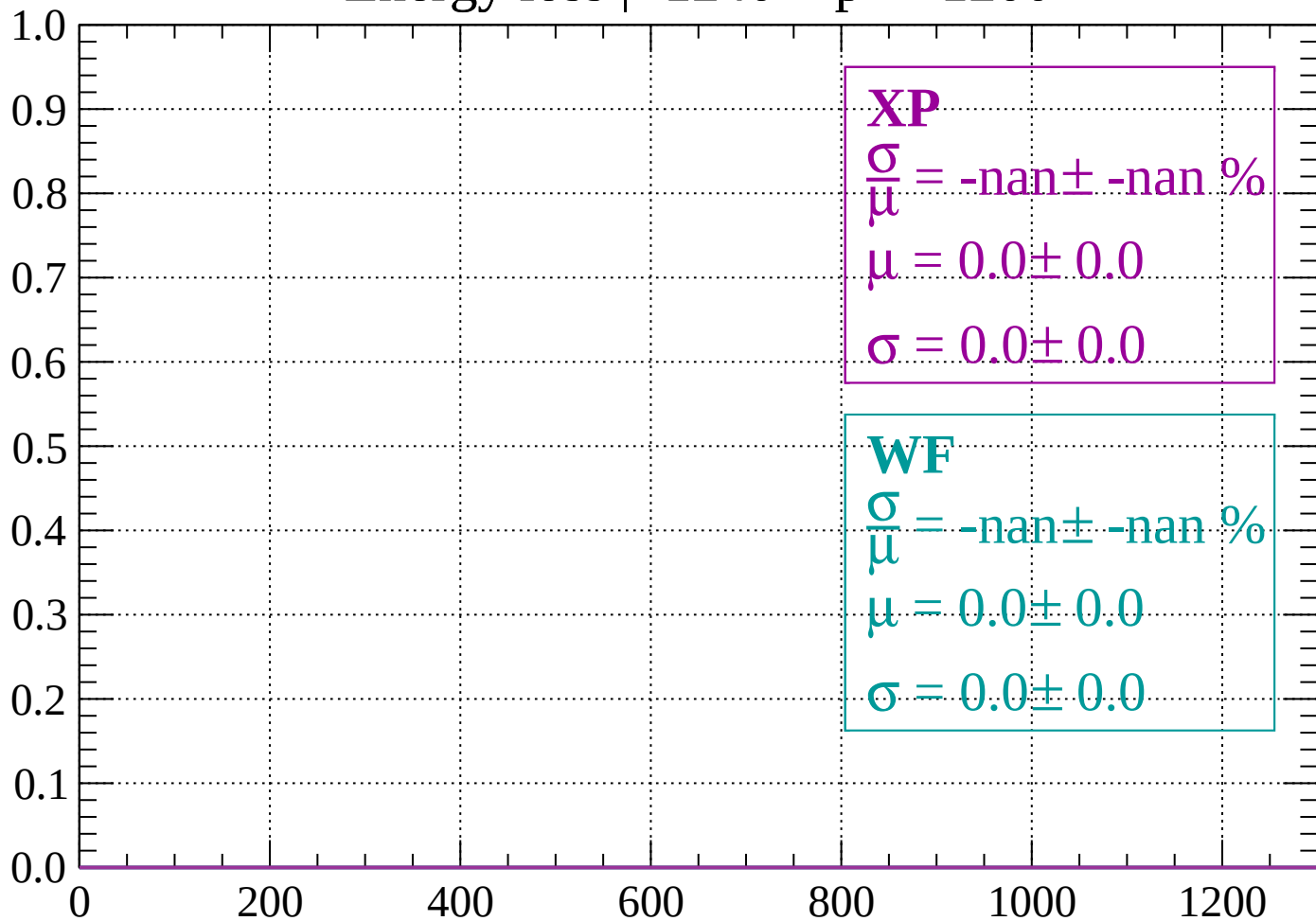
Count



dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

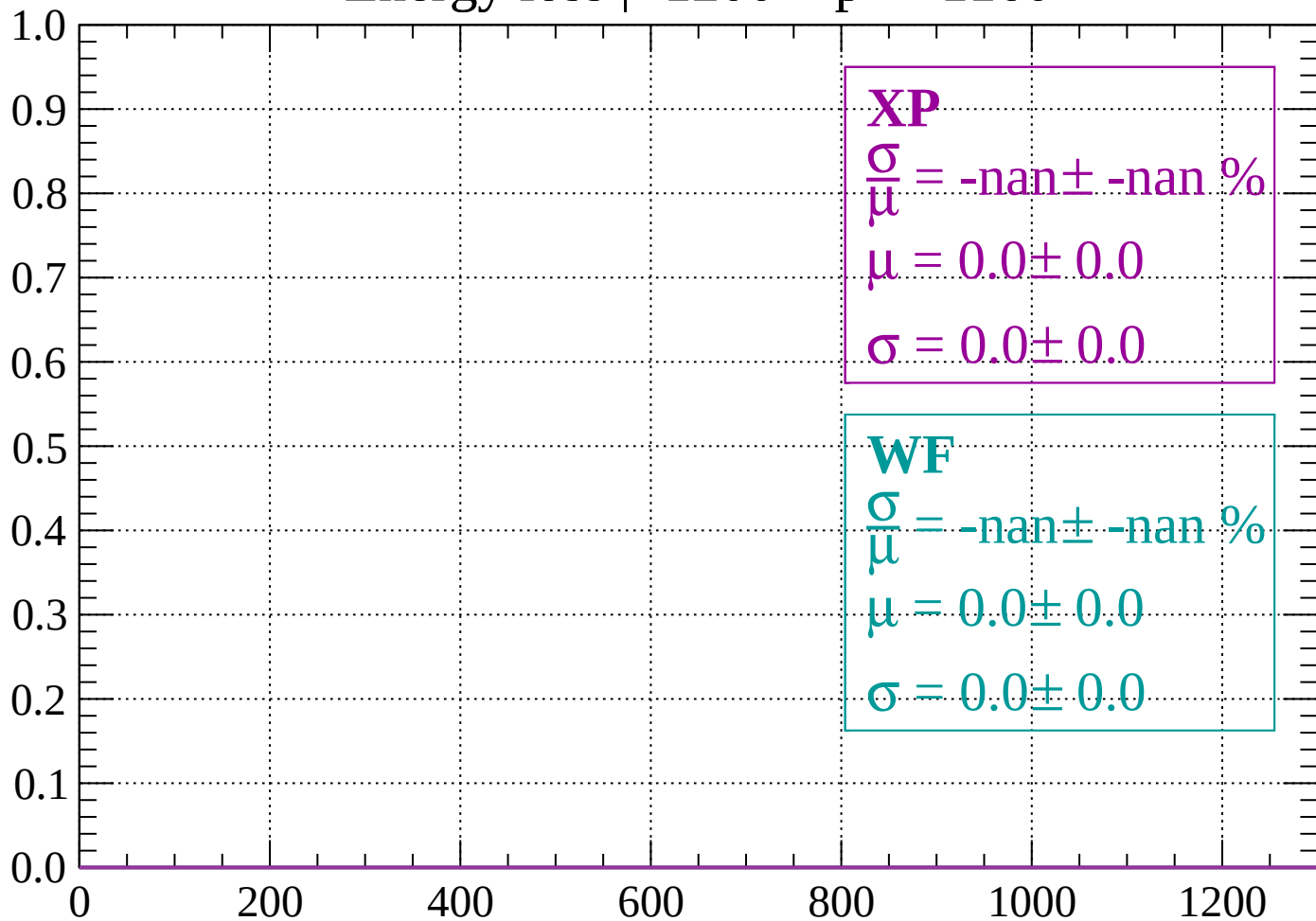
Count



dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1160

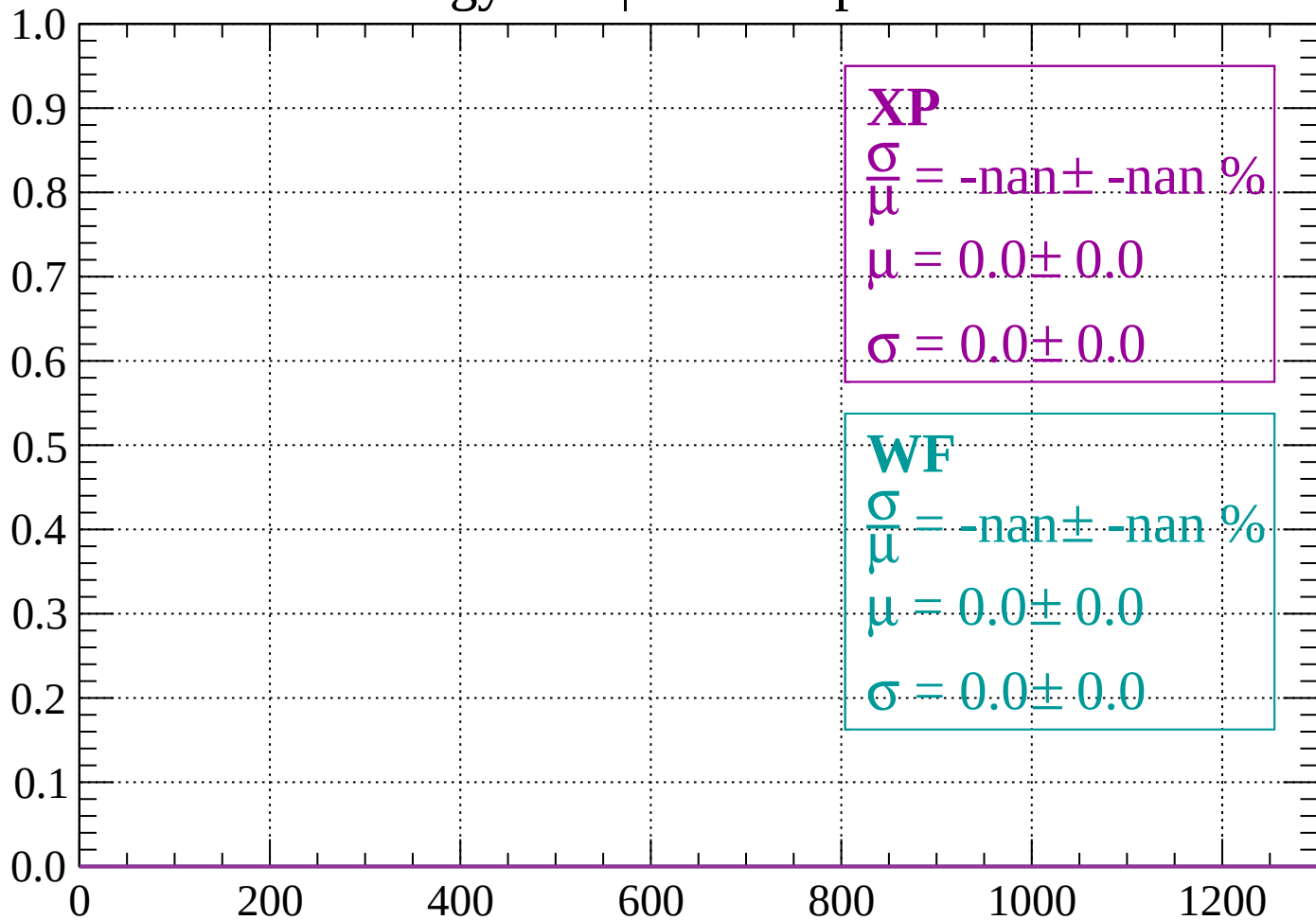
Count



dE/dx (ADC counts/cm)

Energy loss | -1160 < p < -1120

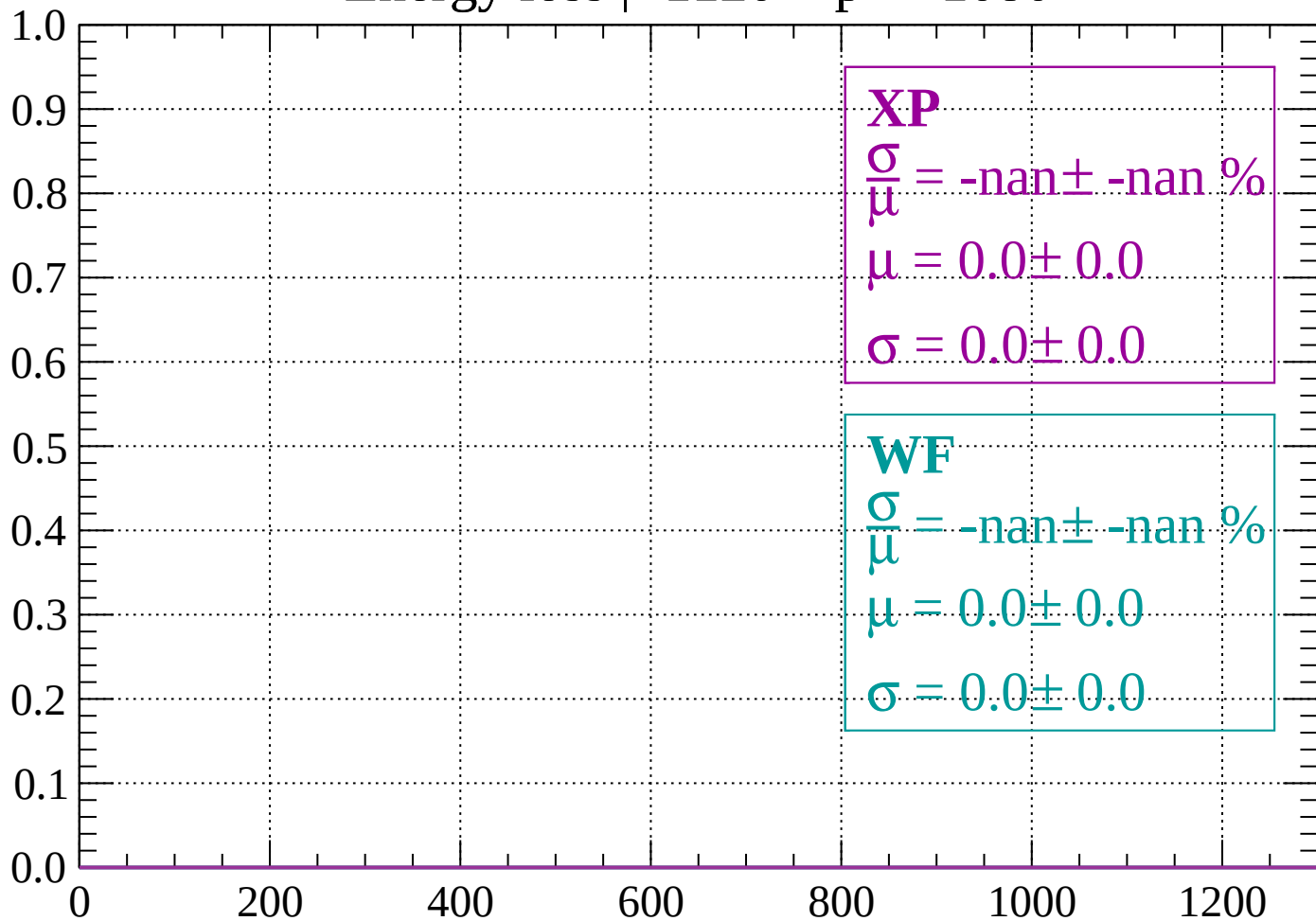
Count



dE/dx (ADC counts/cm)

Energy loss | -1120 < p < -1080

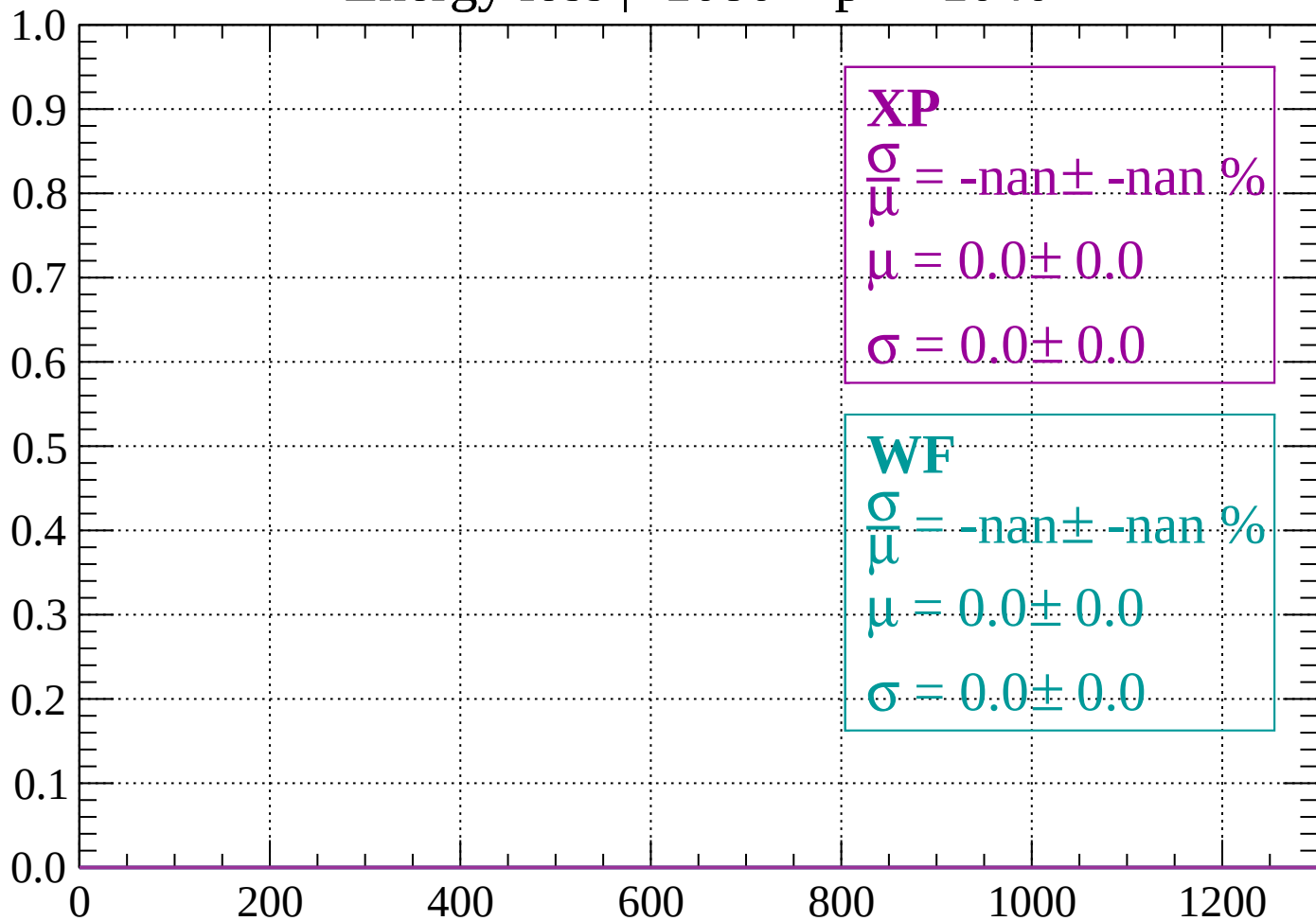
Count



dE/dx (ADC counts/cm)

Energy loss | -1080 < p < -1040

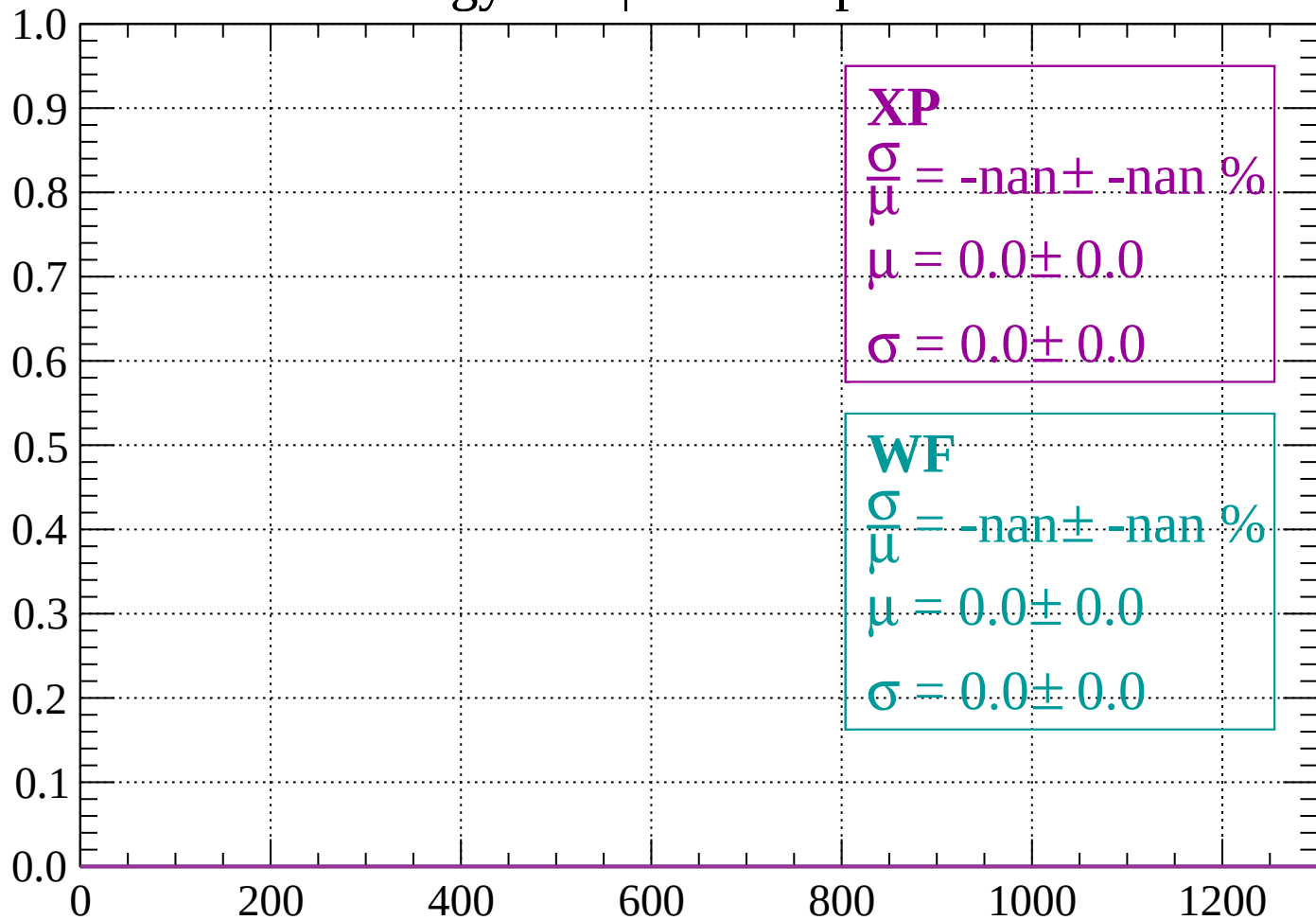
Count



dE/dx (ADC counts/cm)

Energy loss | -1040 < p < -1000

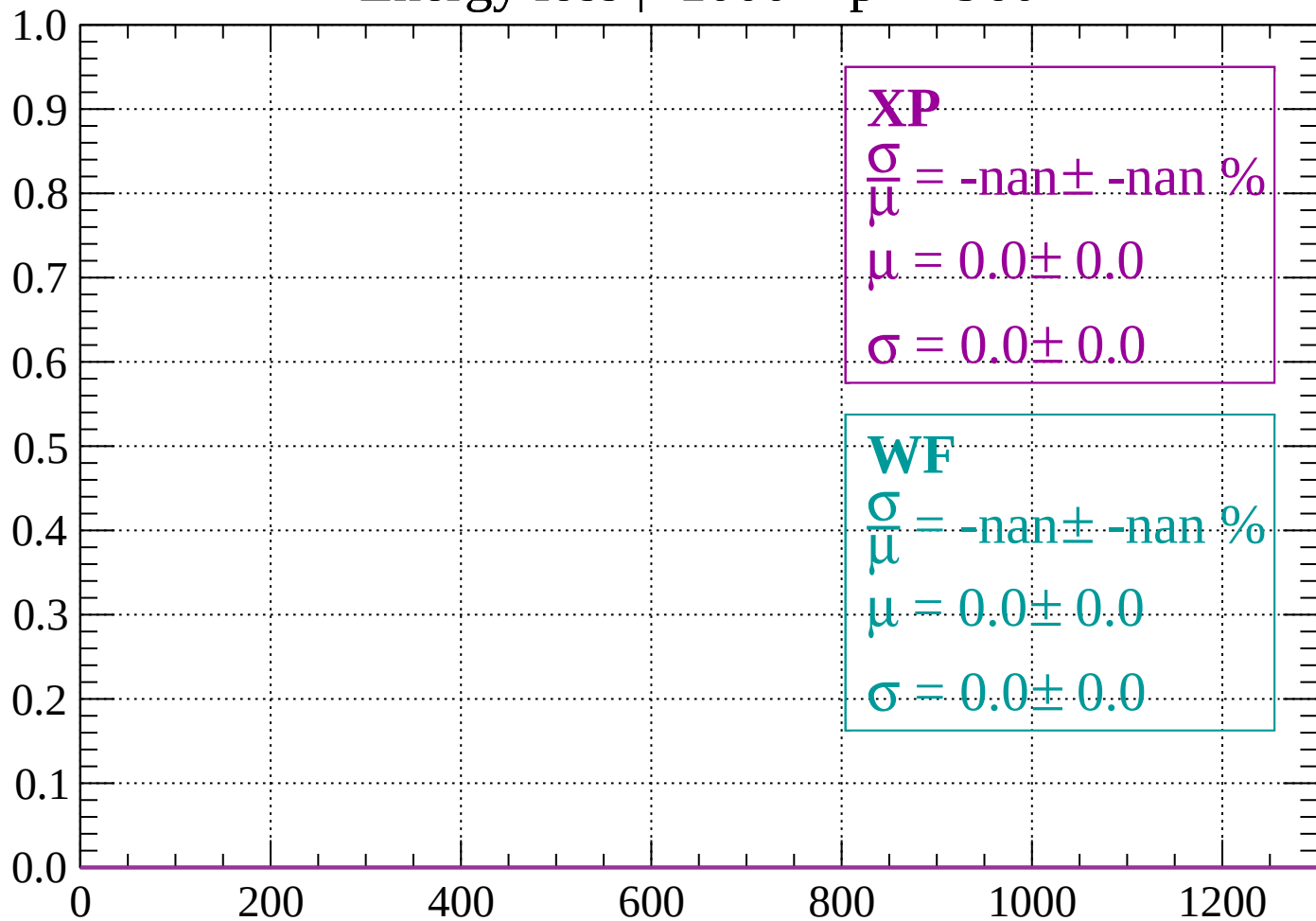
Count



dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

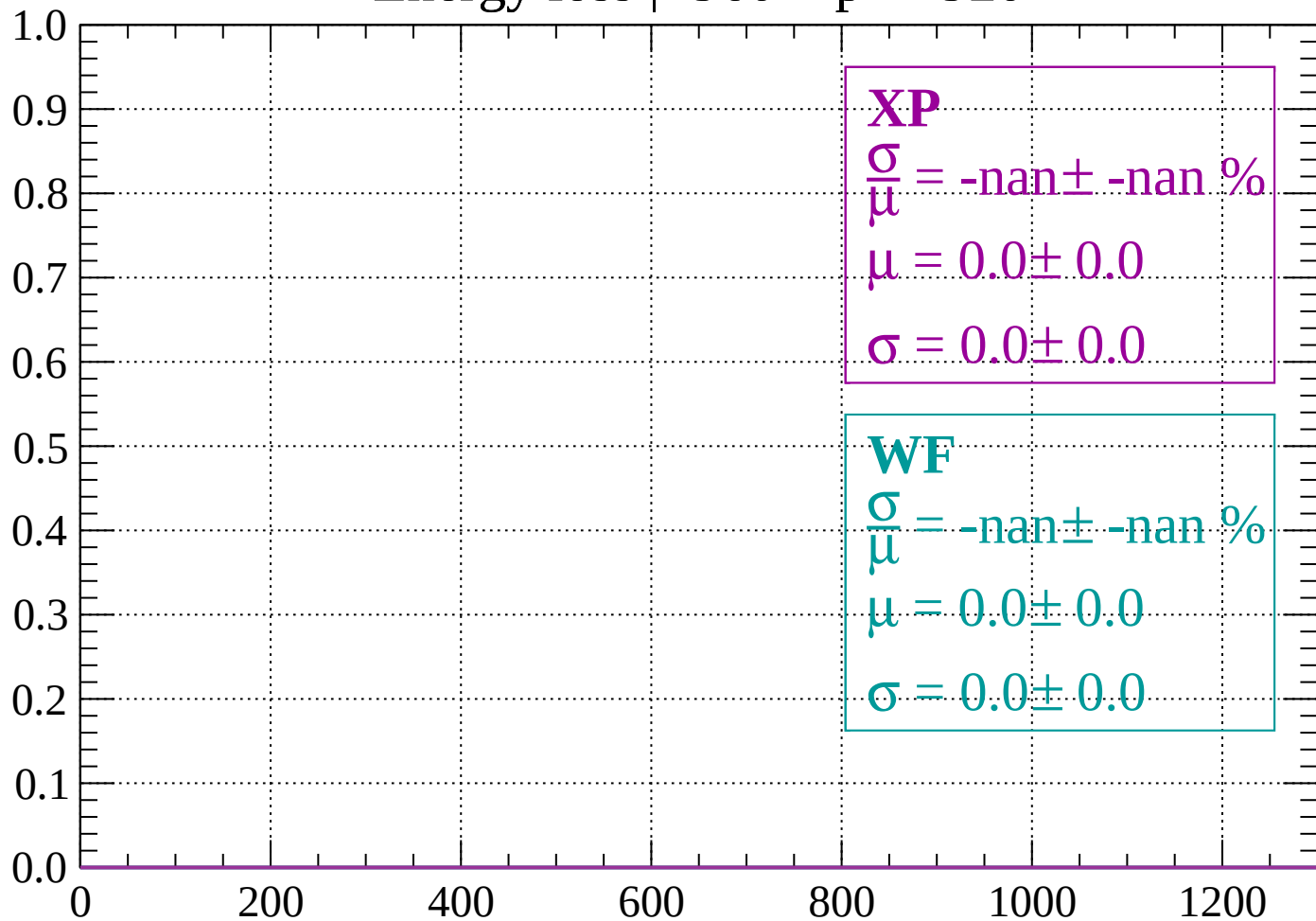
Count



dE/dx (ADC counts/cm)

Energy loss | -960 < p < -920

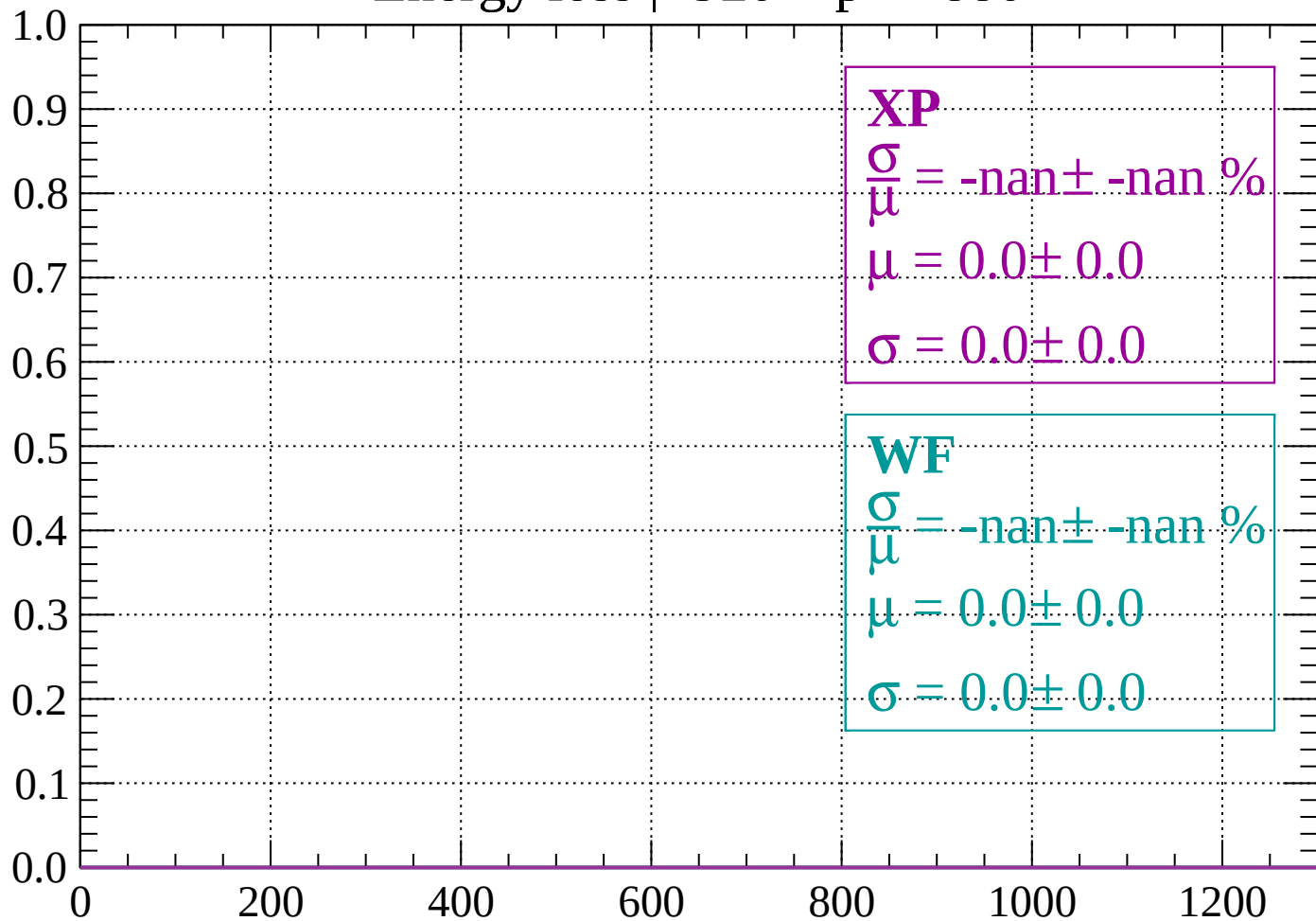
Count



dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

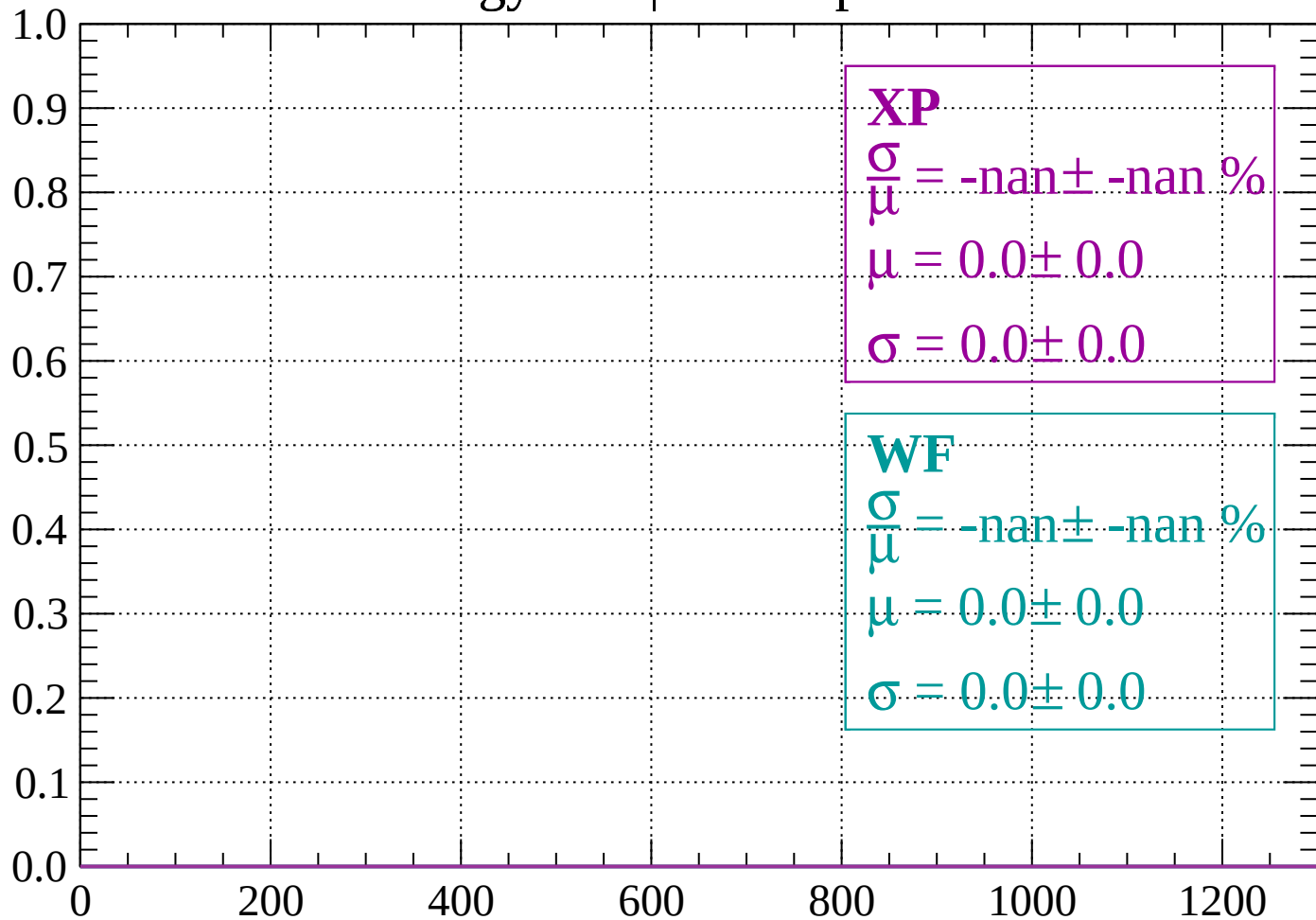
Count



dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

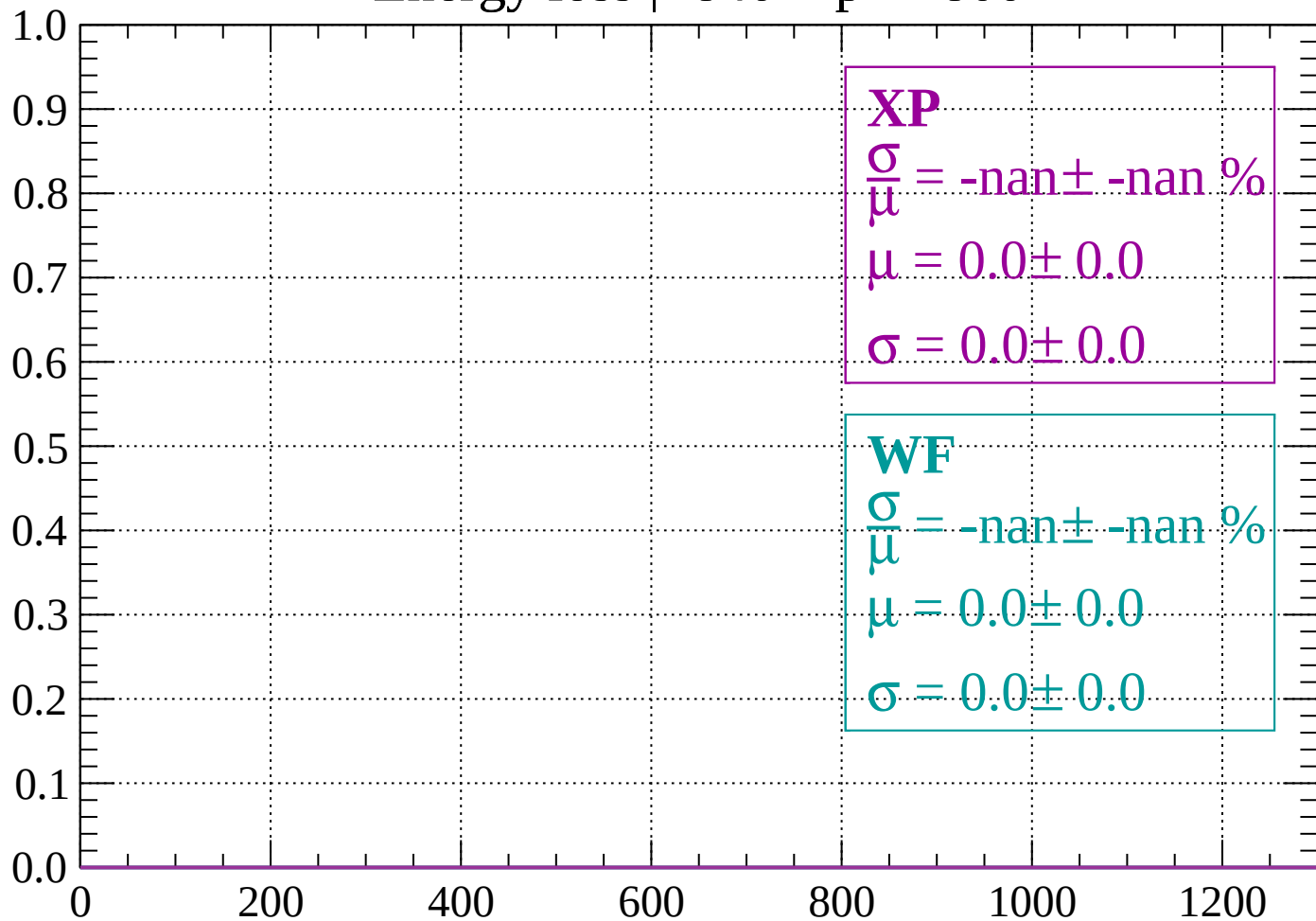
Count



dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

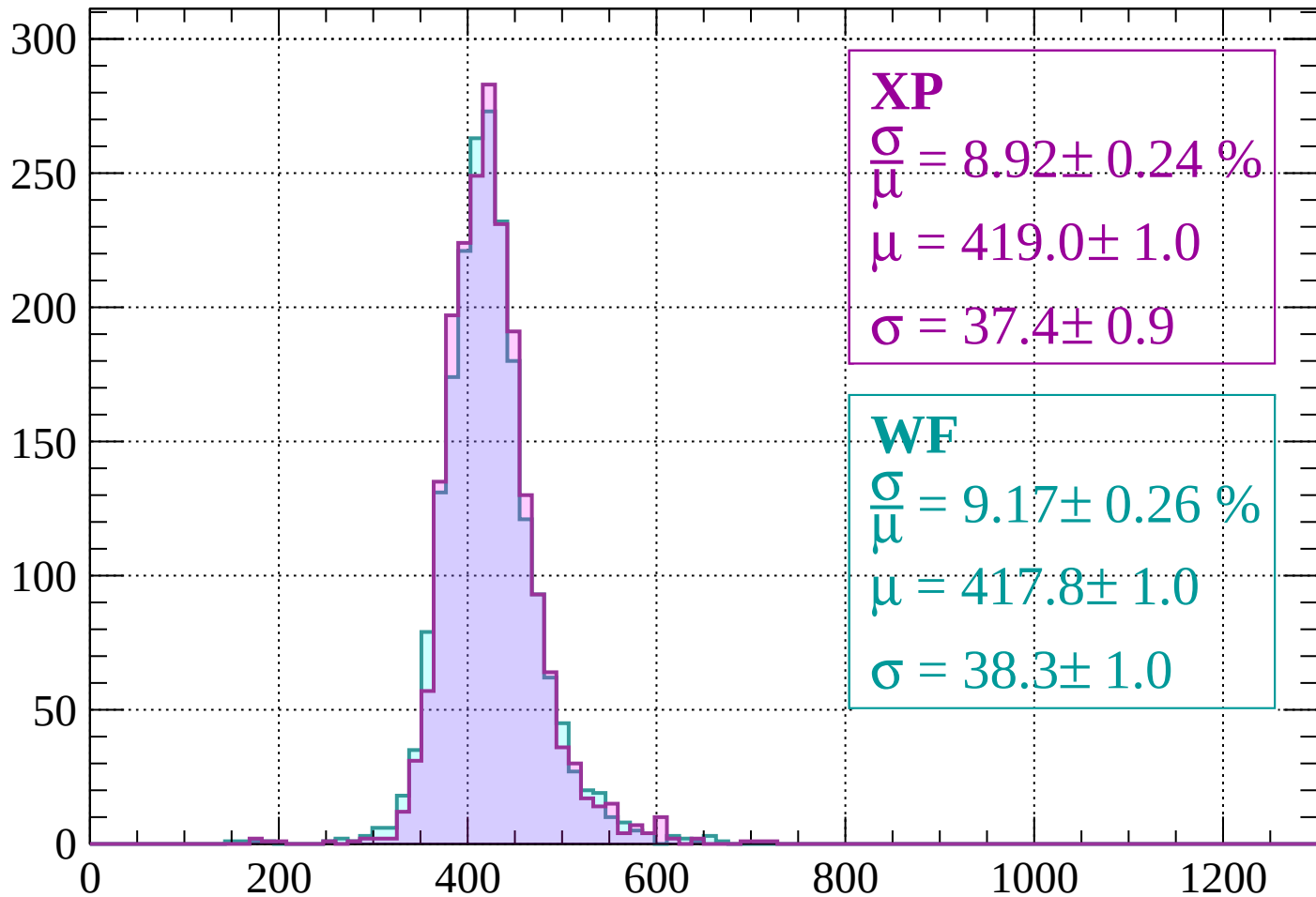
Count



dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 8.92 \pm 0.24 \%$$

$$\mu = 419.0 \pm 1.0$$

$$\sigma = 37.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.17 \pm 0.26 \%$$

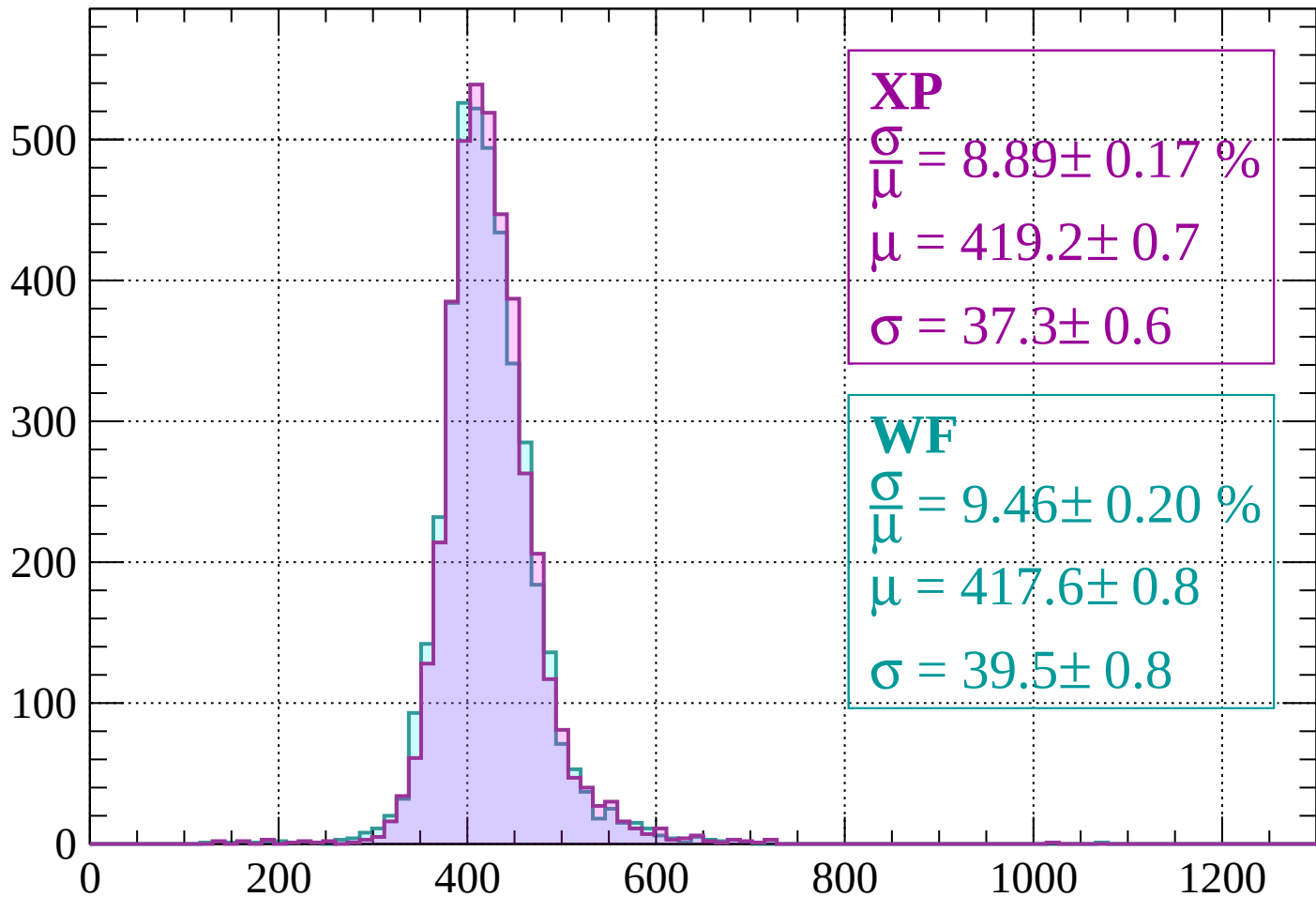
$$\mu = 417.8 \pm 1.0$$

$$\sigma = 38.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 8.89 \pm 0.17 \%$$

$$\mu = 419.2 \pm 0.7$$

$$\sigma = 37.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.46 \pm 0.20 \%$$

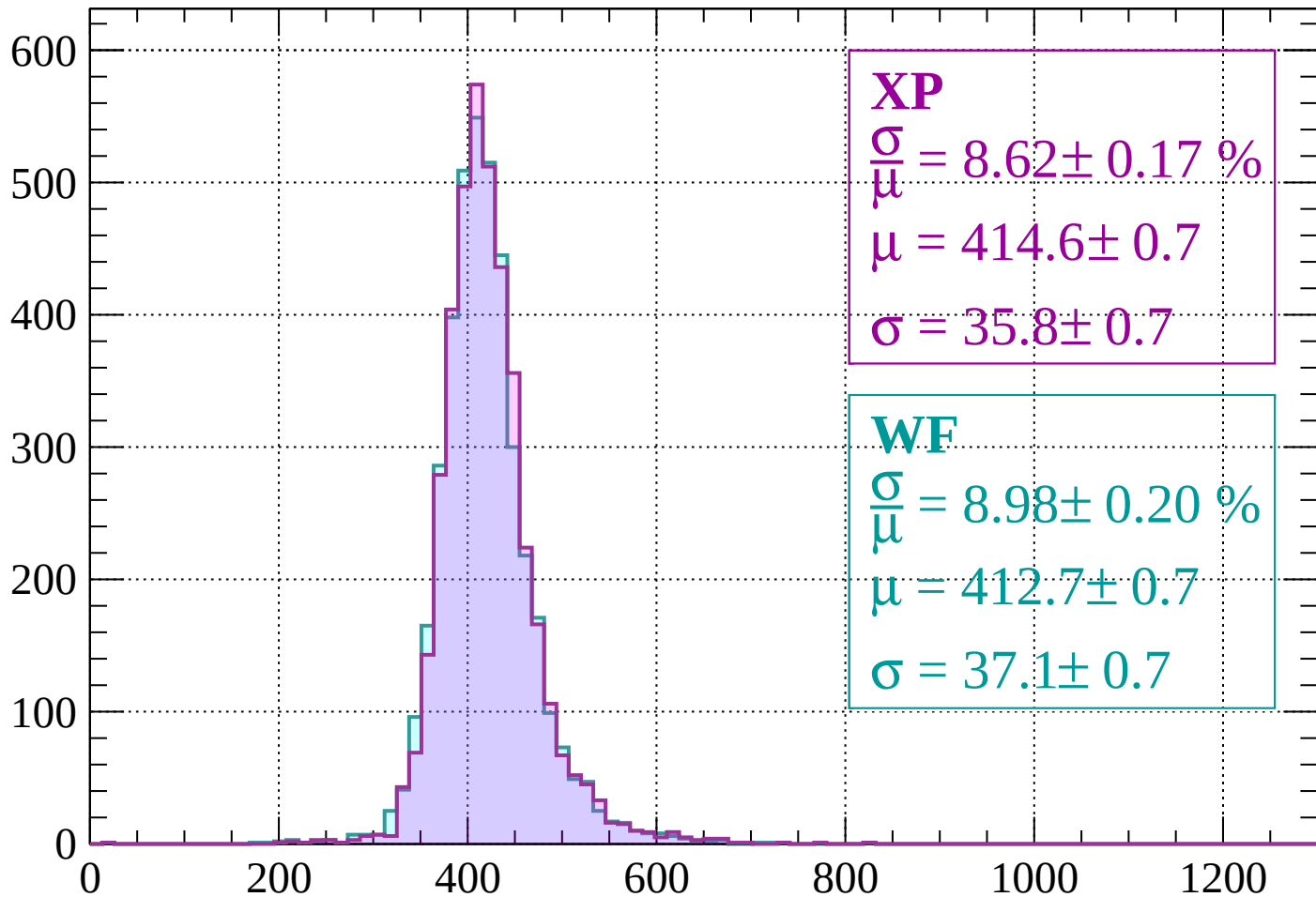
$$\mu = 417.6 \pm 0.8$$

$$\sigma = 39.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 8.62 \pm 0.17 \%$$

$$\mu = 414.6 \pm 0.7$$

$$\sigma = 35.8 \pm 0.7$$

WF

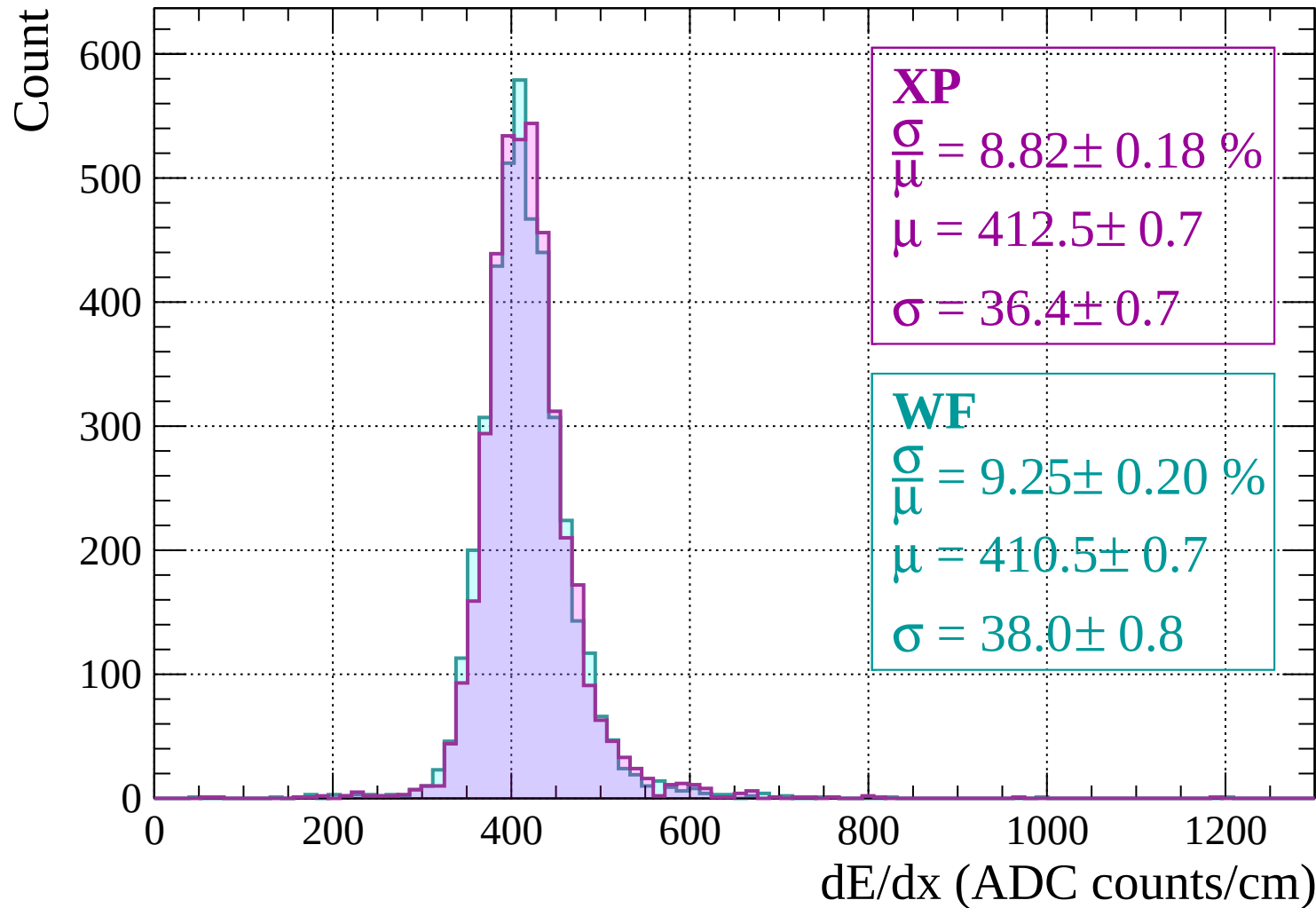
$$\frac{\sigma}{\mu} = 8.98 \pm 0.20 \%$$

$$\mu = 412.7 \pm 0.7$$

$$\sigma = 37.1 \pm 0.7$$

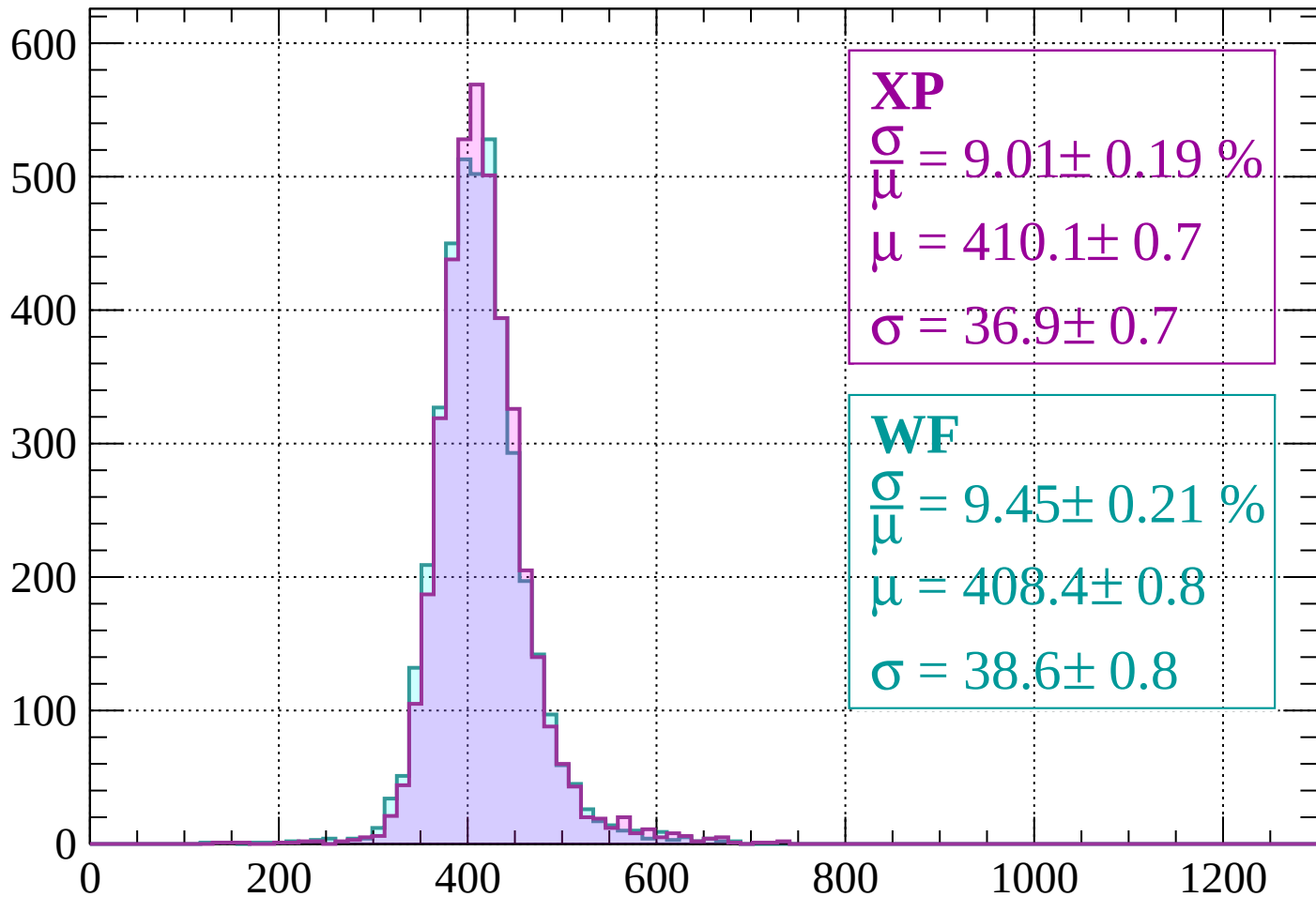
dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$



Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 9.01 \pm 0.19 \%$$

$$\mu = 410.1 \pm 0.7$$

$$\sigma = 36.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.45 \pm 0.21 \%$$

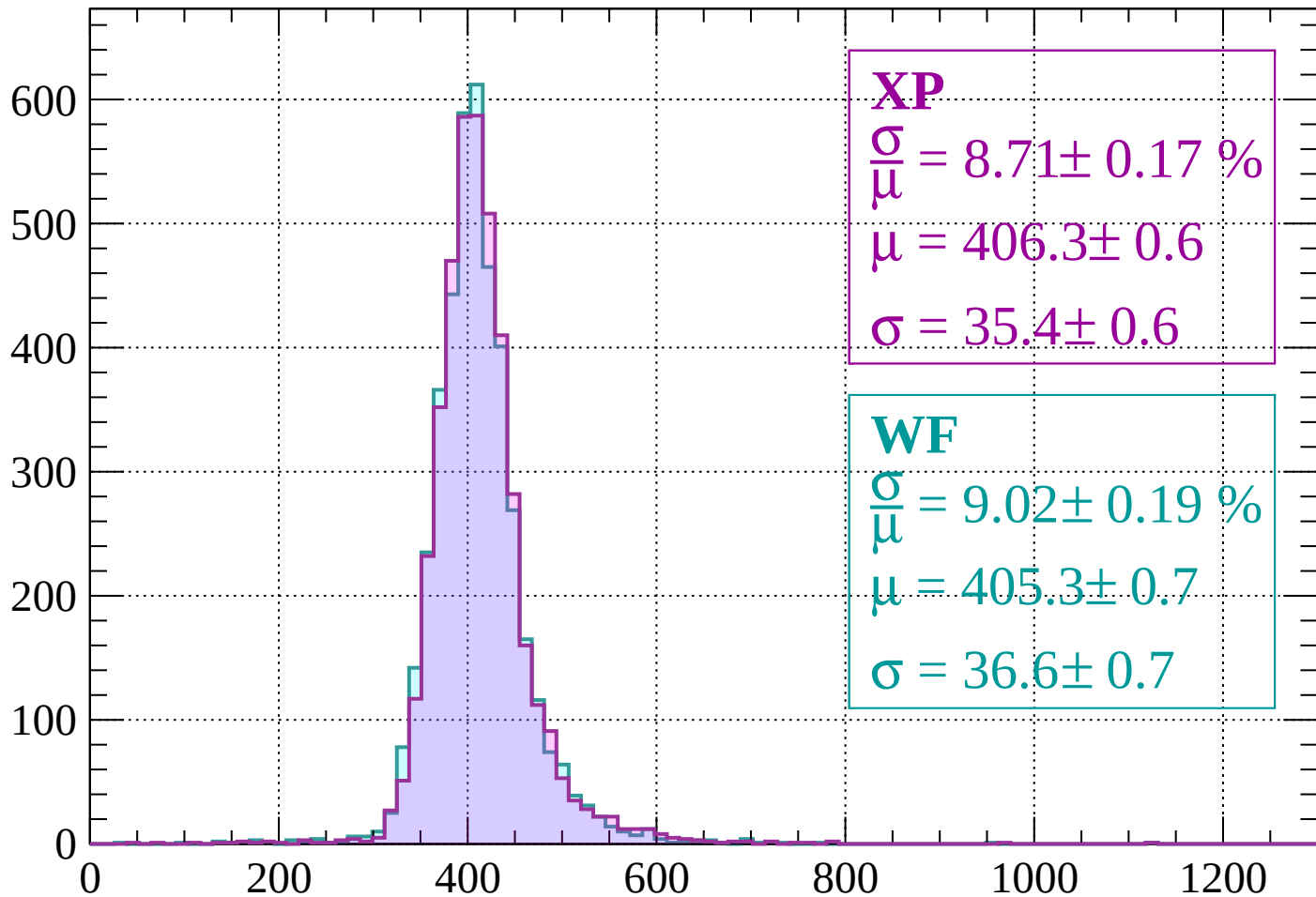
$$\mu = 408.4 \pm 0.8$$

$$\sigma = 38.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 8.71 \pm 0.17 \%$$

$$\mu = 406.3 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

WF

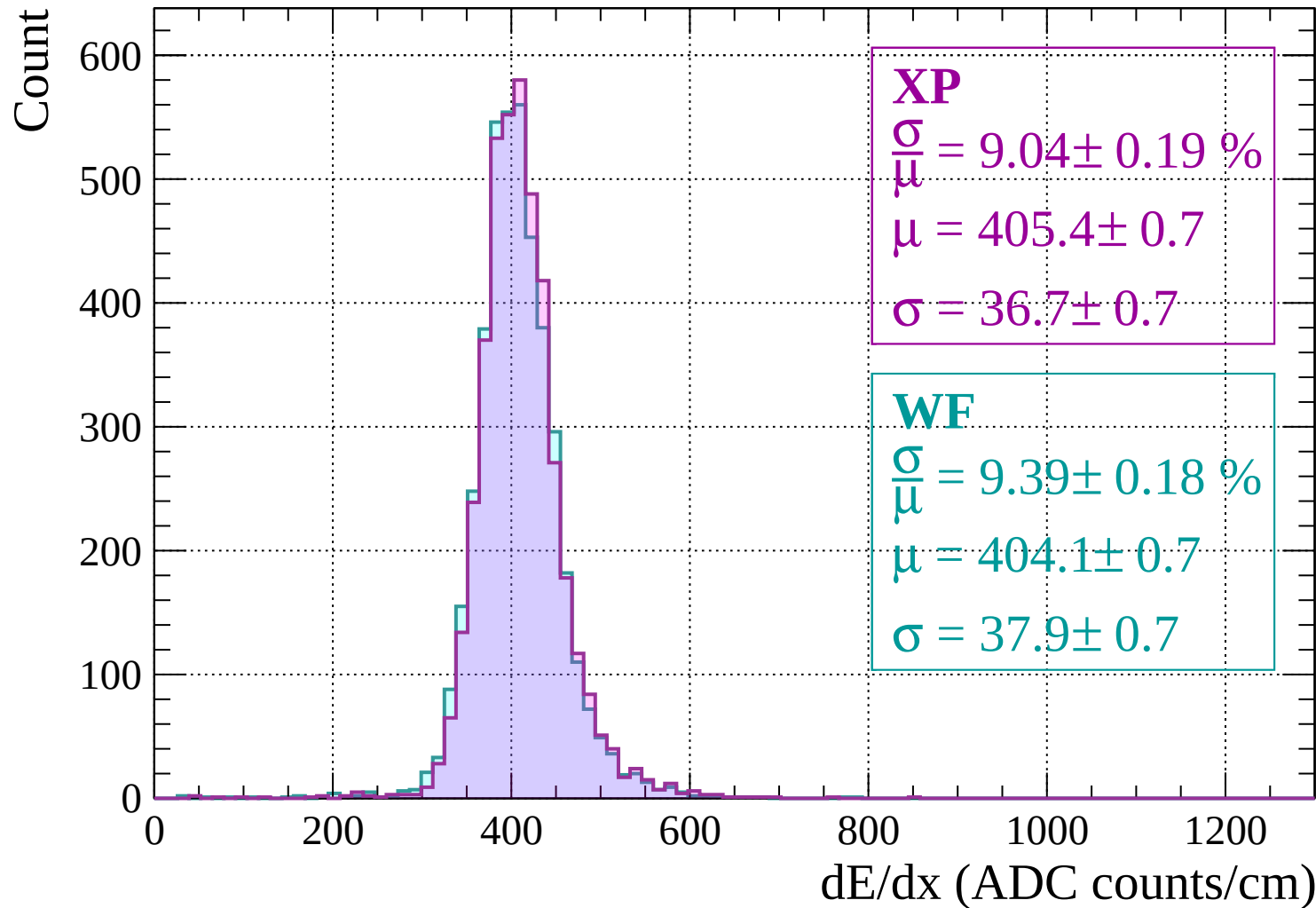
$$\frac{\sigma}{\mu} = 9.02 \pm 0.19 \%$$

$$\mu = 405.3 \pm 0.7$$

$$\sigma = 36.6 \pm 0.7$$

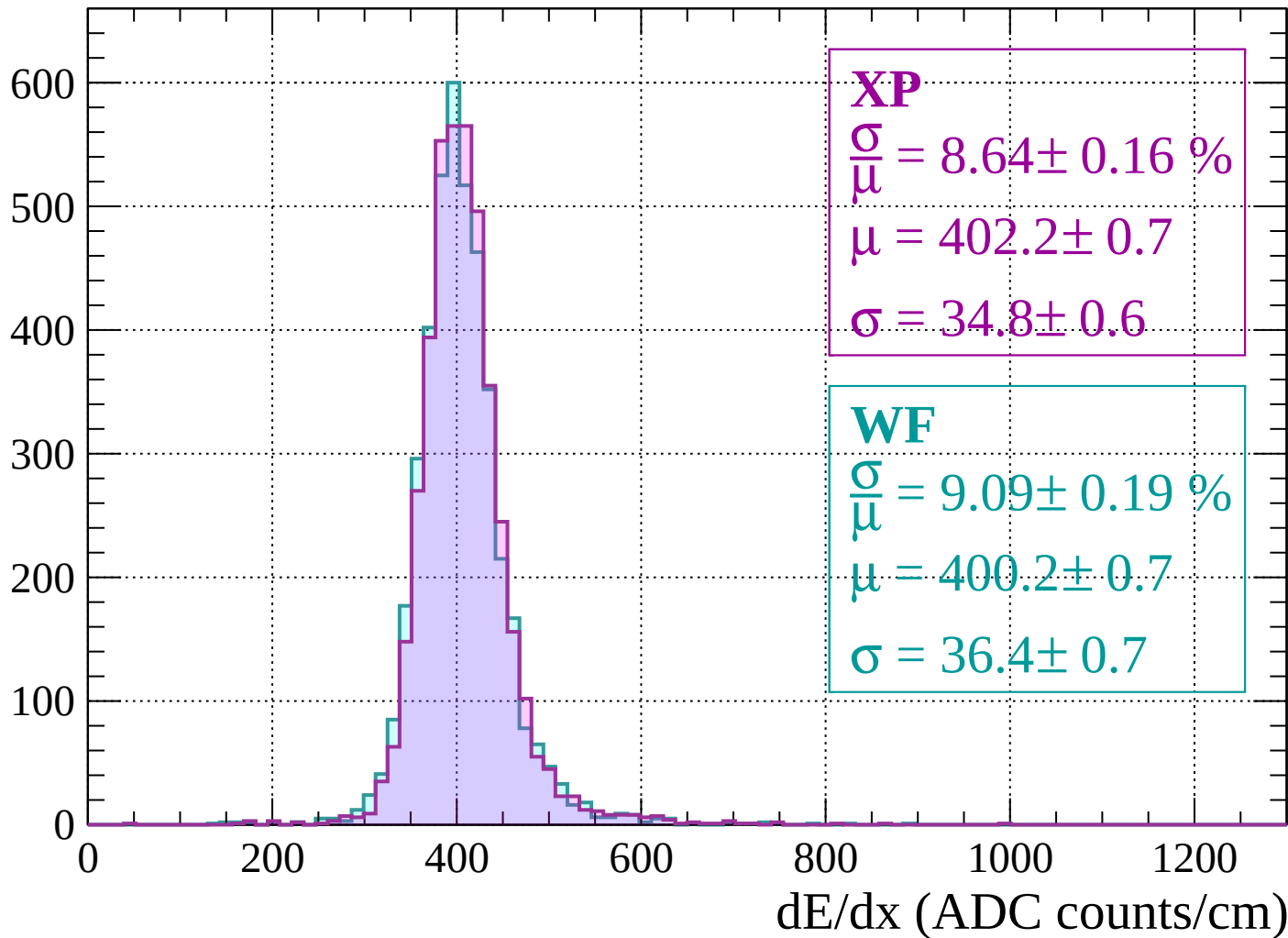
dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$



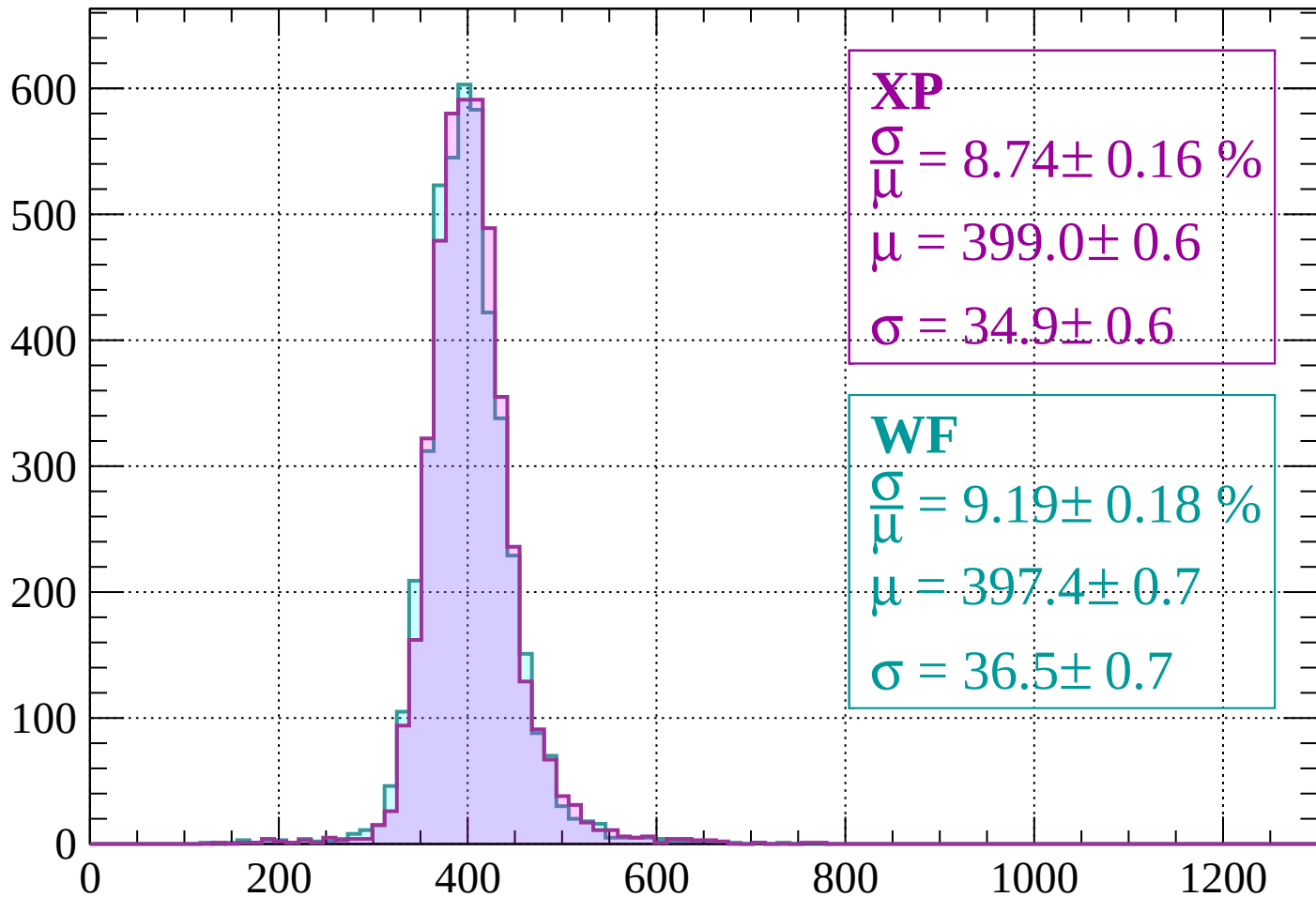
Energy loss | $-520 < p < -480$

Count



Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 8.74 \pm 0.16 \%$$

$$\mu = 399.0 \pm 0.6$$

$$\sigma = 34.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.19 \pm 0.18 \%$$

$$\mu = 397.4 \pm 0.7$$

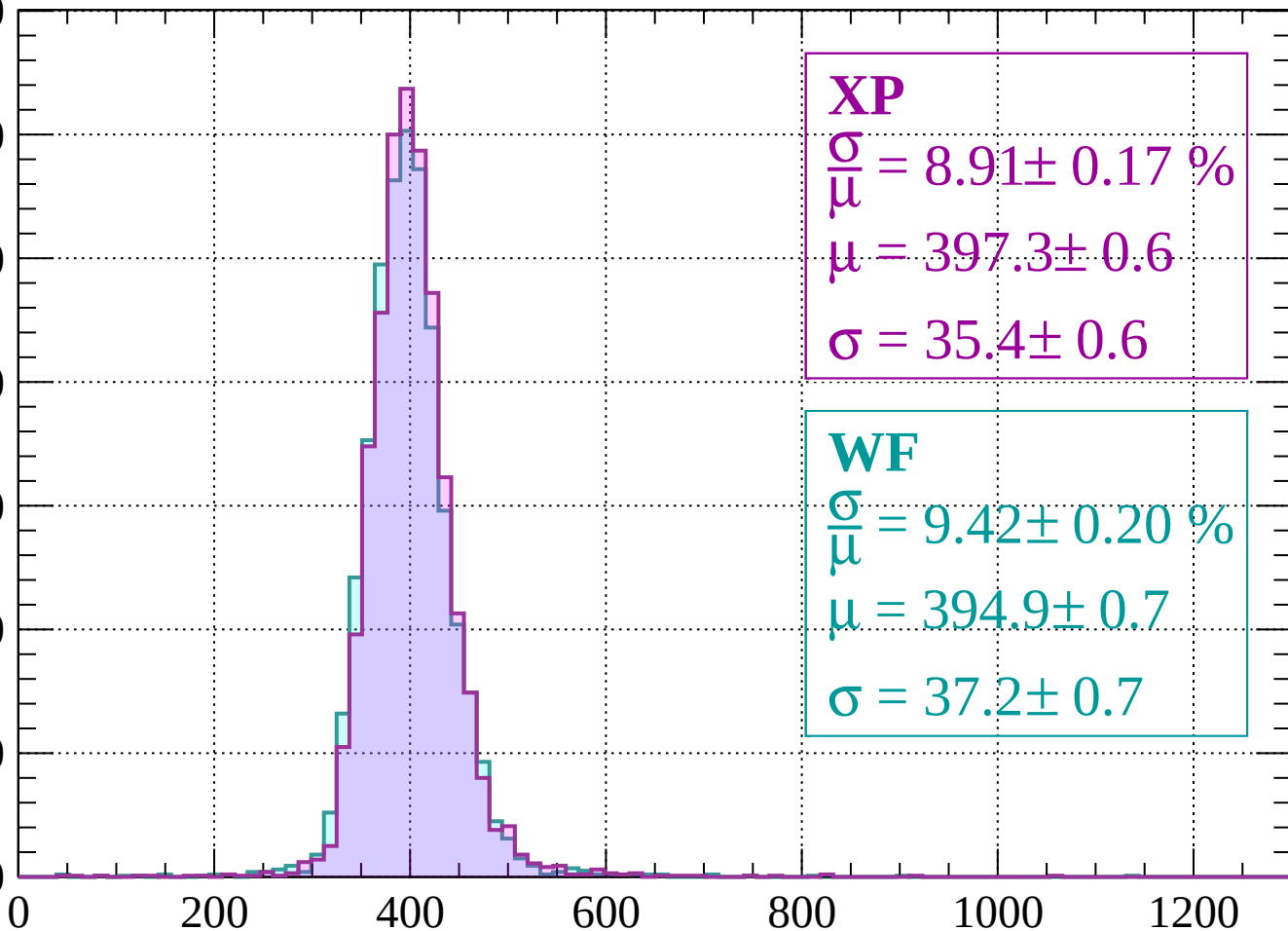
$$\sigma = 36.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 8.91 \pm 0.17 \%$$

$$\mu = 397.3 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.42 \pm 0.20 \%$$

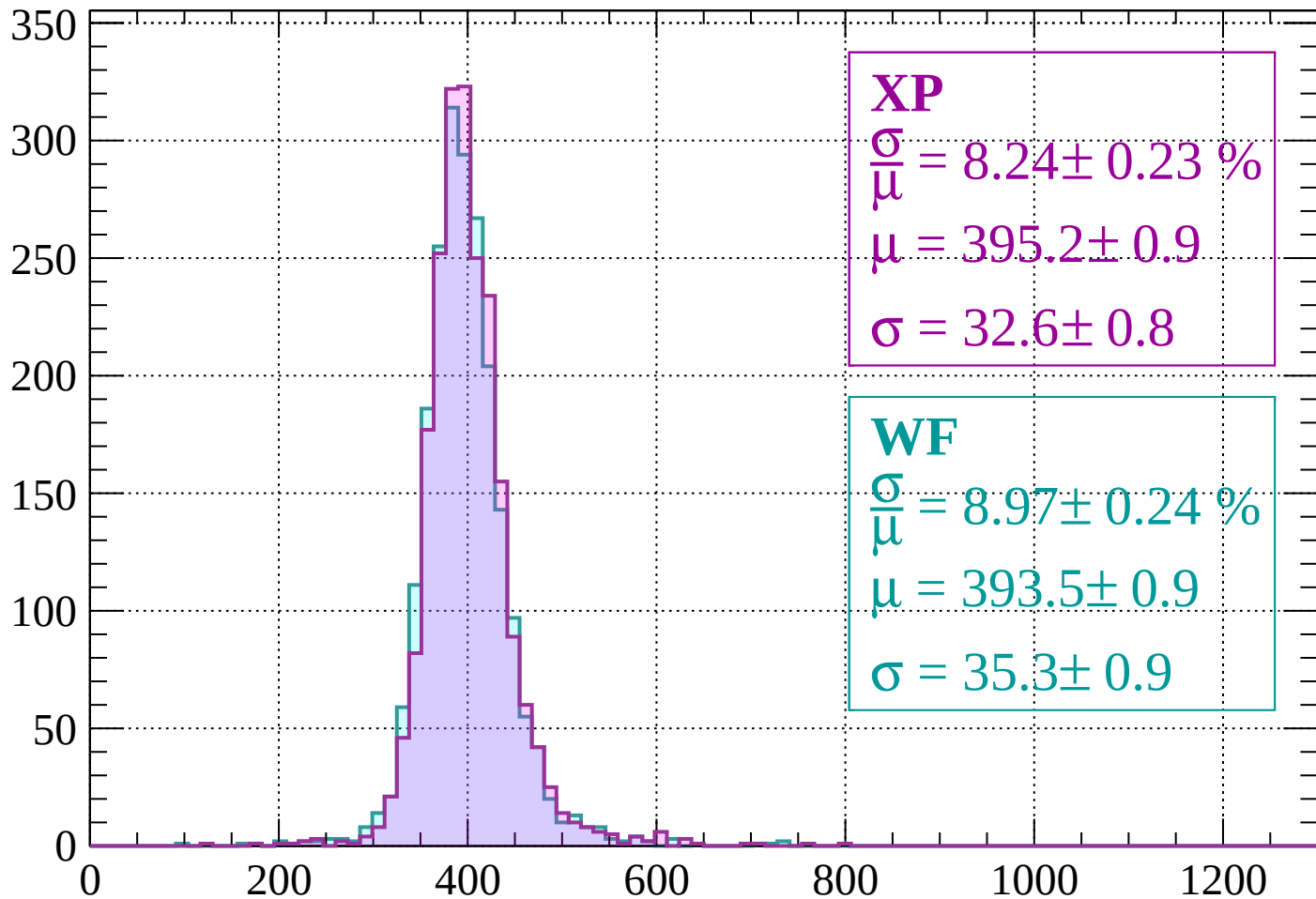
$$\mu = 394.9 \pm 0.7$$

$$\sigma = 37.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 8.24 \pm 0.23 \%$$

$$\mu = 395.2 \pm 0.9$$

$$\sigma = 32.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.97 \pm 0.24 \%$$

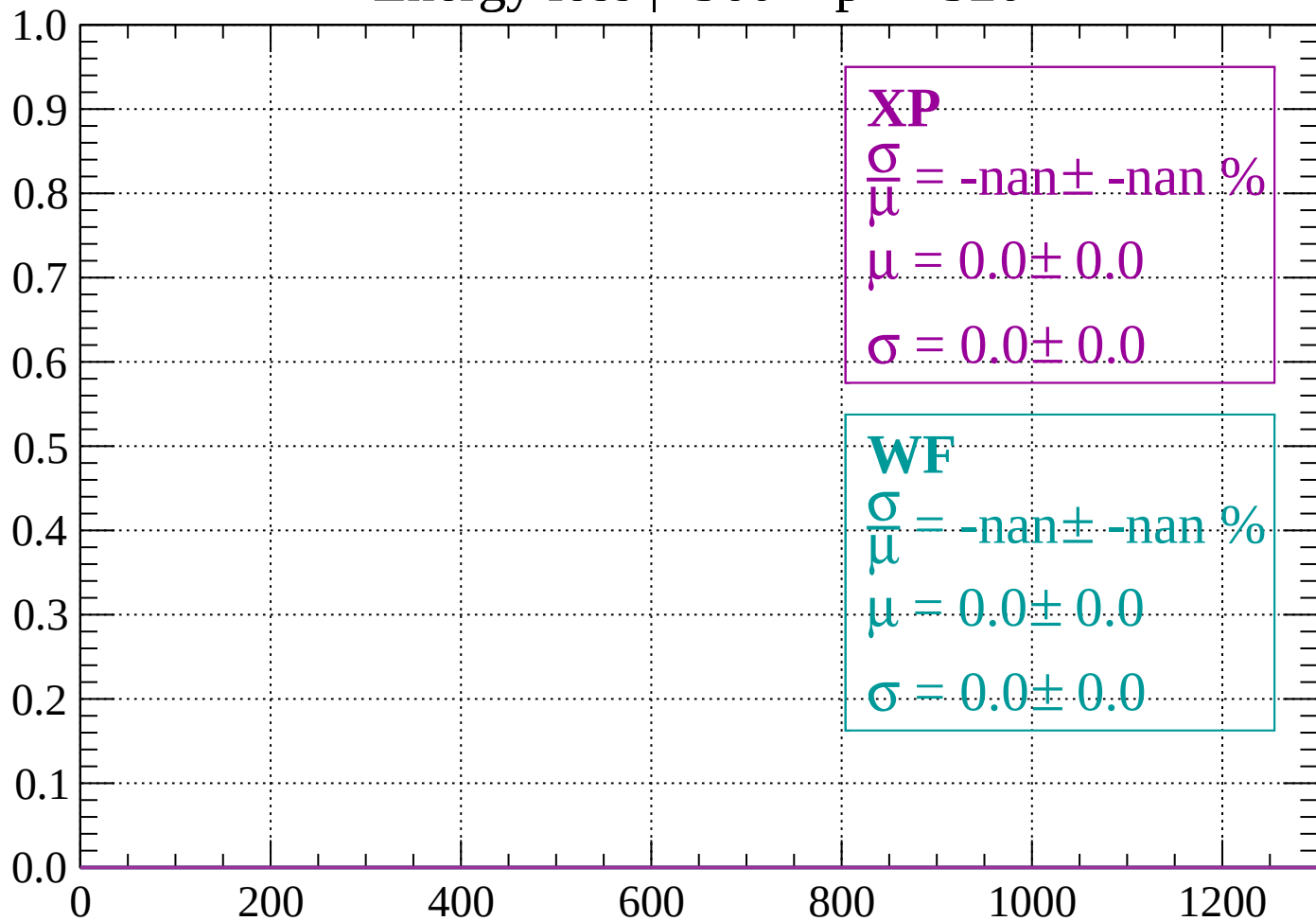
$$\mu = 393.5 \pm 0.9$$

$$\sigma = 35.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

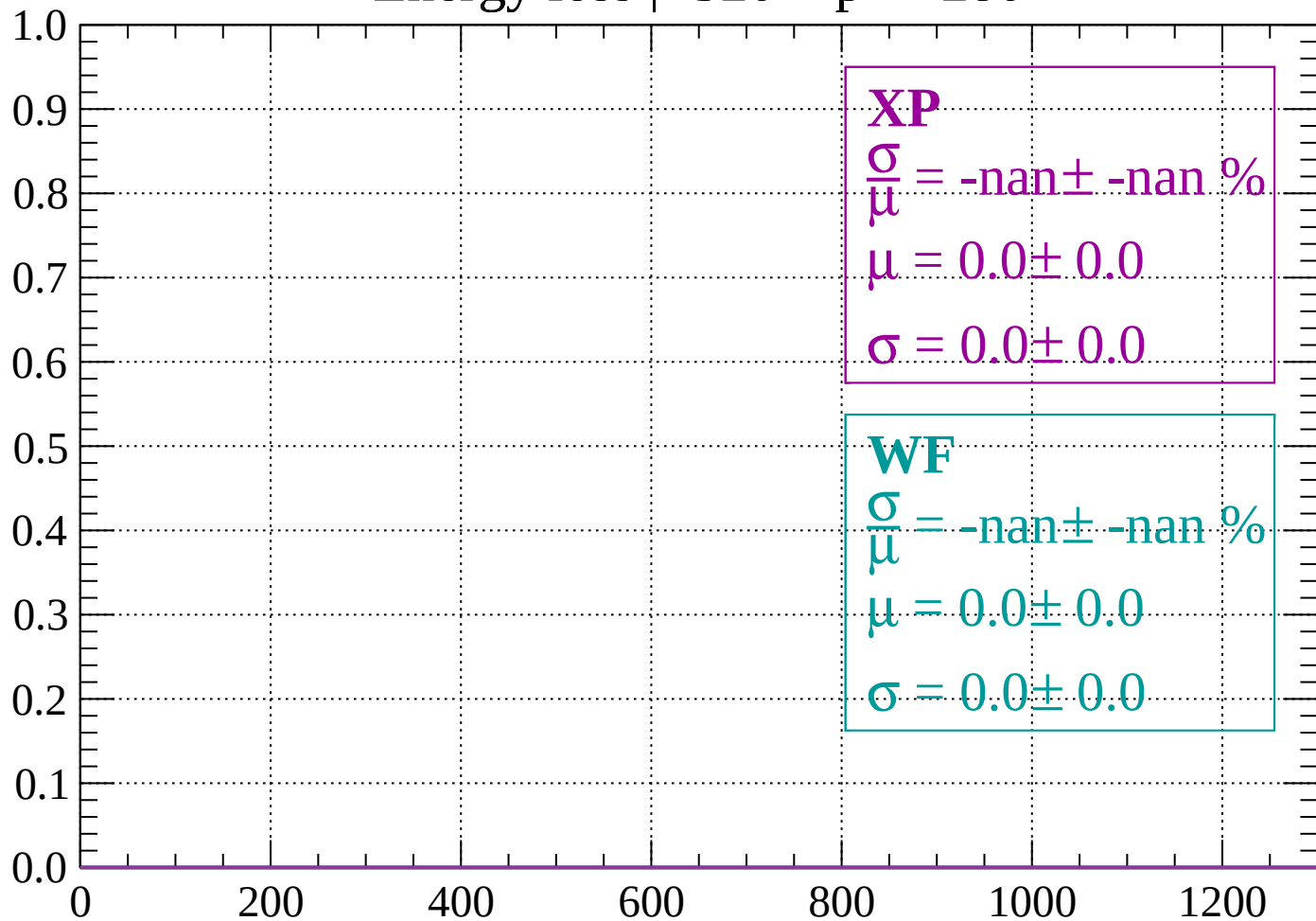
Count



dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

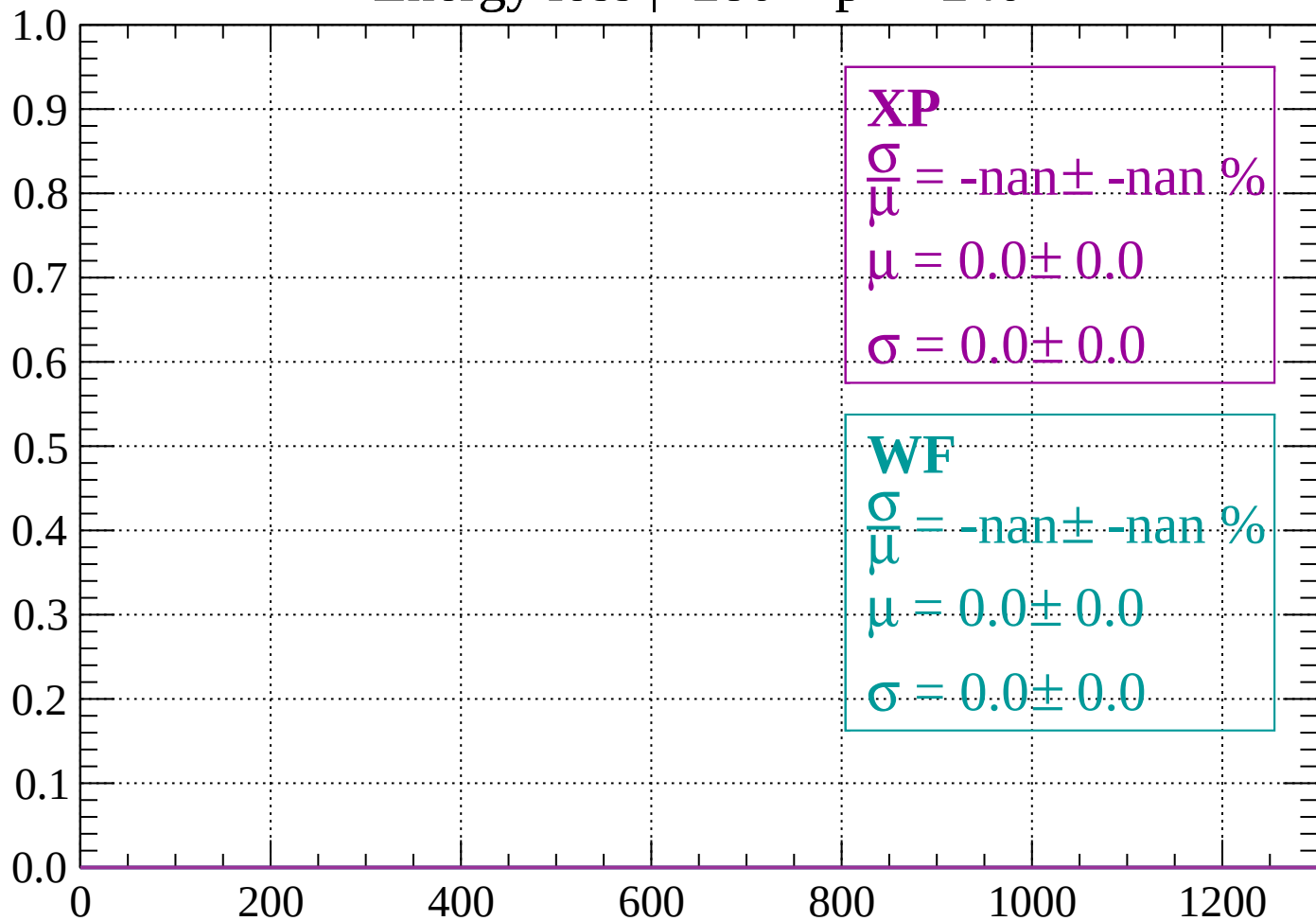
Count



dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

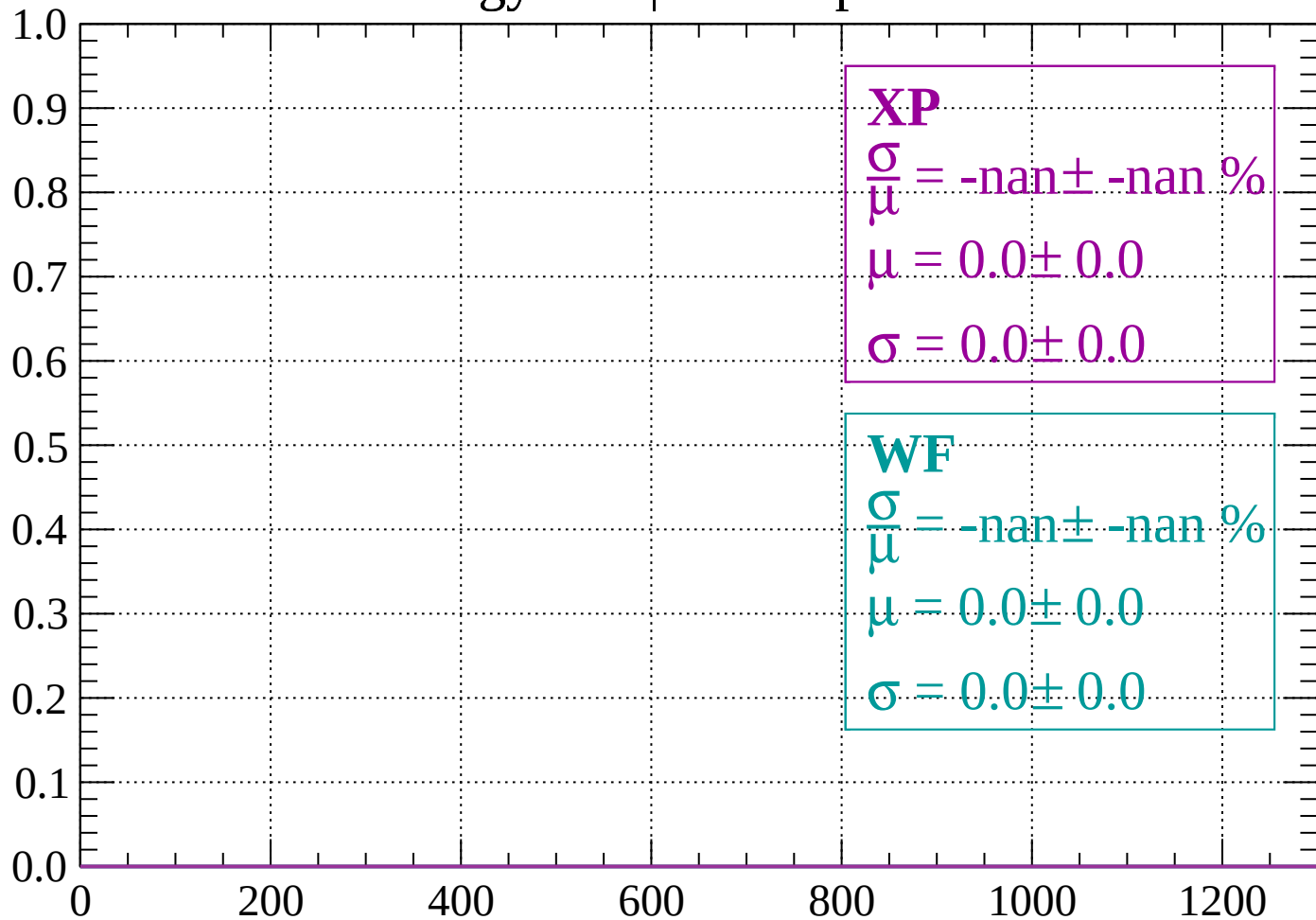
Count



dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

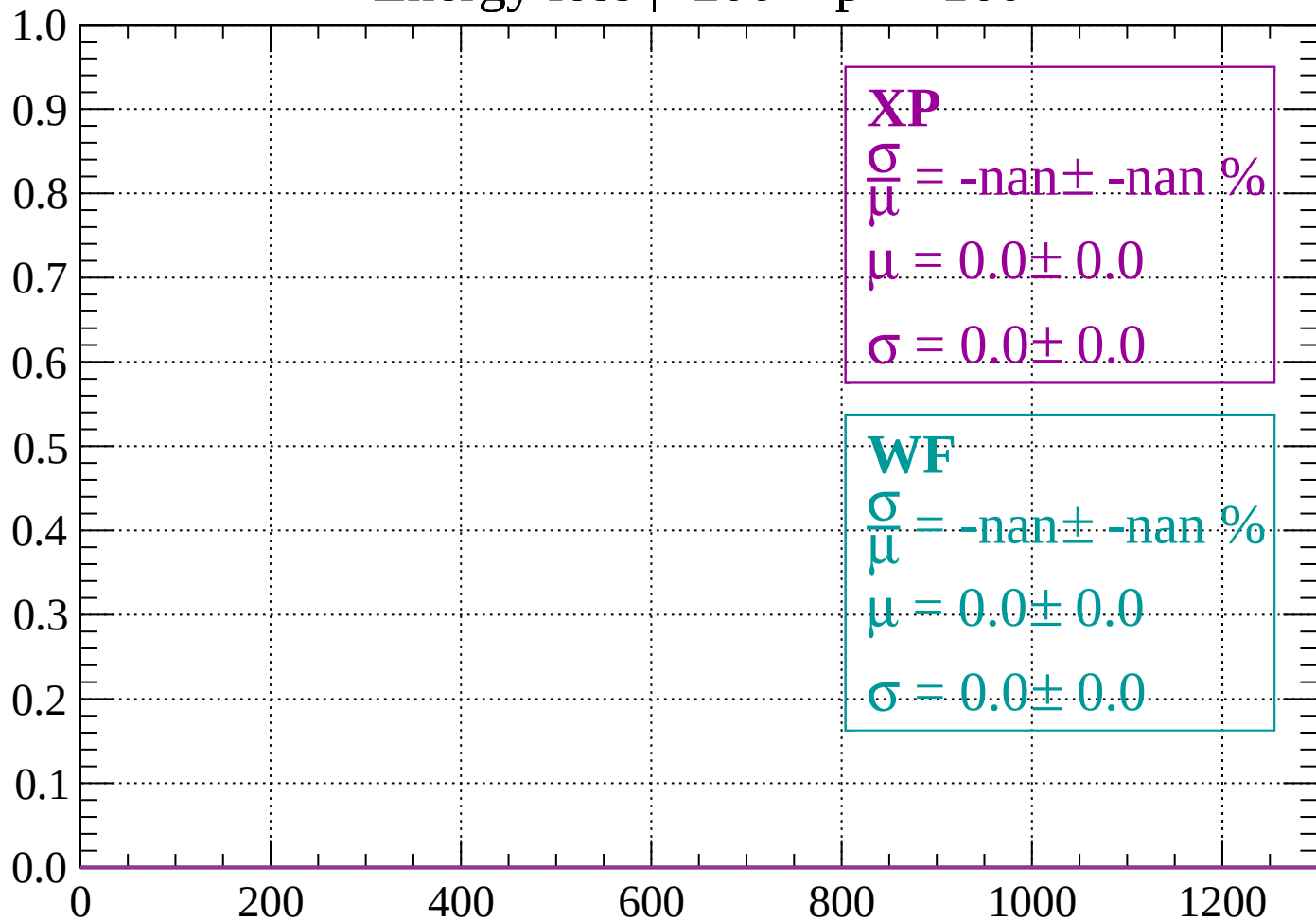
Count



dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

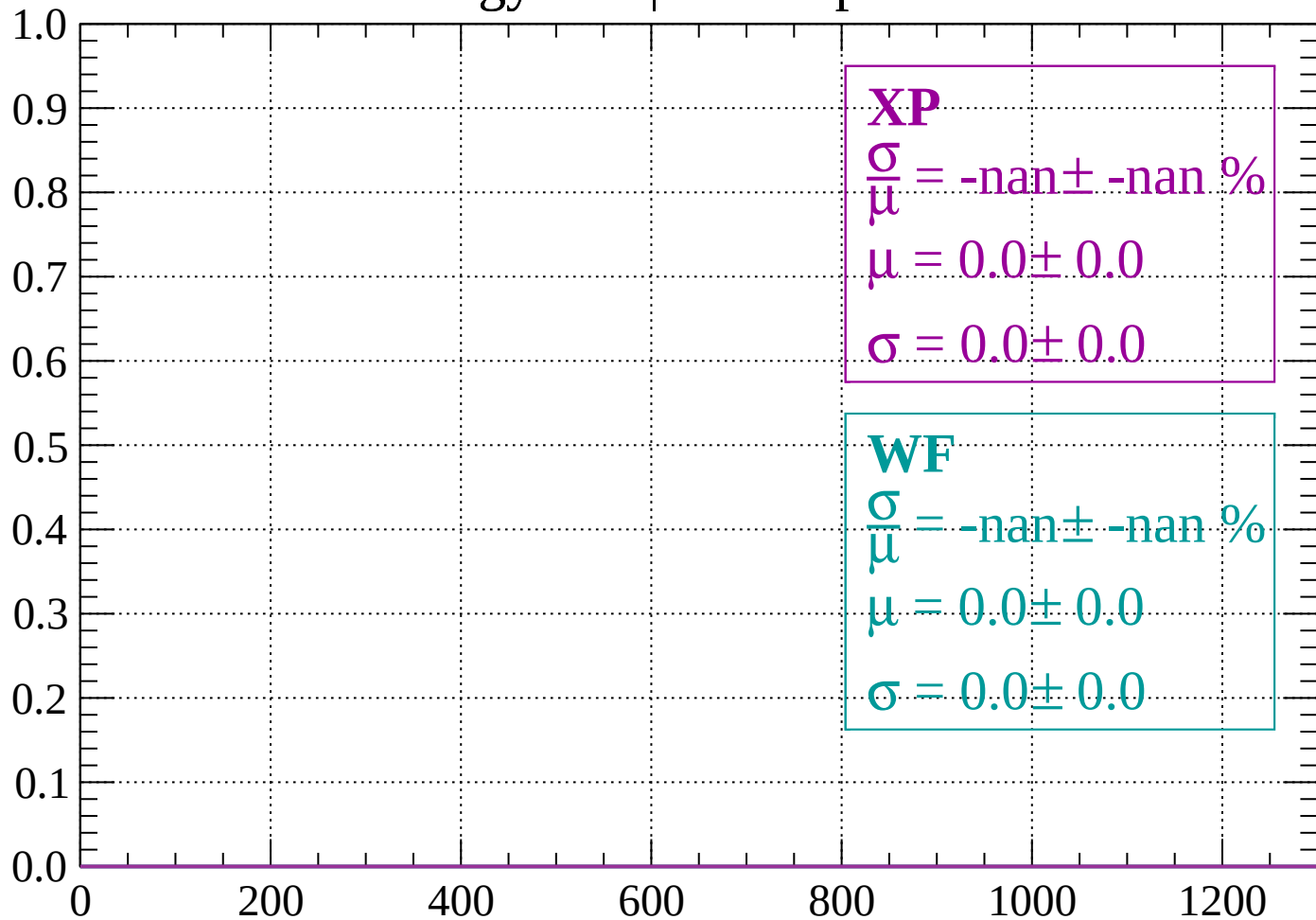
Count



dE/dx (ADC counts/cm)

Energy loss | $-160 < p < -120$

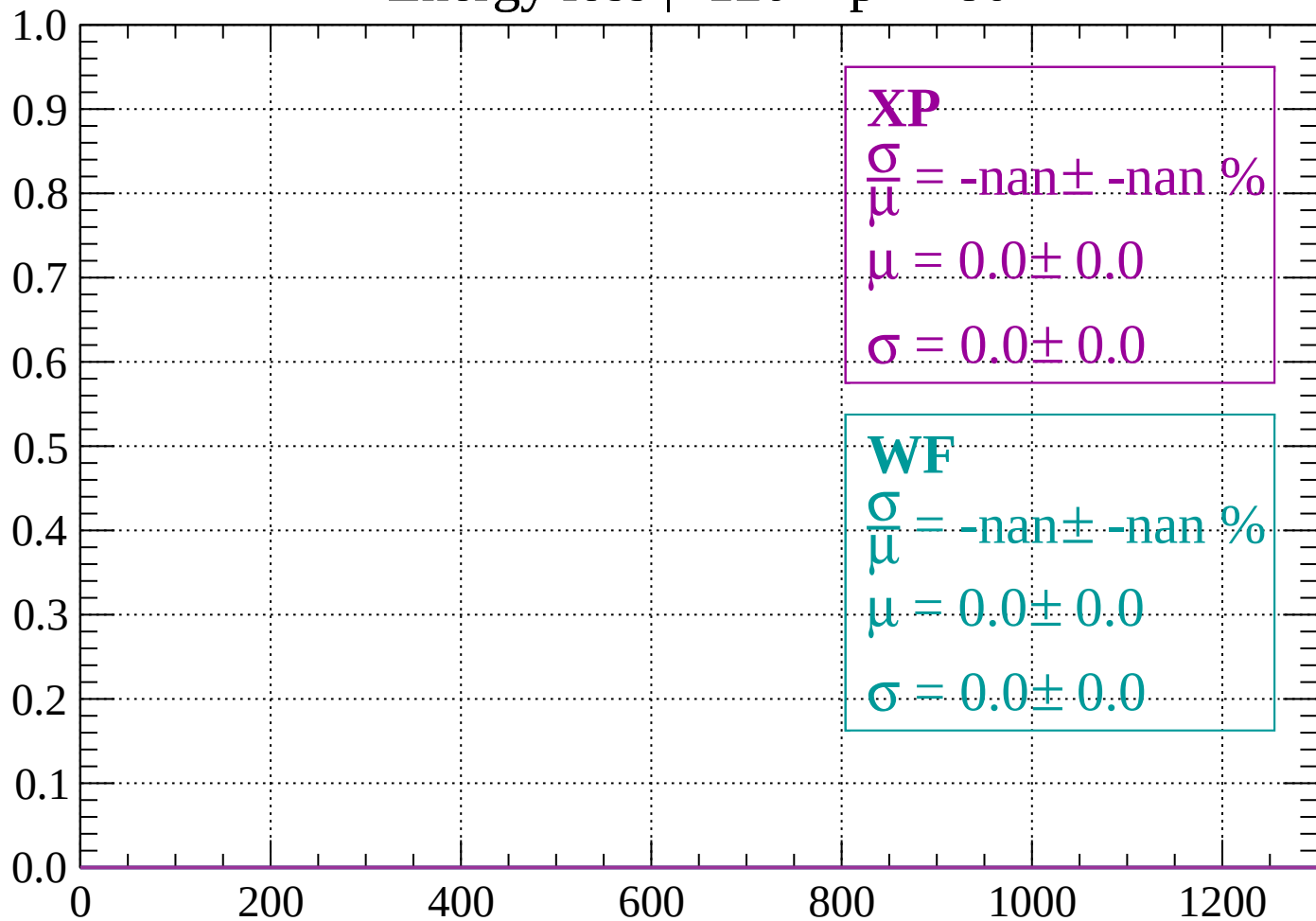
Count



dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -80$

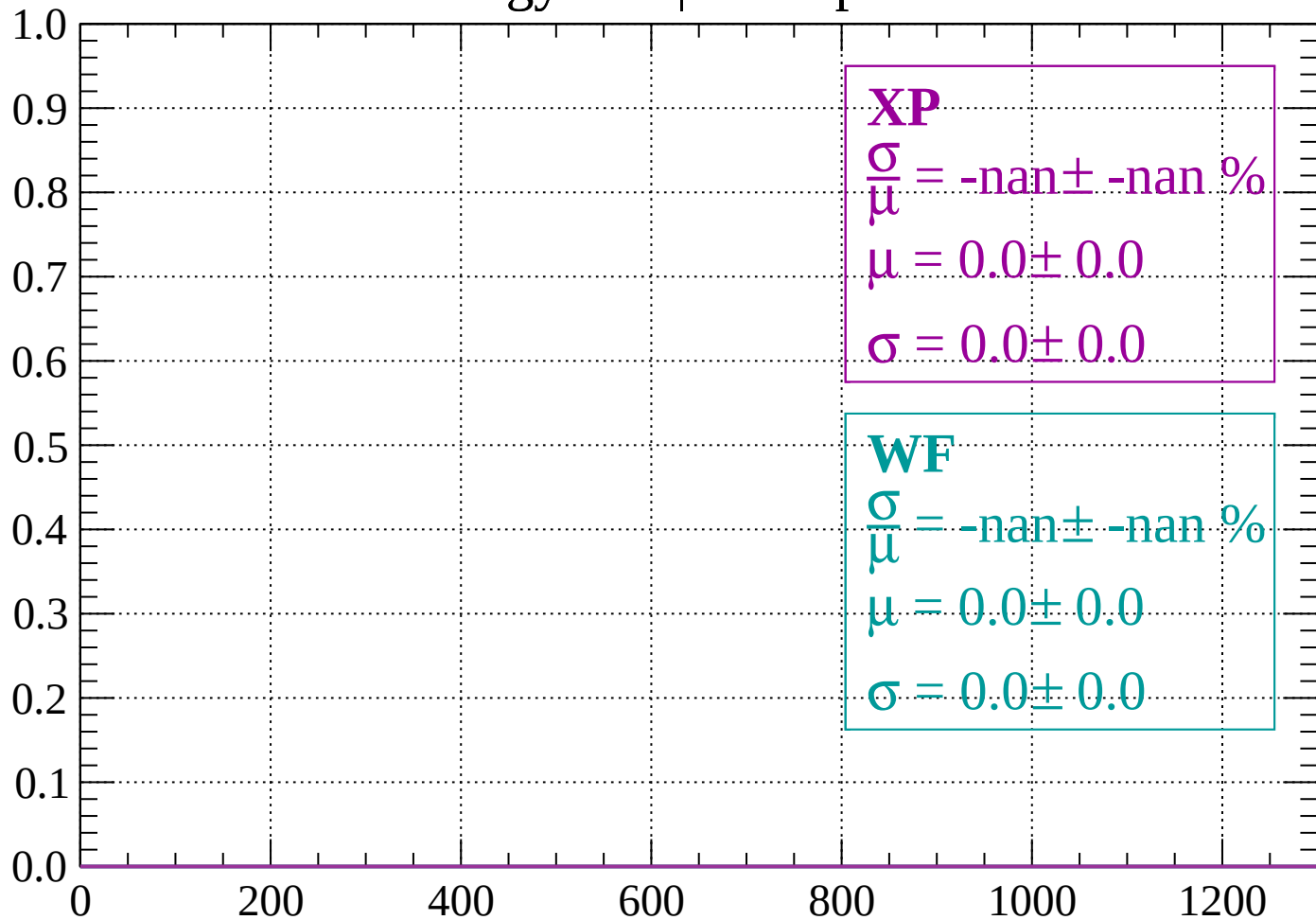
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < p < -40$

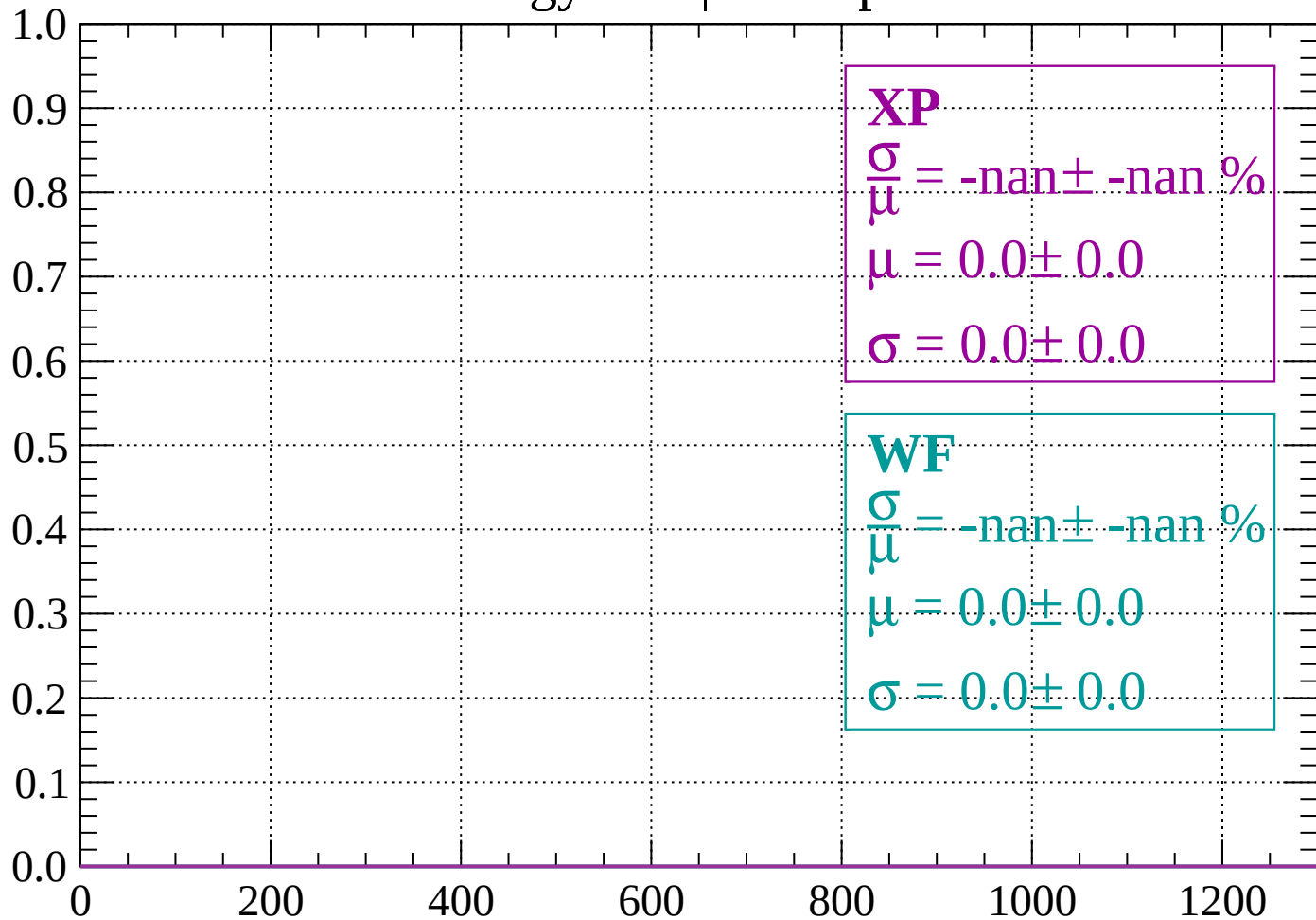
Count



dE/dx (ADC counts/cm)

Energy loss | $-40 < p < 0$

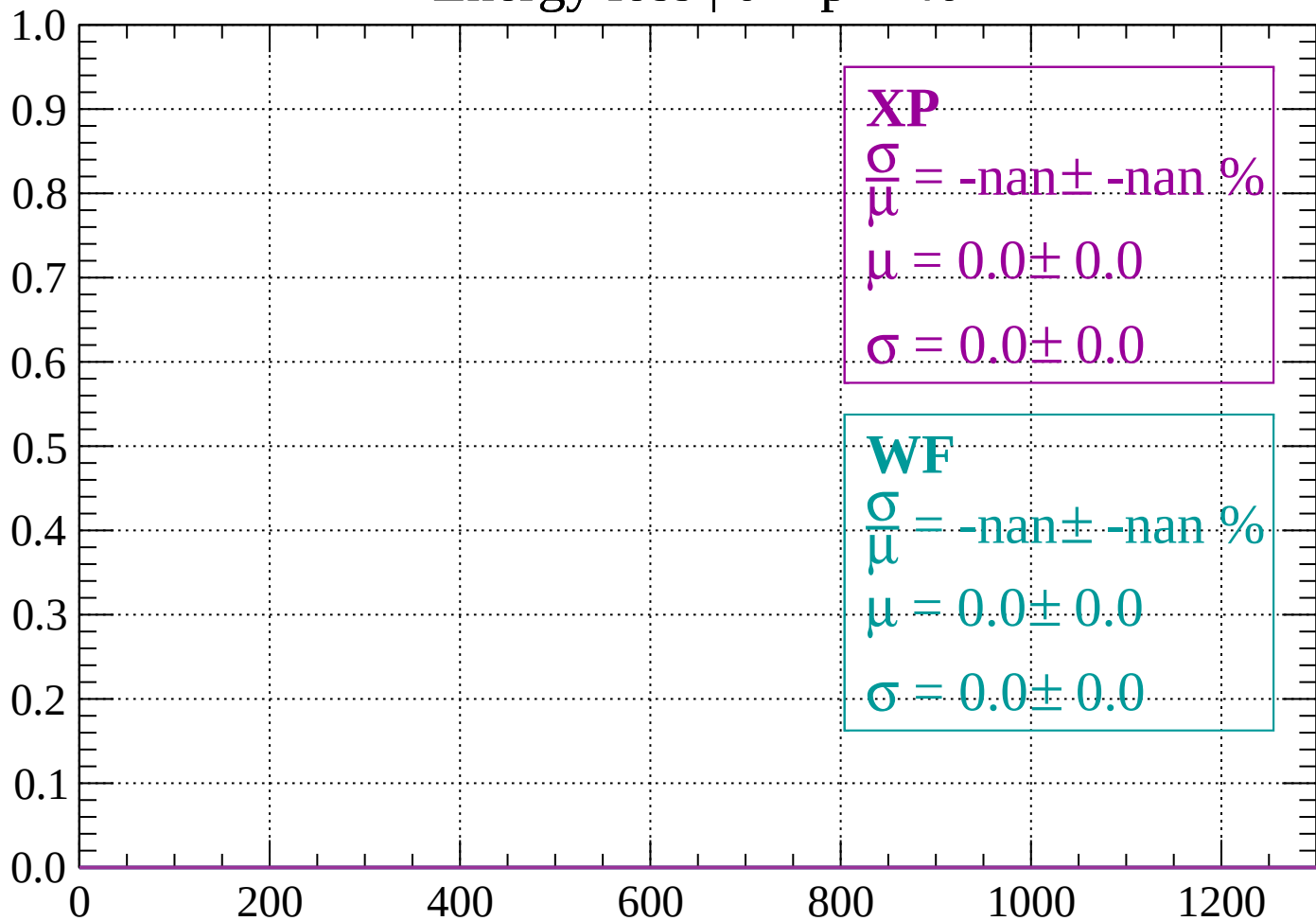
Count



dE/dx (ADC counts/cm)

Energy loss | $0 < p < 40$

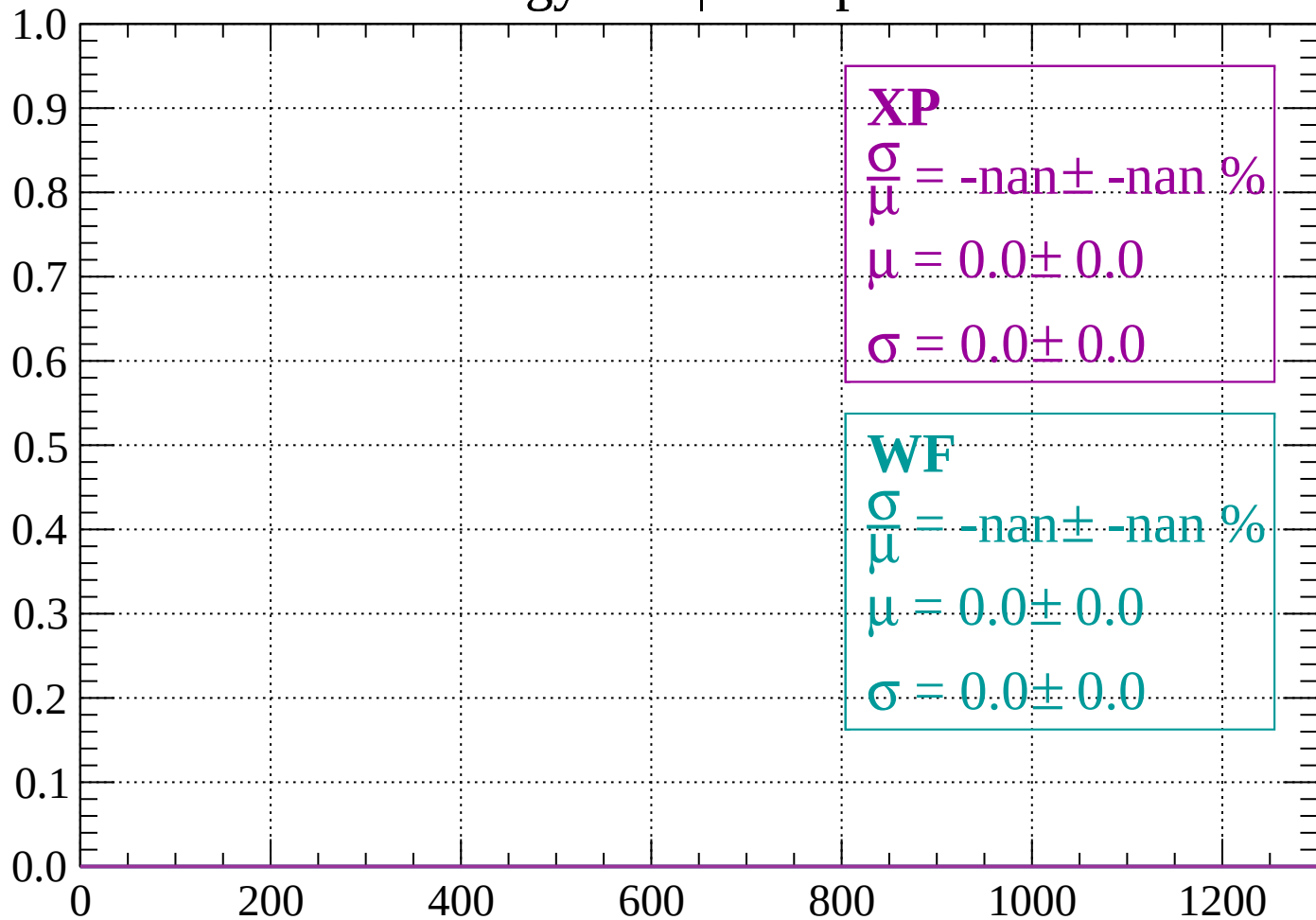
Count



dE/dx (ADC counts/cm)

Energy loss | $40 < p < 80$

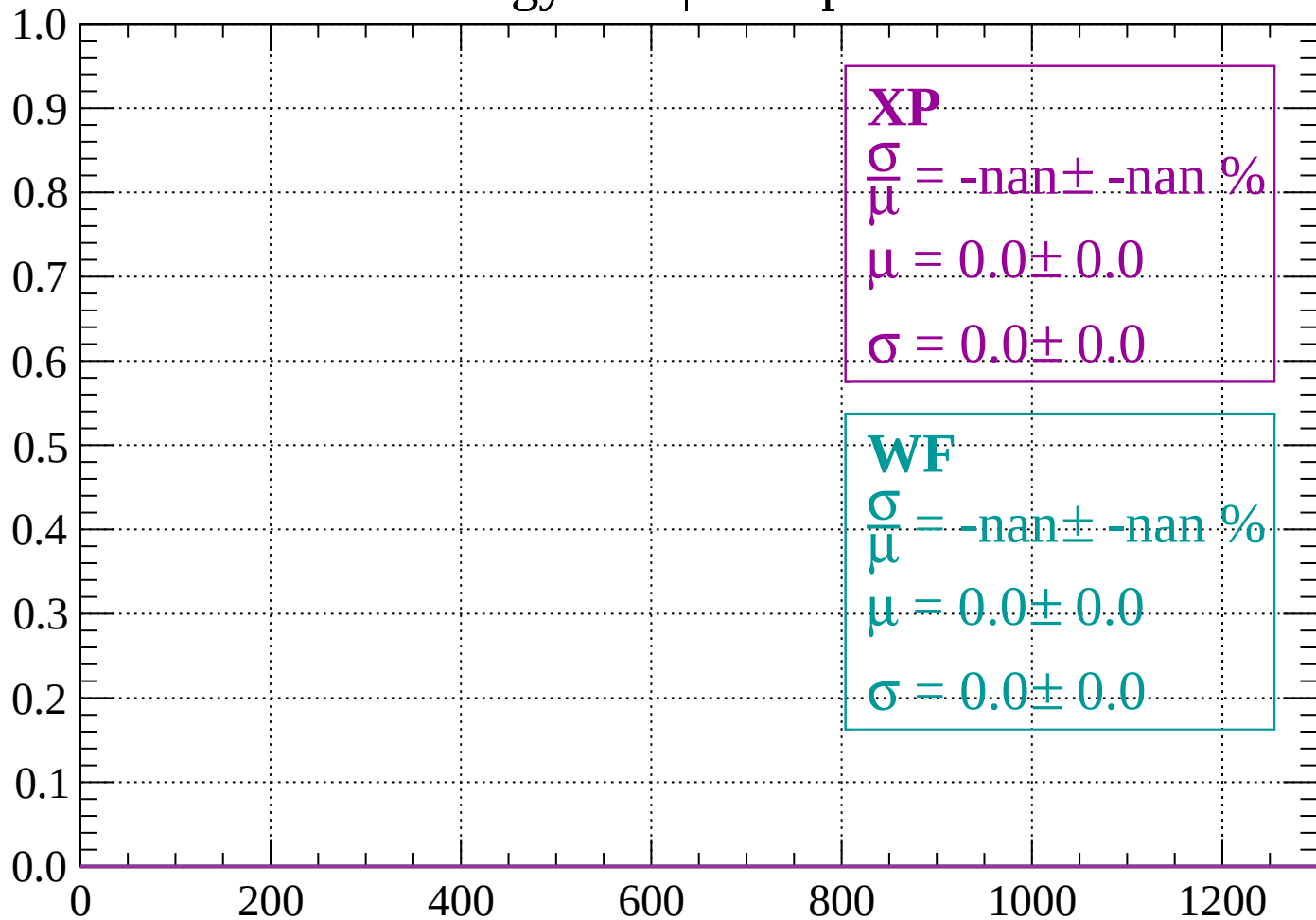
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < p < 120$

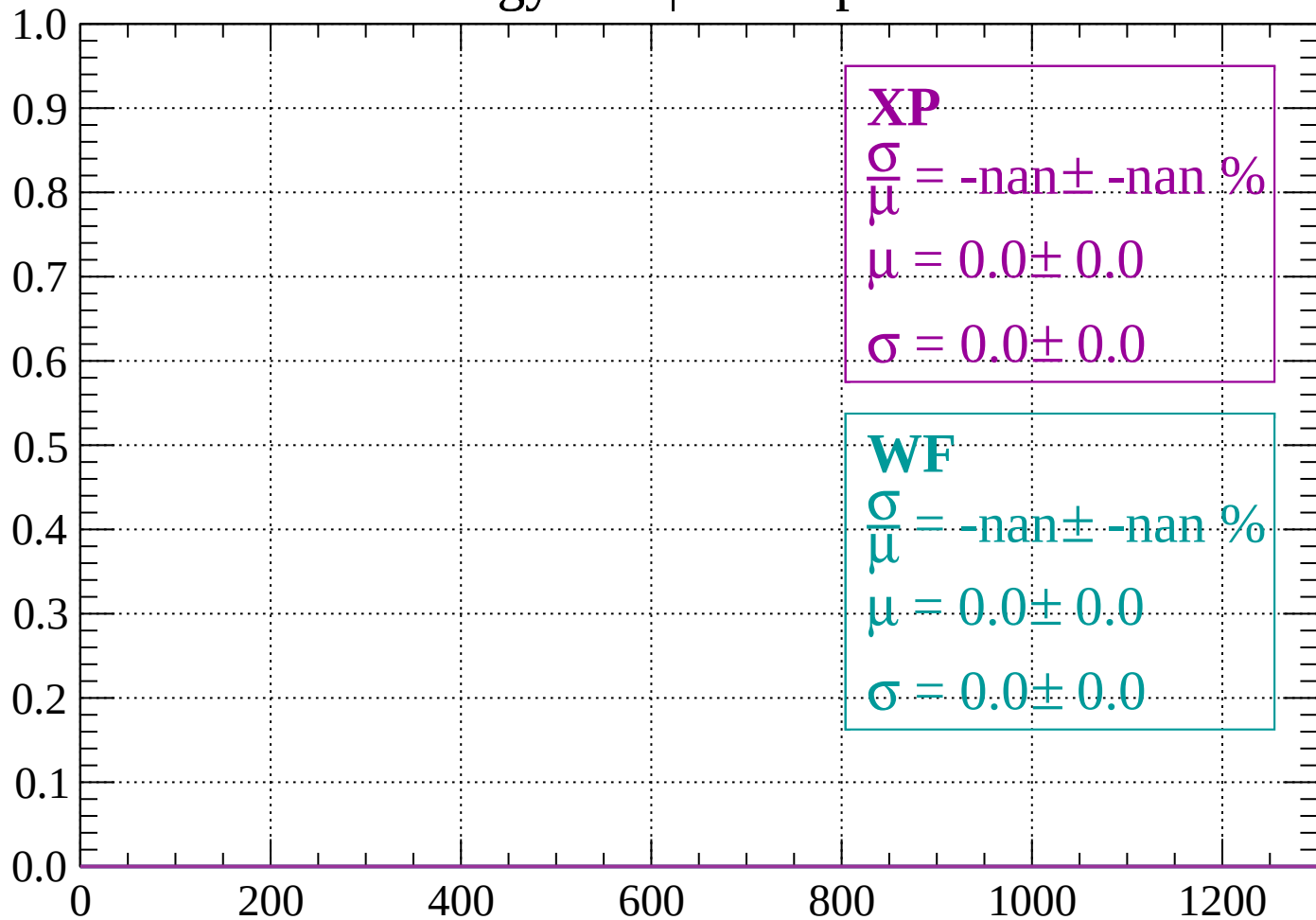
Count



dE/dx (ADC counts/cm)

Energy loss | $120 < p < 160$

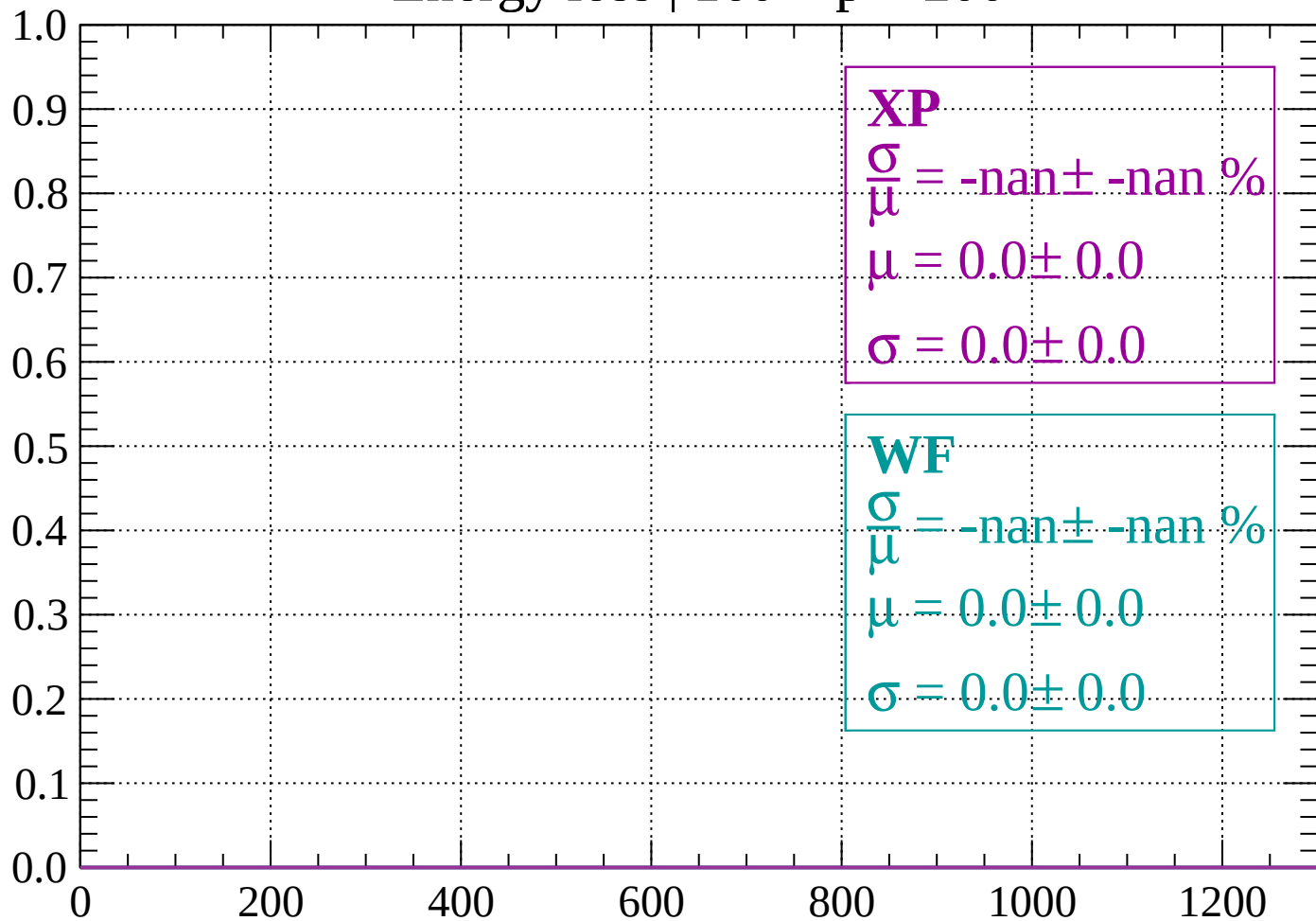
Count



dE/dx (ADC counts/cm)

Energy loss | $160 < p < 200$

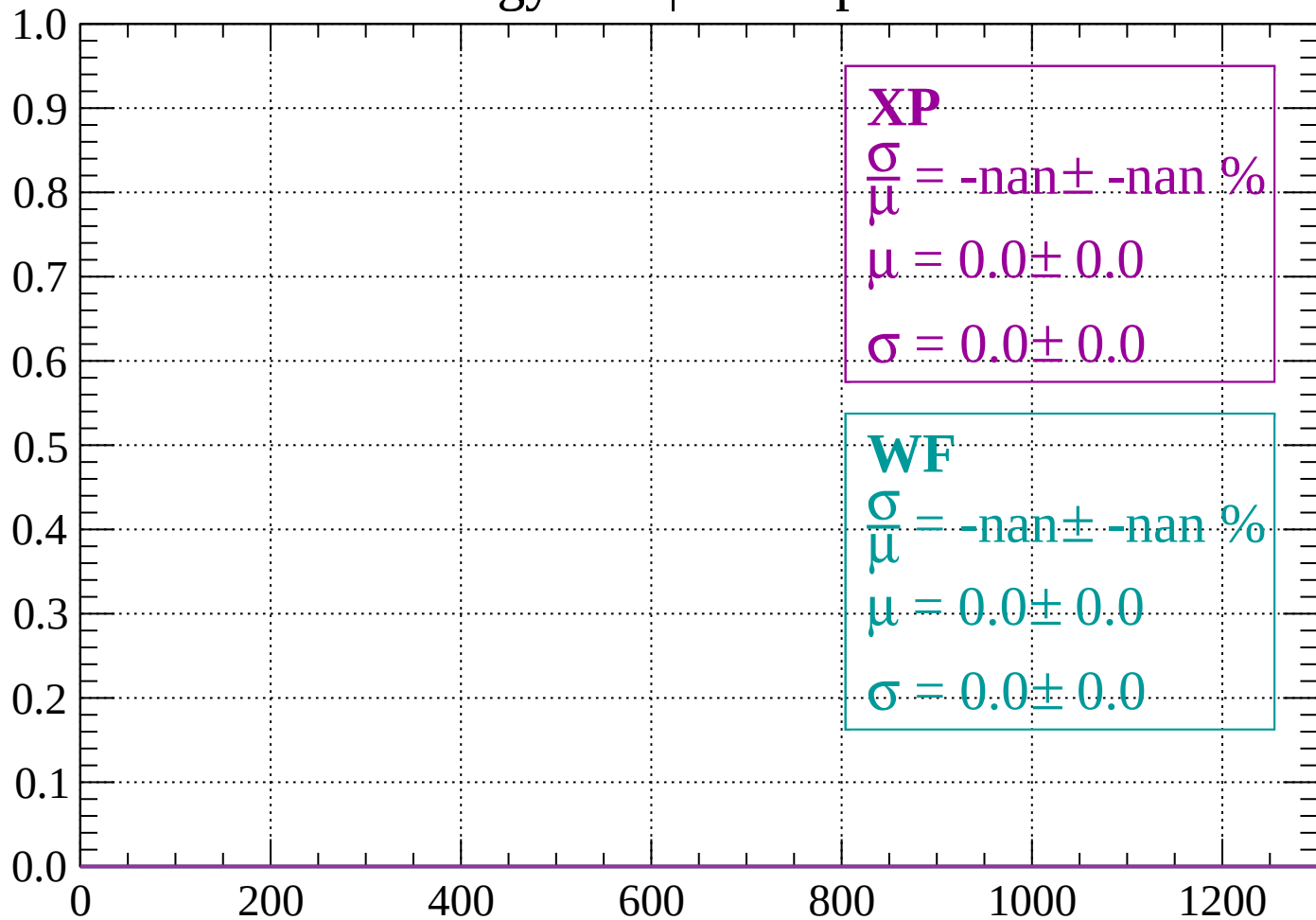
Count



dE/dx (ADC counts/cm)

Energy loss | $200 < p < 240$

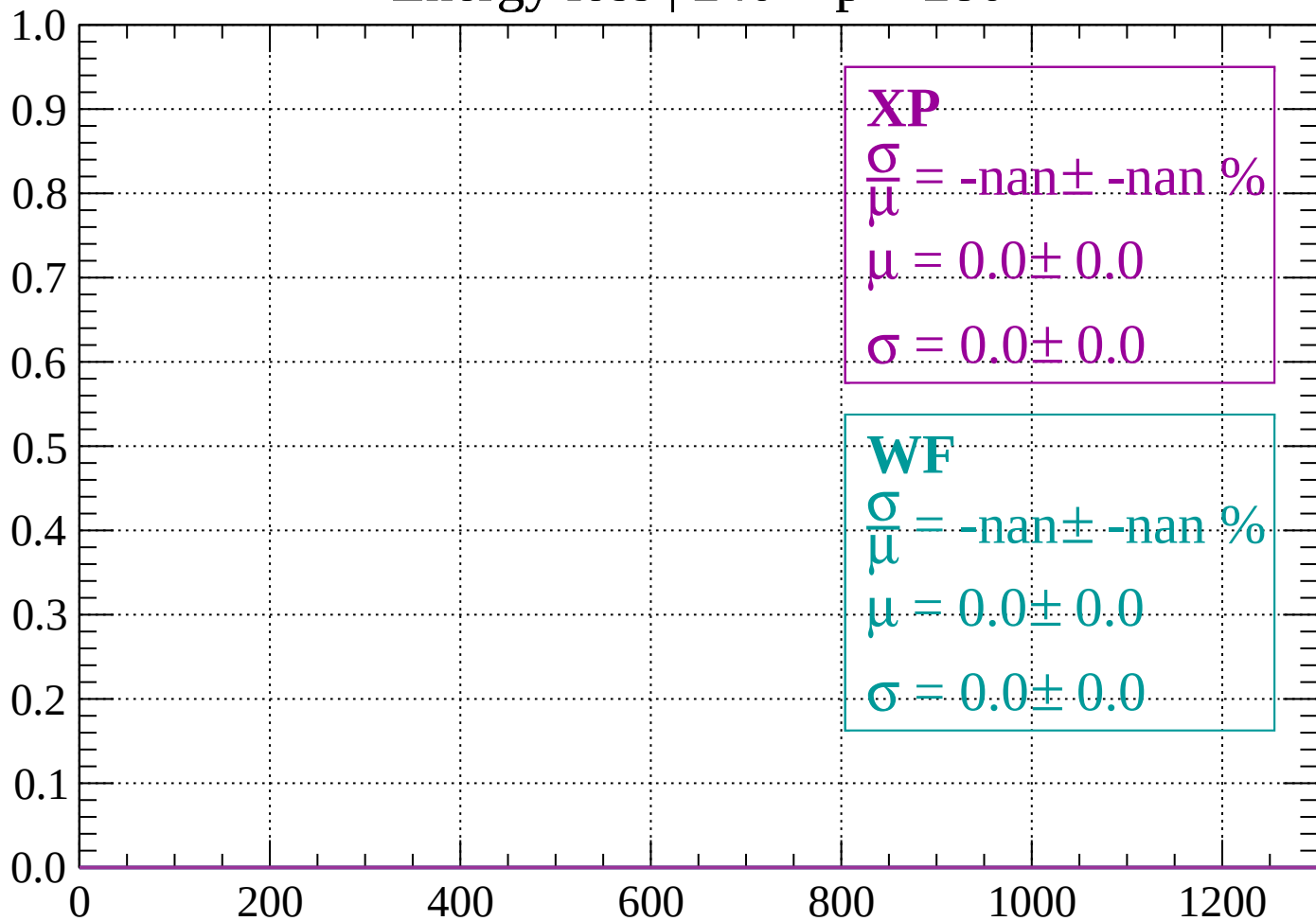
Count



dE/dx (ADC counts/cm)

Energy loss | $240 < p < 280$

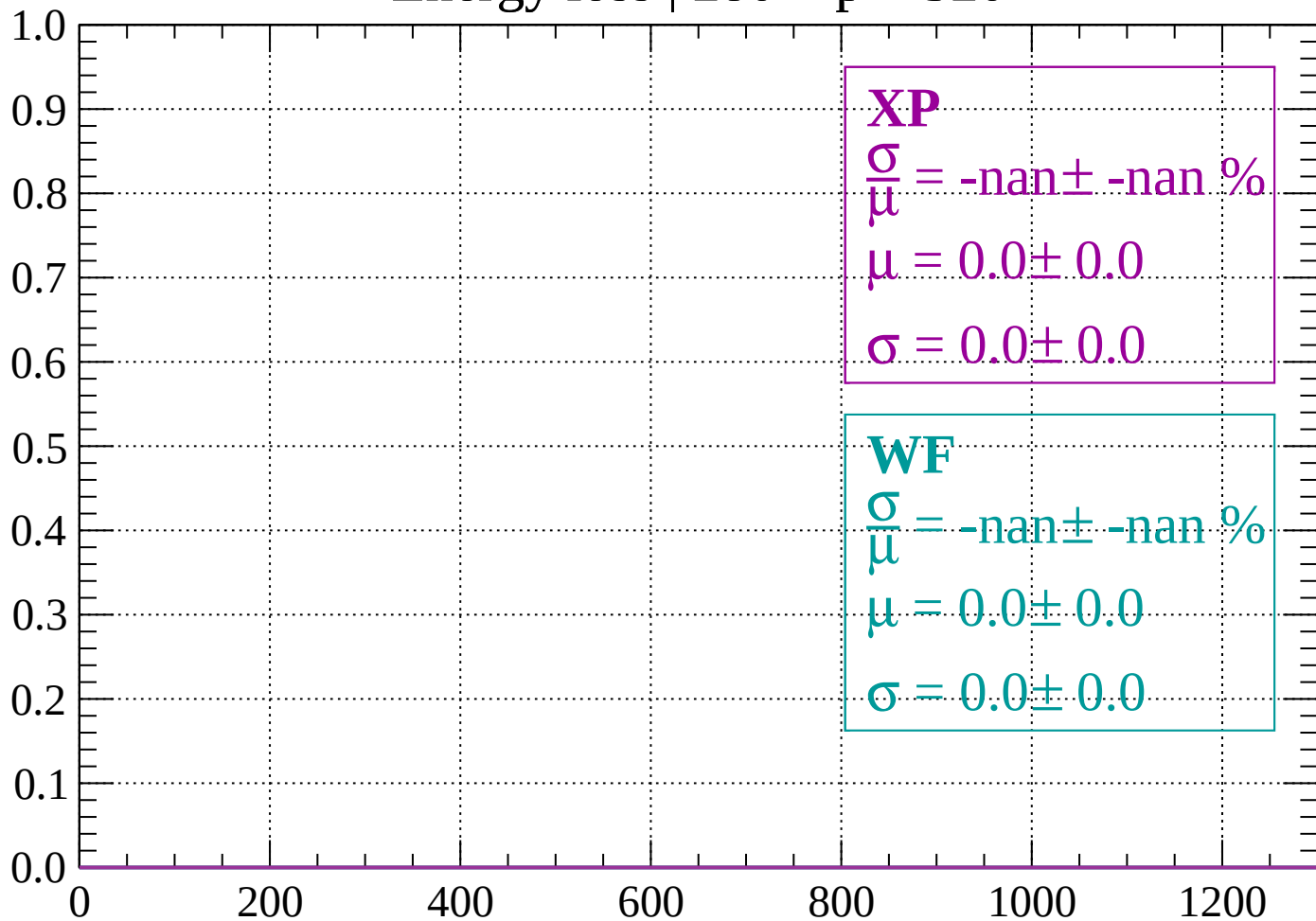
Count



dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

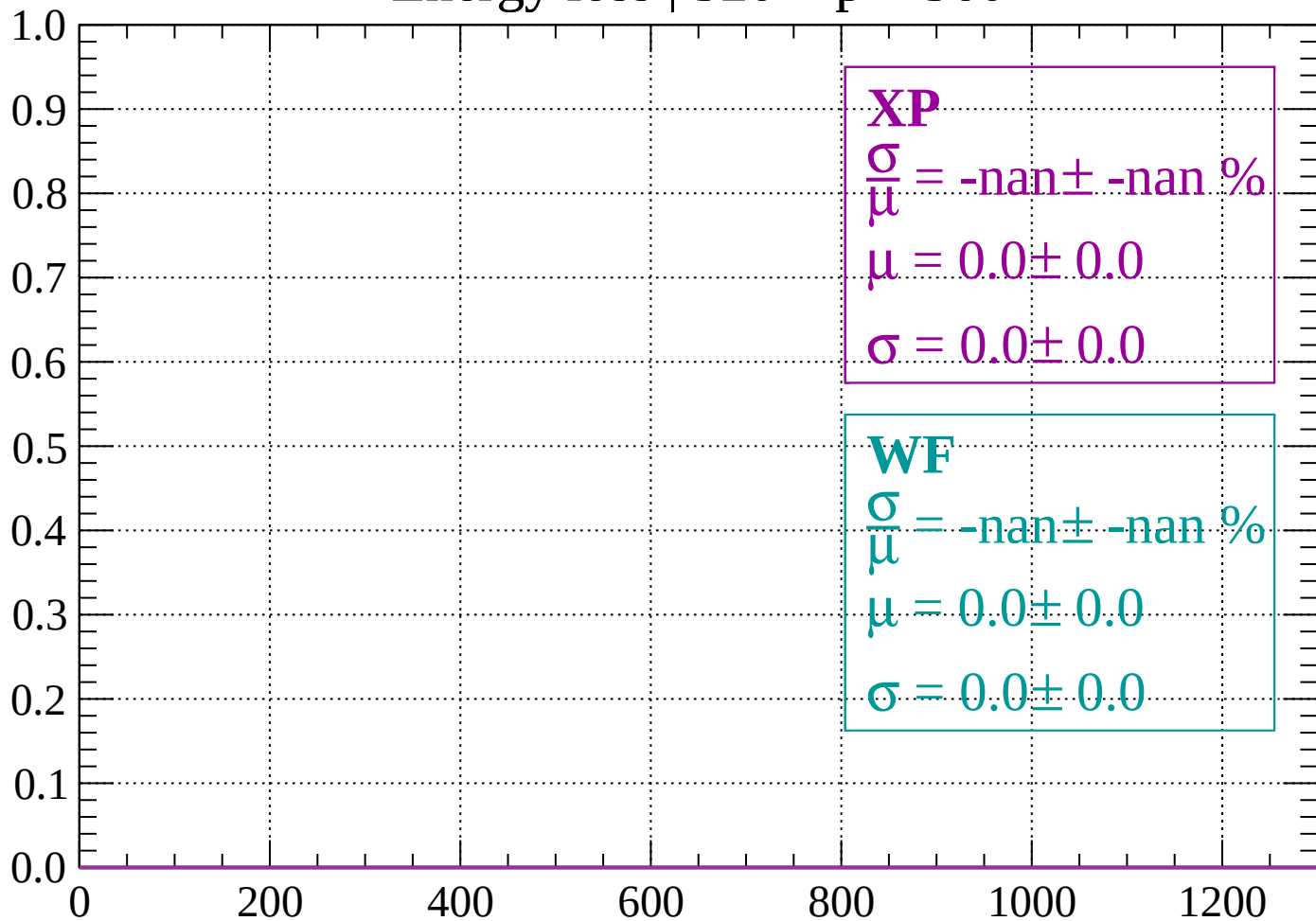
Count



dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

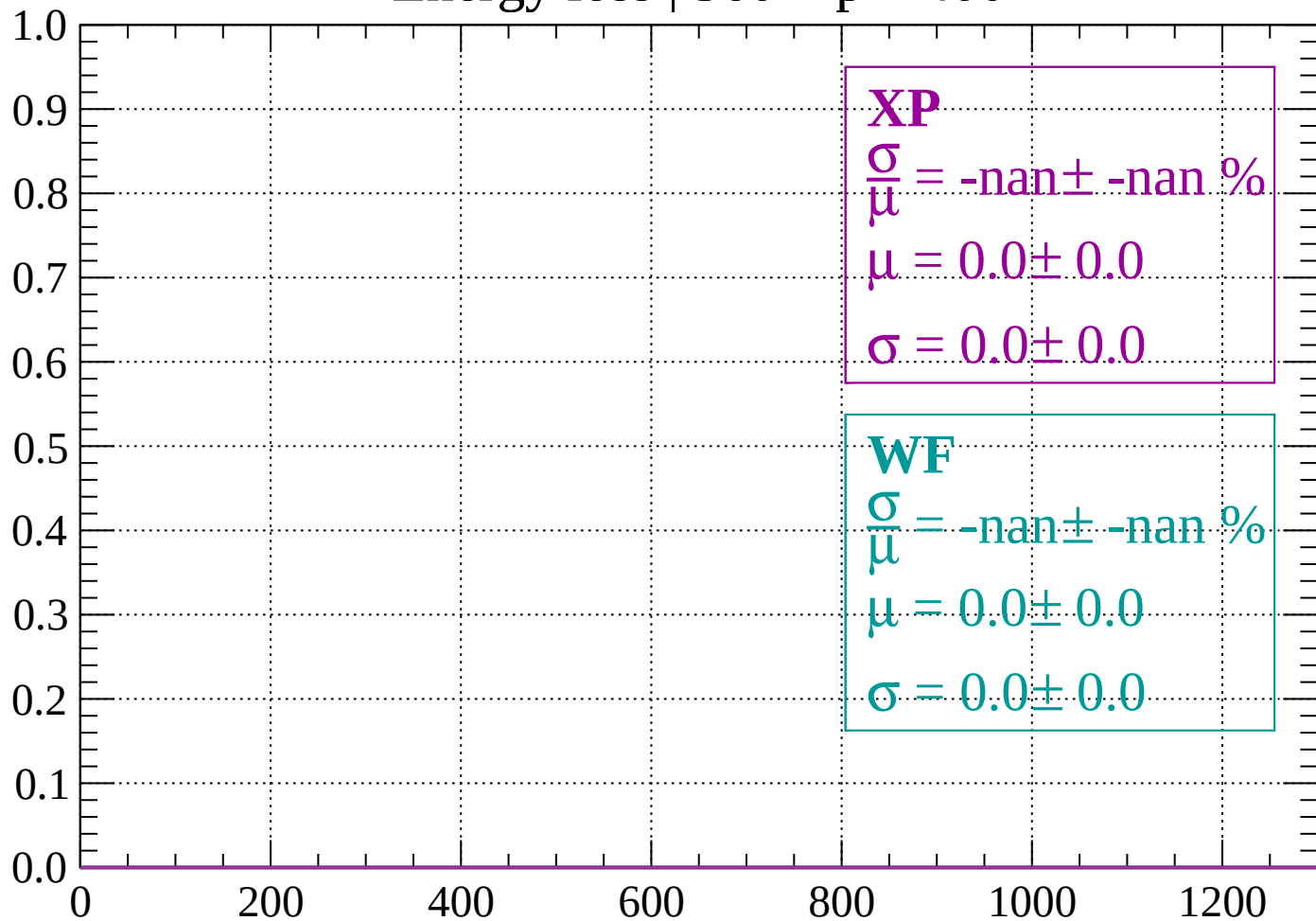
Count



dE/dx (ADC counts/cm)

Energy loss | $360 < p < 400$

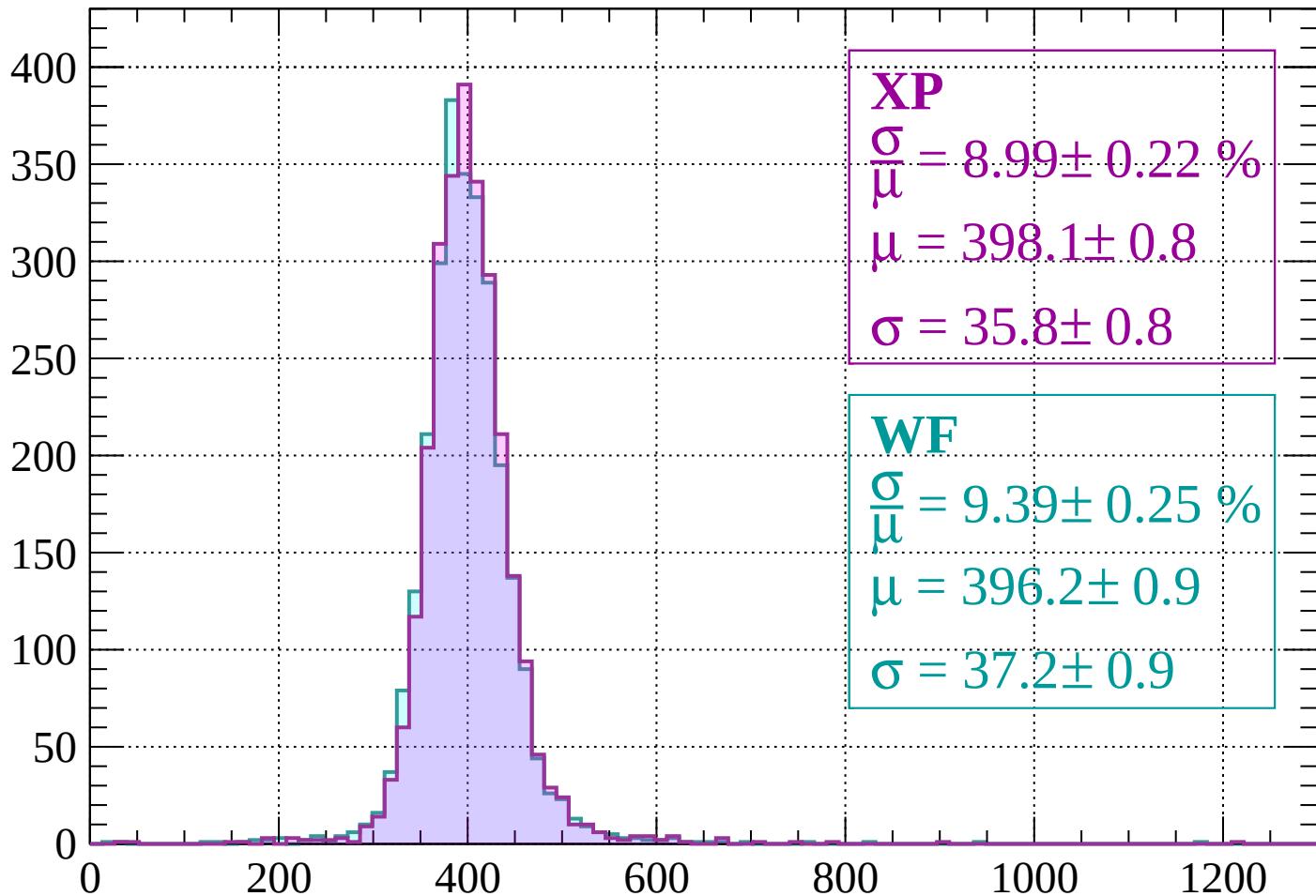
Count



dE/dx (ADC counts/cm)

Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 8.99 \pm 0.22 \%$$

$$\mu = 398.1 \pm 0.8$$

$$\sigma = 35.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.39 \pm 0.25 \%$$

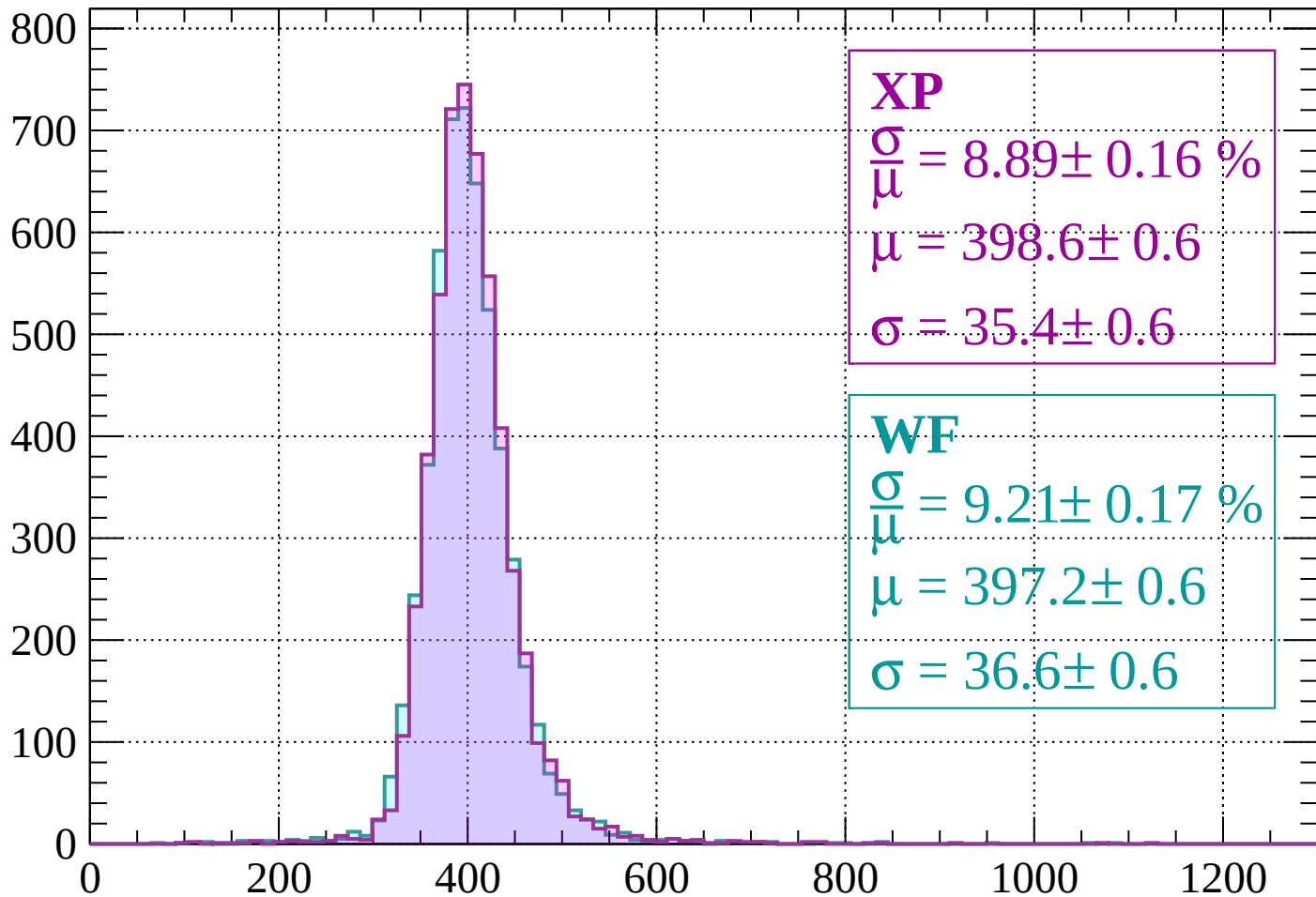
$$\mu = 396.2 \pm 0.9$$

$$\sigma = 37.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 8.89 \pm 0.16 \%$$

$$\mu = 398.6 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.21 \pm 0.17 \%$$

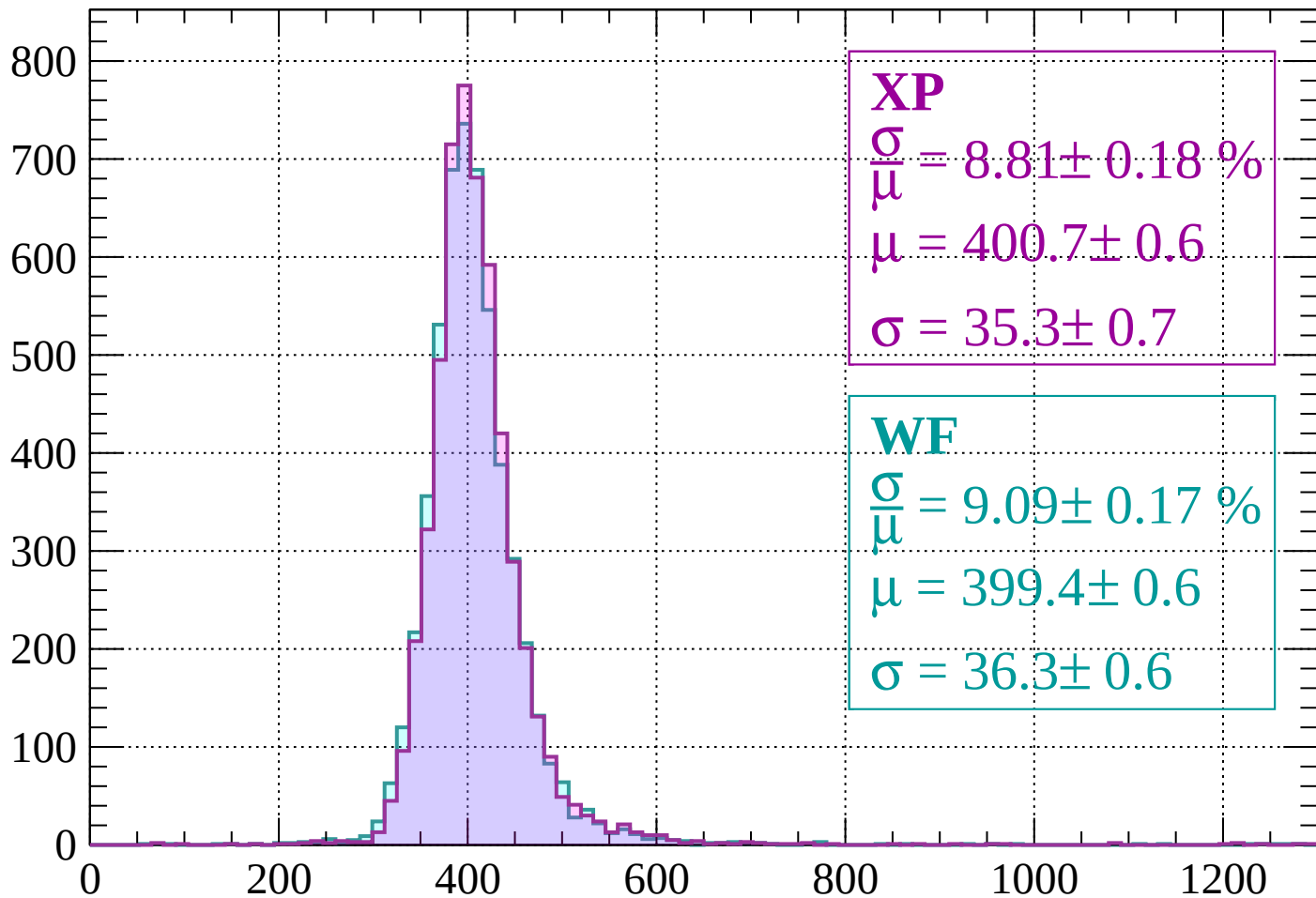
$$\mu = 397.2 \pm 0.6$$

$$\sigma = 36.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 520$

Count



XP

$$\frac{\sigma}{\mu} = 8.81 \pm 0.18 \%$$

$$\mu = 400.7 \pm 0.6$$

$$\sigma = 35.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.09 \pm 0.17 \%$$

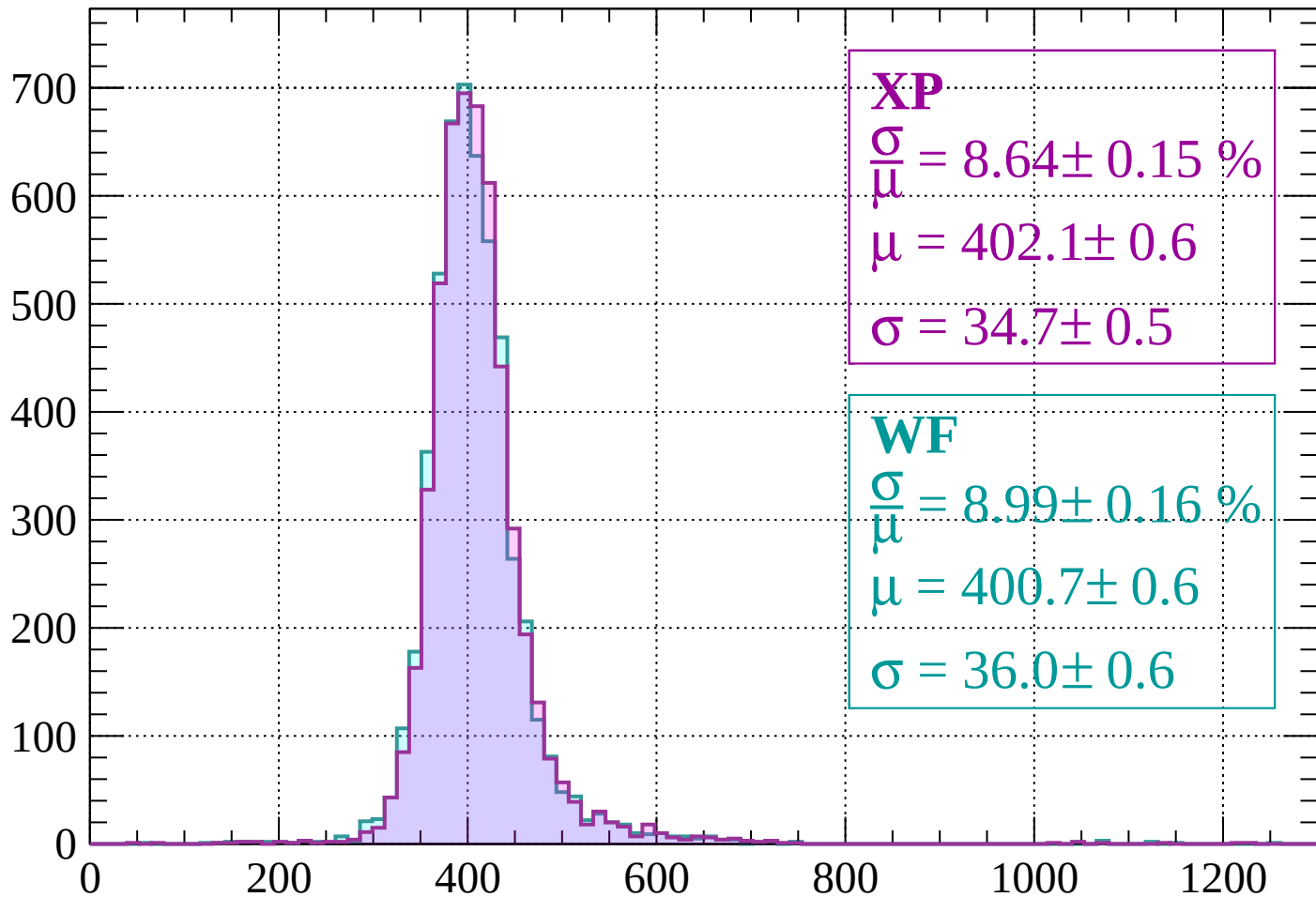
$$\mu = 399.4 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $520 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 8.64 \pm 0.15 \%$$

$$\mu = 402.1 \pm 0.6$$

$$\sigma = 34.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 8.99 \pm 0.16 \%$$

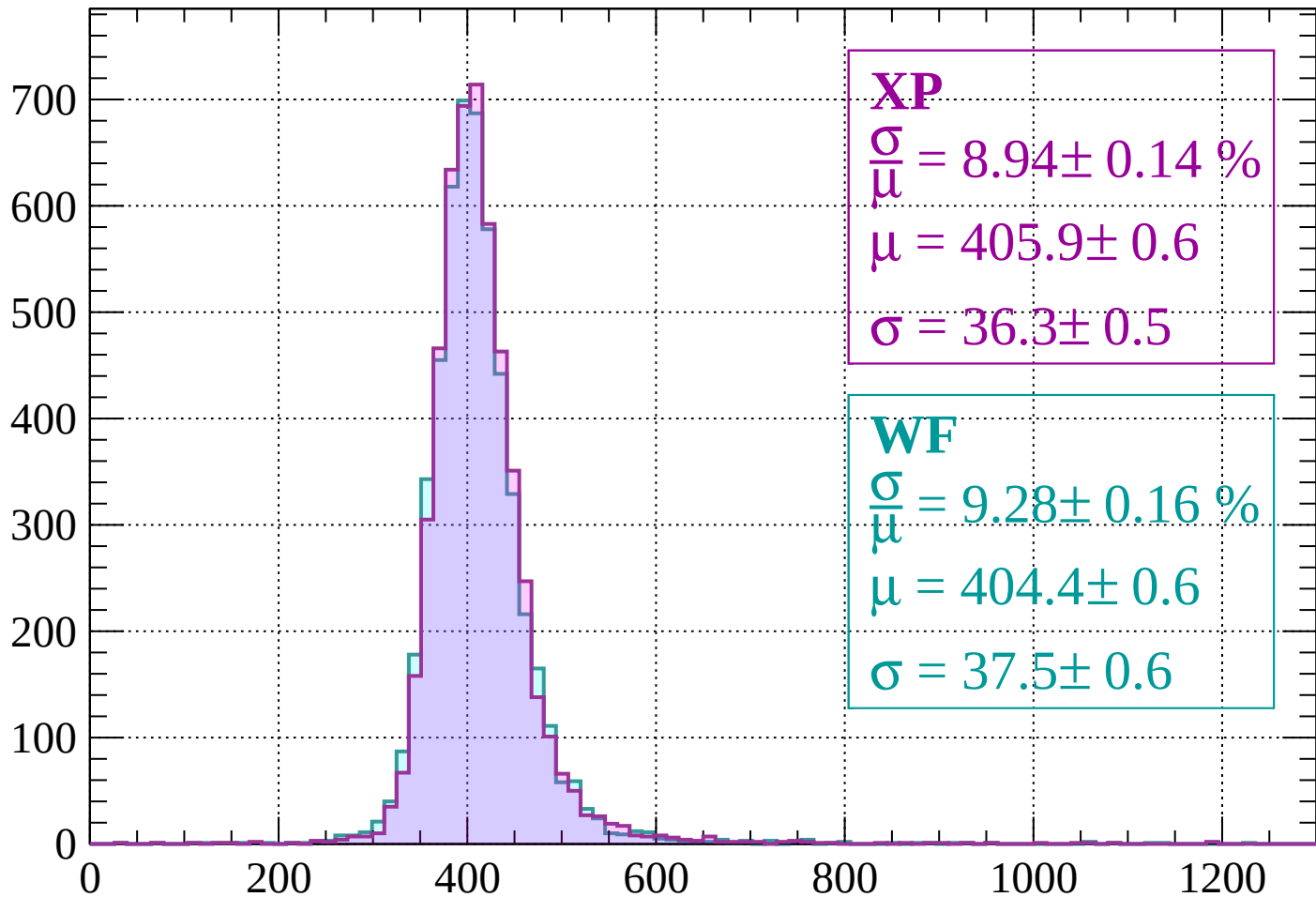
$$\mu = 400.7 \pm 0.6$$

$$\sigma = 36.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 600$

Count



XP

$$\frac{\sigma}{\mu} = 8.94 \pm 0.14 \%$$

$$\mu = 405.9 \pm 0.6$$

$$\sigma = 36.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.28 \pm 0.16 \%$$

$$\mu = 404.4 \pm 0.6$$

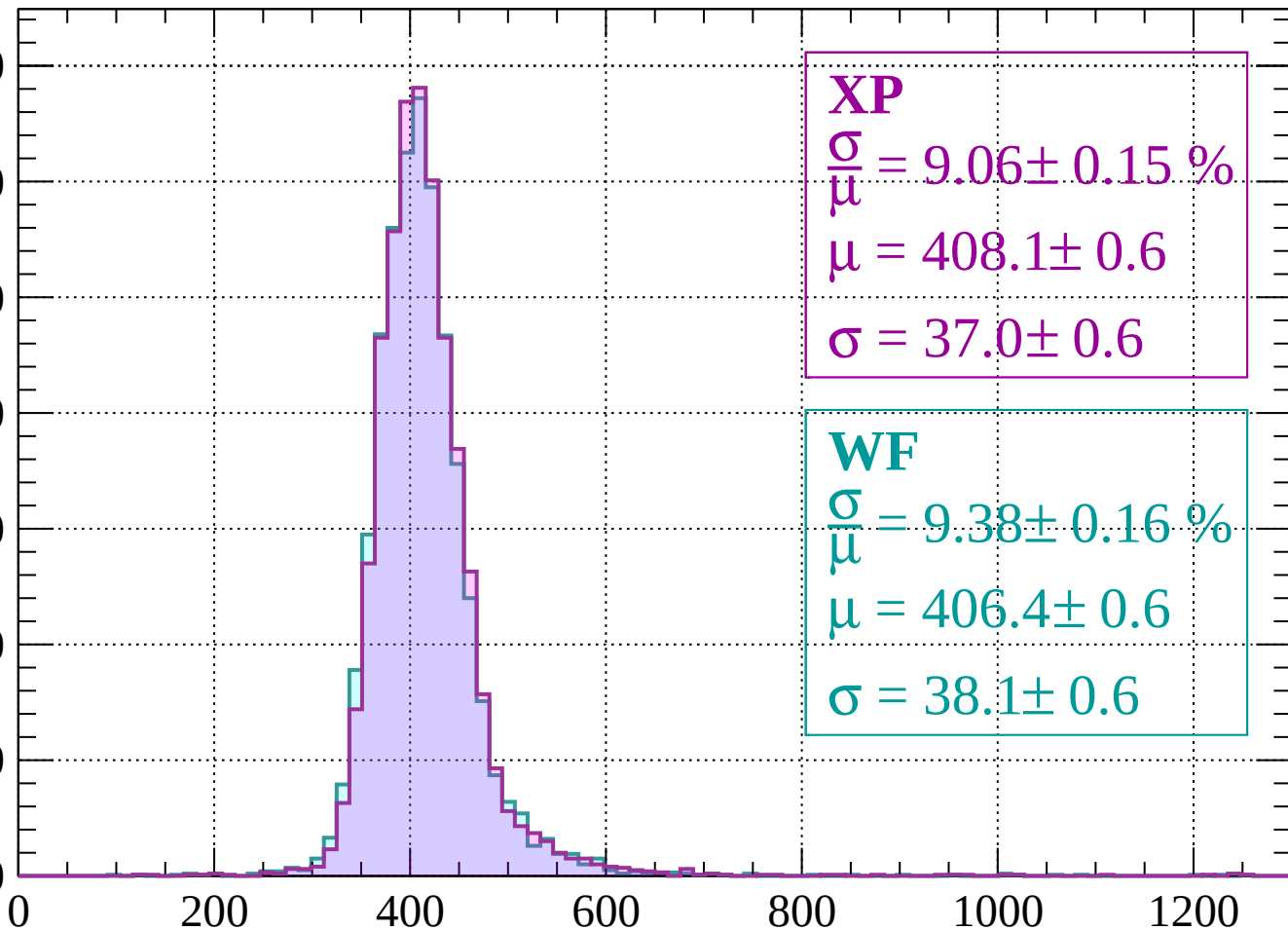
$$\sigma = 37.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 640$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 9.06 \pm 0.15 \%$$

$$\mu = 408.1 \pm 0.6$$

$$\sigma = 37.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.38 \pm 0.16 \%$$

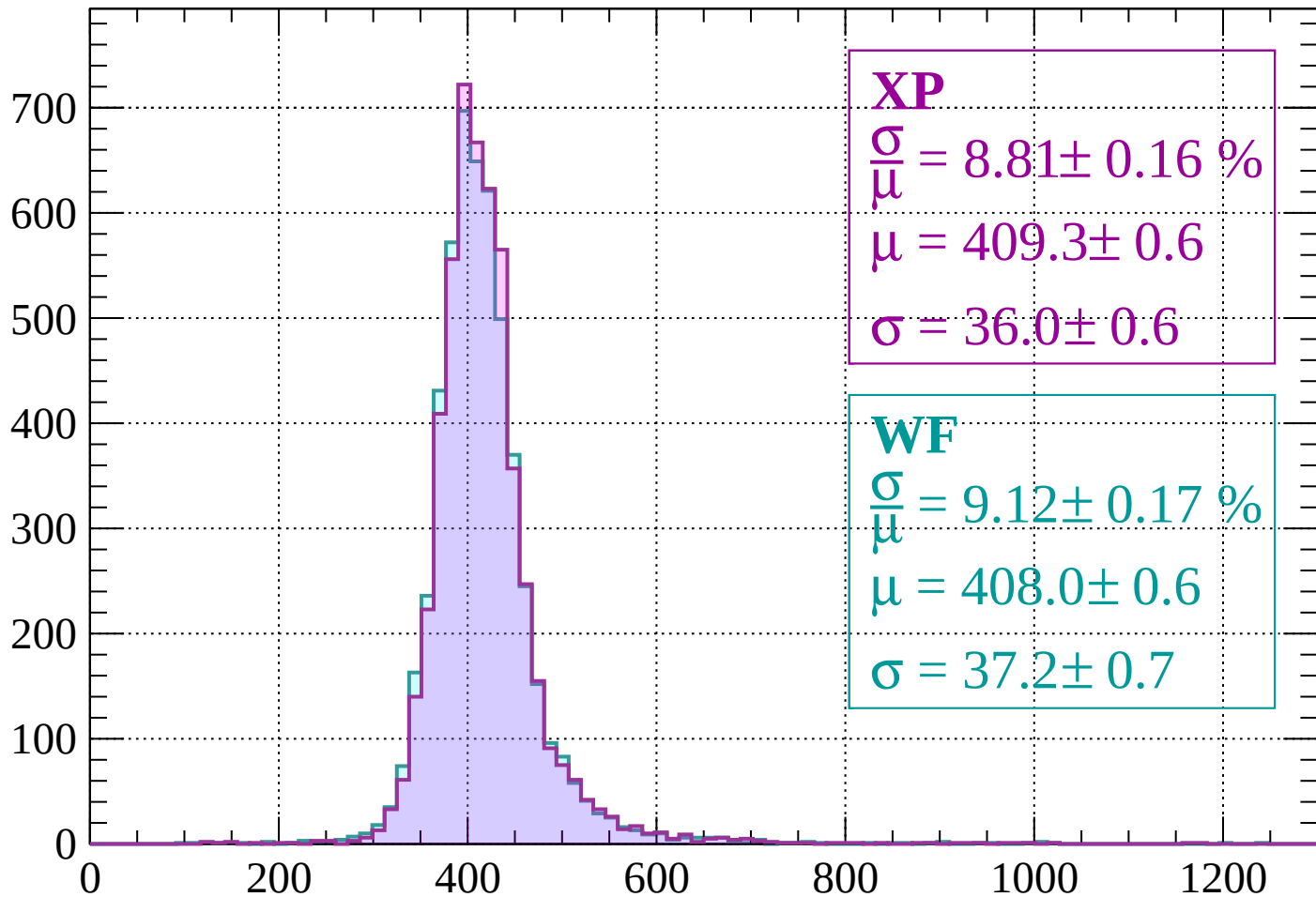
$$\mu = 406.4 \pm 0.6$$

$$\sigma = 38.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 680$

Count



XP

$$\frac{\sigma}{\mu} = 8.81 \pm 0.16 \%$$

$$\mu = 409.3 \pm 0.6$$

$$\sigma = 36.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.12 \pm 0.17 \%$$

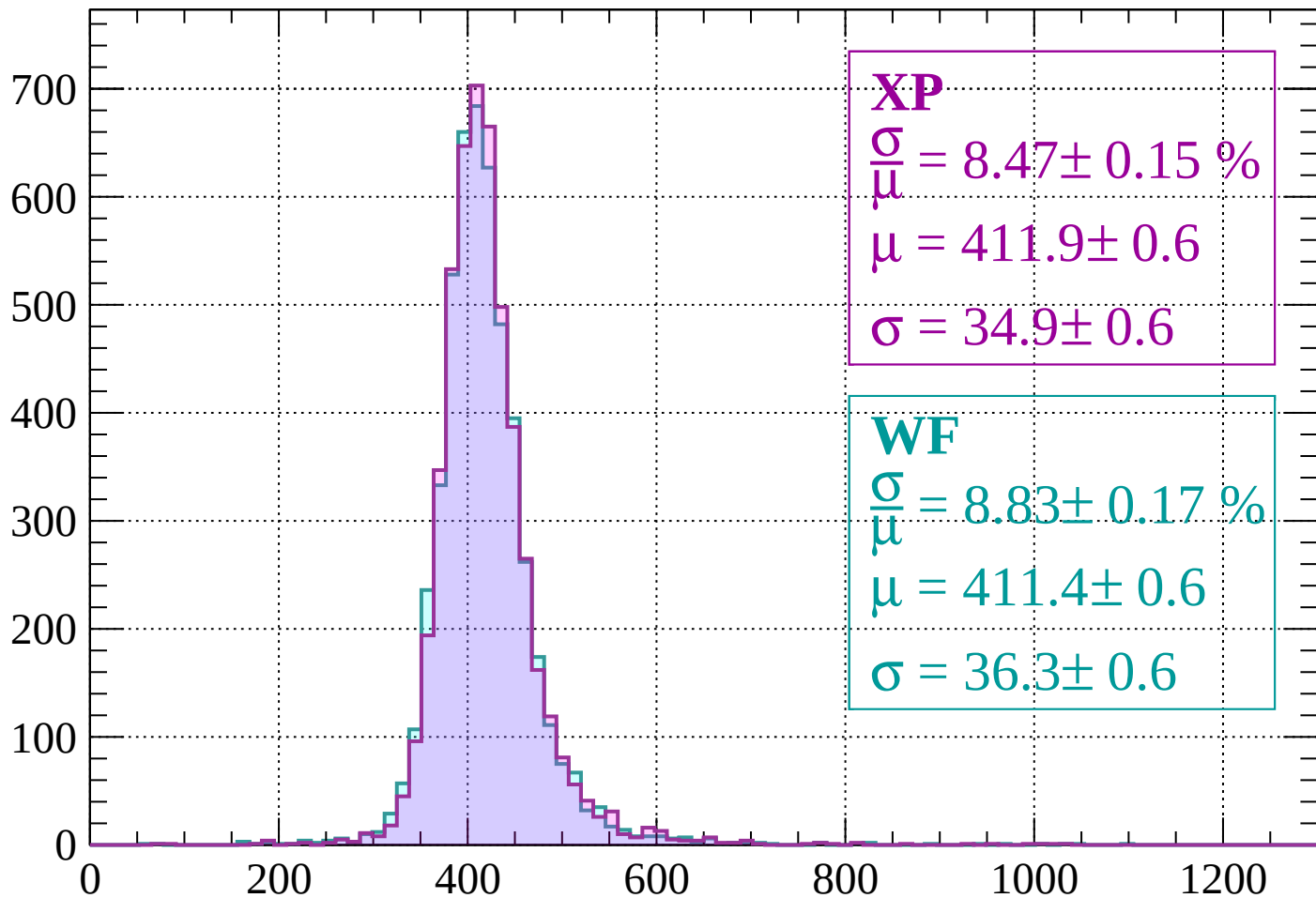
$$\mu = 408.0 \pm 0.6$$

$$\sigma = 37.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $680 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 8.47 \pm 0.15 \%$$

$$\mu = 411.9 \pm 0.6$$

$$\sigma = 34.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.83 \pm 0.17 \%$$

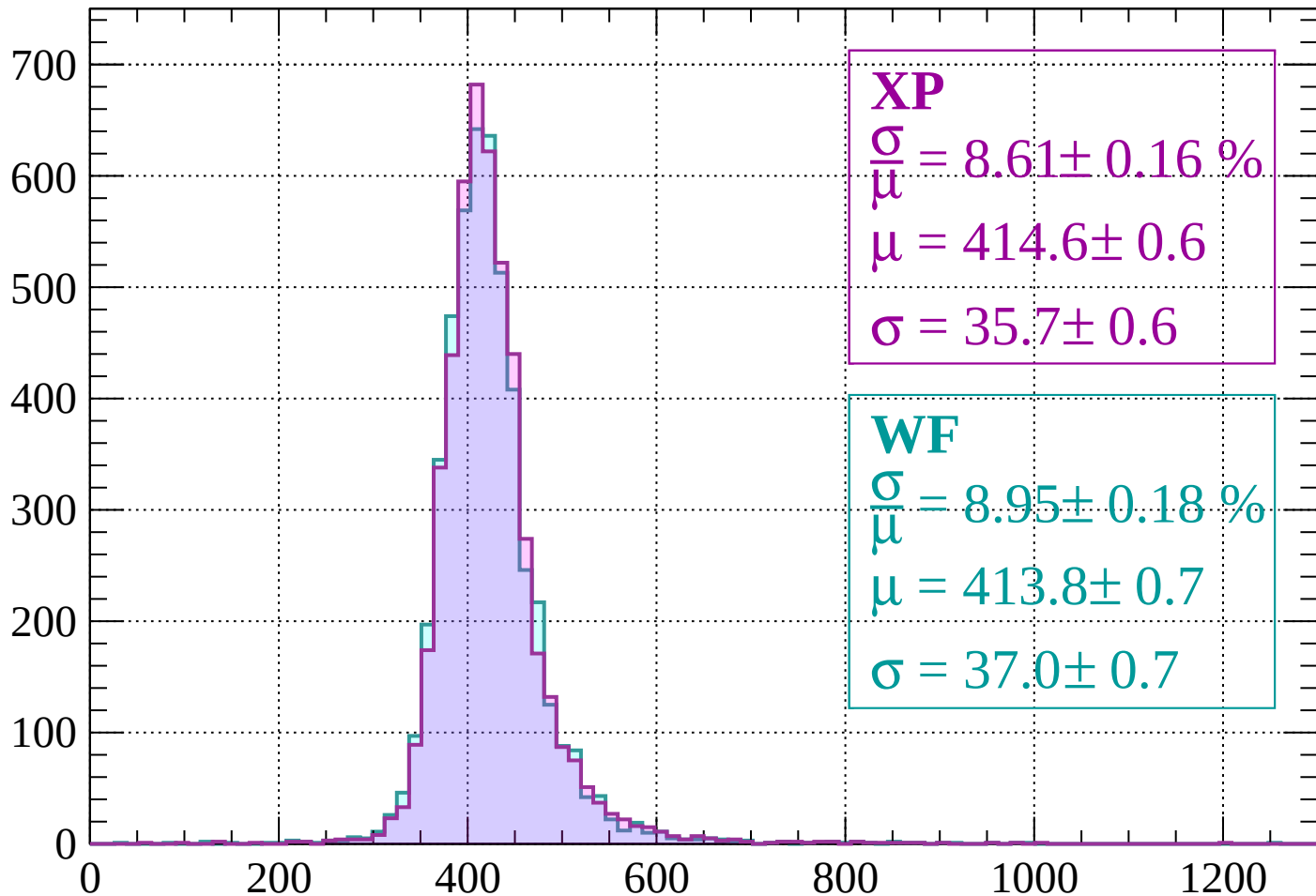
$$\mu = 411.4 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 760$

Count



XP

$$\frac{\sigma}{\mu} = 8.61 \pm 0.16 \%$$

$$\mu = 414.6 \pm 0.6$$

$$\sigma = 35.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.95 \pm 0.18 \%$$

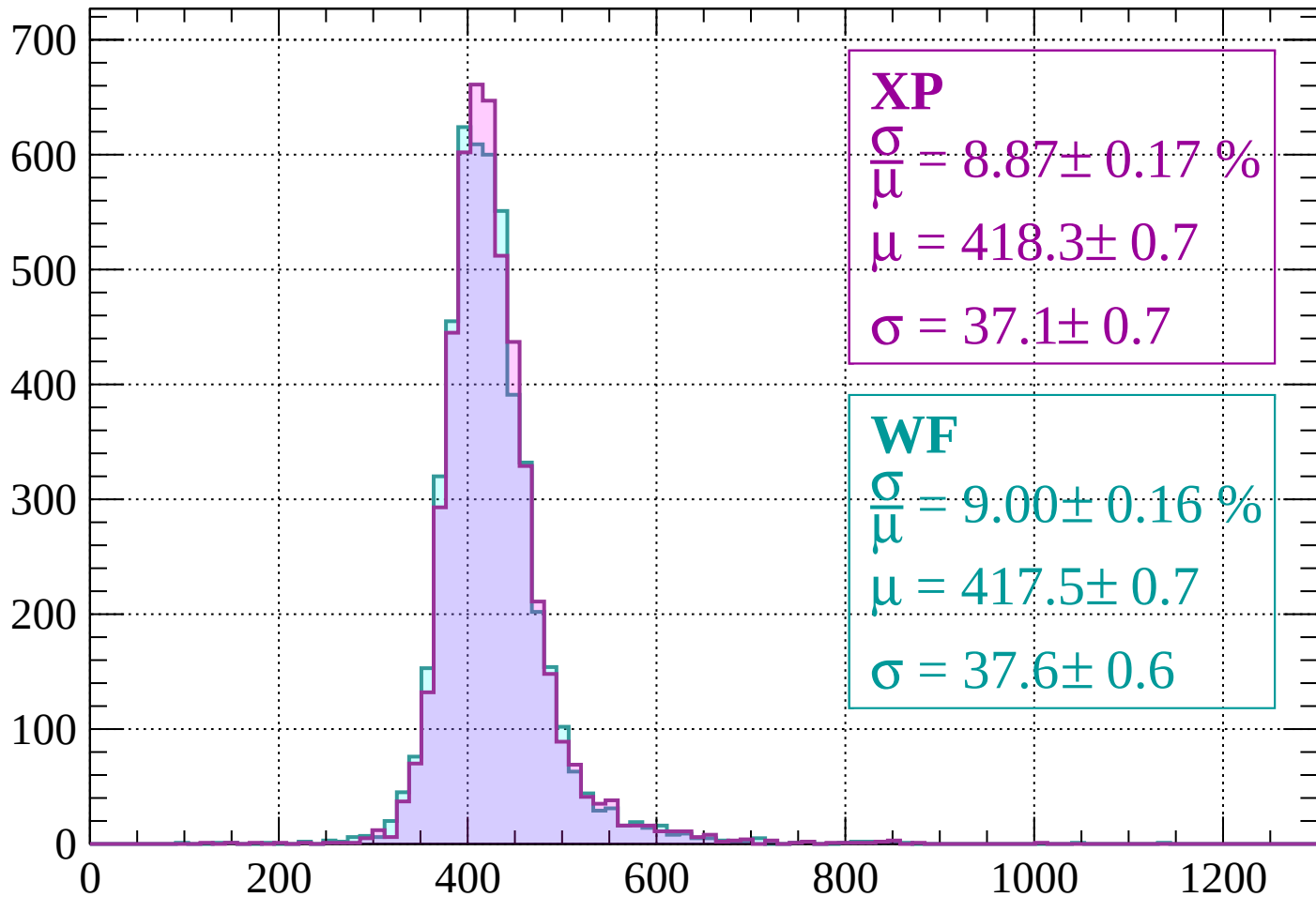
$$\mu = 413.8 \pm 0.7$$

$$\sigma = 37.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.87 \pm 0.17 \%$$

$$\mu = 418.3 \pm 0.7$$

$$\sigma = 37.1 \pm 0.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.00 \pm 0.16 \%$$

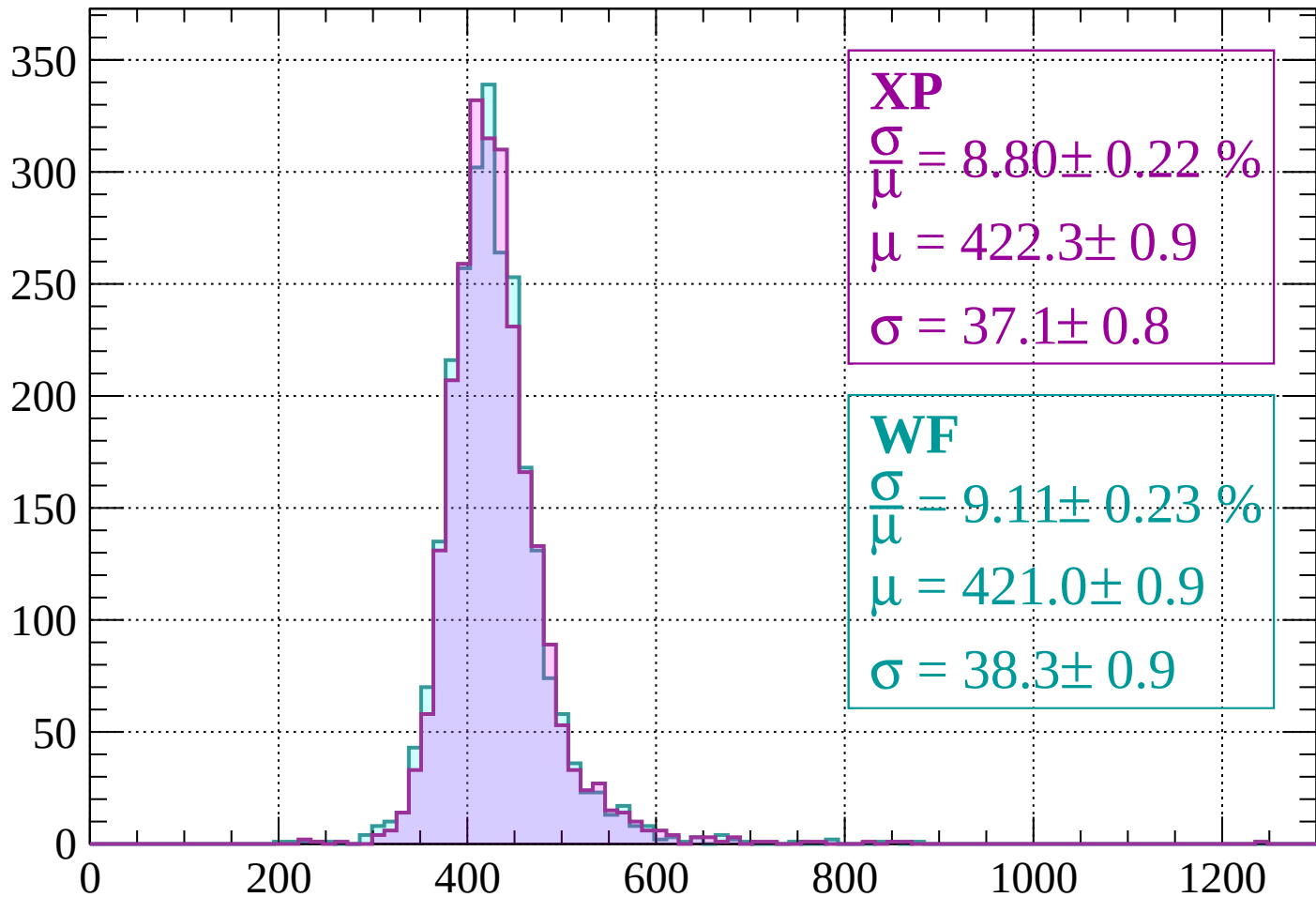
$$\mu = 417.5 \pm 0.7$$

$$\sigma = 37.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 840$

Count



XP

$$\frac{\sigma}{\mu} = 8.80 \pm 0.22 \%$$

$$\mu = 422.3 \pm 0.9$$

$$\sigma = 37.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.11 \pm 0.23 \%$$

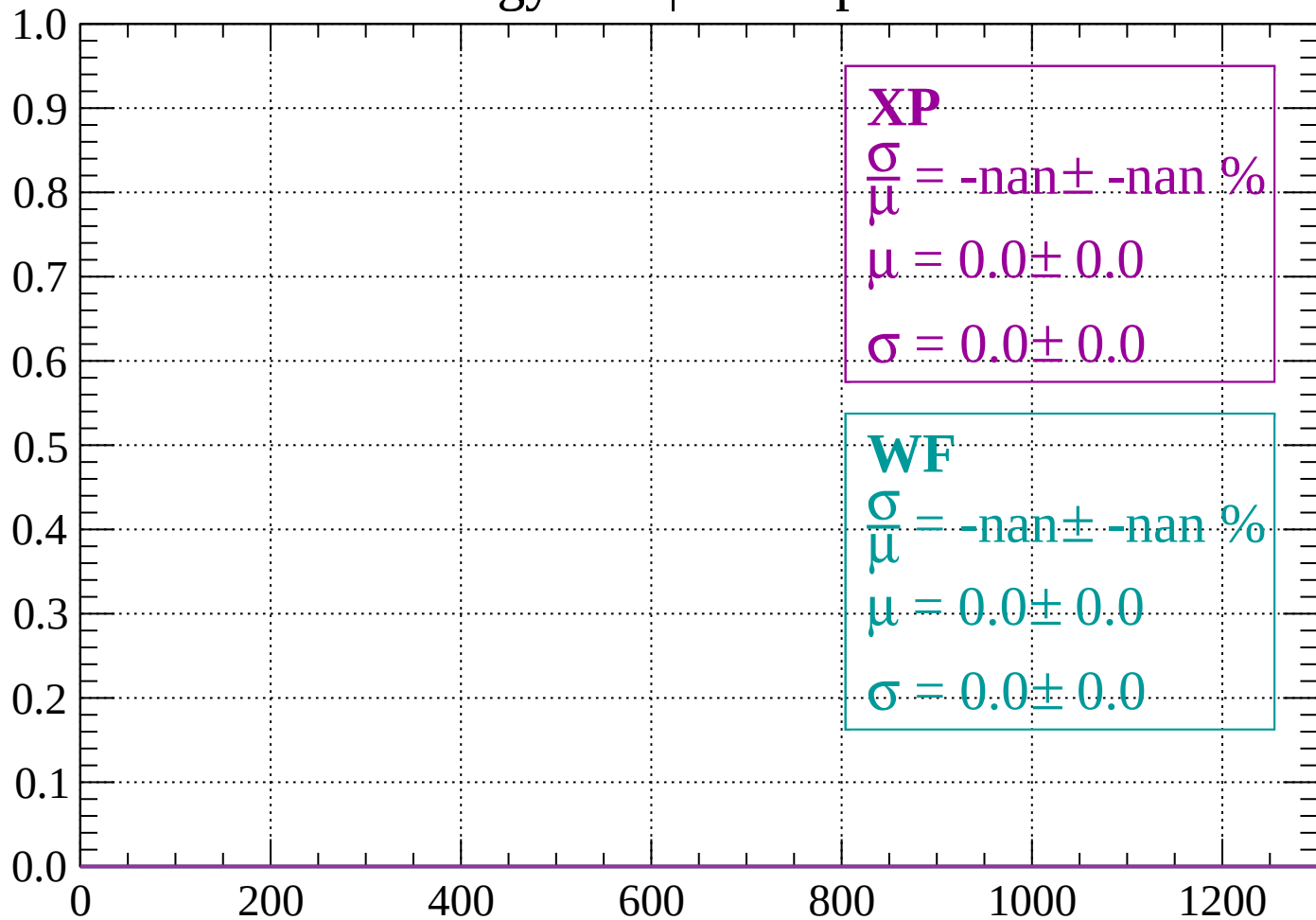
$$\mu = 421.0 \pm 0.9$$

$$\sigma = 38.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 880$

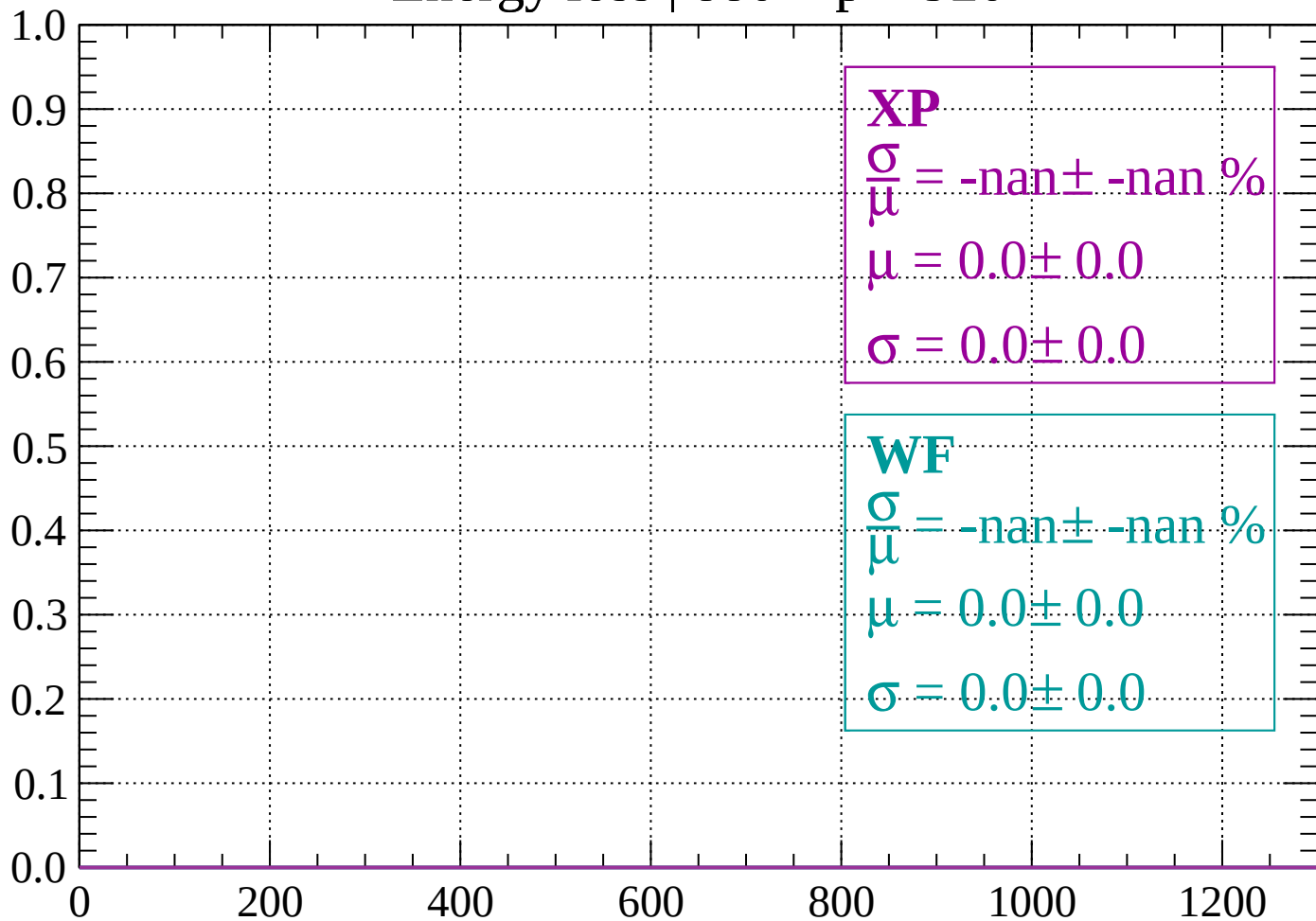
Count



dE/dx (ADC counts/cm)

Energy loss | $880 < p < 920$

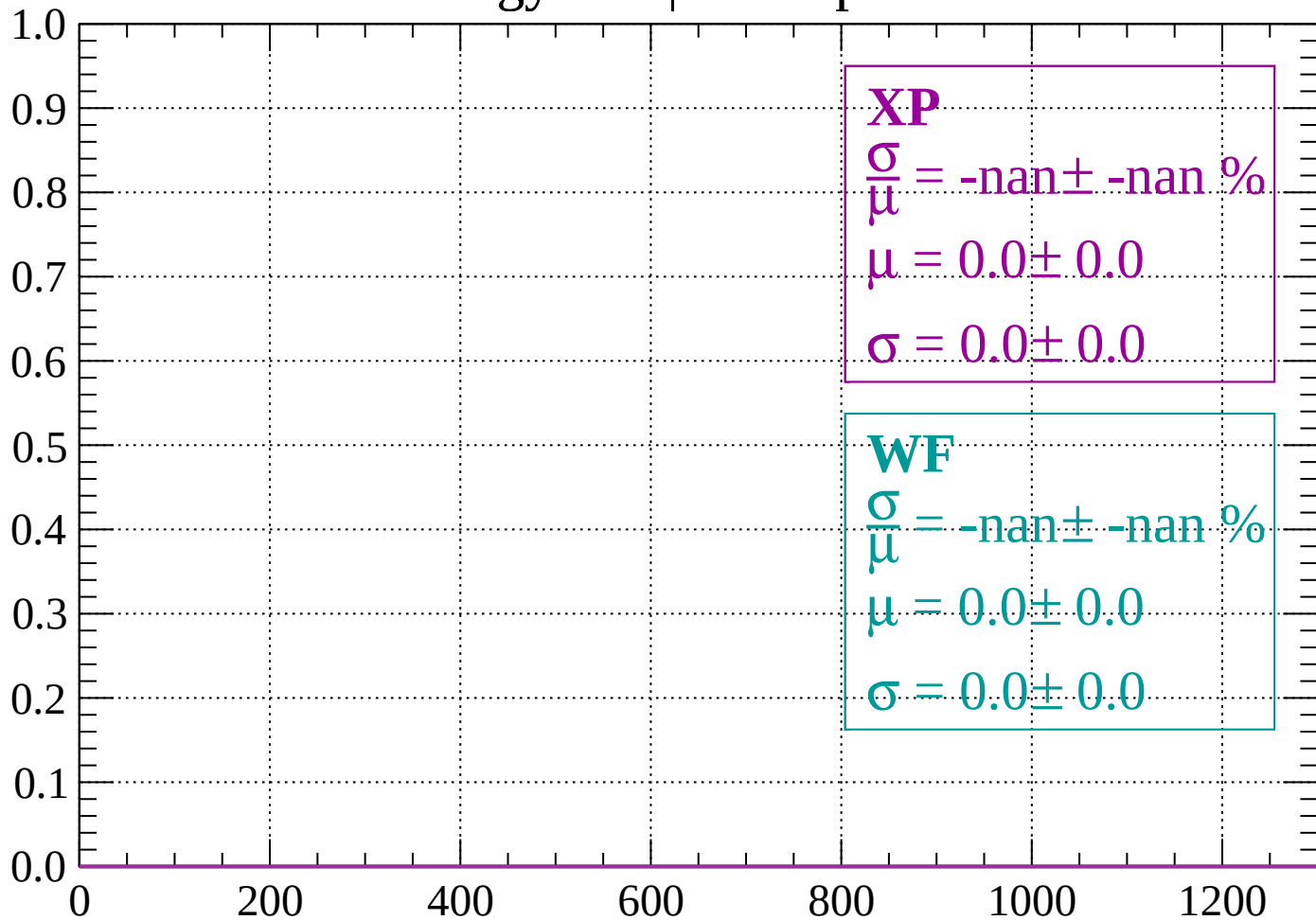
Count



dE/dx (ADC counts/cm)

Energy loss | $920 < p < 960$

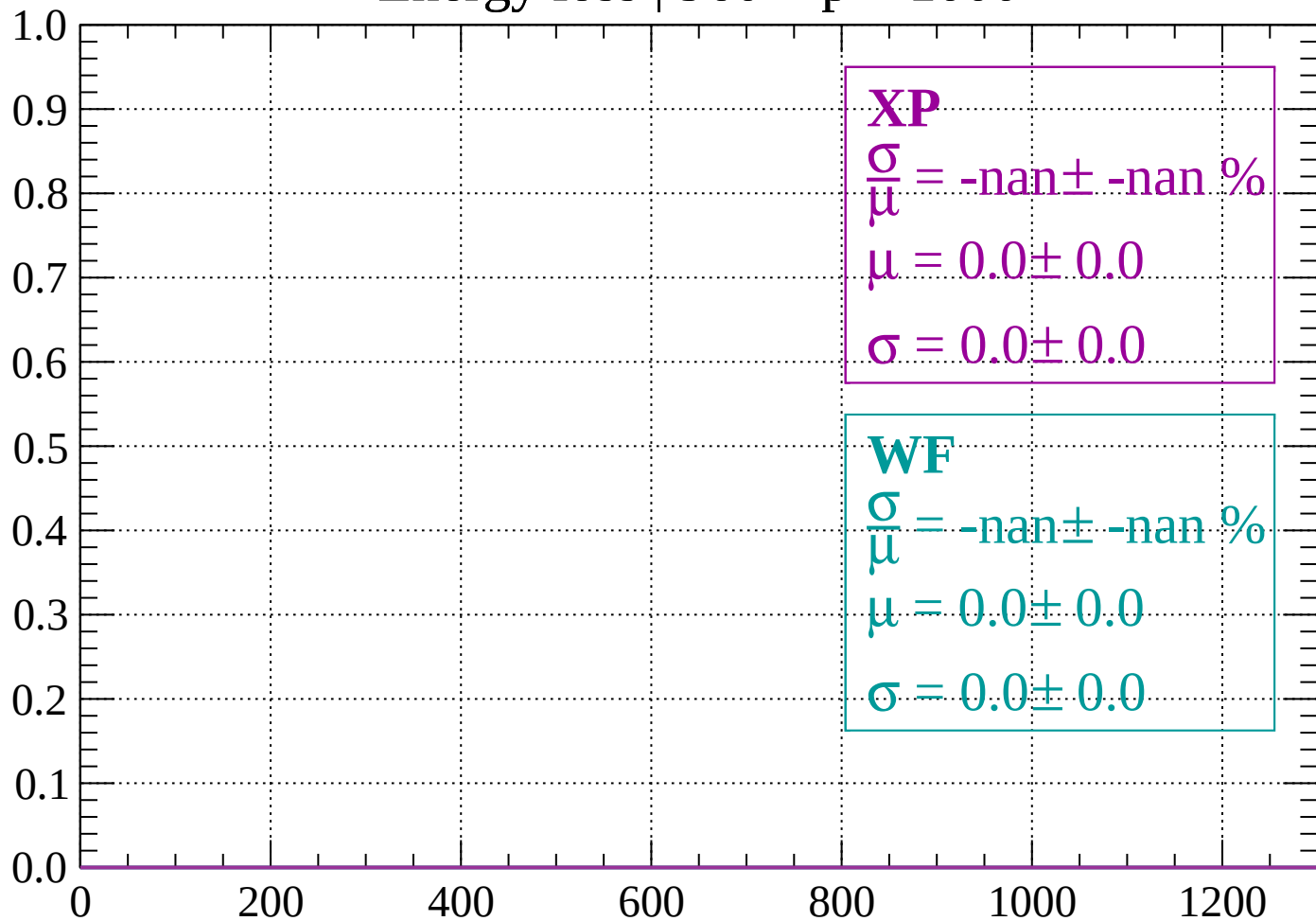
Count



dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1000$

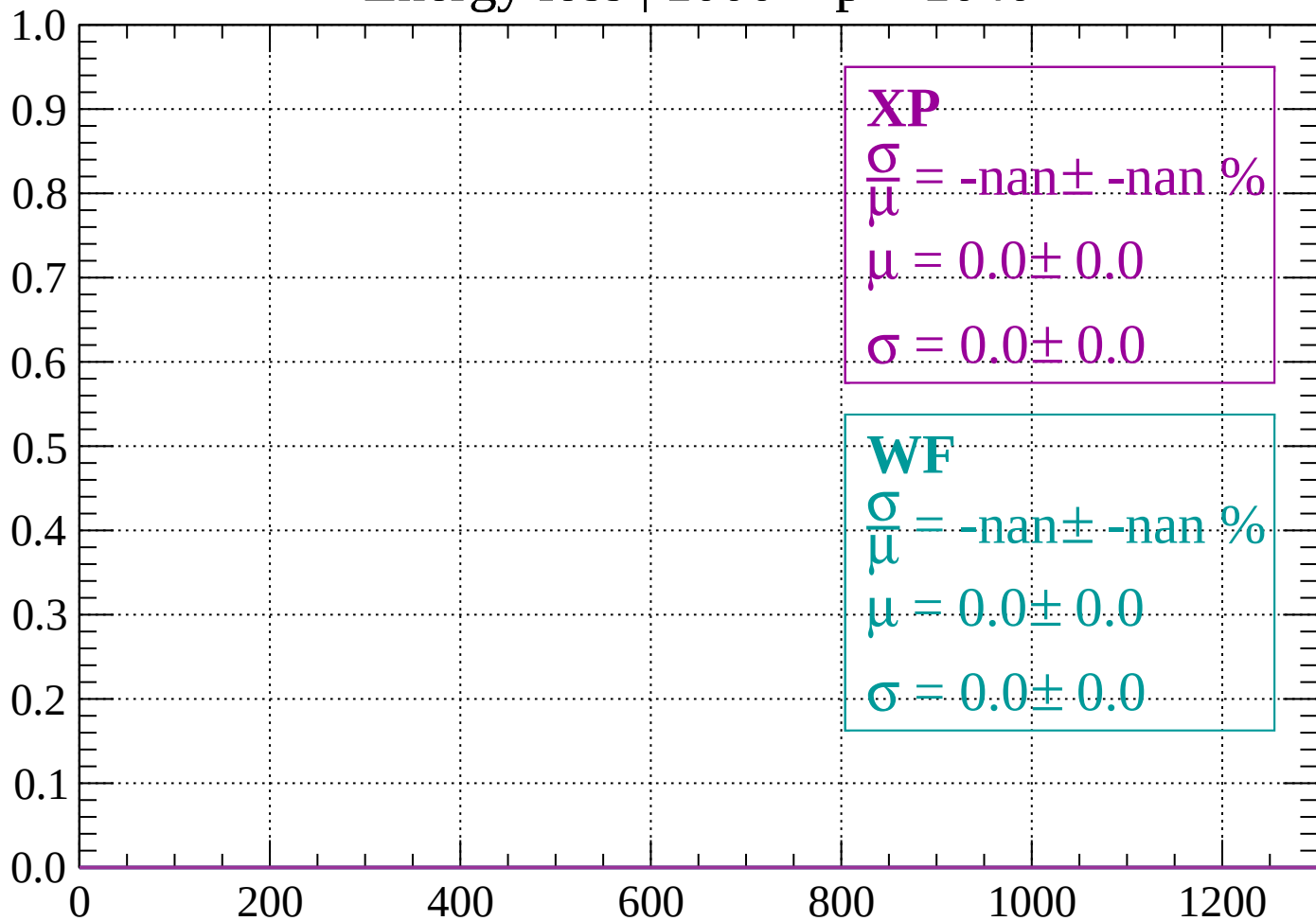
Count



dE/dx (ADC counts/cm)

Energy loss | $1000 < p < 1040$

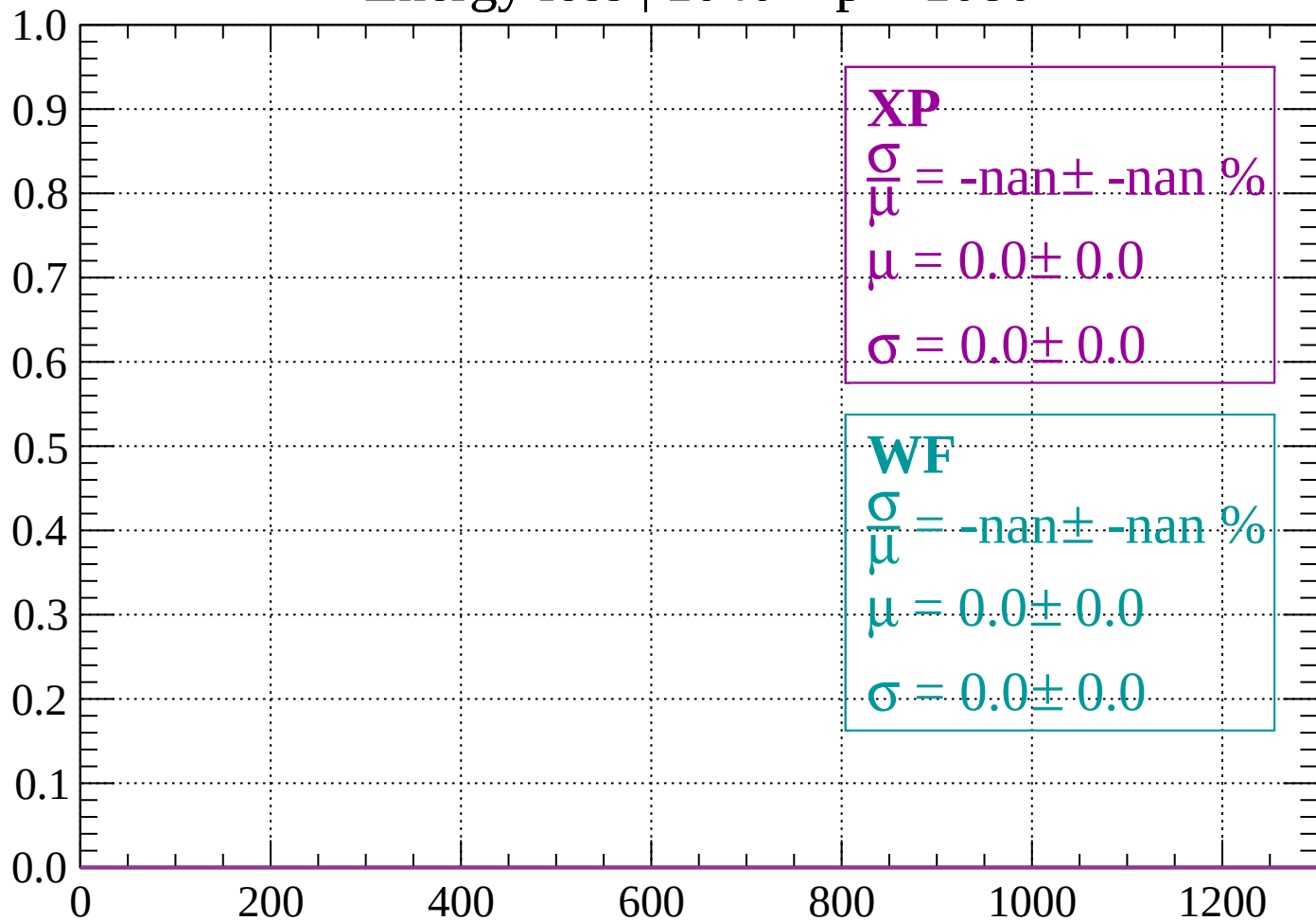
Count



dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1080$

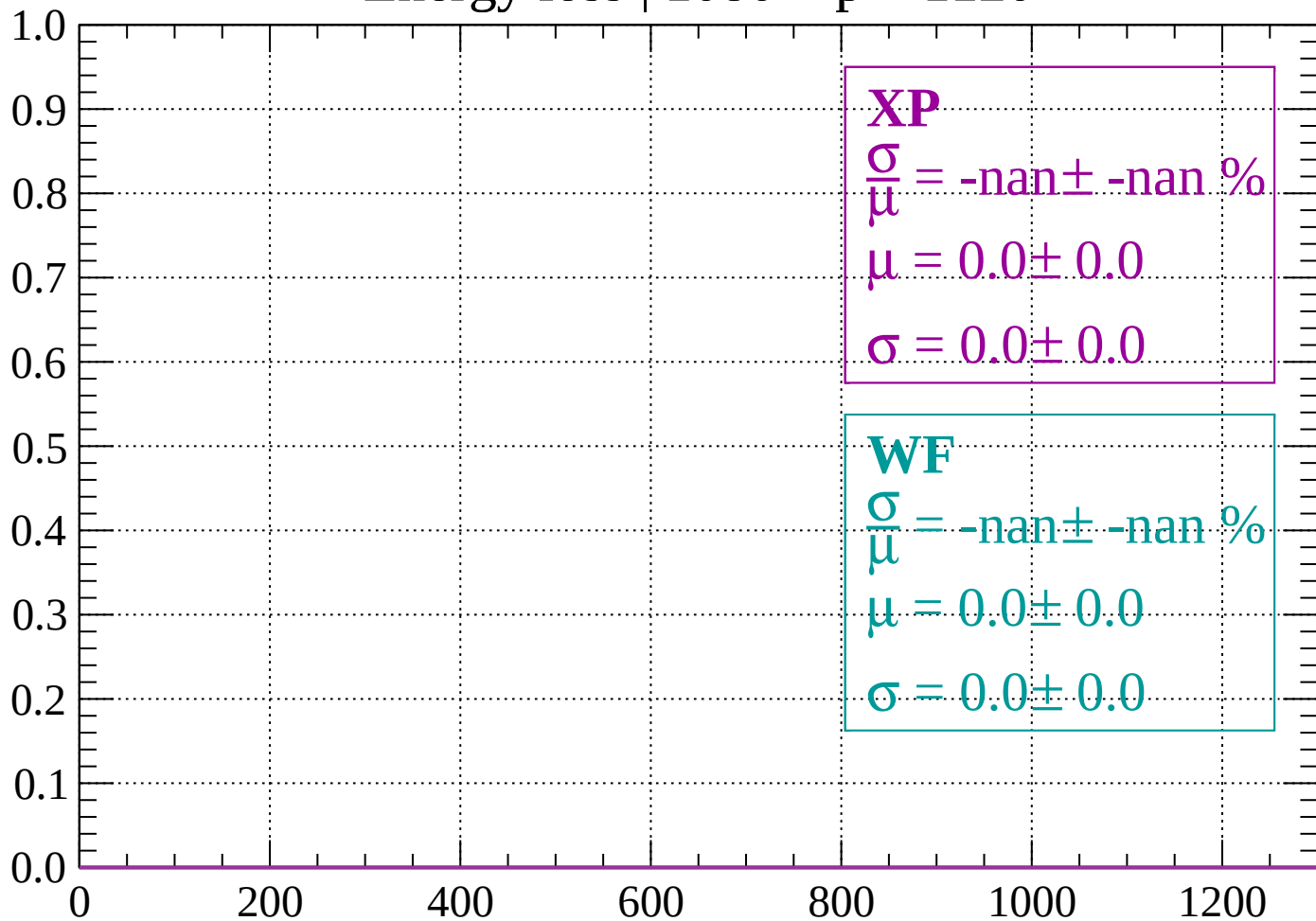
Count



dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1120$

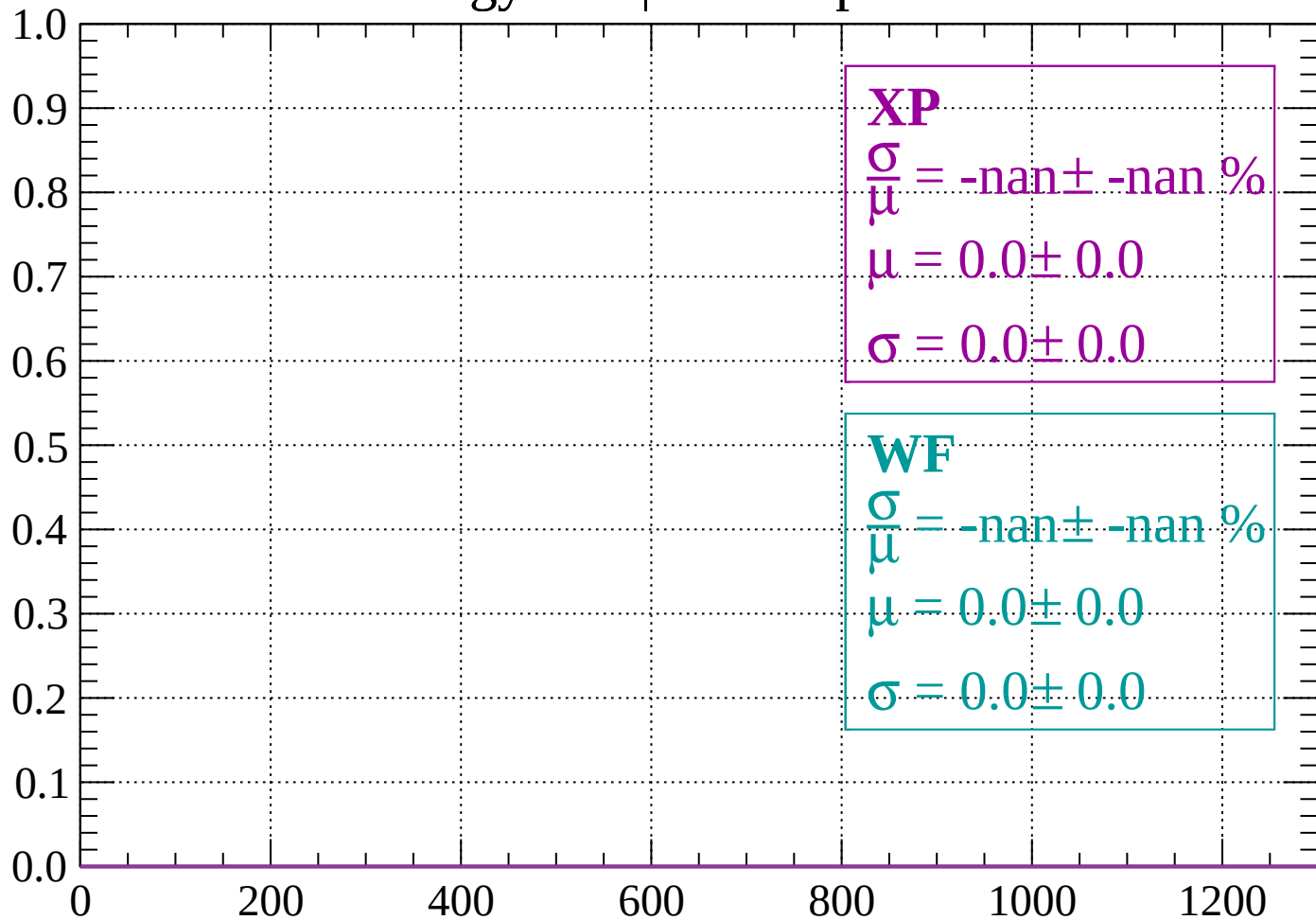
Count



dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1160$

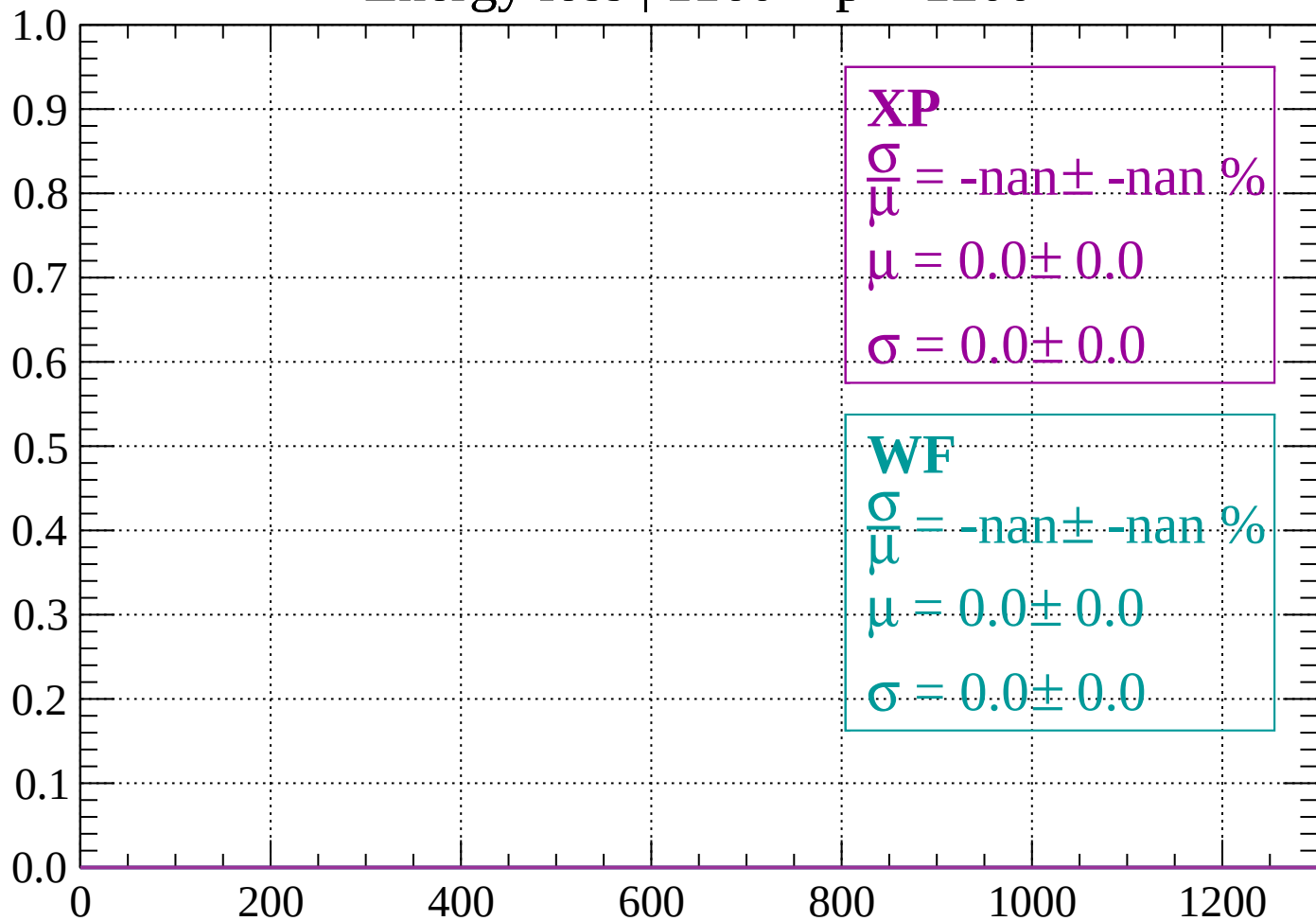
Count



dE/dx (ADC counts/cm)

Energy loss | $1160 < p < 1200$

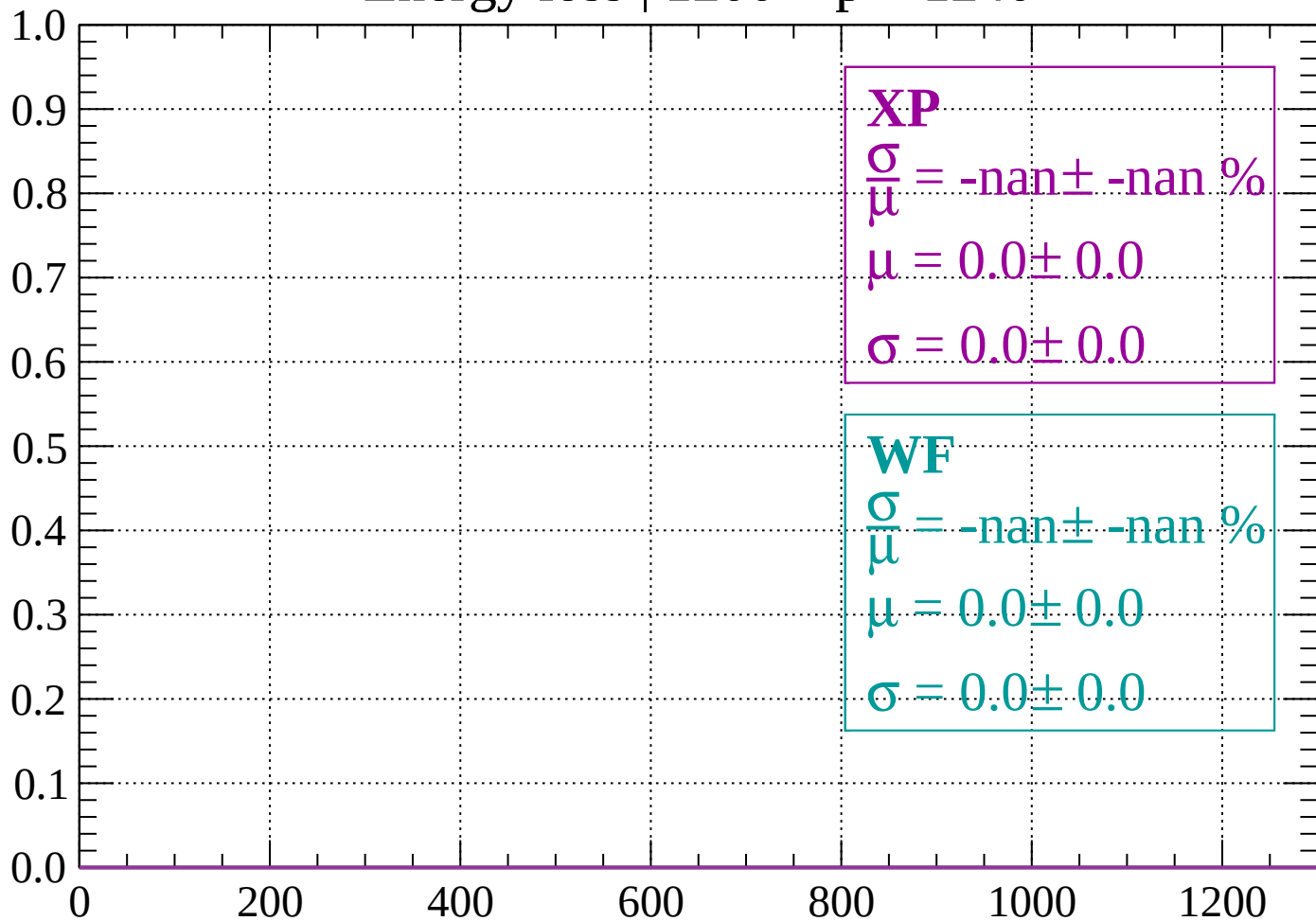
Count



dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1240$

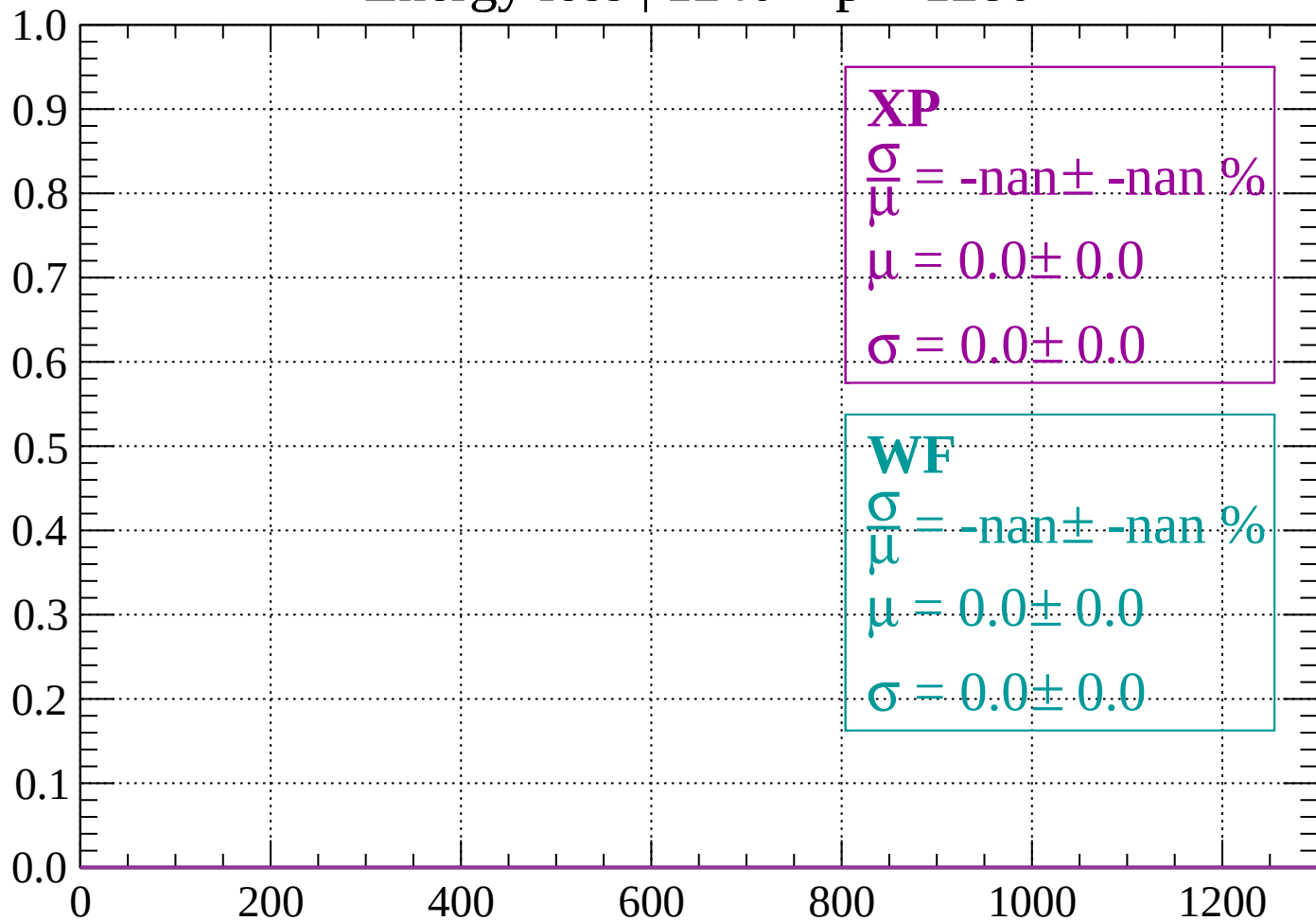
Count



dE/dx (ADC counts/cm)

Energy loss | $1240 < p < 1280$

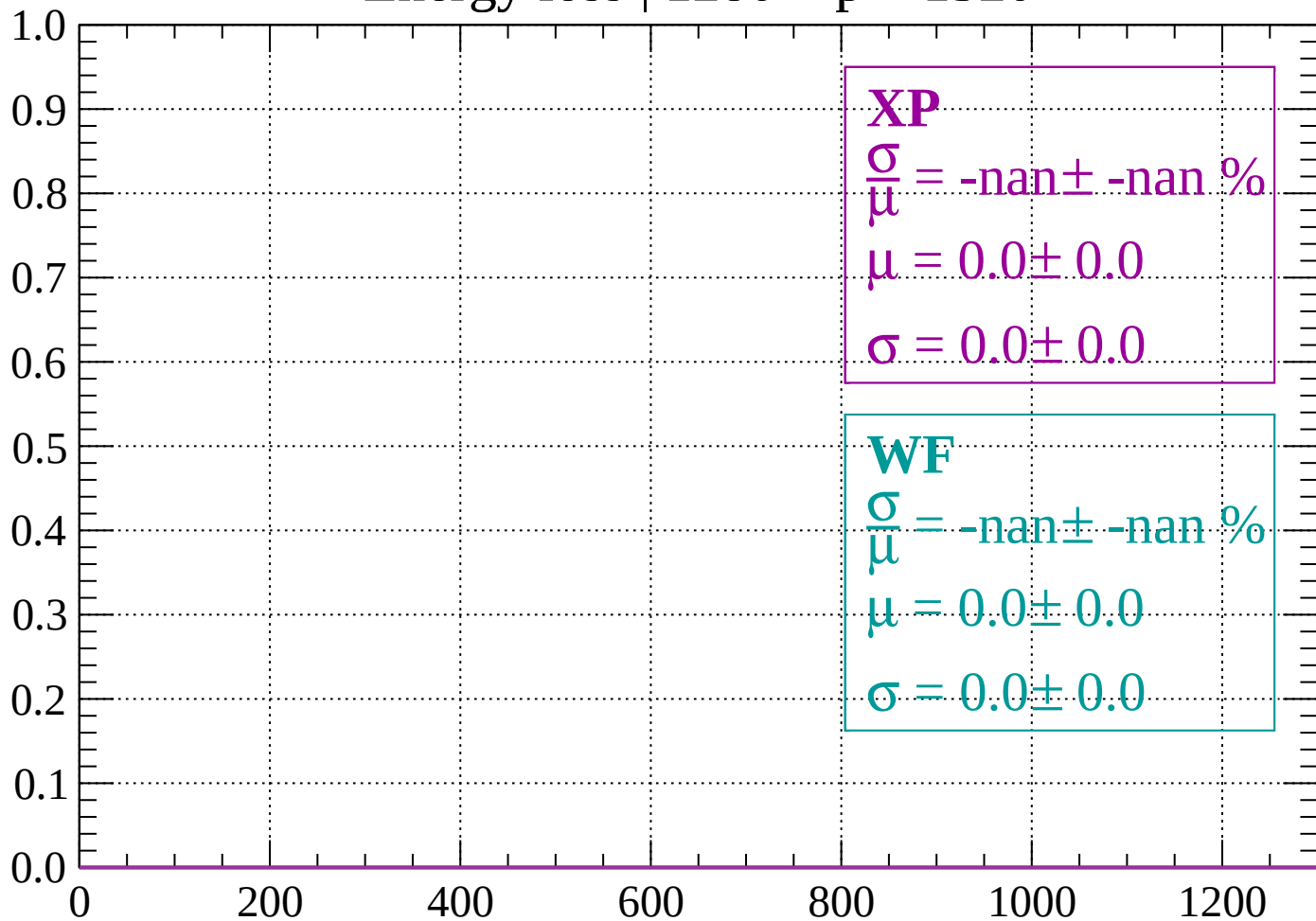
Count



dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1320$

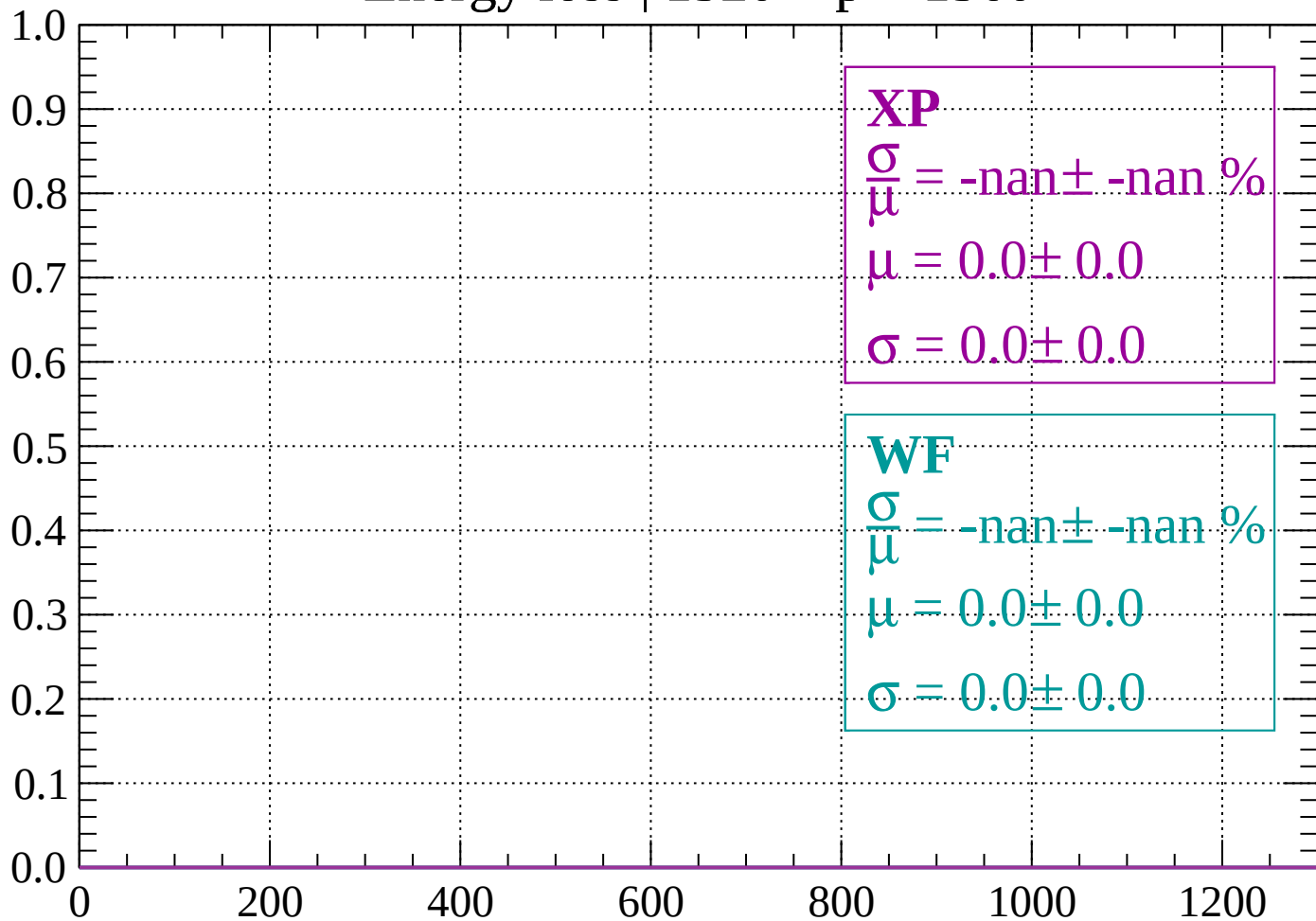
Count



dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1360$

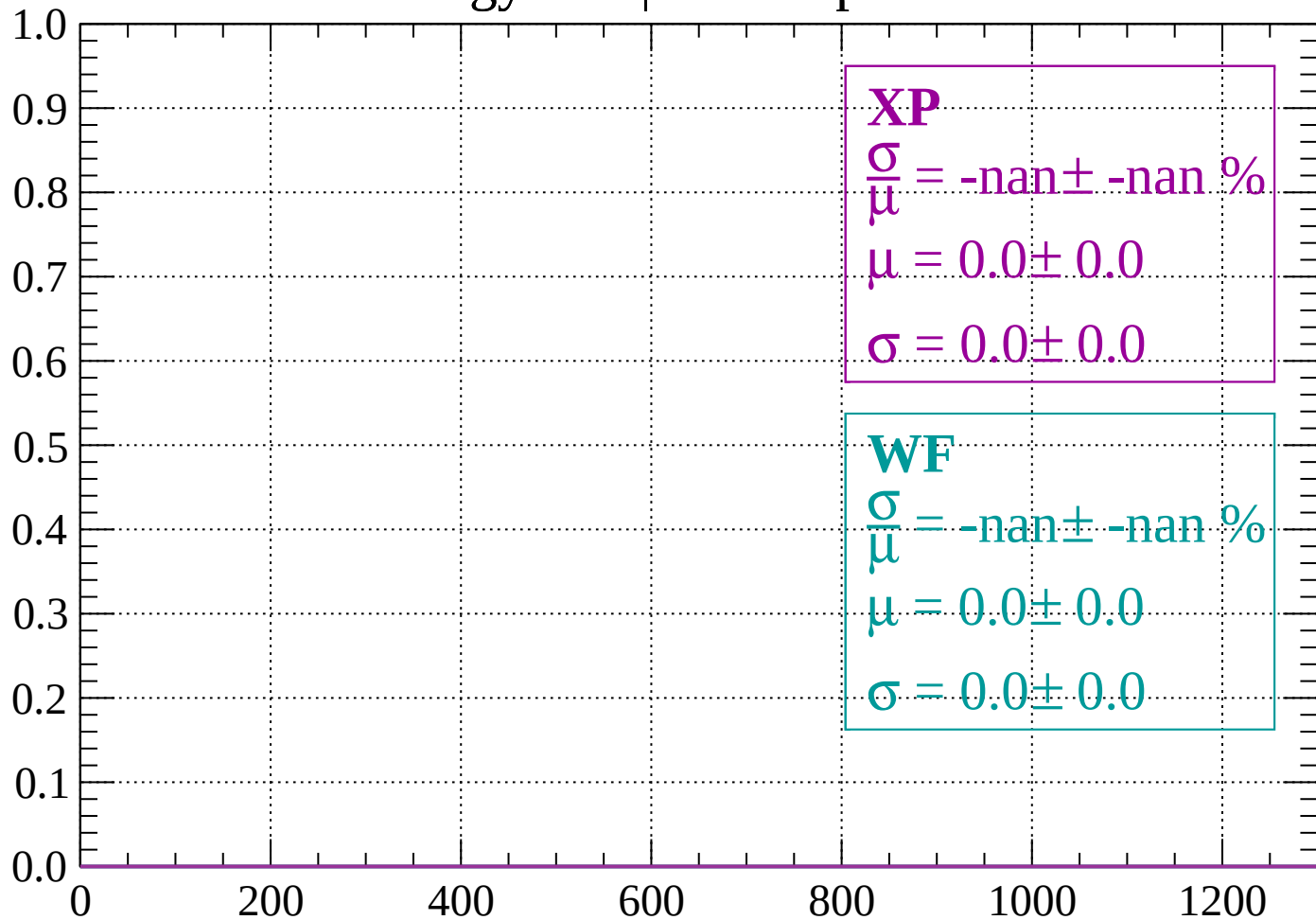
Count



dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1400$

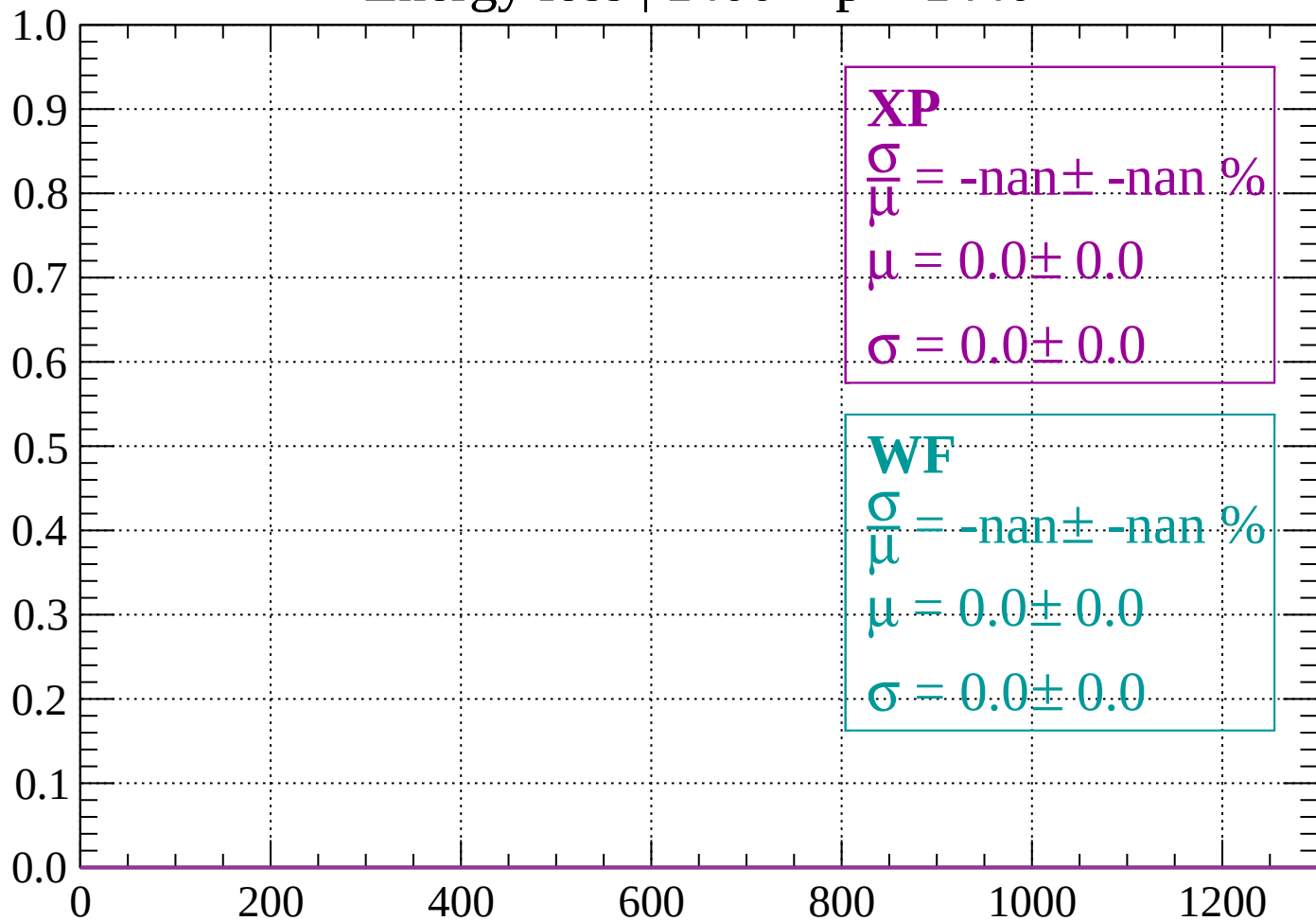
Count



dE/dx (ADC counts/cm)

Energy loss | $1400 < p < 1440$

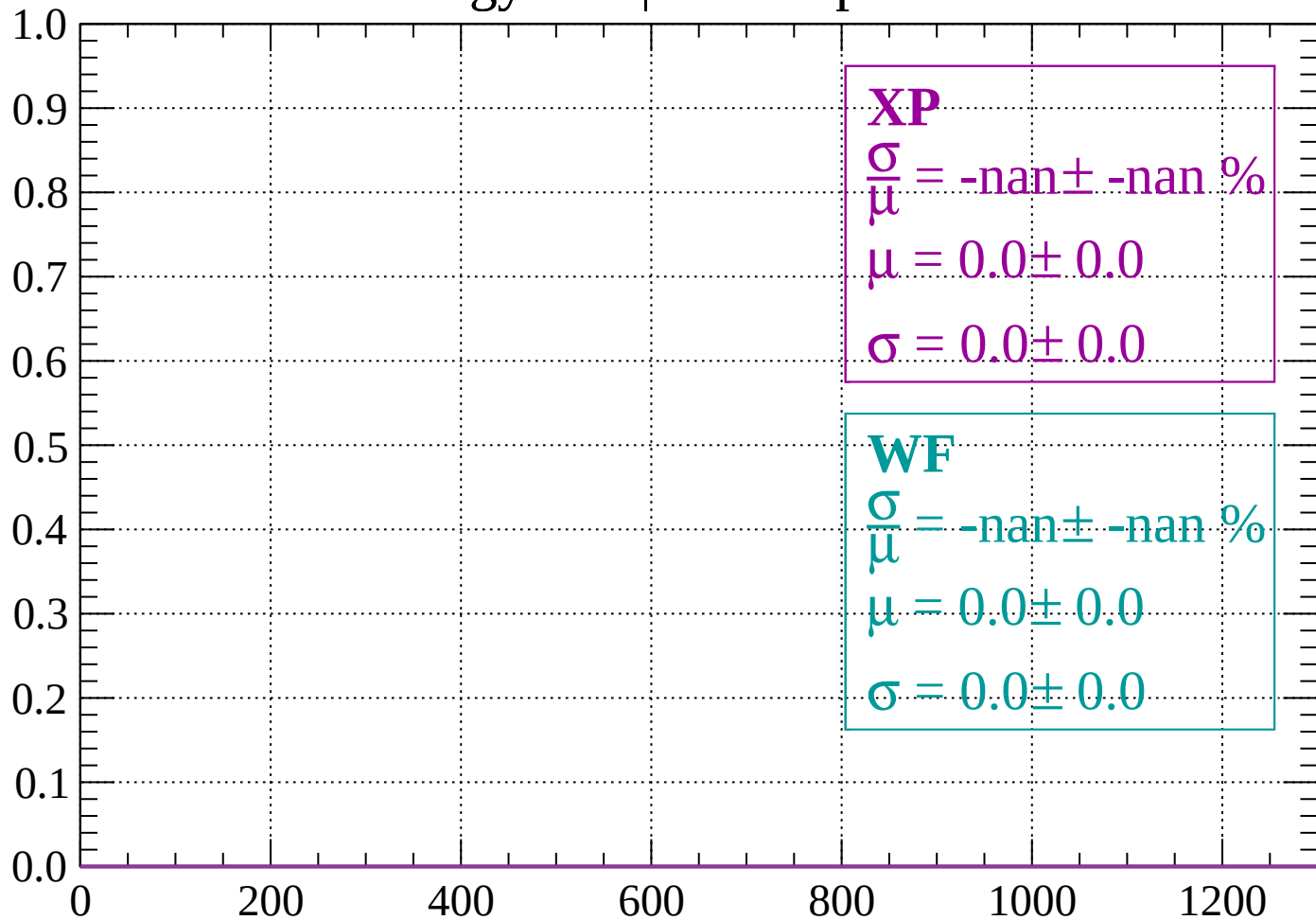
Count



dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1480$

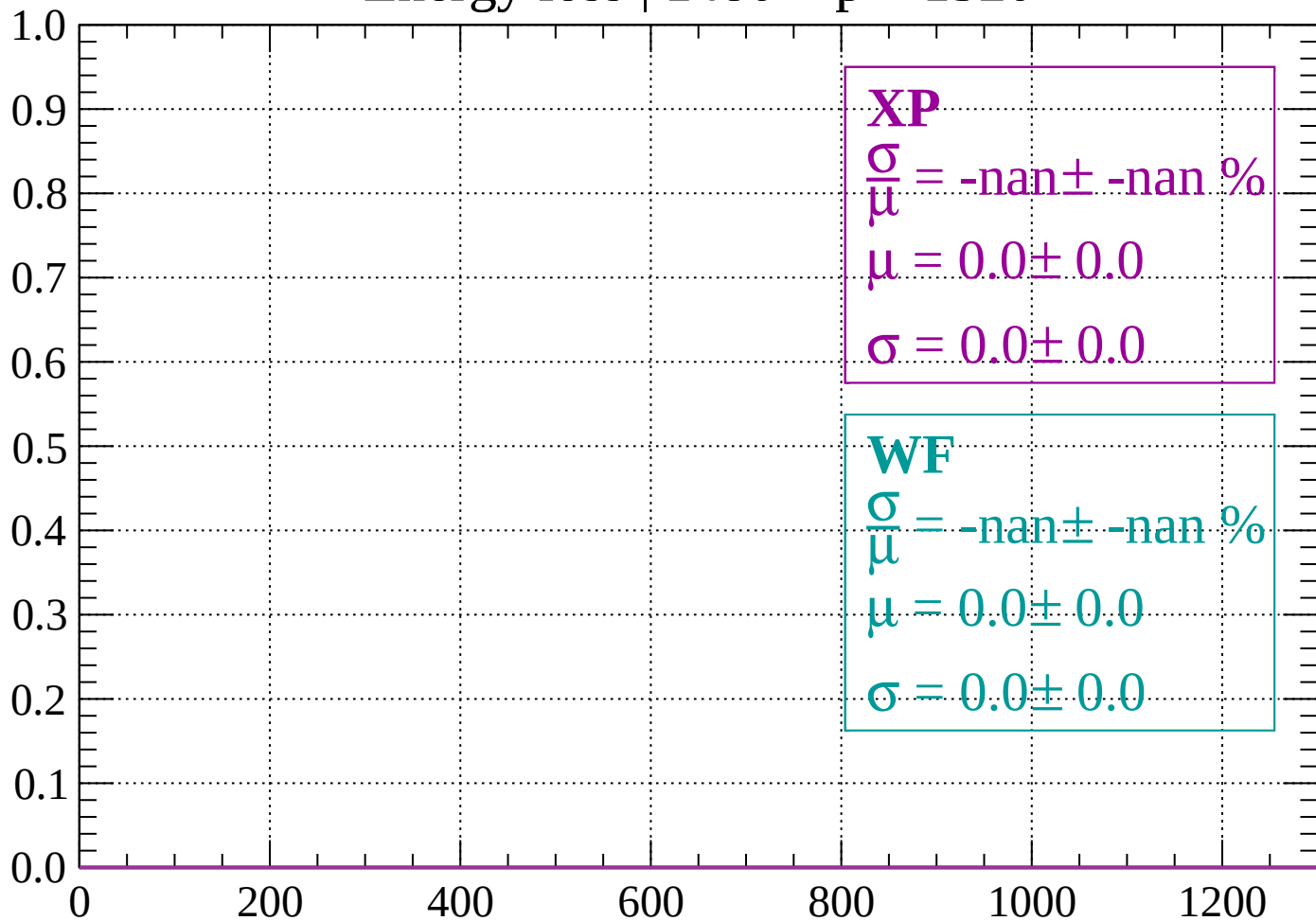
Count



dE/dx (ADC counts/cm)

Energy loss | $1480 < p < 1520$

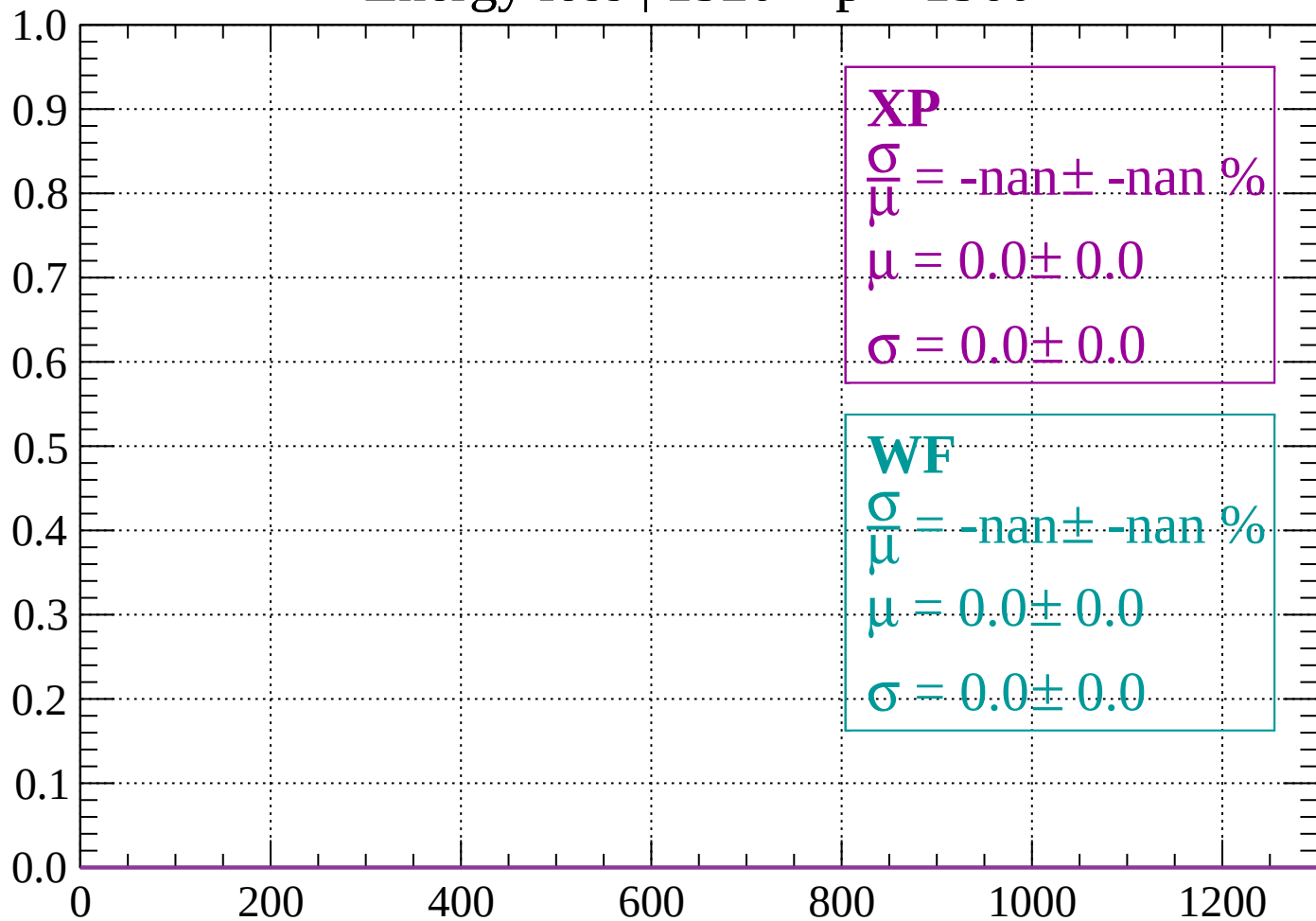
Count



dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1560$

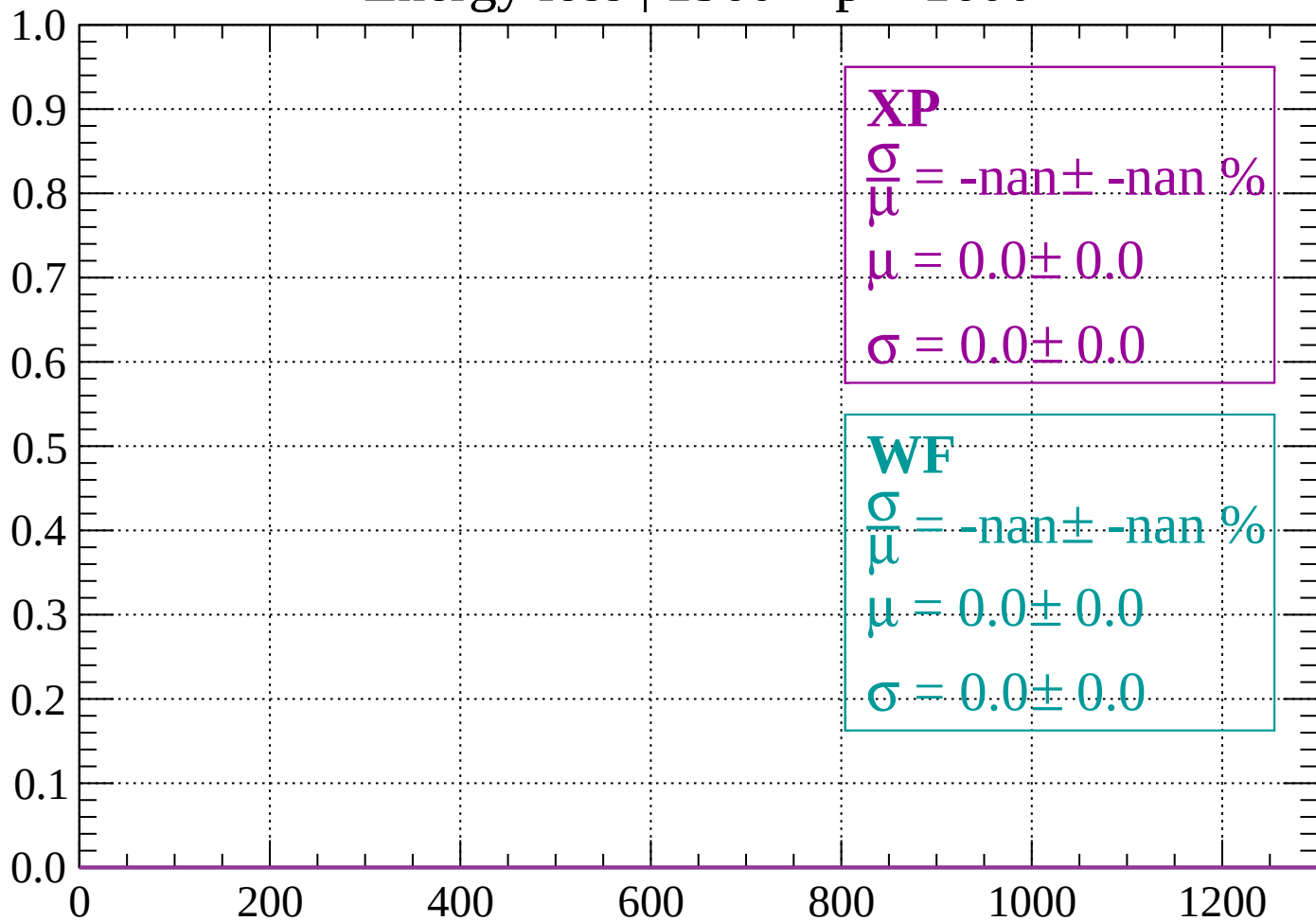
Count



dE/dx (ADC counts/cm)

Energy loss | 1560 < p < 1600

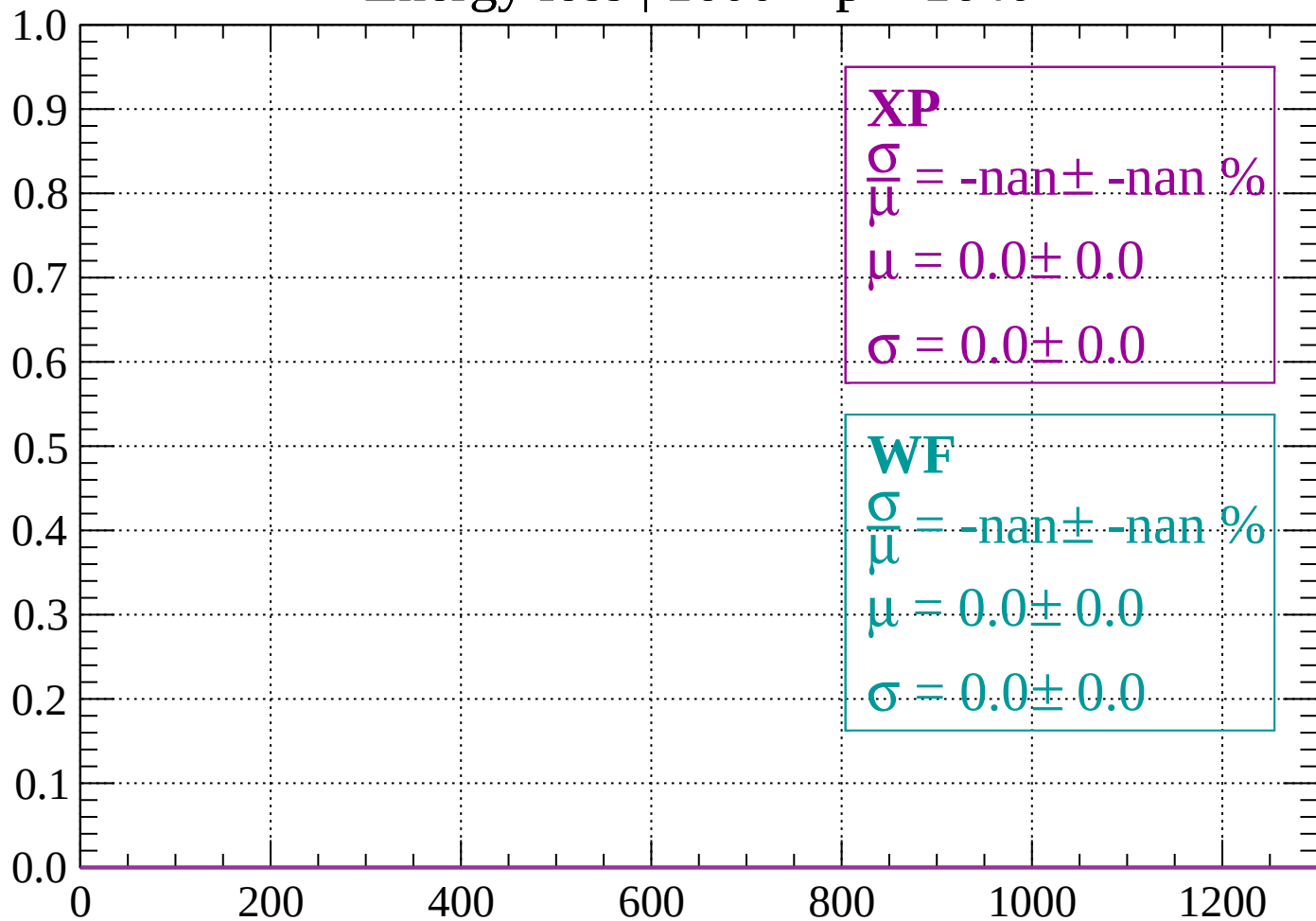
Count



dE/dx (ADC counts/cm)

Energy loss | 1600 < p < 1640

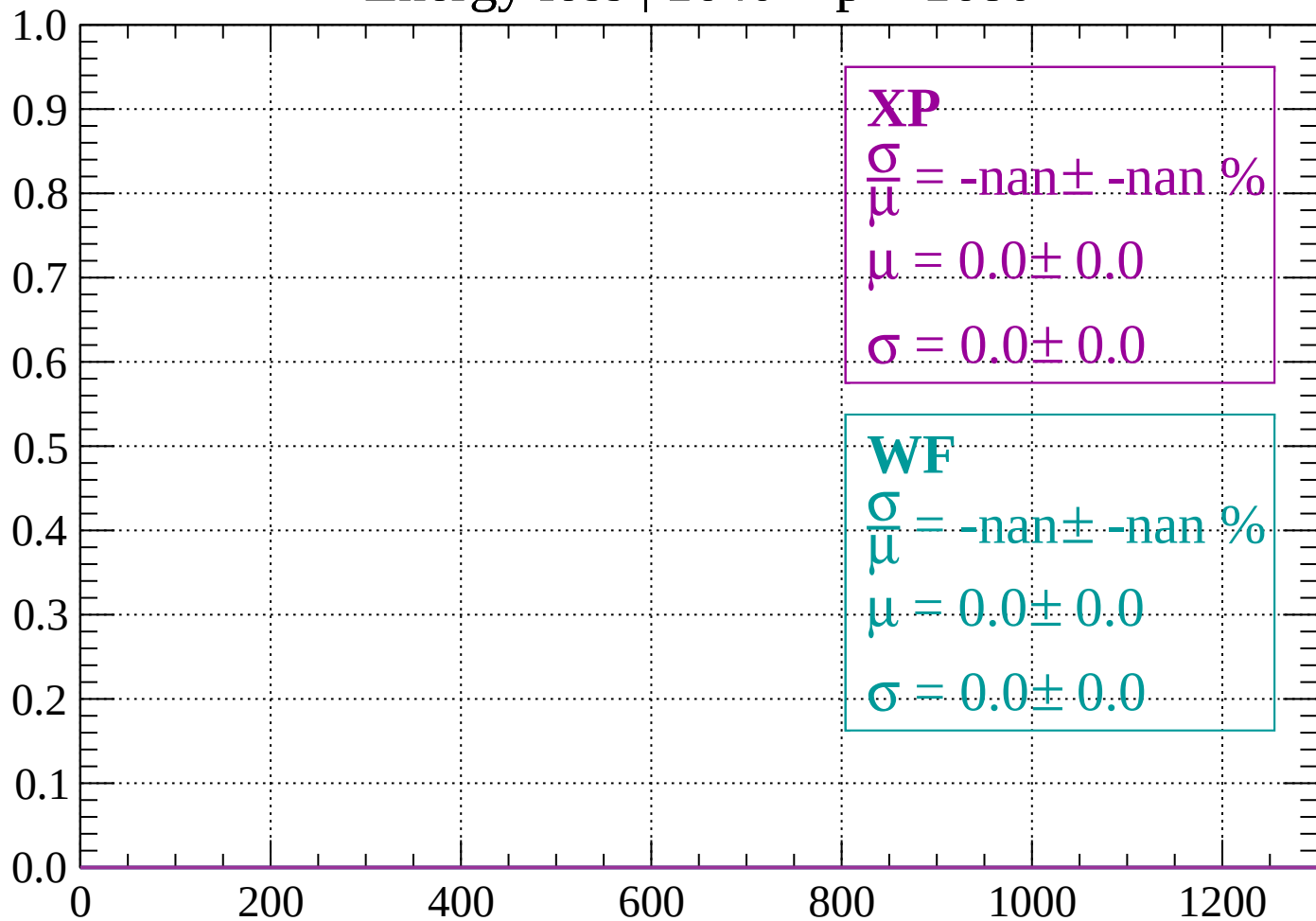
Count



dE/dx (ADC counts/cm)

Energy loss | $1640 < p < 1680$

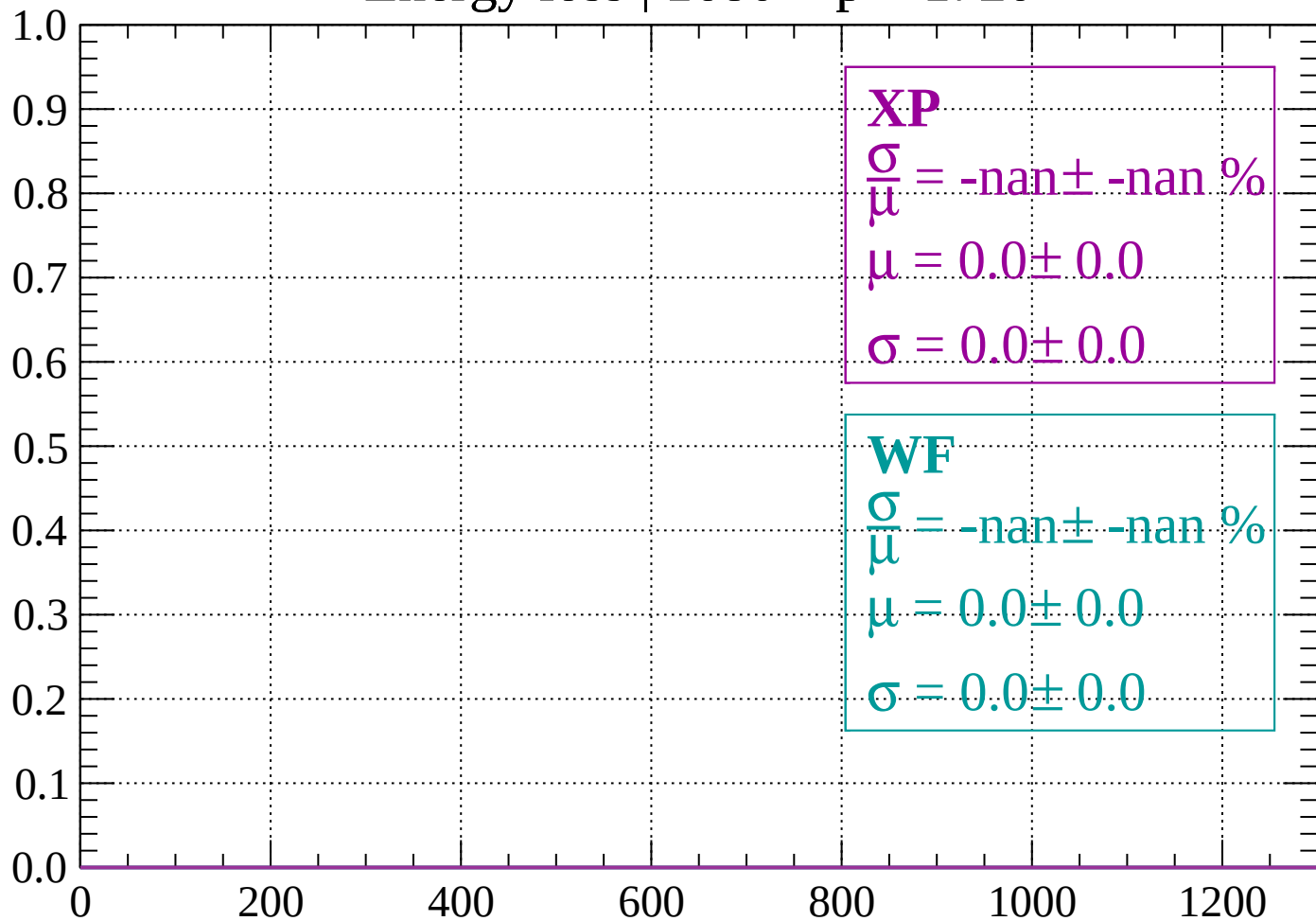
Count



dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1720$

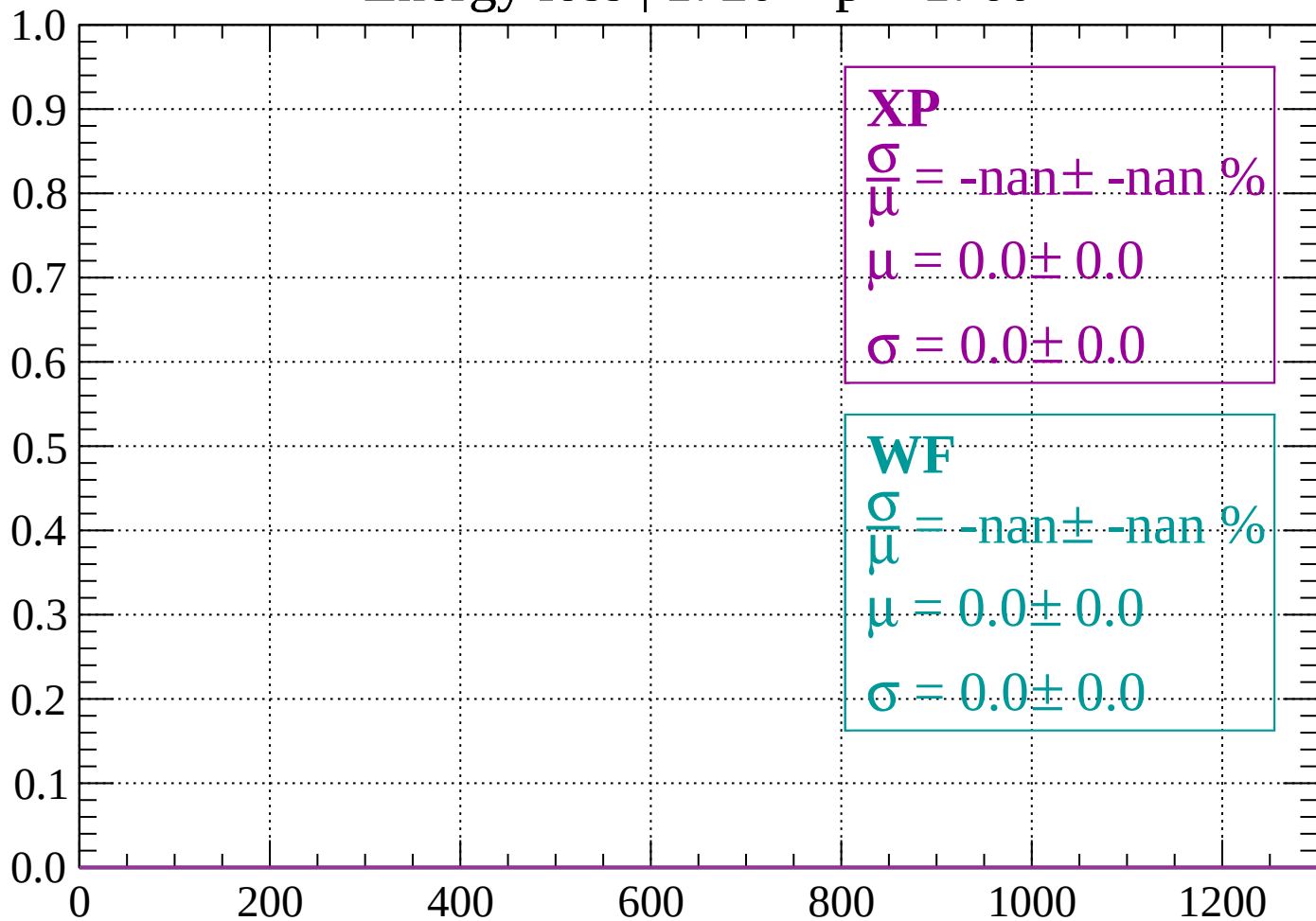
Count



dE/dx (ADC counts/cm)

Energy loss | $1720 < p < 1760$

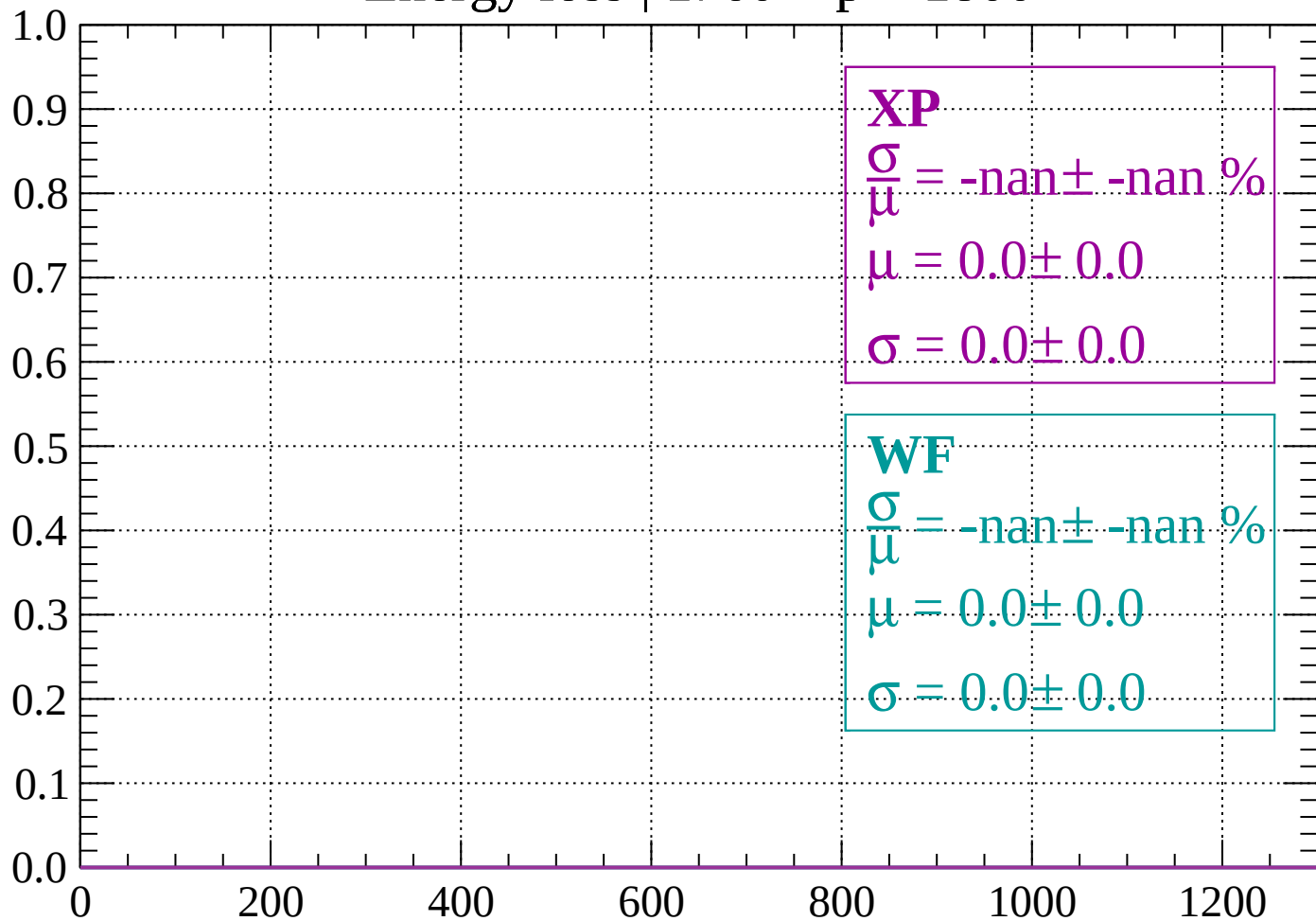
Count



dE/dx (ADC counts/cm)

Energy loss | $1760 < p < 1800$

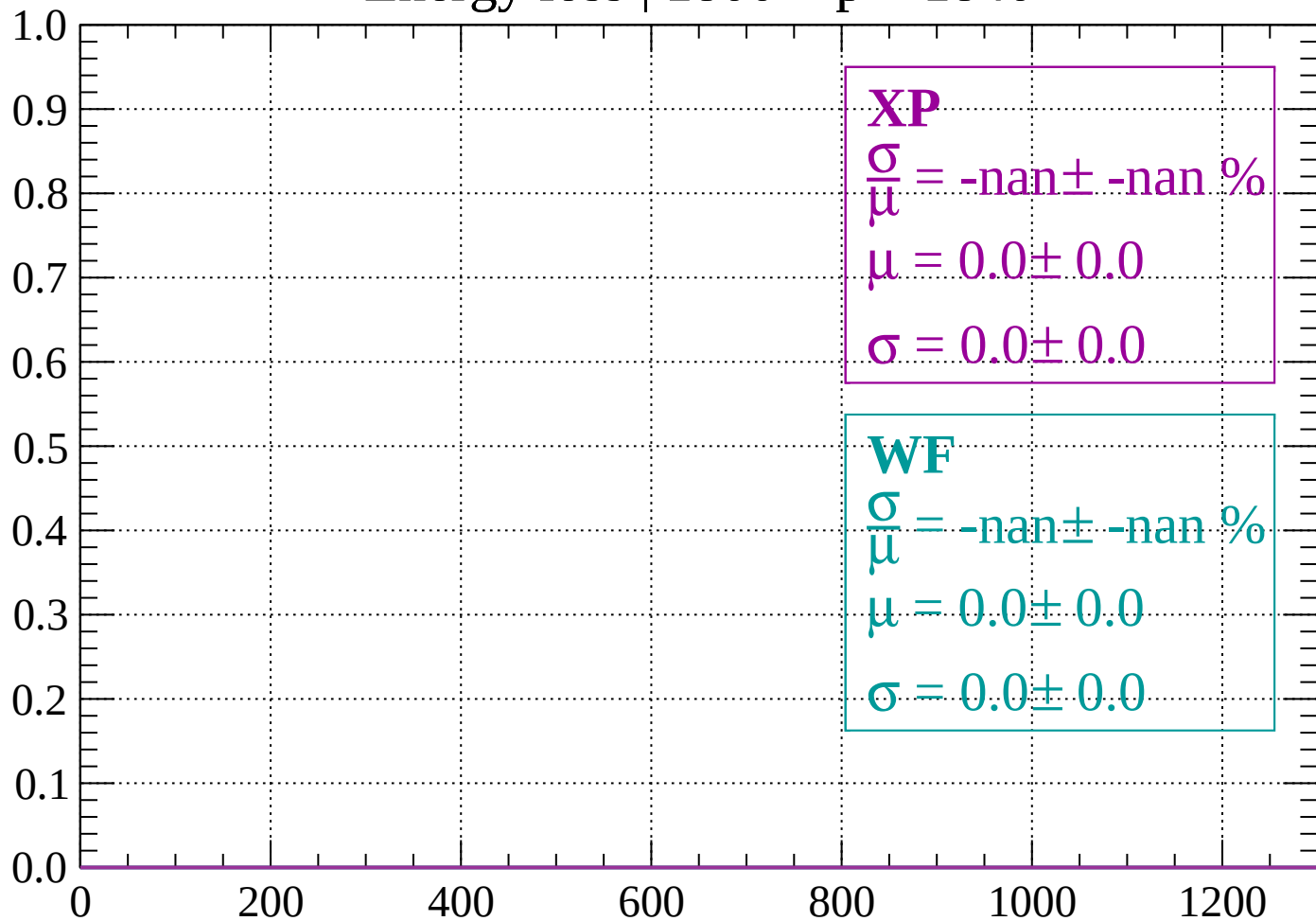
Count



dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1840$

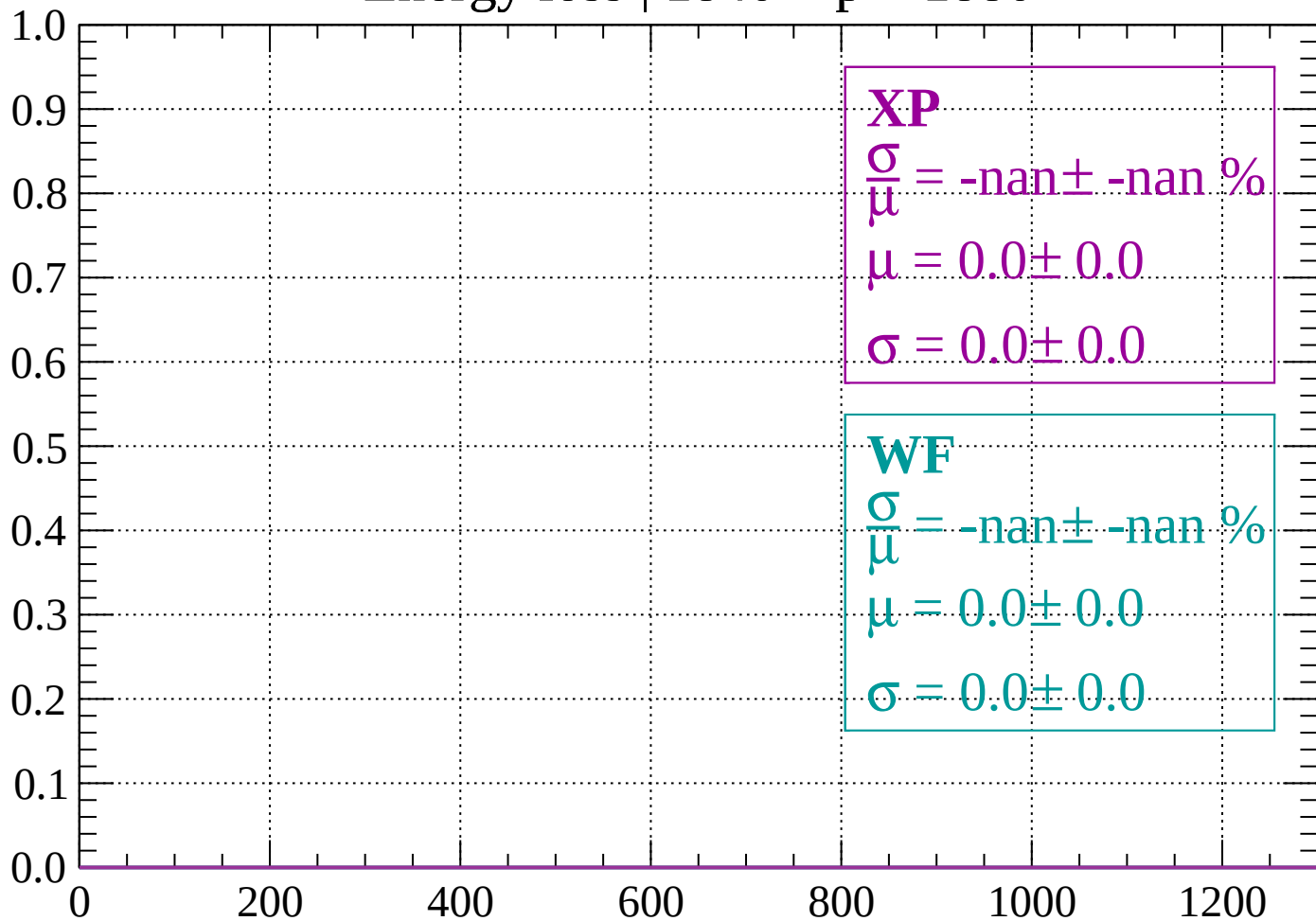
Count



dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1880$

Count



XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

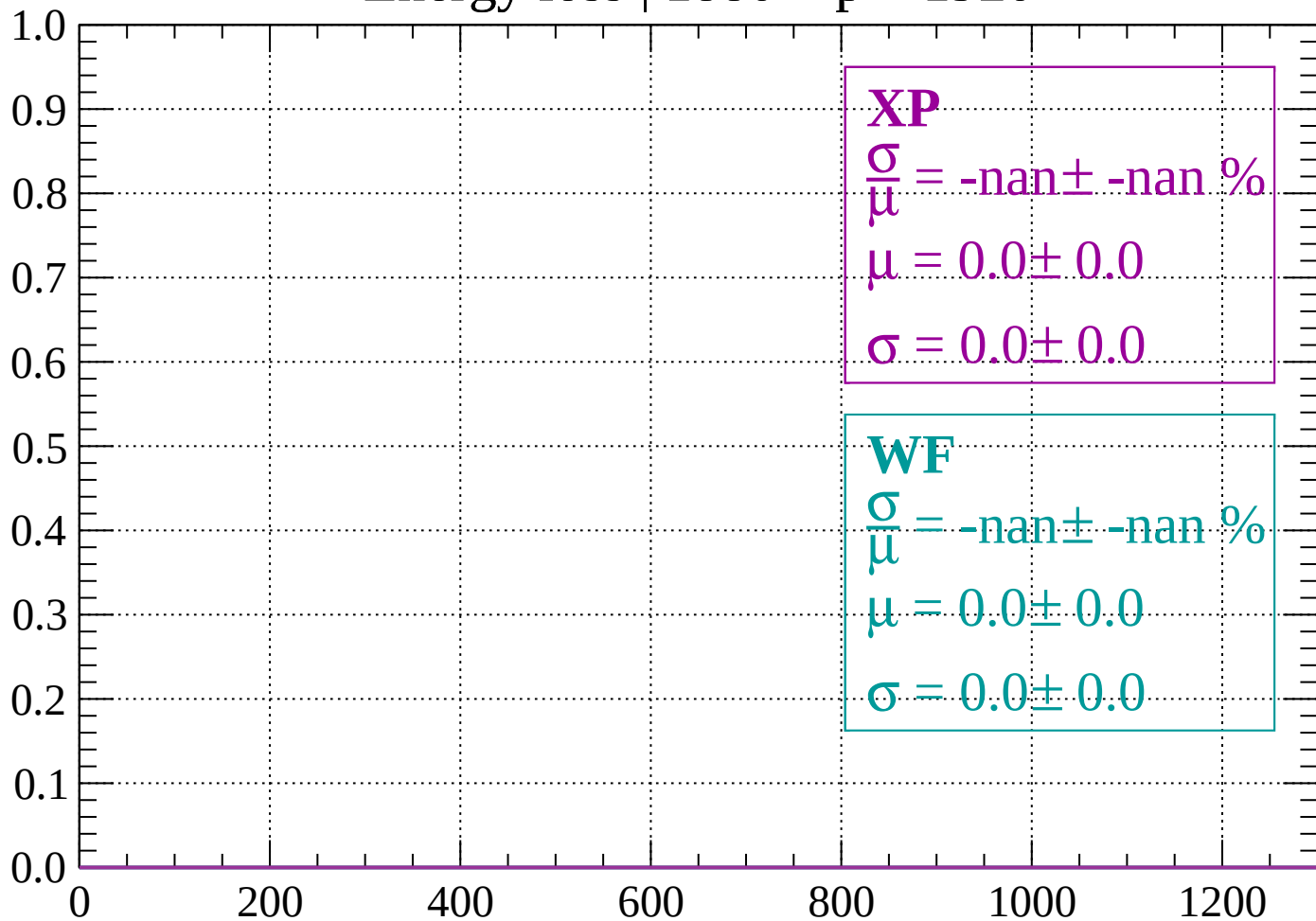
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | 1880 < p < 1920

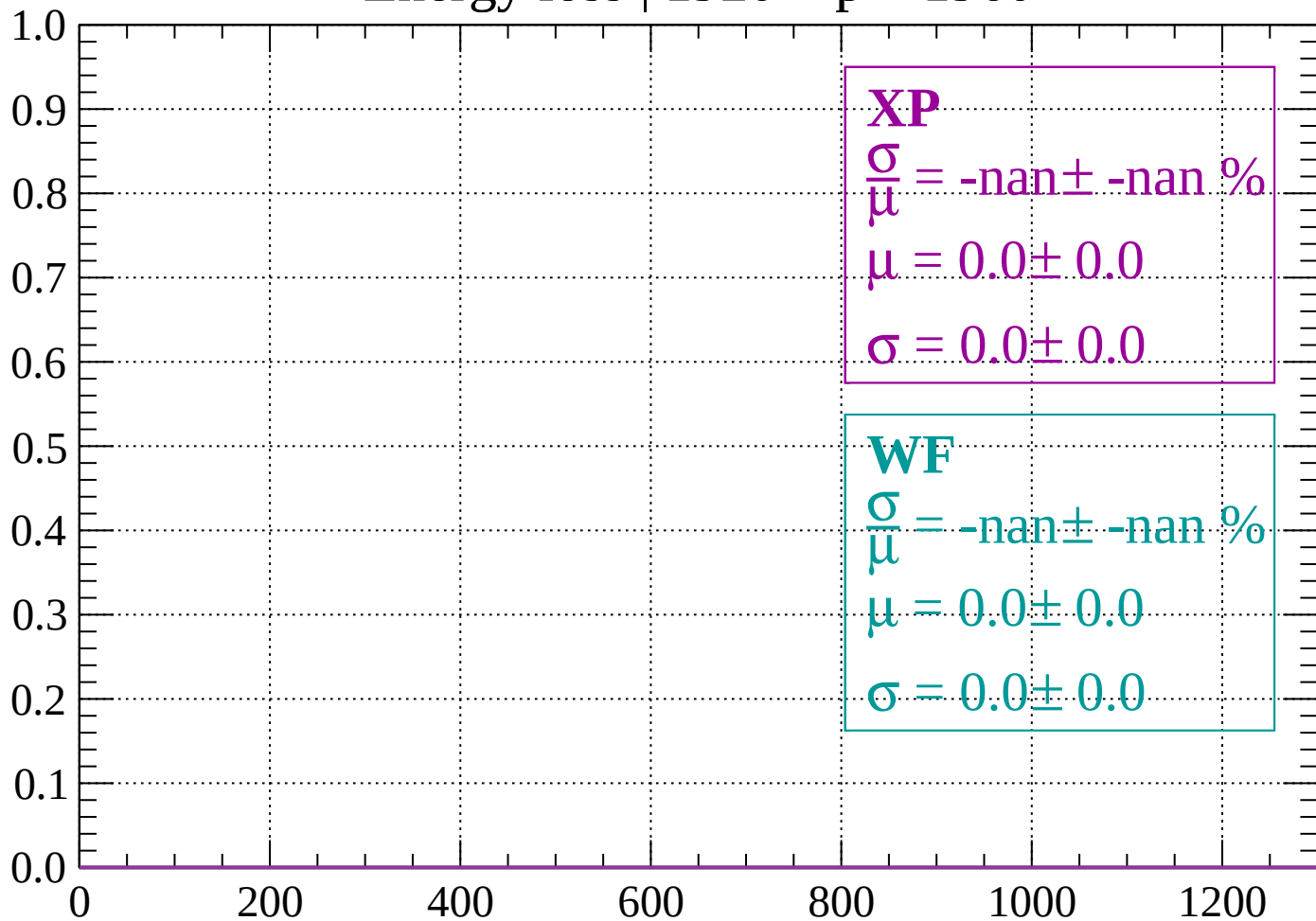
Count



dE/dx (ADC counts/cm)

Energy loss | 1920 < p < 1960

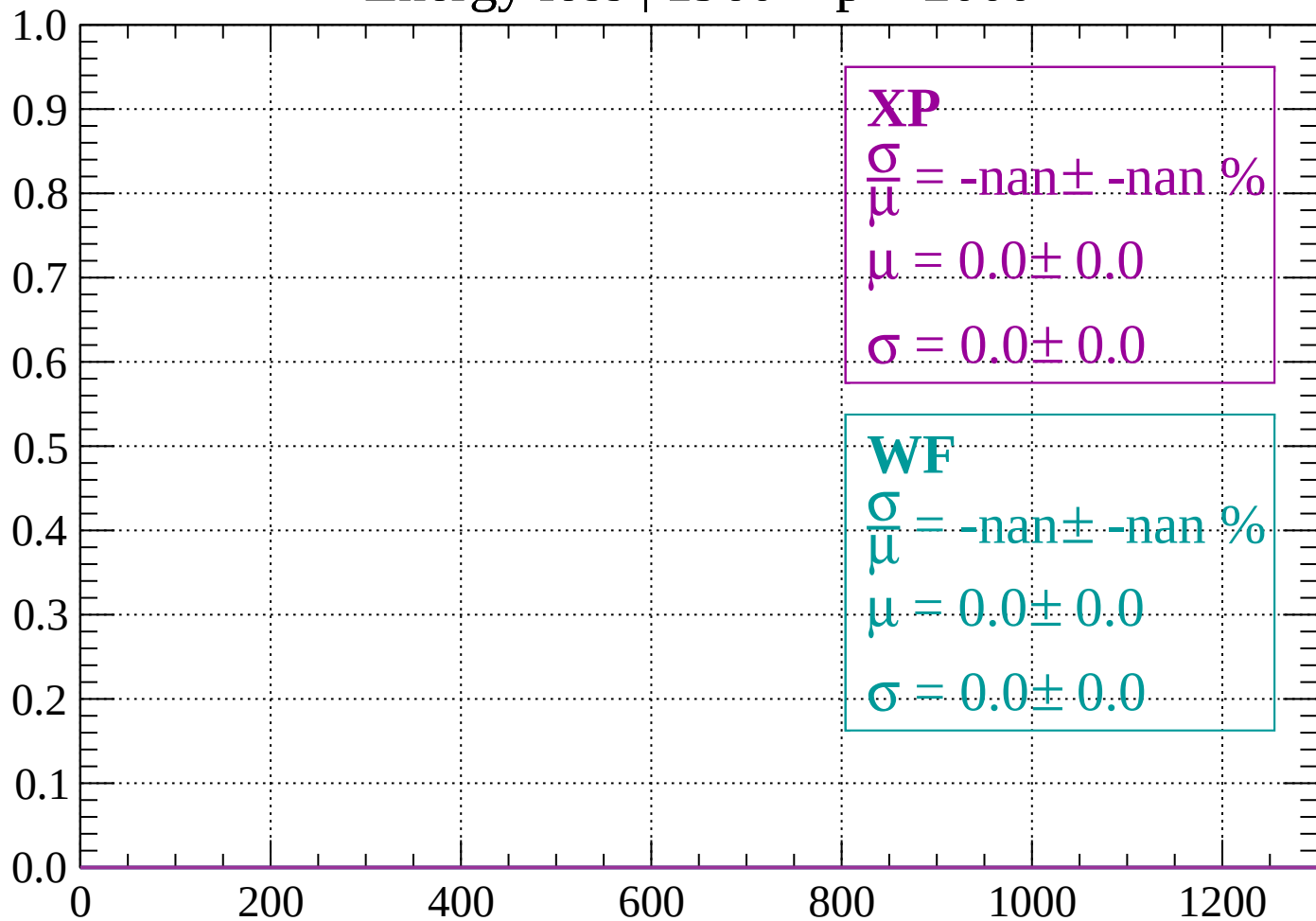
Count



dE/dx (ADC counts/cm)

Energy loss | 1960 < p < 2000

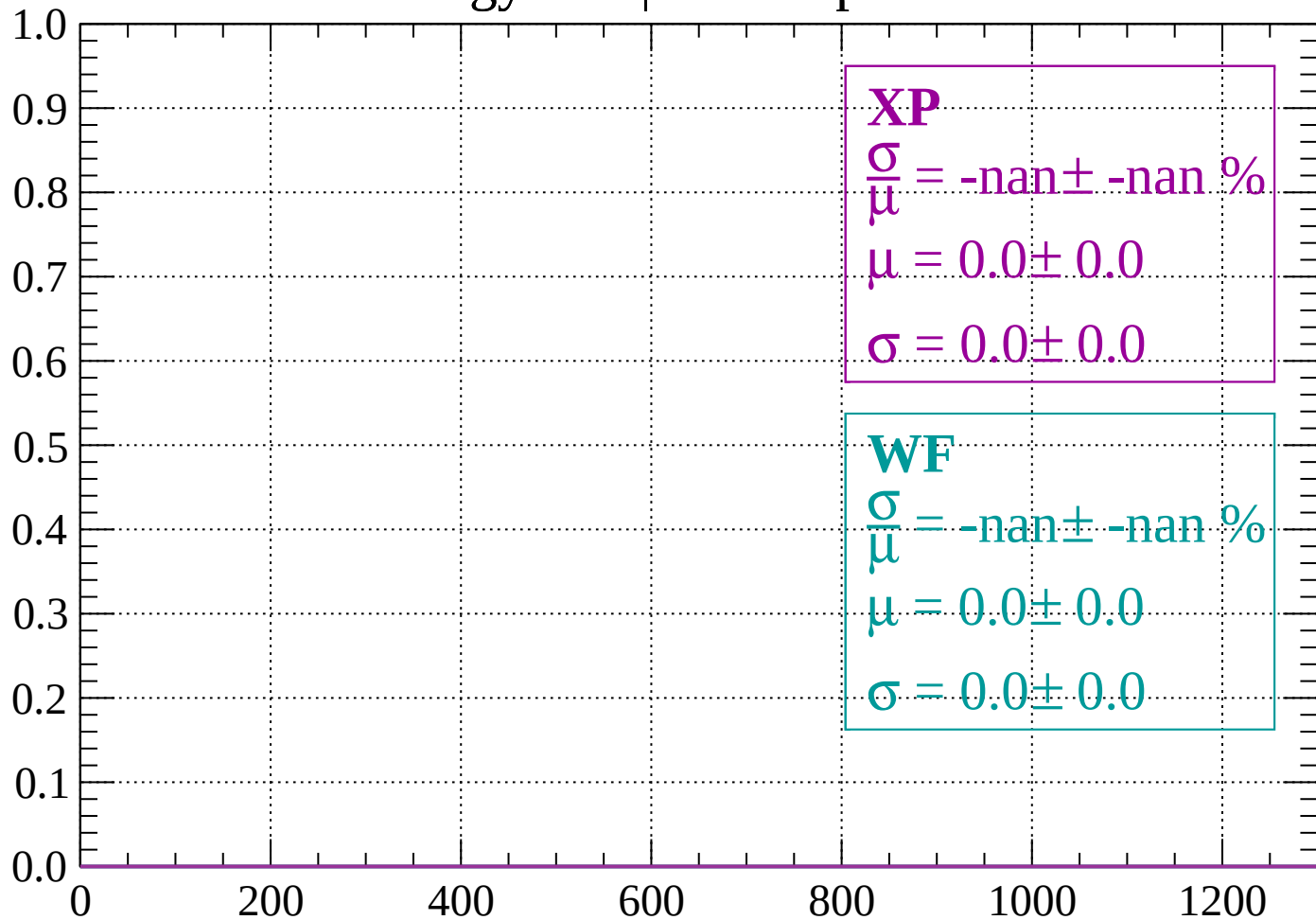
Count



dE/dx (ADC counts/cm)

Energy loss | $2000 < p < 2040$

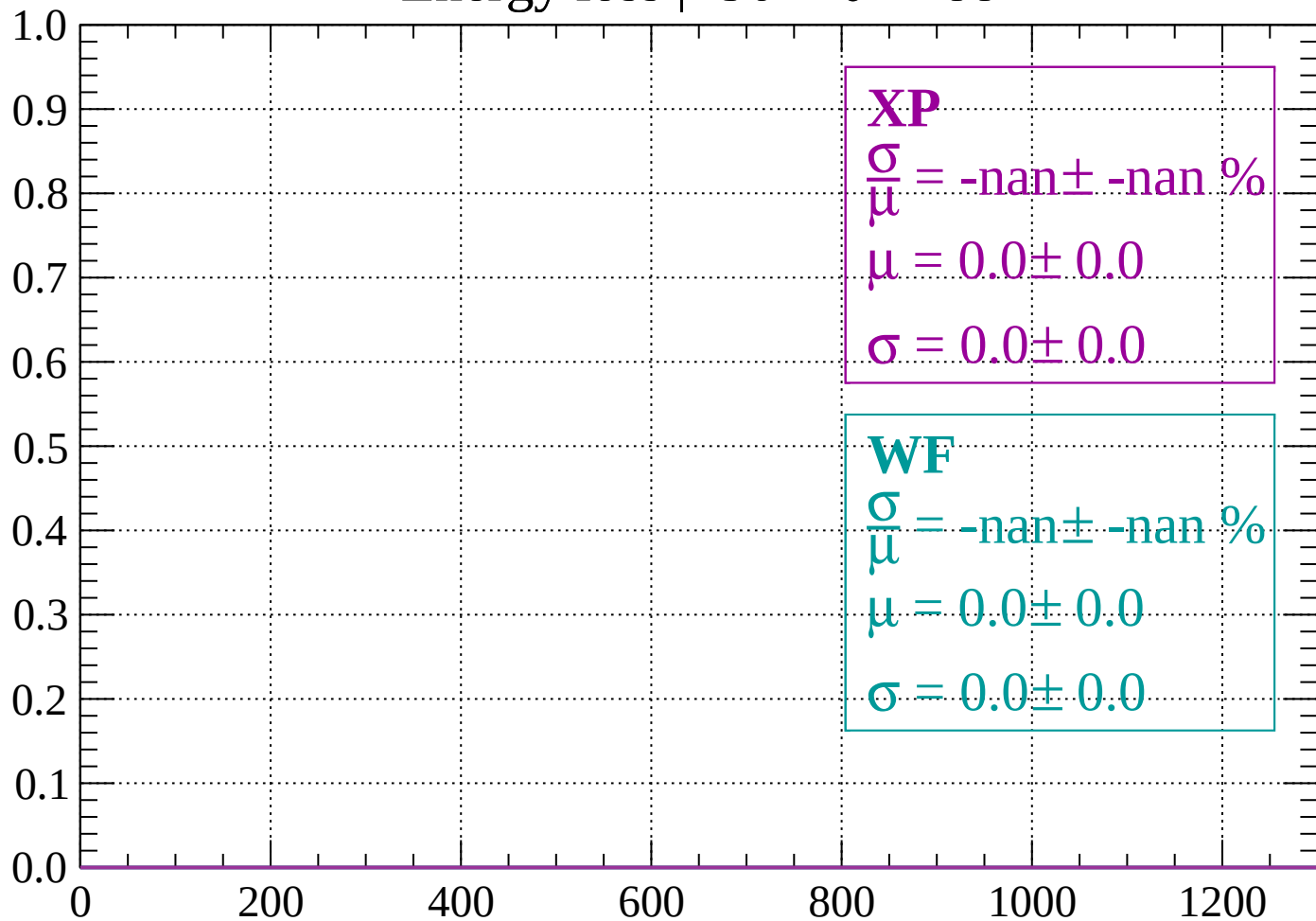
Count



dE/dx (ADC counts/cm)

Energy loss | $-90 < \theta < -88$

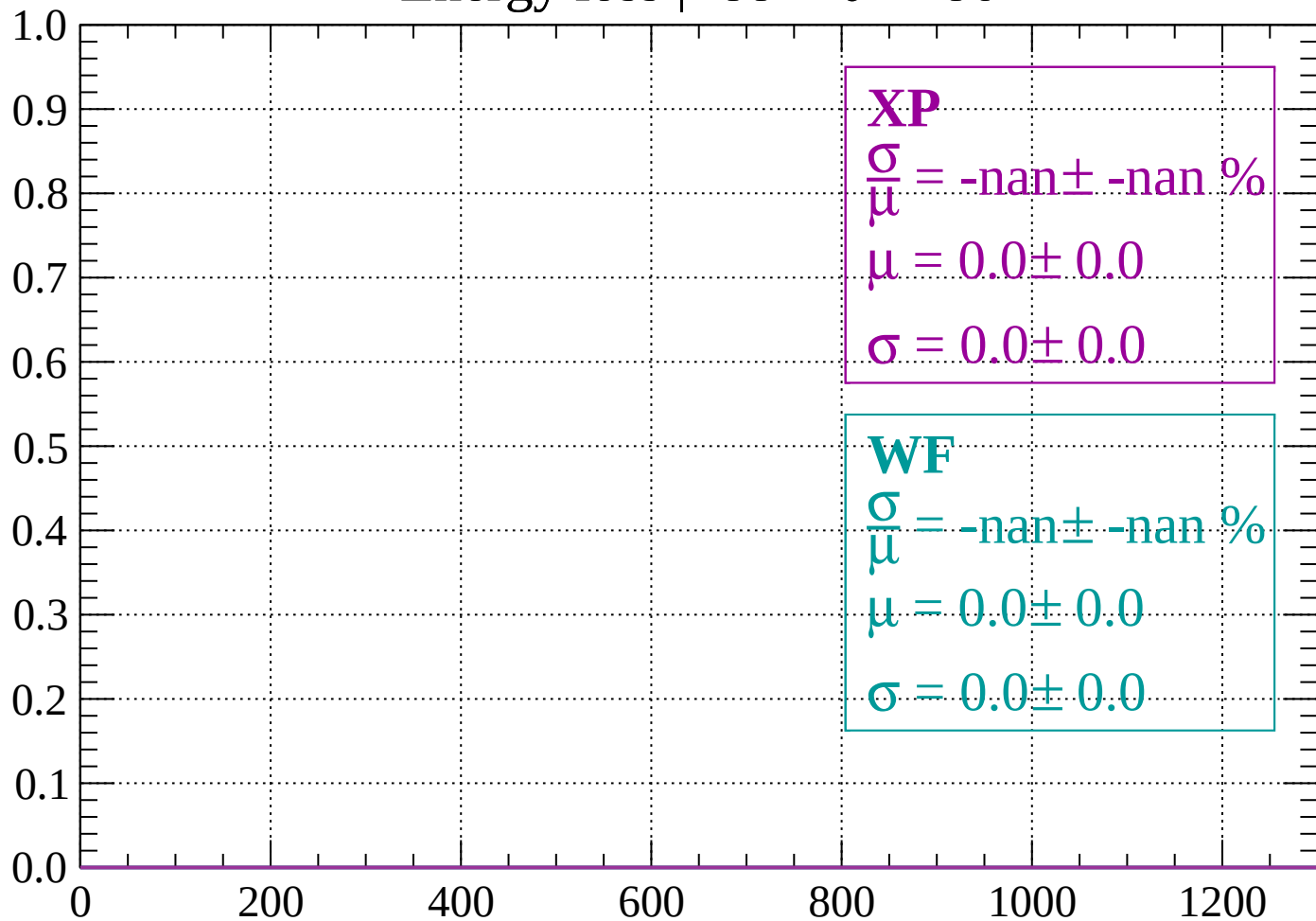
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

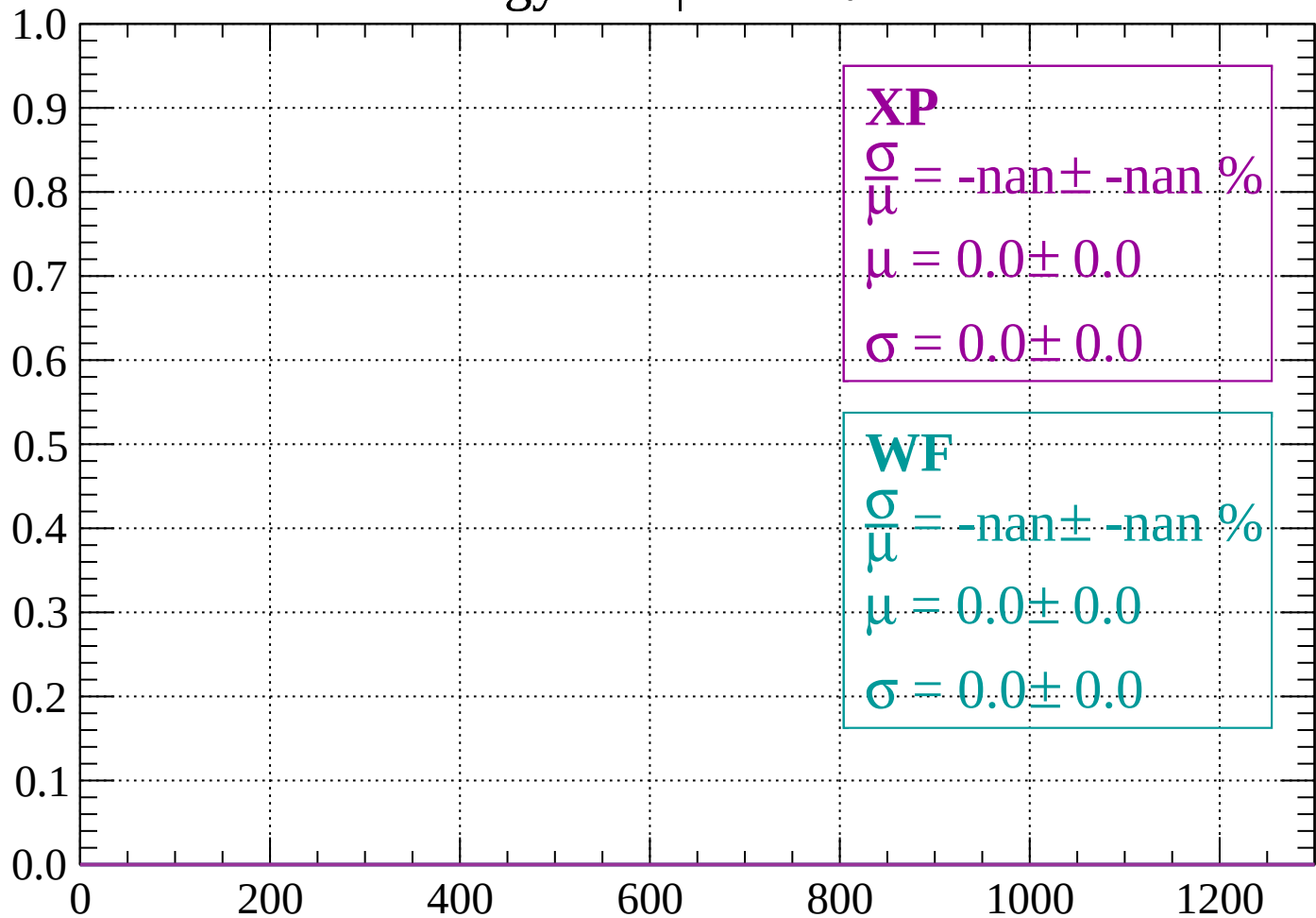
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

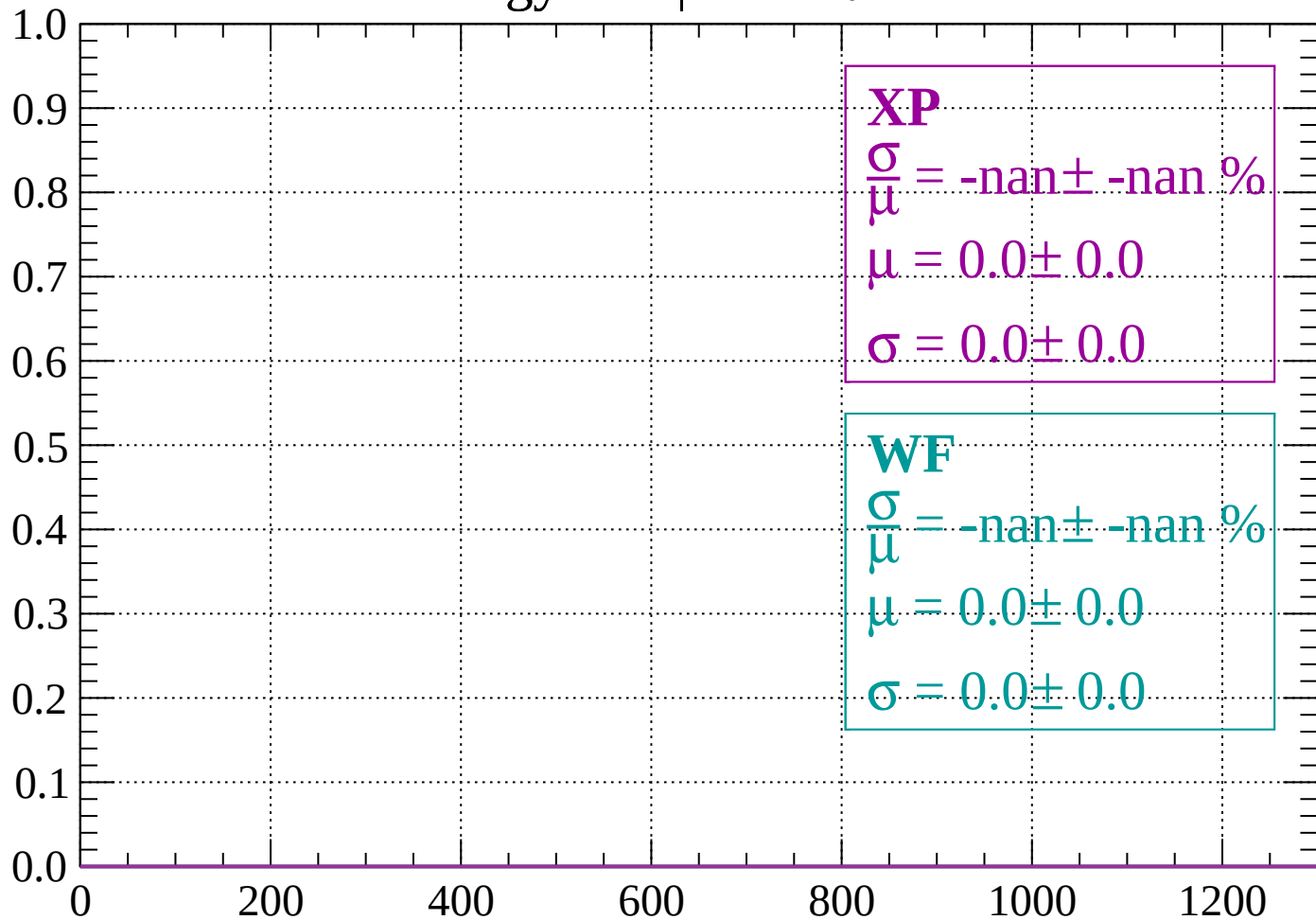
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

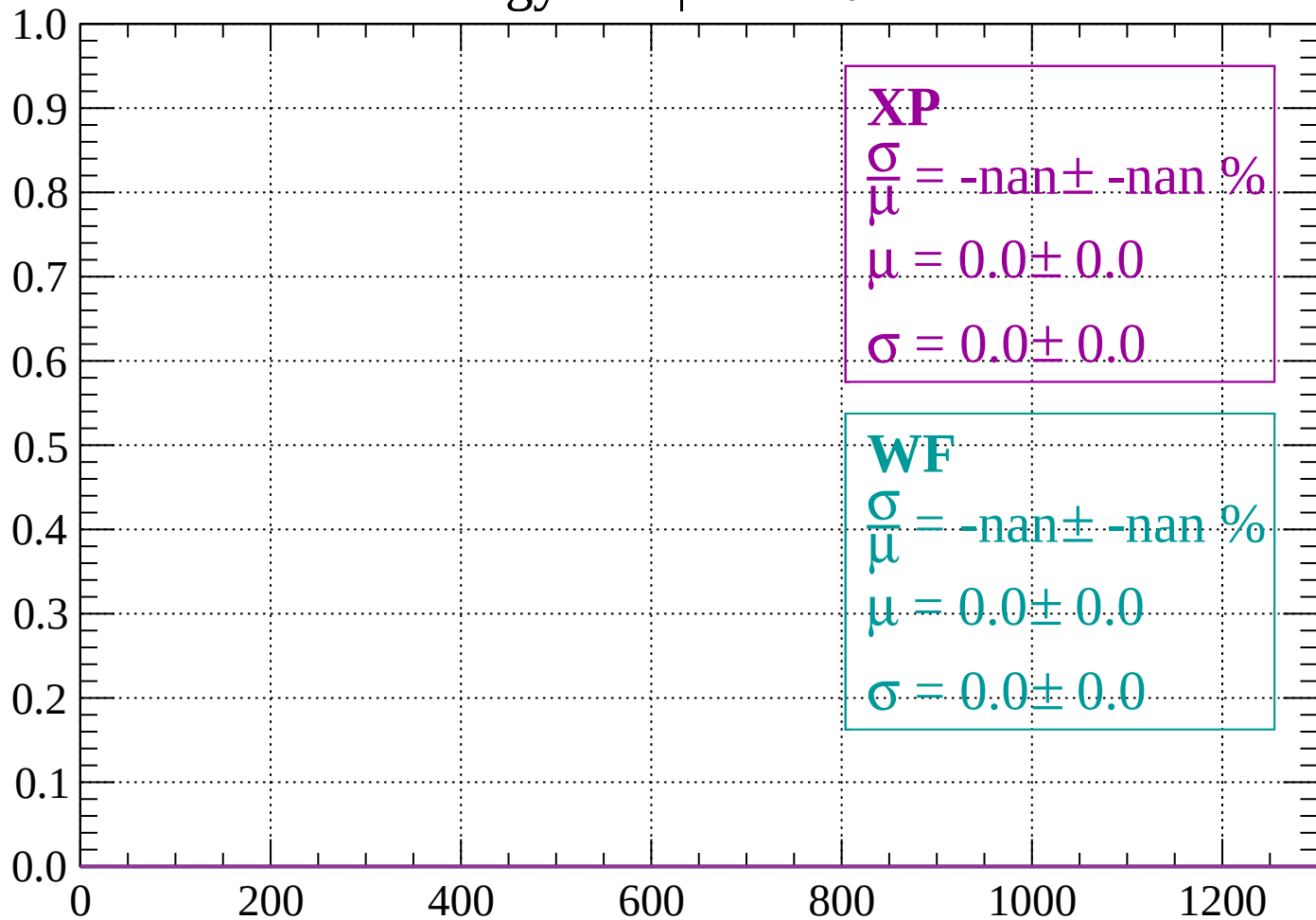
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

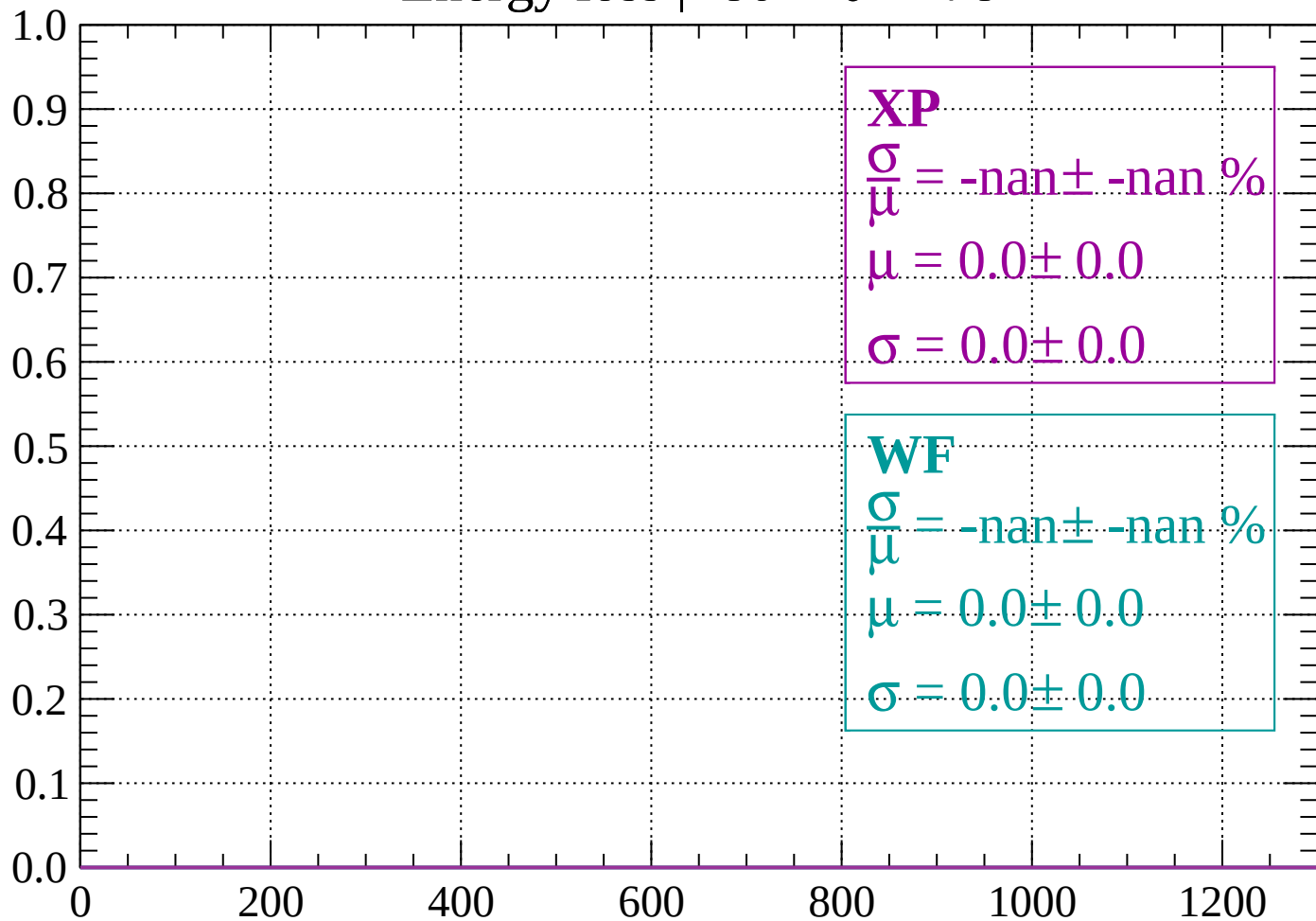
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

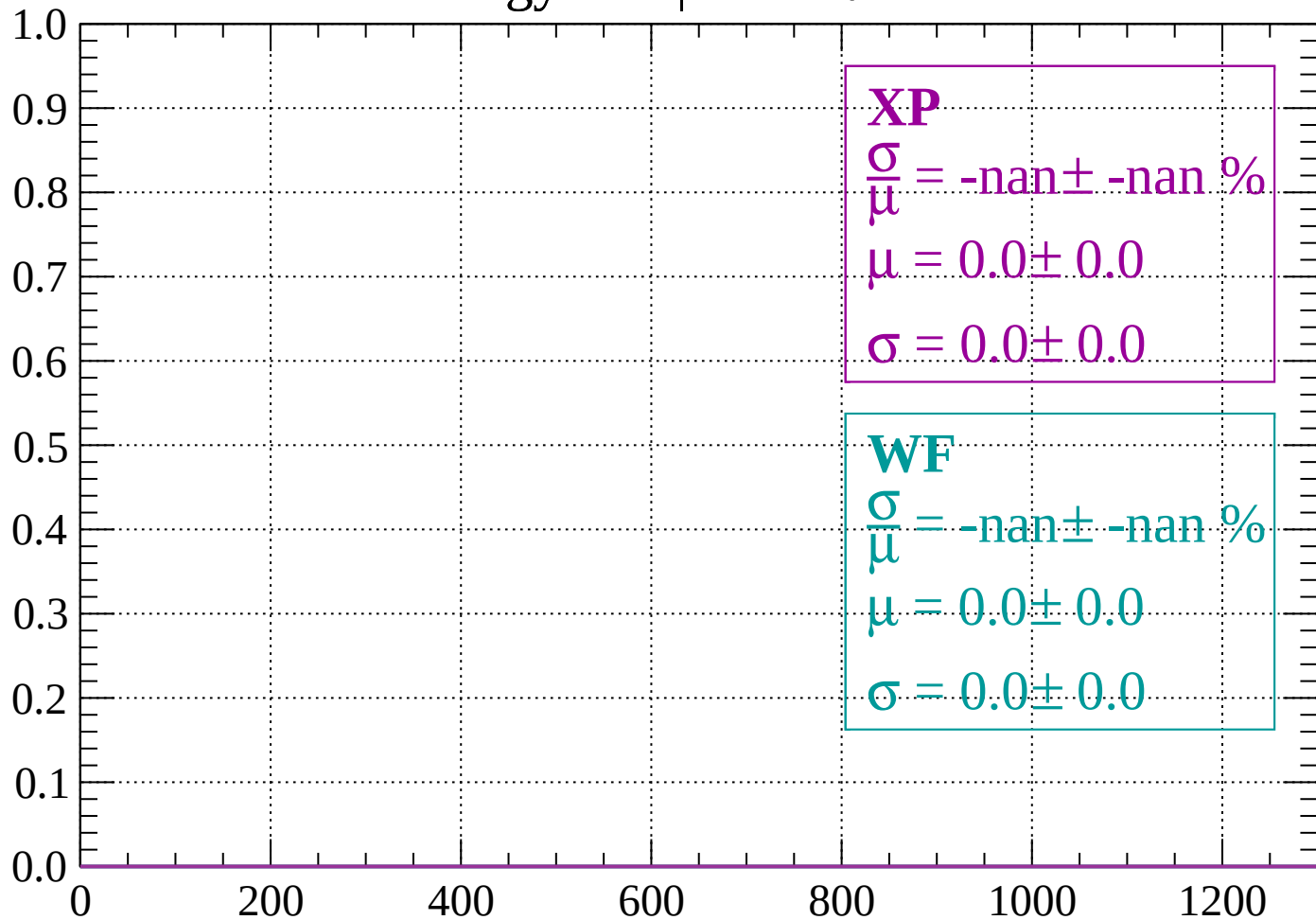
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

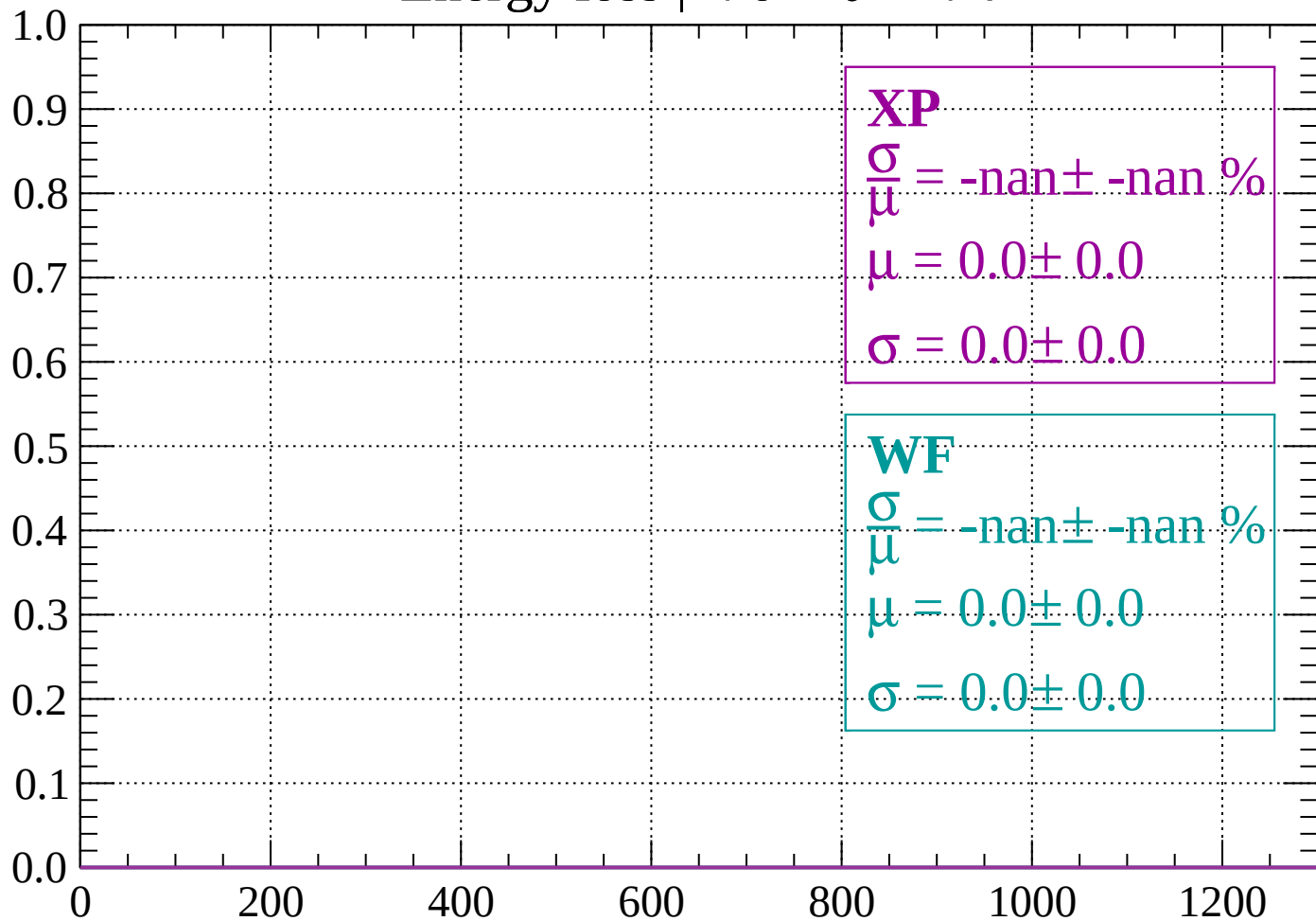
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

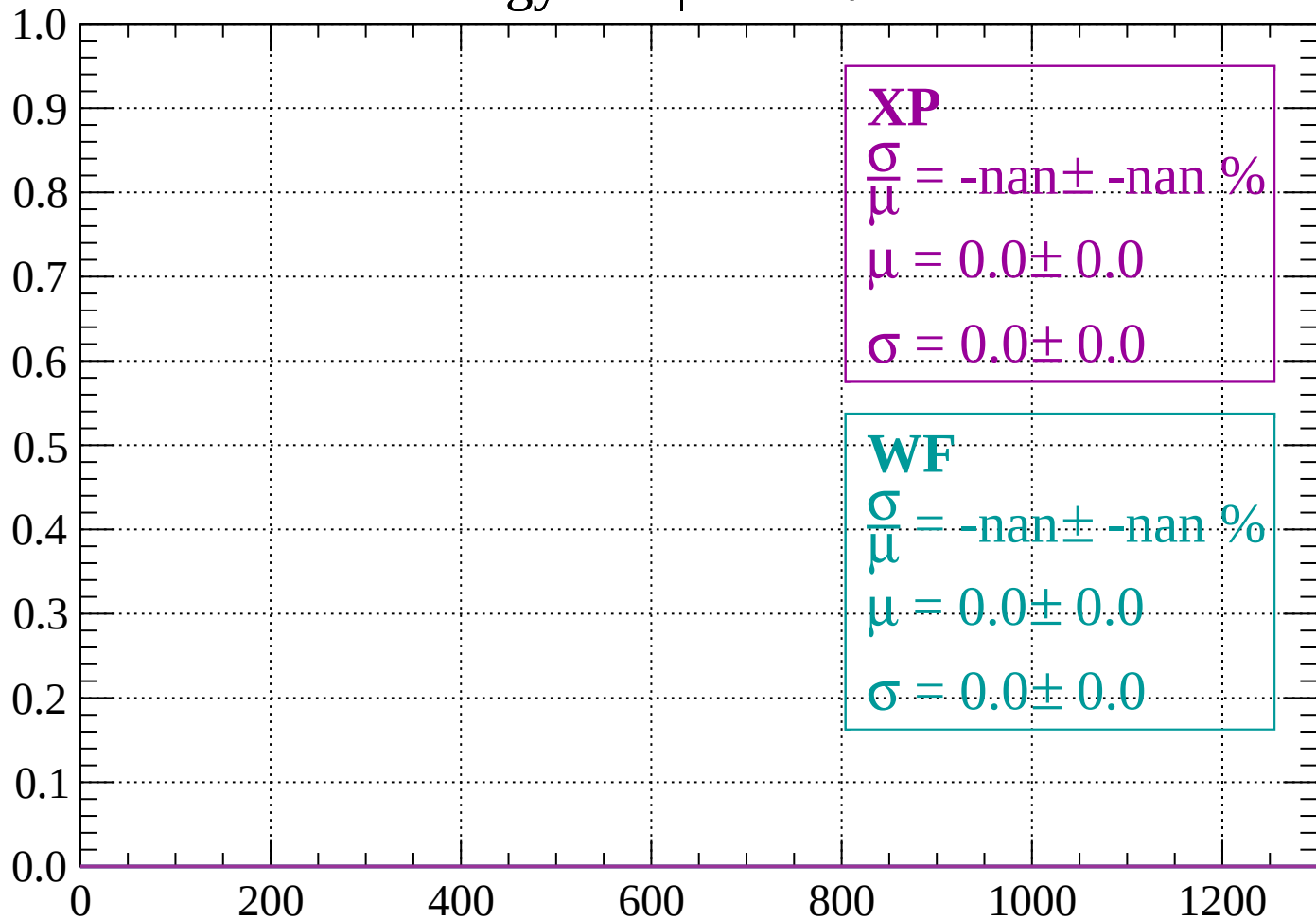
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

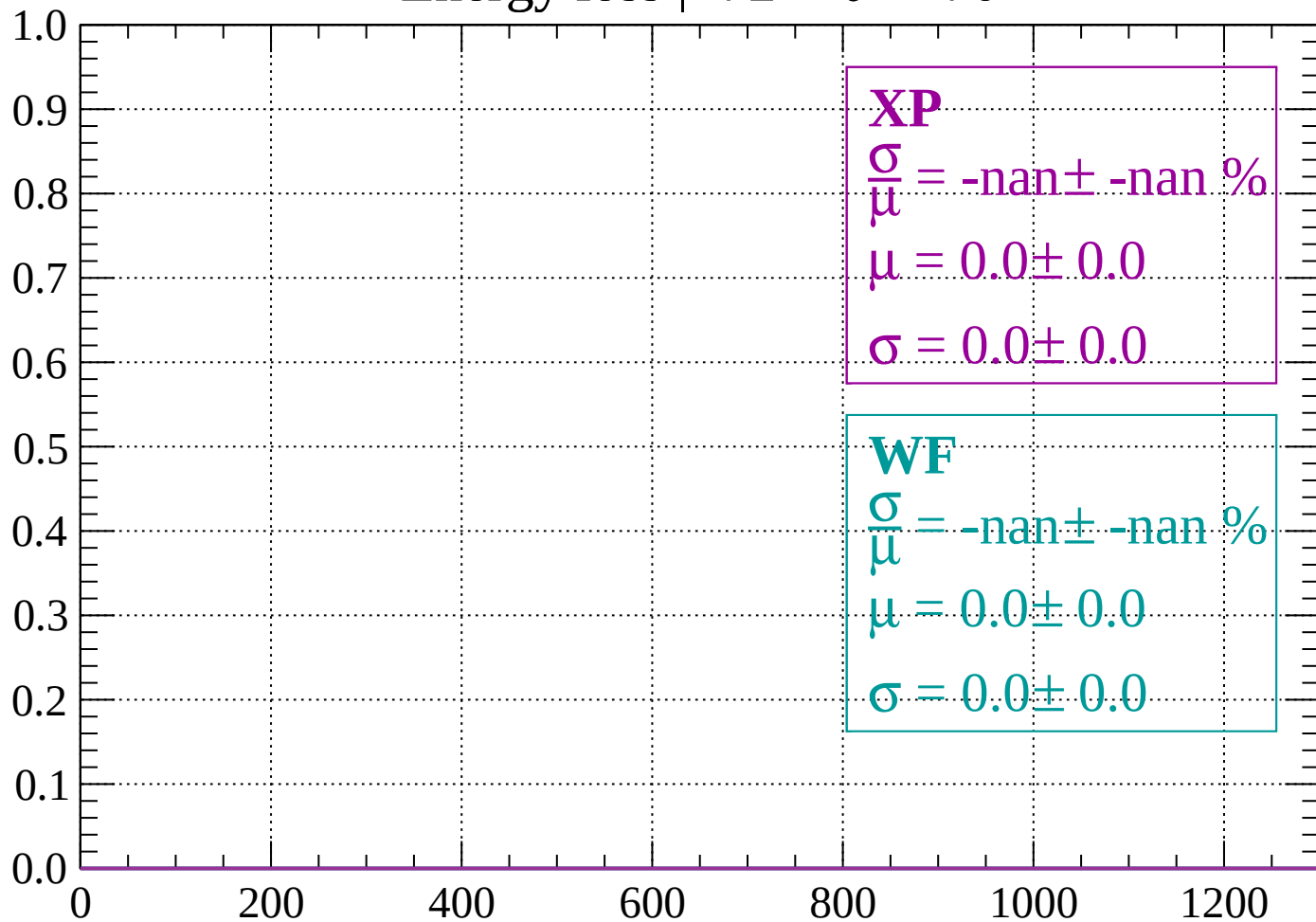
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

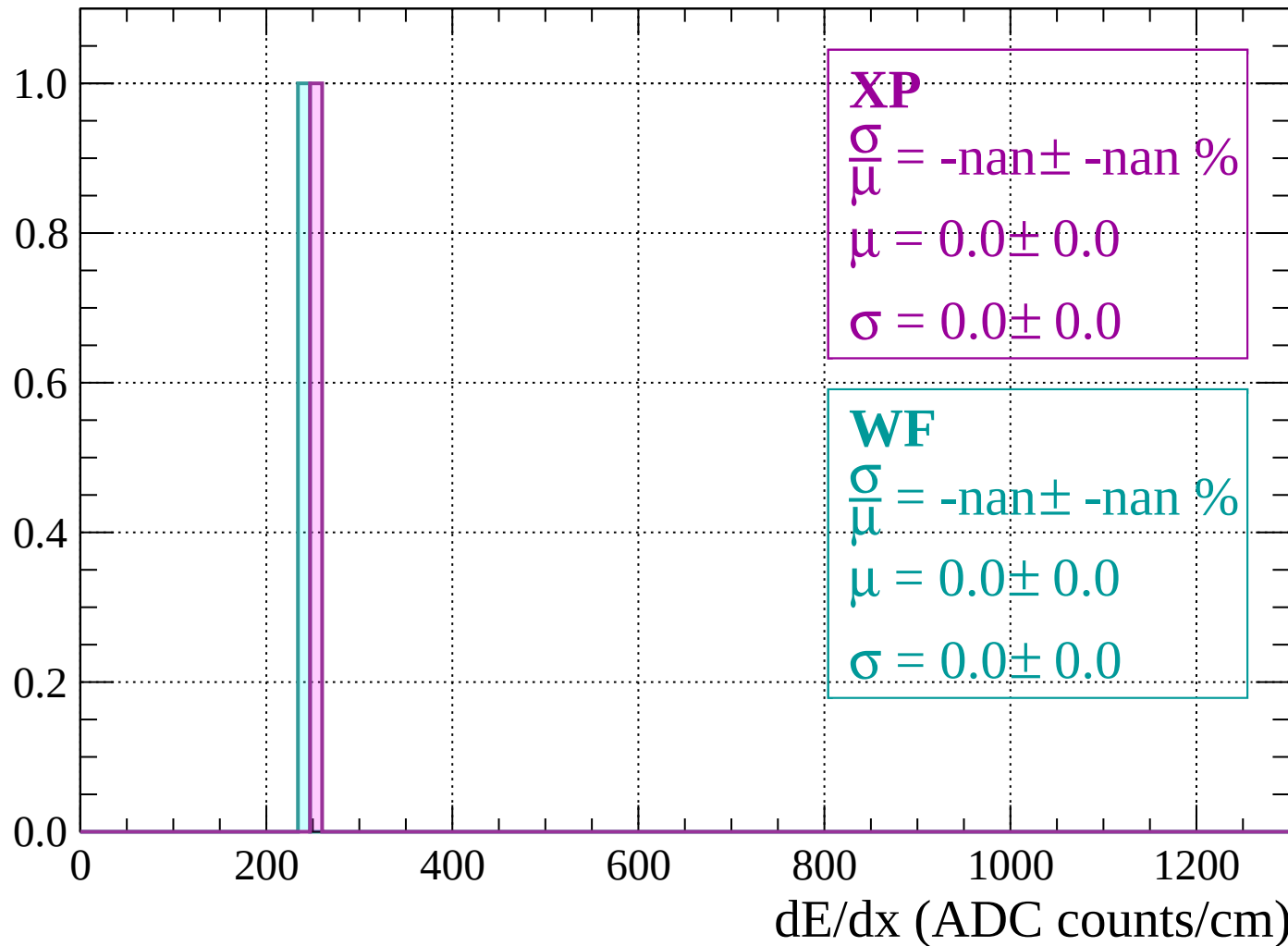
Count



dE/dx (ADC counts/cm)

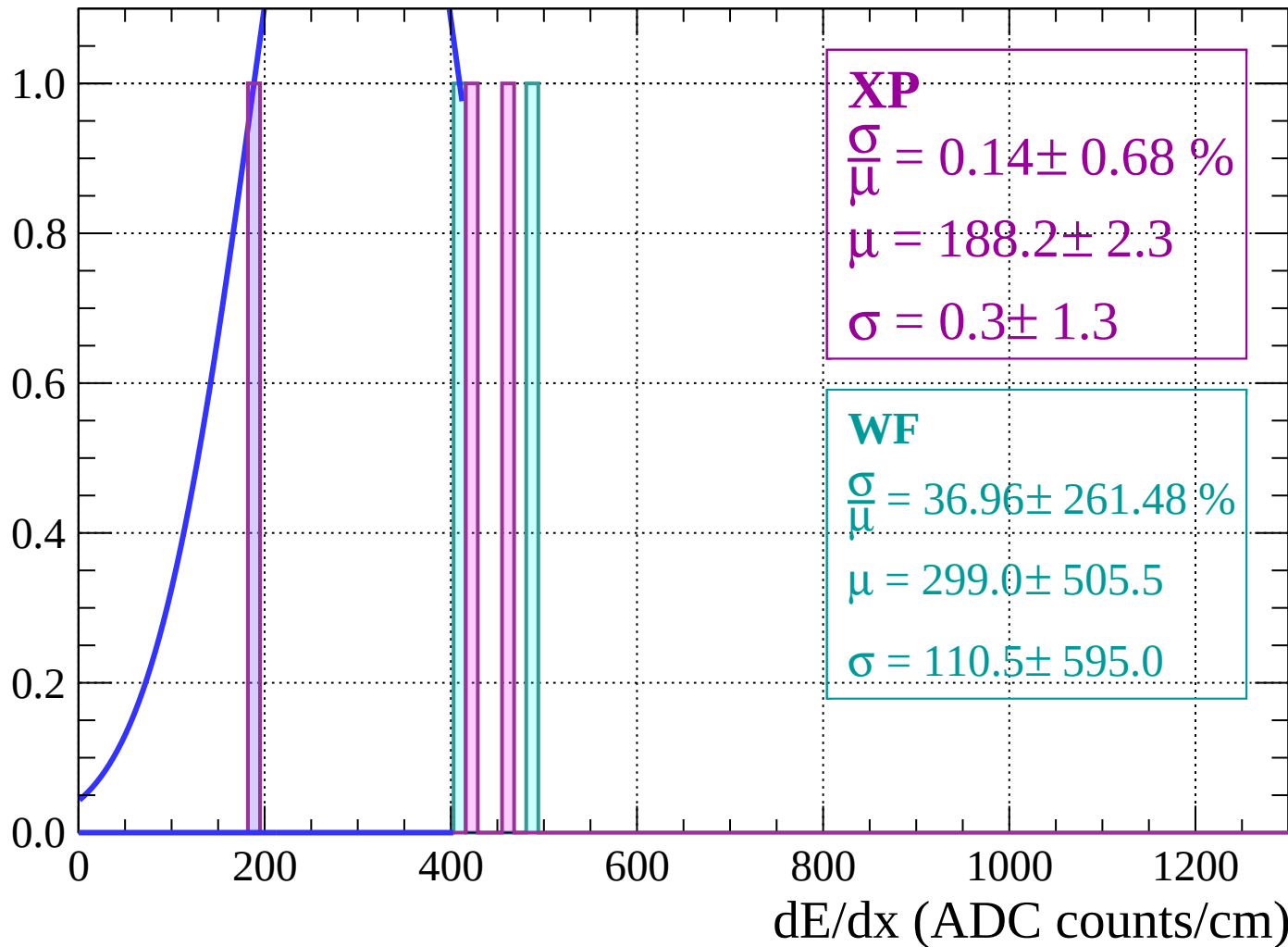
Energy loss | $-70 < \theta < -68$

Count



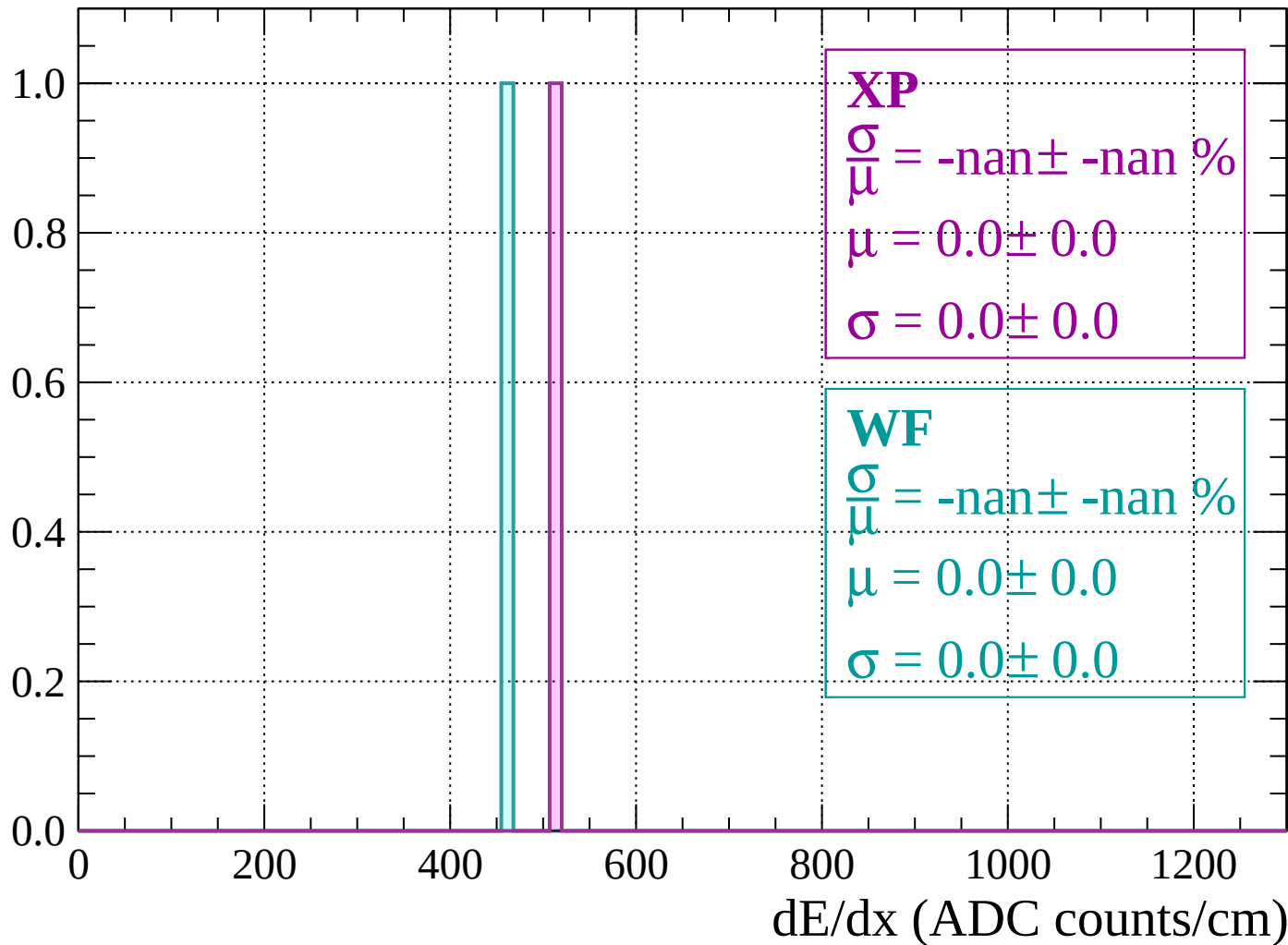
Energy loss | $-68 < \theta < -66$

Count



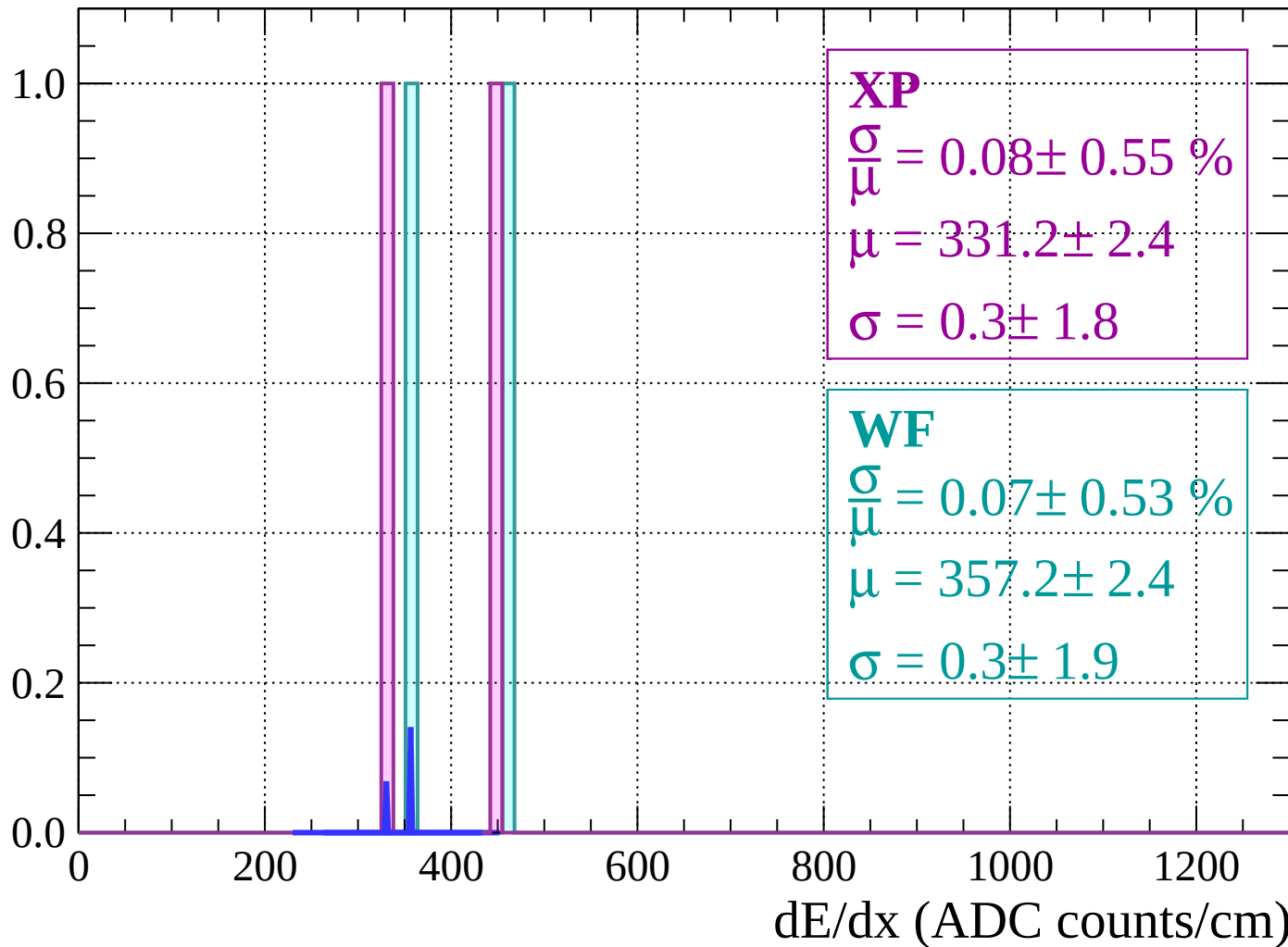
Energy loss | $-66 < \theta < -64$

Count



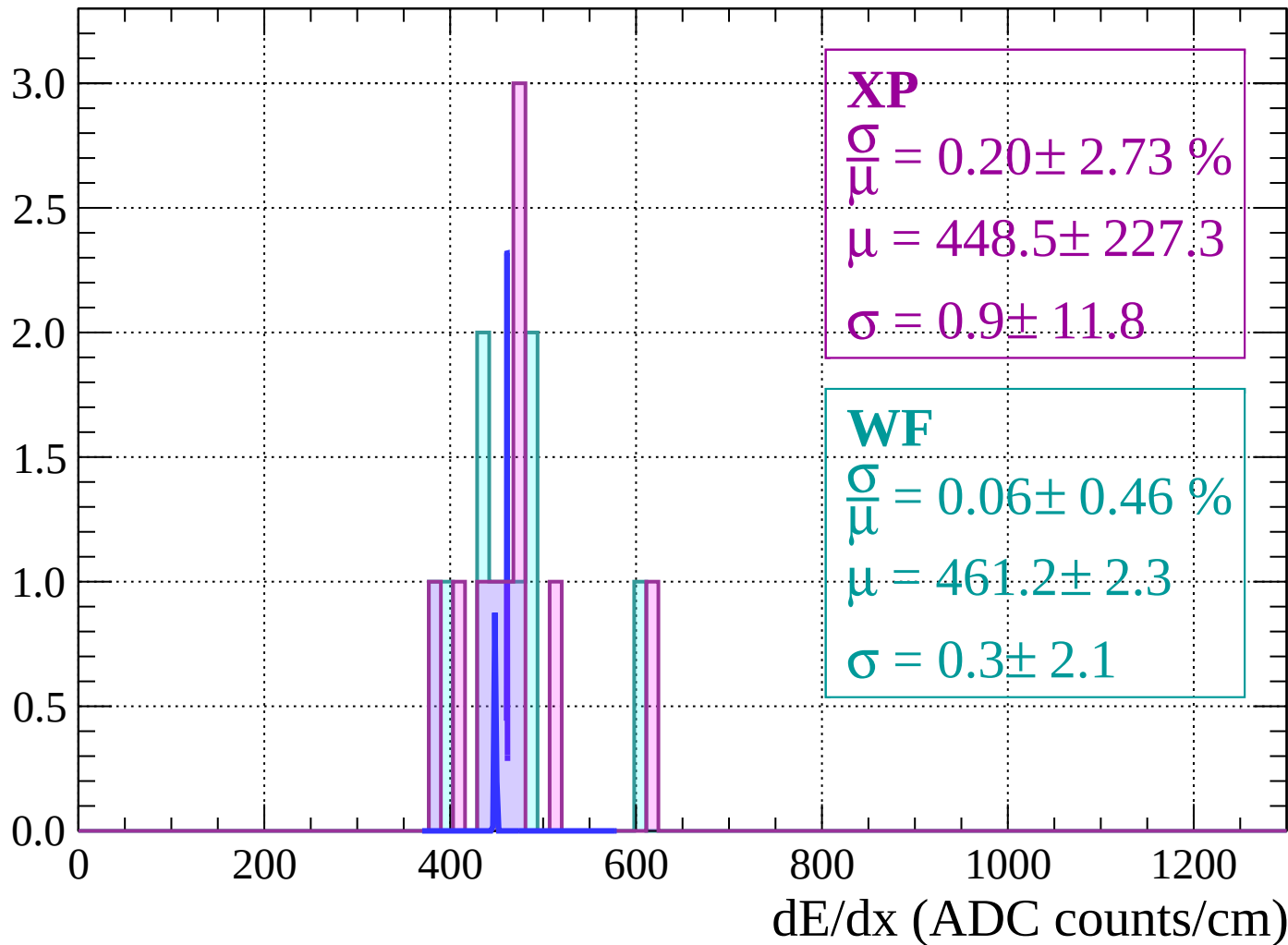
Energy loss | $-64 < \theta < -62$

Count



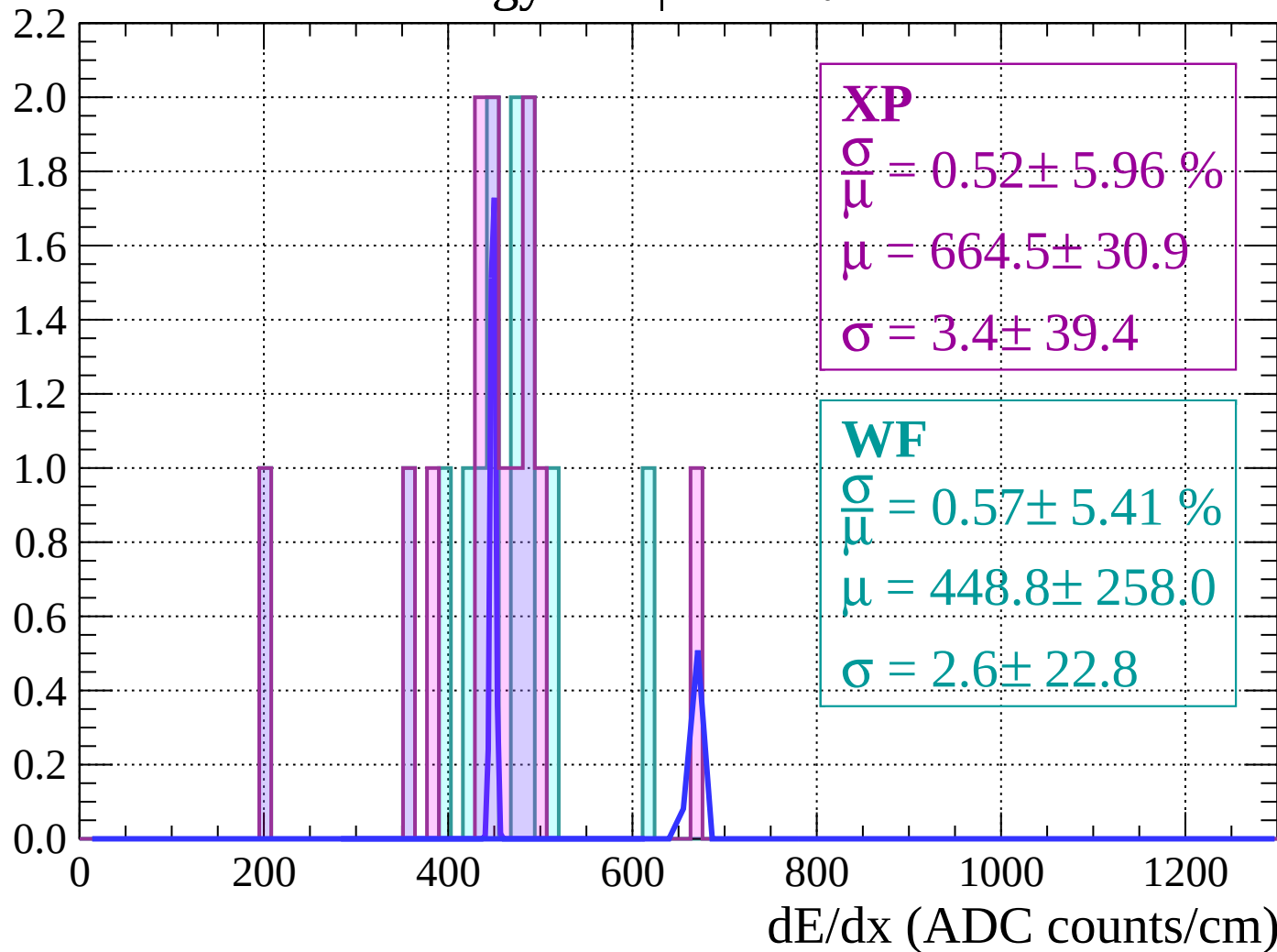
Energy loss | $-62 < \theta < -60$

Count



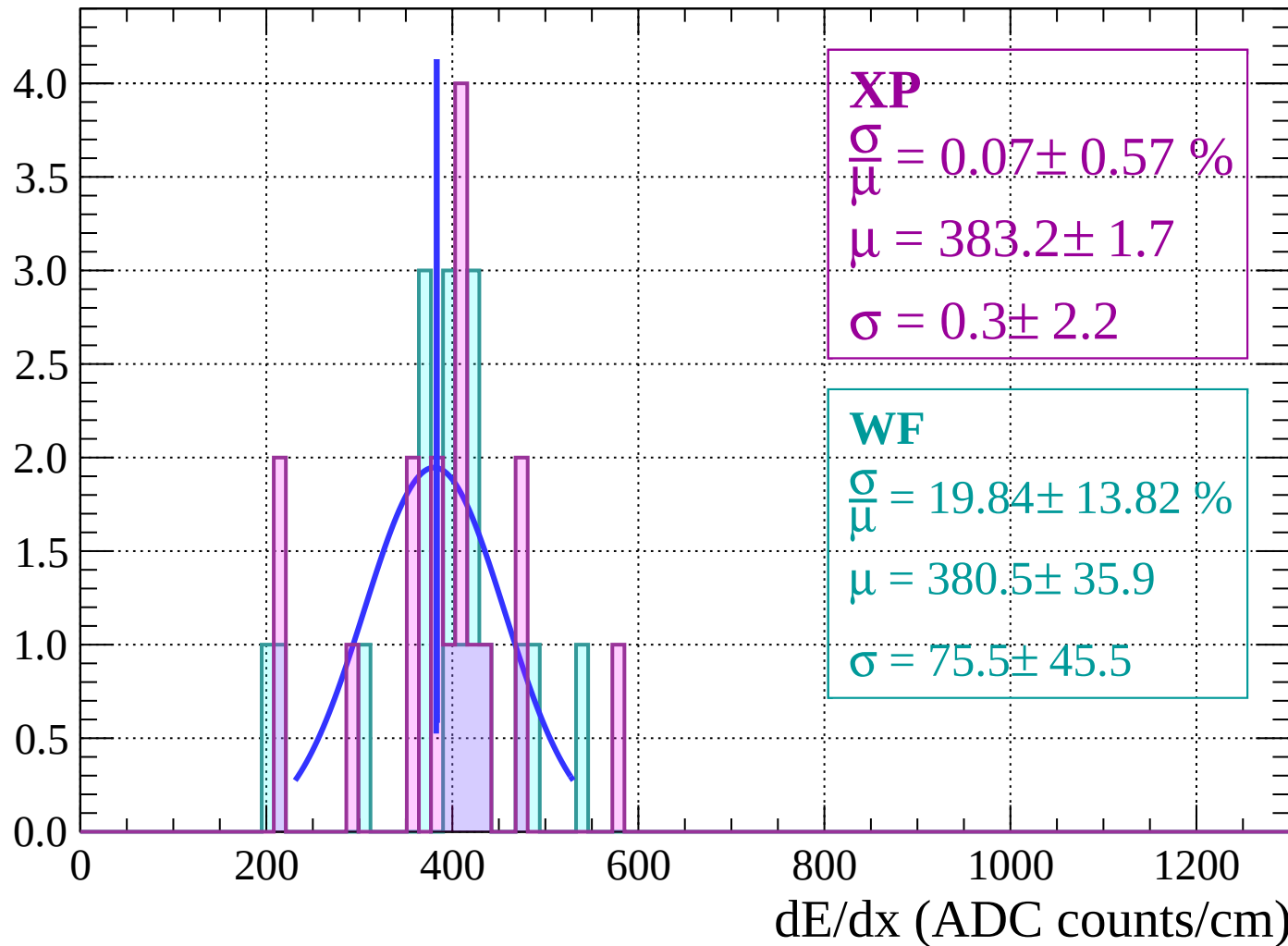
Energy loss | $-60 < \theta < -58$

Count



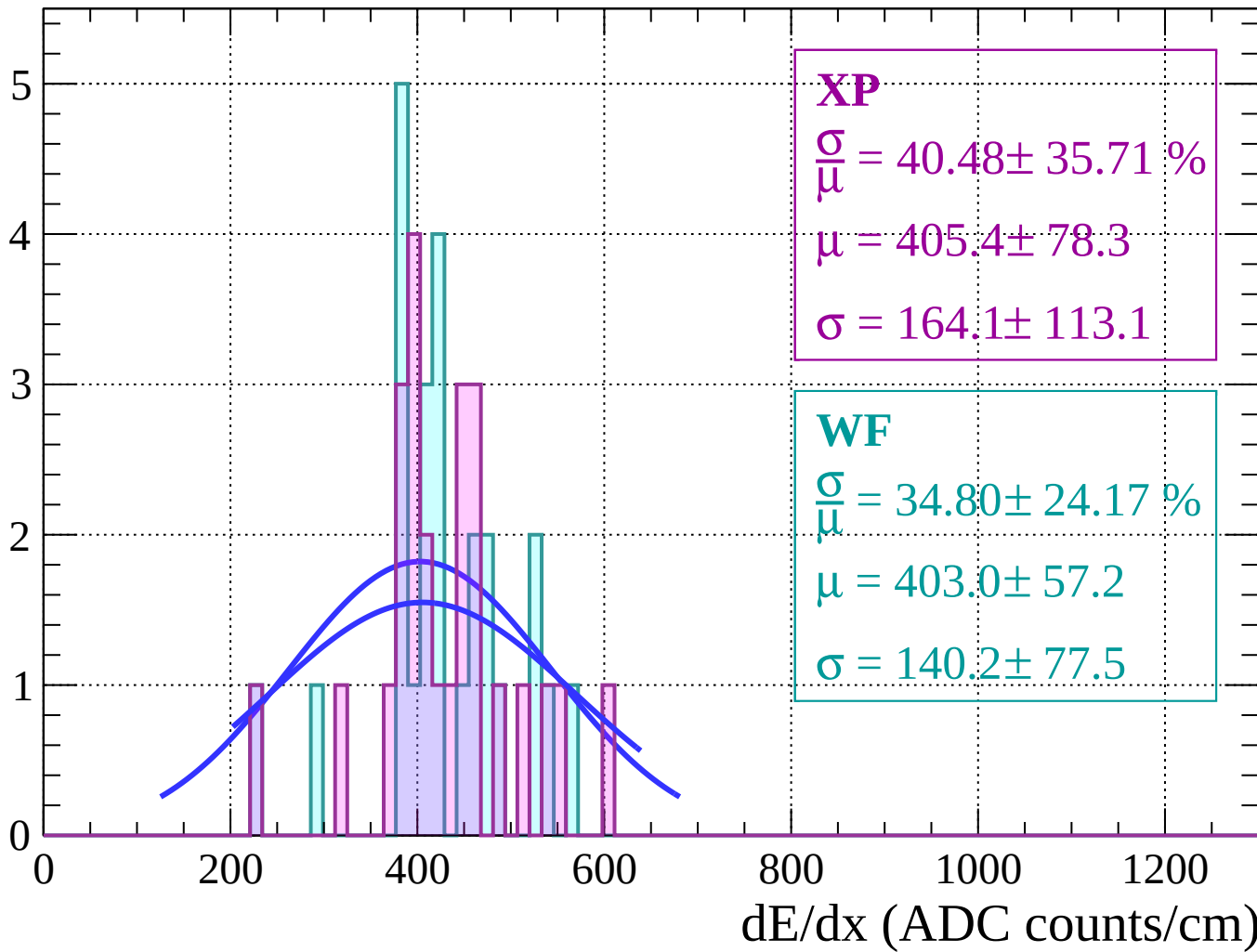
Energy loss | $-58 < \theta < -56$

Count



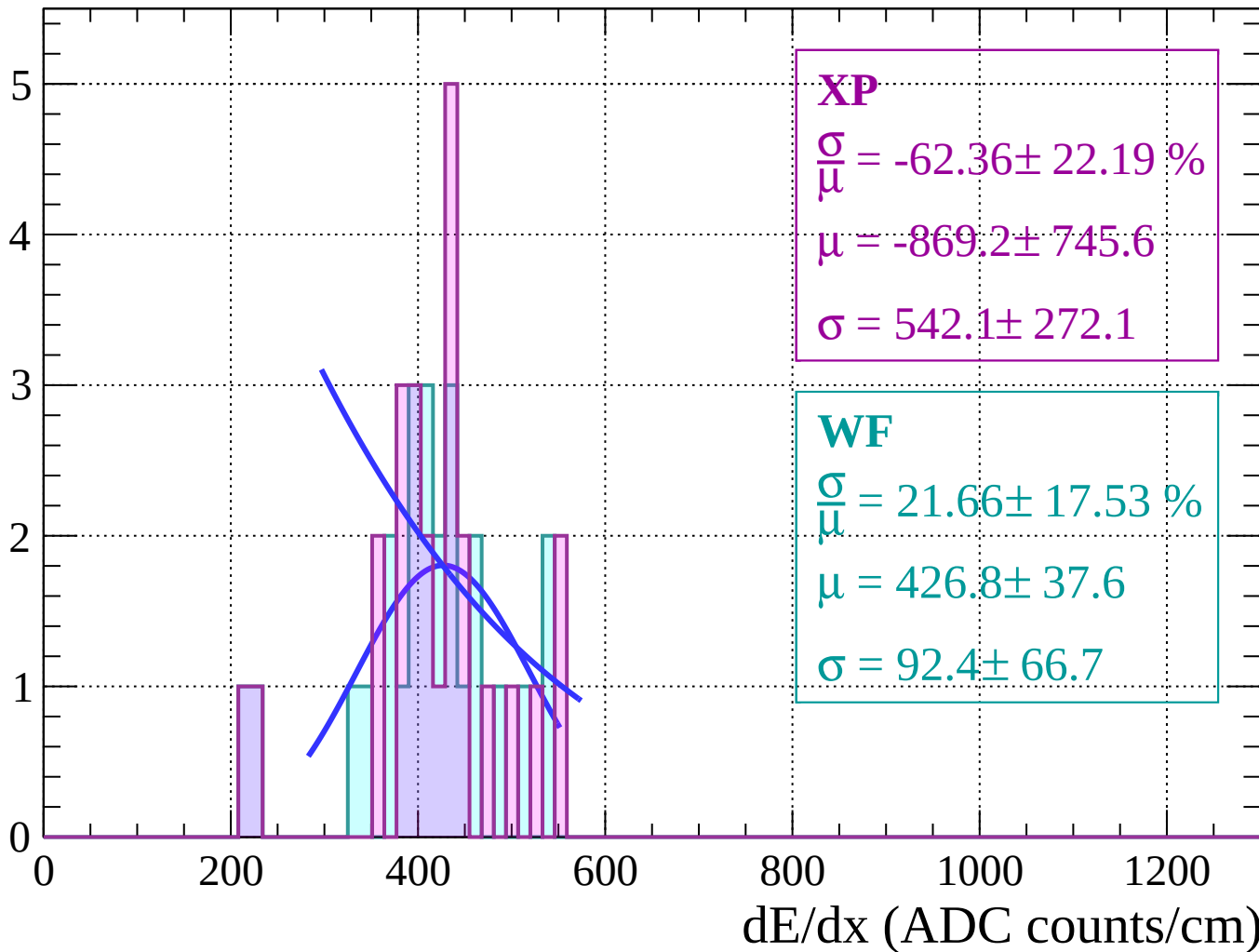
Energy loss | $-56 < \theta < -54$

Count



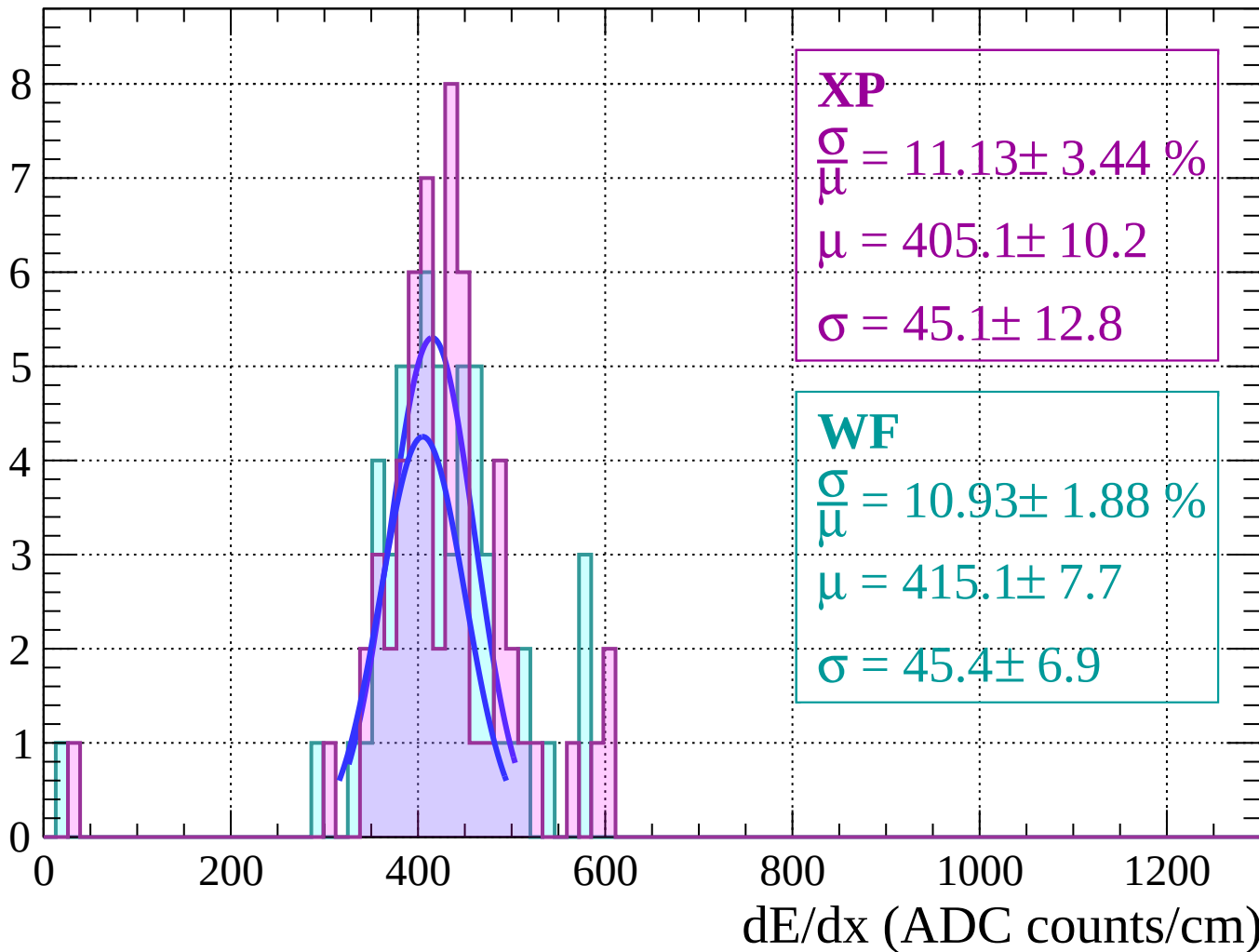
Energy loss | $-54 < \theta < -52$

Count



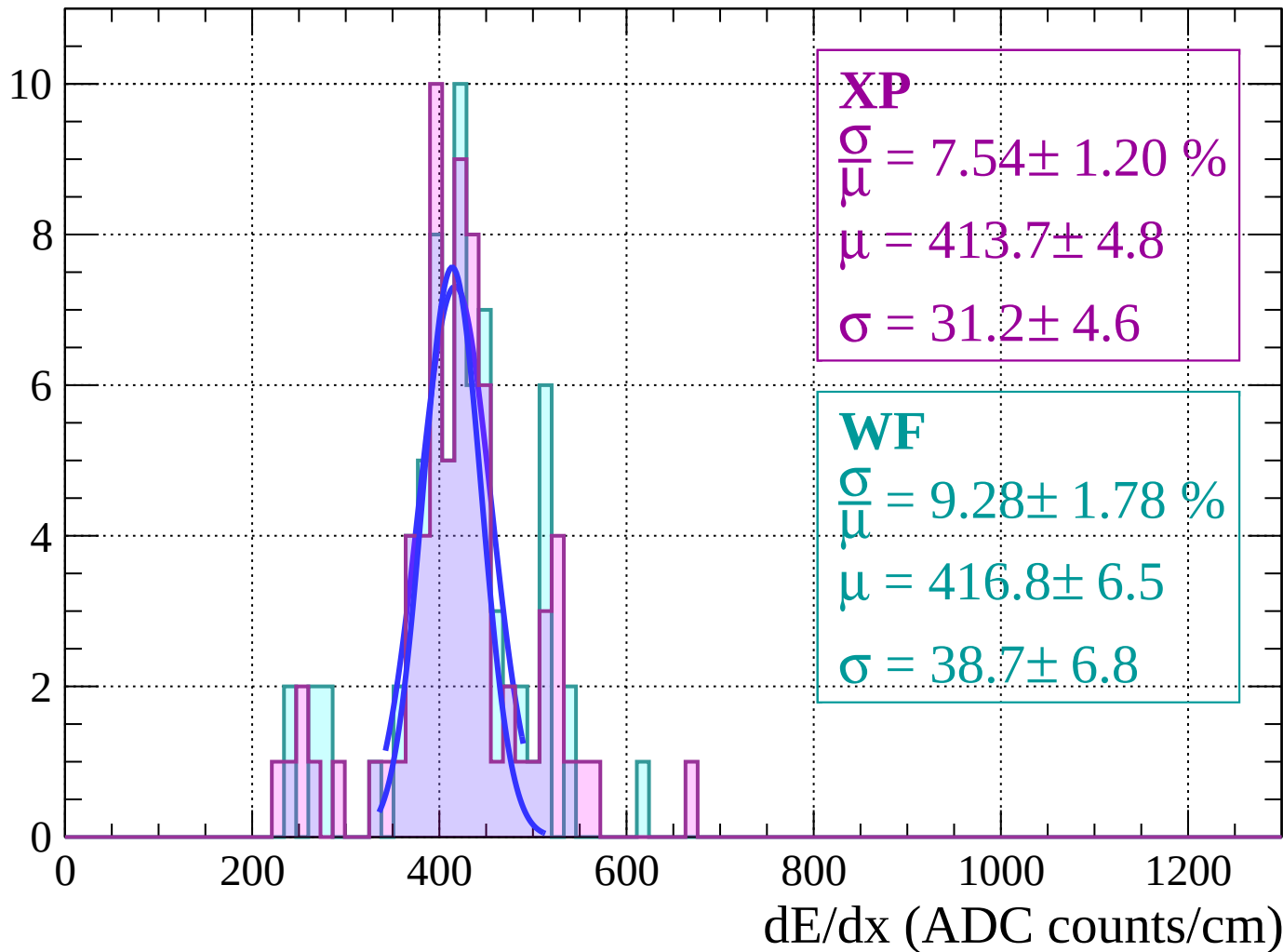
Energy loss | $-52 < \theta < -50$

Count



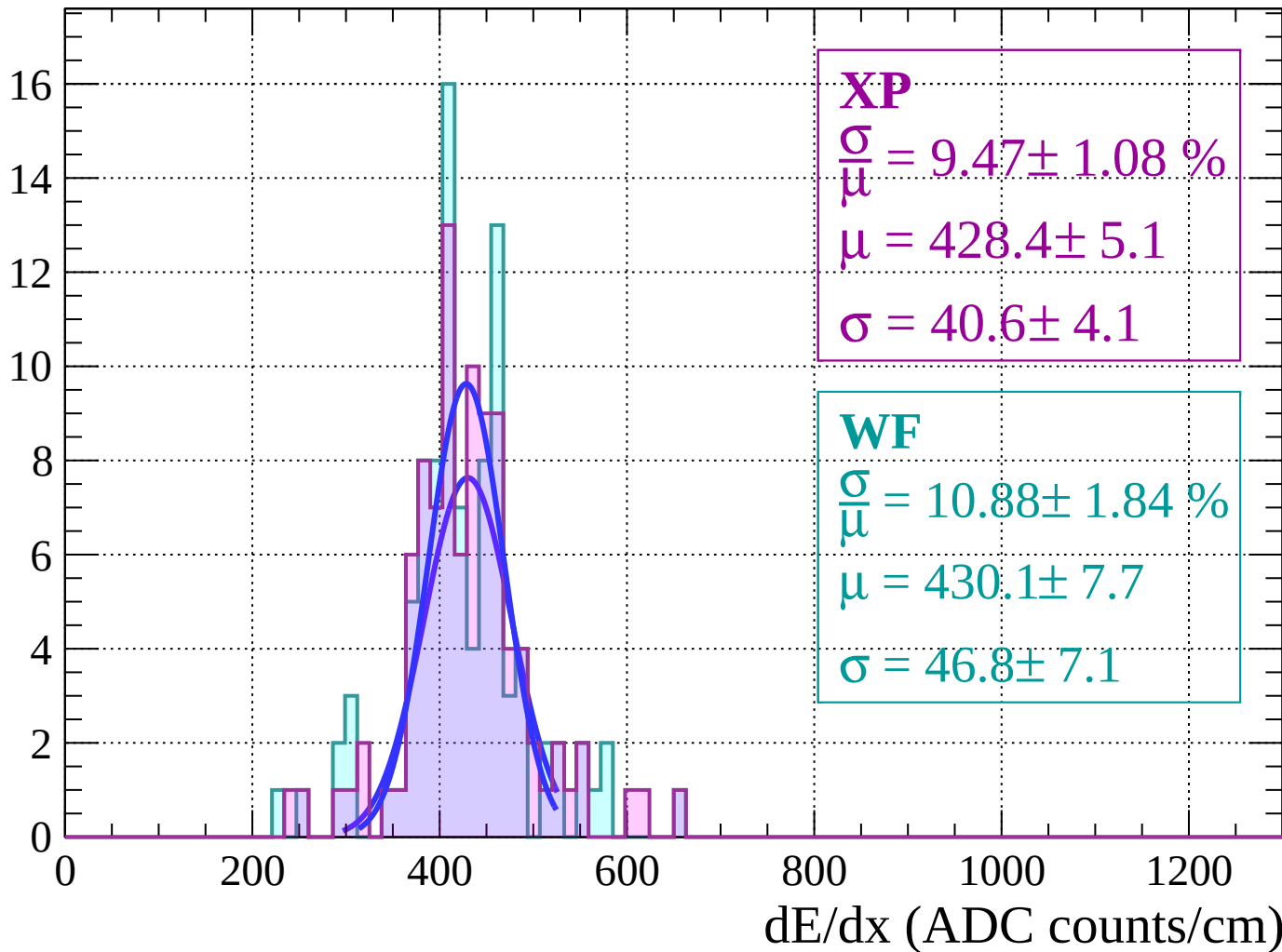
Energy loss | $-50 < \theta < -48$

Count



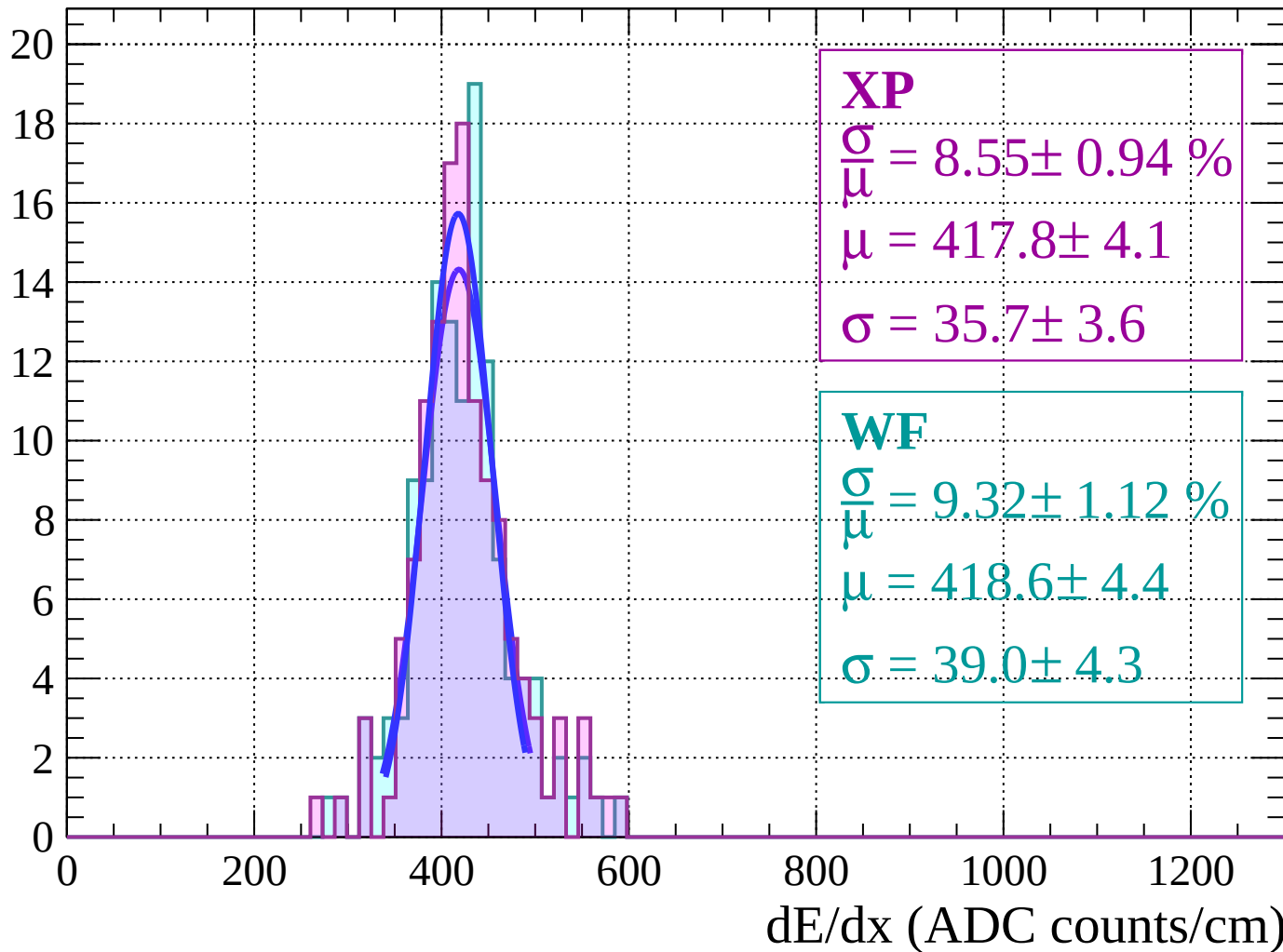
Energy loss | $-48 < \theta < -46$

Count



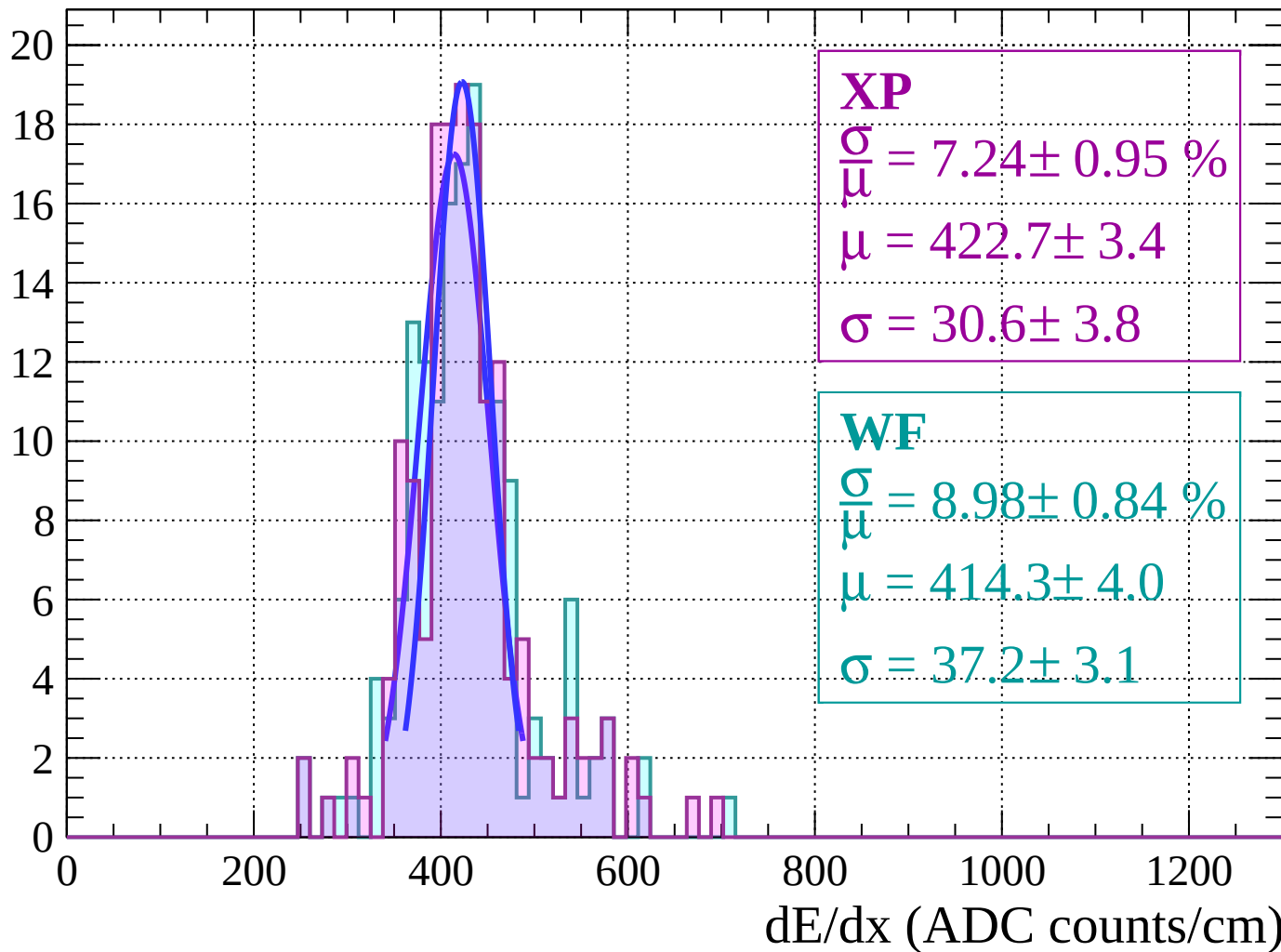
Energy loss | $-46 < \theta < -44$

Count



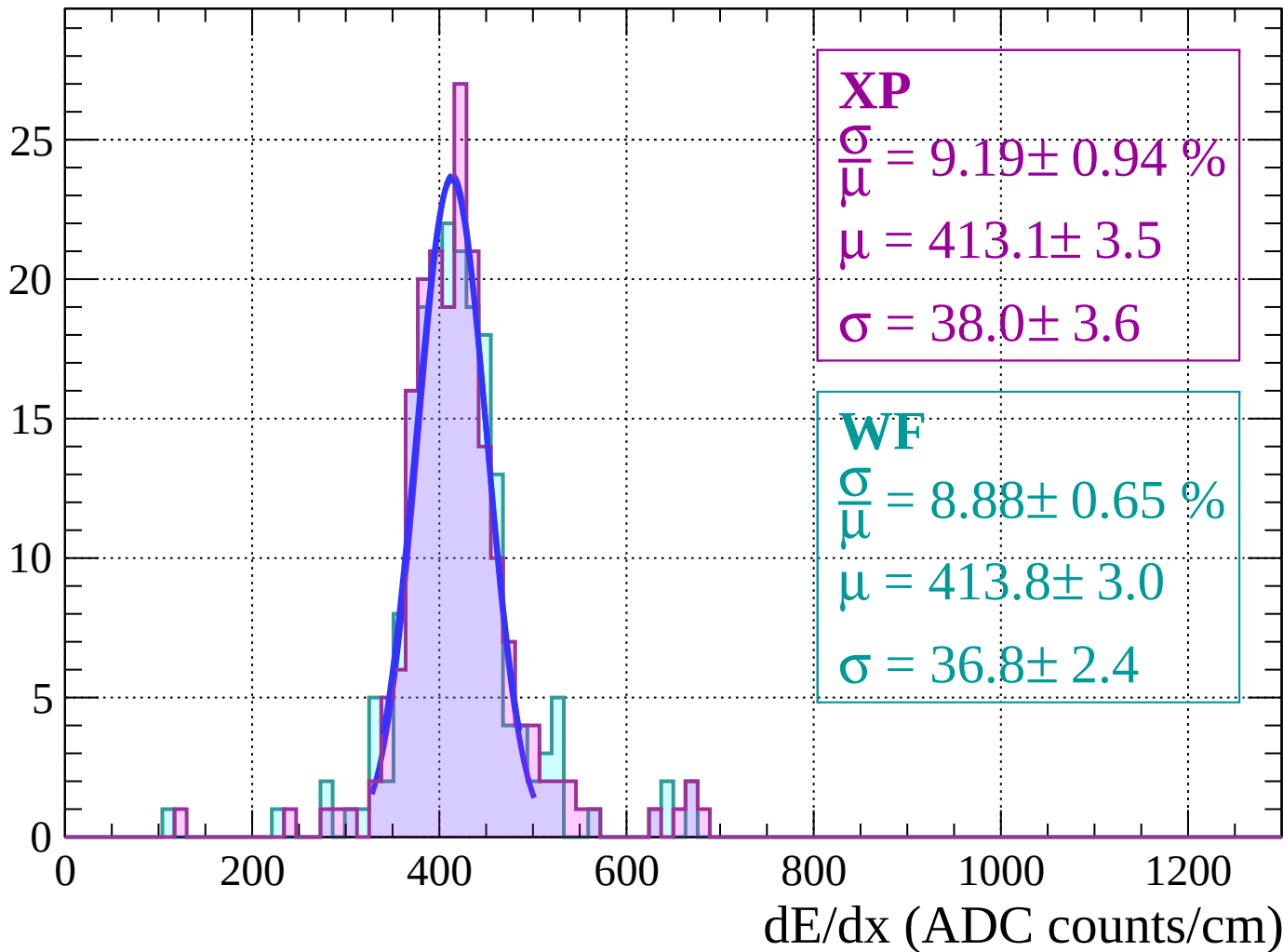
Energy loss $|-44 < \theta < -42$

Count



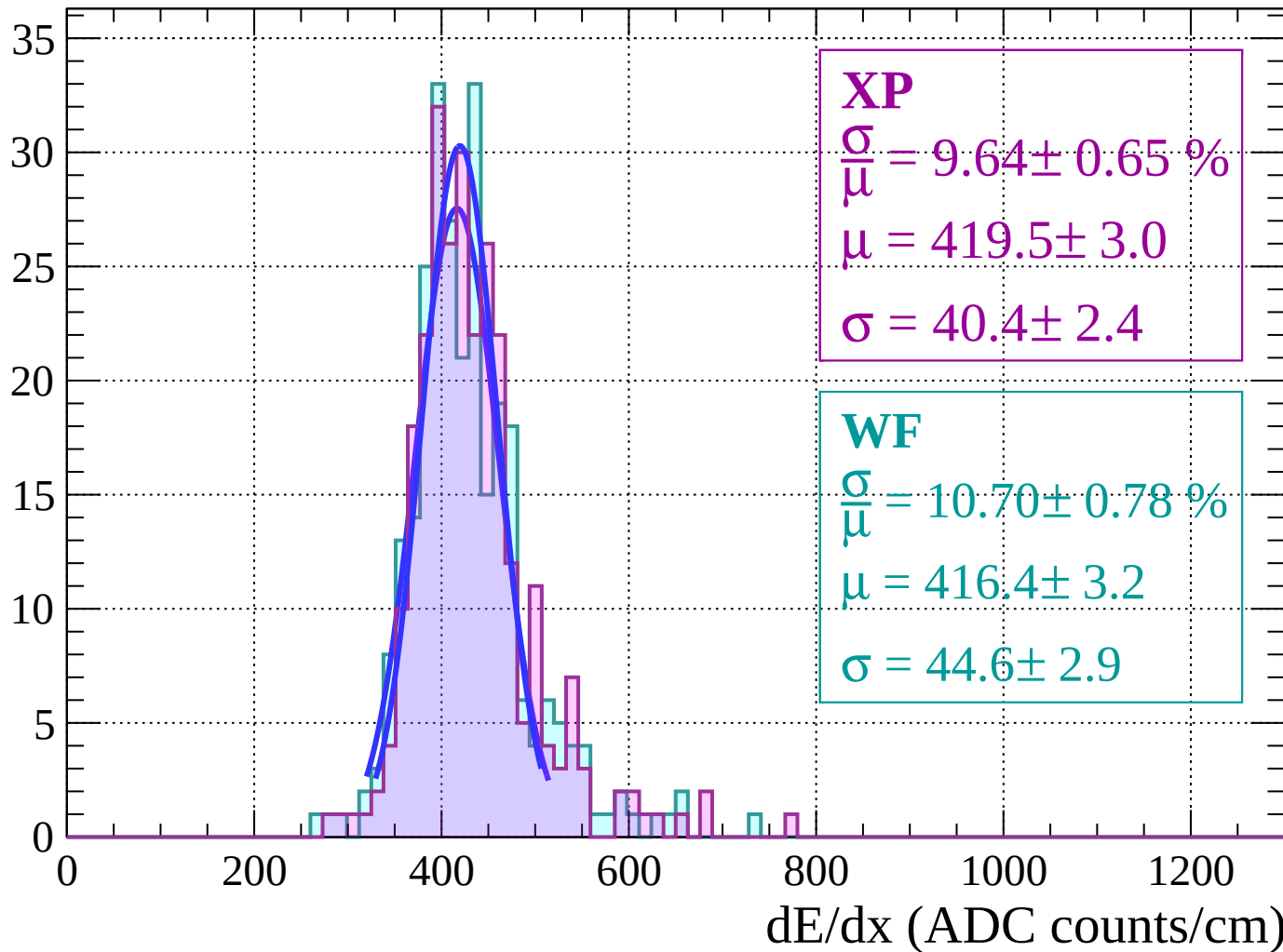
Energy loss | $-42 < \theta < -40$

Count



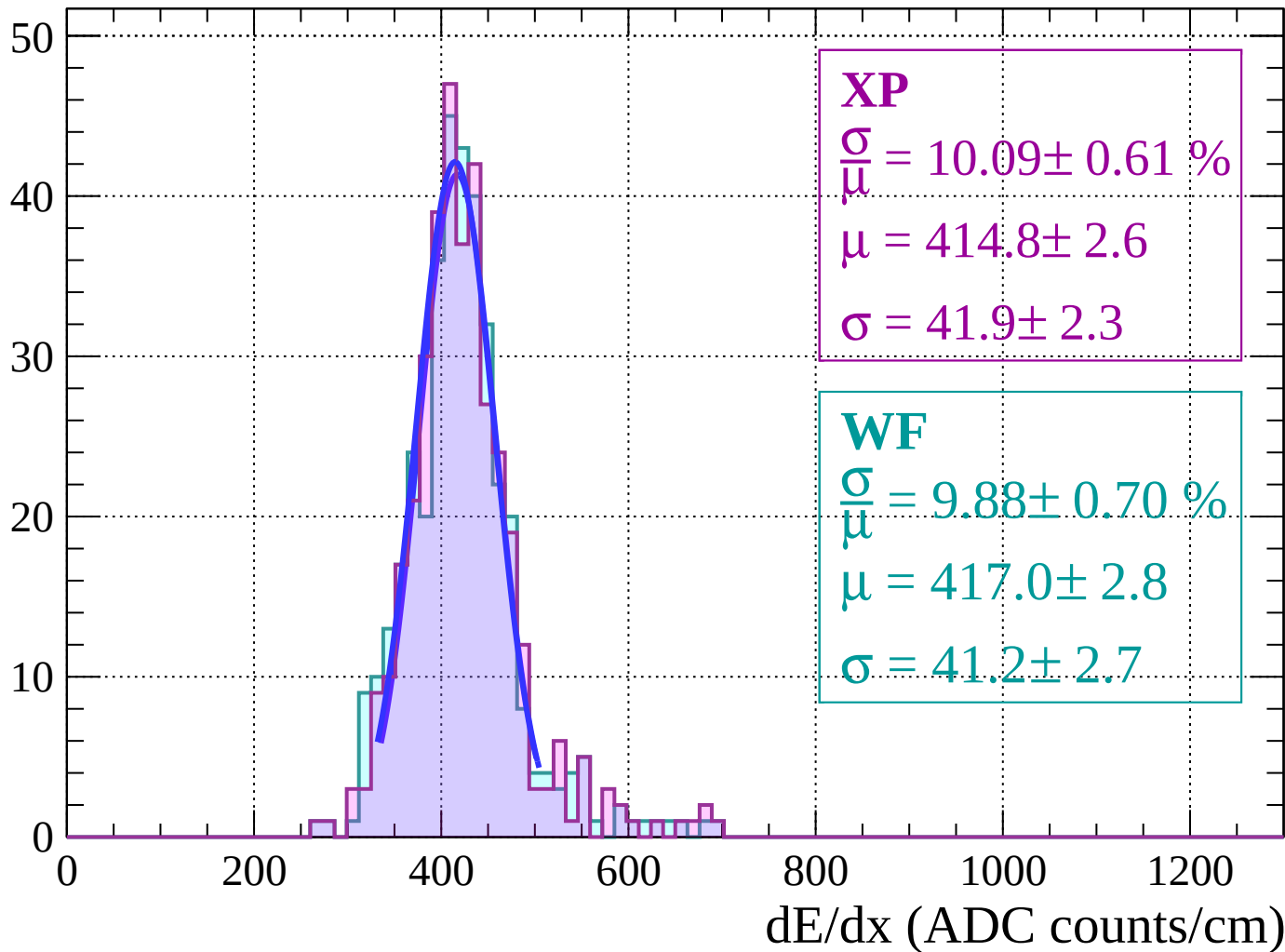
Energy loss | $-40 < \theta < -38$

Count



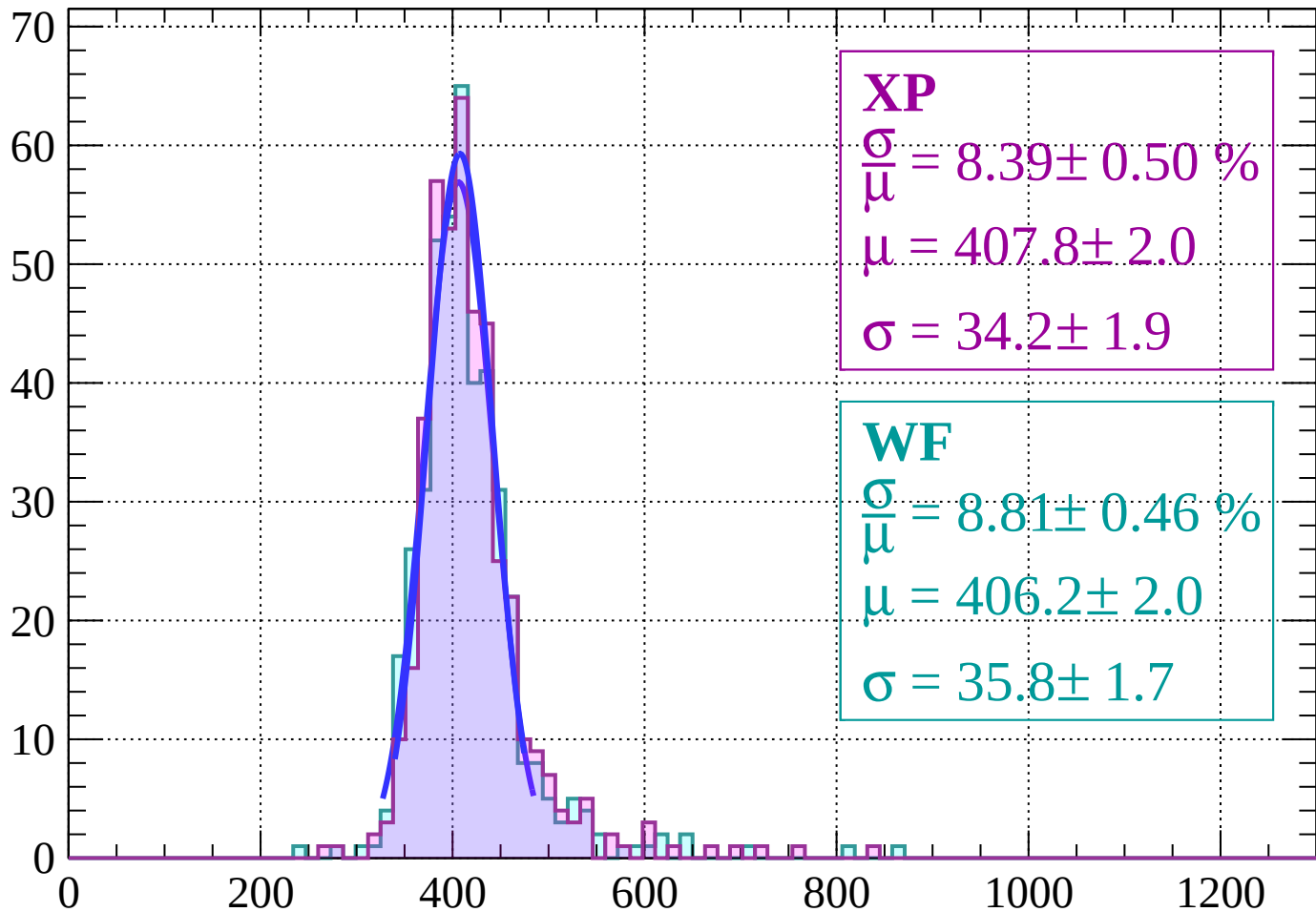
Energy loss | $-38 < \theta < -36$

Count



Energy loss | $-36 < \theta < -34$

Count



XP

$$\frac{\sigma}{\mu} = 8.39 \pm 0.50 \%$$

$$\mu = 407.8 \pm 2.0$$

$$\sigma = 34.2 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 8.81 \pm 0.46 \%$$

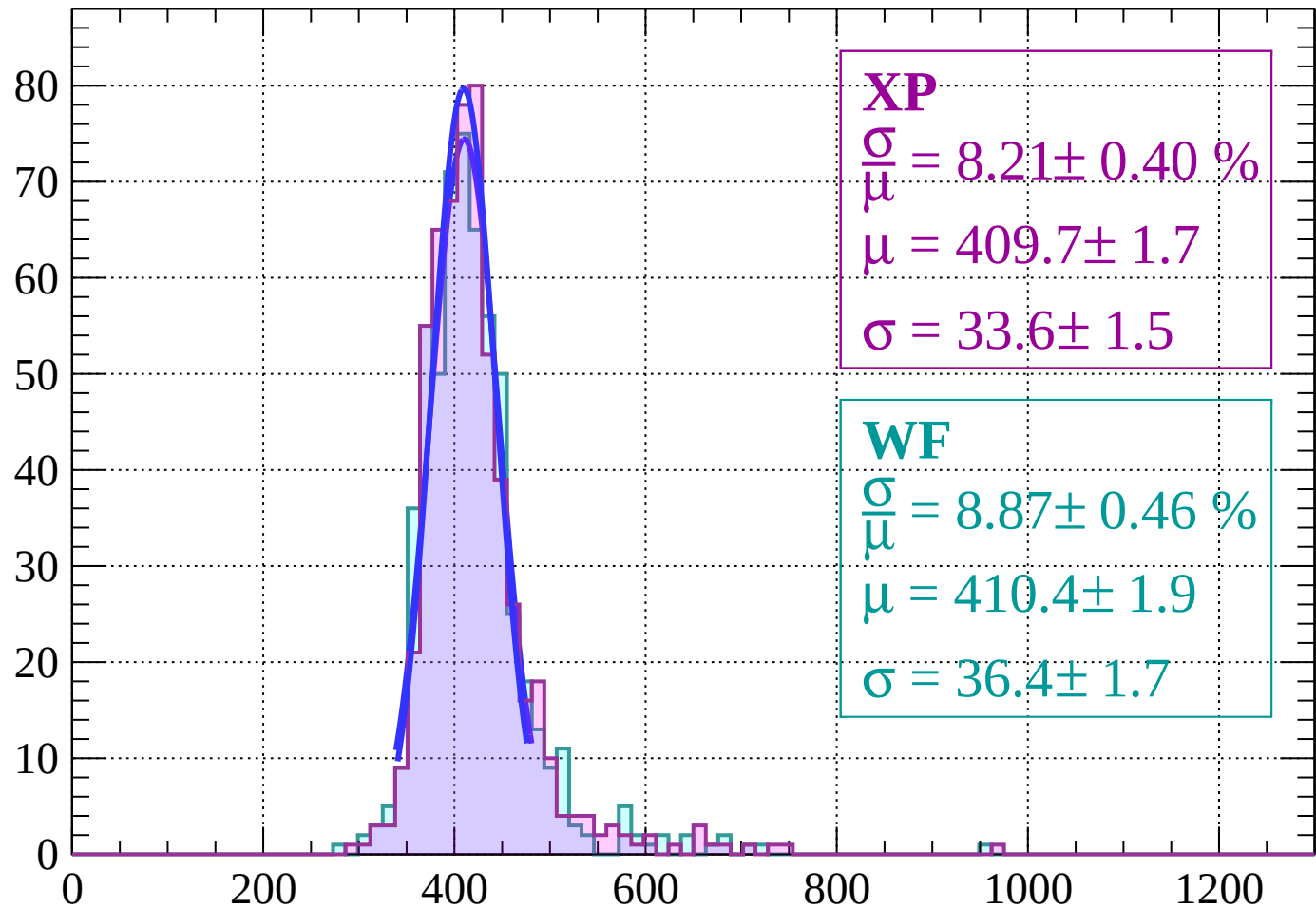
$$\mu = 406.2 \pm 2.0$$

$$\sigma = 35.8 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss $|-34 < \theta < -32$

Count



XP

$$\frac{\sigma}{\mu} = 8.21 \pm 0.40 \%$$

$$\mu = 409.7 \pm 1.7$$

$$\sigma = 33.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.87 \pm 0.46 \%$$

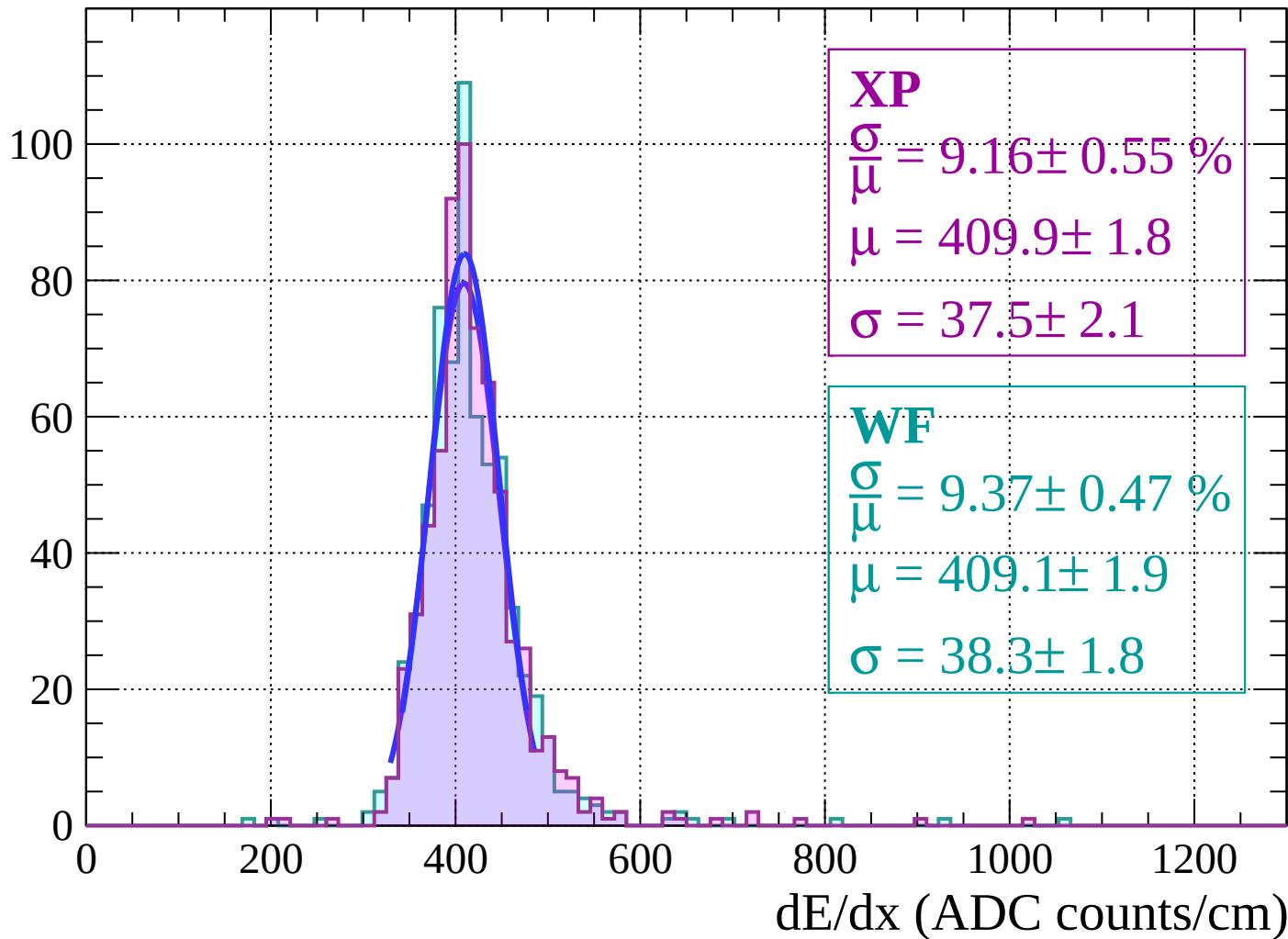
$$\mu = 410.4 \pm 1.9$$

$$\sigma = 36.4 \pm 1.7$$

dE/dx (ADC counts/cm)

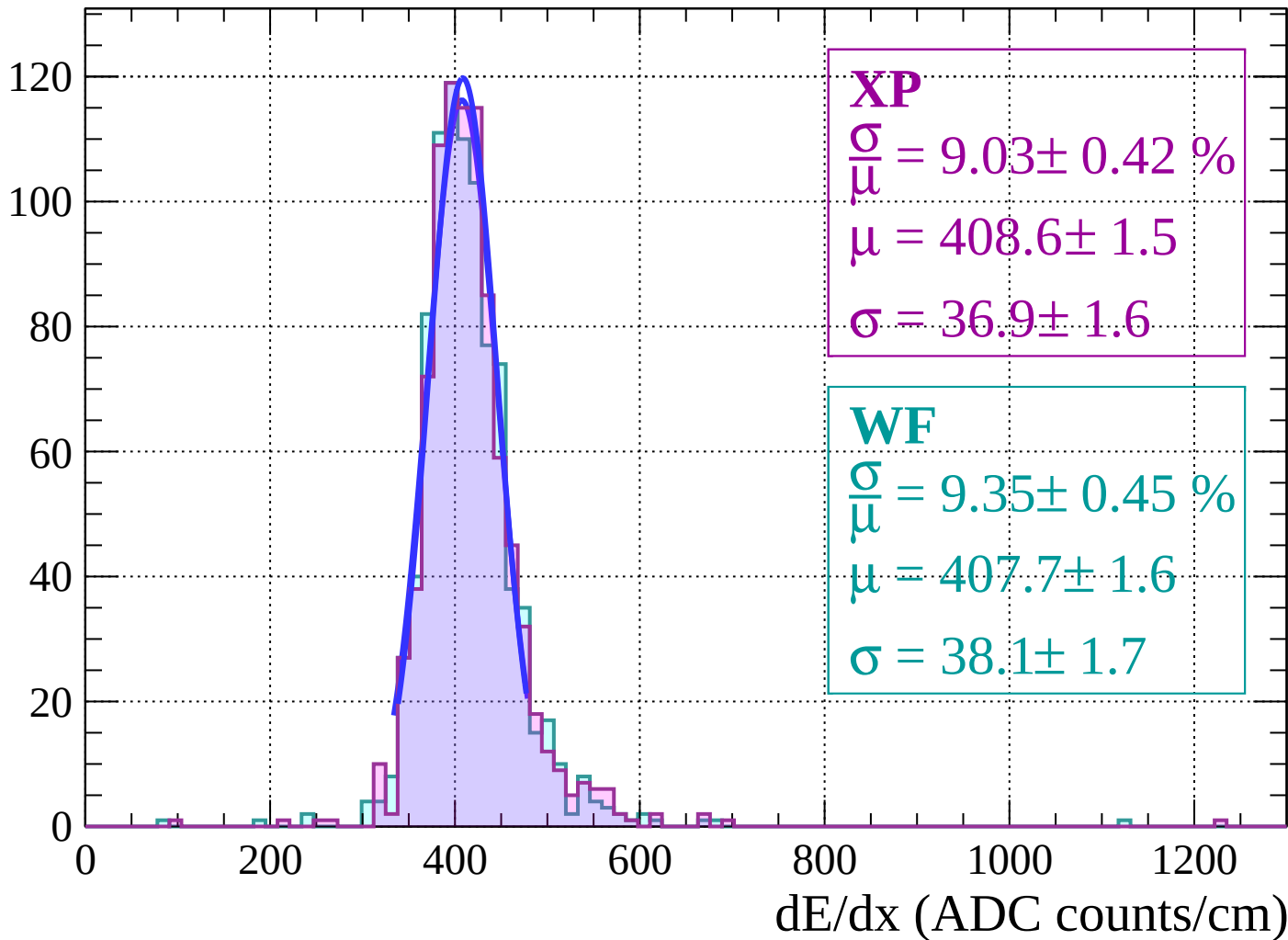
Energy loss | $-32 < \theta < -30$

Count



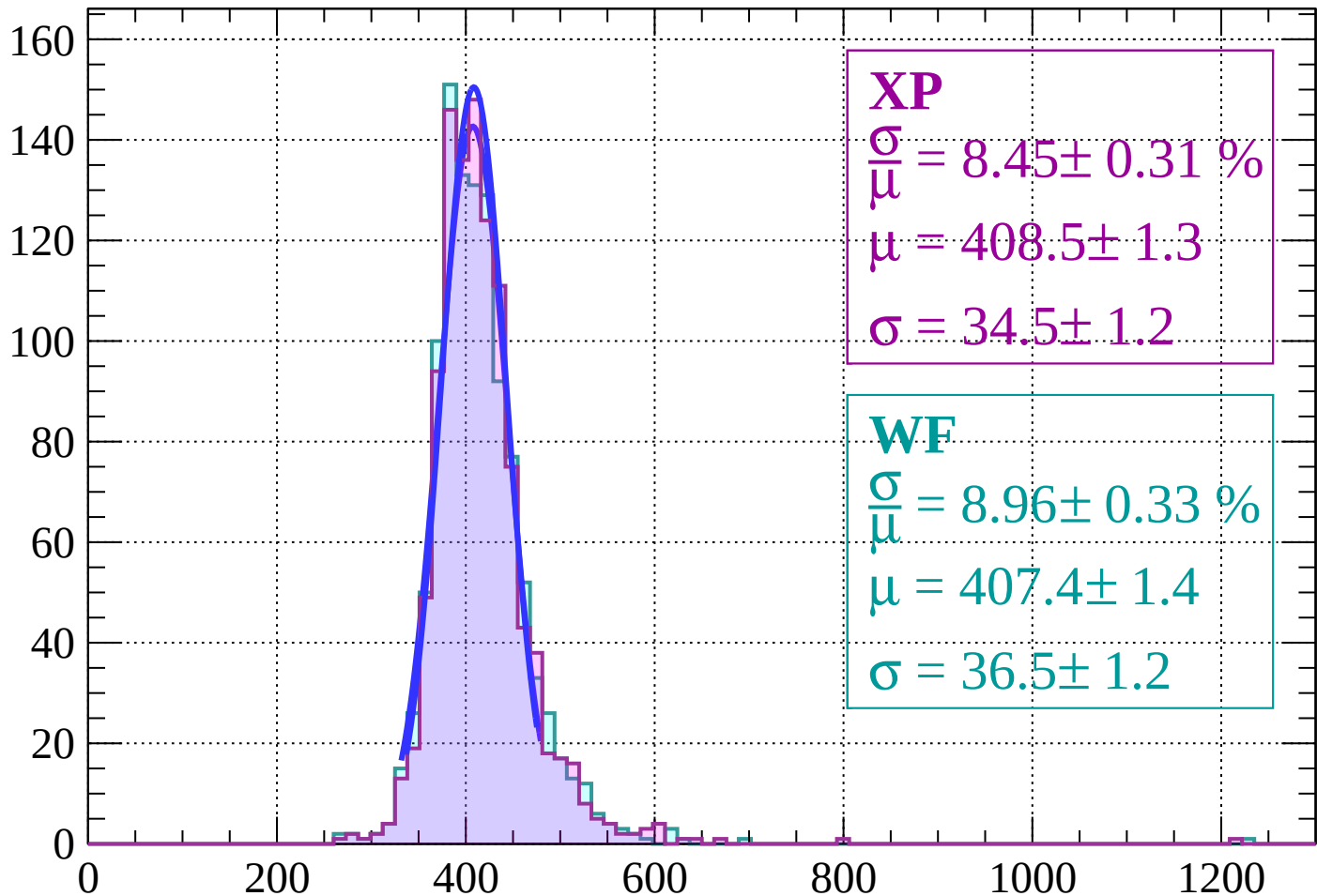
Energy loss | $-30 < \theta < -28$

Count



Energy loss | $-28 < \theta < -26$

Count



XP

$$\frac{\sigma}{\mu} = 8.45 \pm 0.31 \%$$

$$\mu = 408.5 \pm 1.3$$

$$\sigma = 34.5 \pm 1.2$$

WF

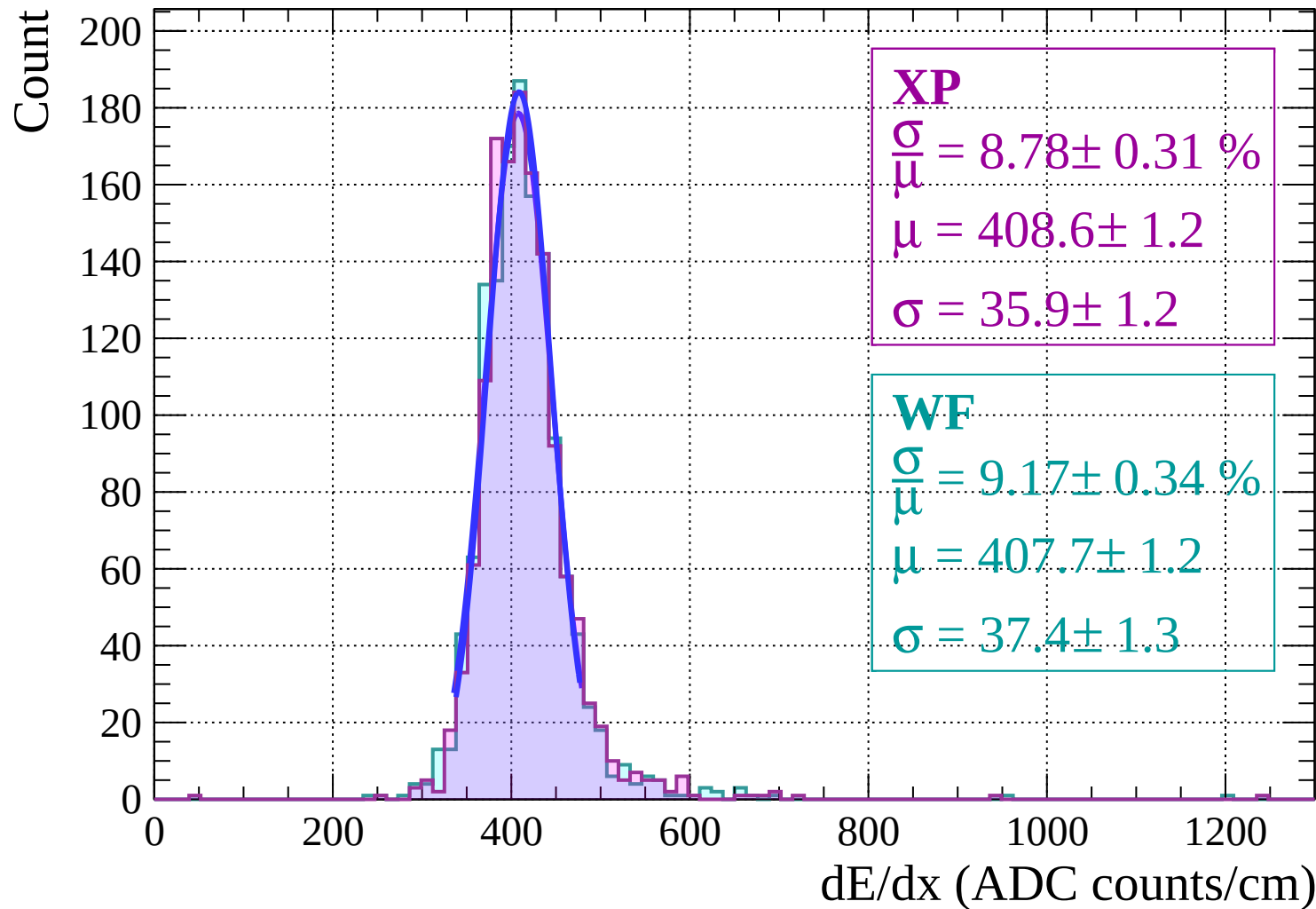
$$\frac{\sigma}{\mu} = 8.96 \pm 0.33 \%$$

$$\mu = 407.4 \pm 1.4$$

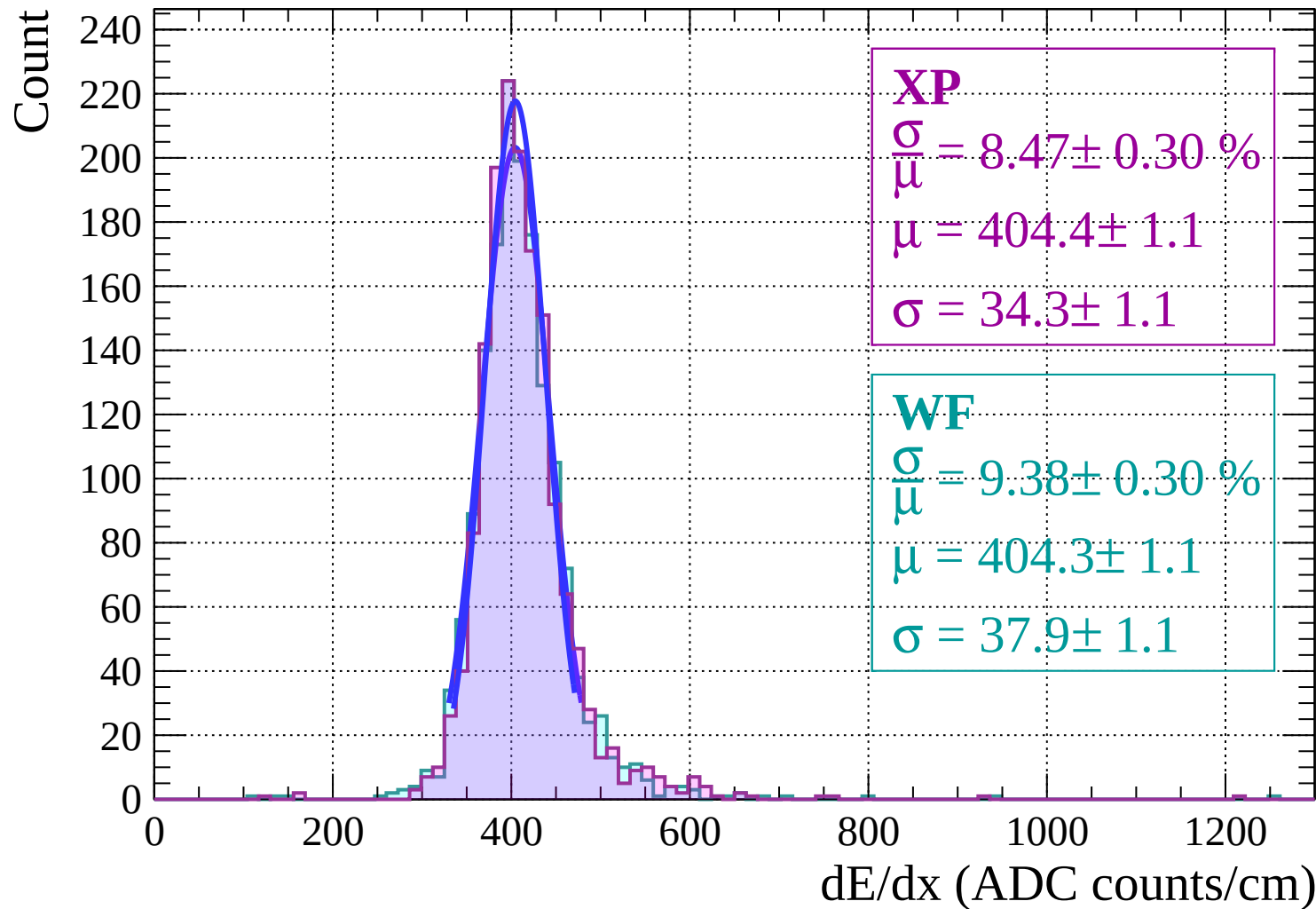
$$\sigma = 36.5 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-26 < \theta < -24$

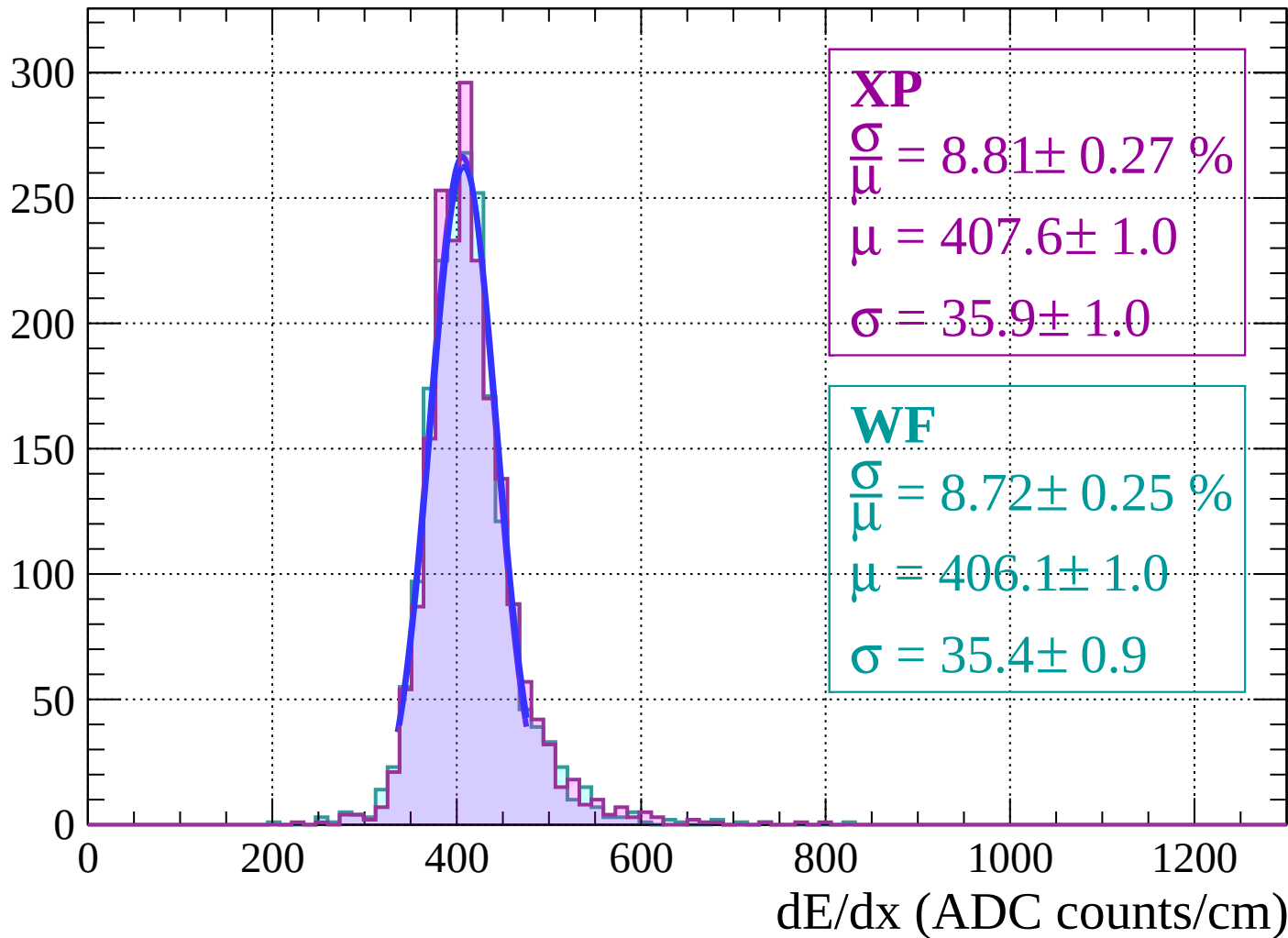


Energy loss | $-24 < \theta < -22$



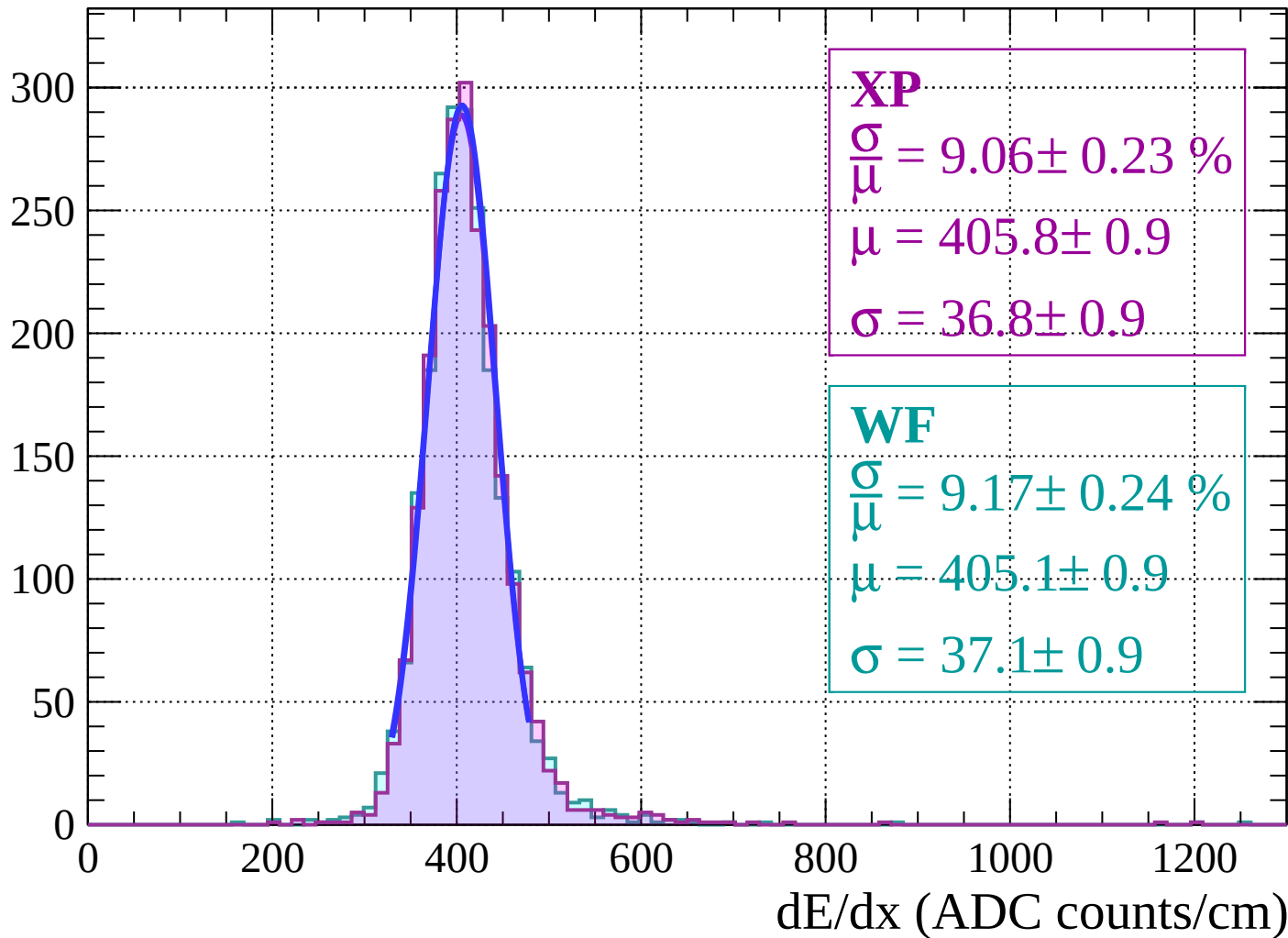
Energy loss | $-22 < \theta < -20$

Count



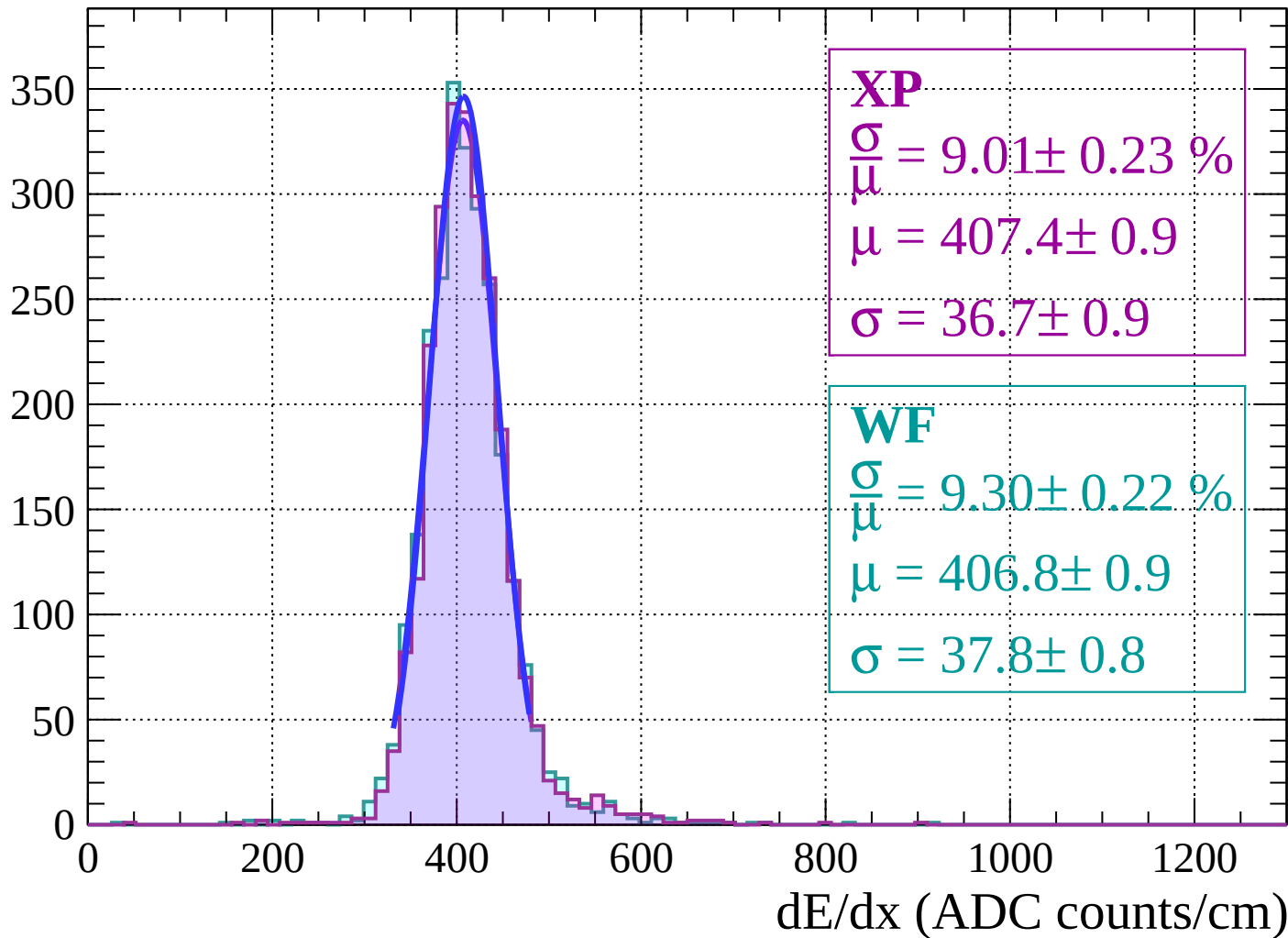
Energy loss | $-20 < \theta < -18$

Count



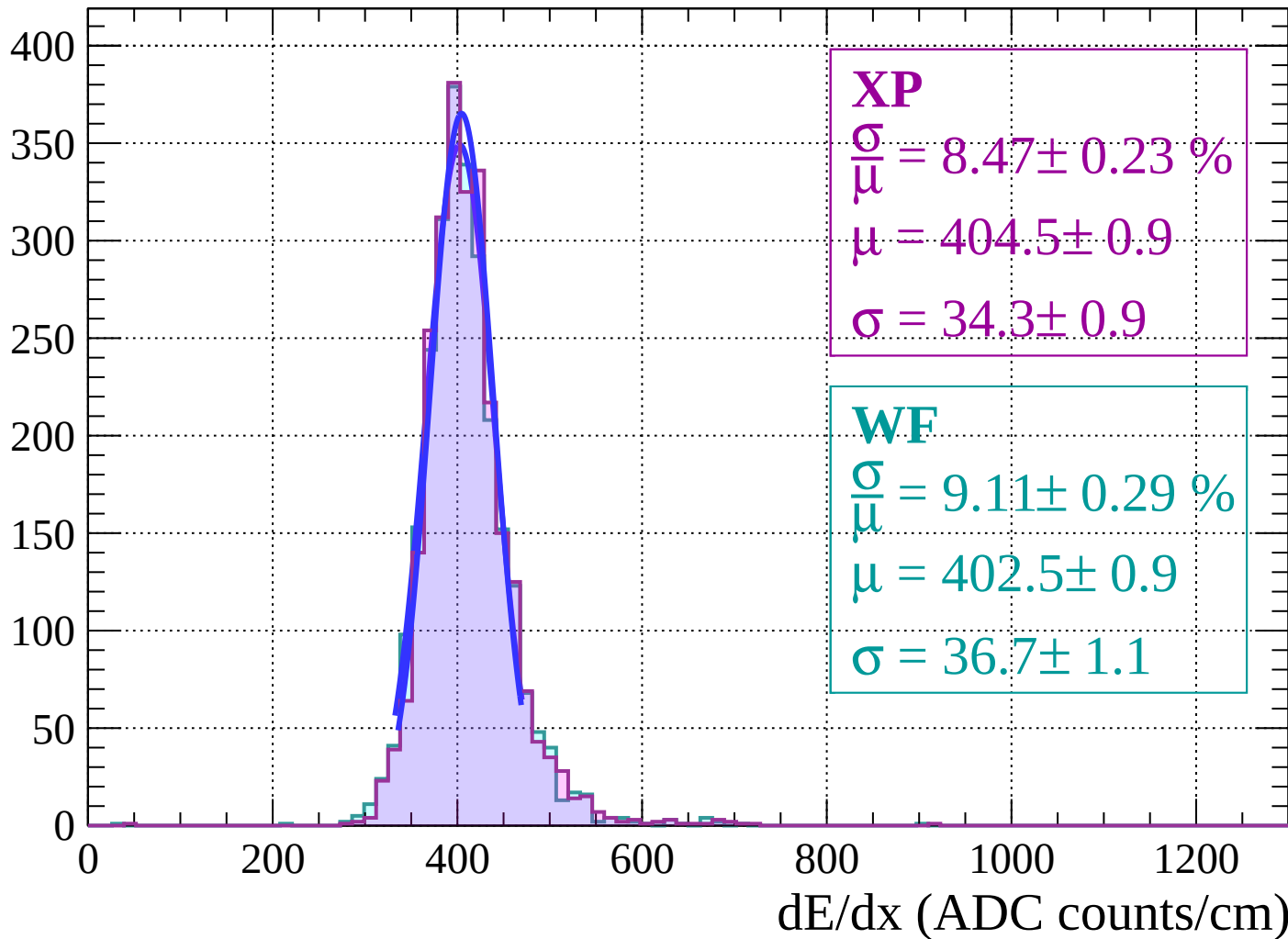
Energy loss $|-18 < \theta < -16$

Count

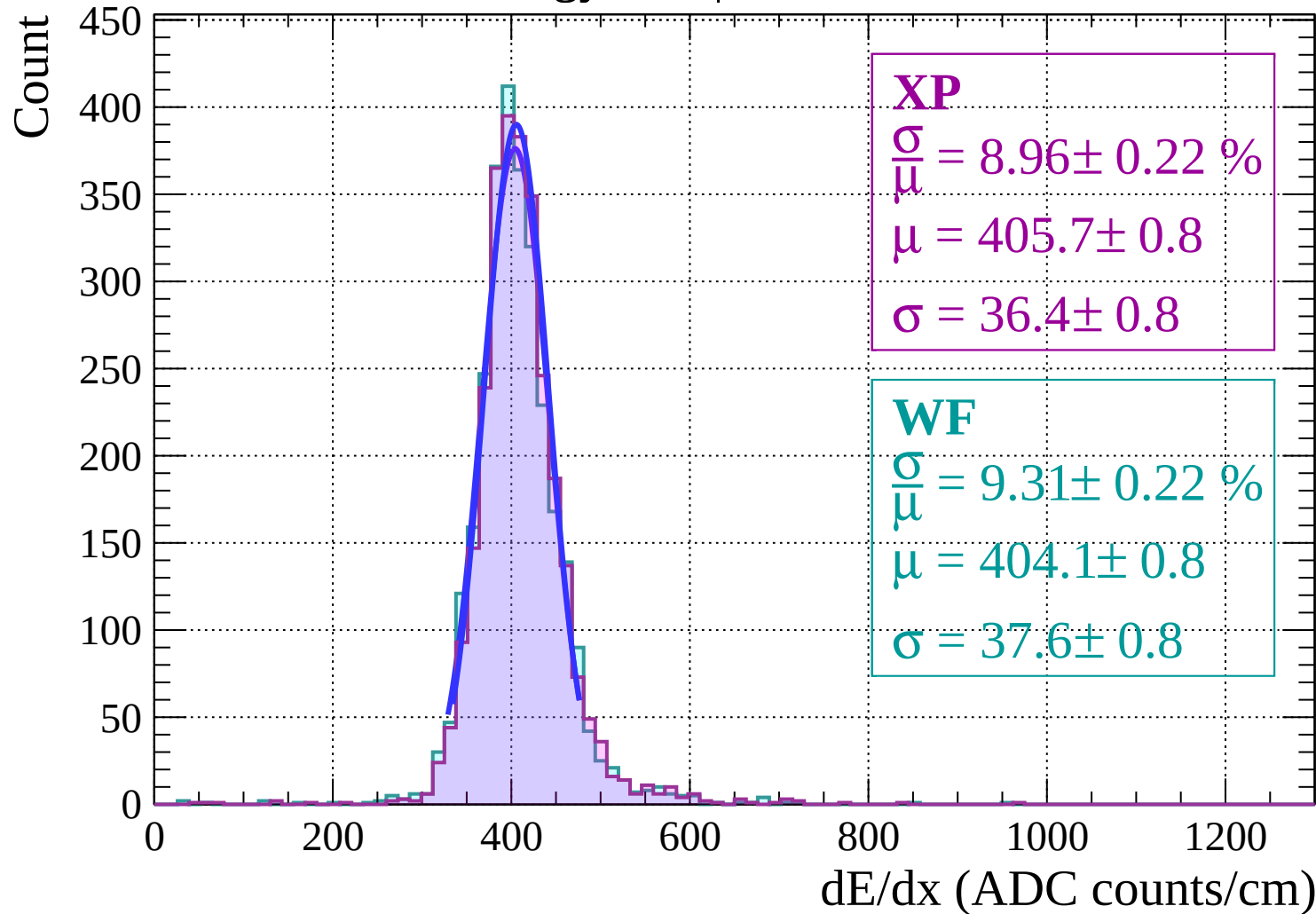


Energy loss | $-16 < \theta < -14$

Count

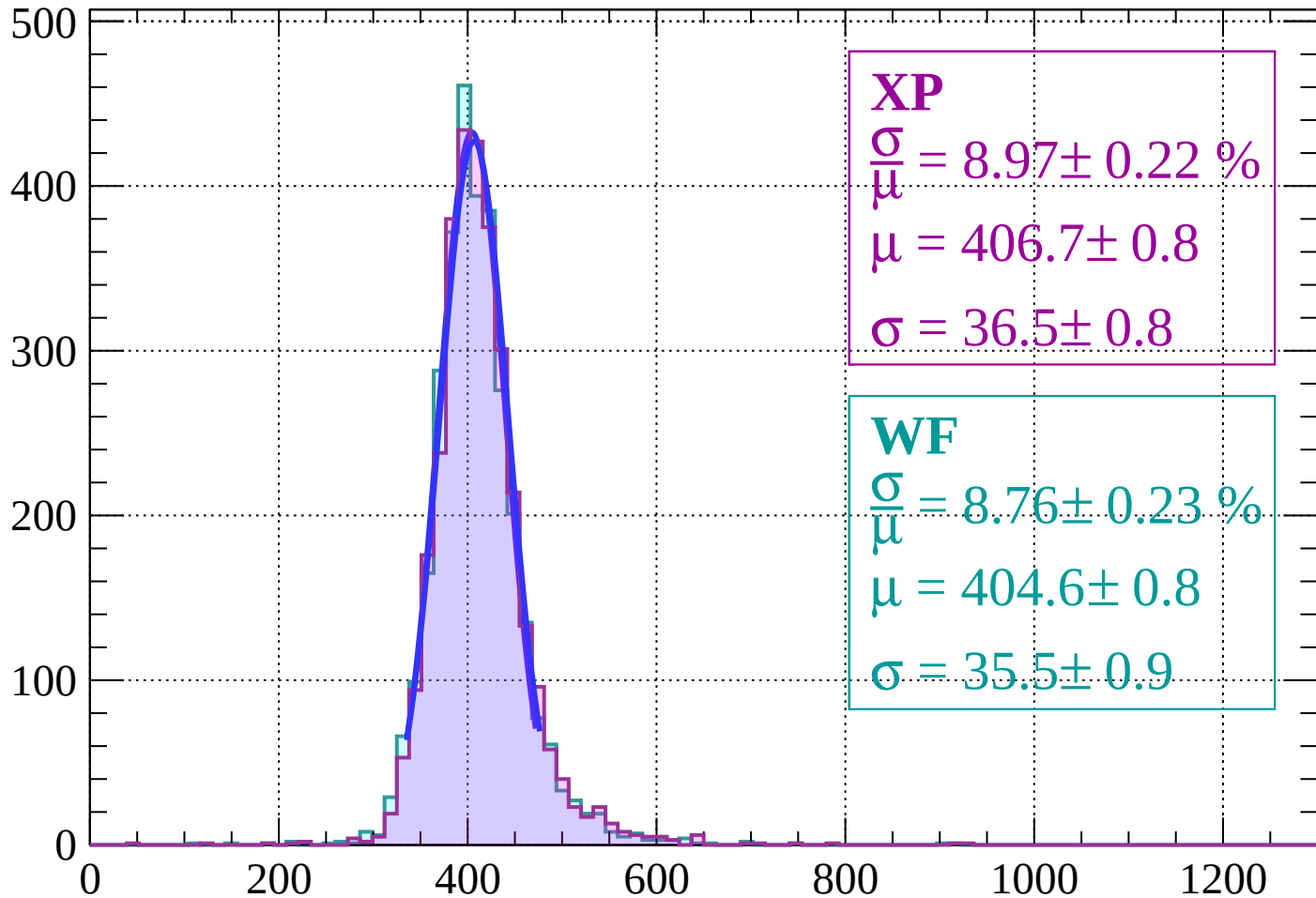


Energy loss | $-14 < \theta < -12$



Energy loss | $-12 < \theta < -10$

Count



XP

$$\frac{\sigma}{\mu} = 8.97 \pm 0.22 \%$$

$$\mu = 406.7 \pm 0.8$$

$$\sigma = 36.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.76 \pm 0.23 \%$$

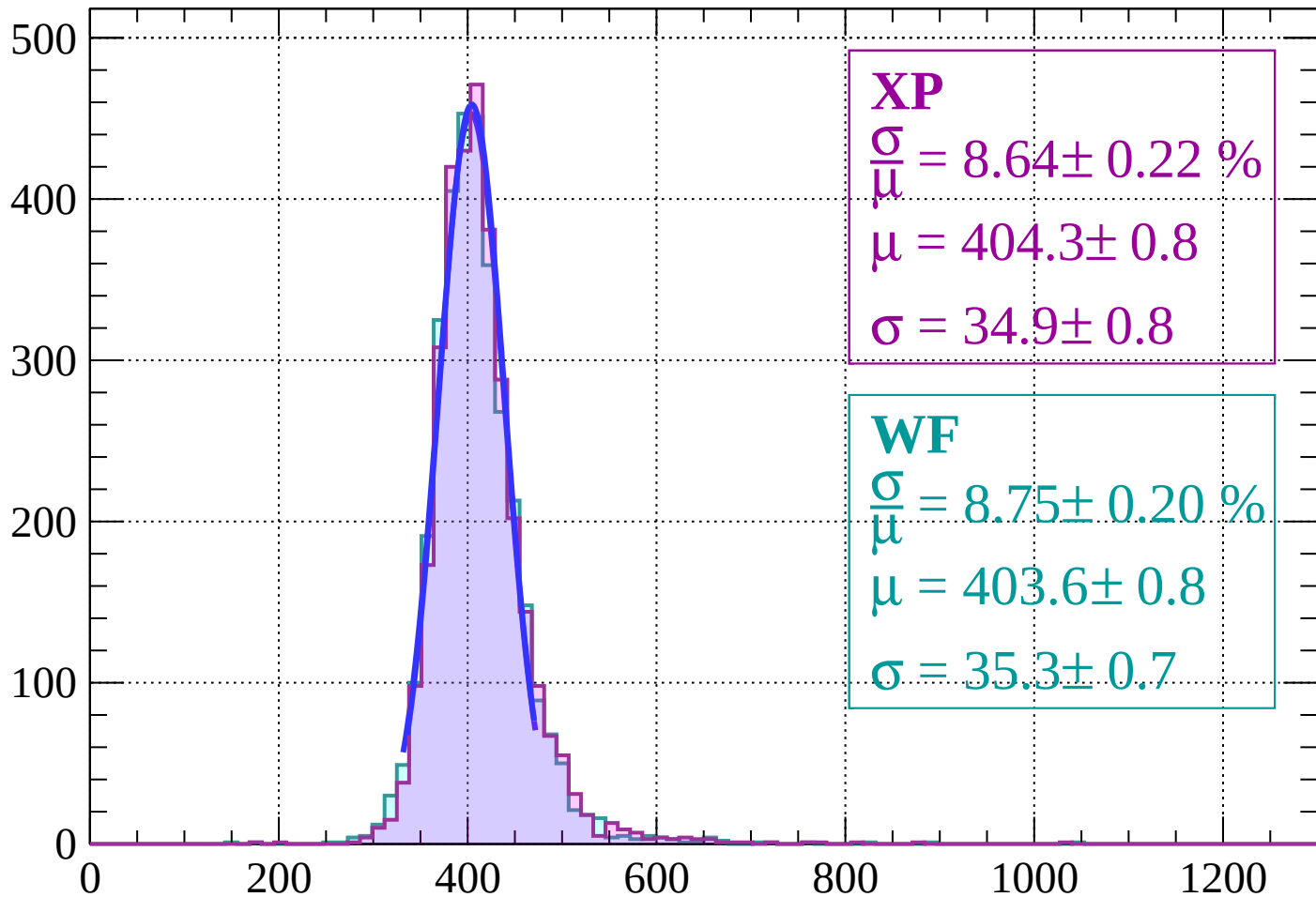
$$\mu = 404.6 \pm 0.8$$

$$\sigma = 35.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-10 < \theta < -8$

Count



XP

$$\frac{\sigma}{\mu} = 8.64 \pm 0.22 \%$$

$$\mu = 404.3 \pm 0.8$$

$$\sigma = 34.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.75 \pm 0.20 \%$$

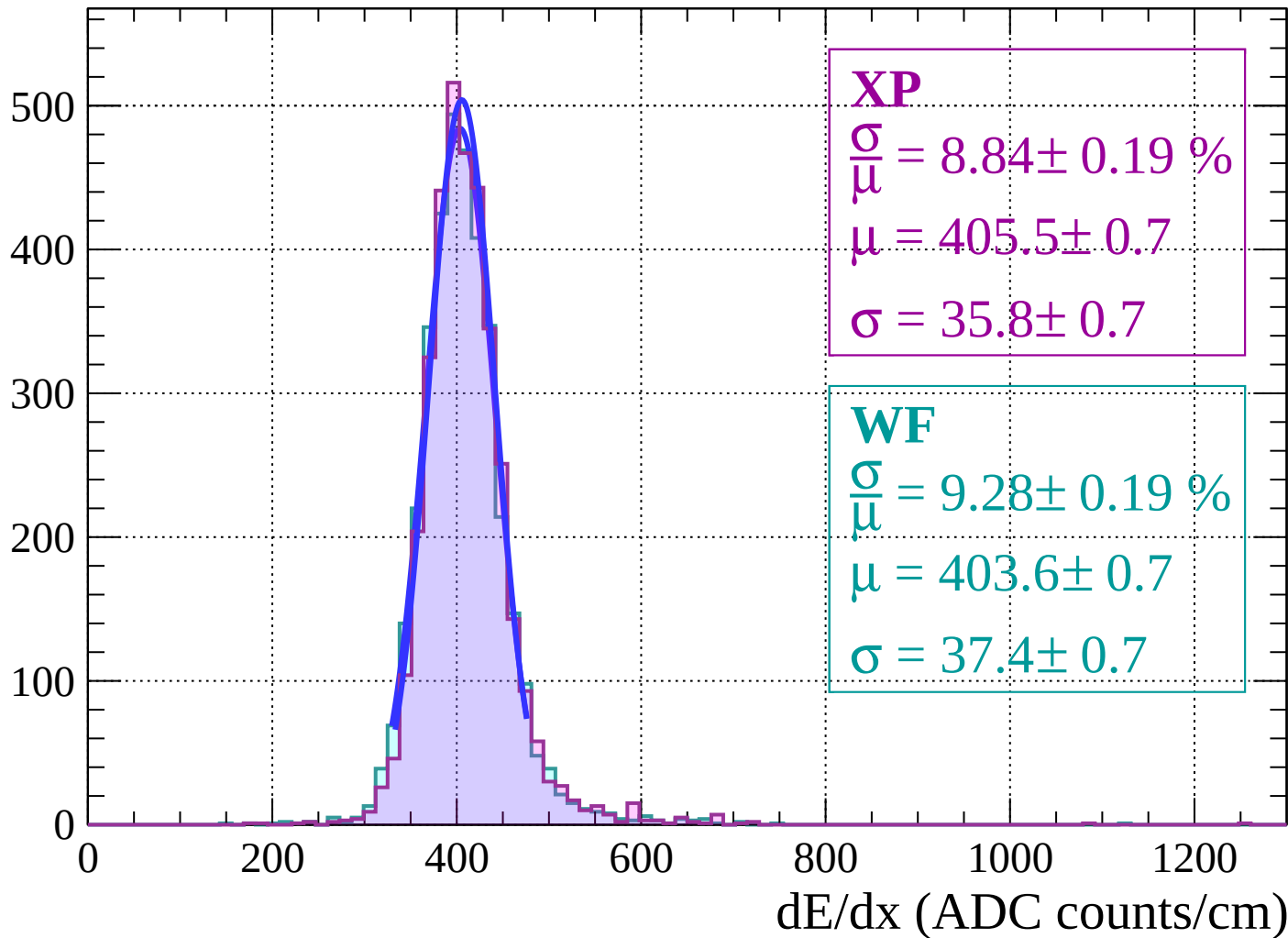
$$\mu = 403.6 \pm 0.8$$

$$\sigma = 35.3 \pm 0.7$$

dE/dx (ADC counts/cm)

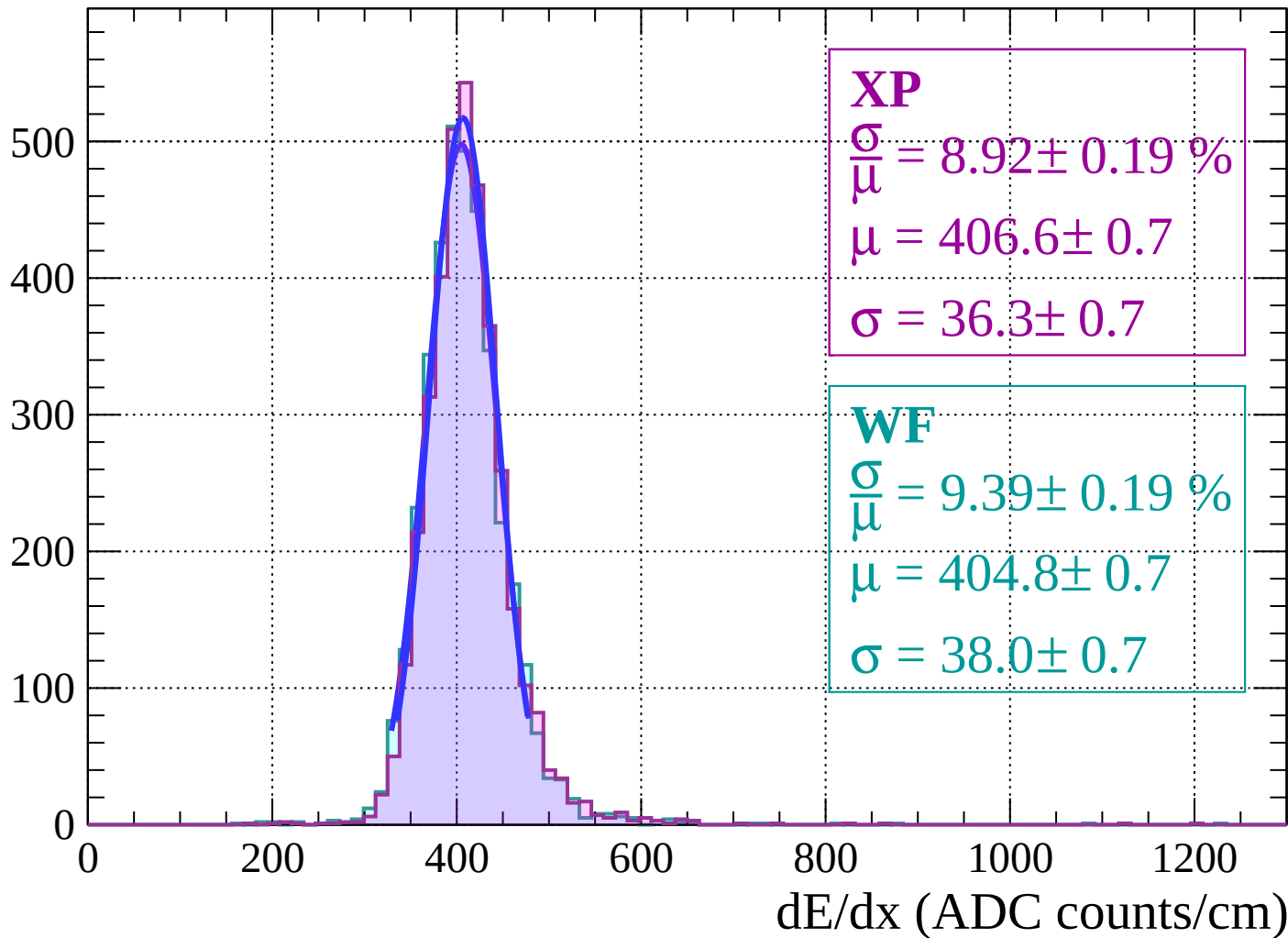
Energy loss | $-8 < \theta < -6$

Count



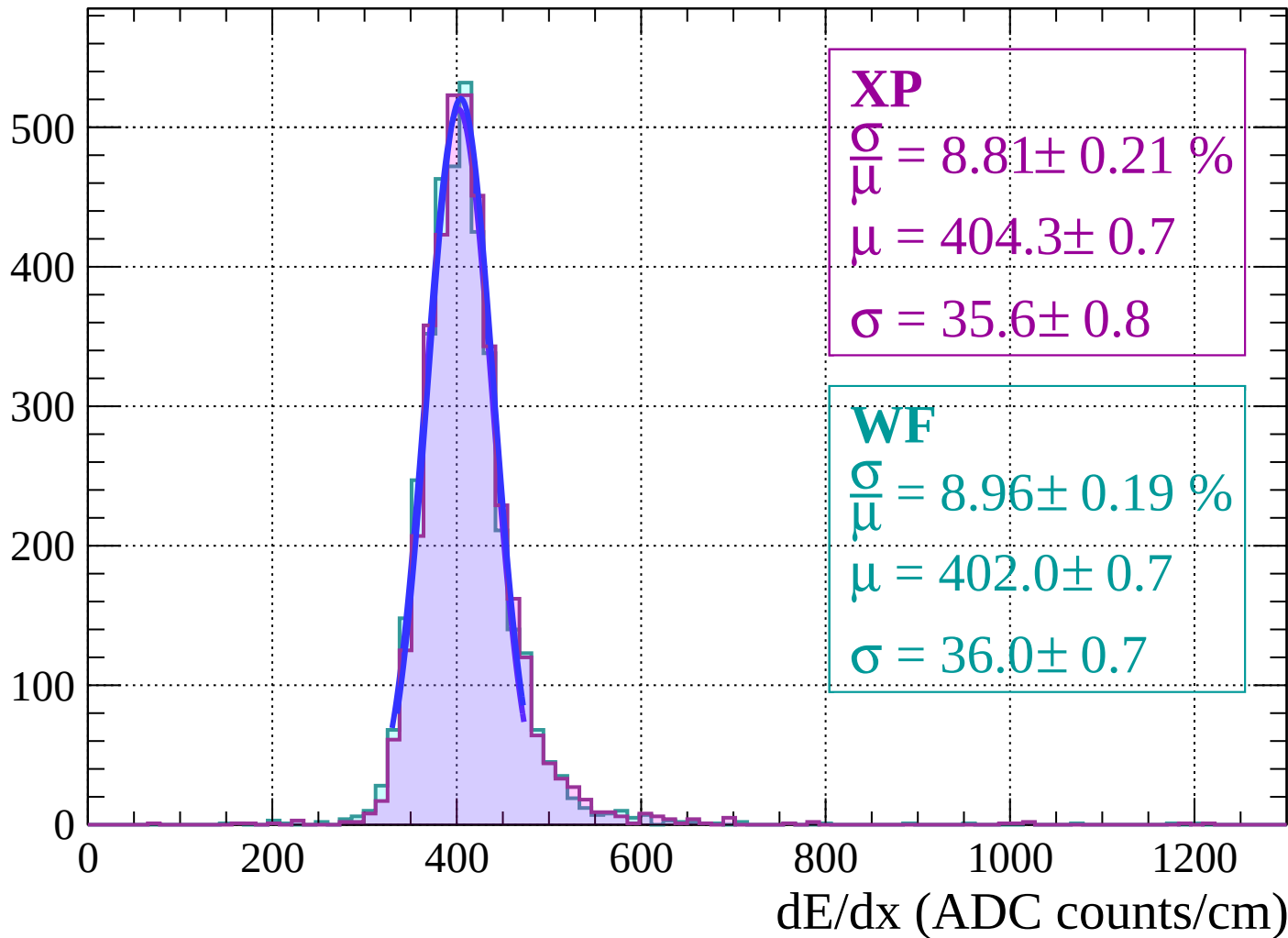
Energy loss | $-6 < \theta < -4$

Count



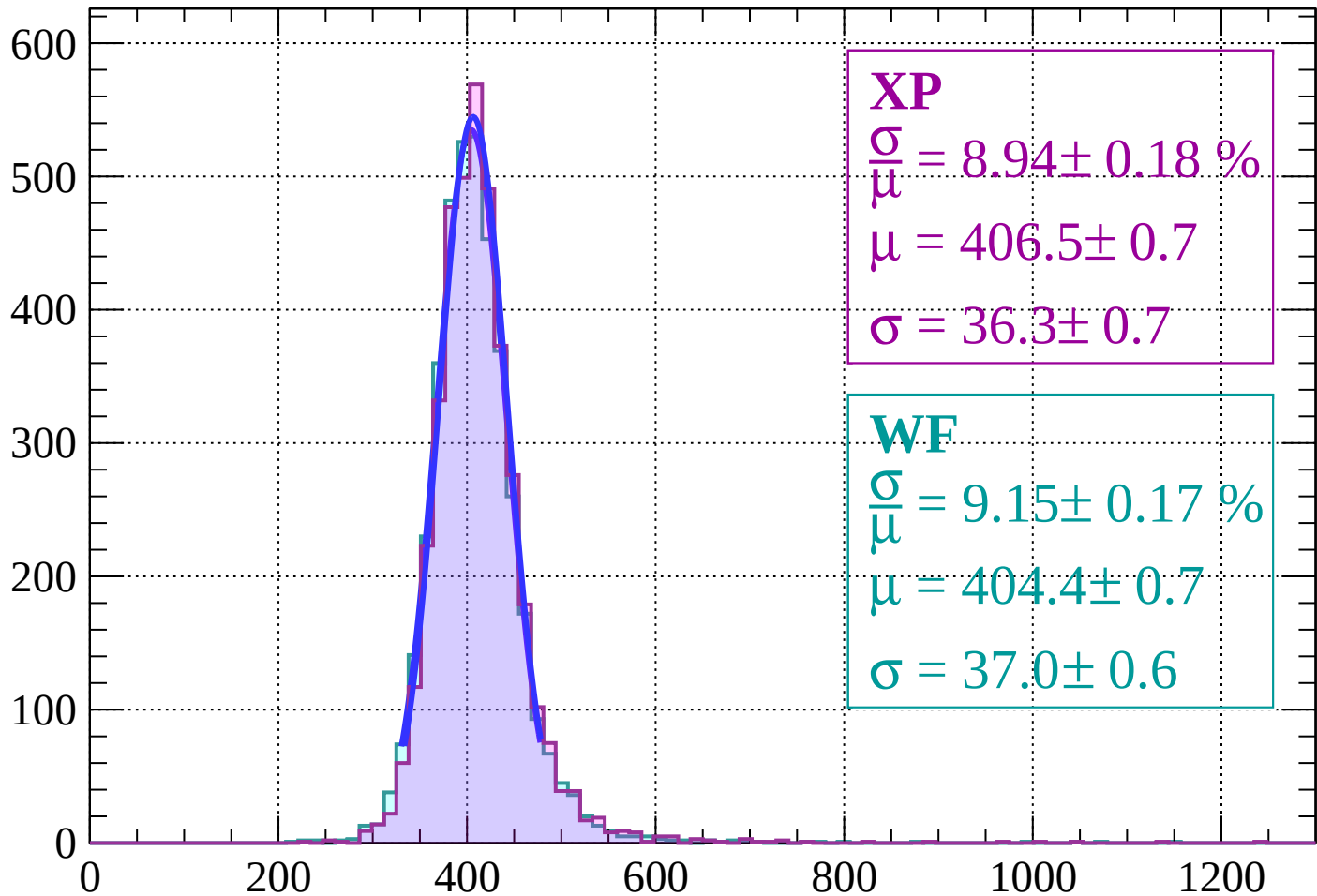
Energy loss | $-4 < \theta < -2$

Count



Energy loss | $-2 < \theta < 0$

Count



XP

$$\frac{\sigma}{\mu} = 8.94 \pm 0.18 \%$$

$$\mu = 406.5 \pm 0.7$$

$$\sigma = 36.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.15 \pm 0.17 \%$$

$$\mu = 404.4 \pm 0.7$$

$$\sigma = 37.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $0 < \theta < 2$

Count

1200

1000

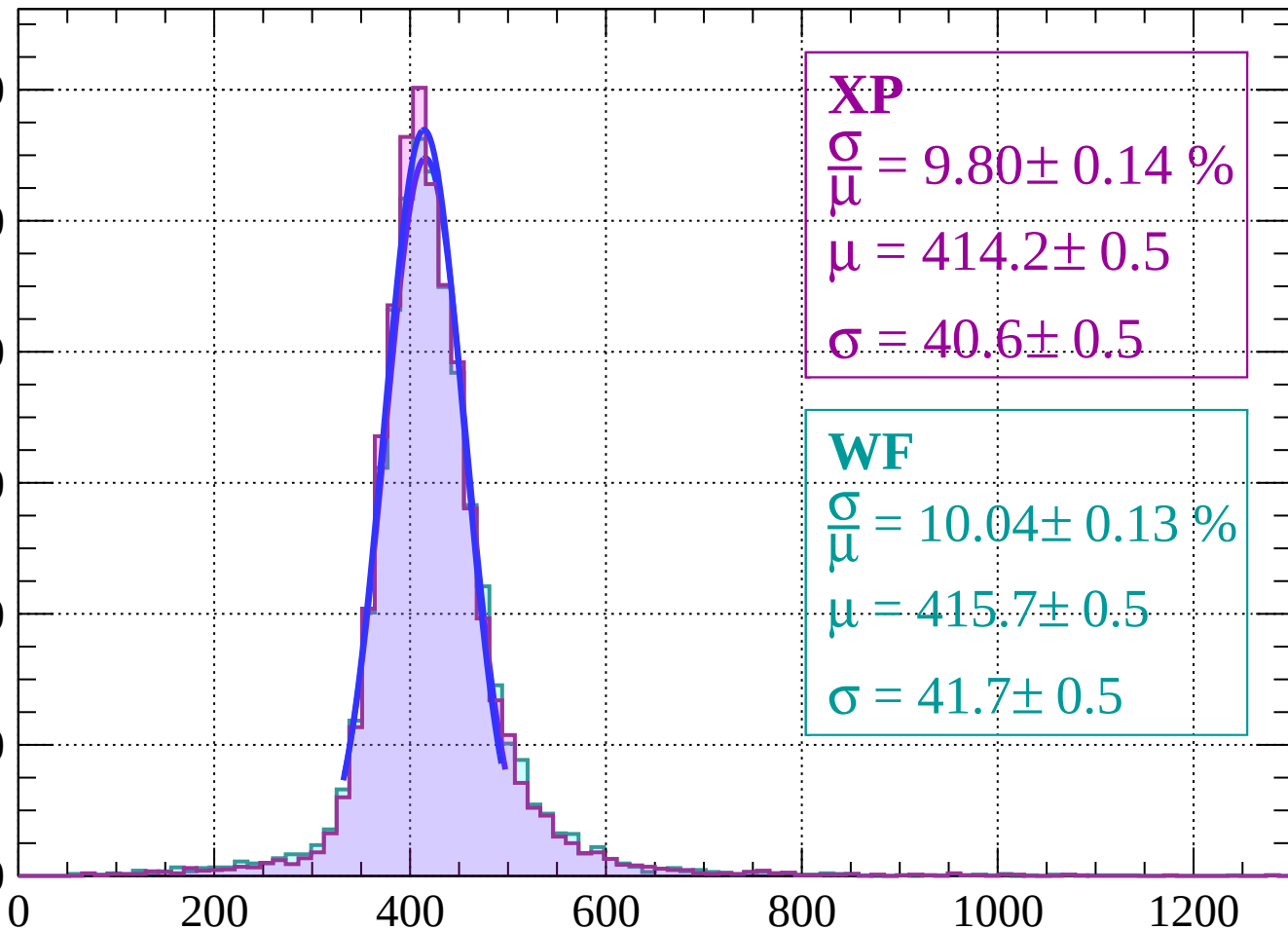
800

600

400

200

0



XP

$$\frac{\sigma}{\bar{\mu}} = 9.80 \pm 0.14 \%$$

$$\mu = 414.2 \pm 0.5$$

$$\sigma = 40.6 \pm 0.5$$

WF

$$\frac{\sigma}{\bar{\mu}} = 10.04 \pm 0.13 \%$$

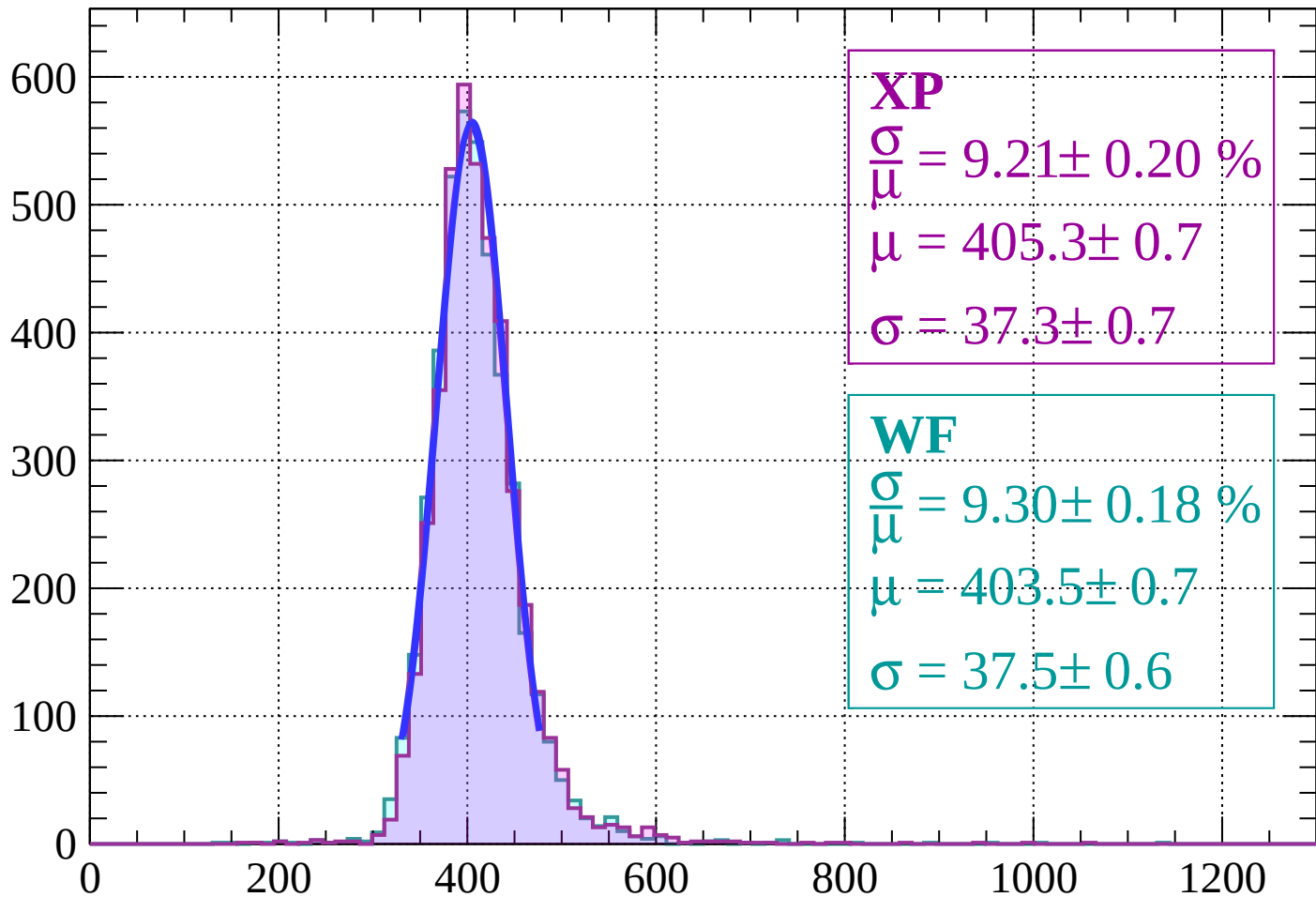
$$\mu = 415.7 \pm 0.5$$

$$\sigma = 41.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $2 < \theta < 4$

Count



XP

$$\frac{\sigma}{\mu} = 9.21 \pm 0.20 \%$$

$$\mu = 405.3 \pm 0.7$$

$$\sigma = 37.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.30 \pm 0.18 \%$$

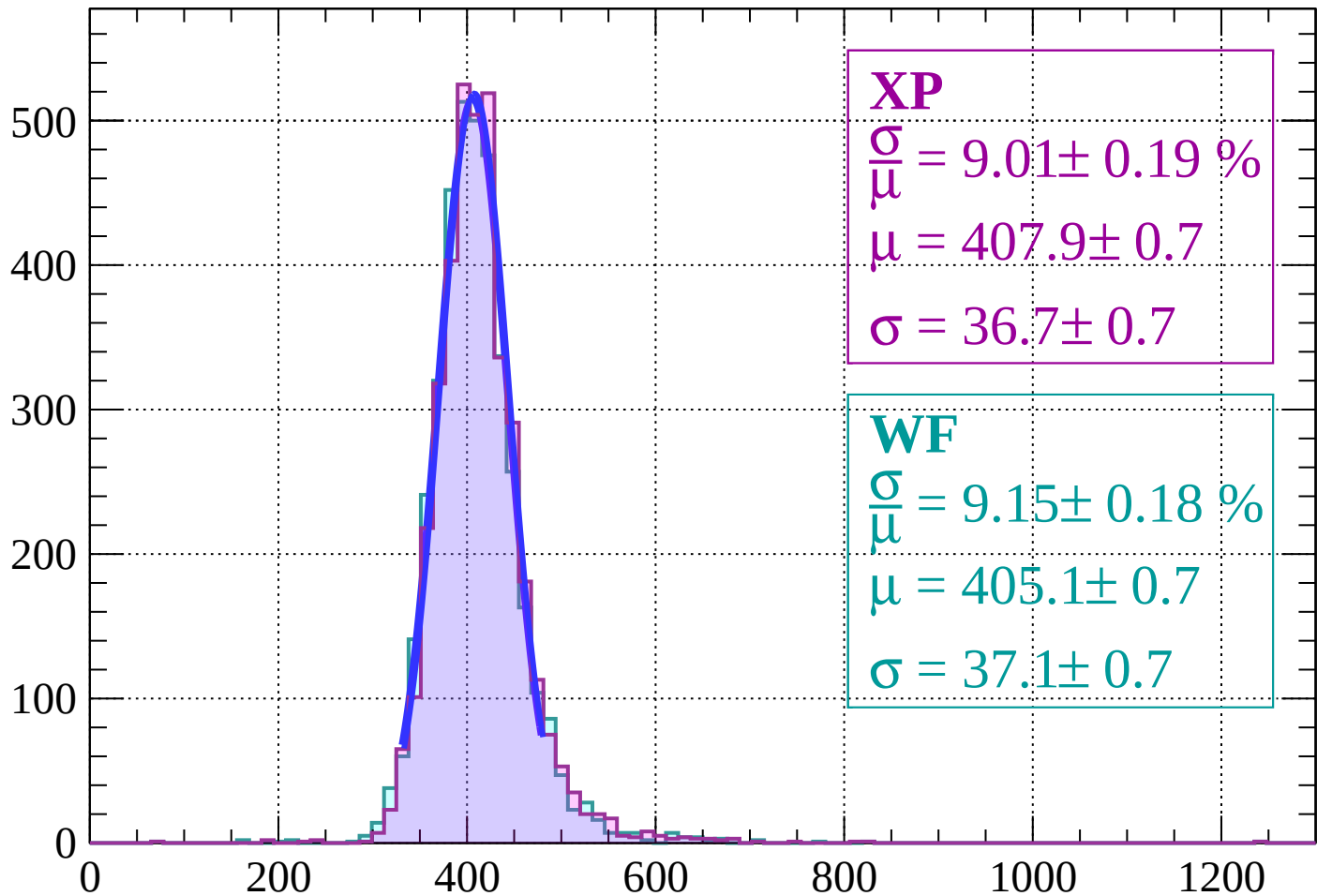
$$\mu = 403.5 \pm 0.7$$

$$\sigma = 37.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $4 < \theta < 6$

Count



XP

$$\frac{\sigma}{\mu} = 9.01 \pm 0.19 \%$$

$$\mu = 407.9 \pm 0.7$$

$$\sigma = 36.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.15 \pm 0.18 \%$$

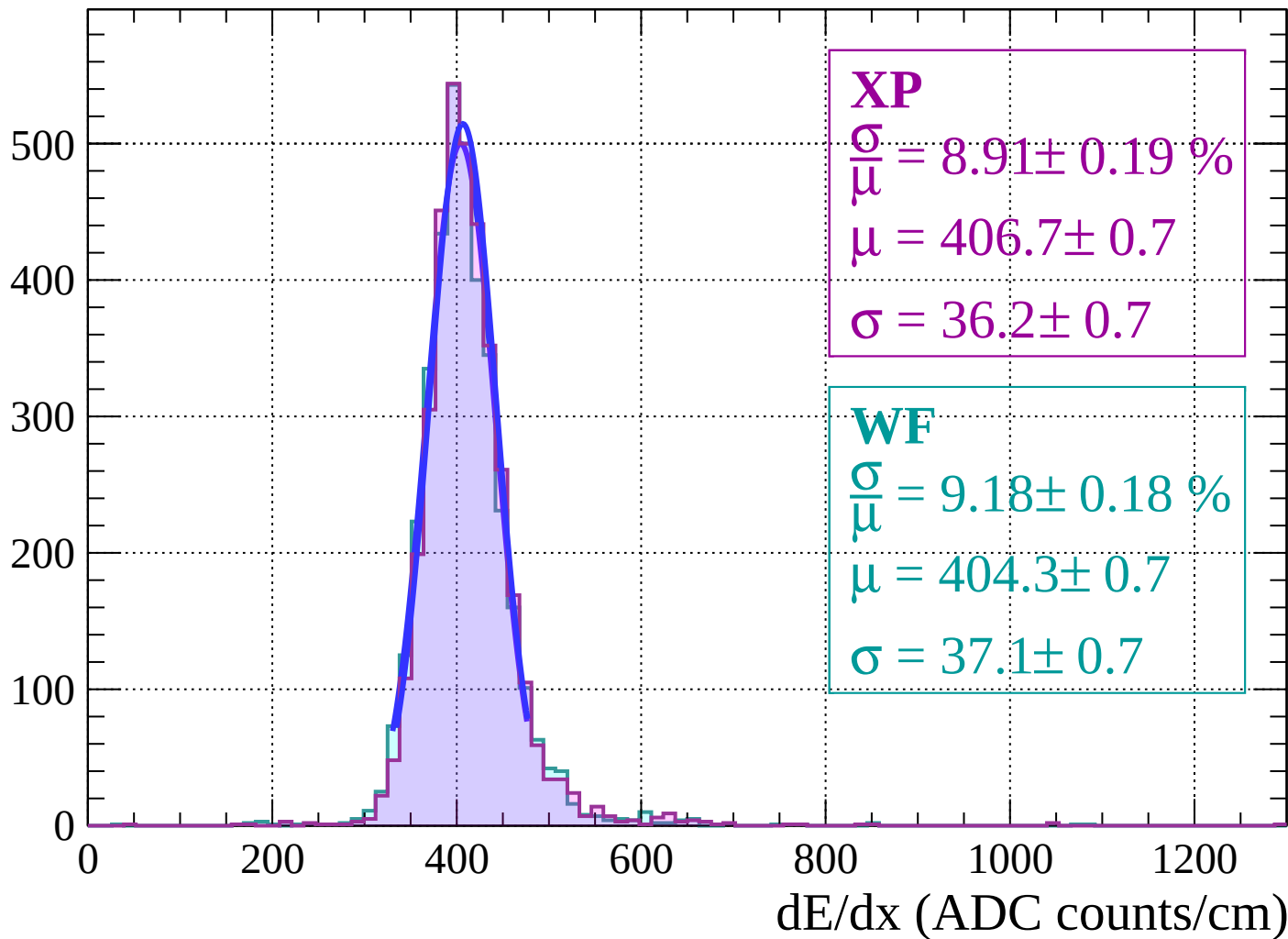
$$\mu = 405.1 \pm 0.7$$

$$\sigma = 37.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 8$

Count



Energy loss | $8 < \theta < 10$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 8.68 \pm 0.16 \%$$

$$\mu = 406.9 \pm 0.7$$

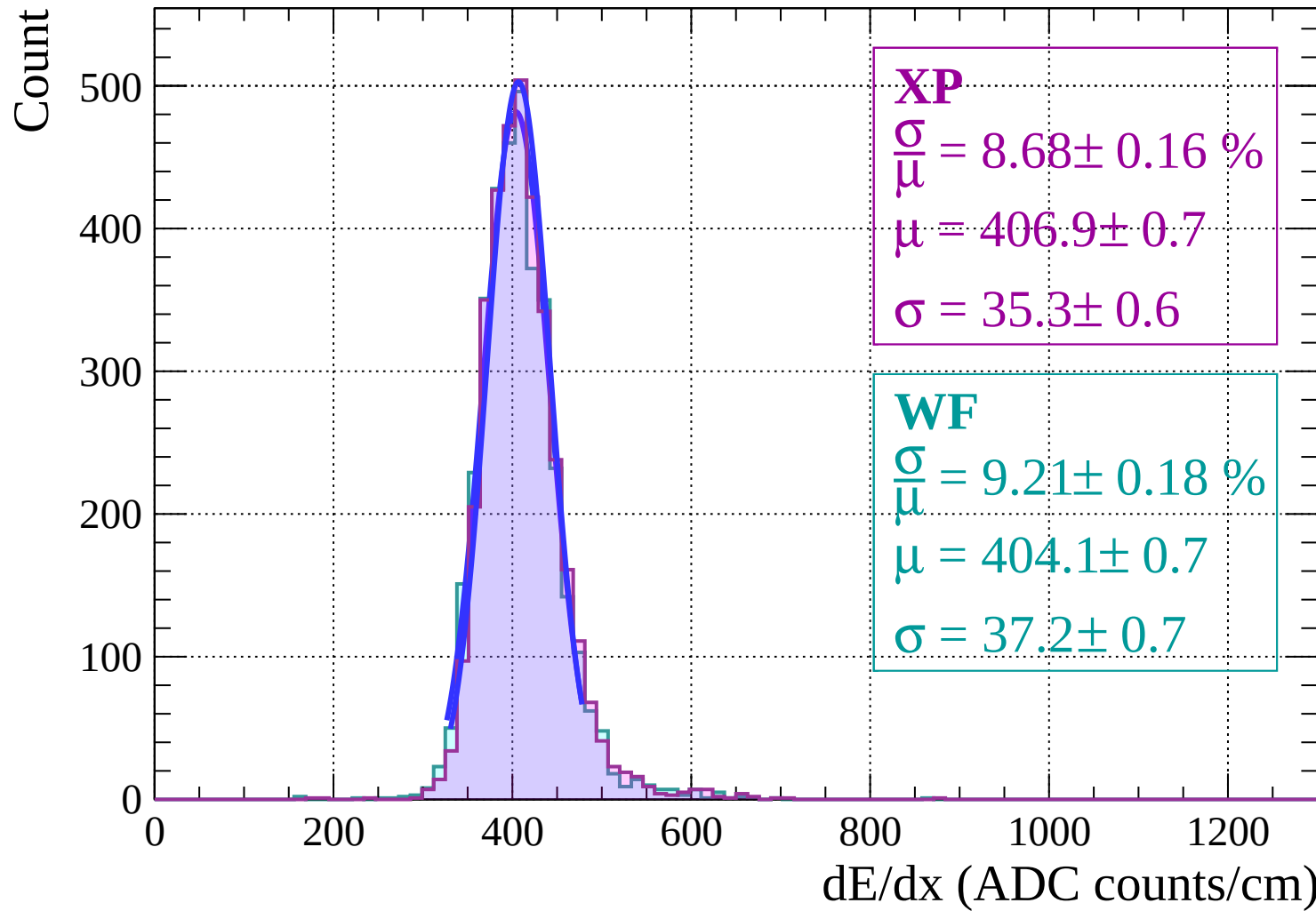
$$\sigma = 35.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.21 \pm 0.18 \%$$

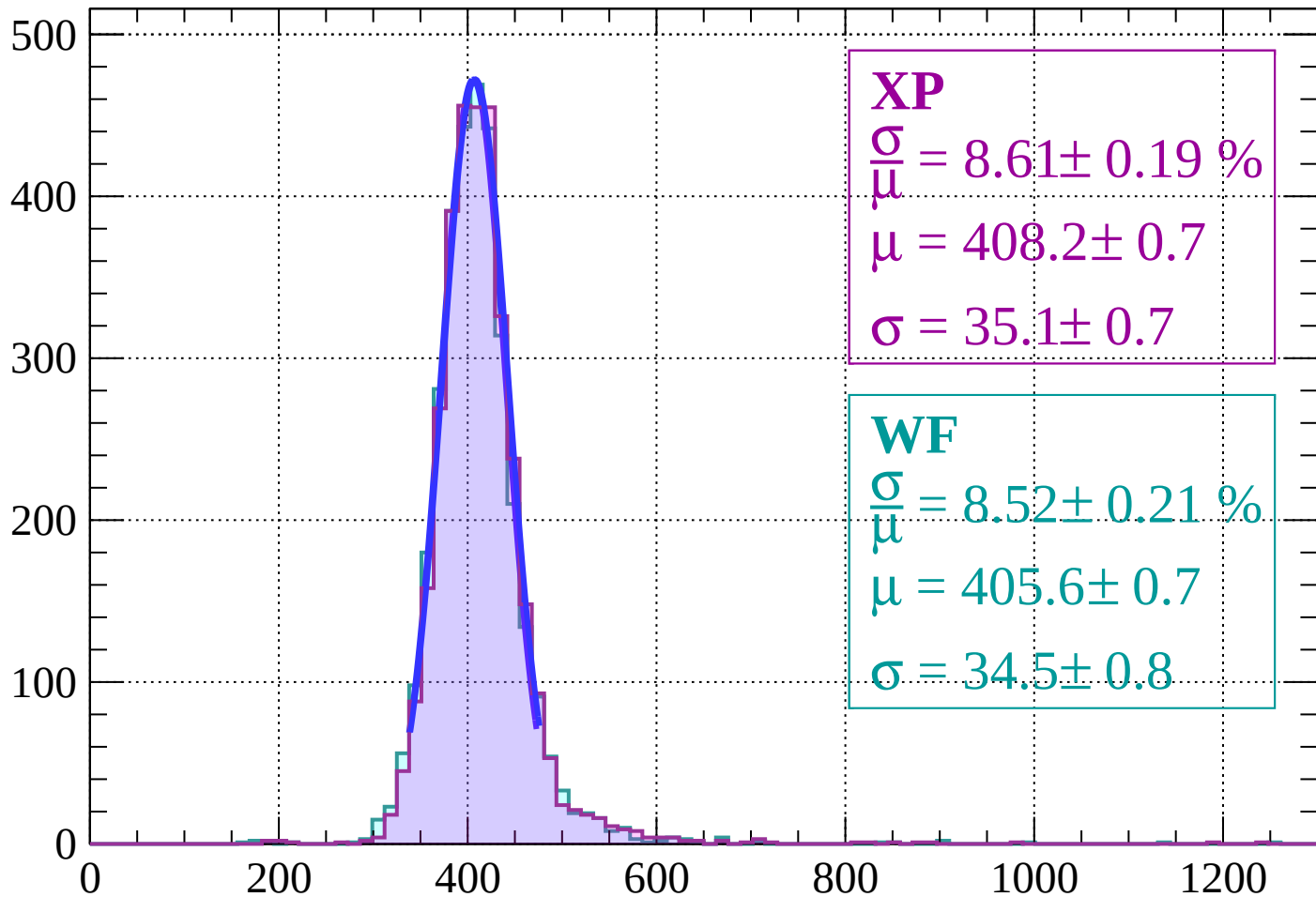
$$\mu = 404.1 \pm 0.7$$

$$\sigma = 37.2 \pm 0.7$$



Energy loss | $10 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 8.61 \pm 0.19 \%$$

$$\mu = 408.2 \pm 0.7$$

$$\sigma = 35.1 \pm 0.7$$

WF

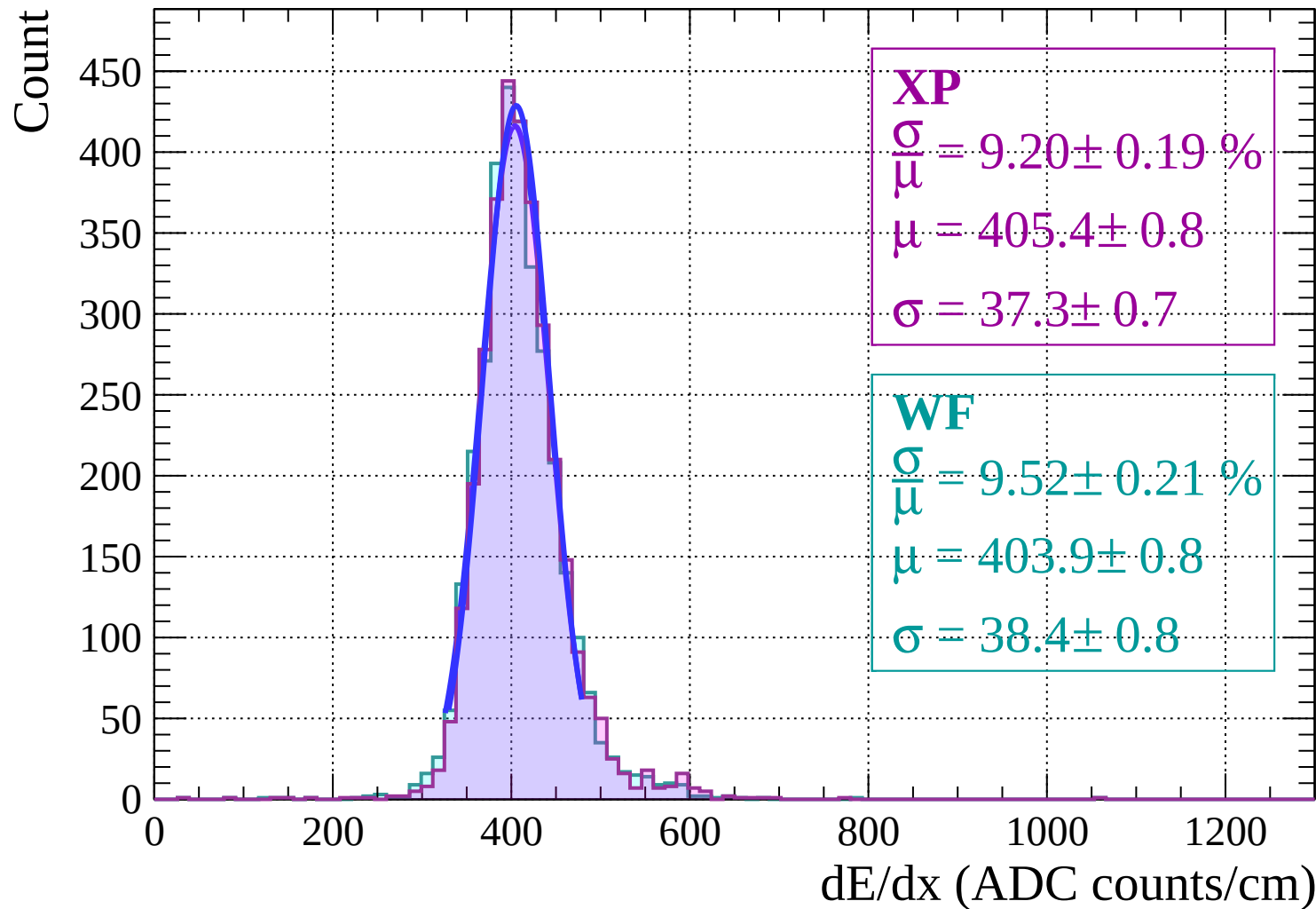
$$\frac{\sigma}{\mu} = 8.52 \pm 0.21 \%$$

$$\mu = 405.6 \pm 0.7$$

$$\sigma = 34.5 \pm 0.8$$

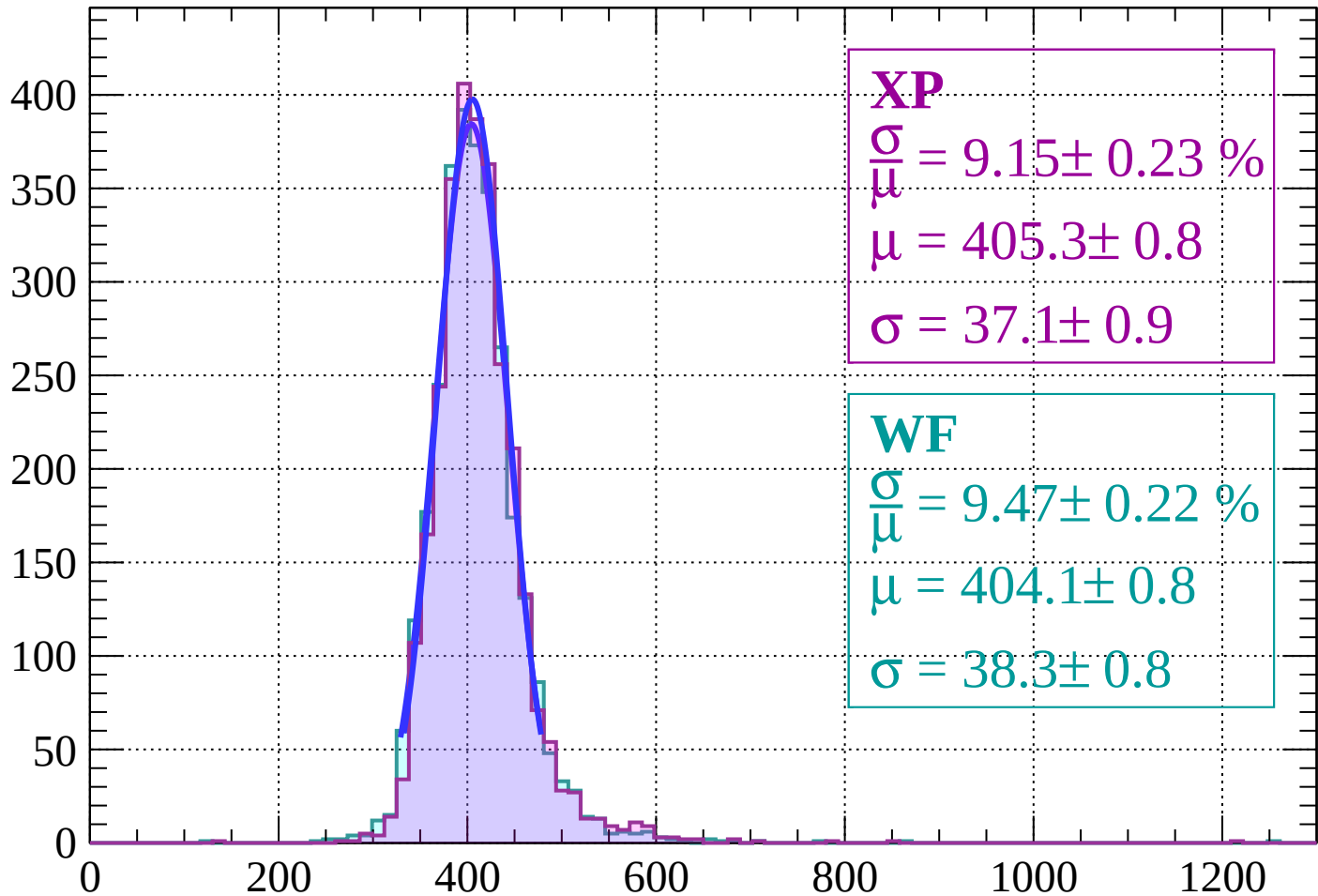
dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 14$



Energy loss | $14 < \theta < 16$

Count



XP

$$\frac{\sigma}{\mu} = 9.15 \pm 0.23 \%$$

$$\mu = 405.3 \pm 0.8$$

$$\sigma = 37.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.47 \pm 0.22 \%$$

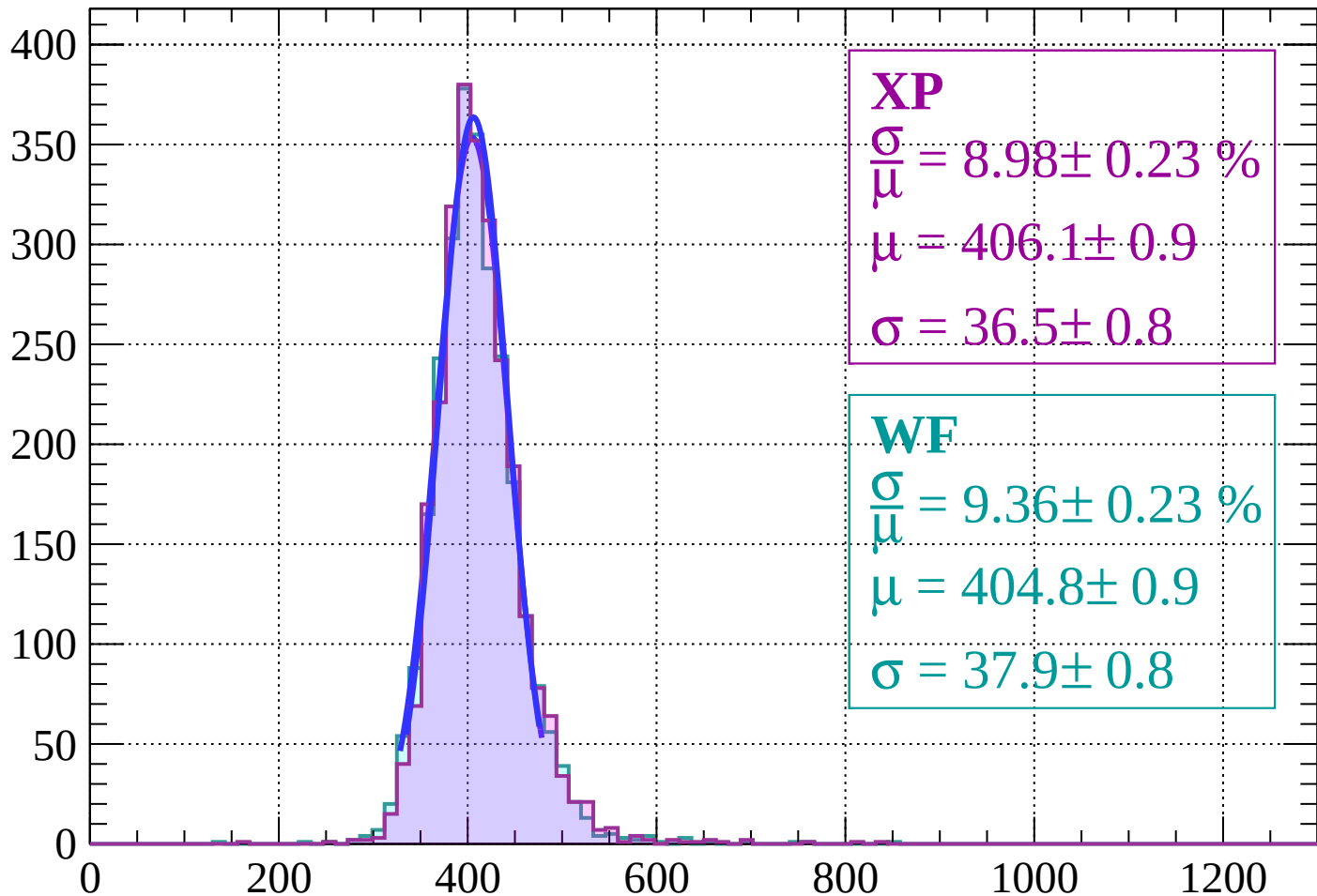
$$\mu = 404.1 \pm 0.8$$

$$\sigma = 38.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $16 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 8.98 \pm 0.23 \%$$

$$\mu = 406.1 \pm 0.9$$

$$\sigma = 36.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.36 \pm 0.23 \%$$

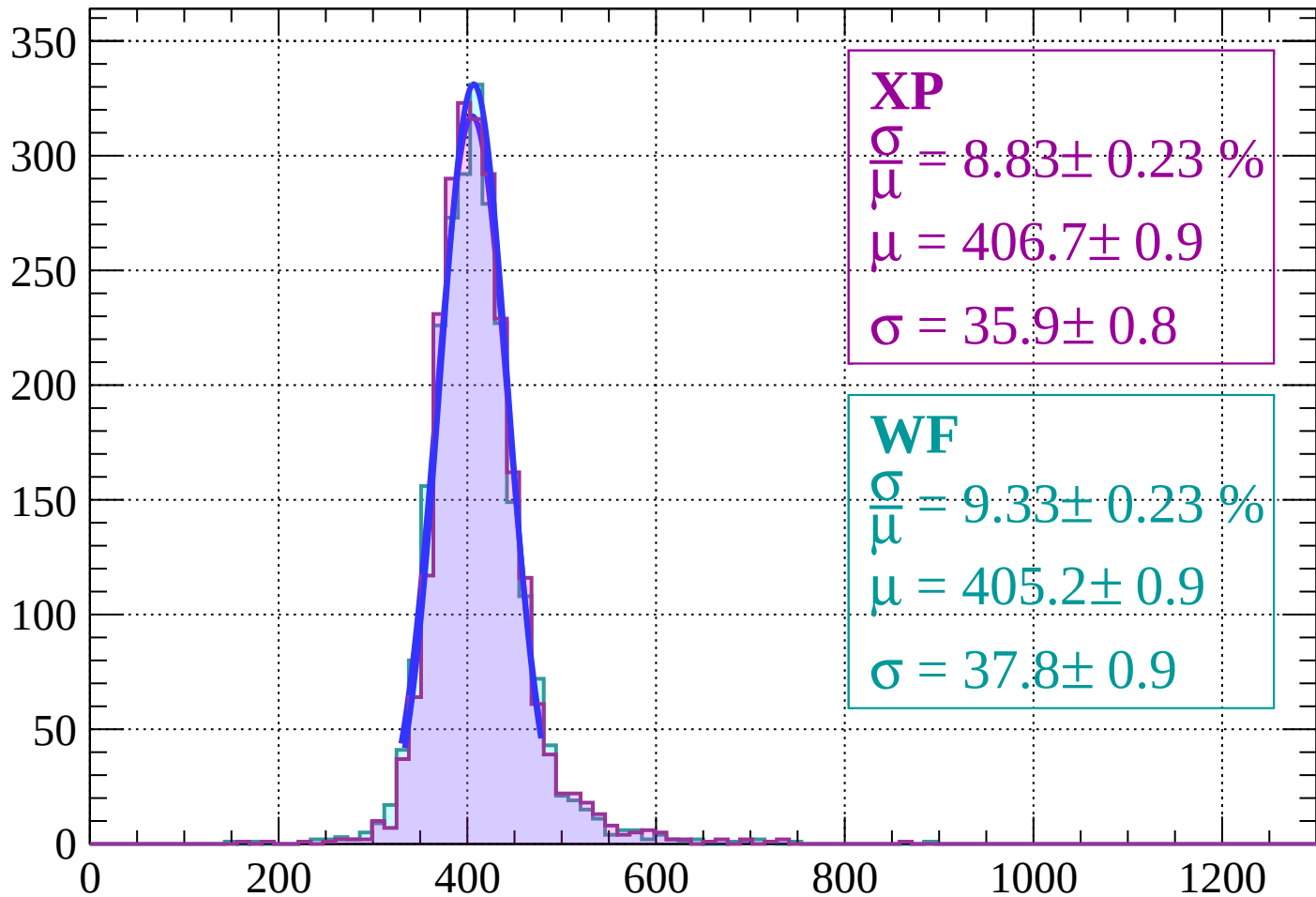
$$\mu = 404.8 \pm 0.9$$

$$\sigma = 37.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 20$

Count



XP

$$\frac{\sigma}{\mu} = 8.83 \pm 0.23 \%$$

$$\mu = 406.7 \pm 0.9$$

$$\sigma = 35.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.33 \pm 0.23 \%$$

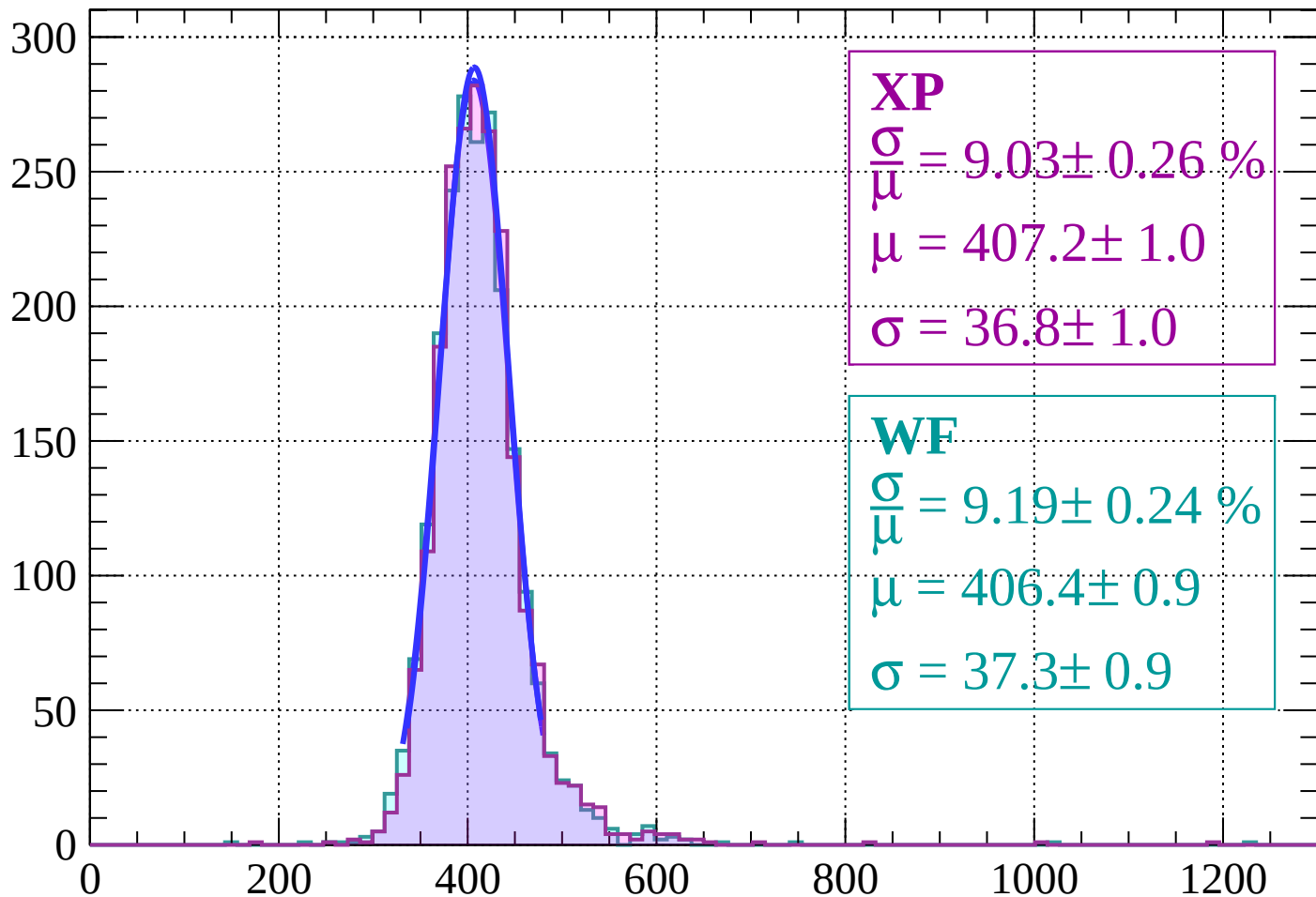
$$\mu = 405.2 \pm 0.9$$

$$\sigma = 37.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $20 < \theta < 22$

Count



XP

$$\frac{\sigma}{\mu} = 9.03 \pm 0.26 \%$$

$$\mu = 407.2 \pm 1.0$$

$$\sigma = 36.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.19 \pm 0.24 \%$$

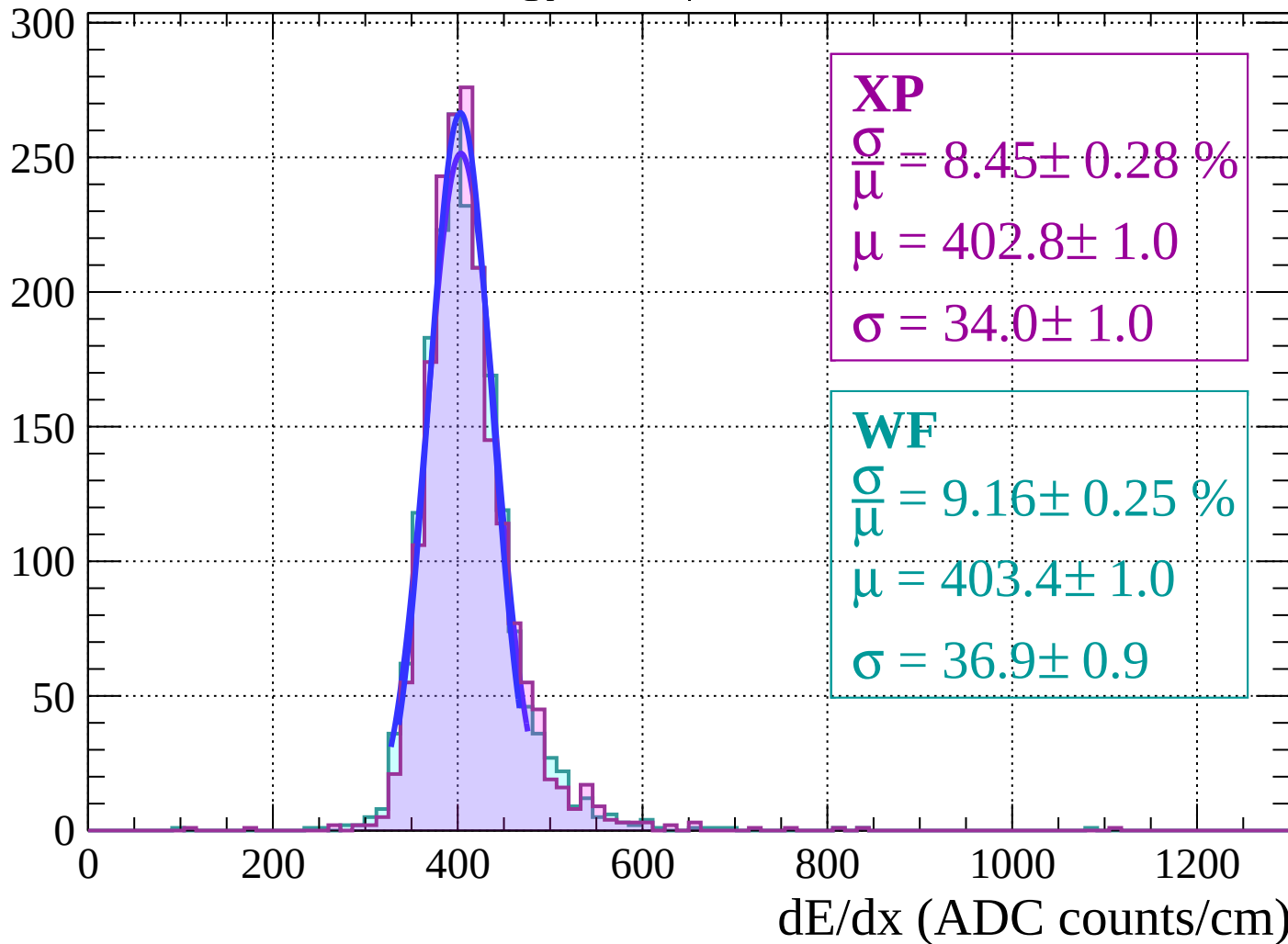
$$\mu = 406.4 \pm 0.9$$

$$\sigma = 37.3 \pm 0.9$$

dE/dx (ADC counts/cm)

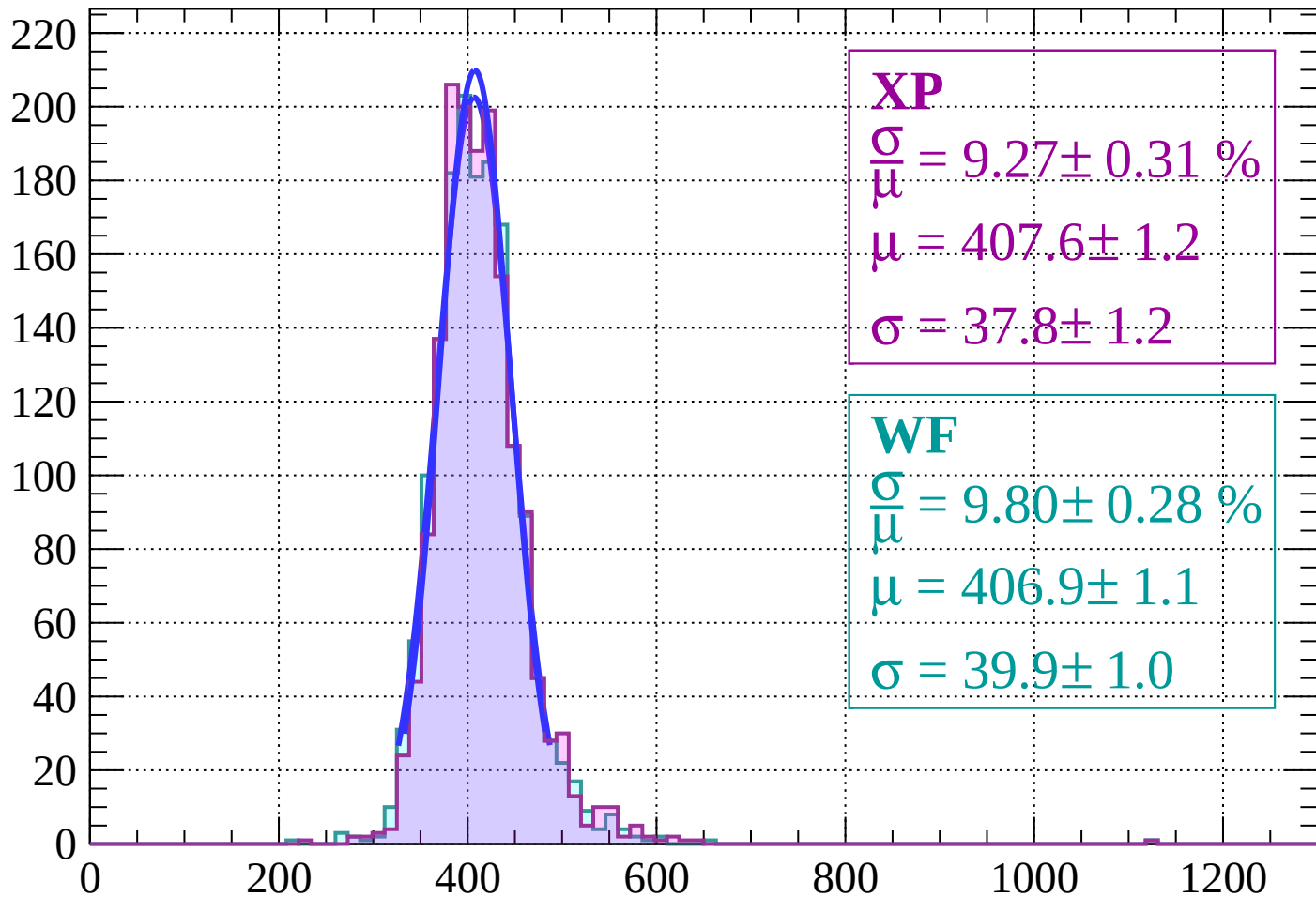
Energy loss | $22 < \theta < 24$

Count



Energy loss | $24 < \theta < 26$

Count



XP

$$\frac{\sigma}{\mu} = 9.27 \pm 0.31 \%$$

$$\mu = 407.6 \pm 1.2$$

$$\sigma = 37.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.80 \pm 0.28 \%$$

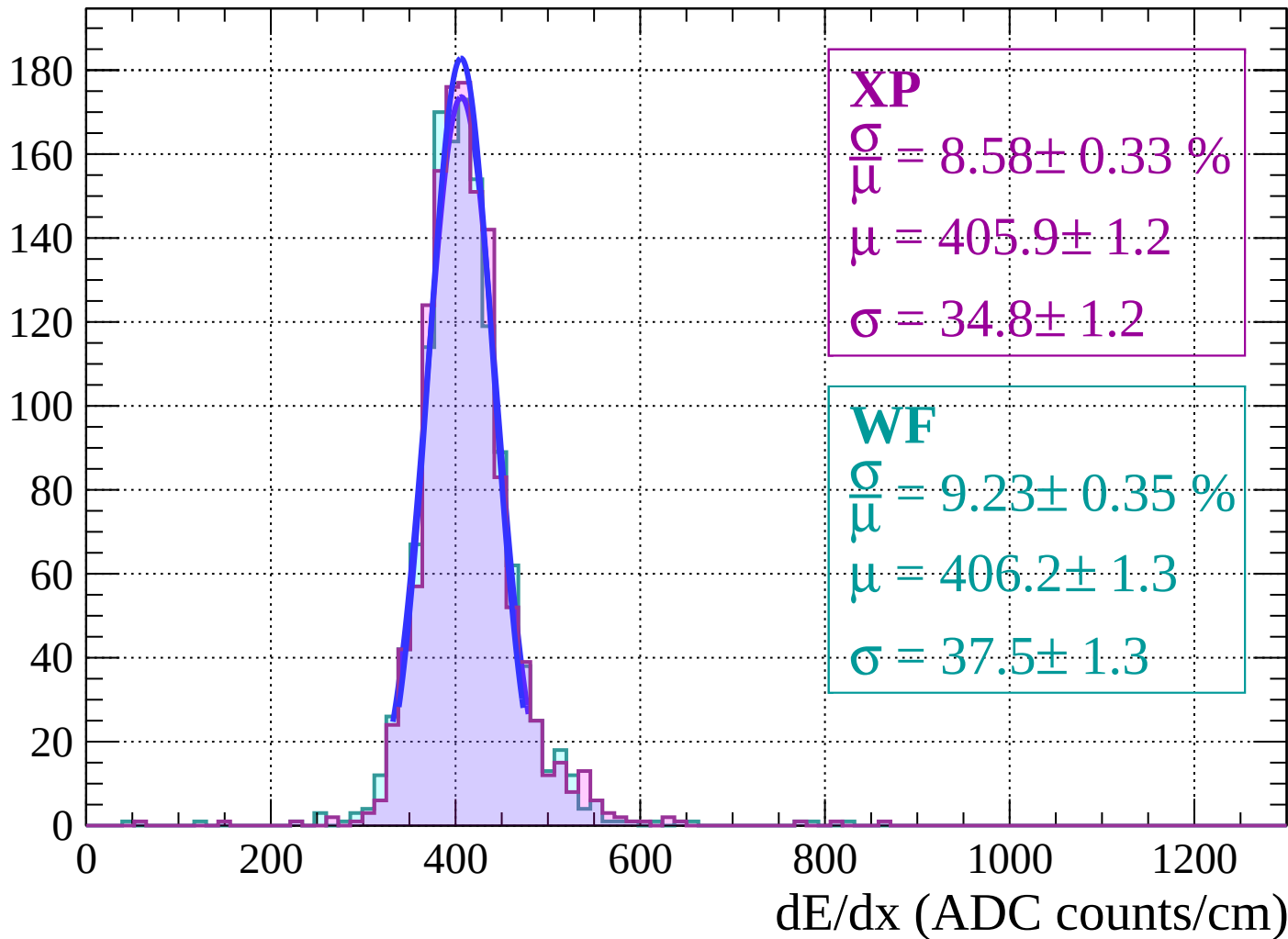
$$\mu = 406.9 \pm 1.1$$

$$\sigma = 39.9 \pm 1.0$$

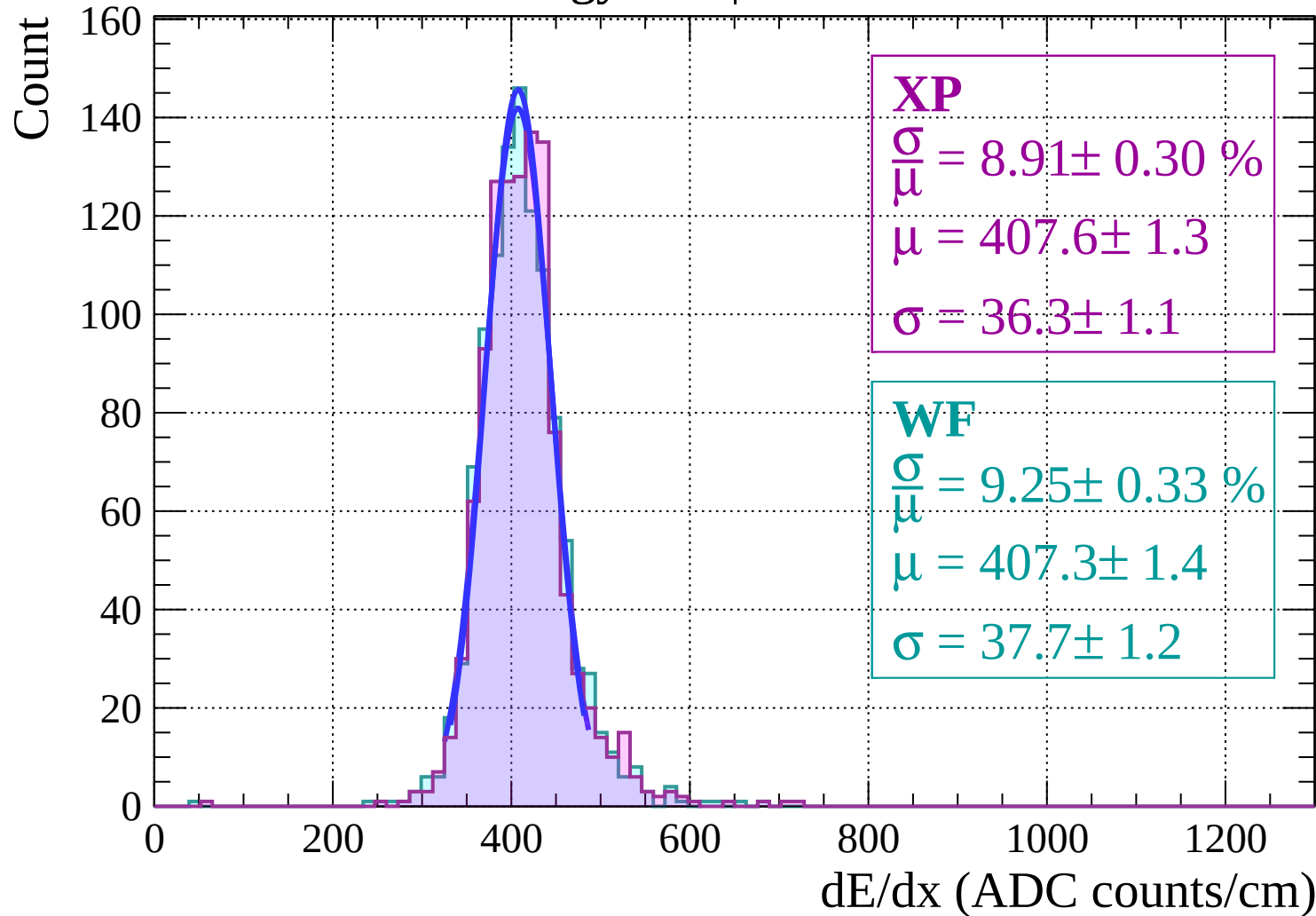
dE/dx (ADC counts/cm)

Energy loss | $26 < \theta < 28$

Count

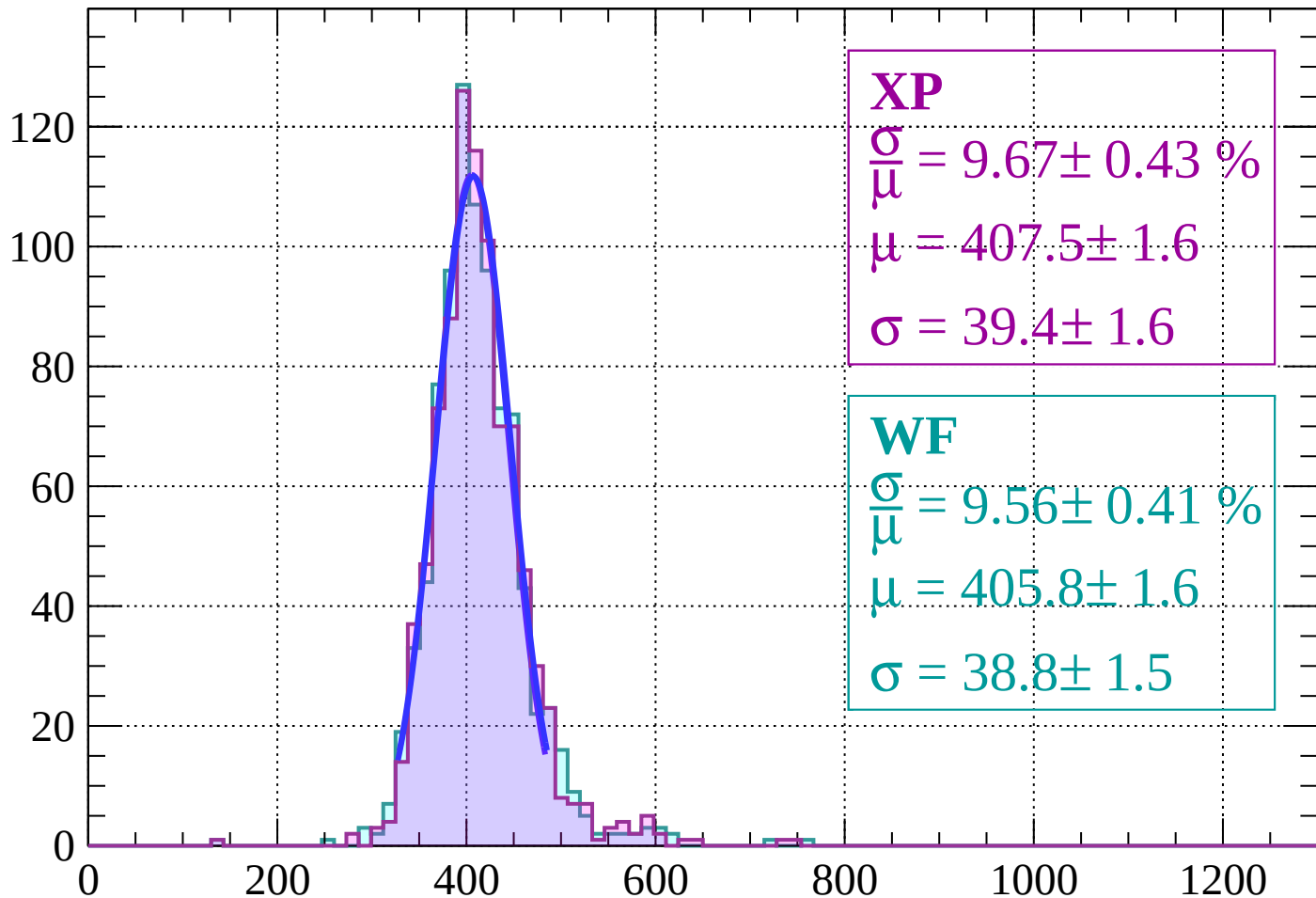


Energy loss | $28 < \theta < 30$



Energy loss | $30 < \theta < 32$

Count



XP

$$\frac{\sigma}{\mu} = 9.67 \pm 0.43 \%$$

$$\mu = 407.5 \pm 1.6$$

$$\sigma = 39.4 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.41 \%$$

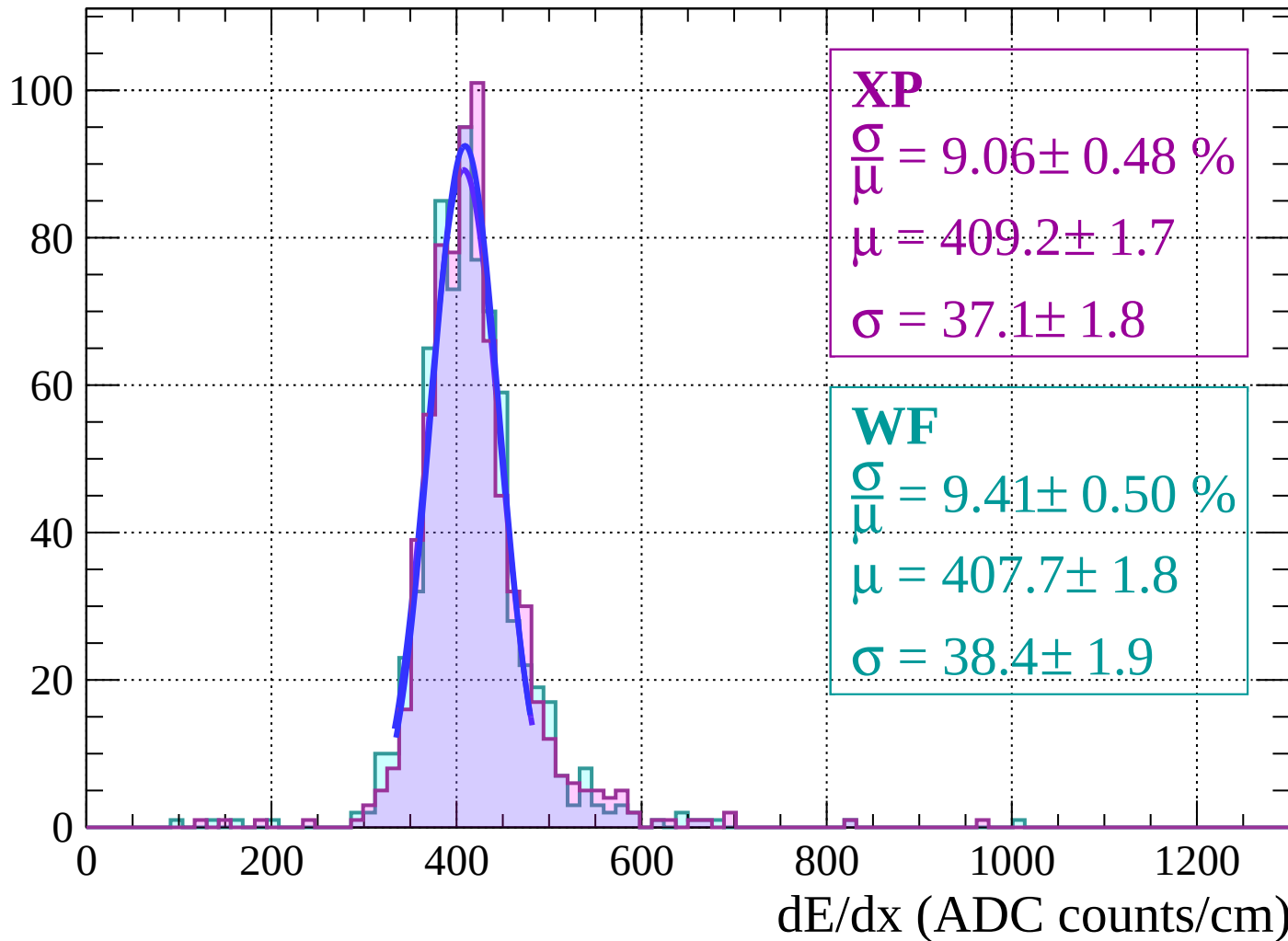
$$\mu = 405.8 \pm 1.6$$

$$\sigma = 38.8 \pm 1.5$$

dE/dx (ADC counts/cm)

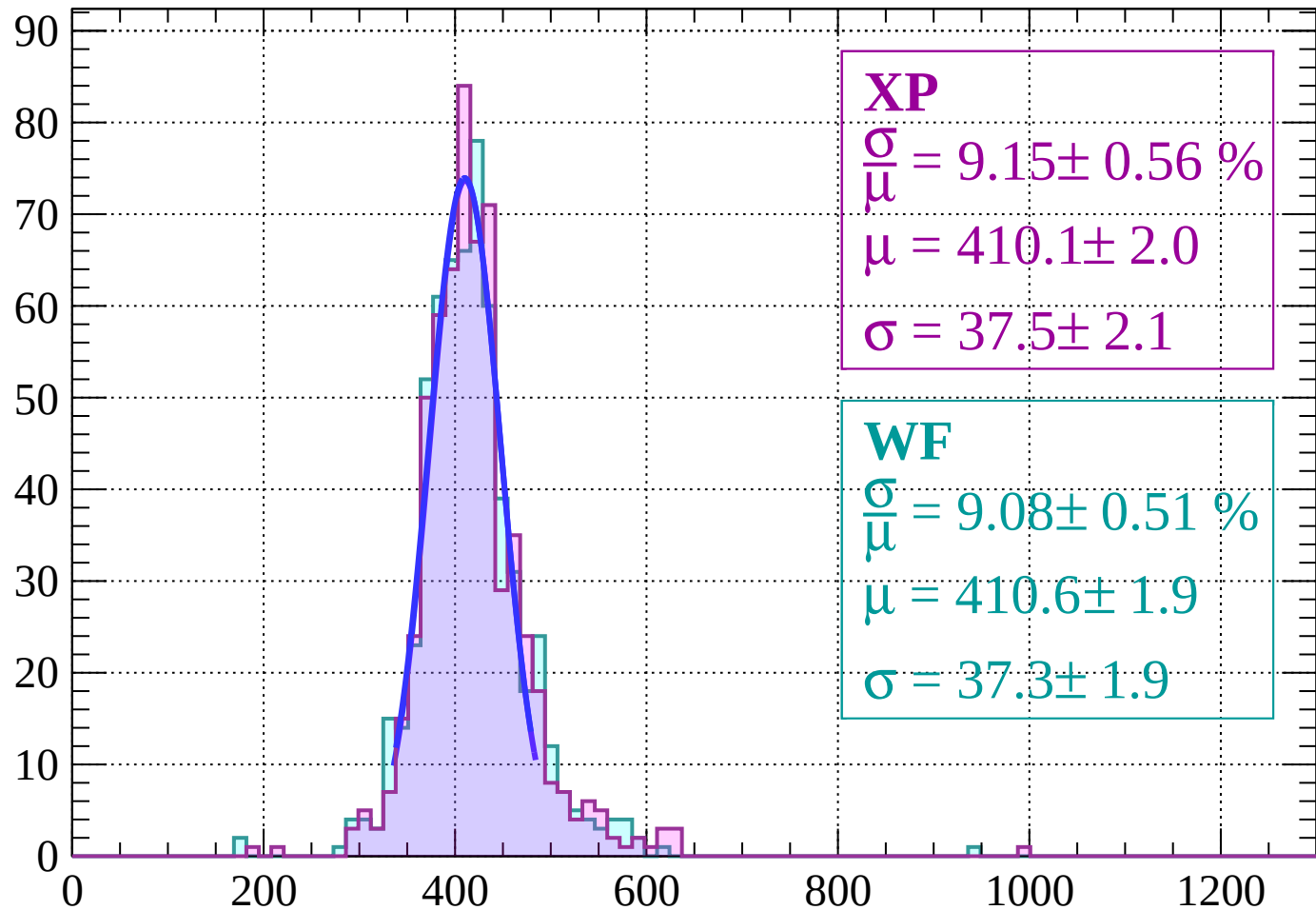
Energy loss | $32 < \theta < 34$

Count



Energy loss | $34 < \theta < 36$

Count



XP

$$\frac{\sigma}{\mu} = 9.15 \pm 0.56 \%$$

$$\mu = 410.1 \pm 2.0$$

$$\sigma = 37.5 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 9.08 \pm 0.51 \%$$

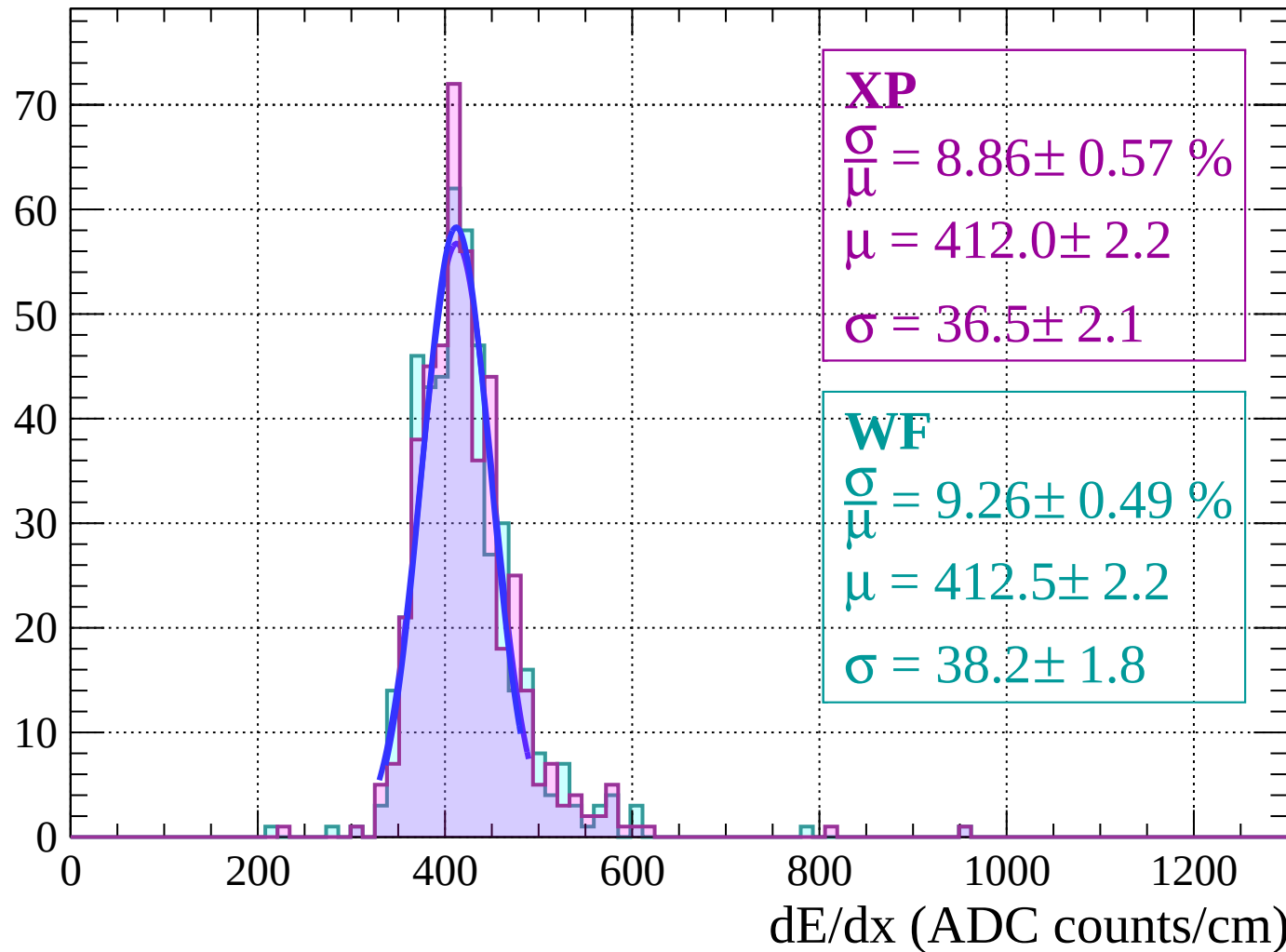
$$\mu = 410.6 \pm 1.9$$

$$\sigma = 37.3 \pm 1.9$$

dE/dx (ADC counts/cm)

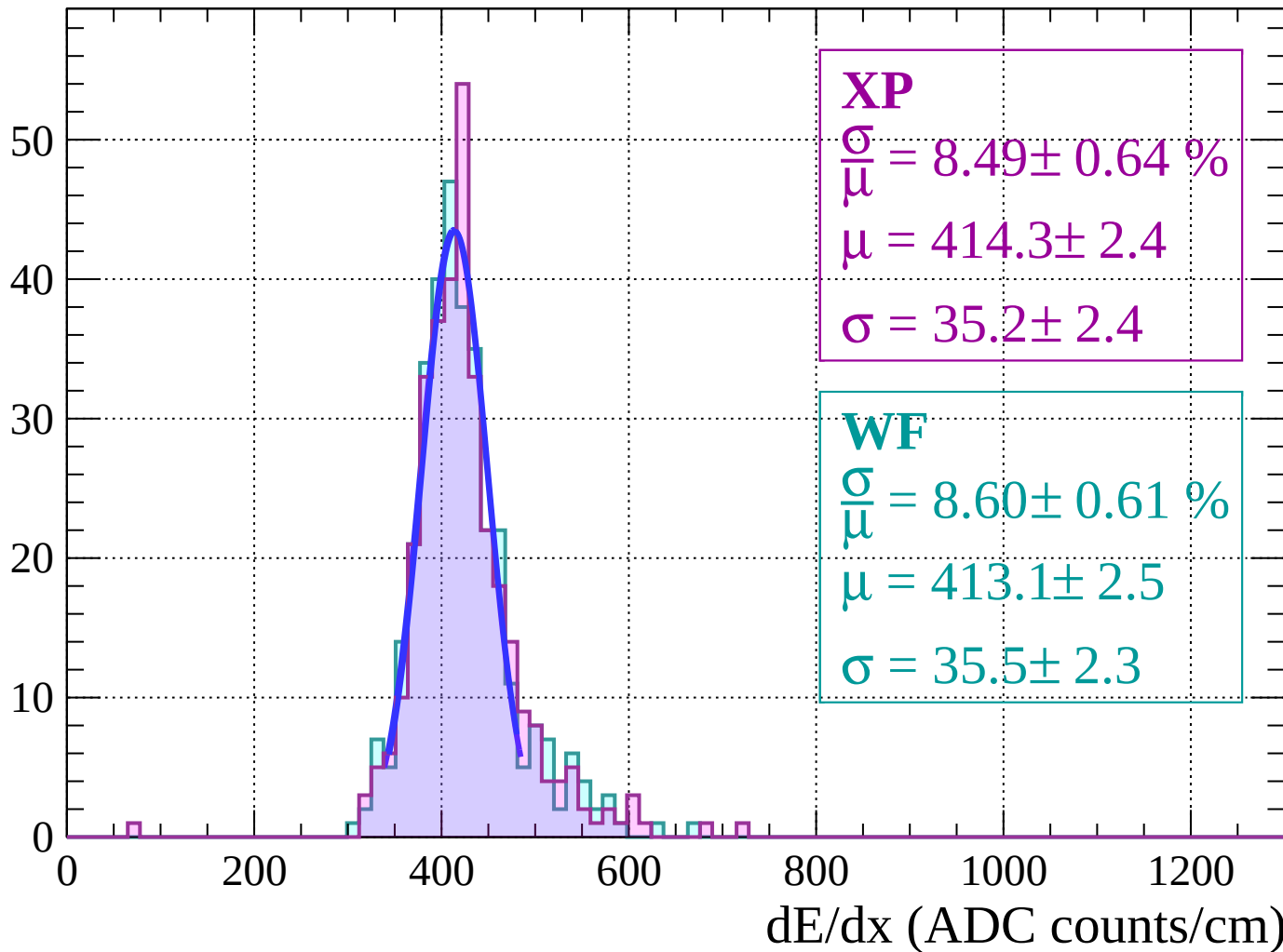
Energy loss | $36 < \theta < 38$

Count



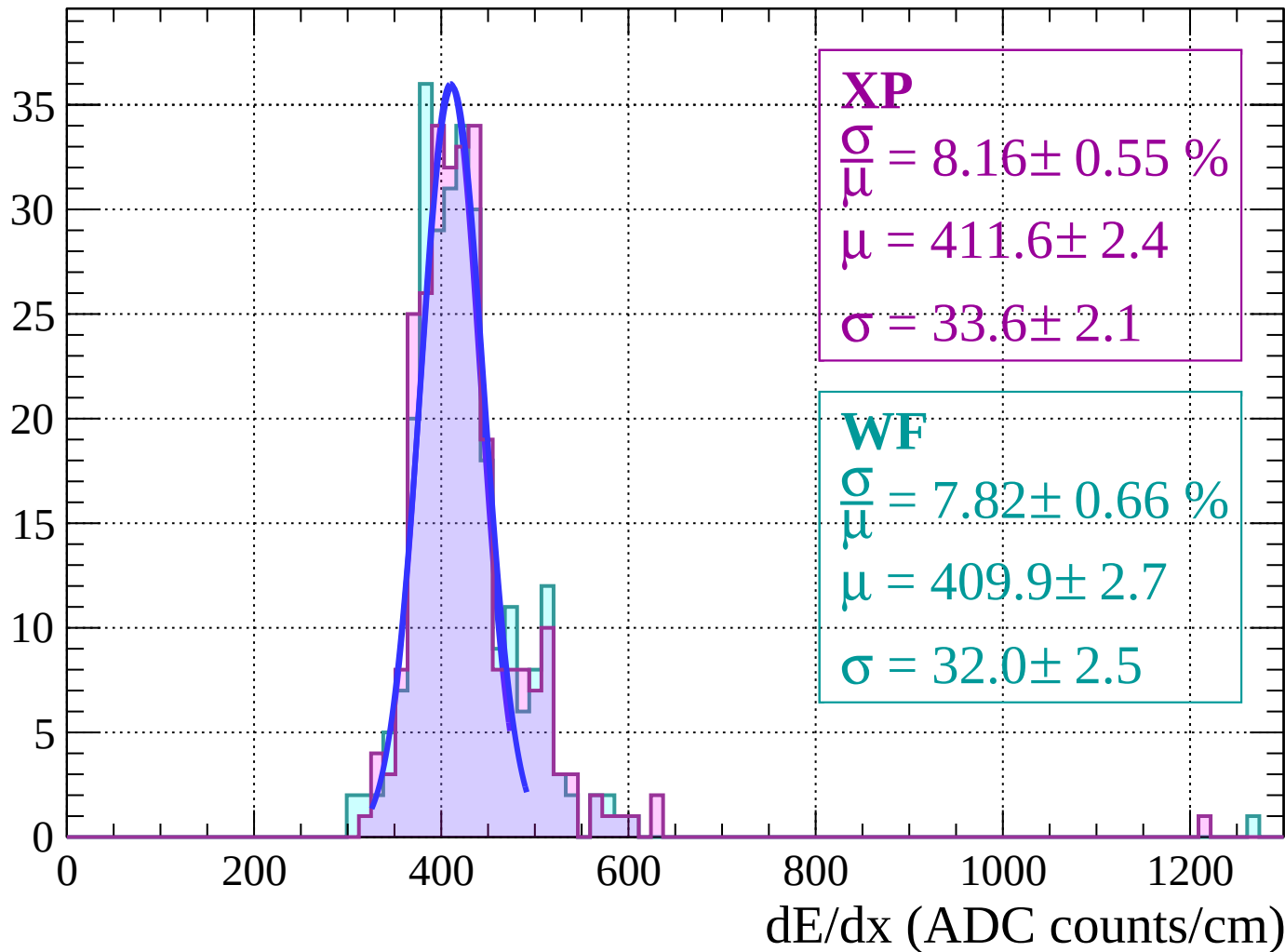
Energy loss | $38 < \theta < 40$

Count



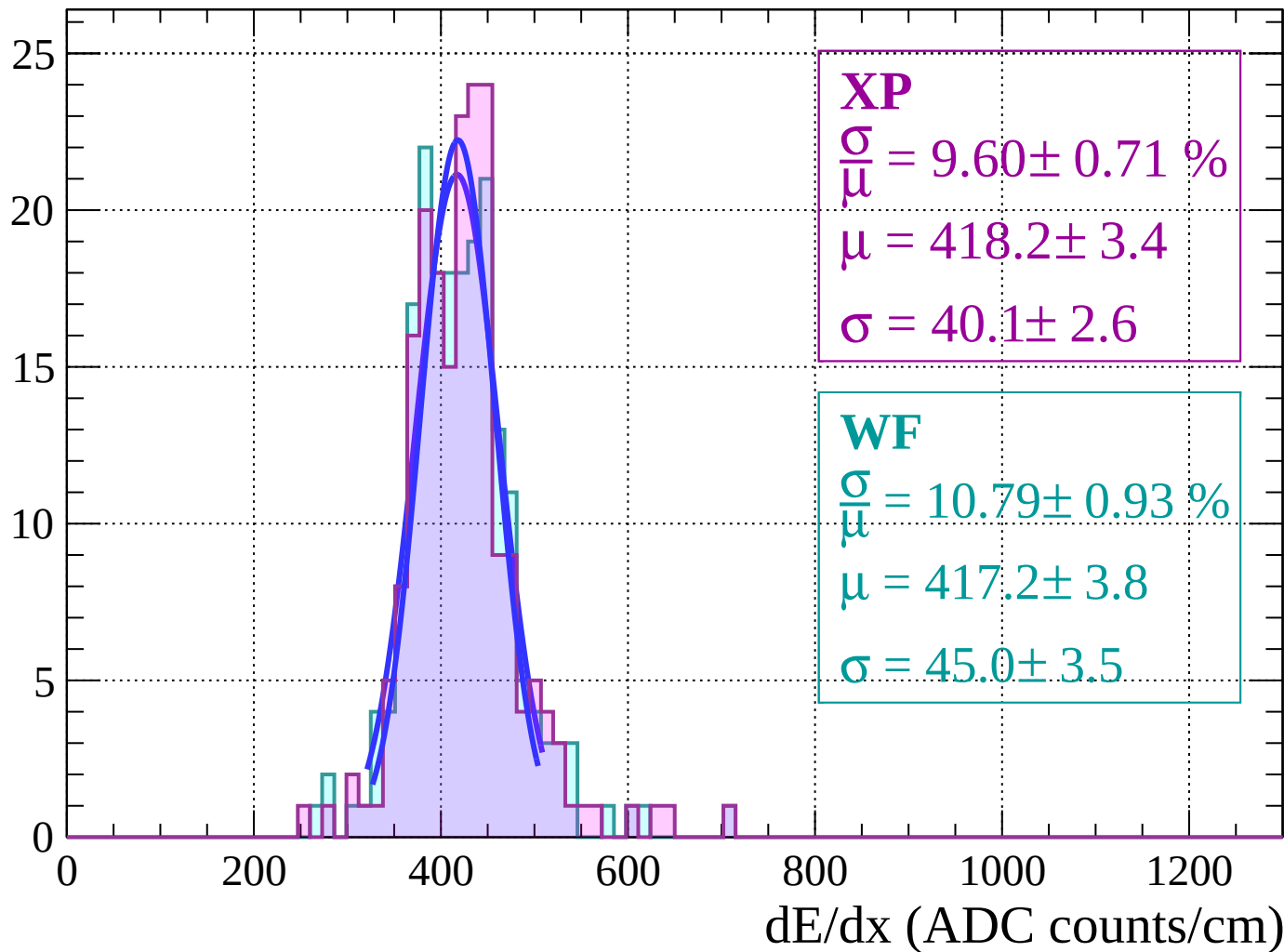
Energy loss | $40 < \theta < 42$

Count



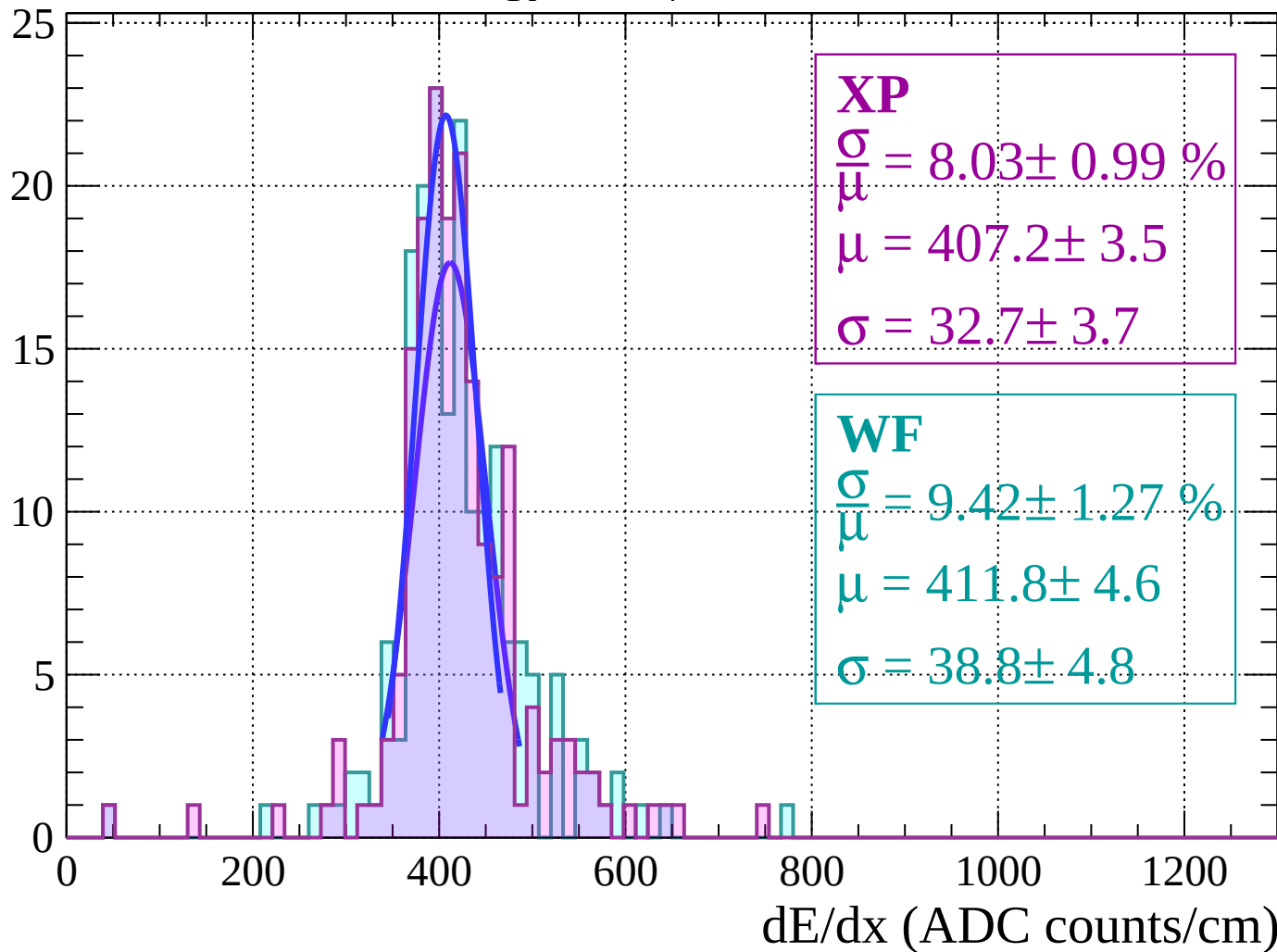
Energy loss | $42 < \theta < 44$

Count



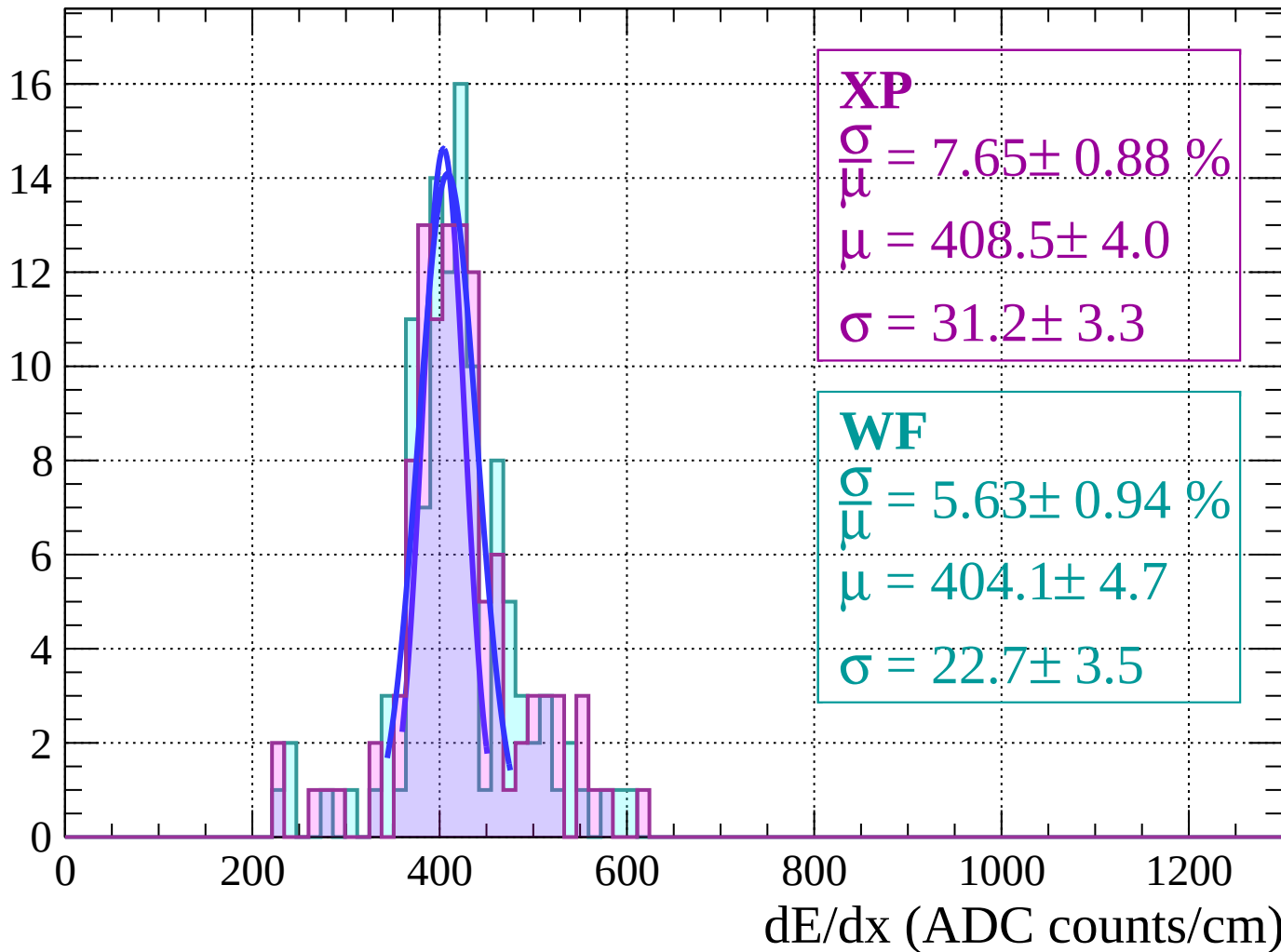
Energy loss | $44 < \theta < 46$

Count



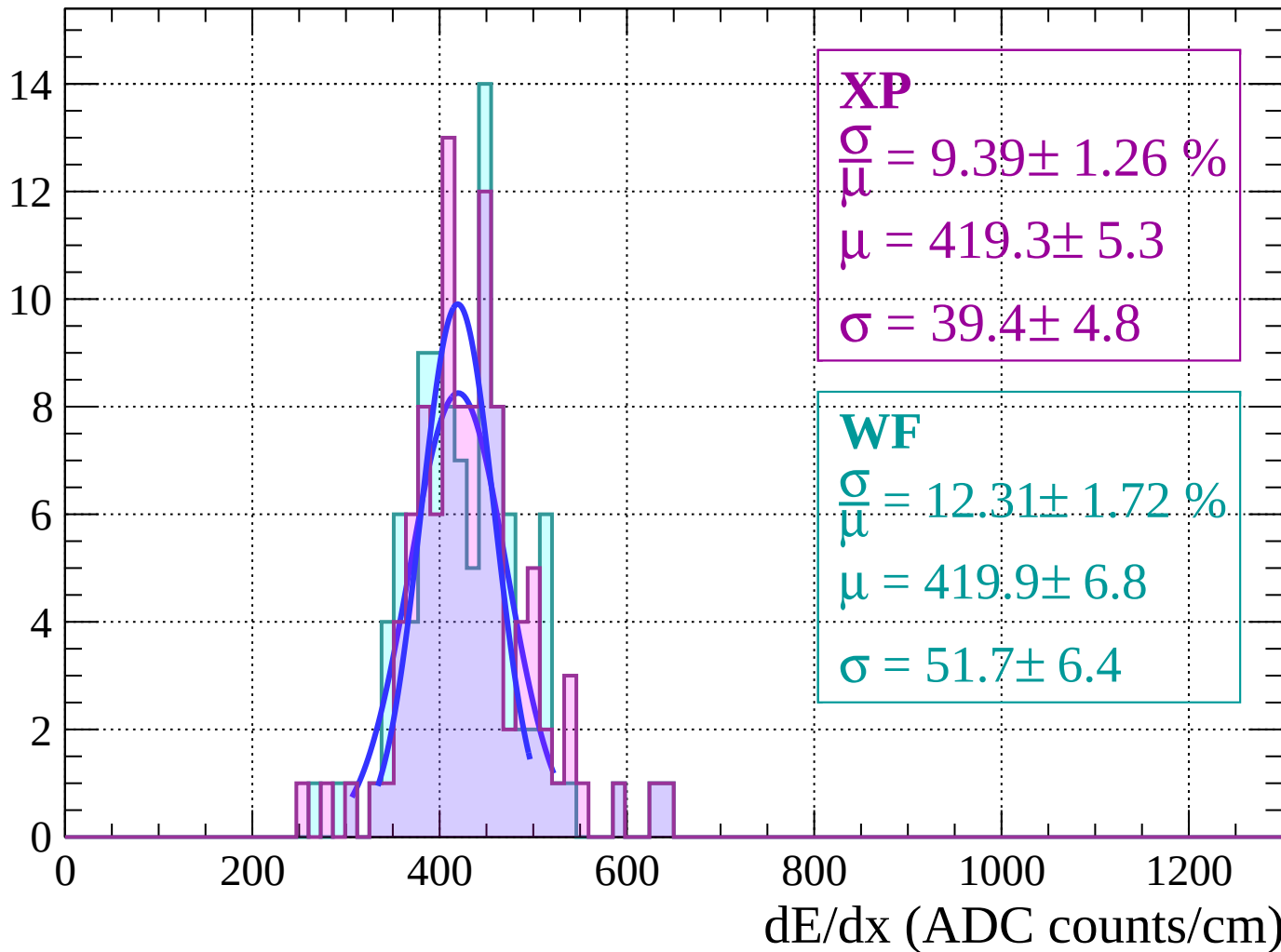
Energy loss | $46 < \theta < 48$

Count



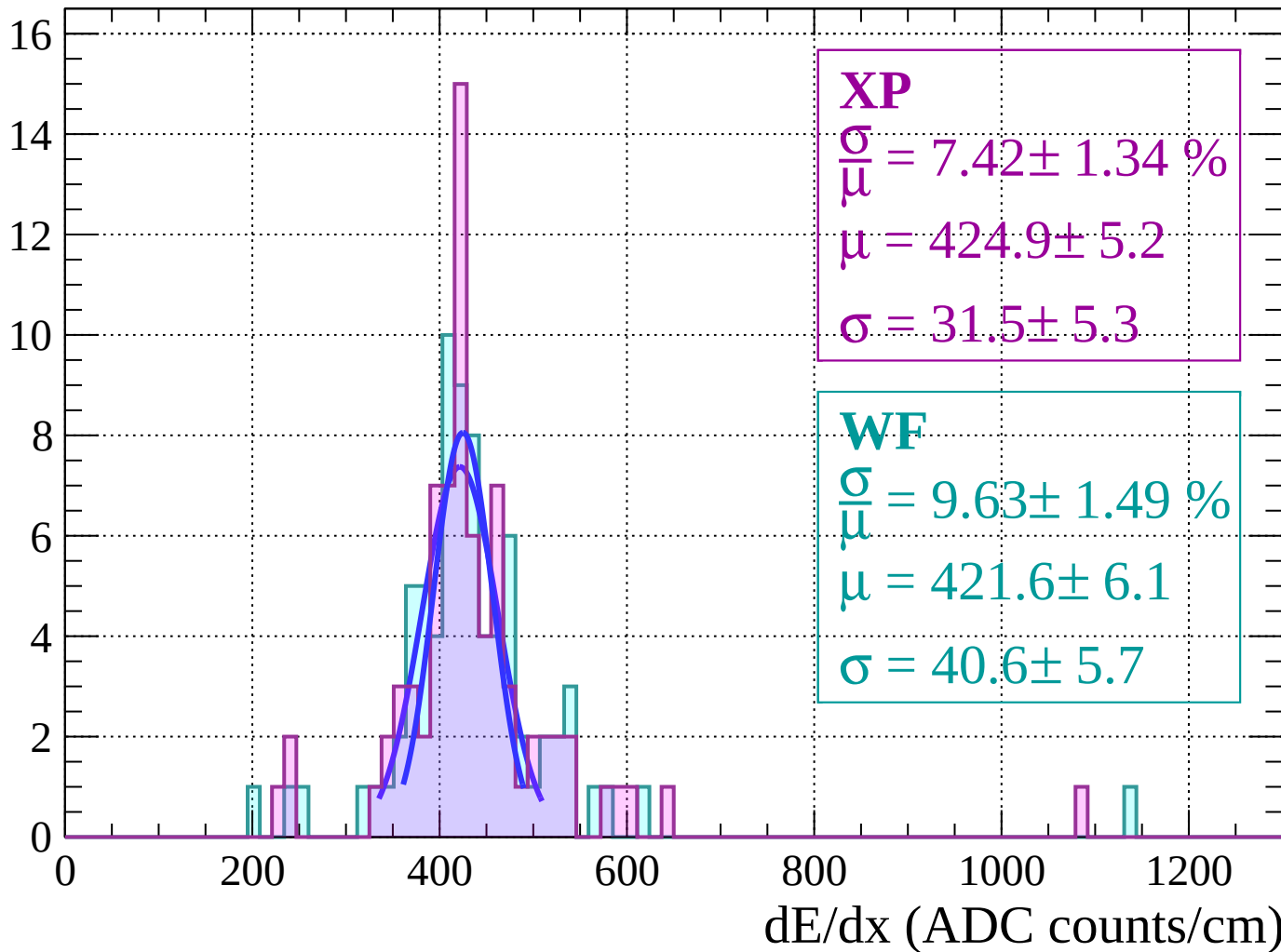
Energy loss | $48 < \theta < 50$

Count



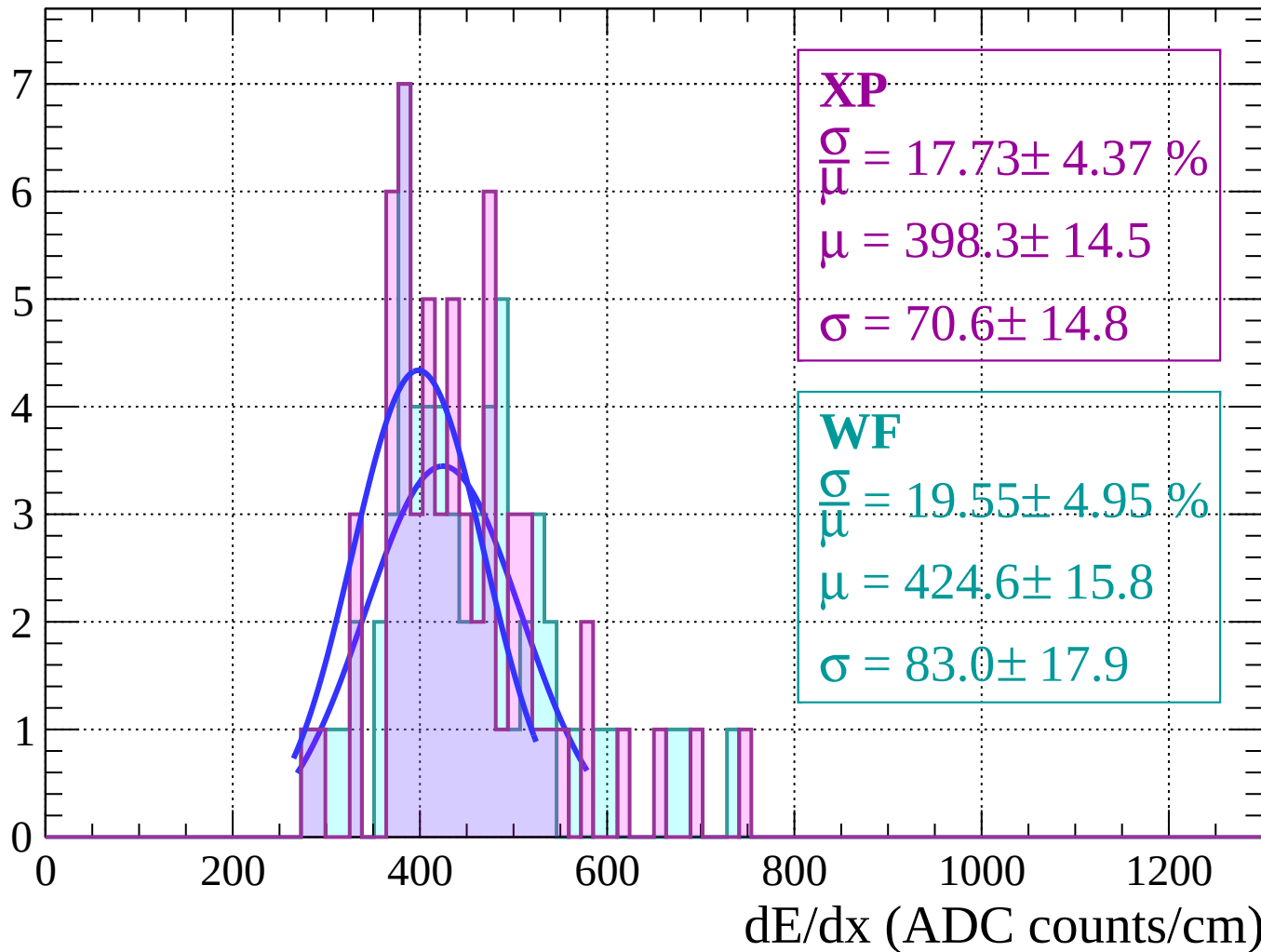
Energy loss | $50 < \theta < 52$

Count



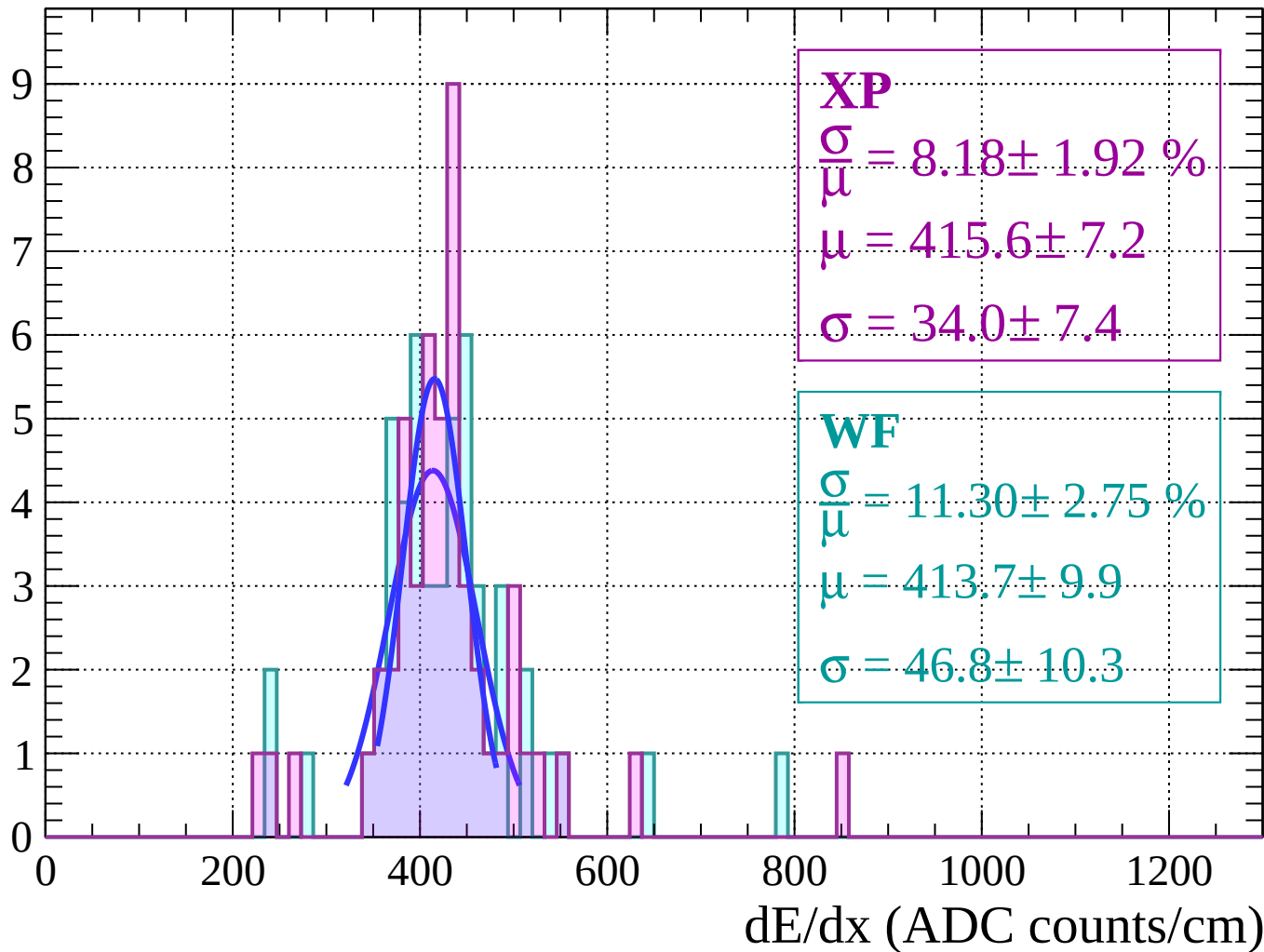
Energy loss | $52 < \theta < 54$

Count



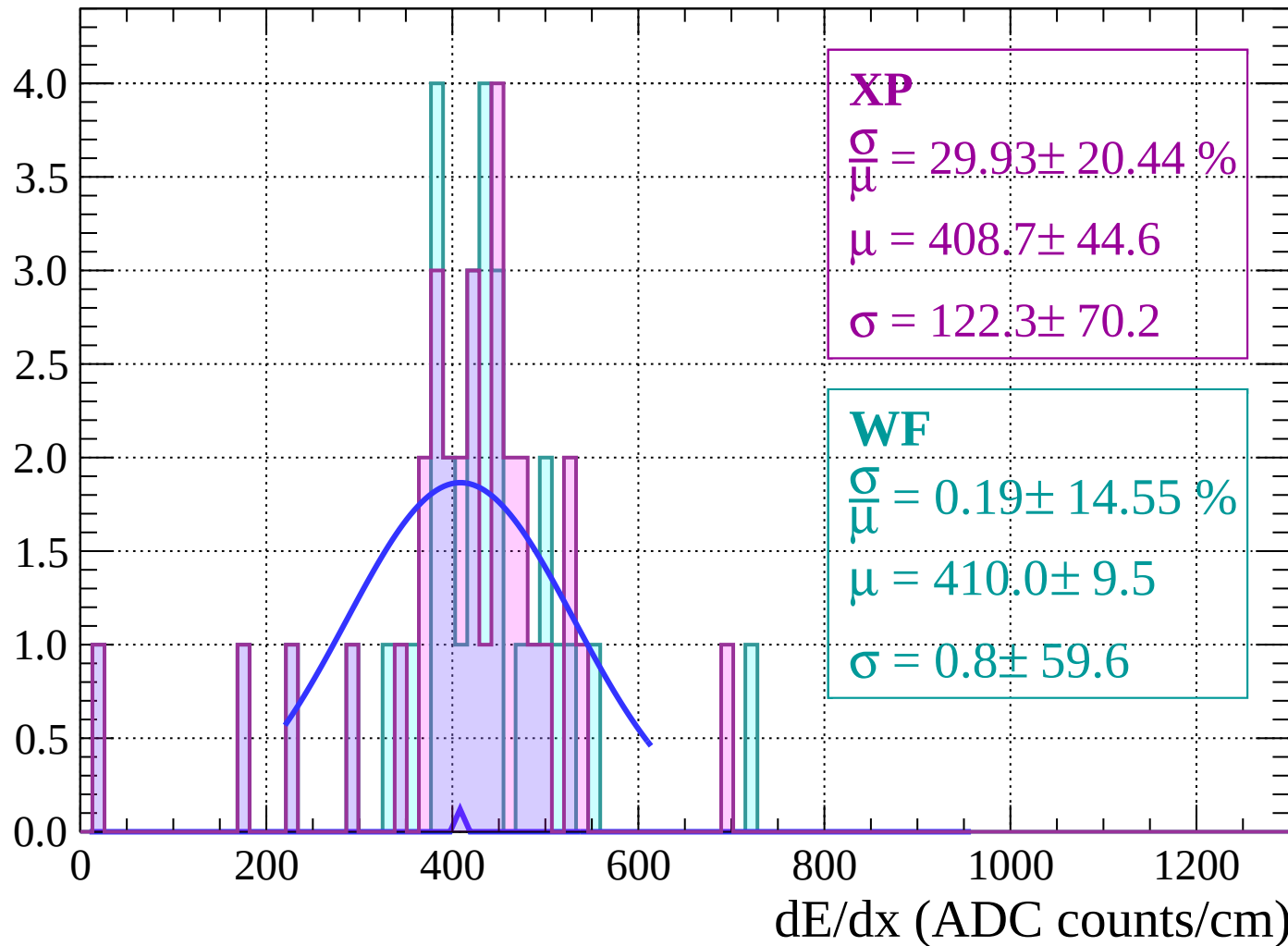
Energy loss | $54 < \theta < 56$

Count



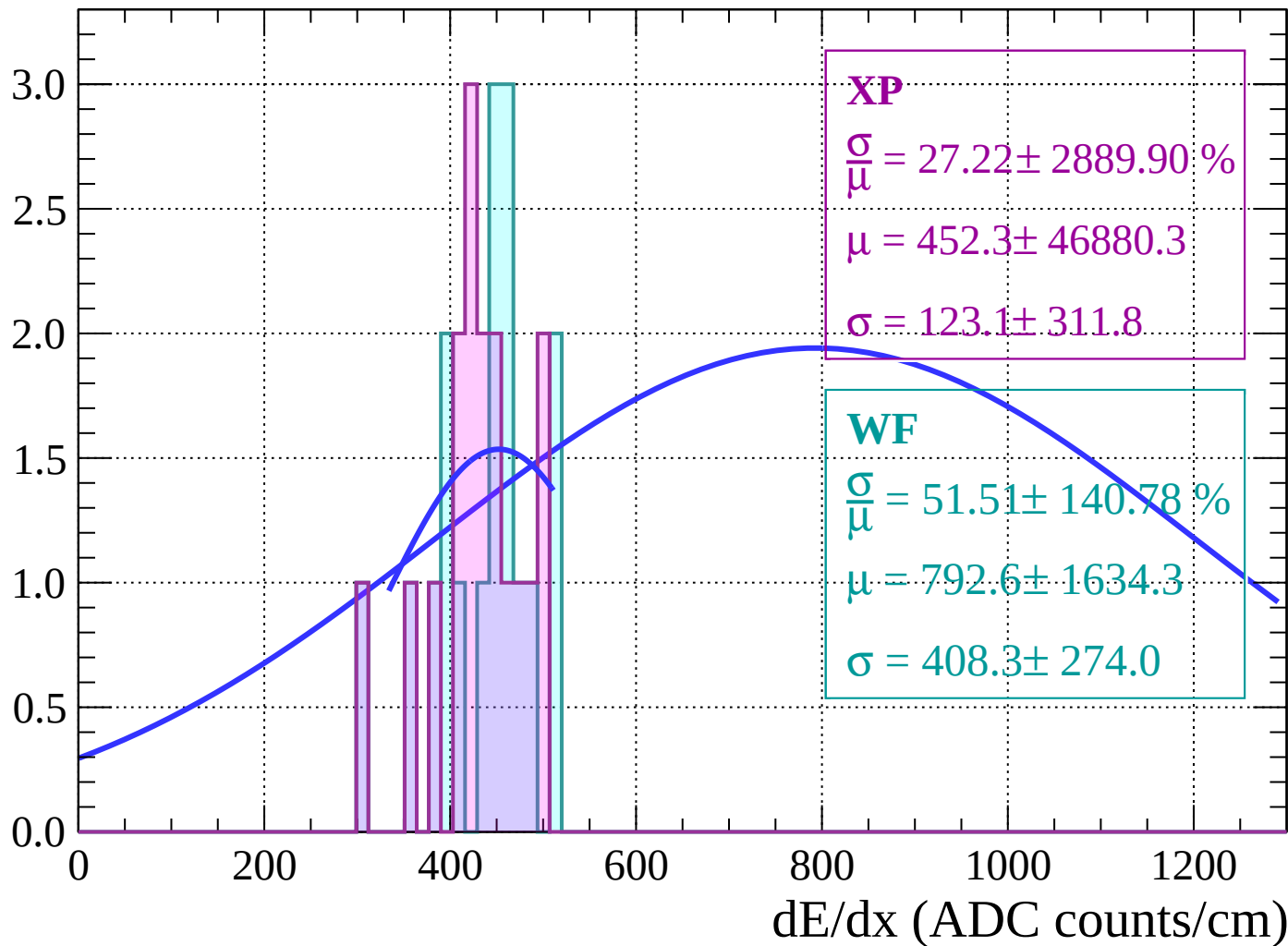
Energy loss | $56 < \theta < 58$

Count



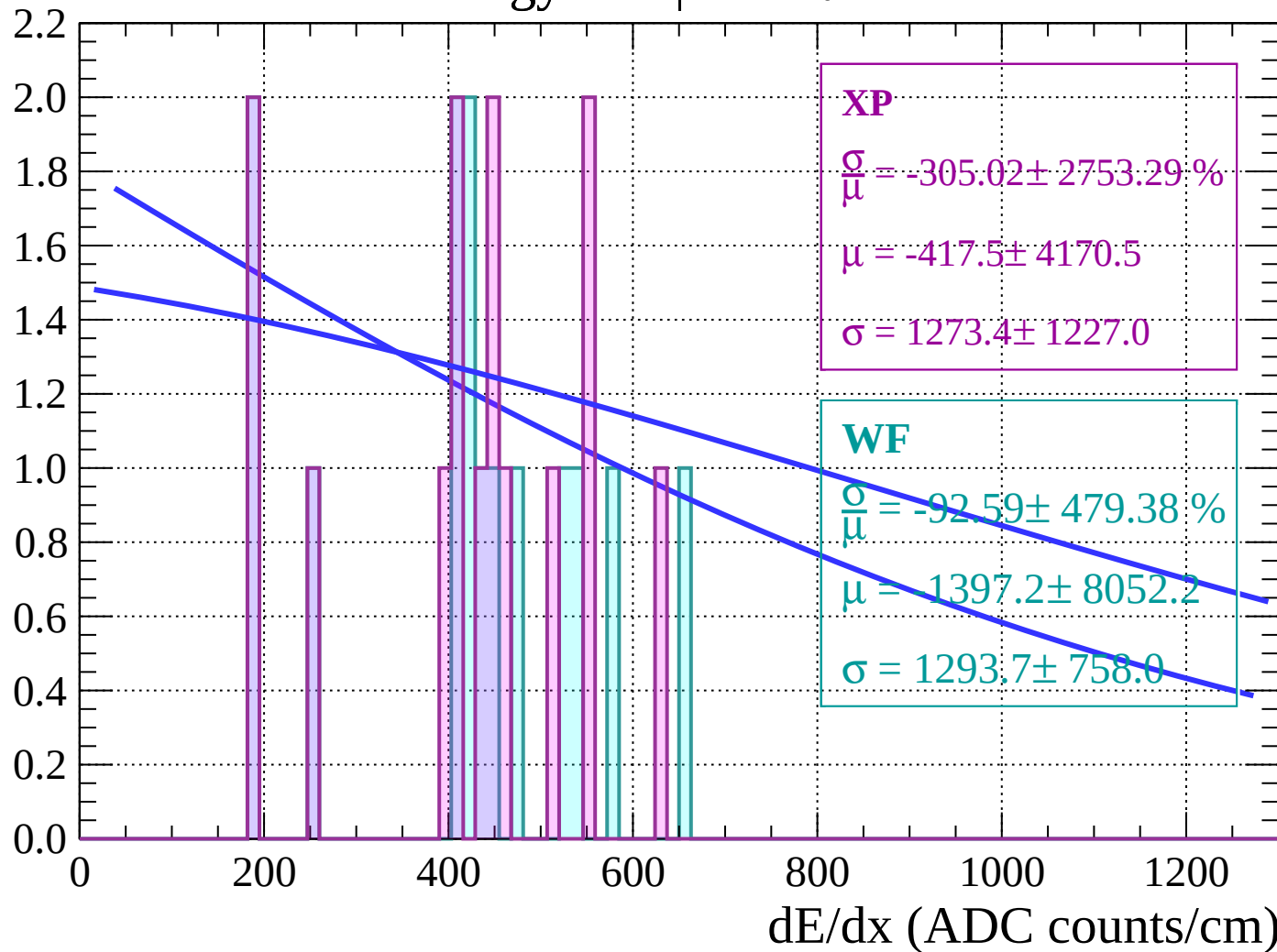
Energy loss | $58 < \theta < 60$

Count



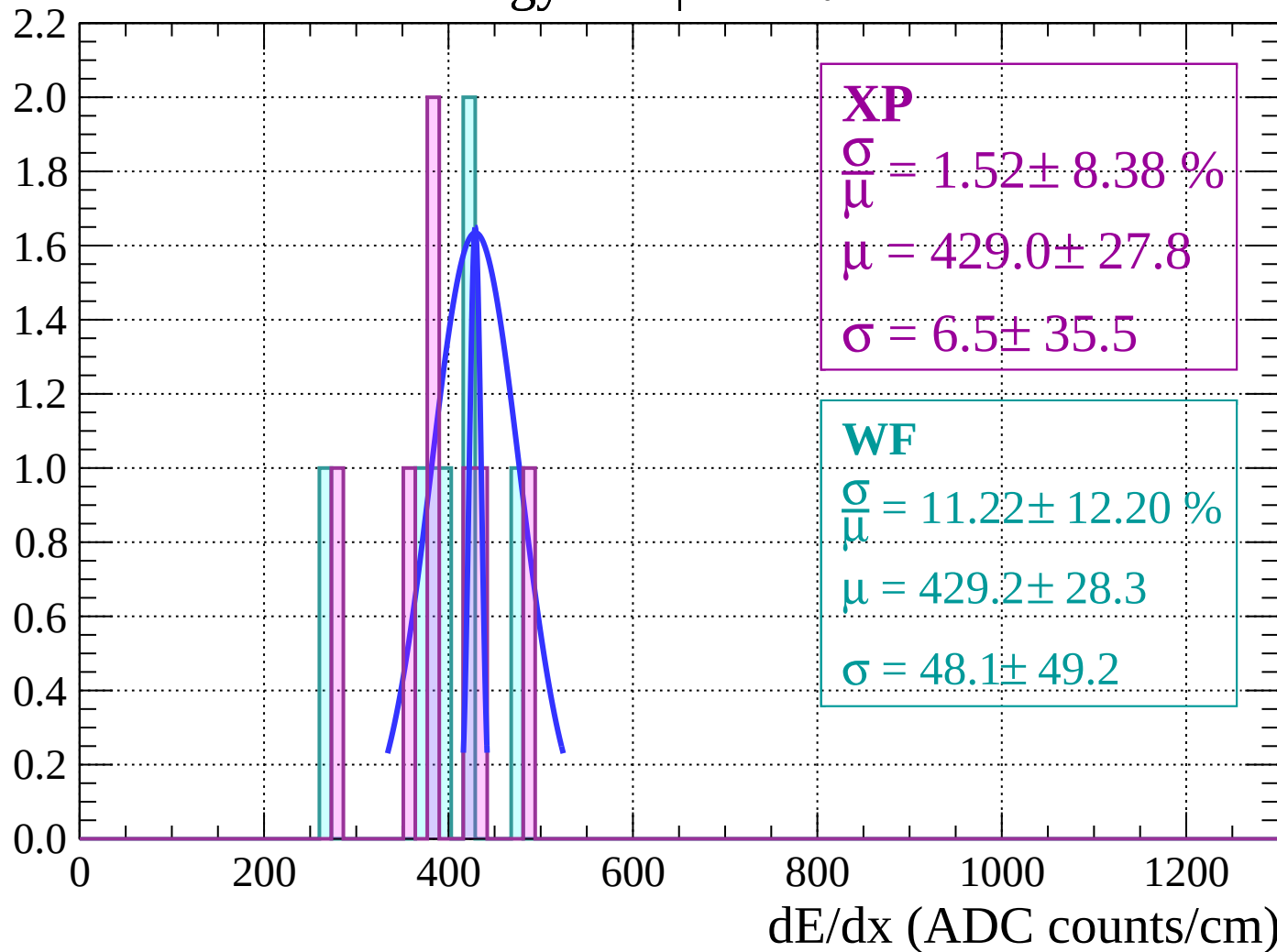
Energy loss | $60 < \theta < 62$

Count



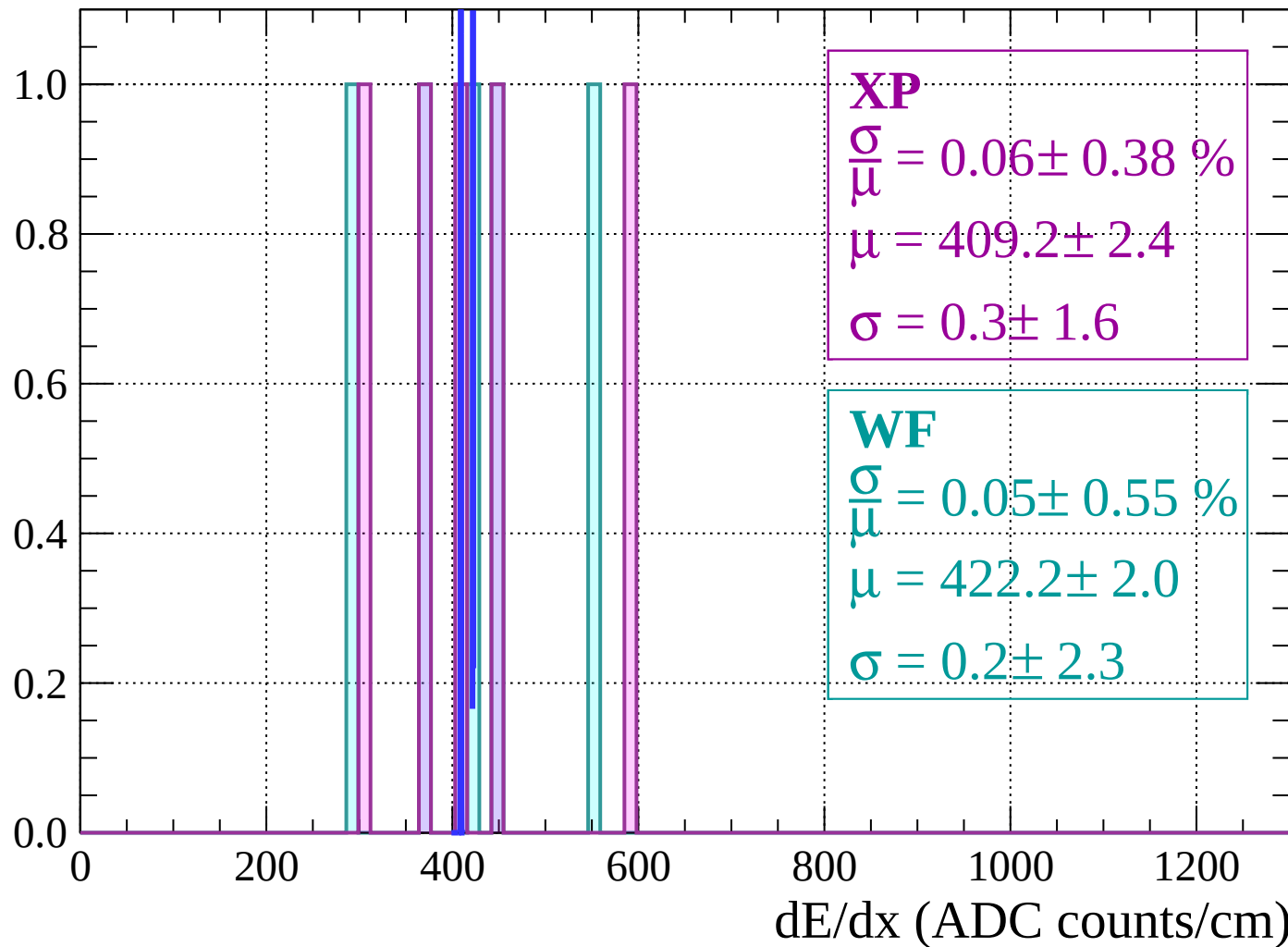
Energy loss | $62 < \theta < 64$

Count



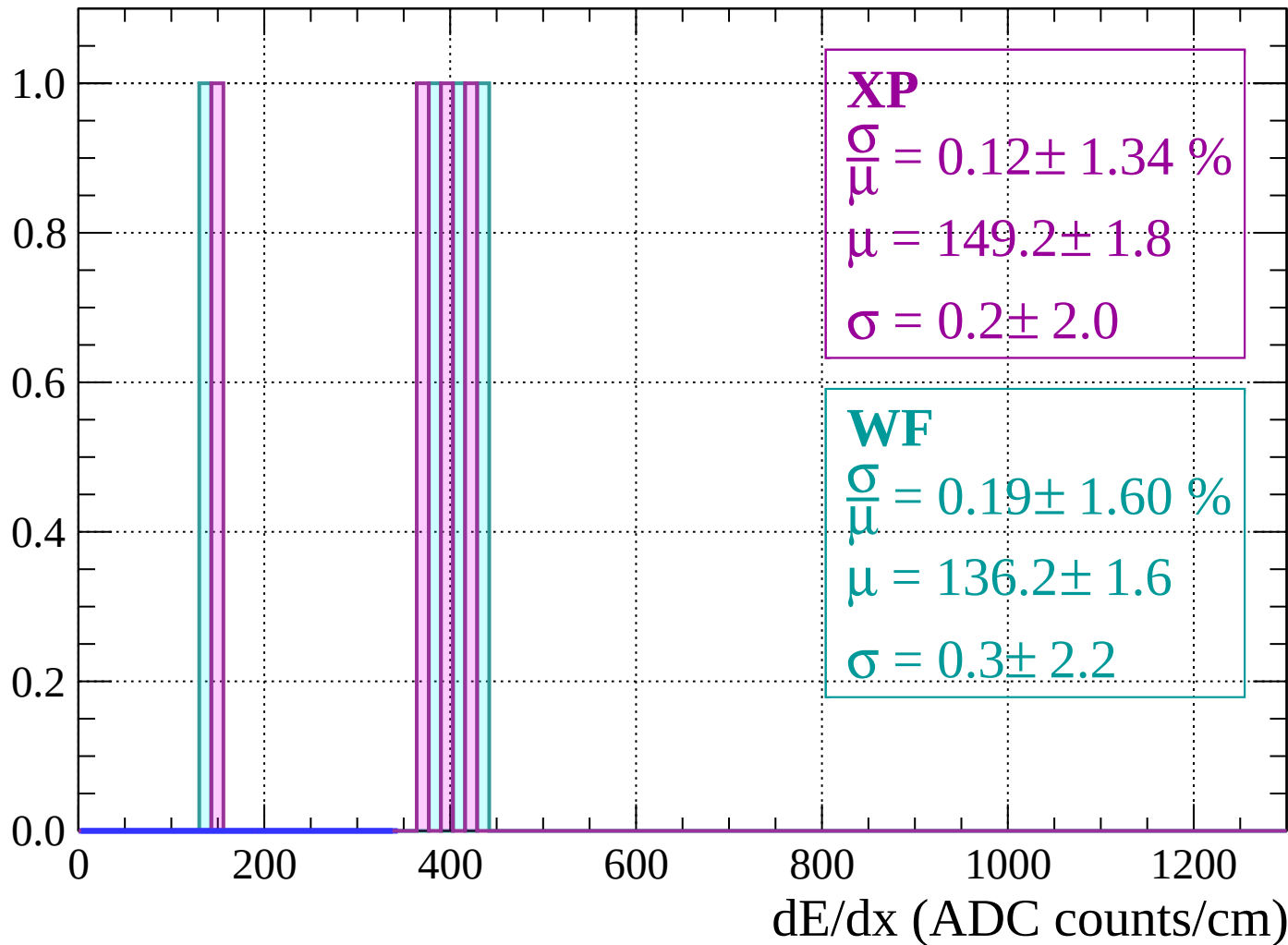
Energy loss | $64 < \theta < 66$

Count



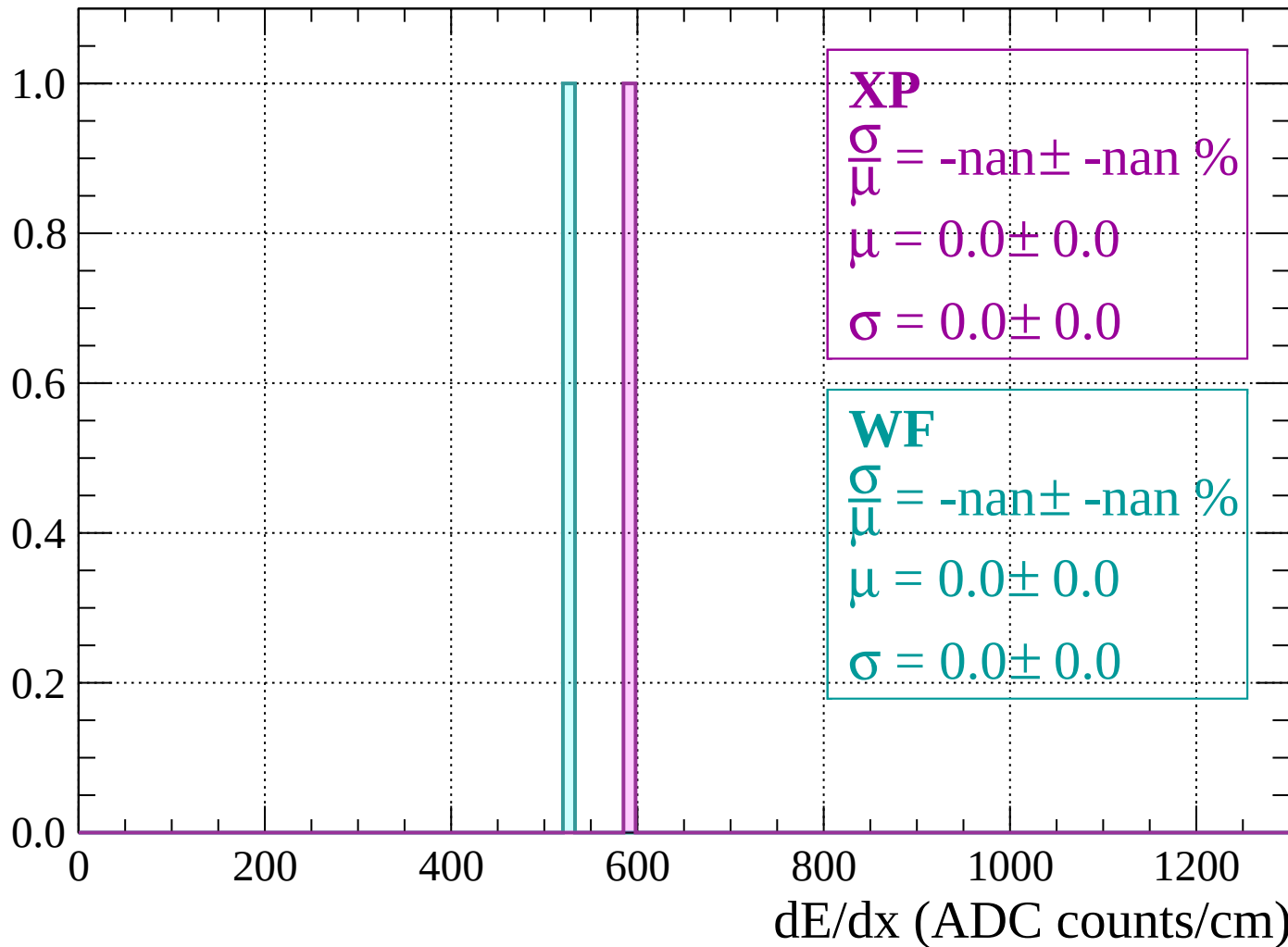
Energy loss | $66 < \theta < 68$

Count



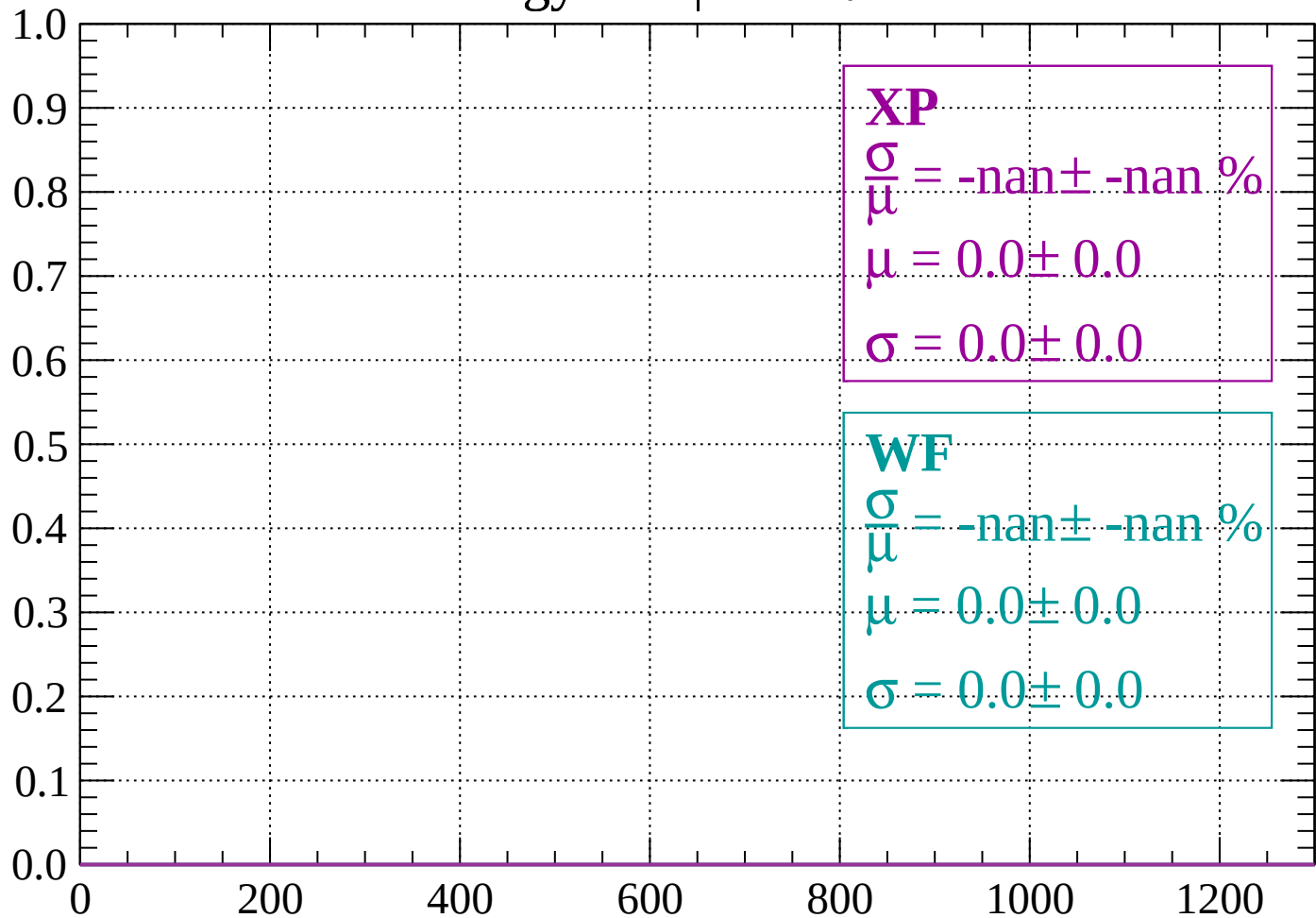
Energy loss | $68 < \theta < 70$

Count



Energy loss | $70 < \theta < 72$

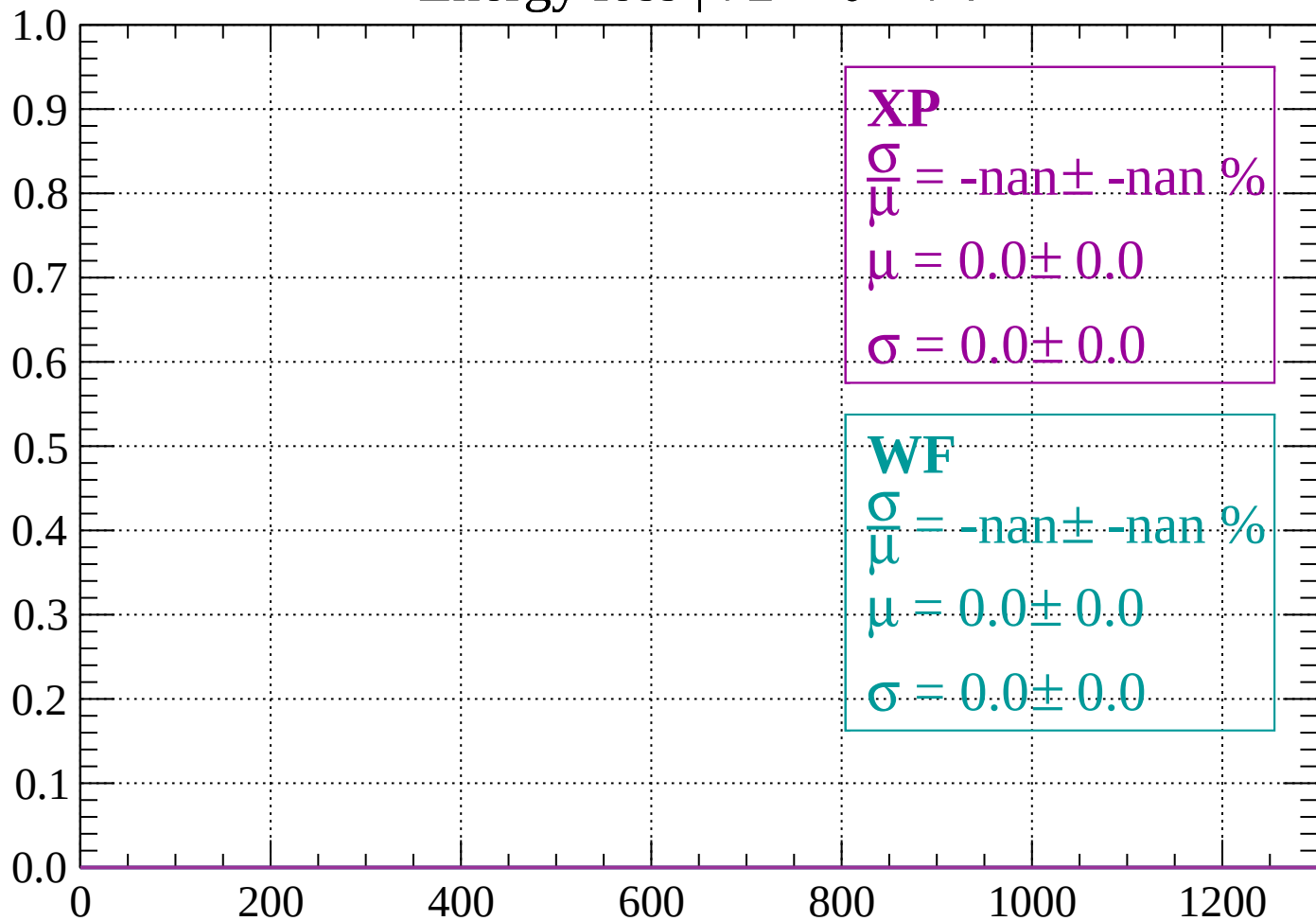
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 74$

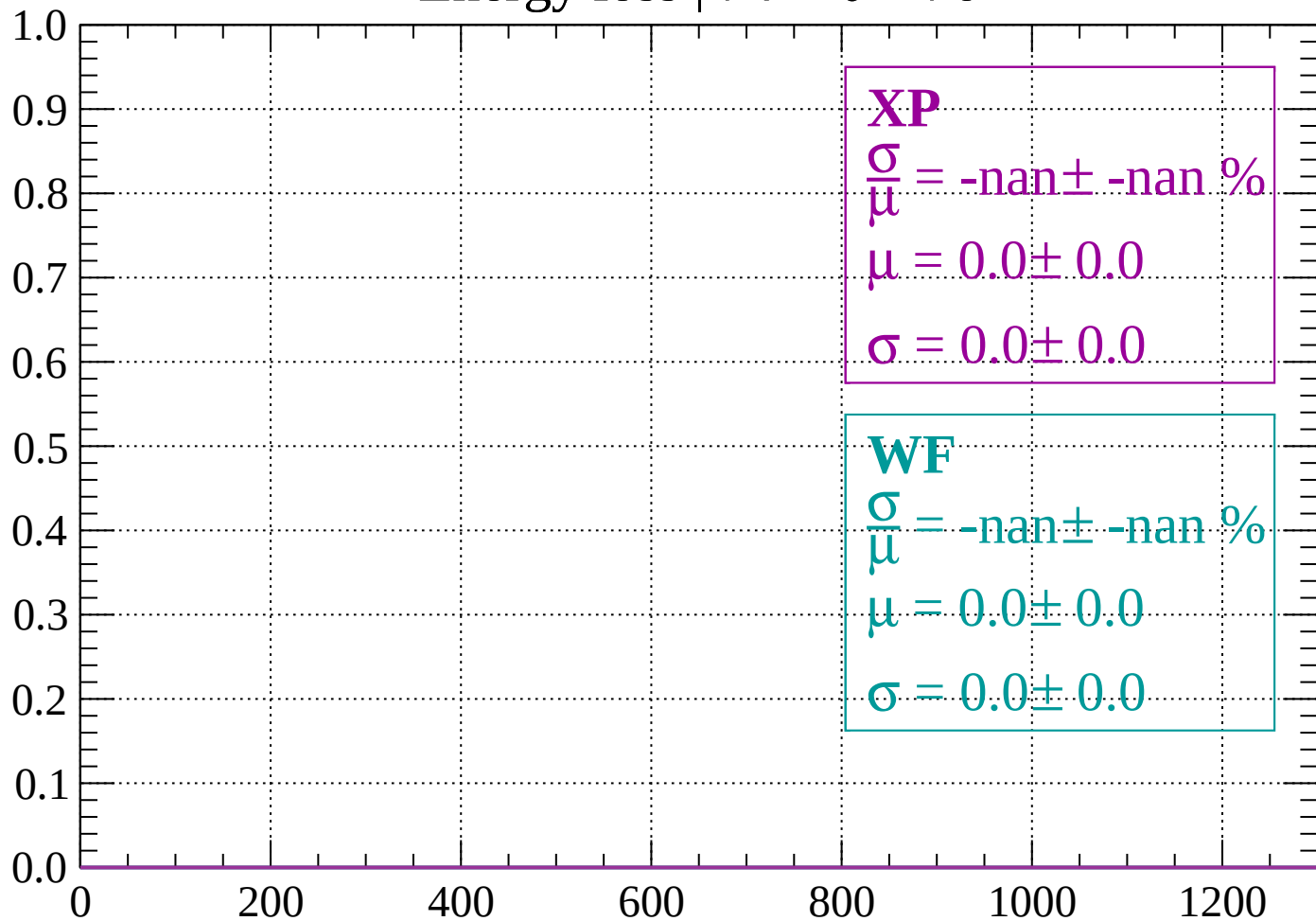
Count



dE/dx (ADC counts/cm)

Energy loss | $74 < \theta < 76$

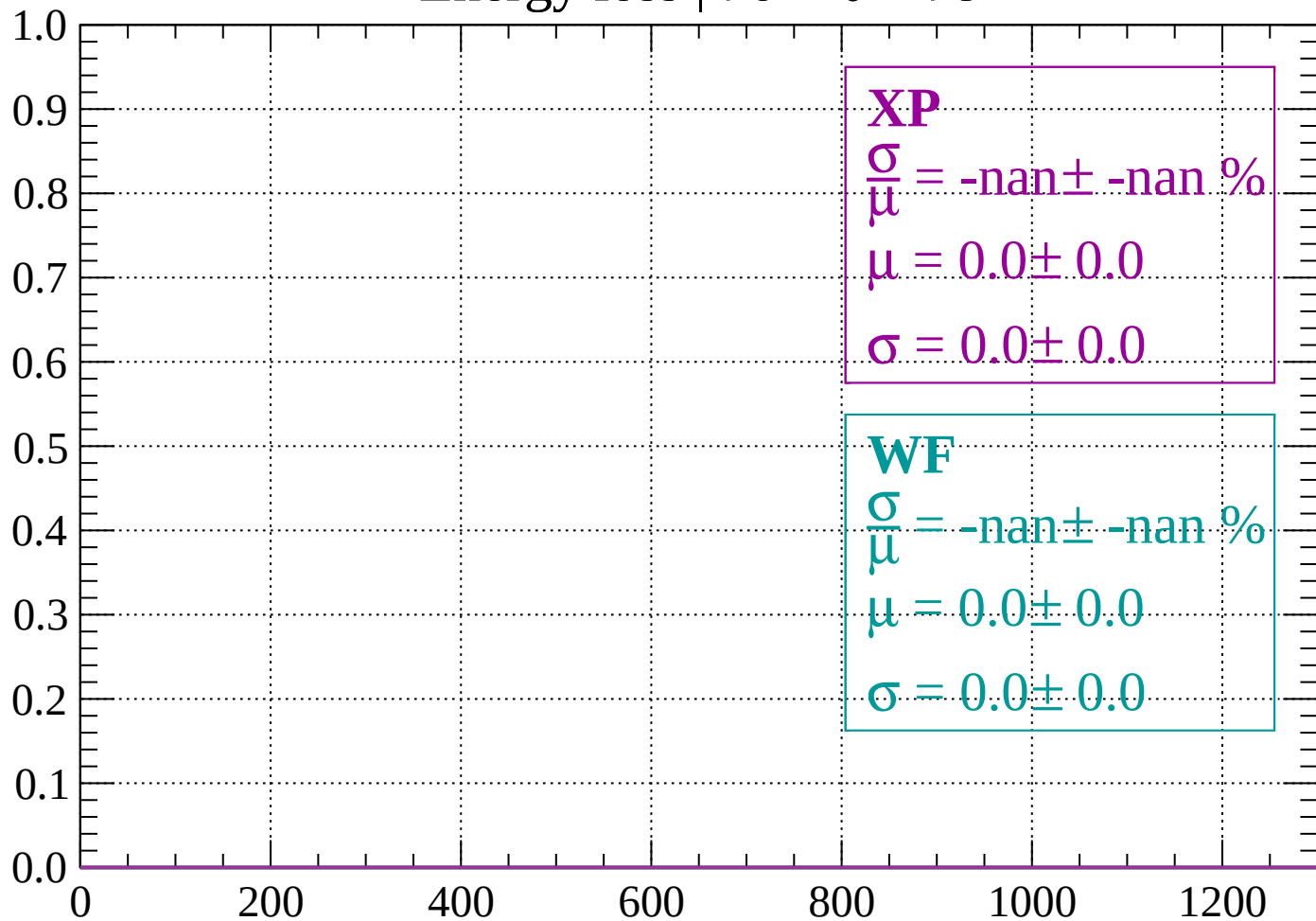
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

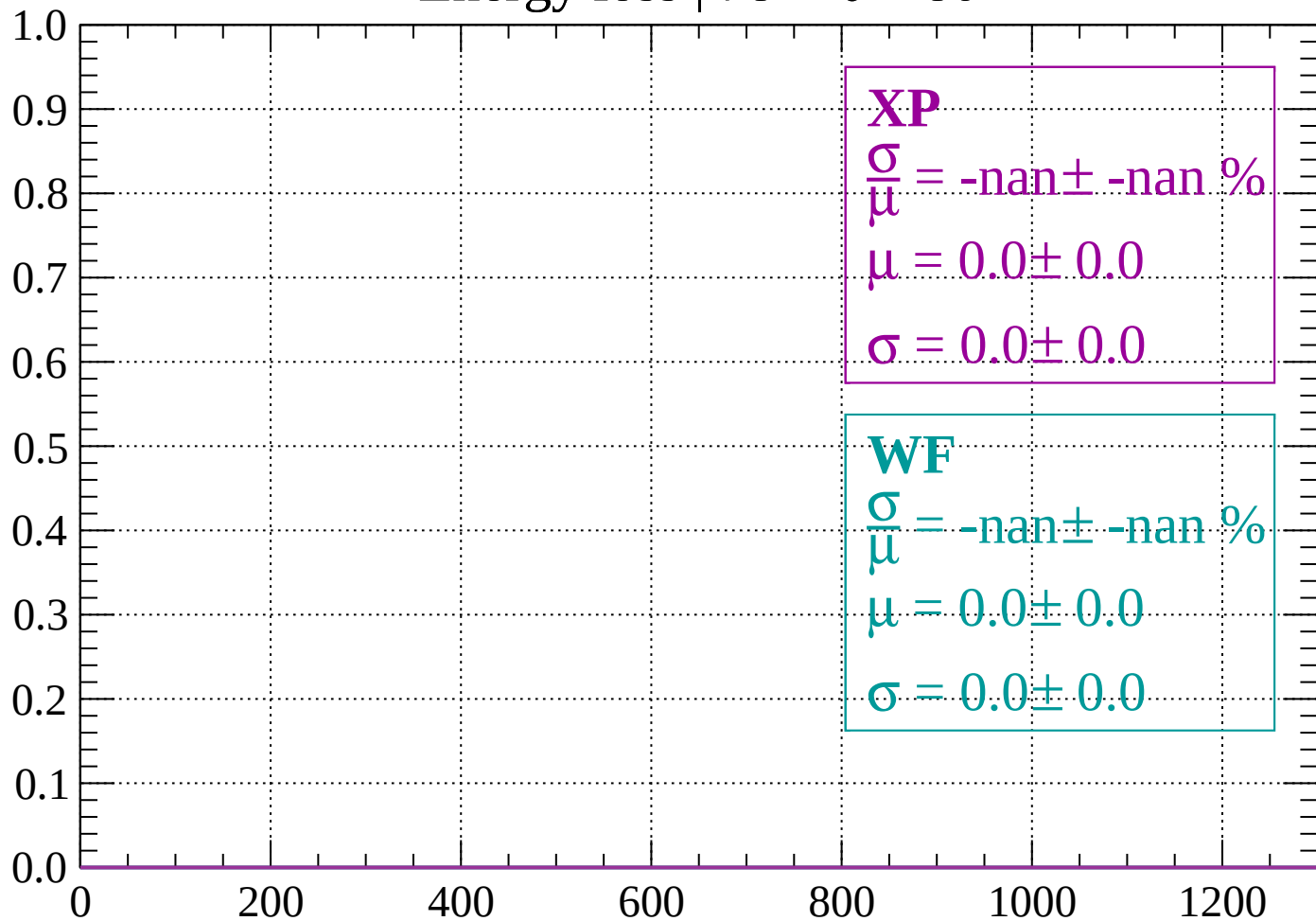
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

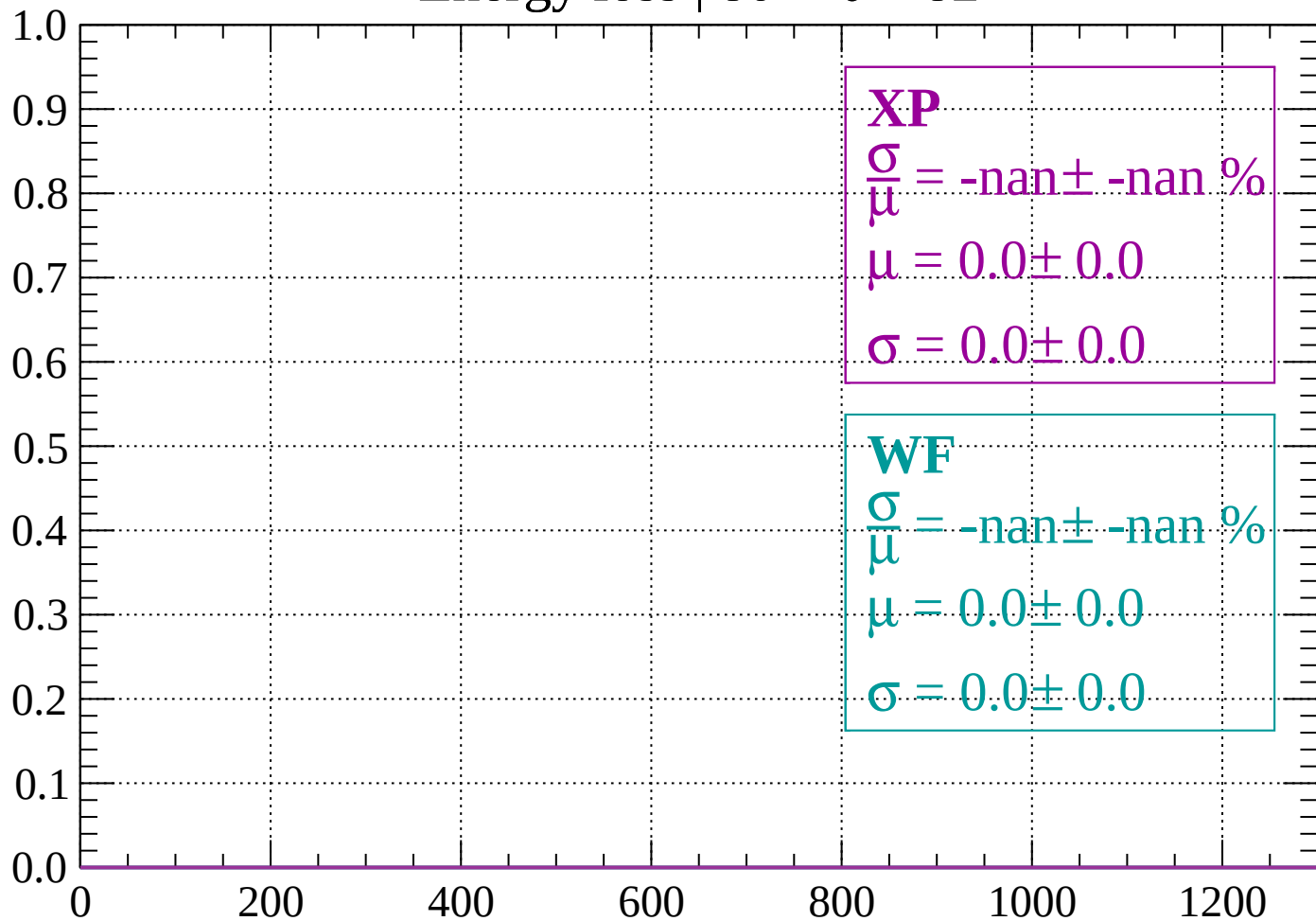
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

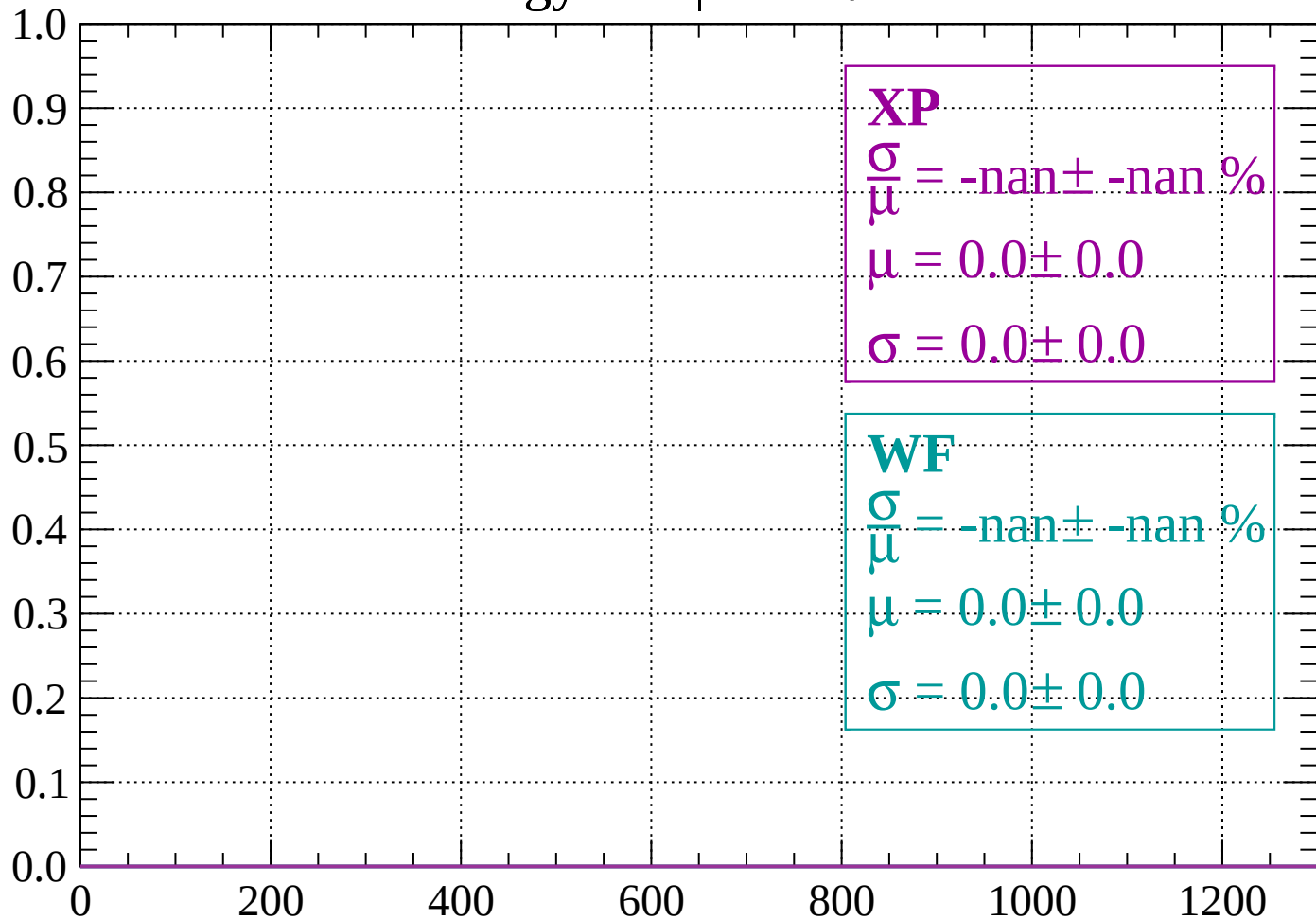
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

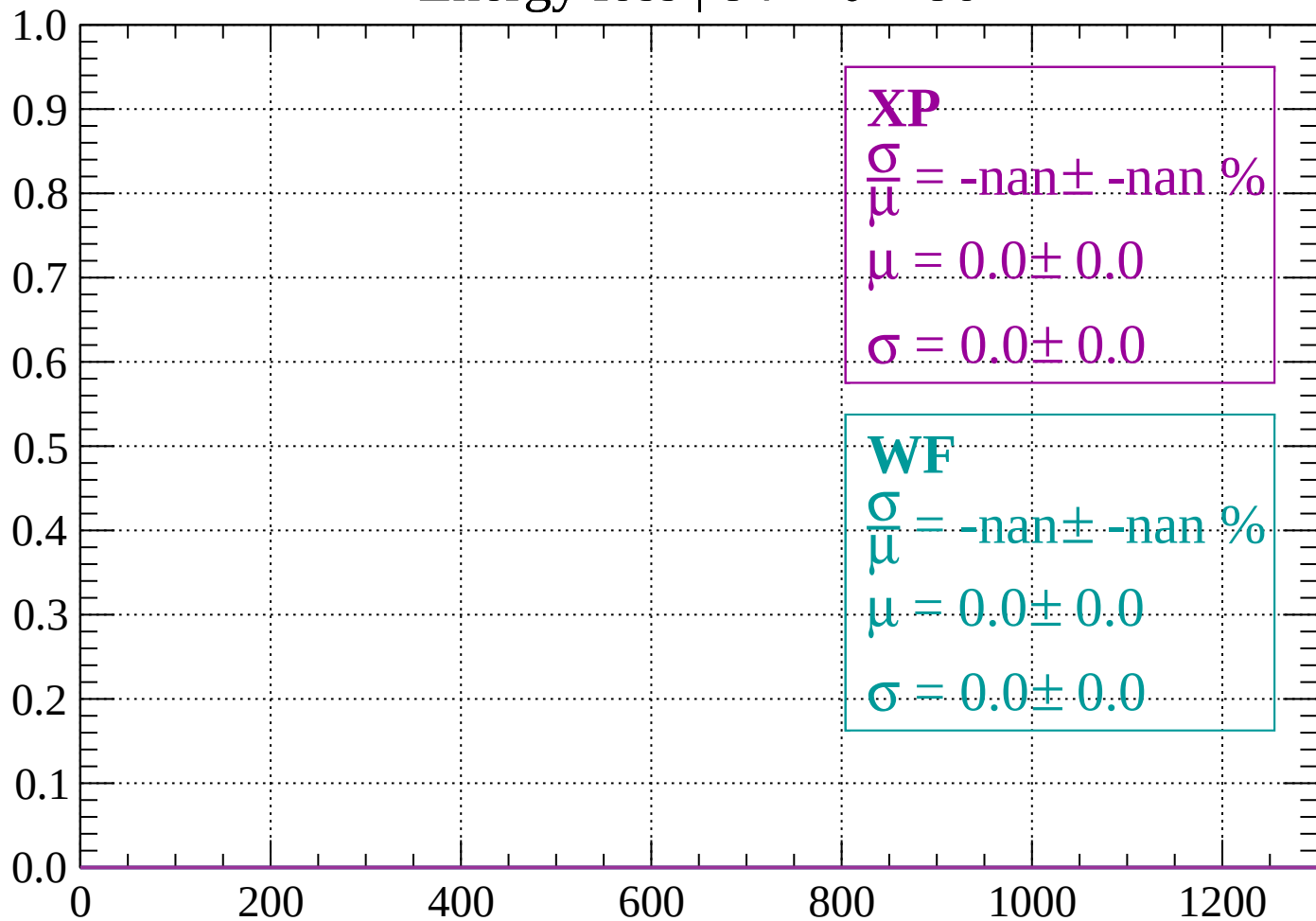
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

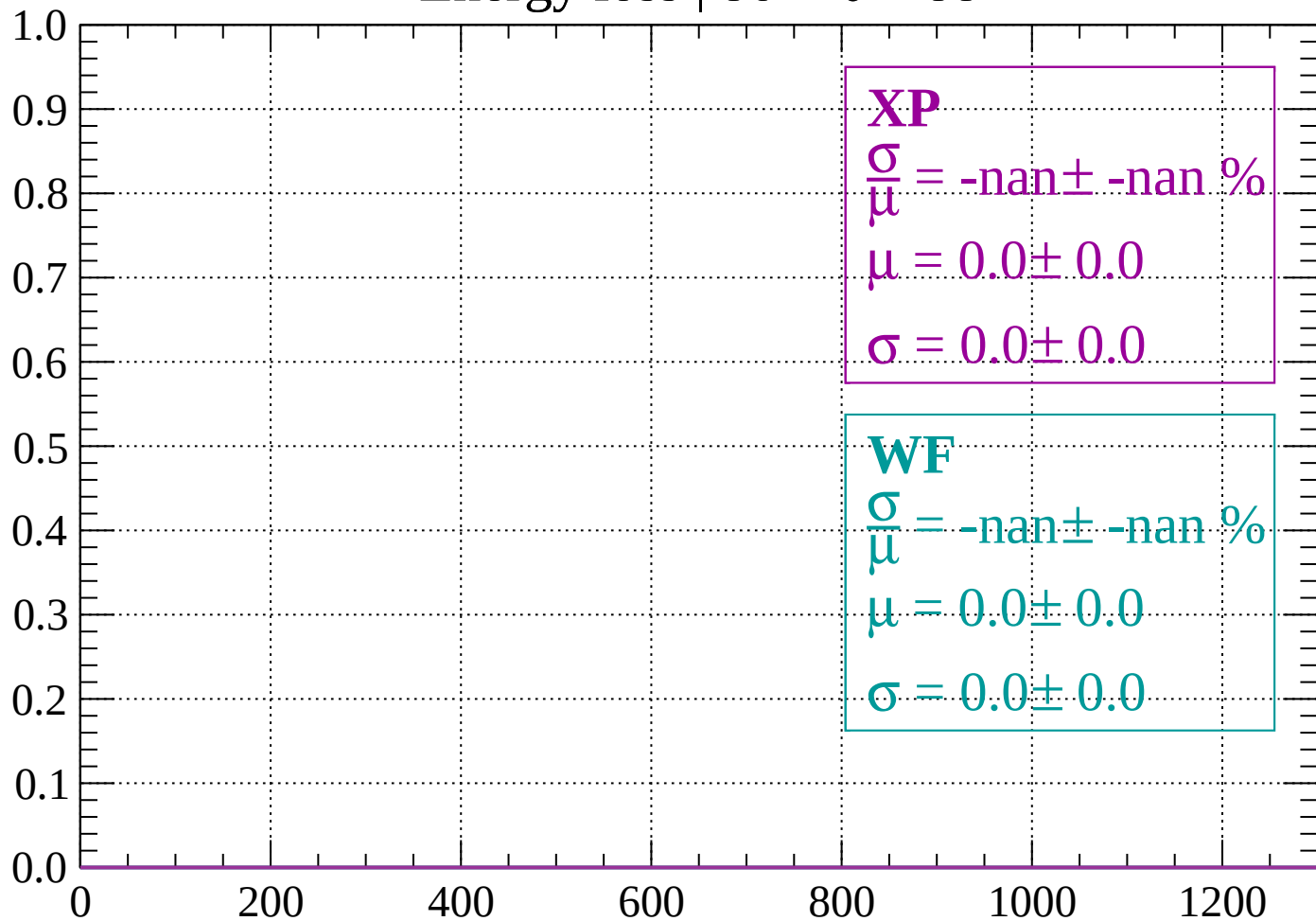
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

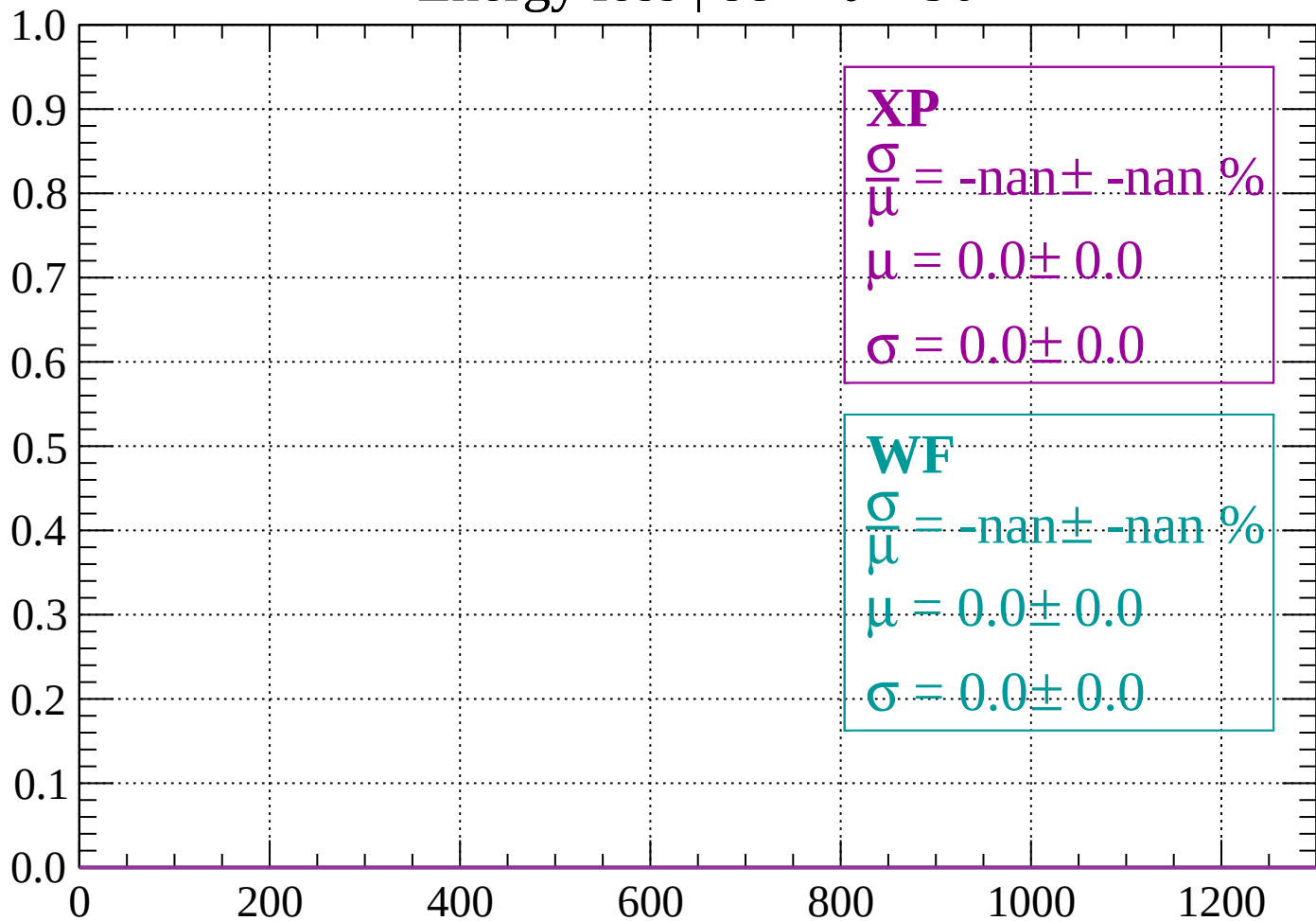
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

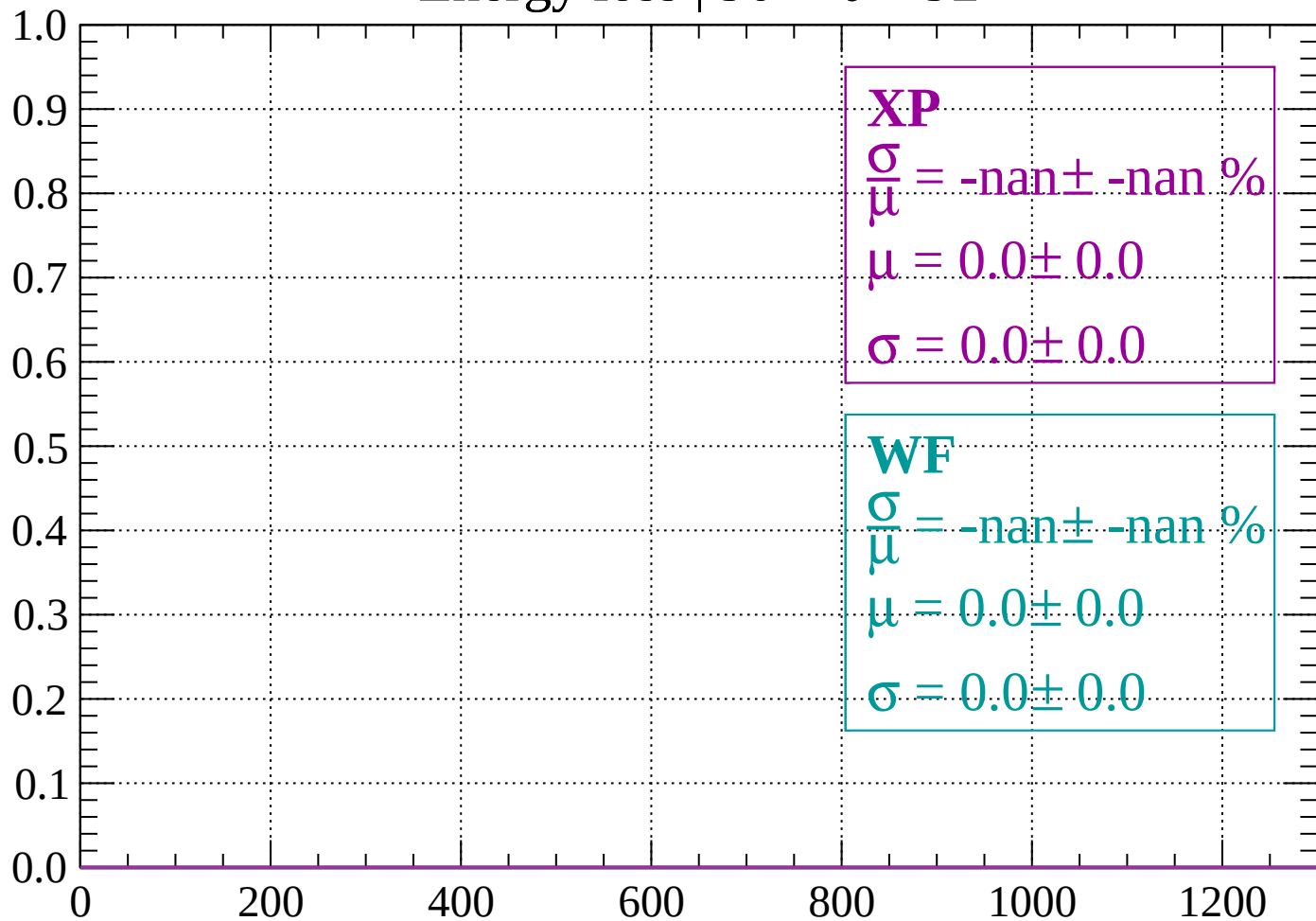
Count



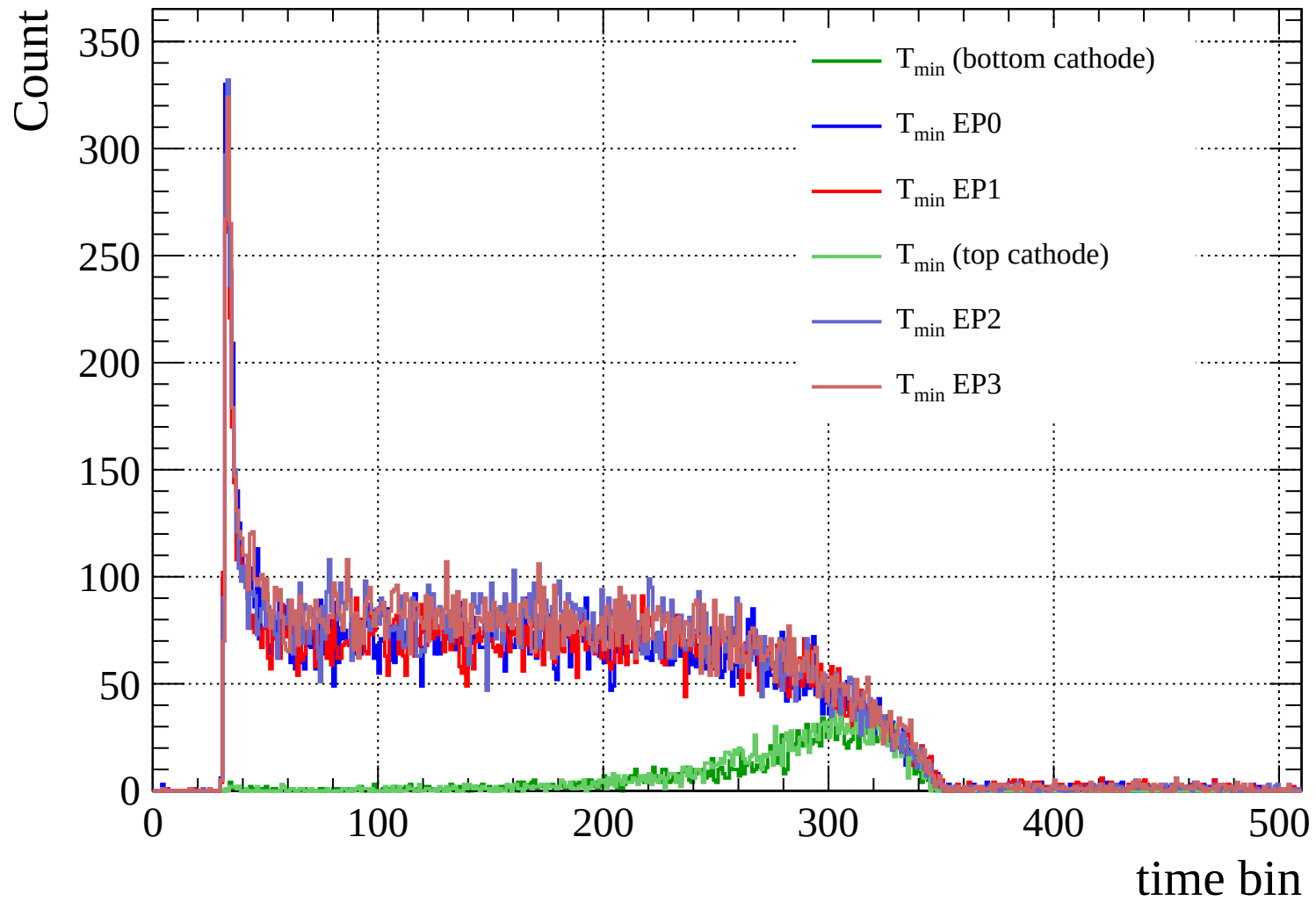
dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)



Count

